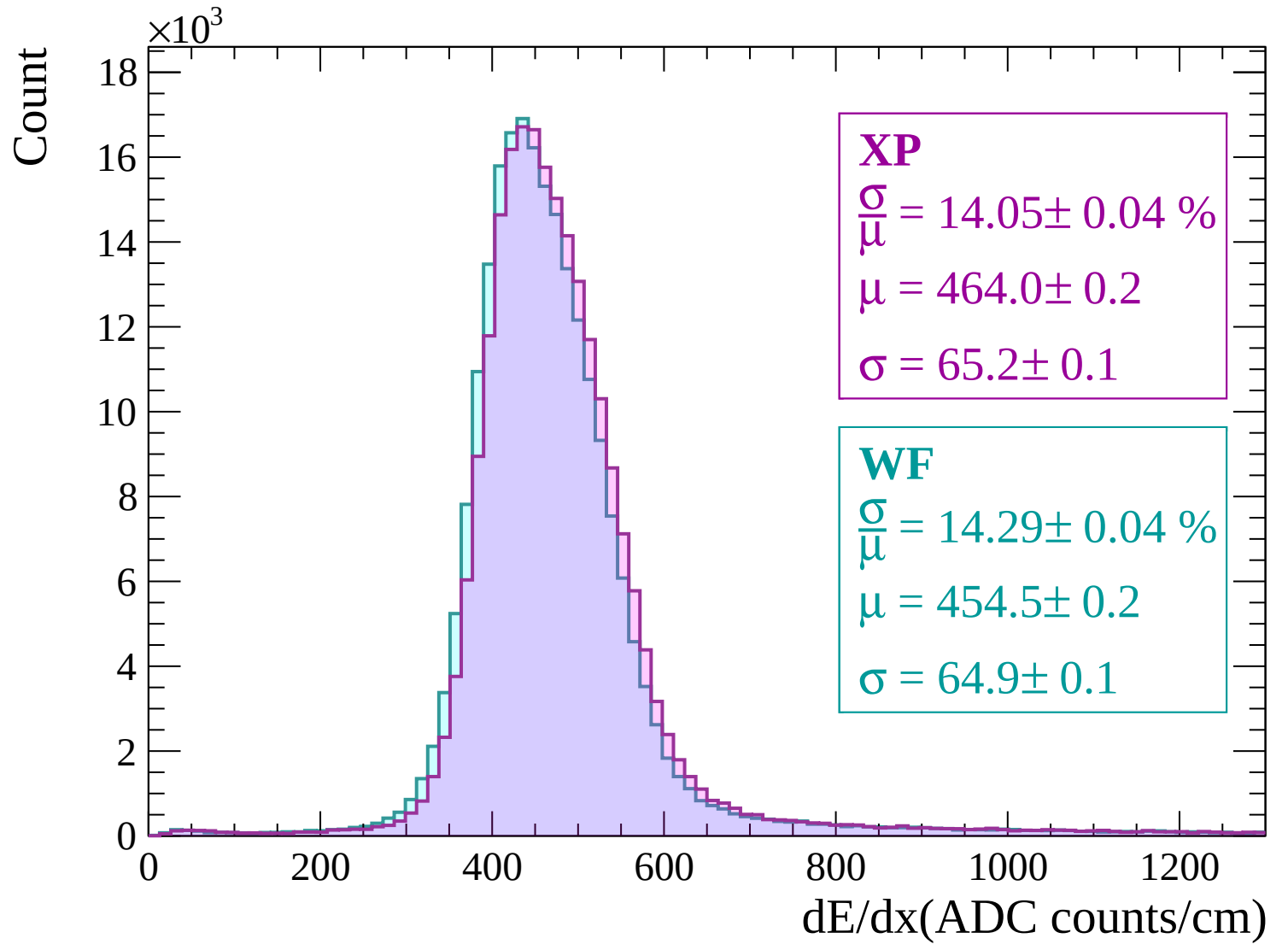
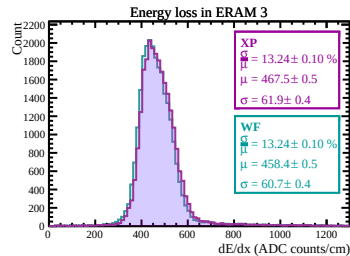
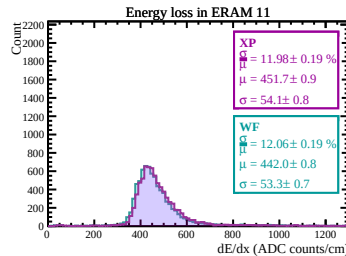
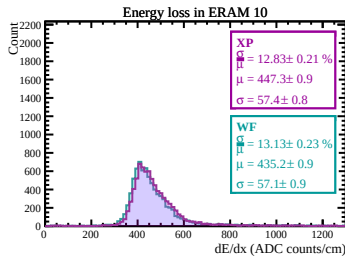
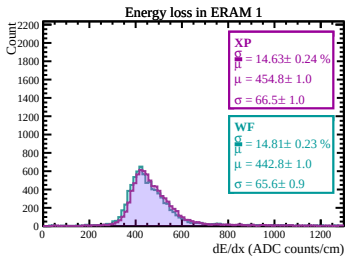
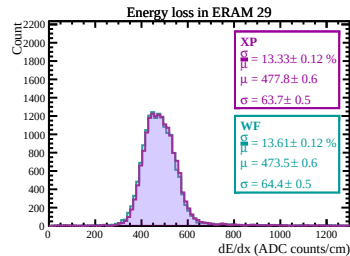
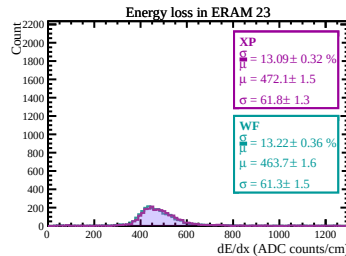
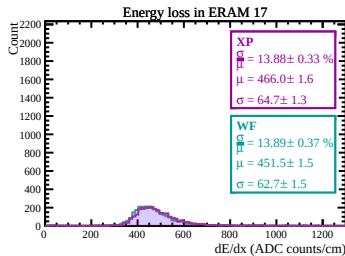
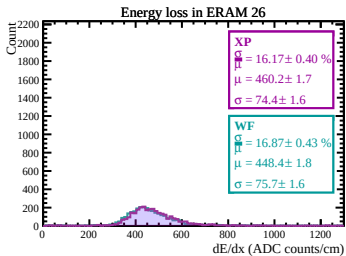
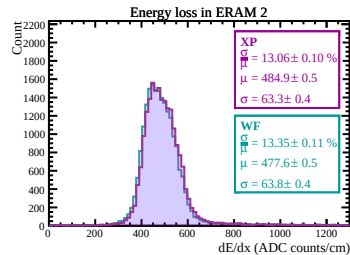
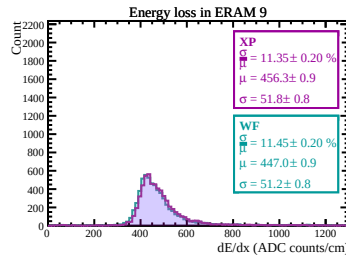
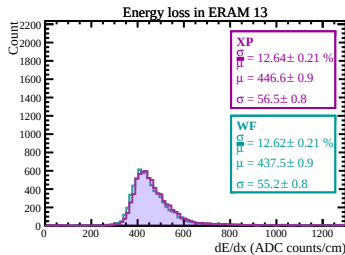
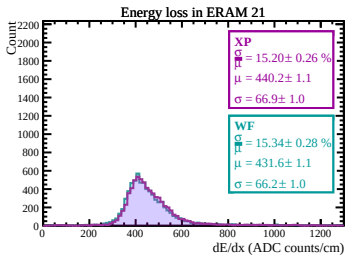
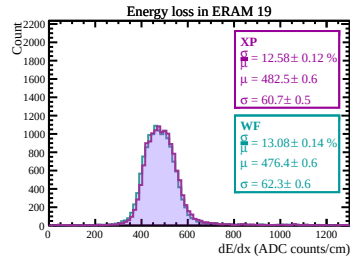
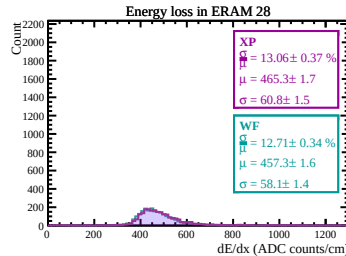
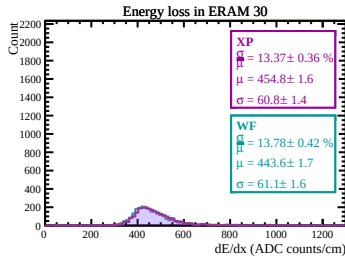
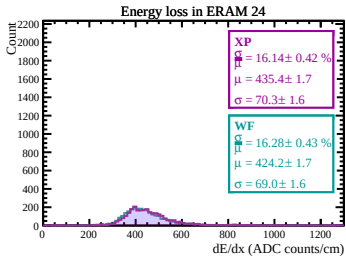
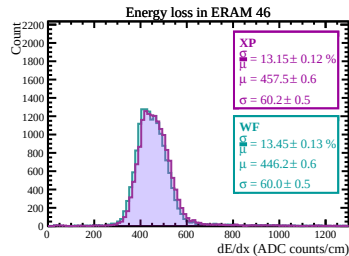
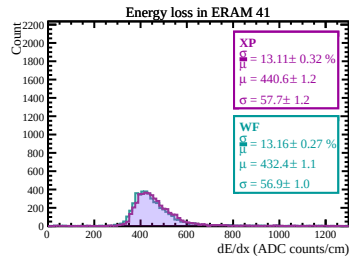
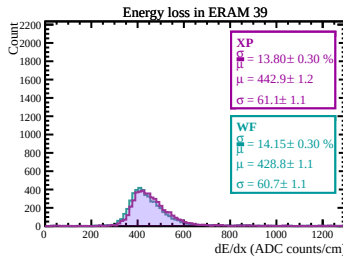
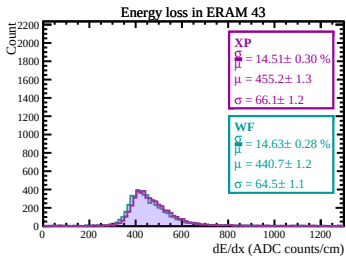
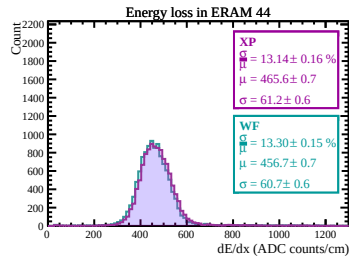
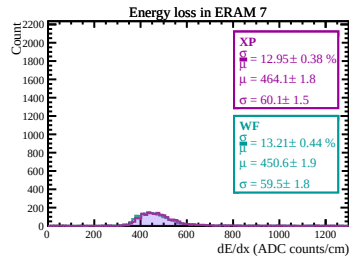
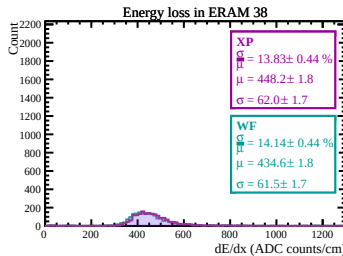
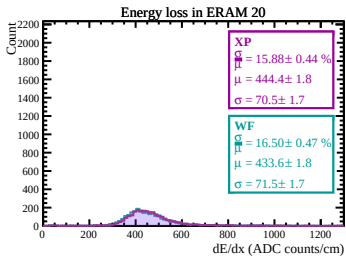
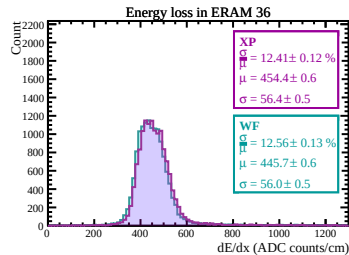
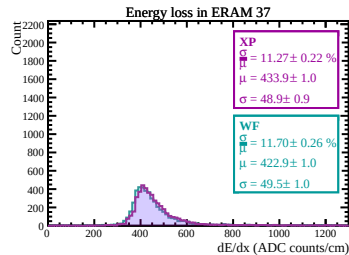
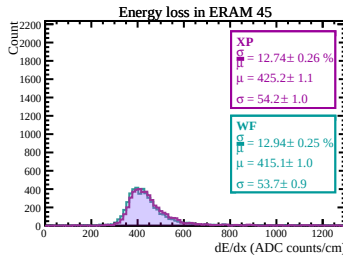
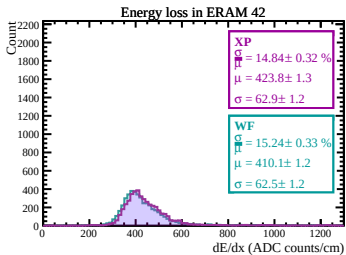
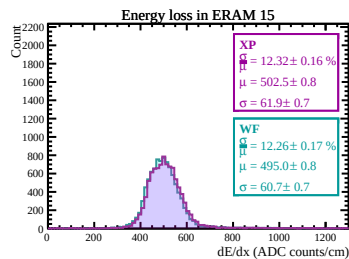
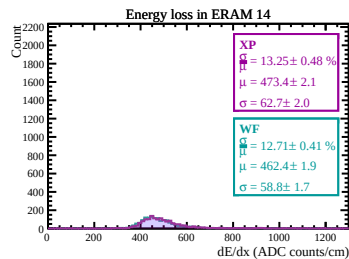
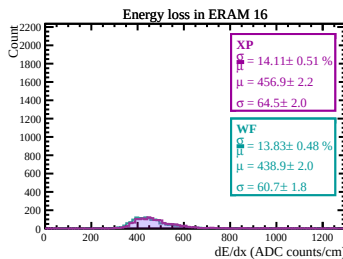
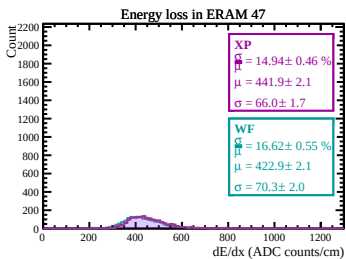
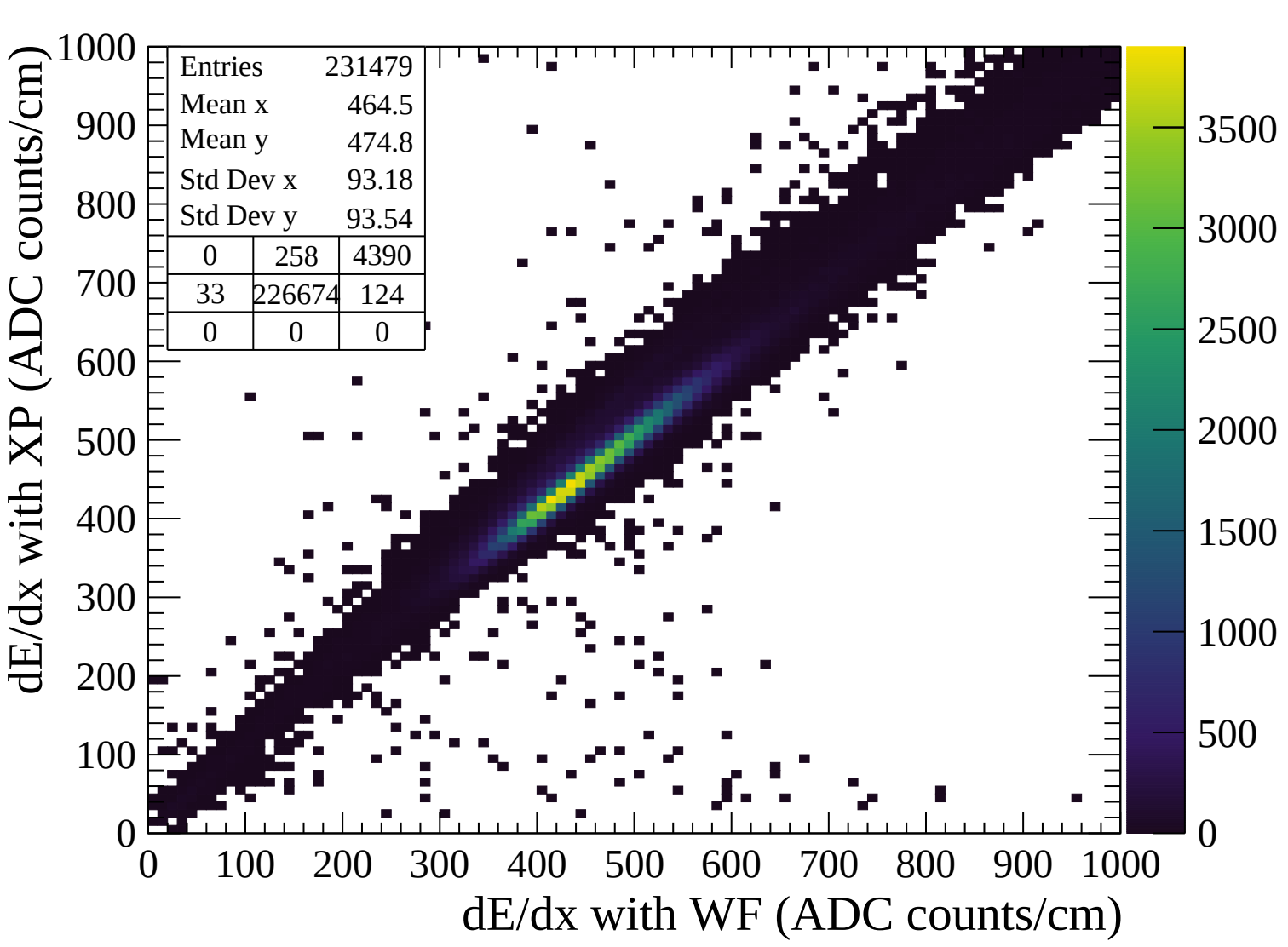
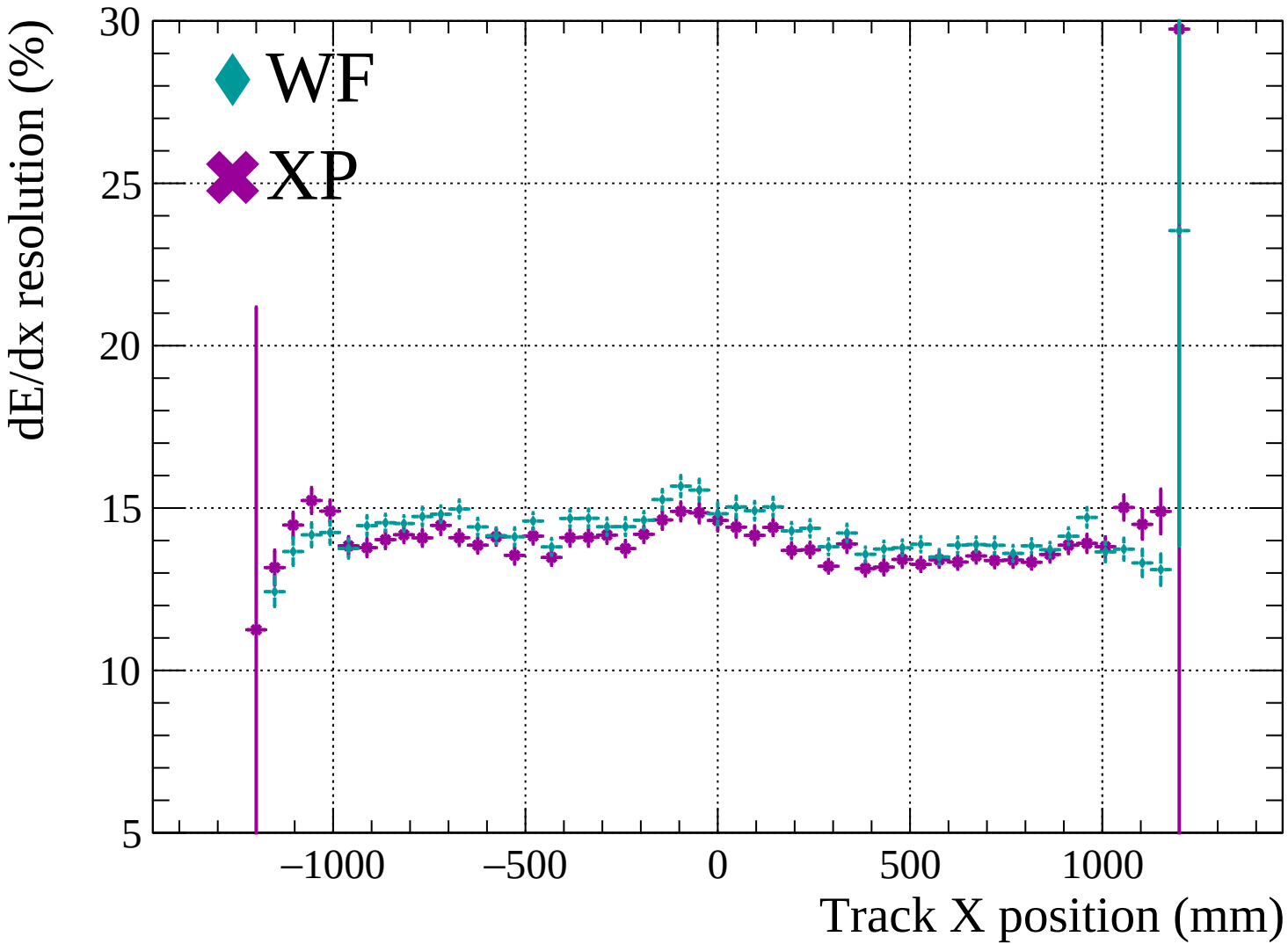


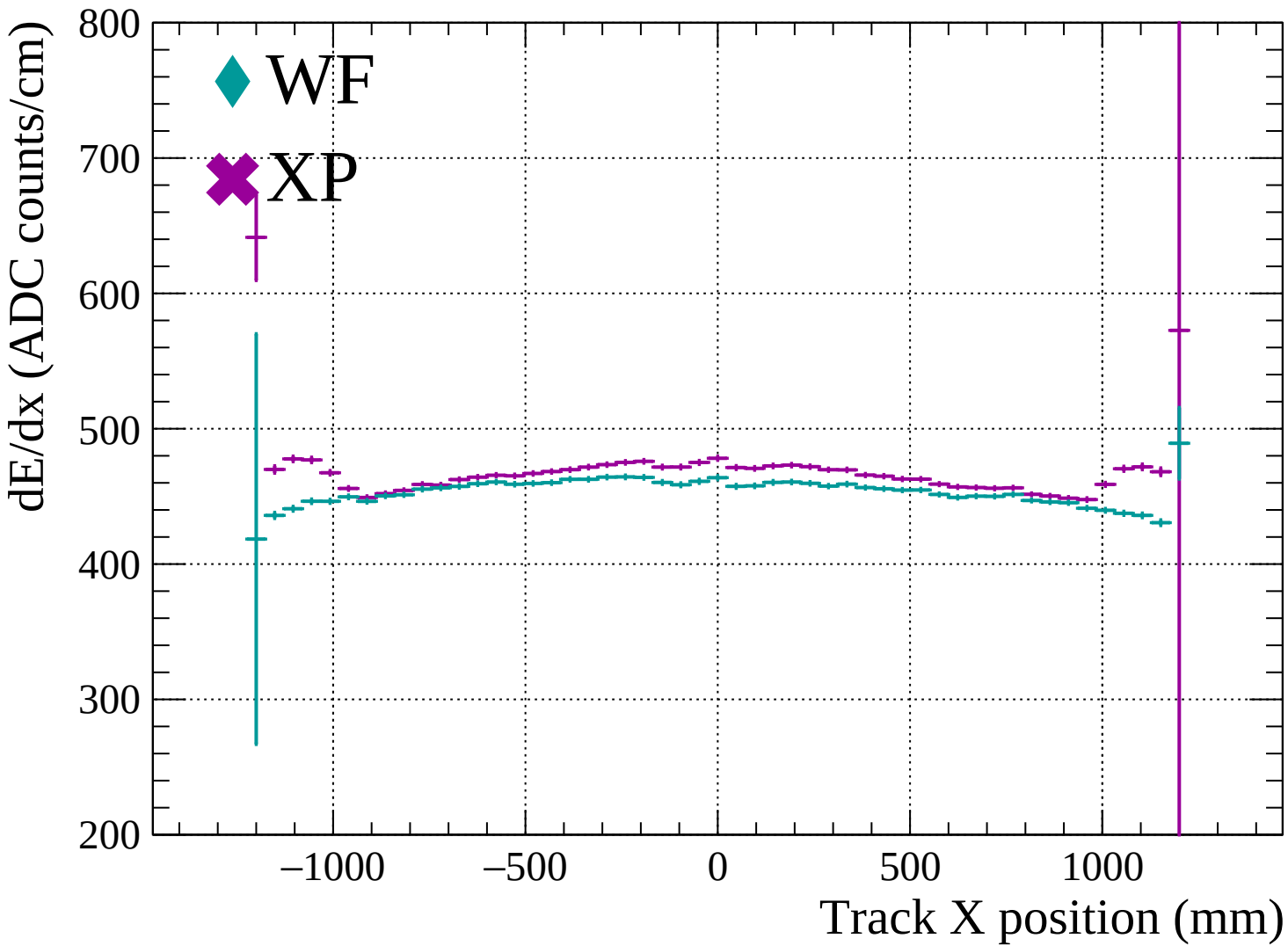


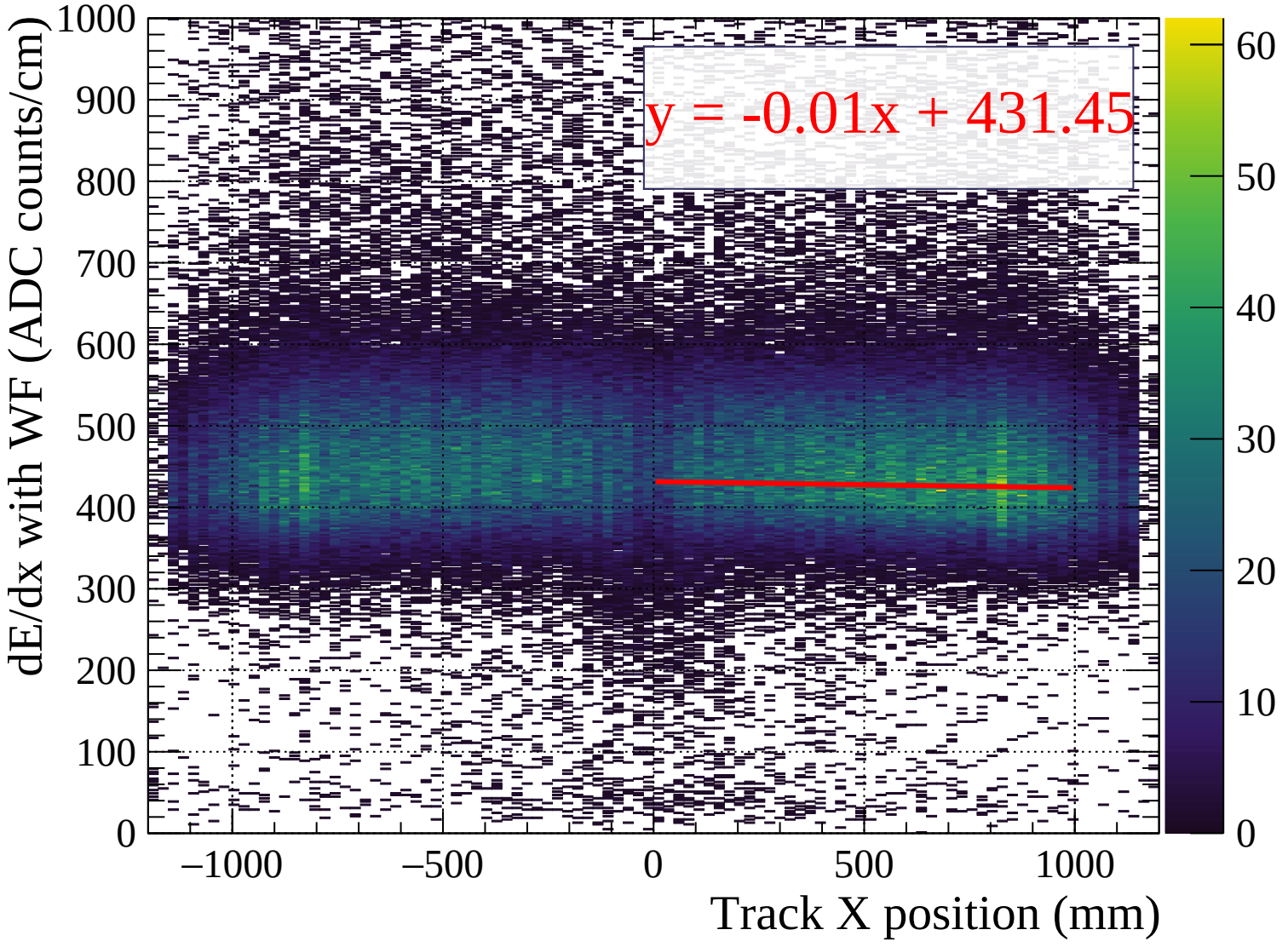


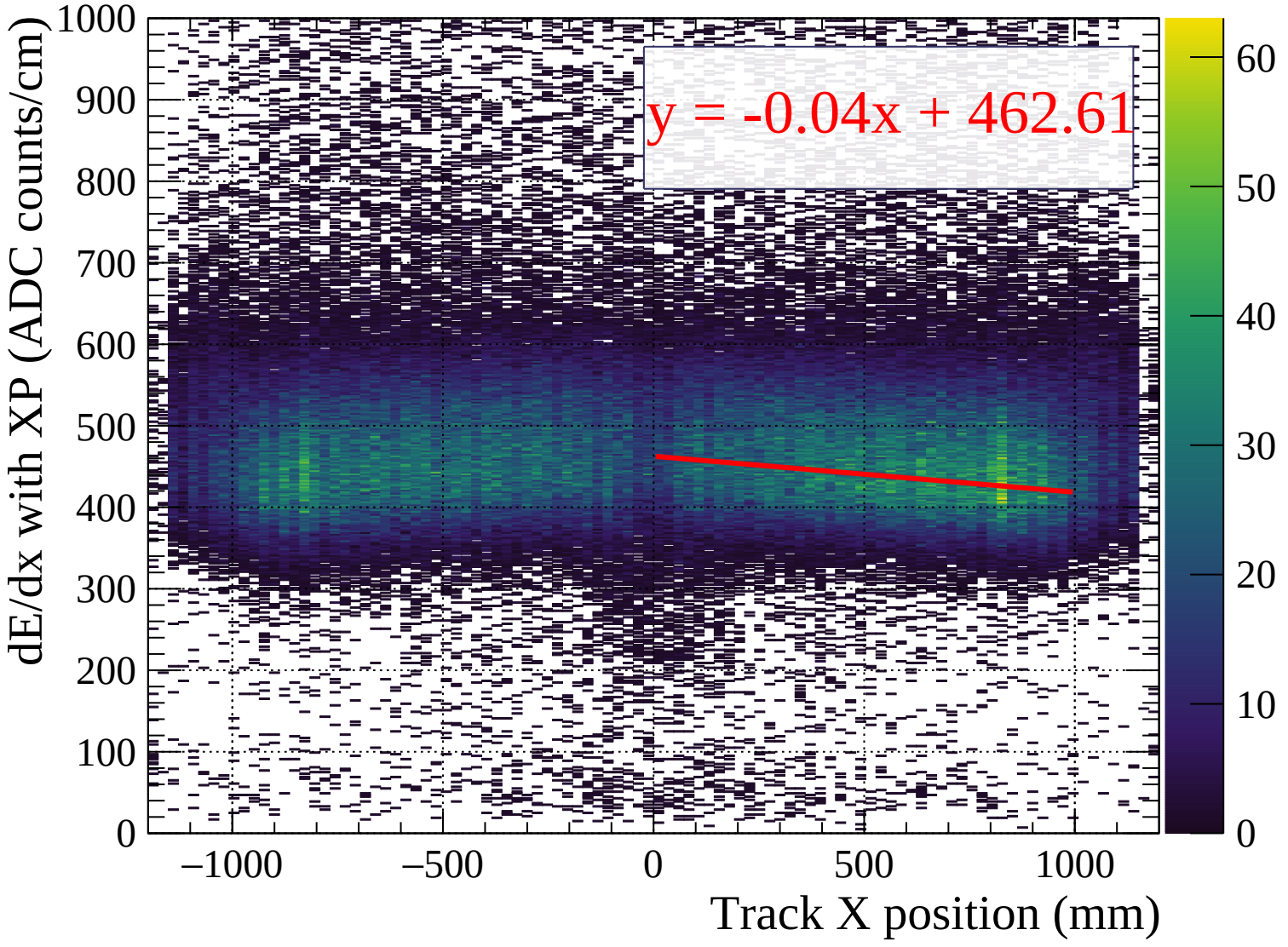


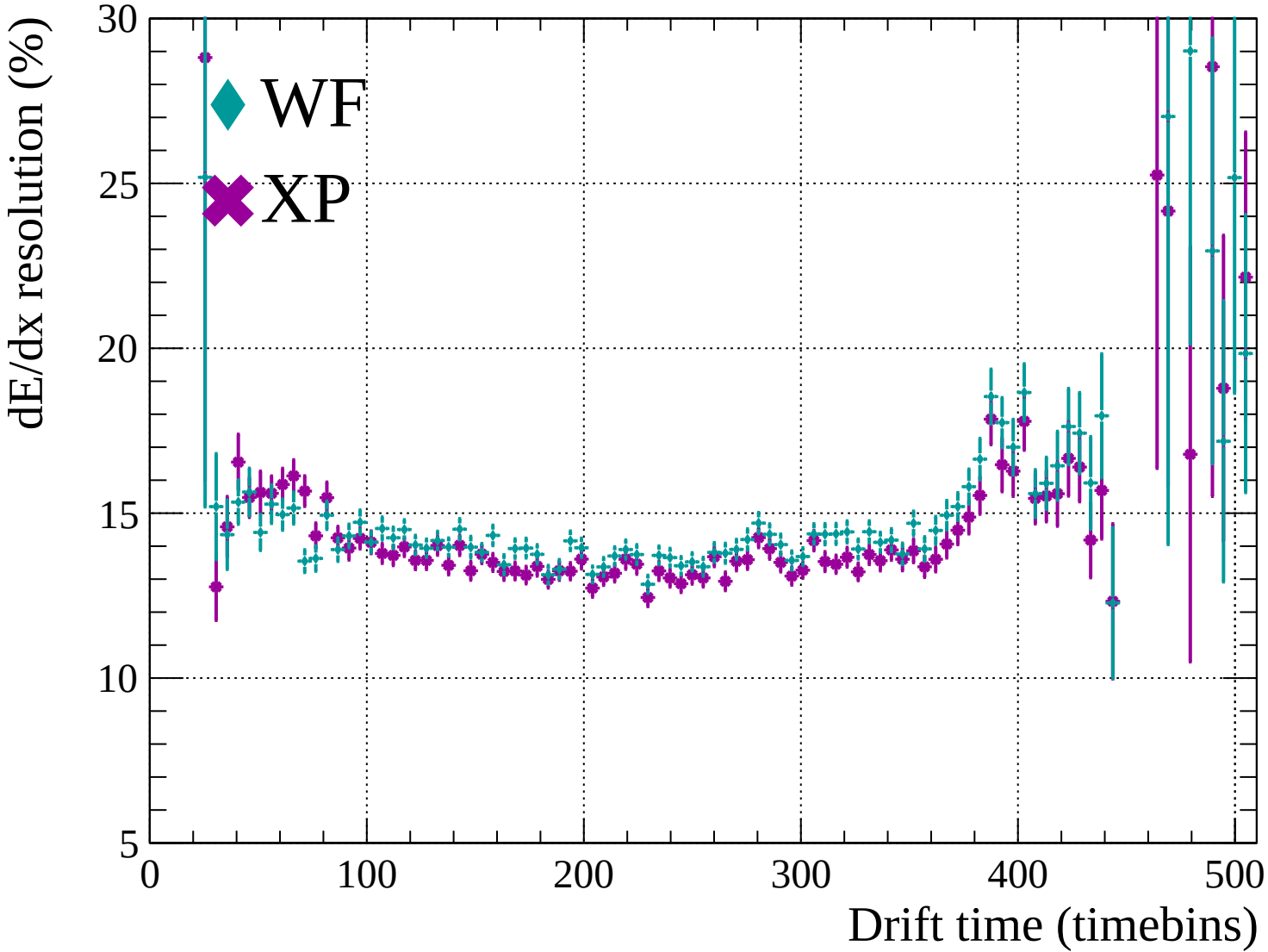


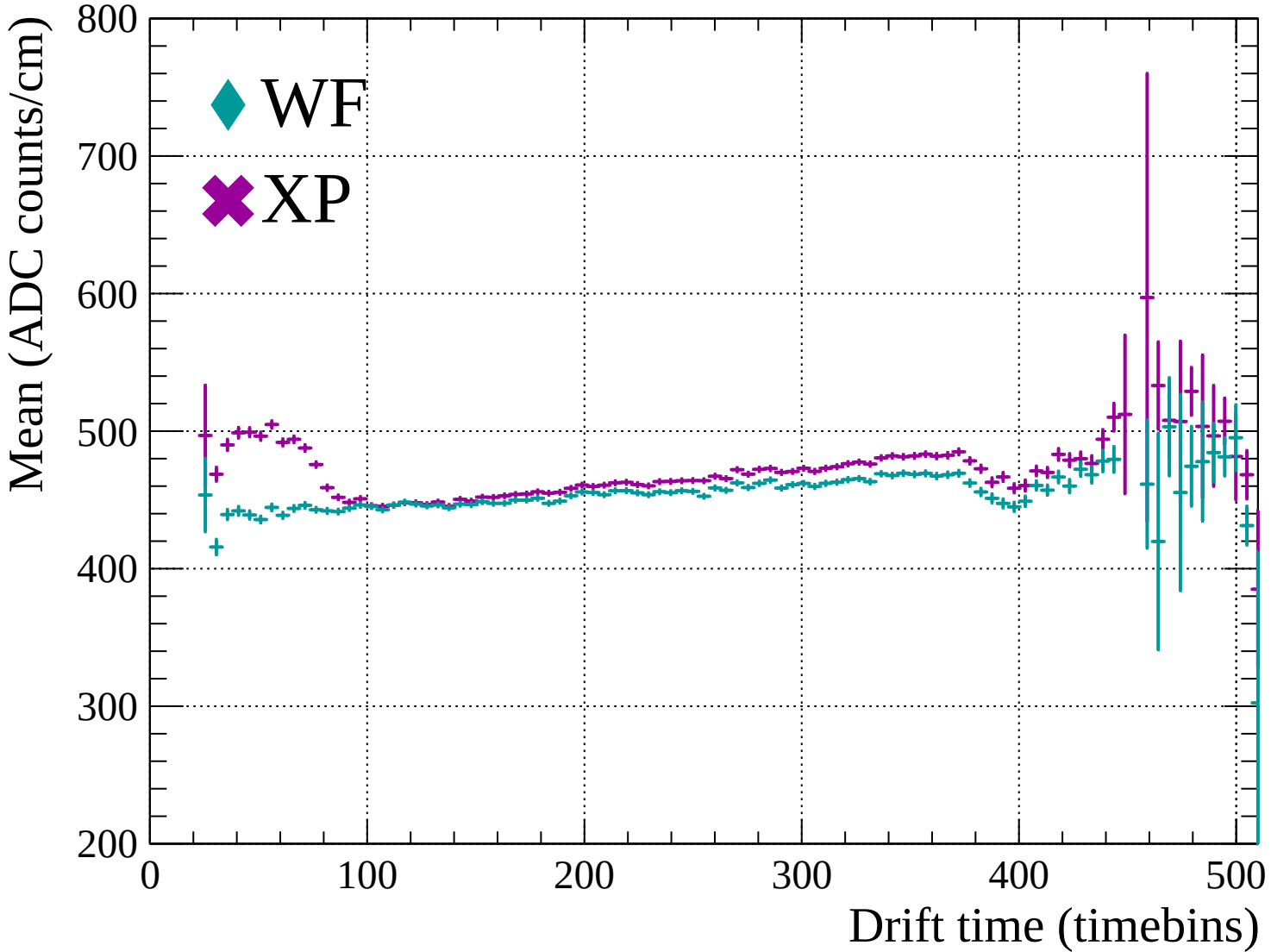


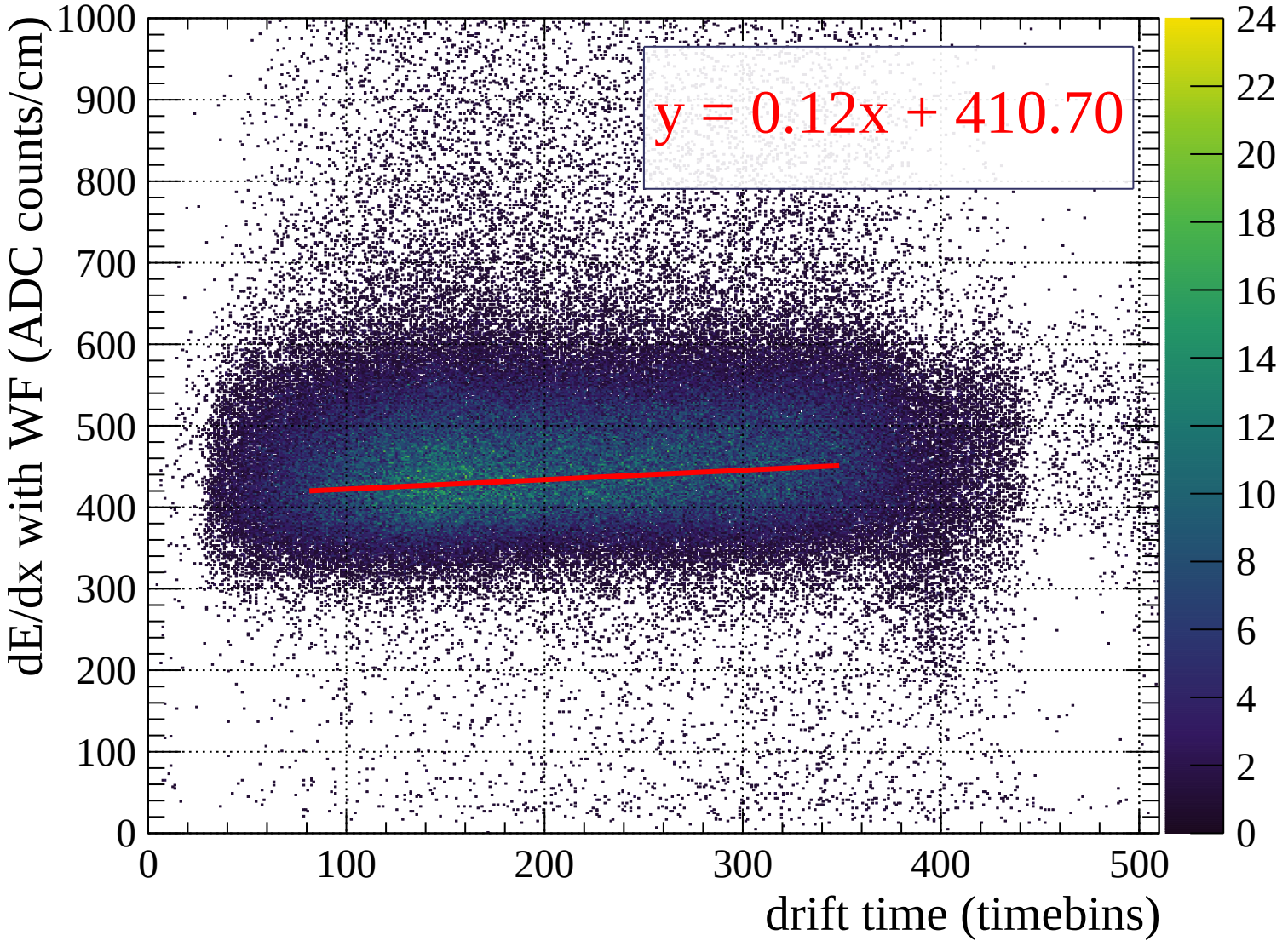


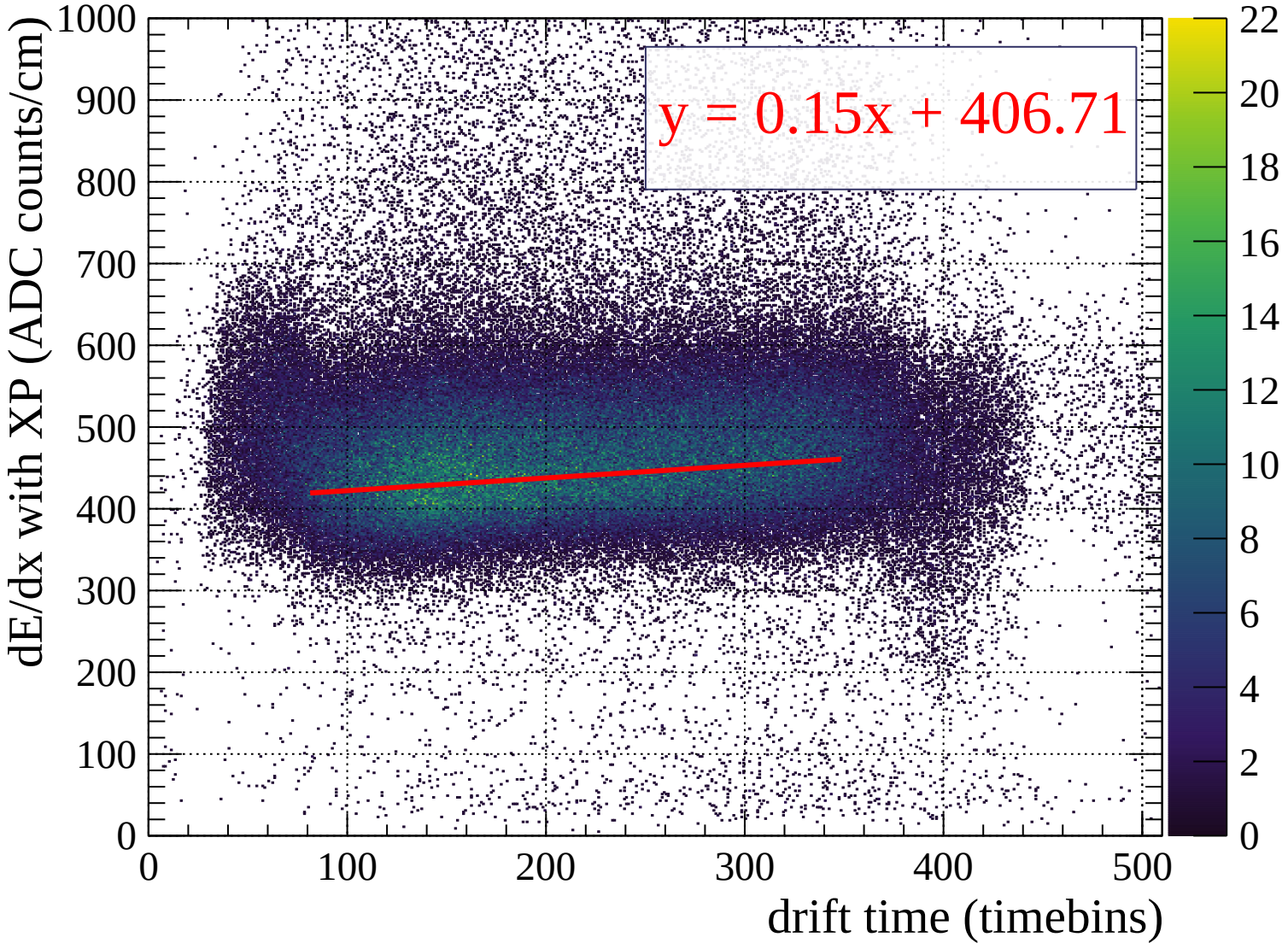


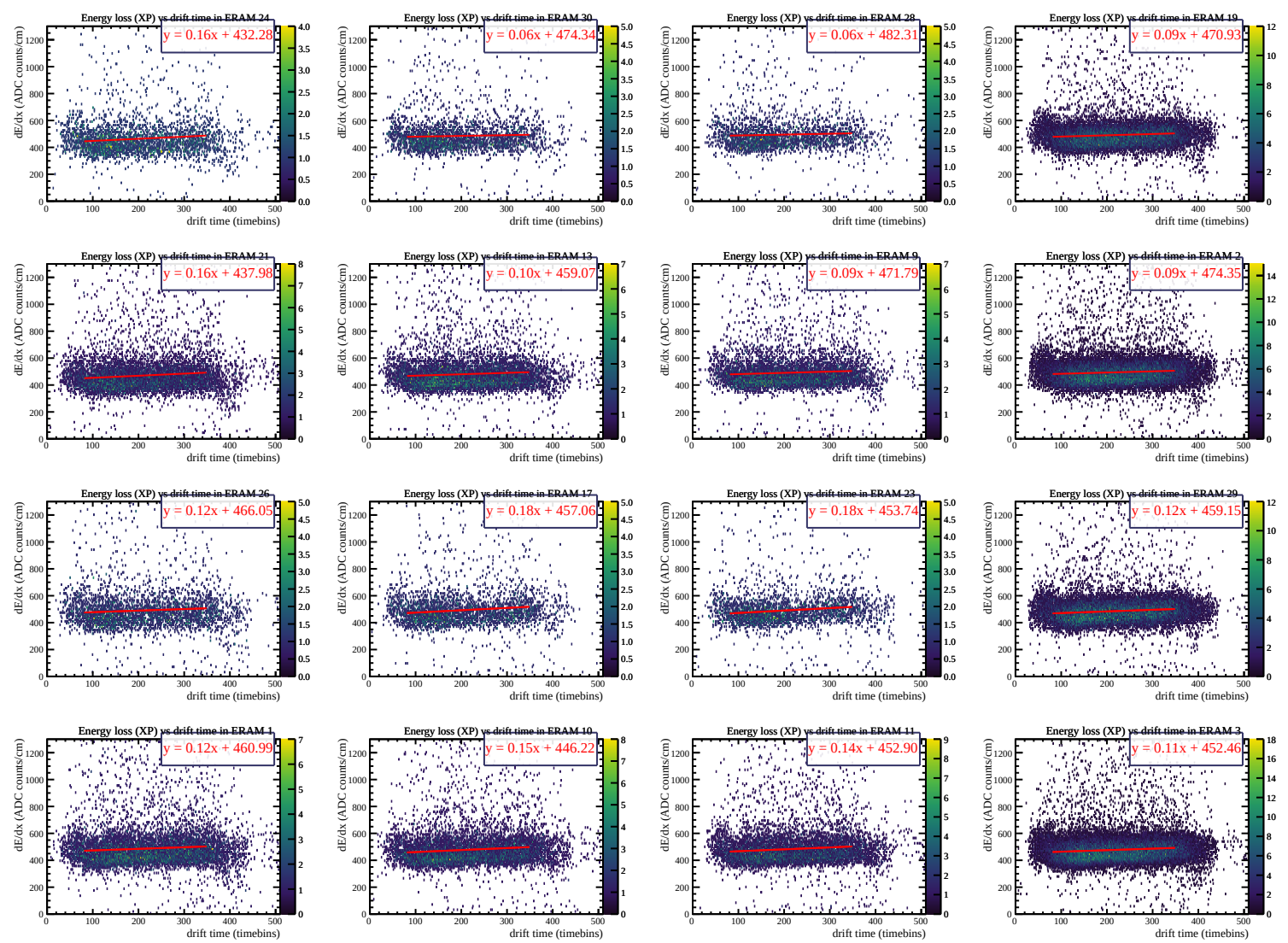


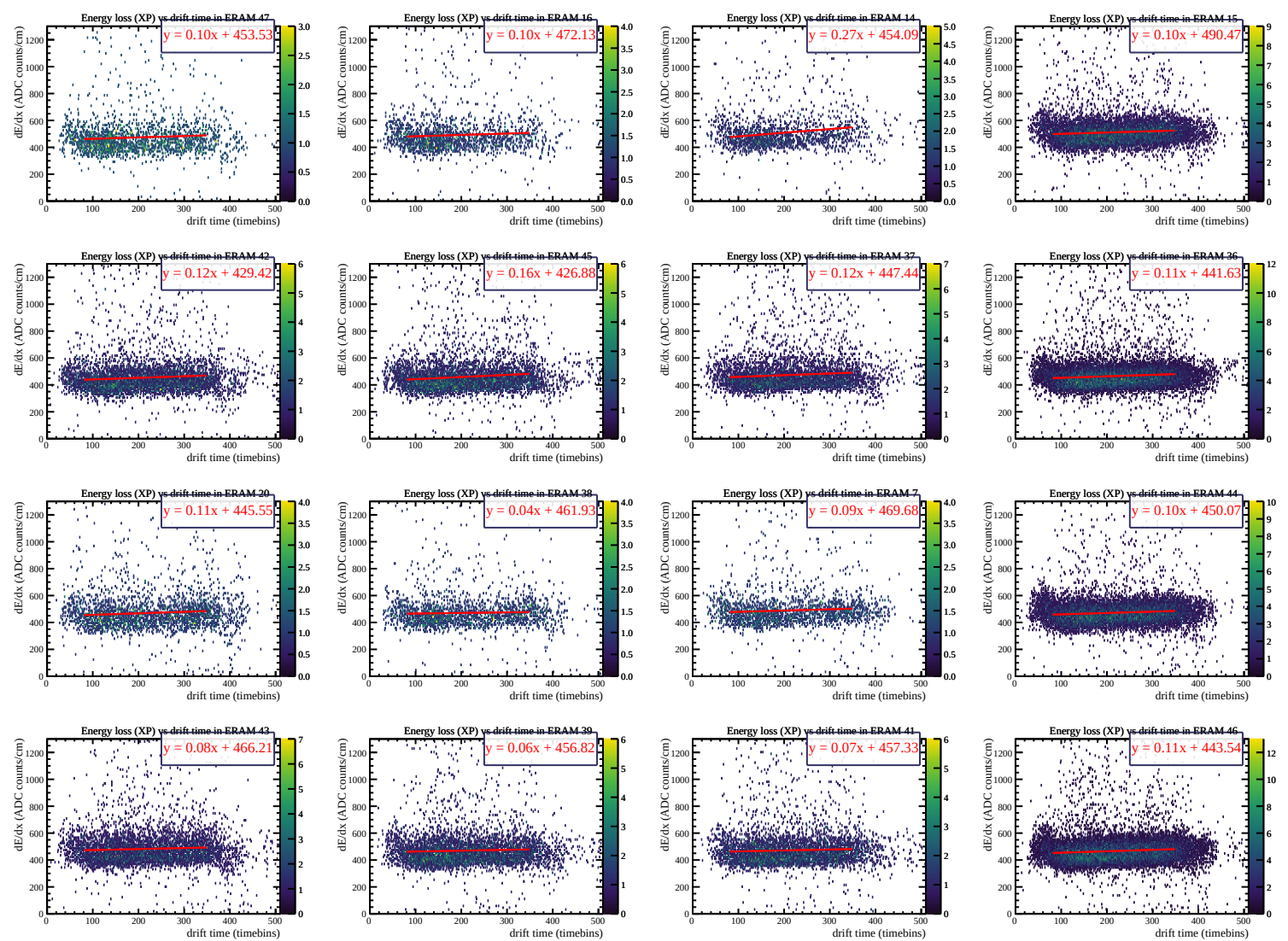


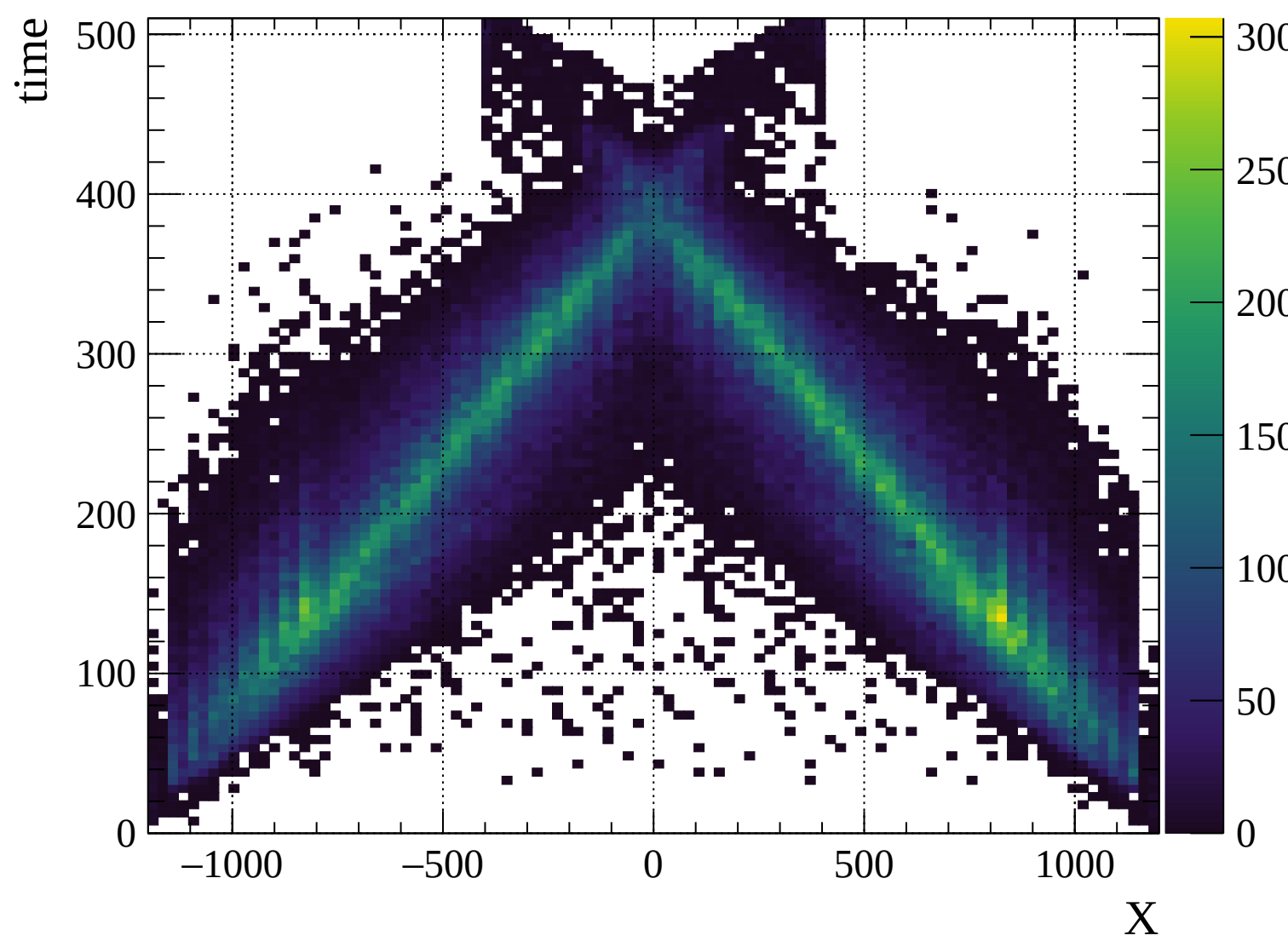


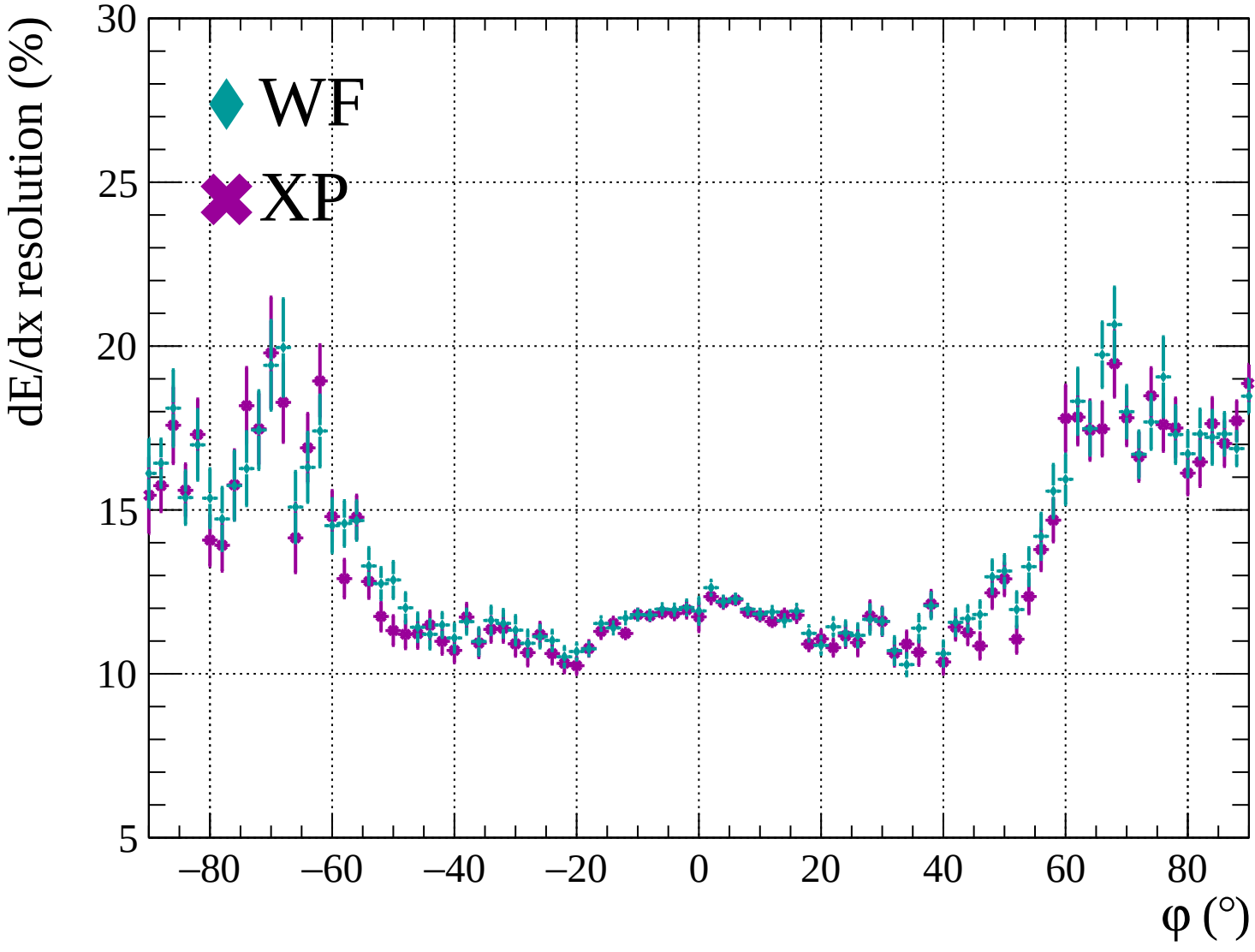


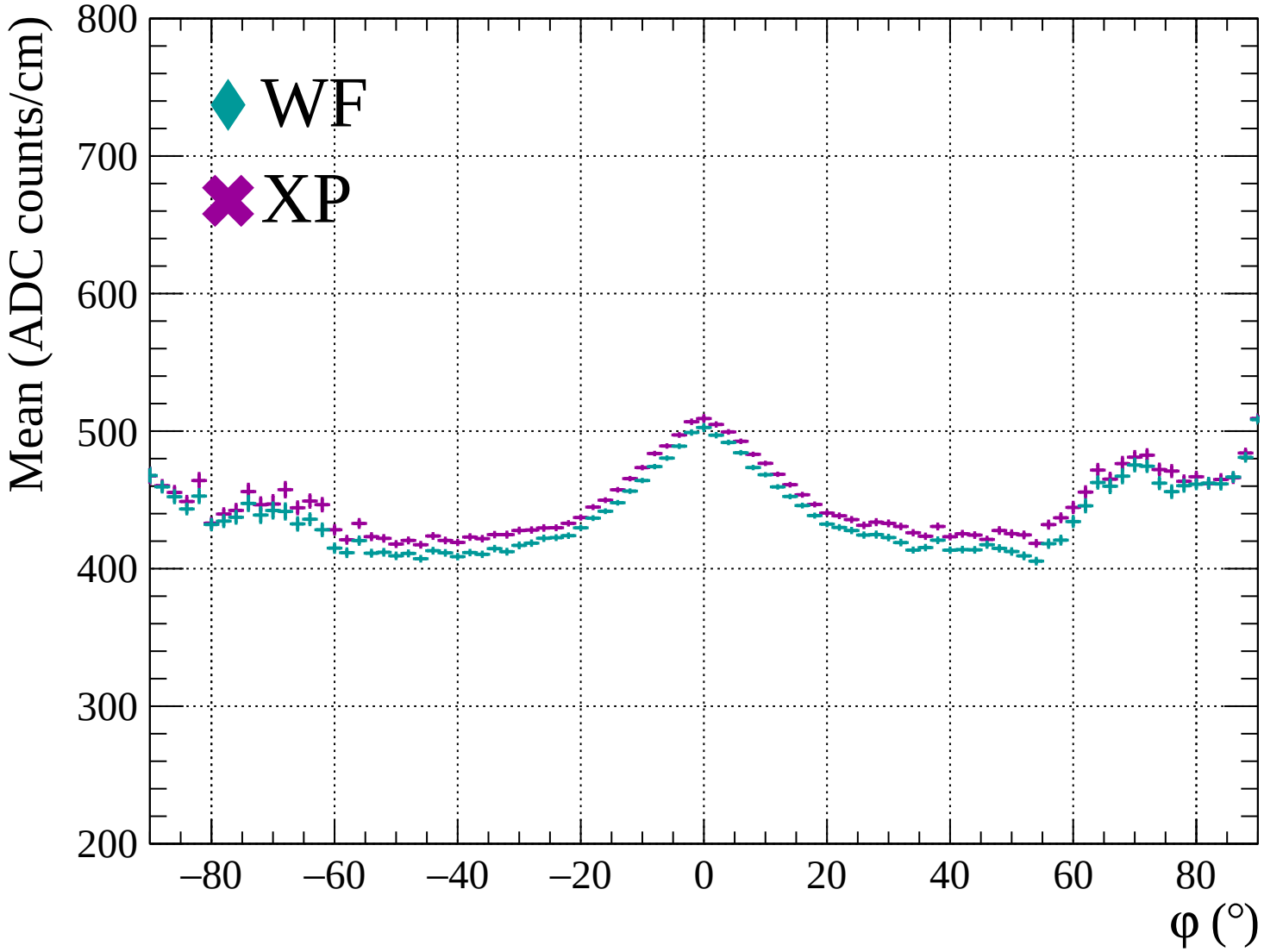


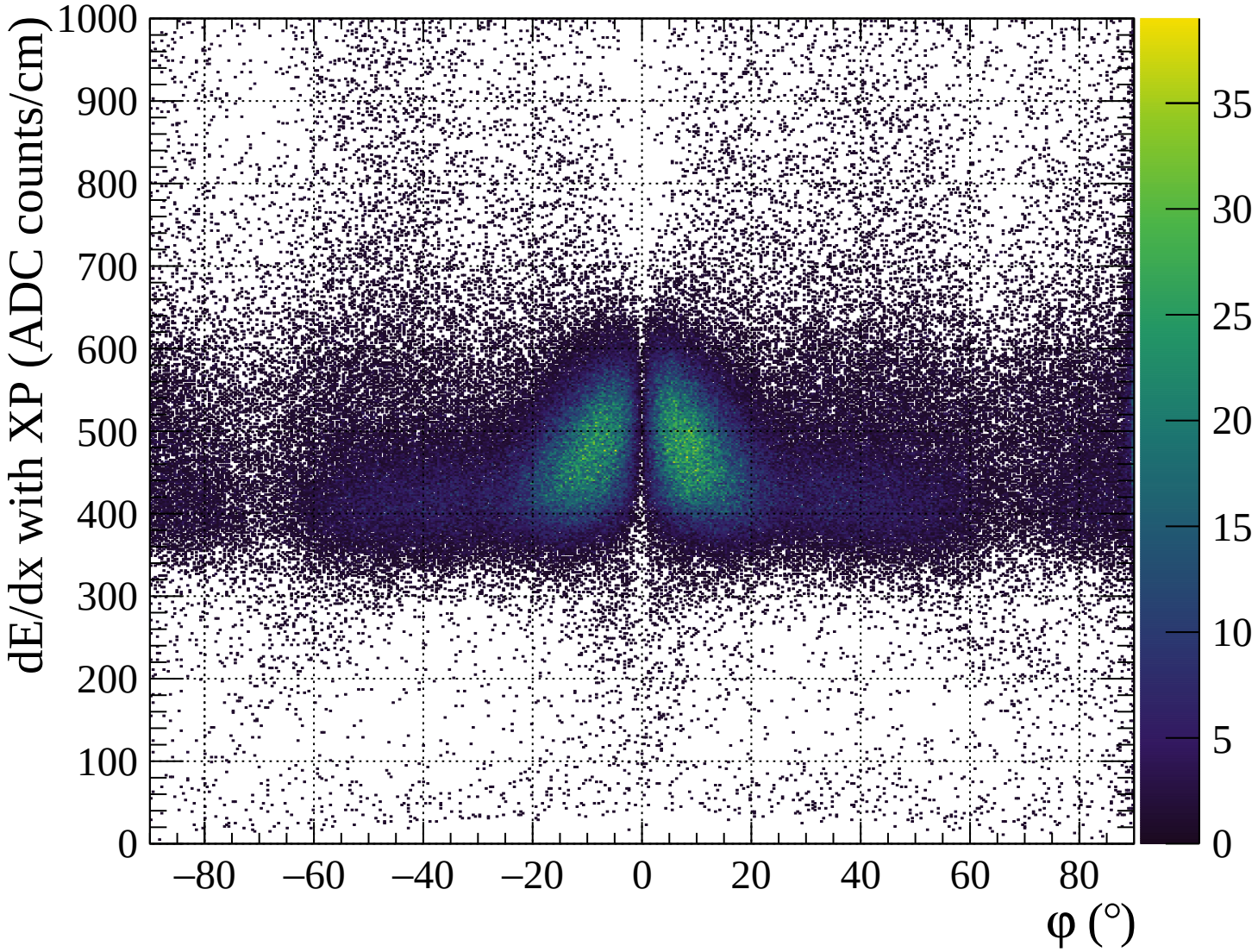


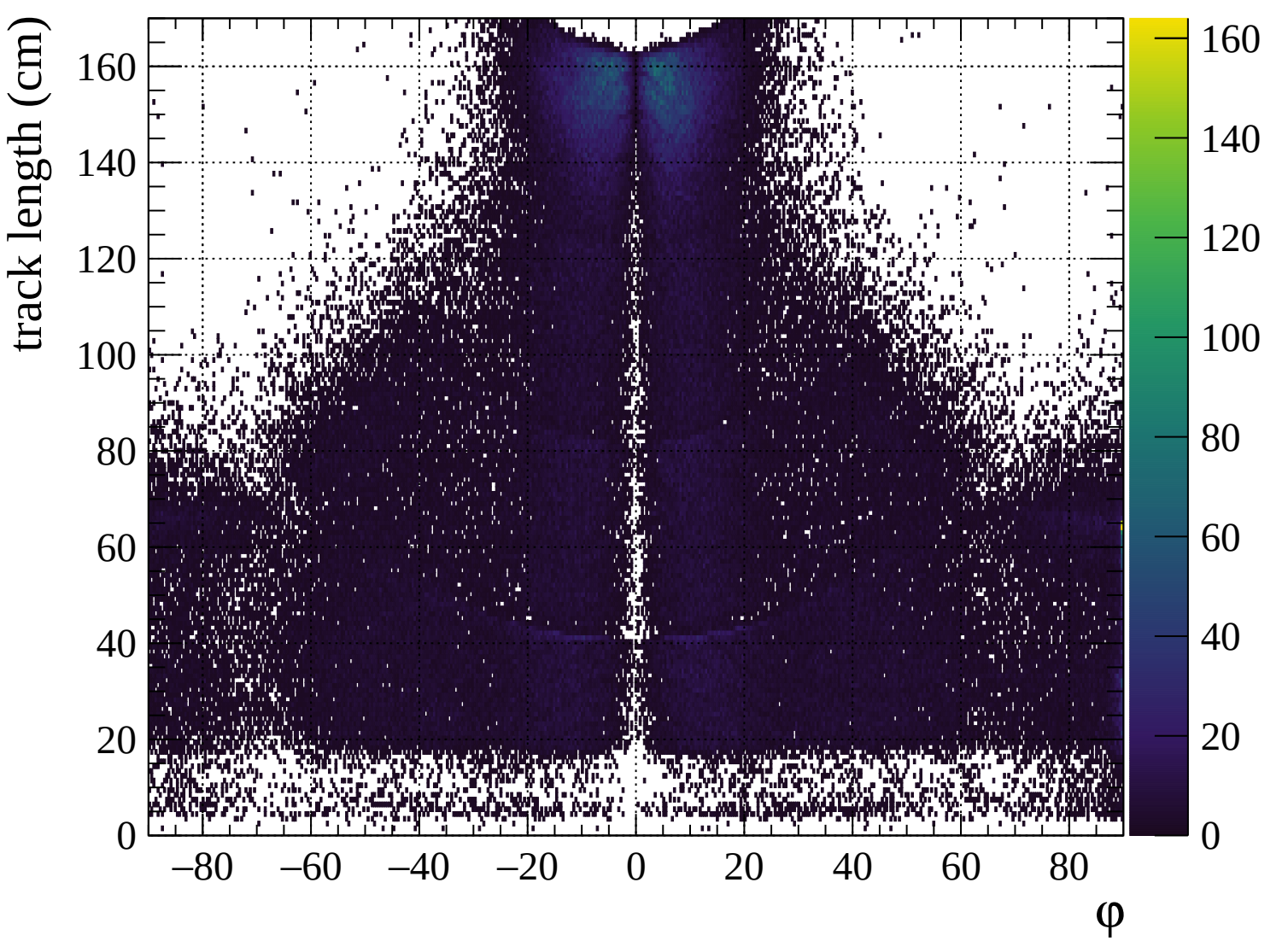


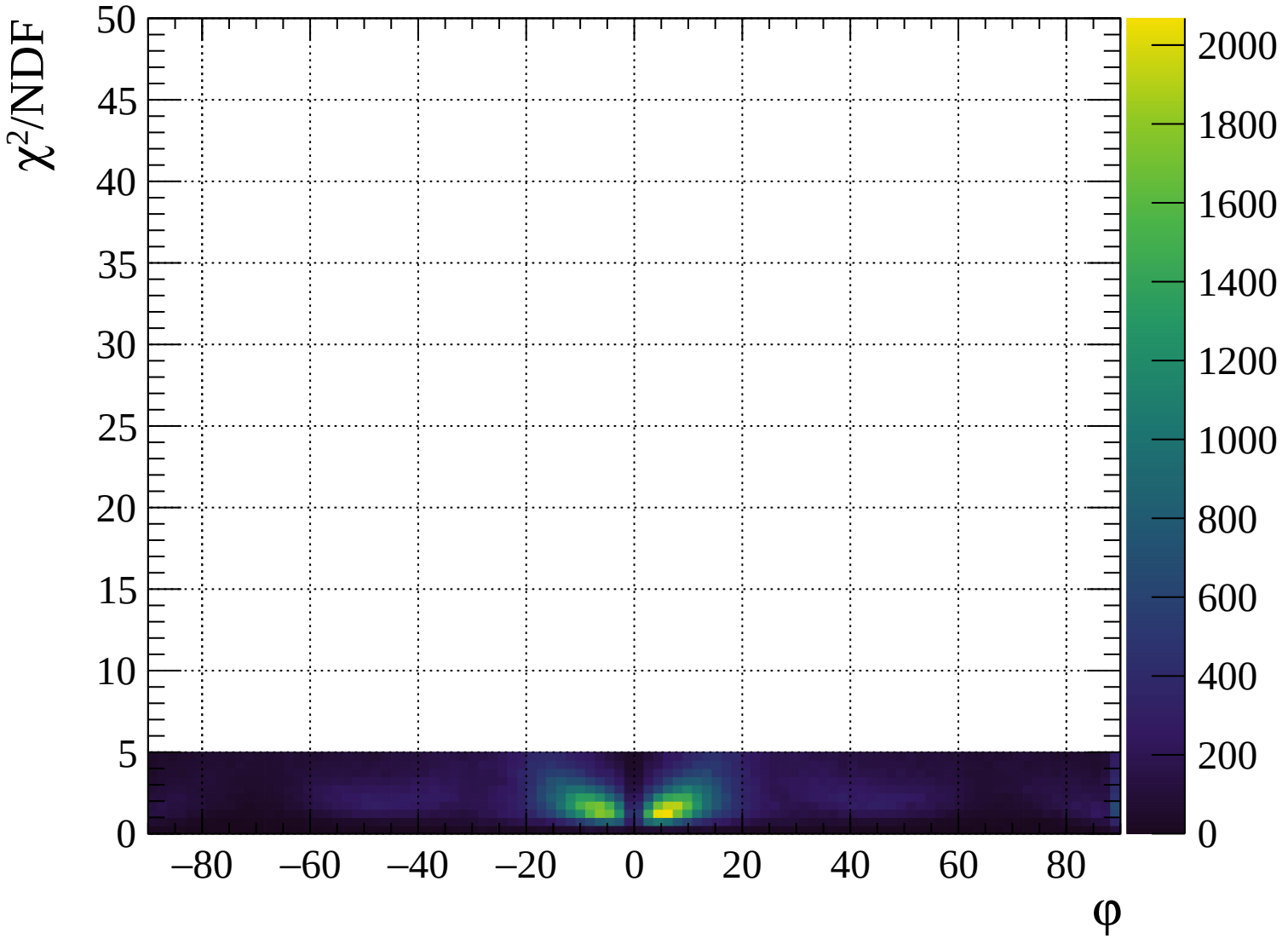


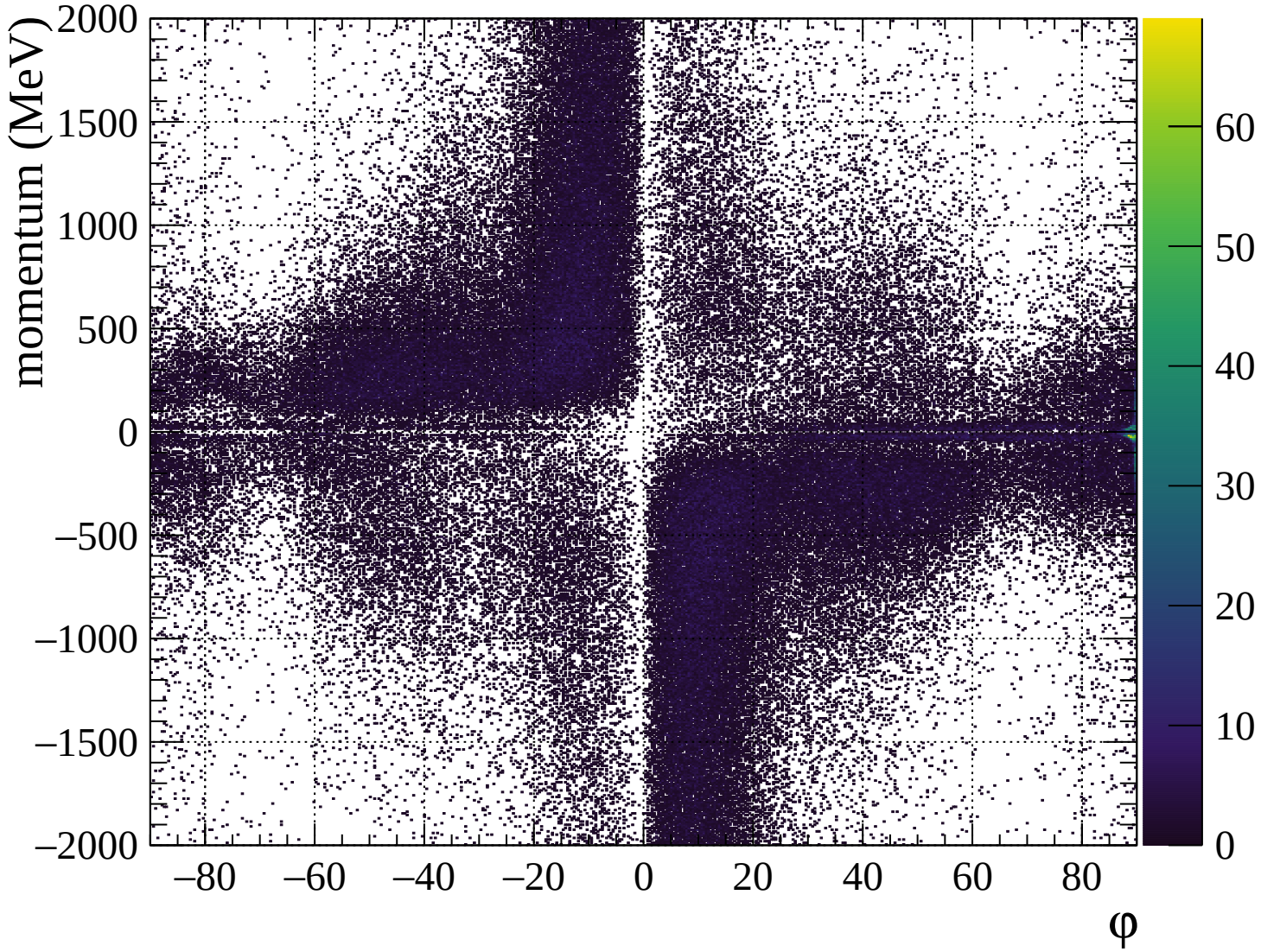


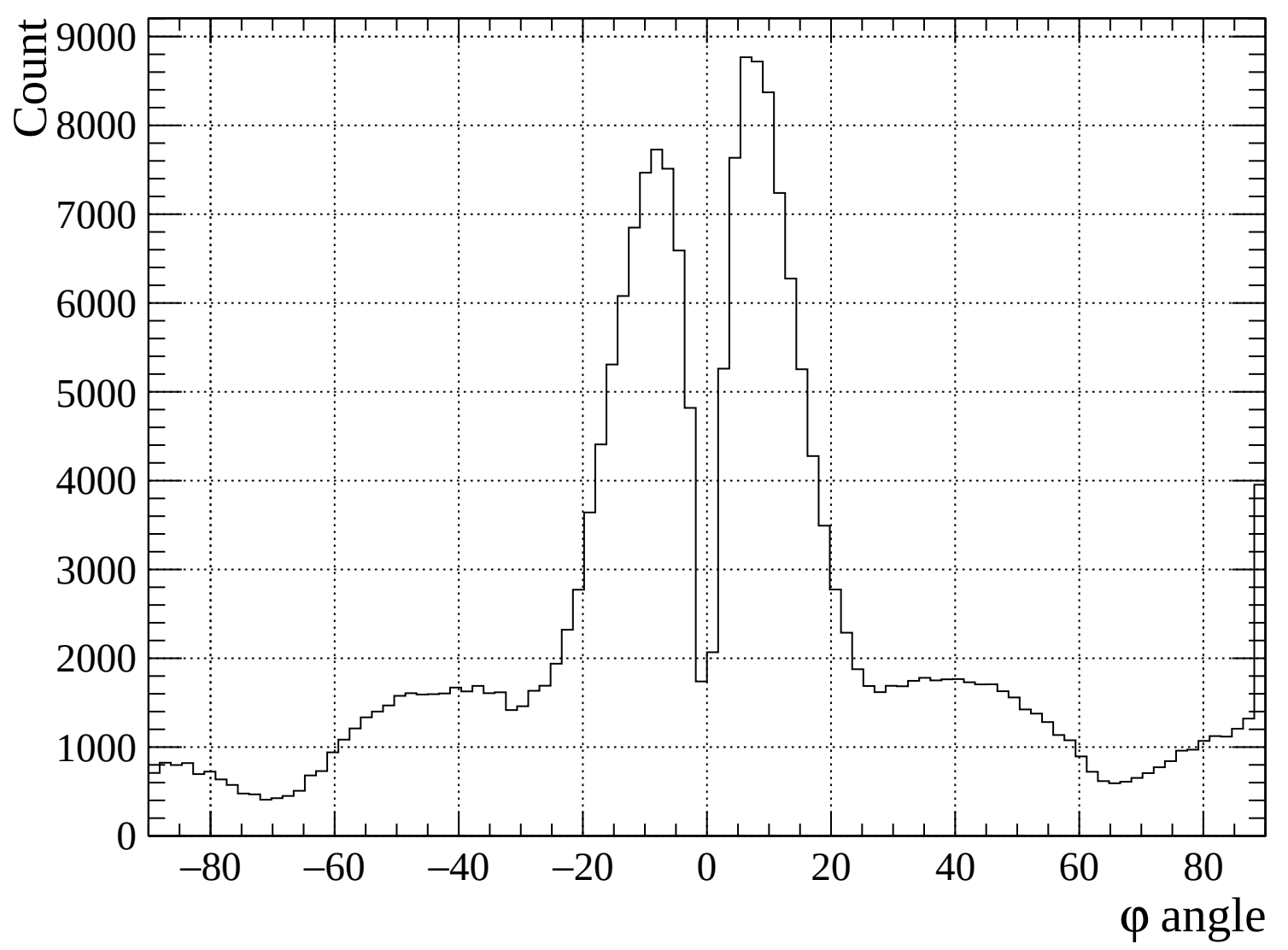


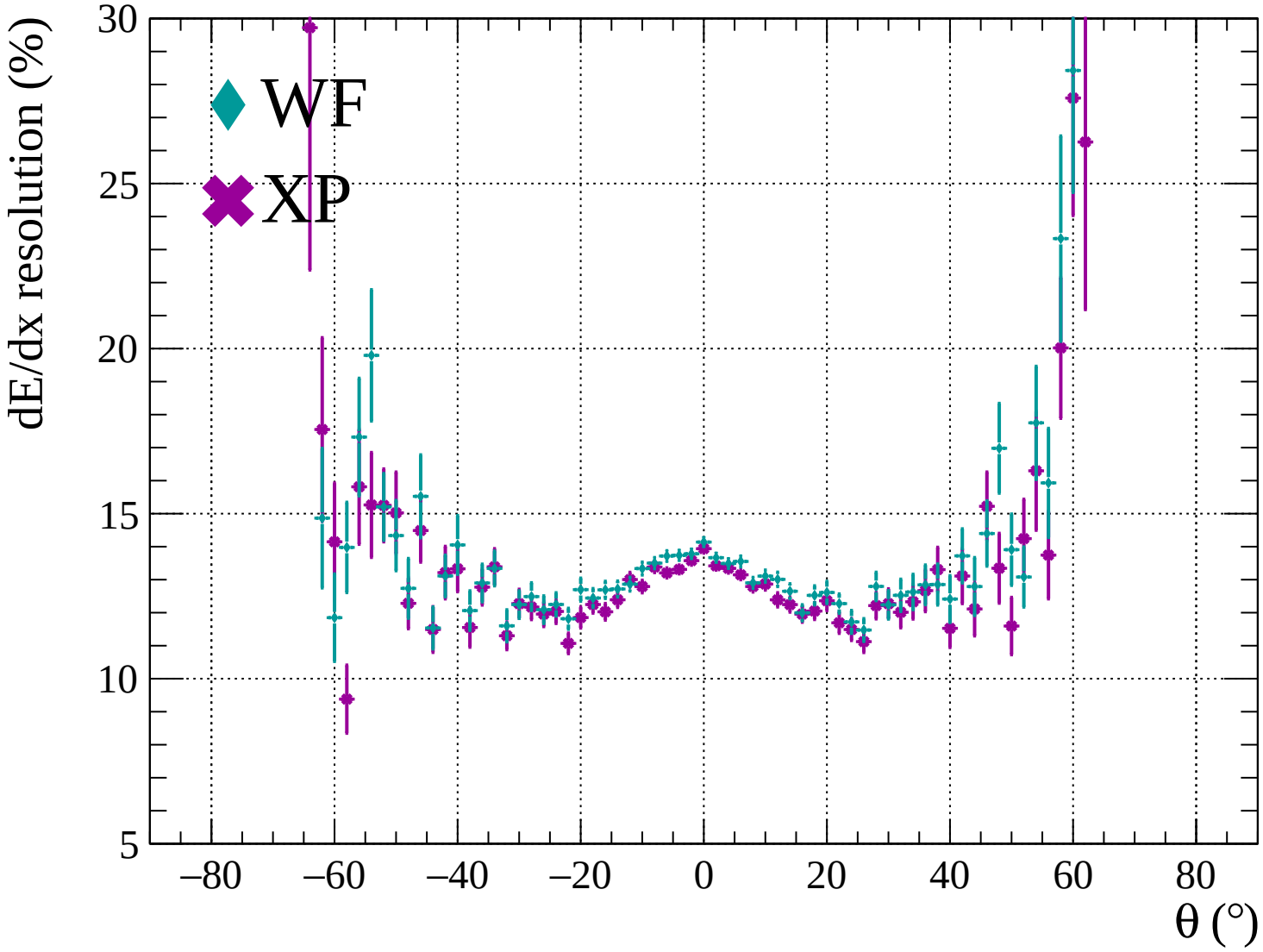


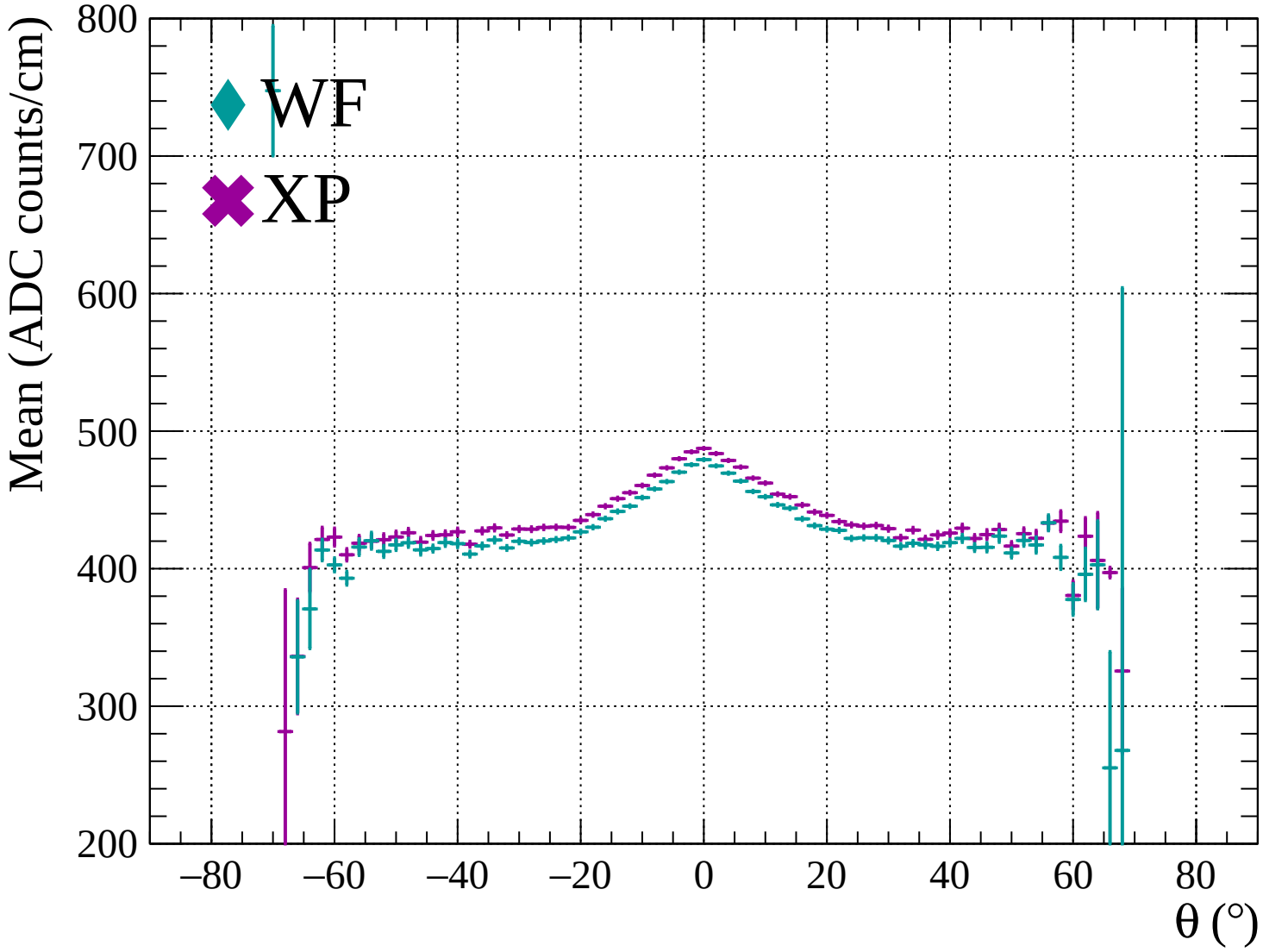


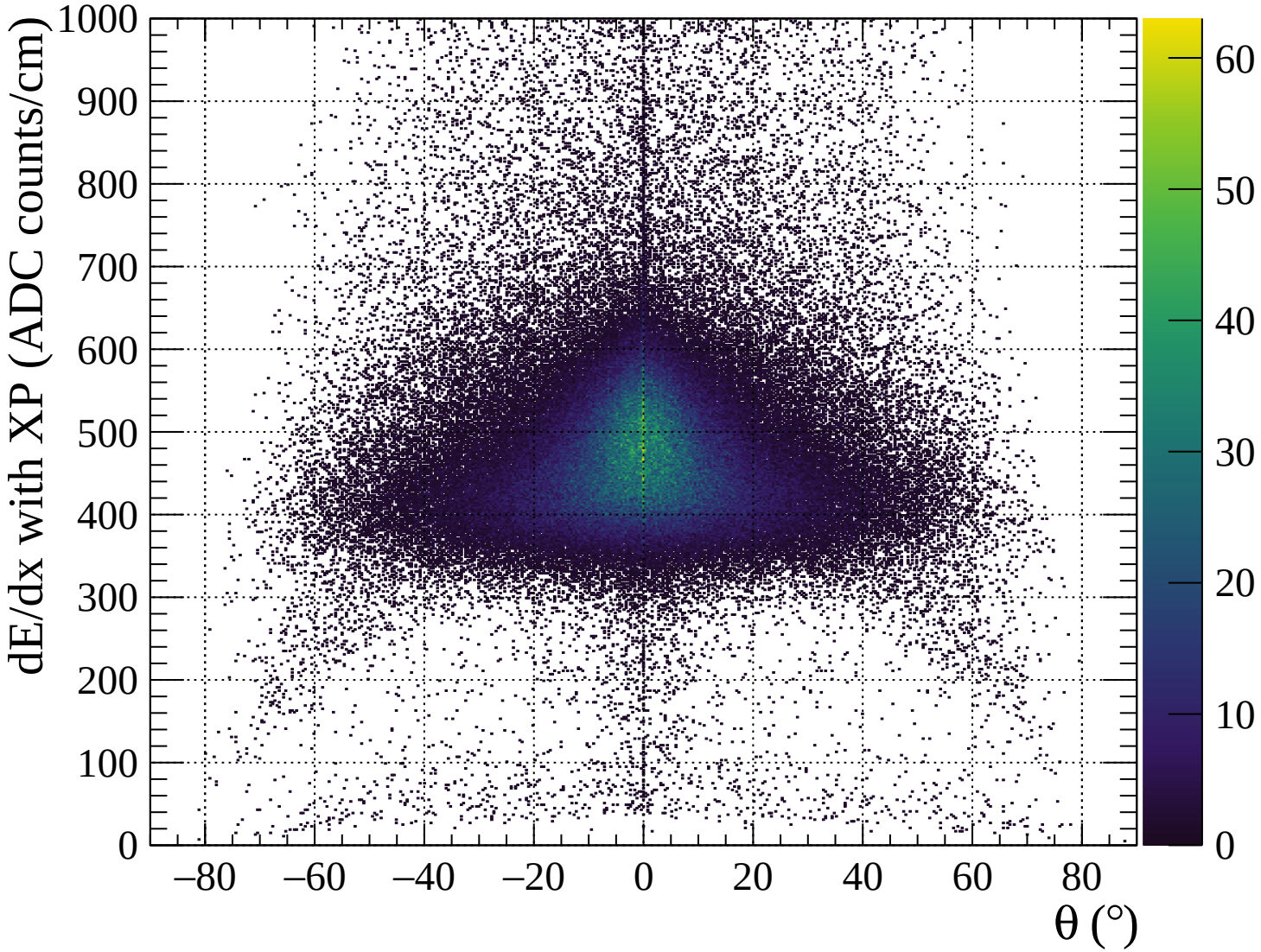


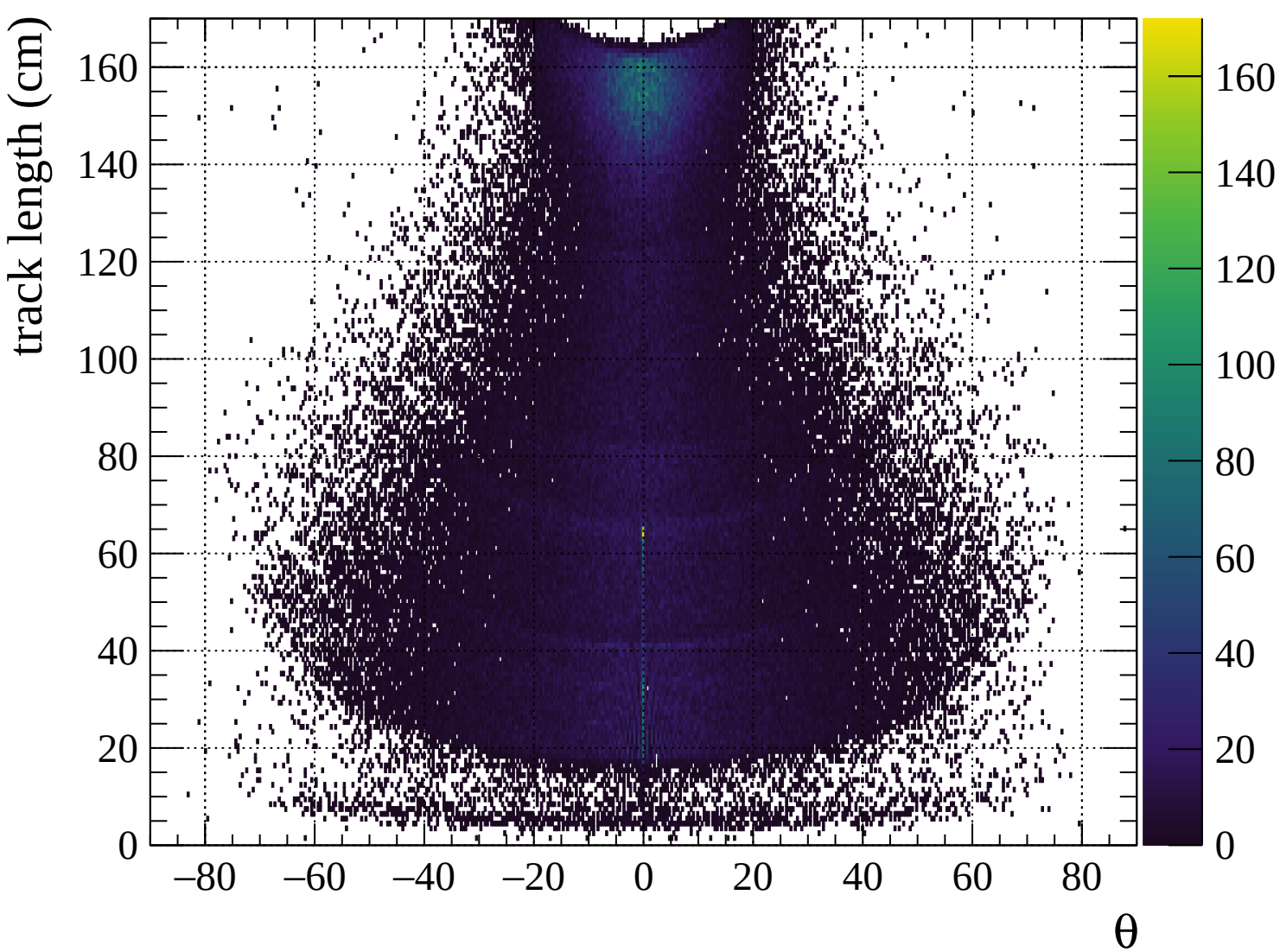


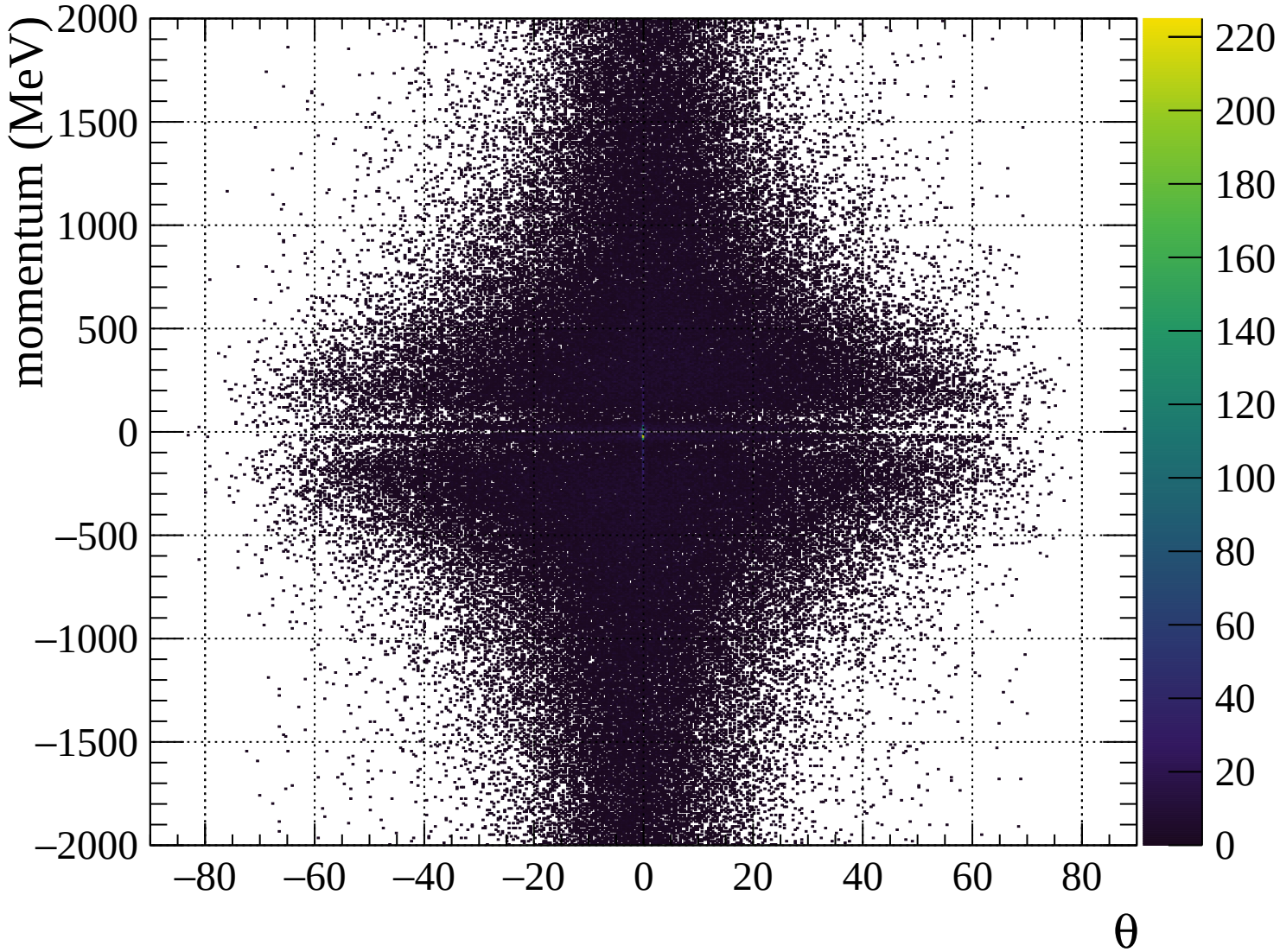


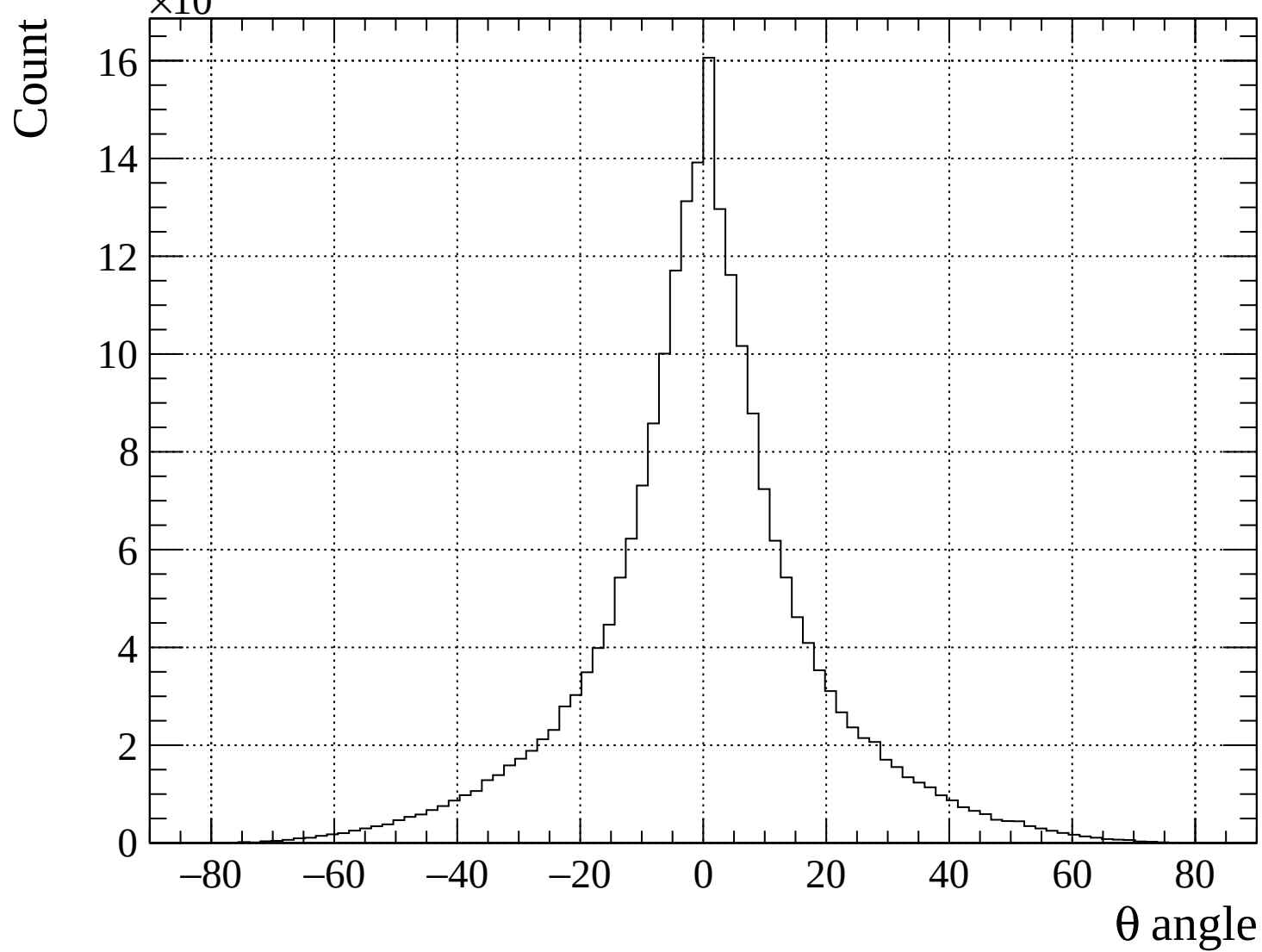


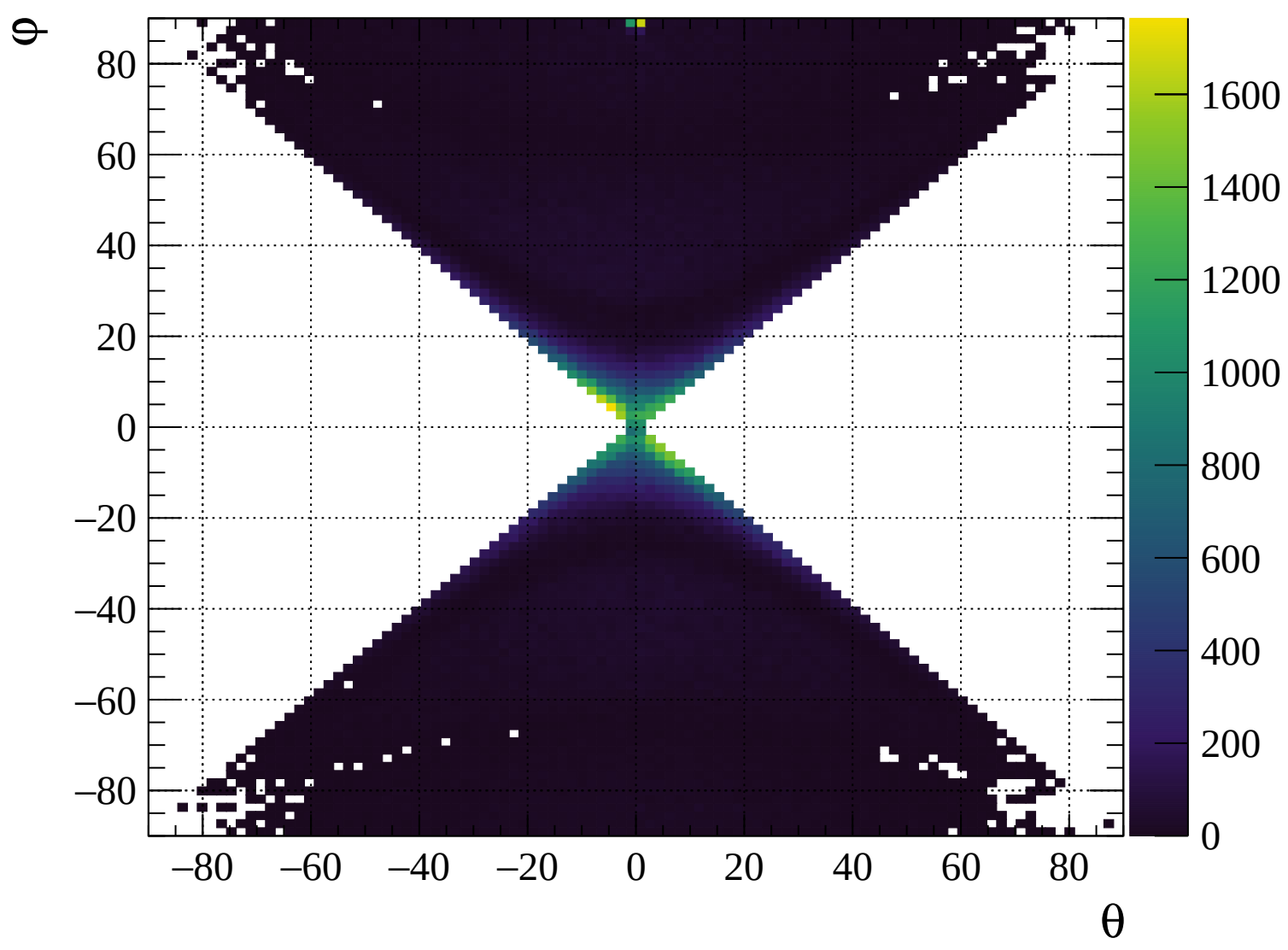


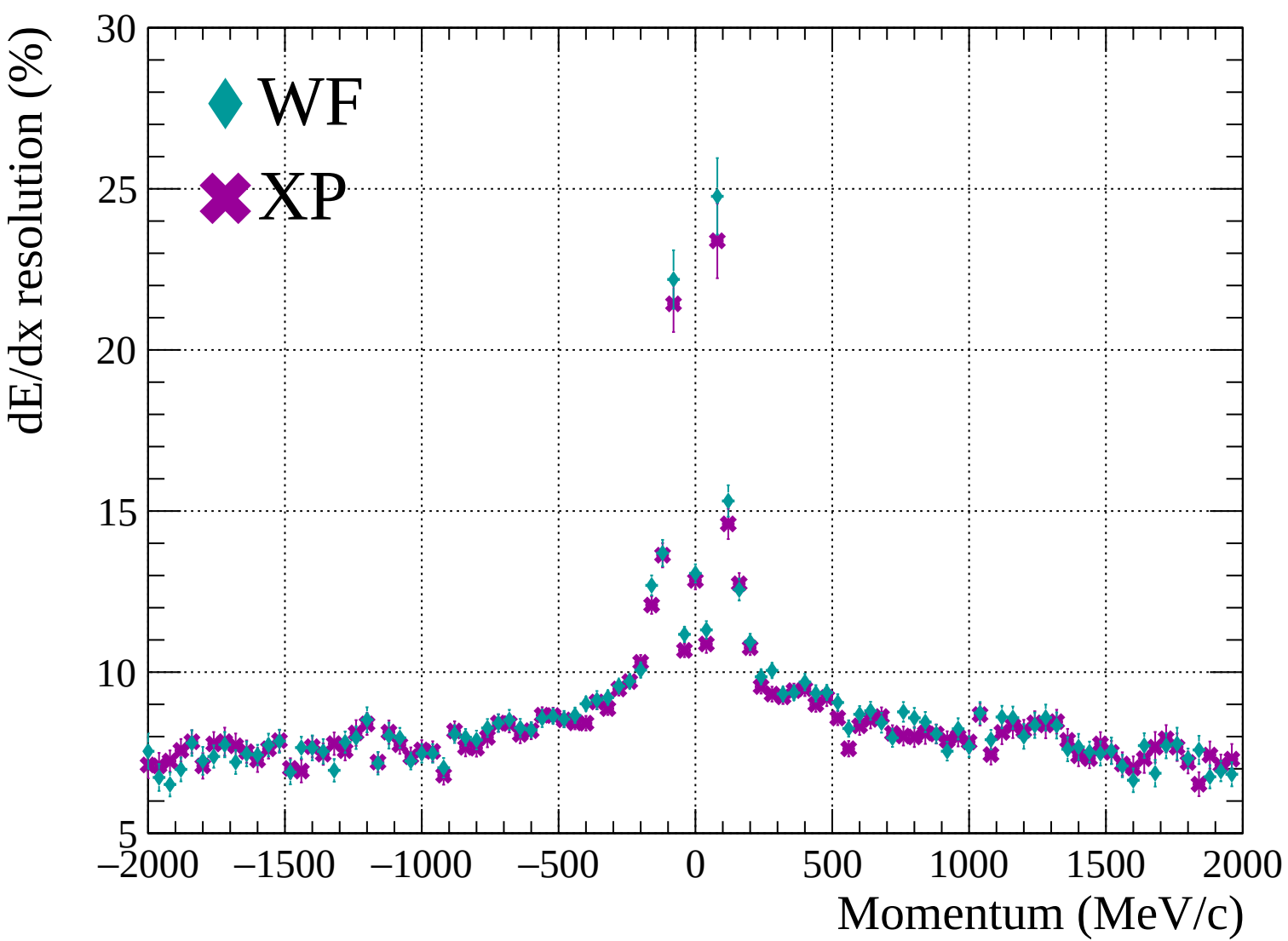


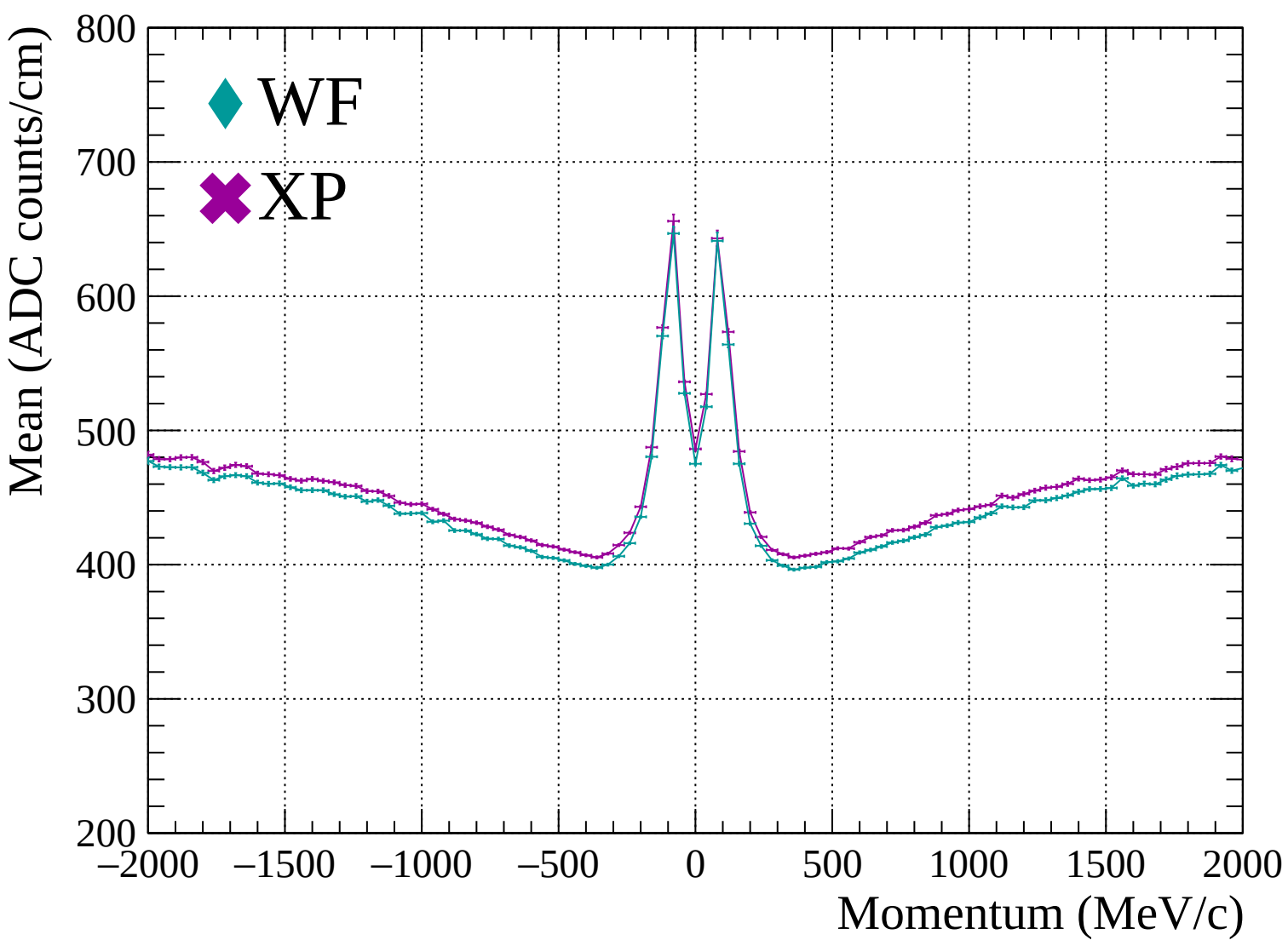


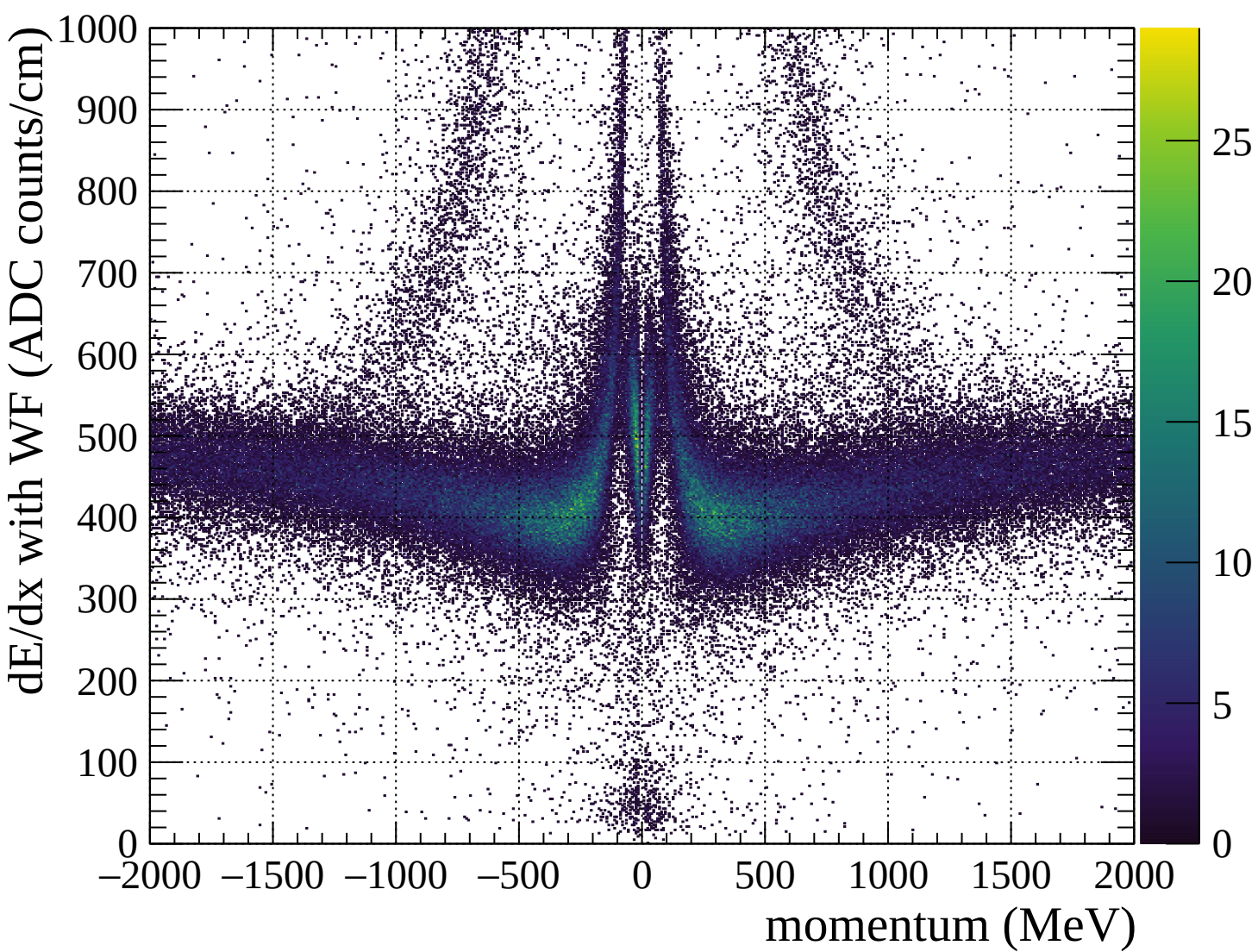


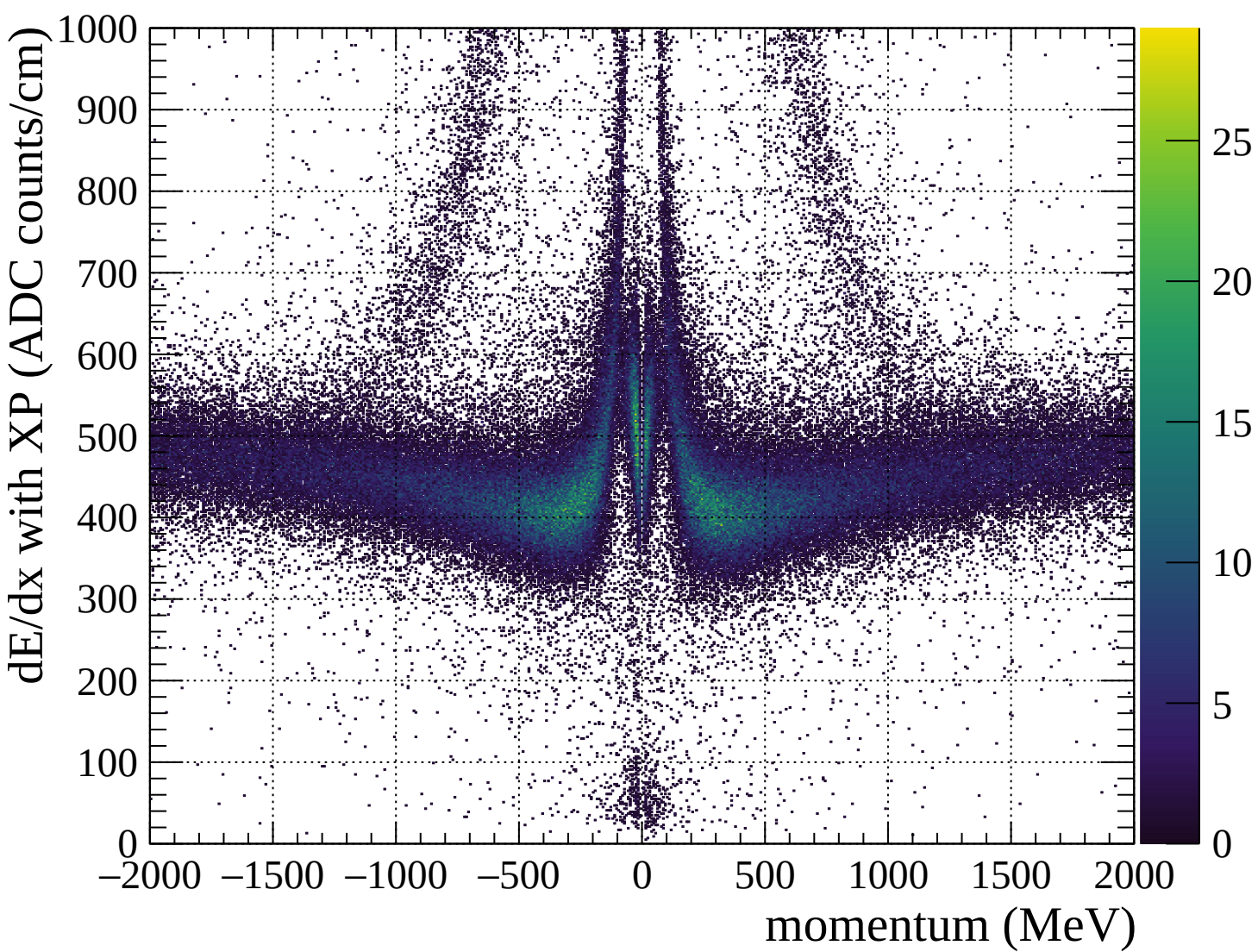


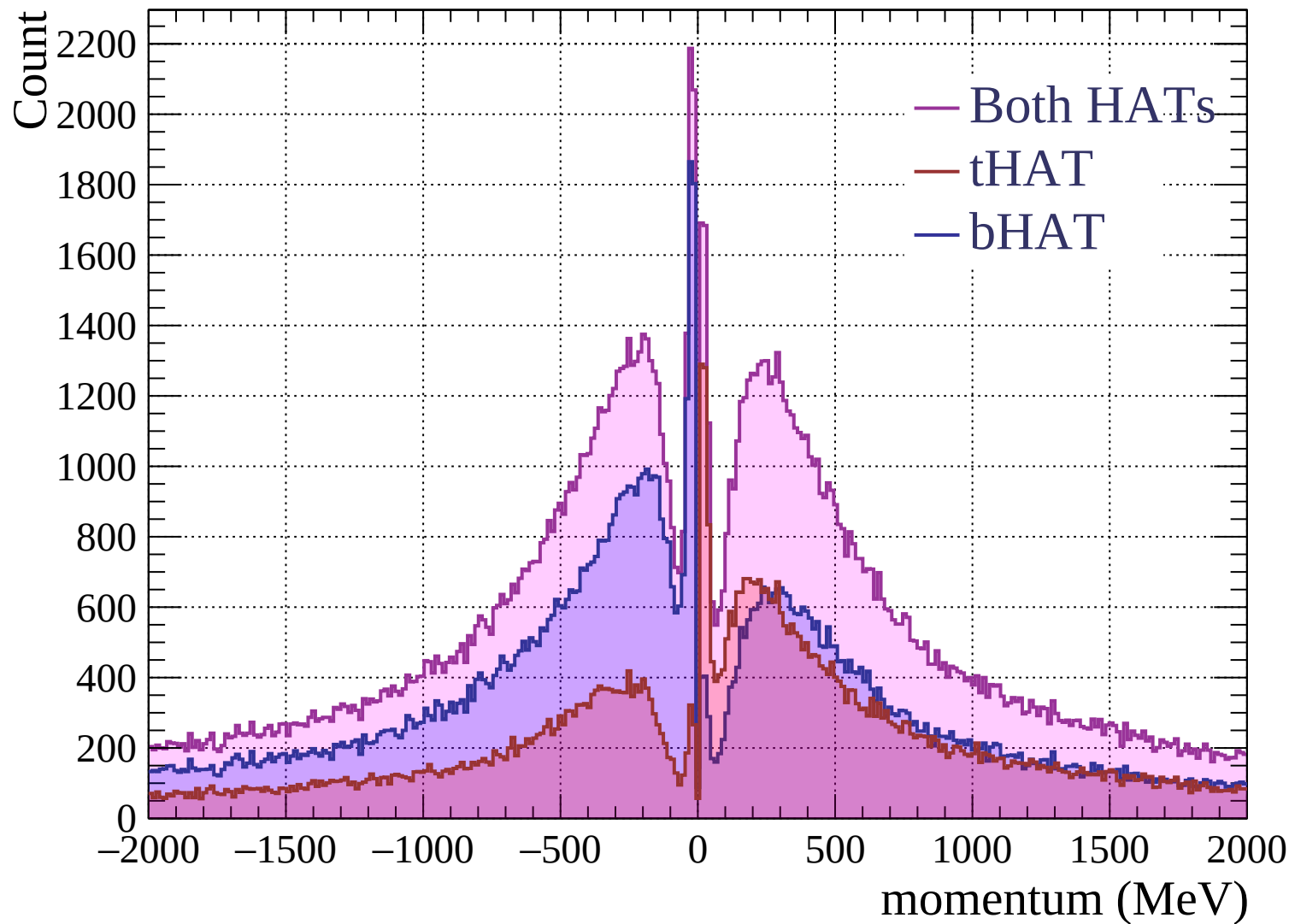


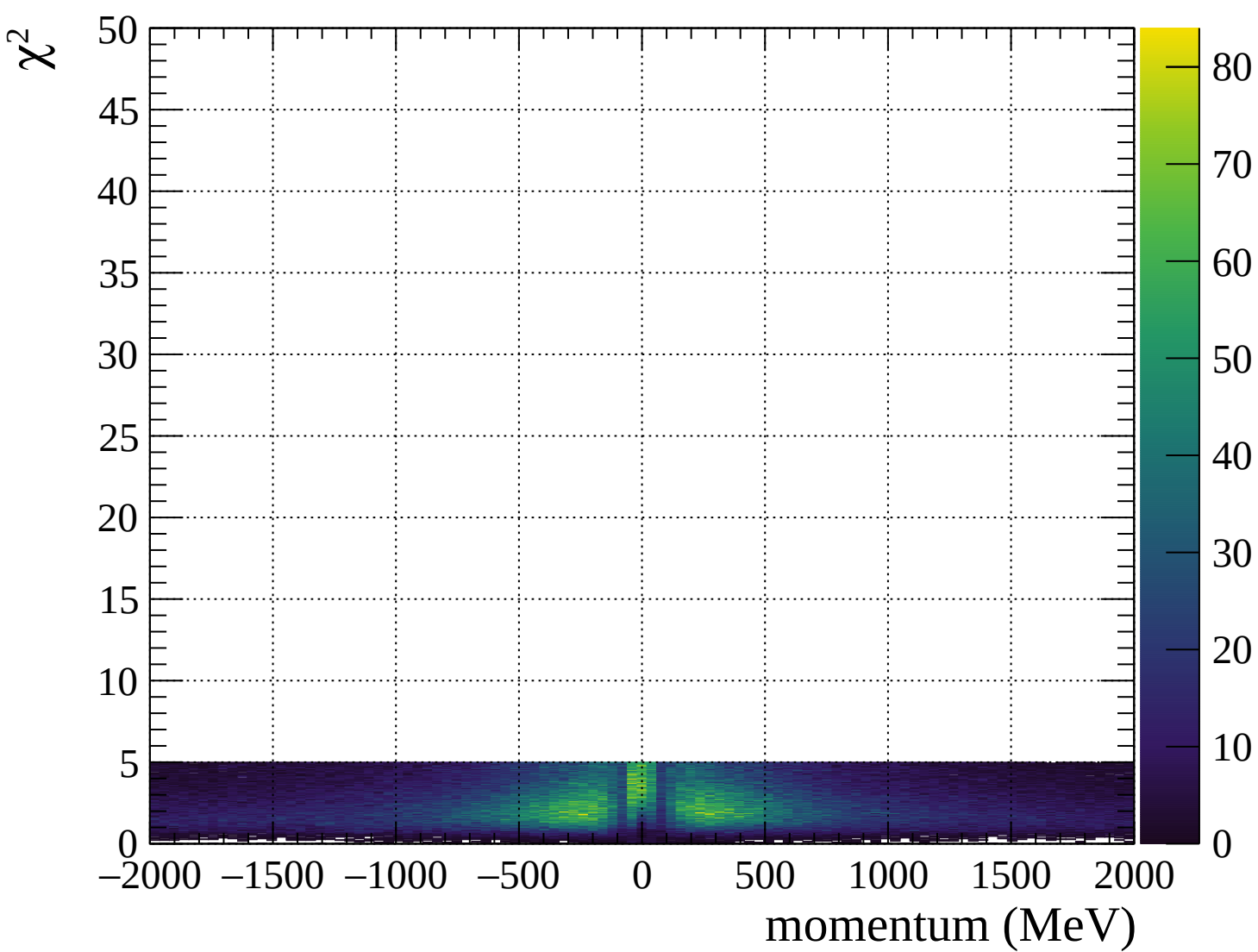




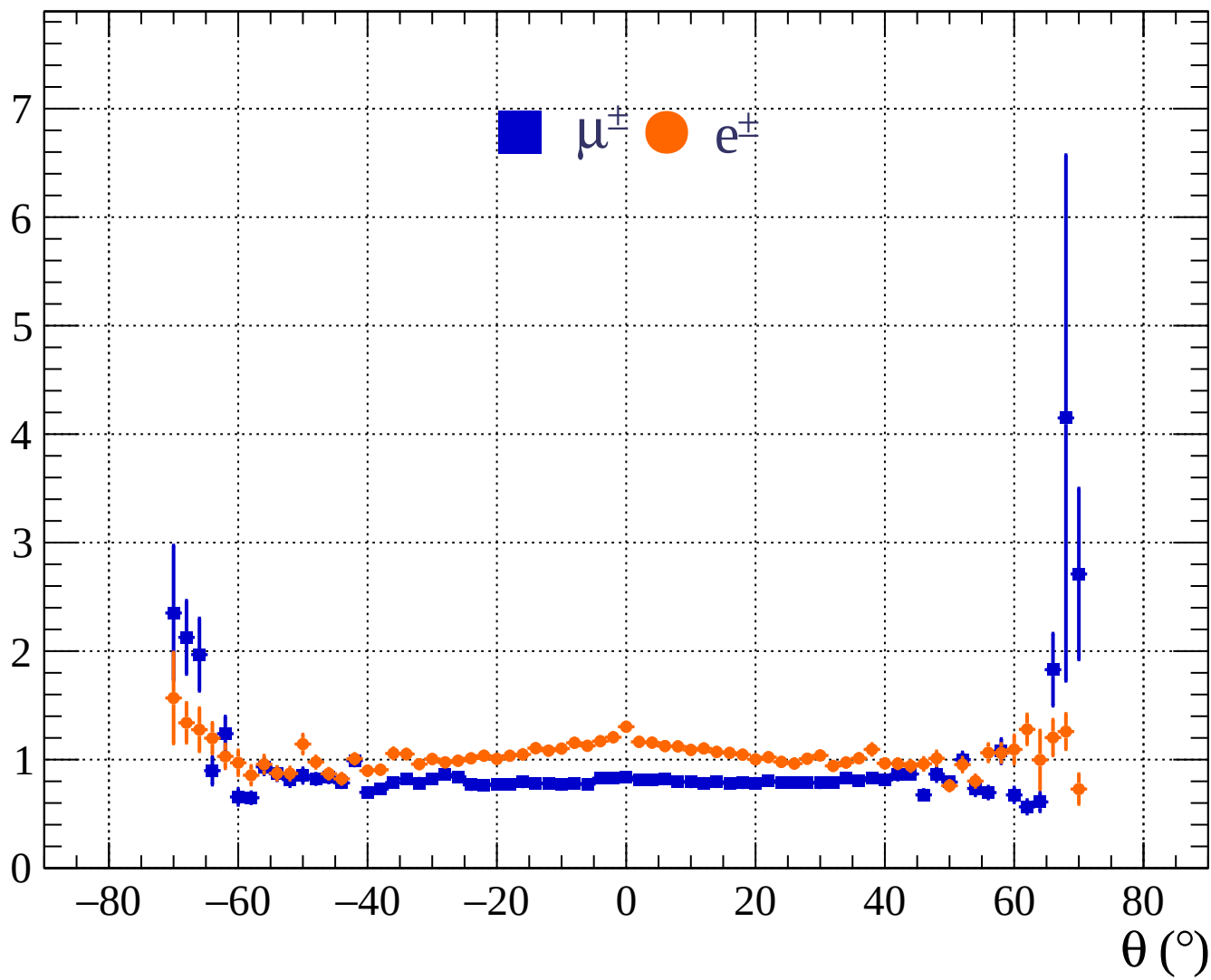




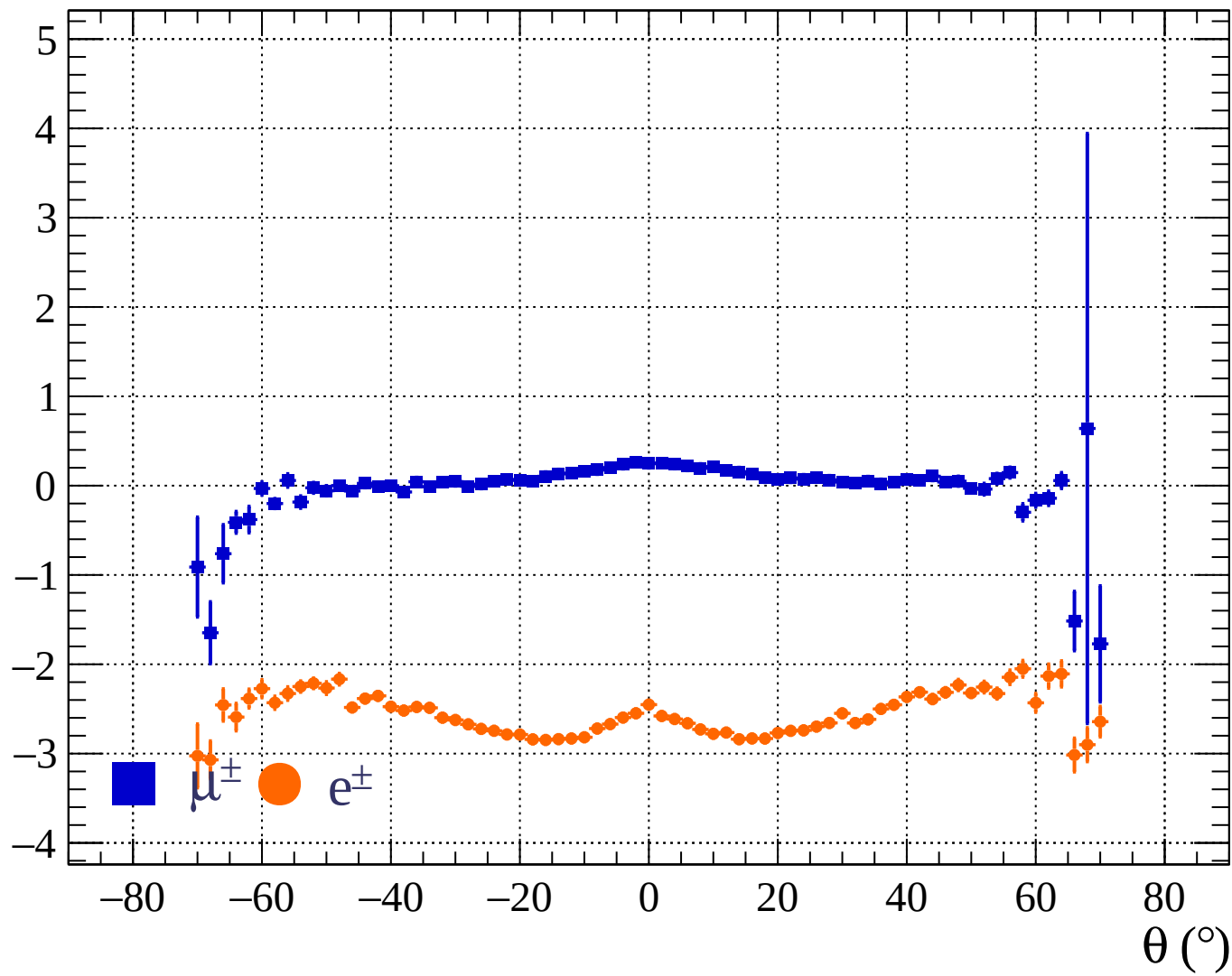




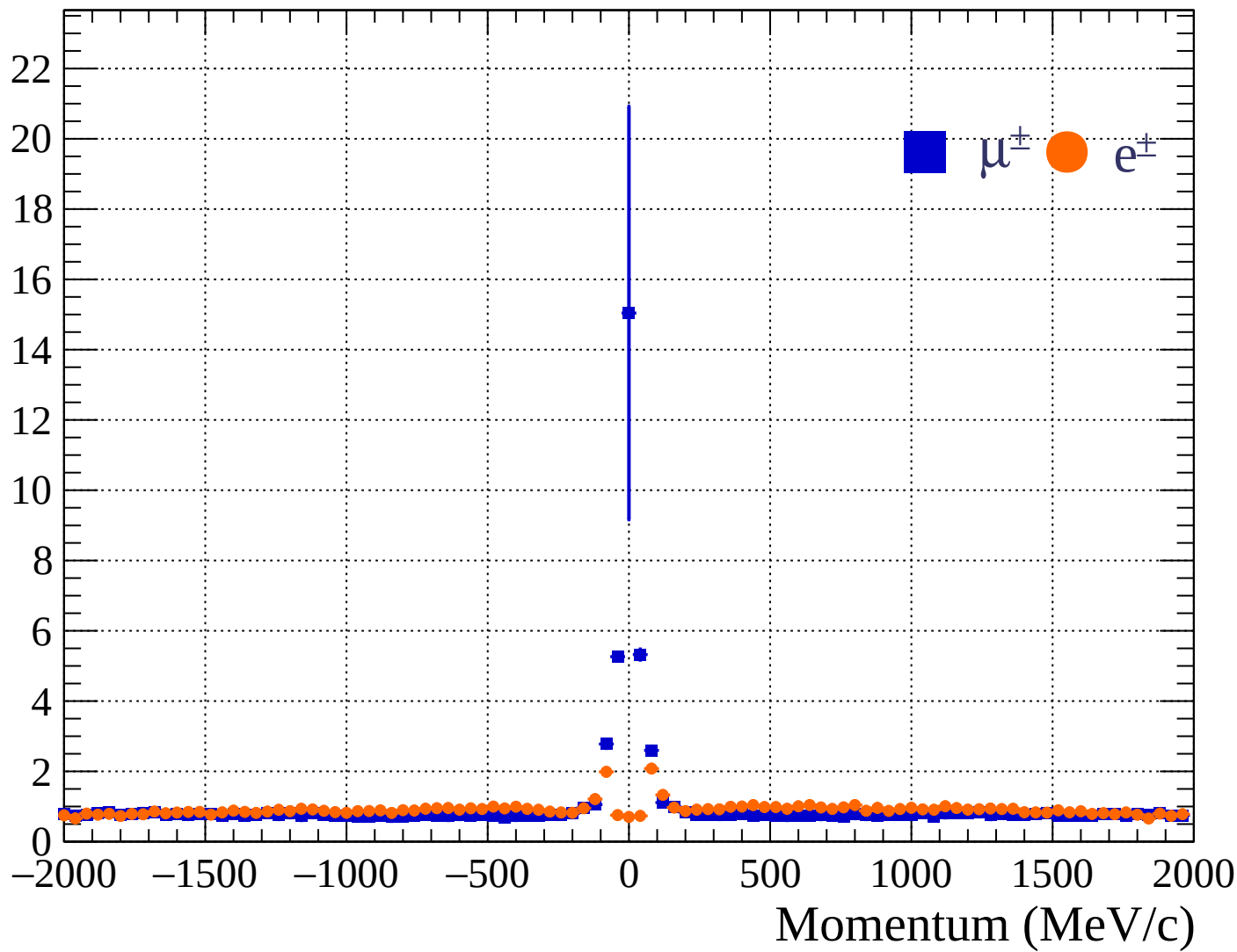
Pulls standard deviation



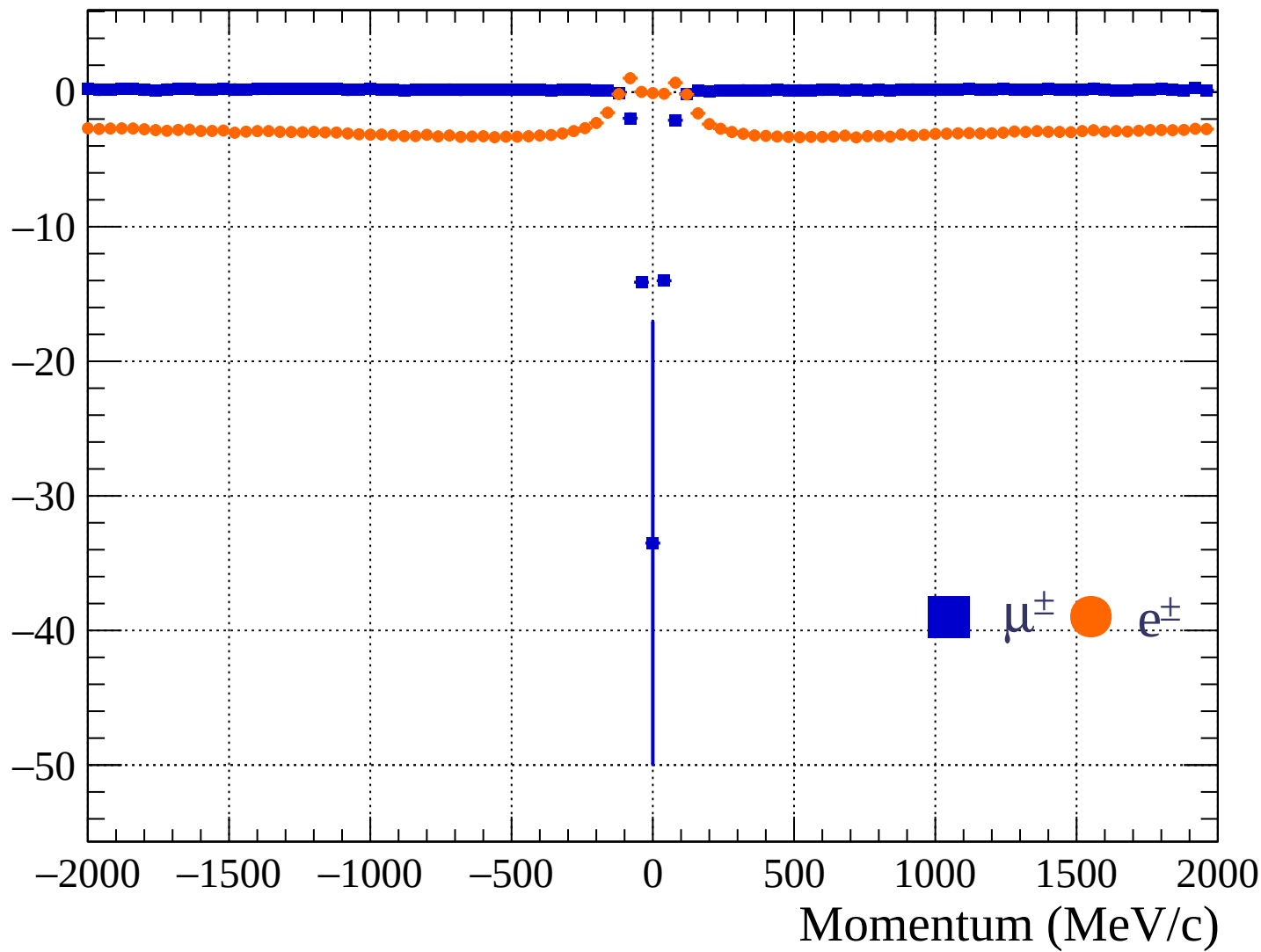
Pulls mean

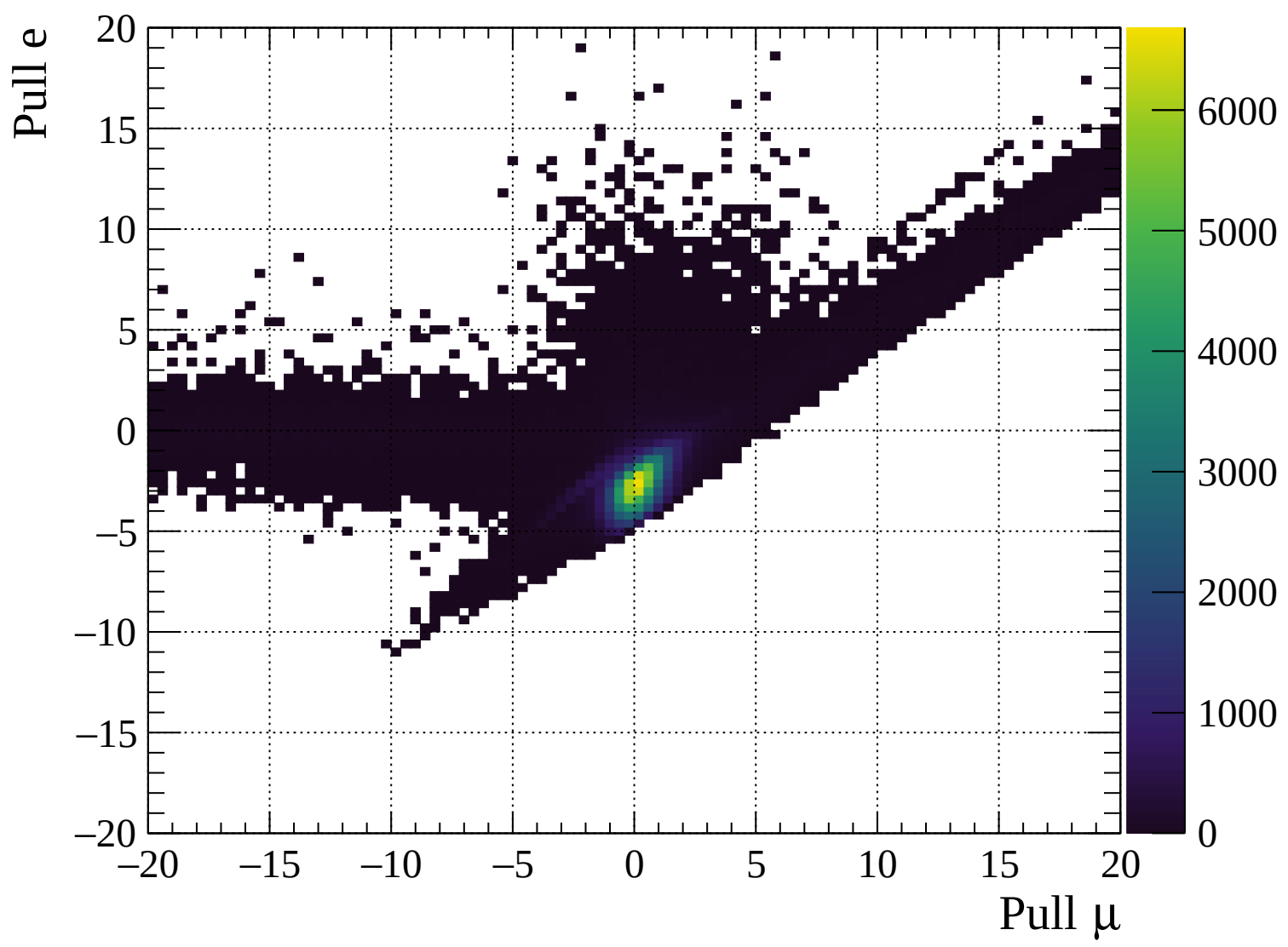


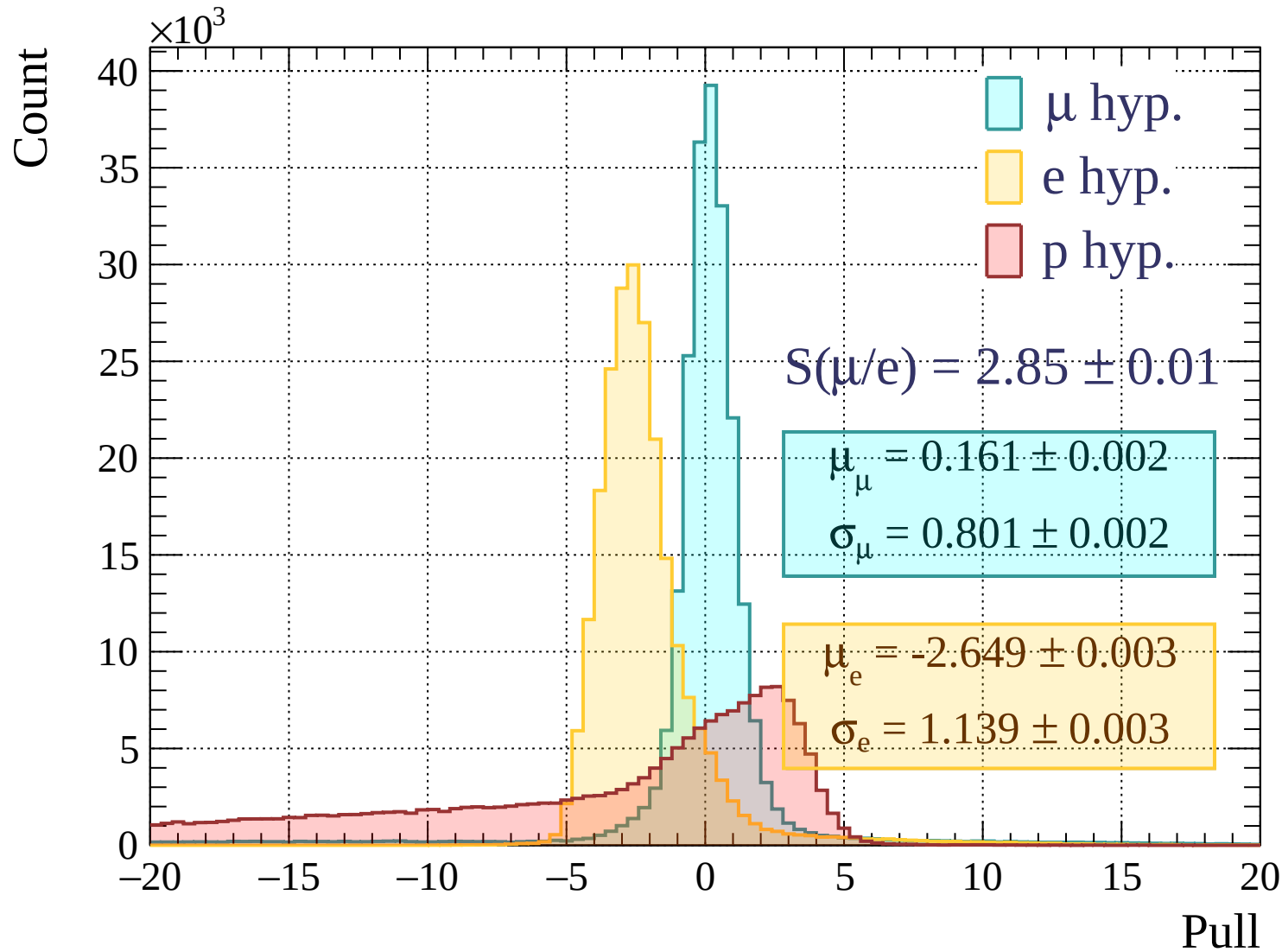
Pulls standard deviation

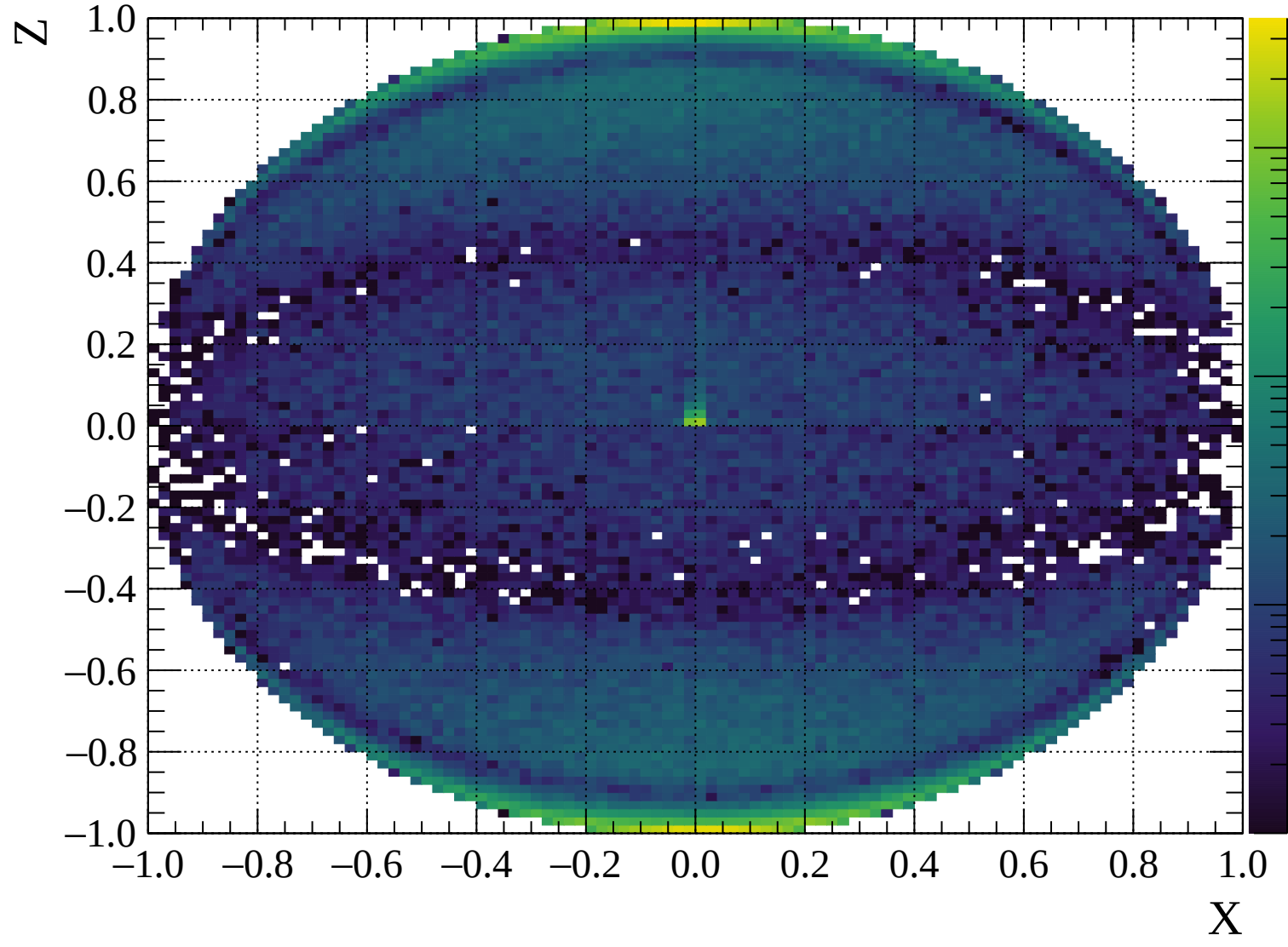


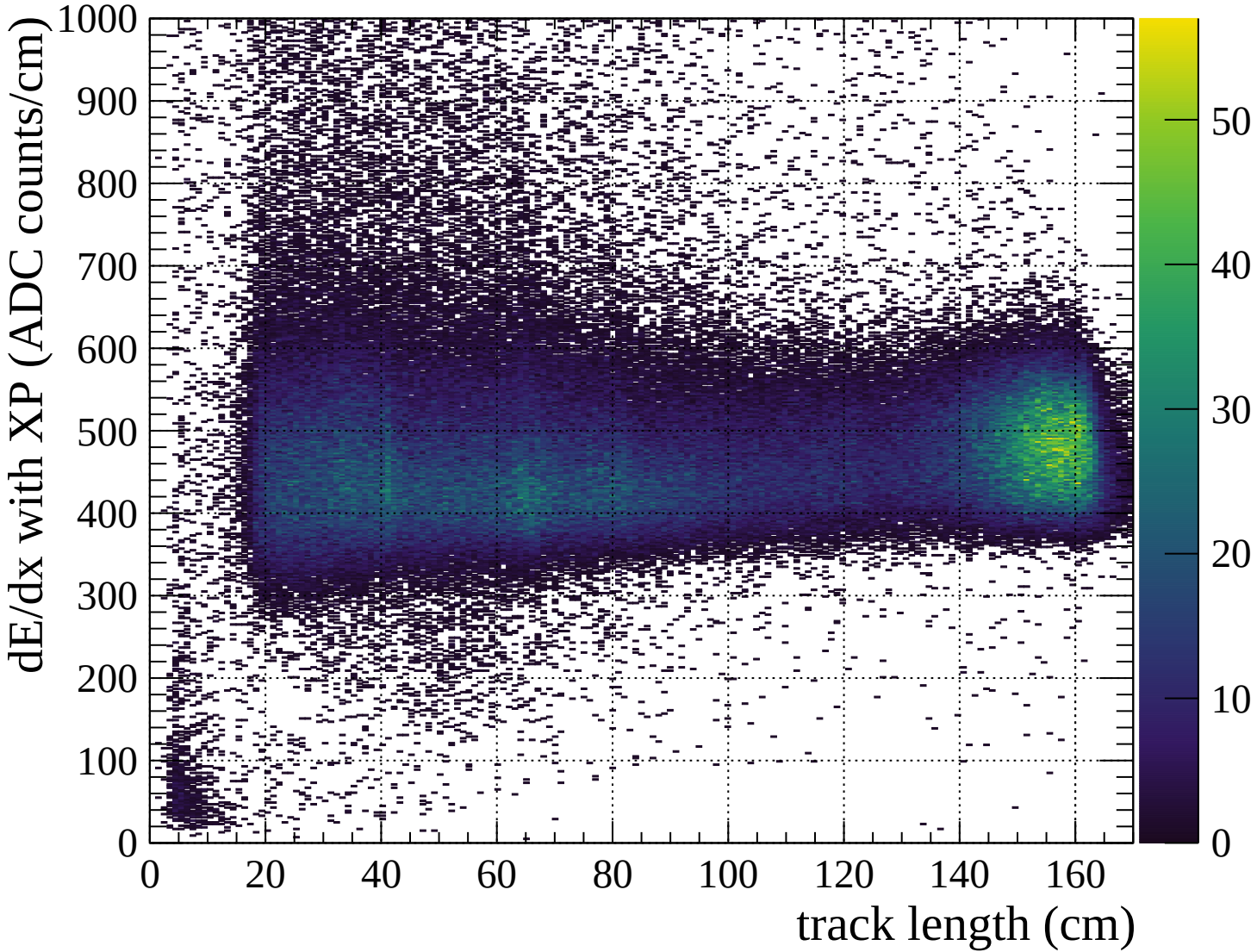
Pulls mean

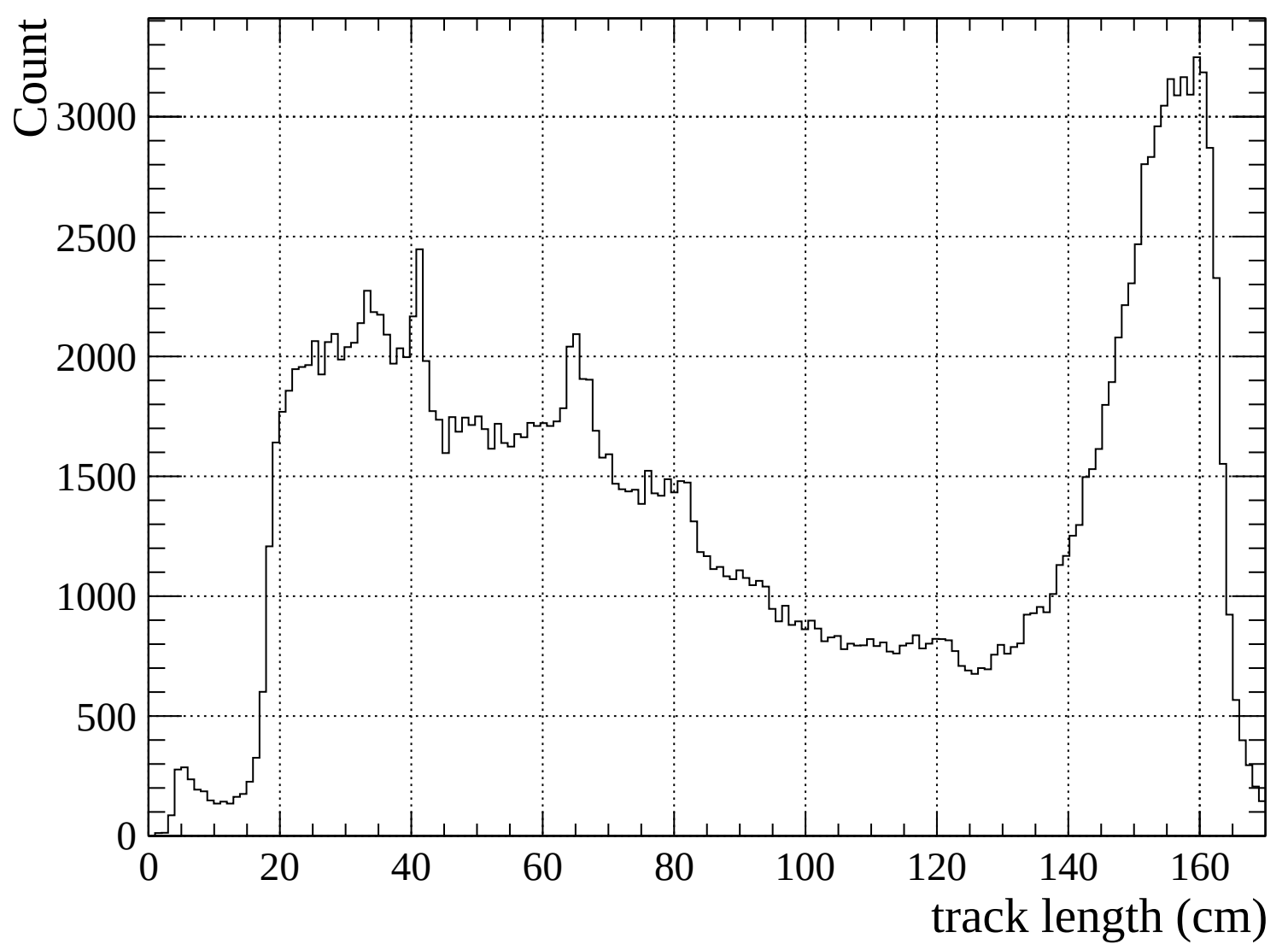


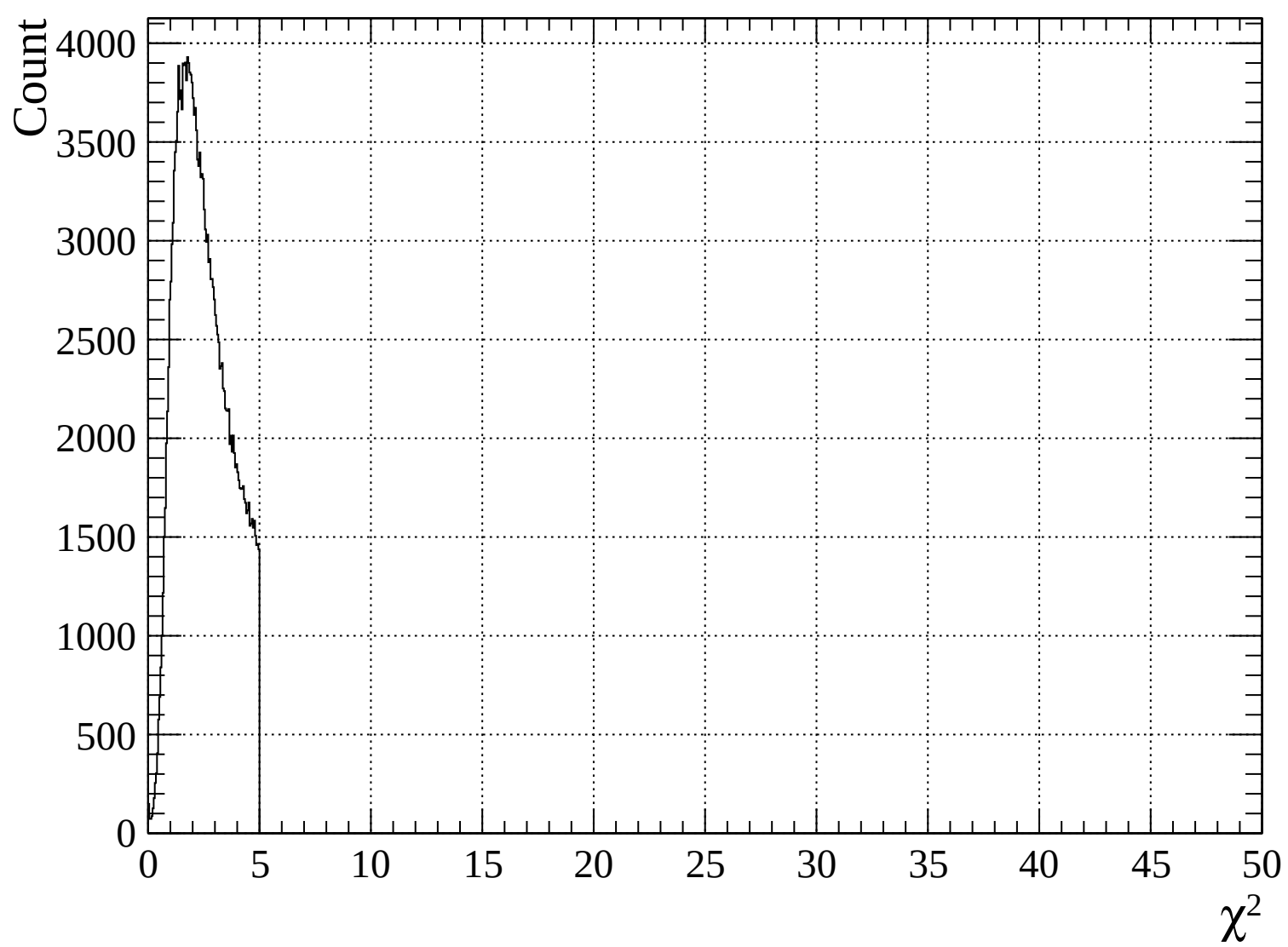


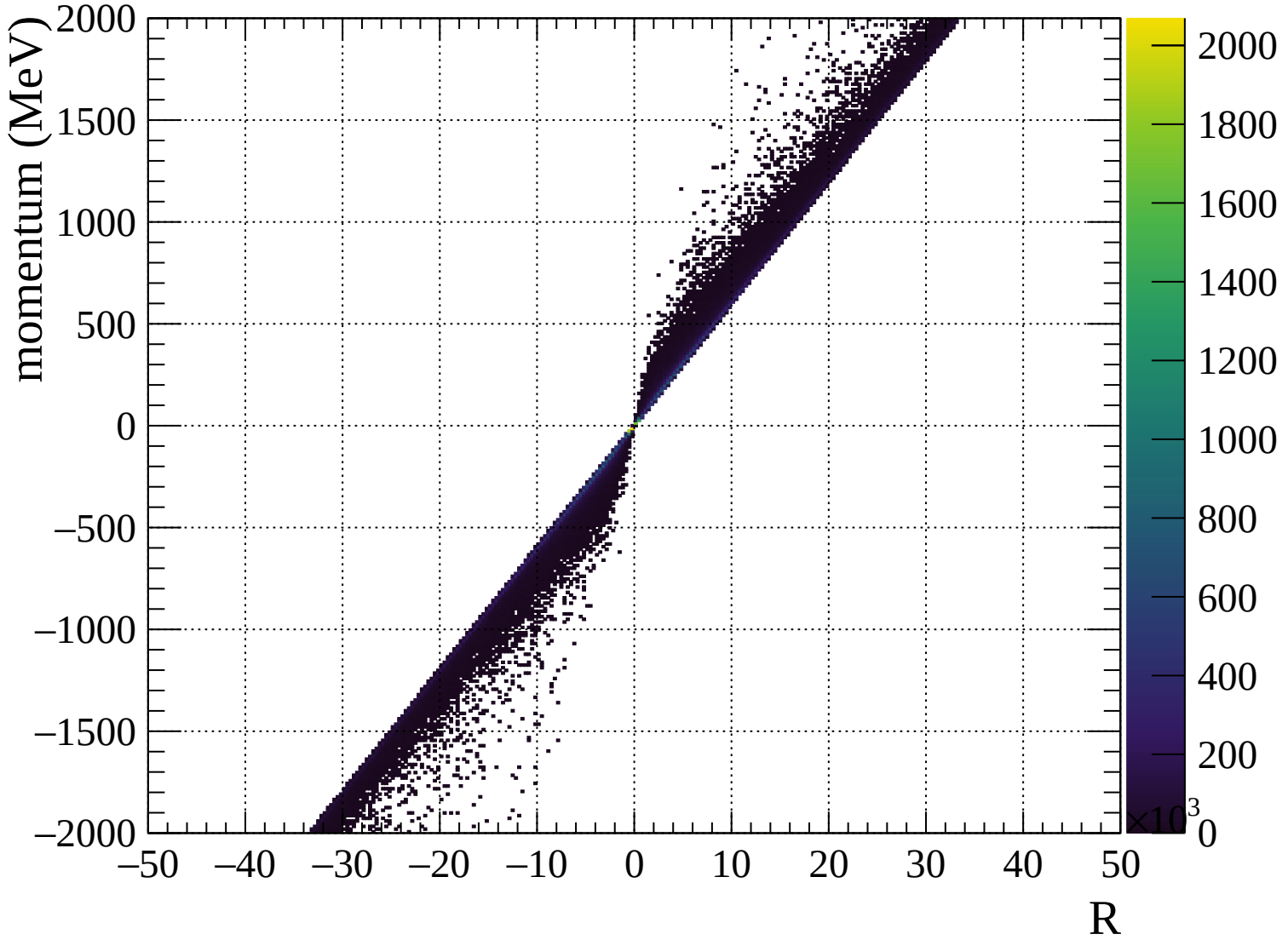


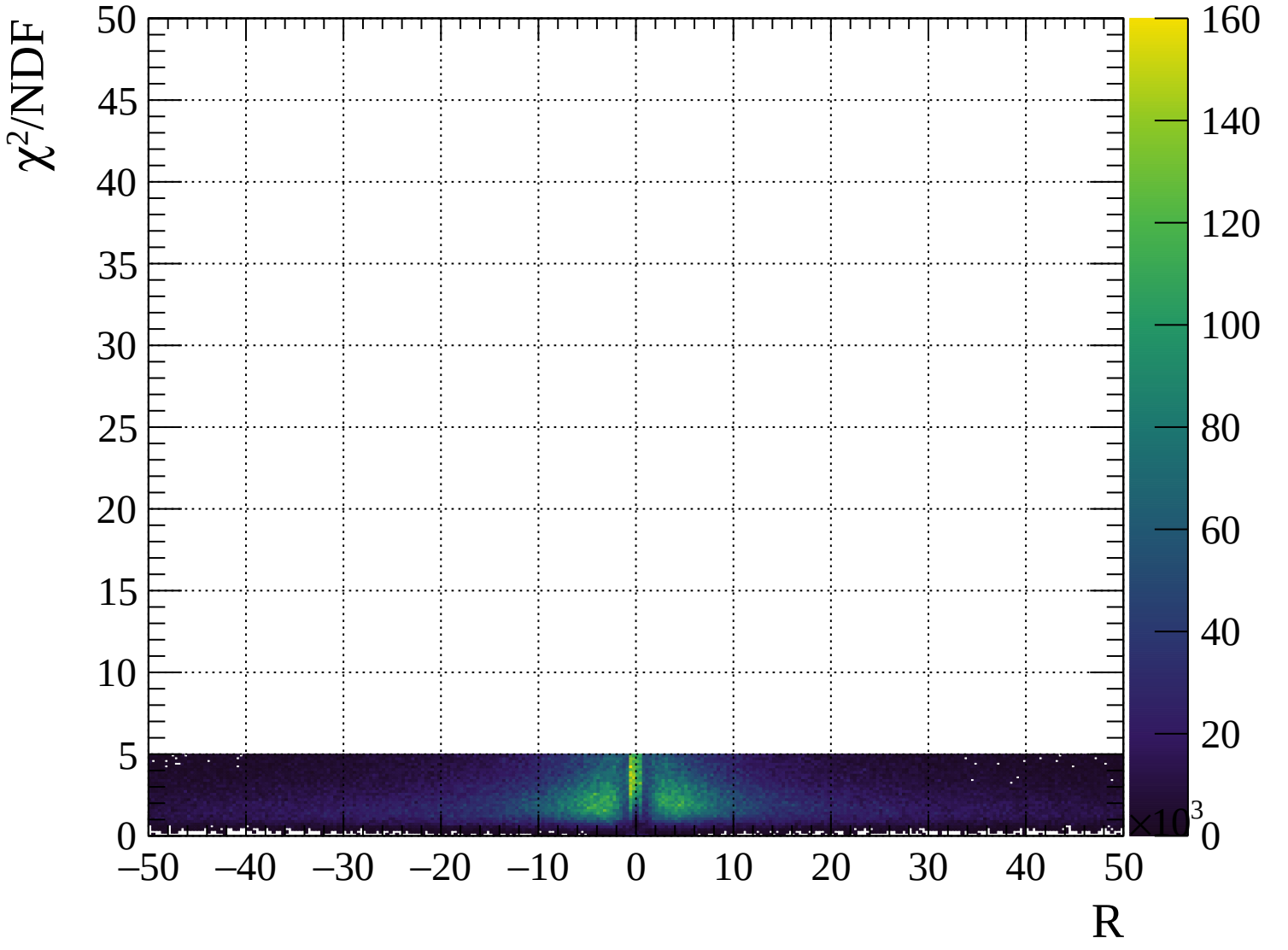






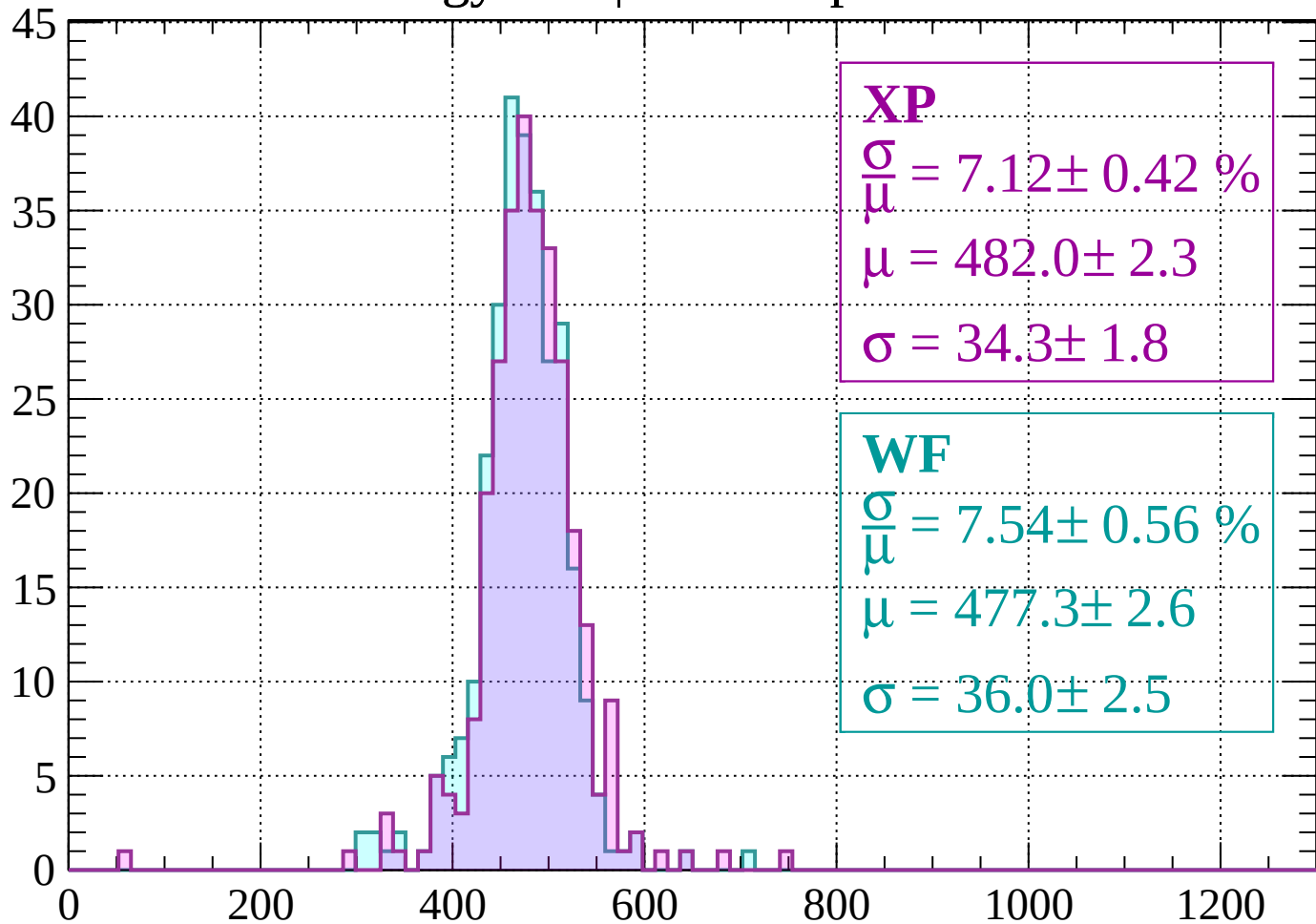






Energy loss | $-2000 < p < -1960$

Count



XP

$$\frac{\sigma}{\mu} = 7.12 \pm 0.42 \%$$

$$\mu = 482.0 \pm 2.3$$

$$\sigma = 34.3 \pm 1.8$$

WF

$$\frac{\sigma}{\mu} = 7.54 \pm 0.56 \%$$

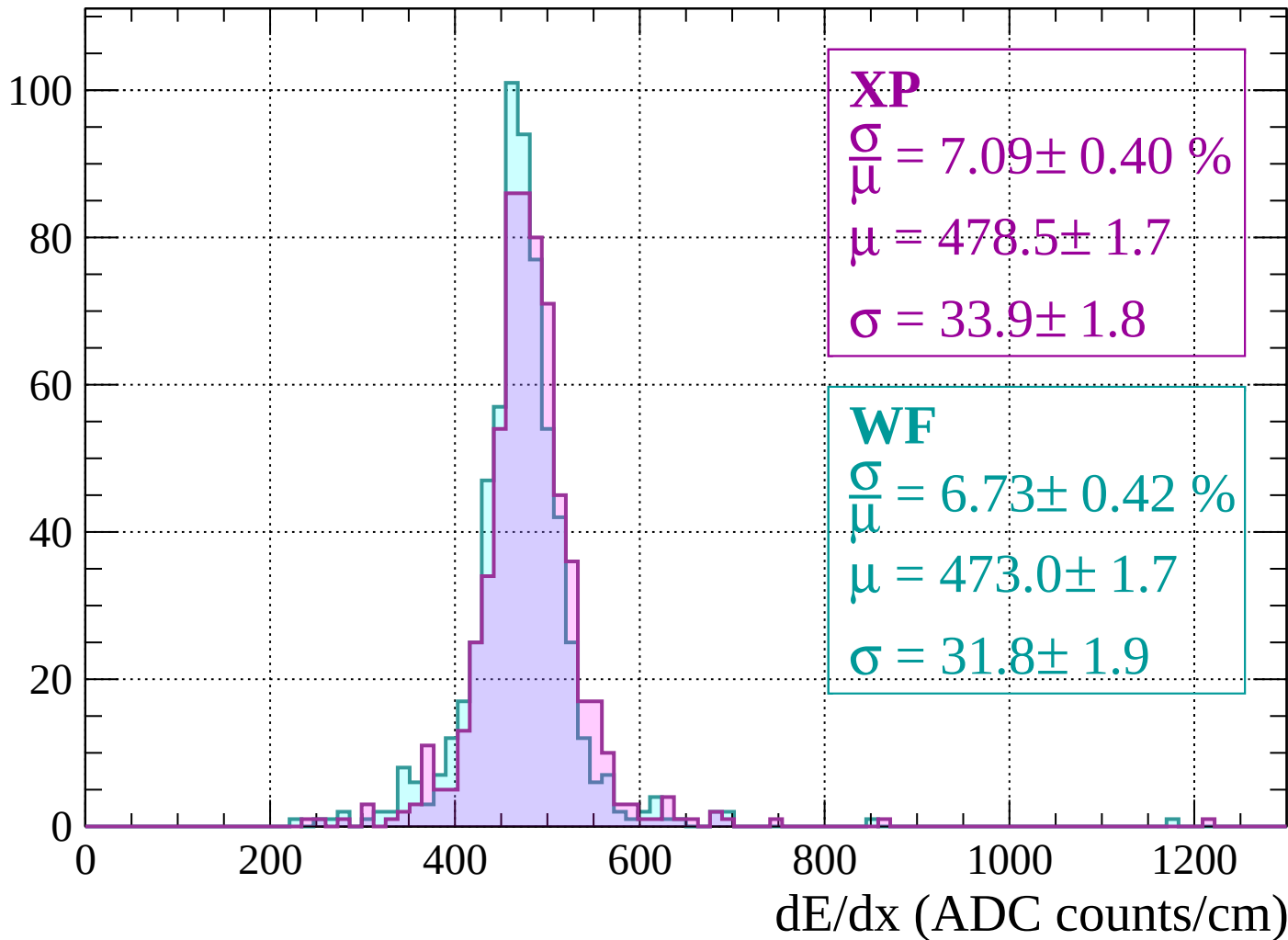
$$\mu = 477.3 \pm 2.6$$

$$\sigma = 36.0 \pm 2.5$$

dE/dx (ADC counts/cm)

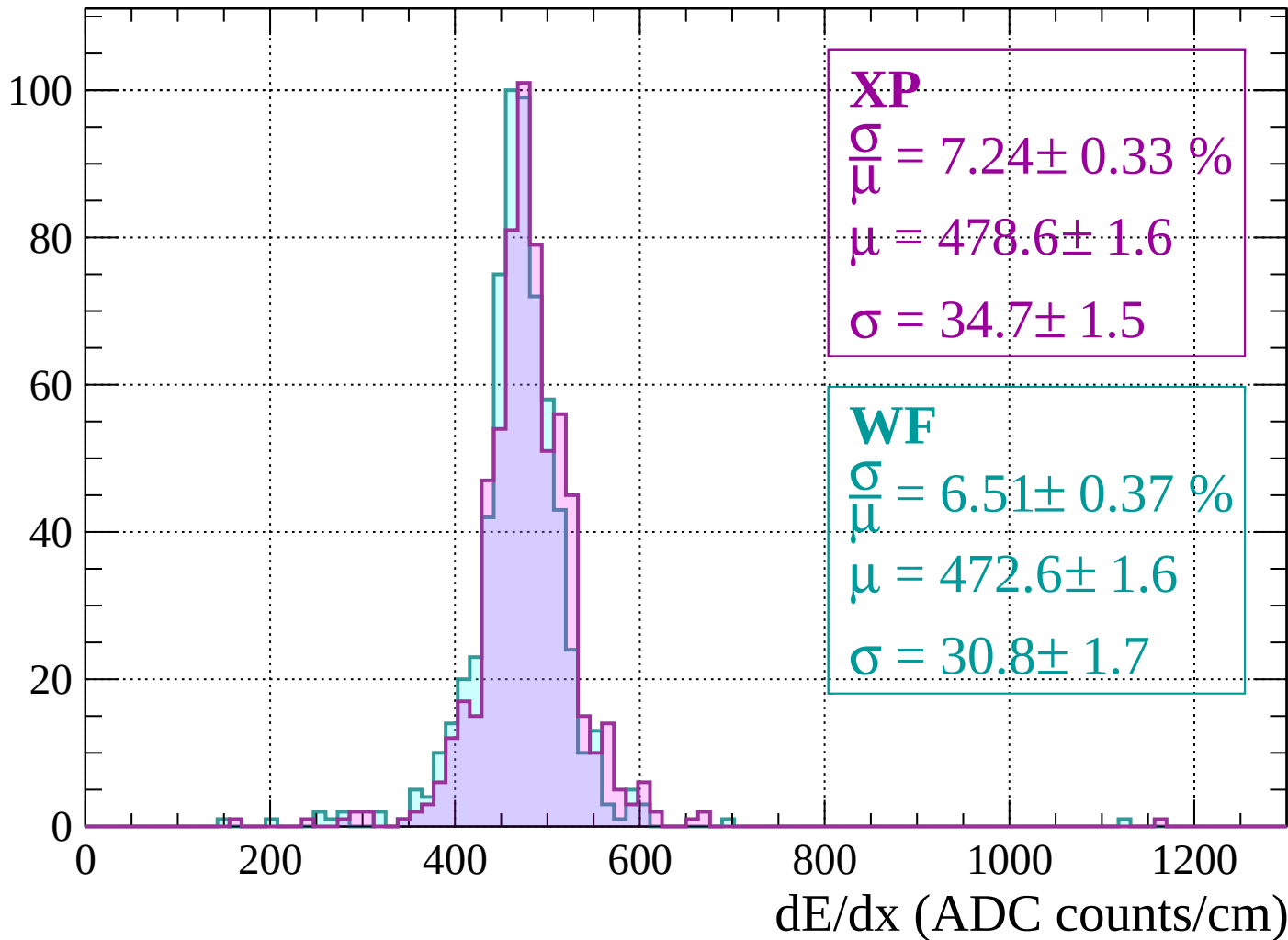
Energy loss | -1960 < p < -1920

Count



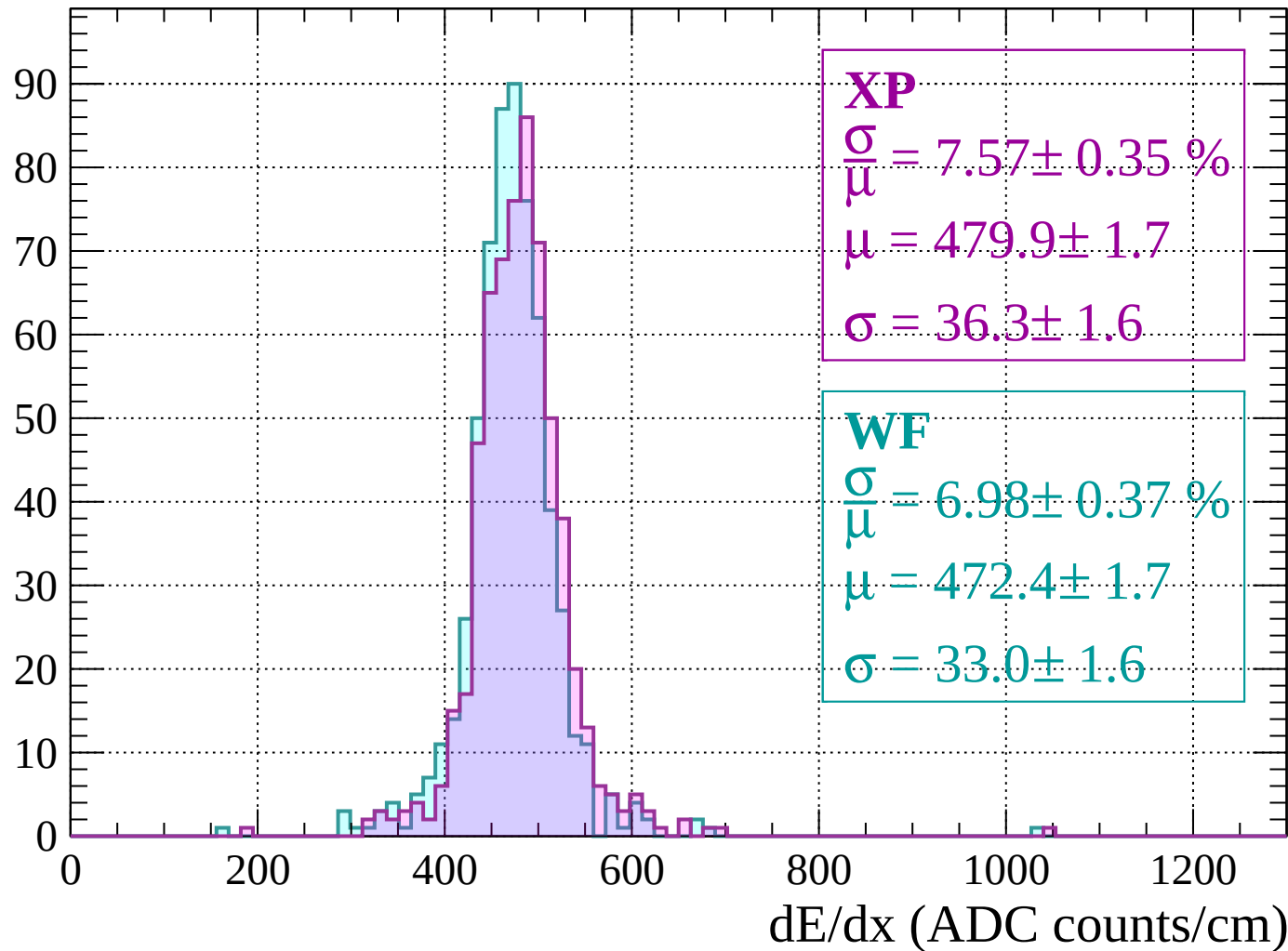
Energy loss | $-1920 < p < -1880$

Count



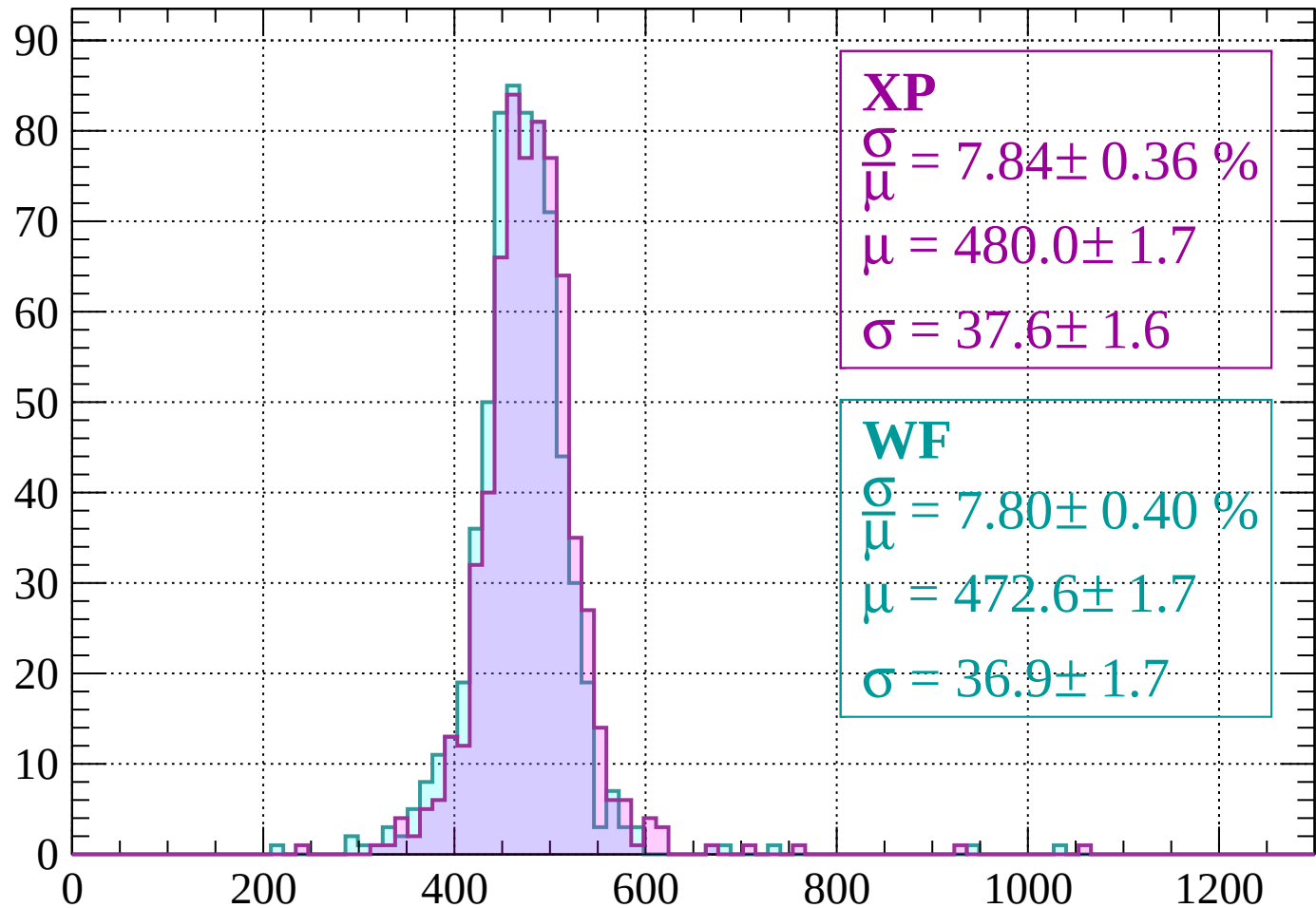
Energy loss | $-1880 < p < -1840$

Count



Energy loss | $-1840 < p < -1800$

Count



XP

$$\frac{\sigma}{\mu} = 7.84 \pm 0.36 \%$$

$$\mu = 480.0 \pm 1.7$$

$$\sigma = 37.6 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 7.80 \pm 0.40 \%$$

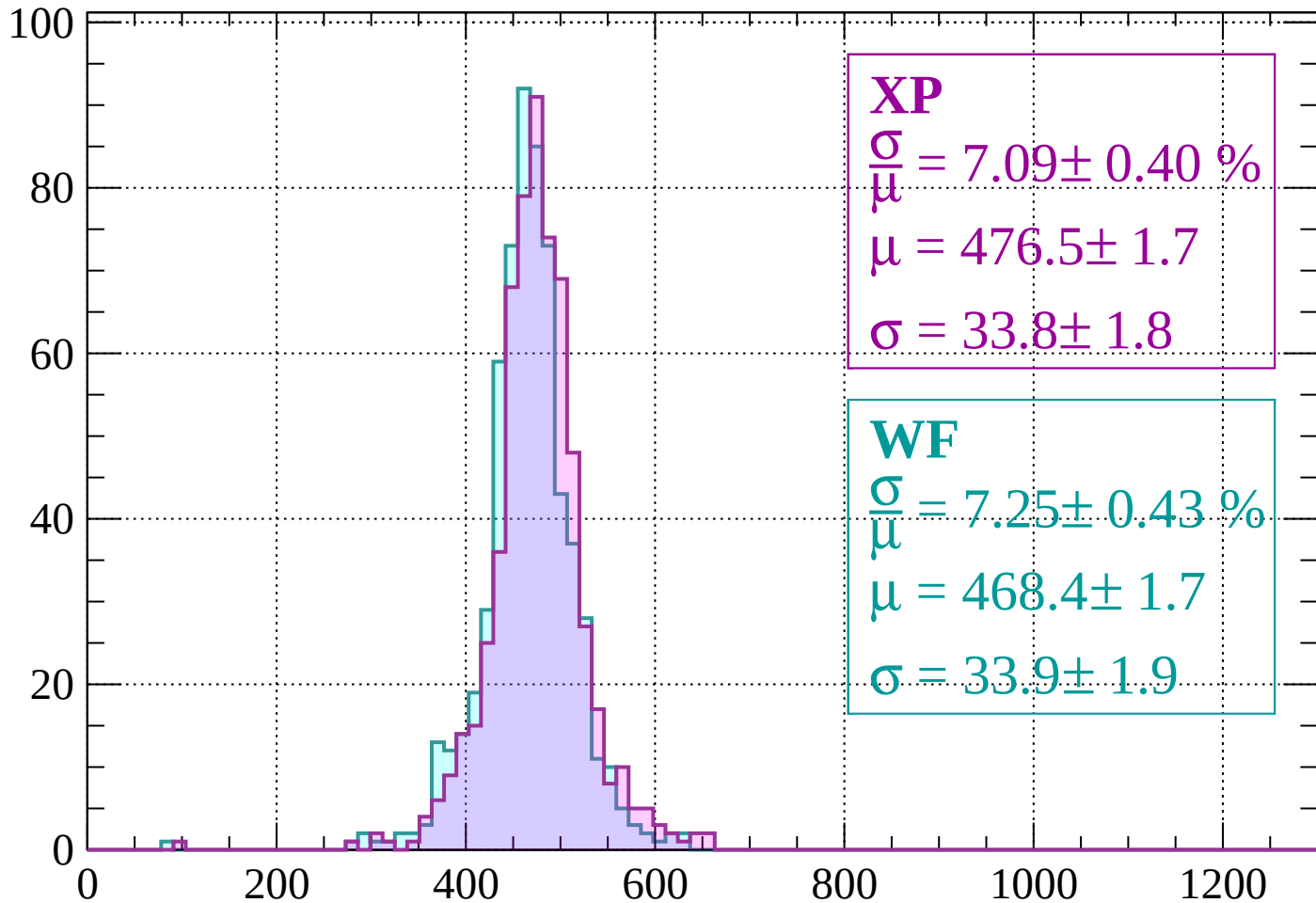
$$\mu = 472.6 \pm 1.7$$

$$\sigma = 36.9 \pm 1.7$$

dE/dx (ADC counts/cm)

Energy loss | $-1800 < p < -1760$

Count



XP

$$\frac{\sigma}{\mu} = 7.09 \pm 0.40 \%$$

$$\mu = 476.5 \pm 1.7$$

$$\sigma = 33.8 \pm 1.8$$

WF

$$\frac{\sigma}{\mu} = 7.25 \pm 0.43 \%$$

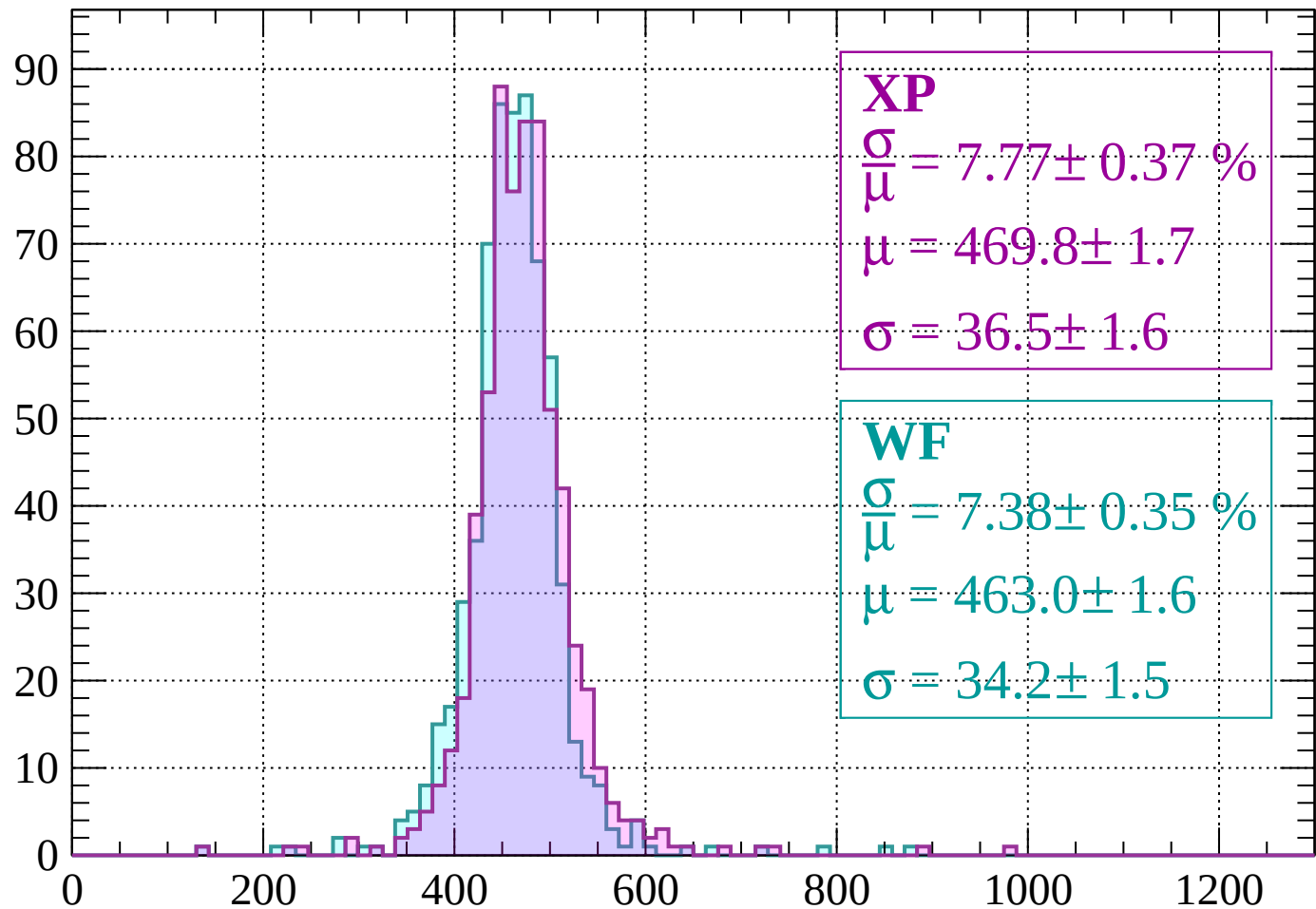
$$\mu = 468.4 \pm 1.7$$

$$\sigma = 33.9 \pm 1.9$$

dE/dx (ADC counts/cm)

Energy loss | -1760 < p < -1720

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 7.77 \pm 0.37 \%$$

$$\mu = 469.8 \pm 1.7$$

$$\sigma = 36.5 \pm 1.6$$

WF

$$\frac{\sigma}{\bar{\mu}} = 7.38 \pm 0.35 \%$$

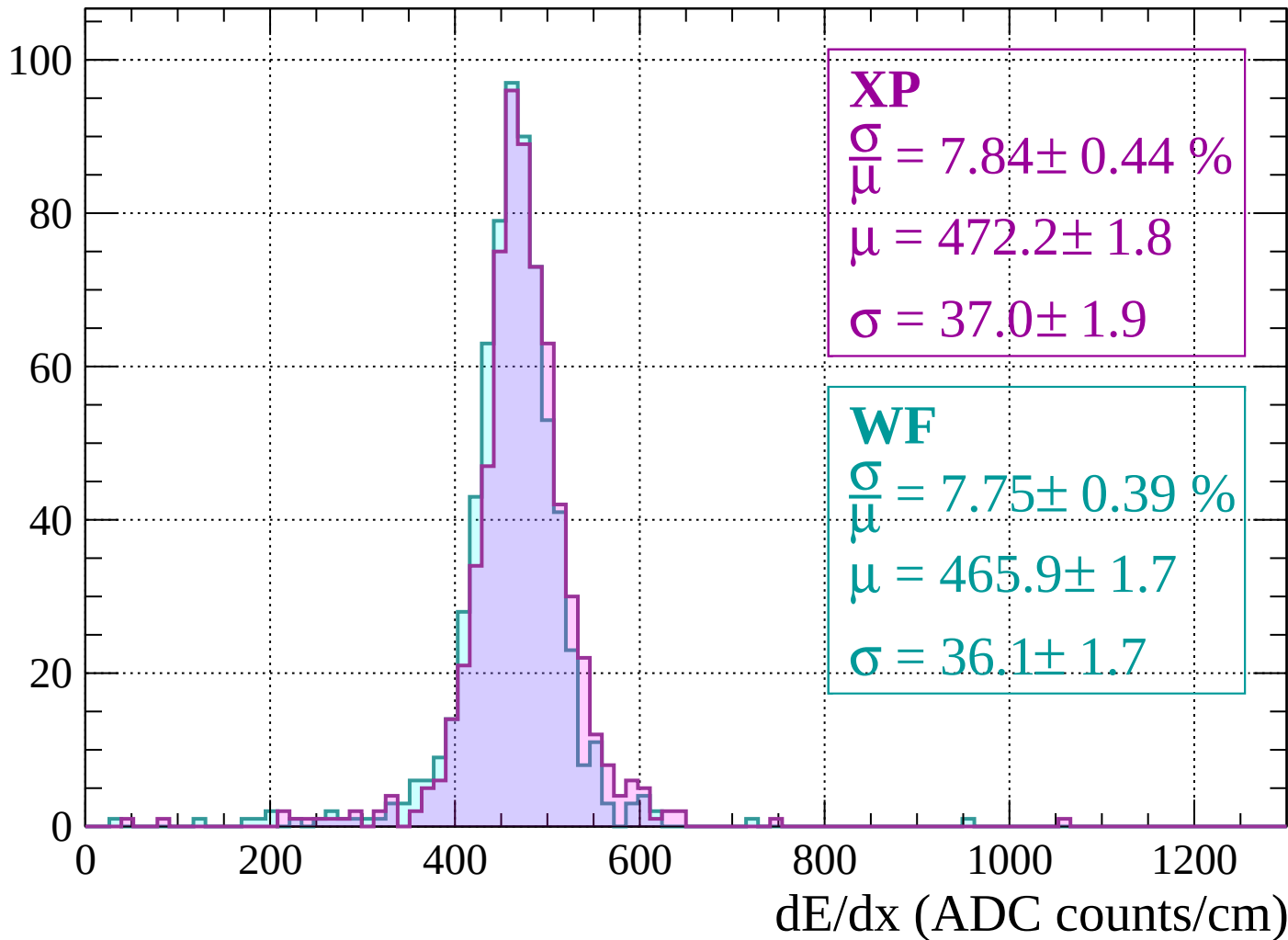
$$\mu = 463.0 \pm 1.6$$

$$\sigma = 34.2 \pm 1.5$$

dE/dx (ADC counts/cm)

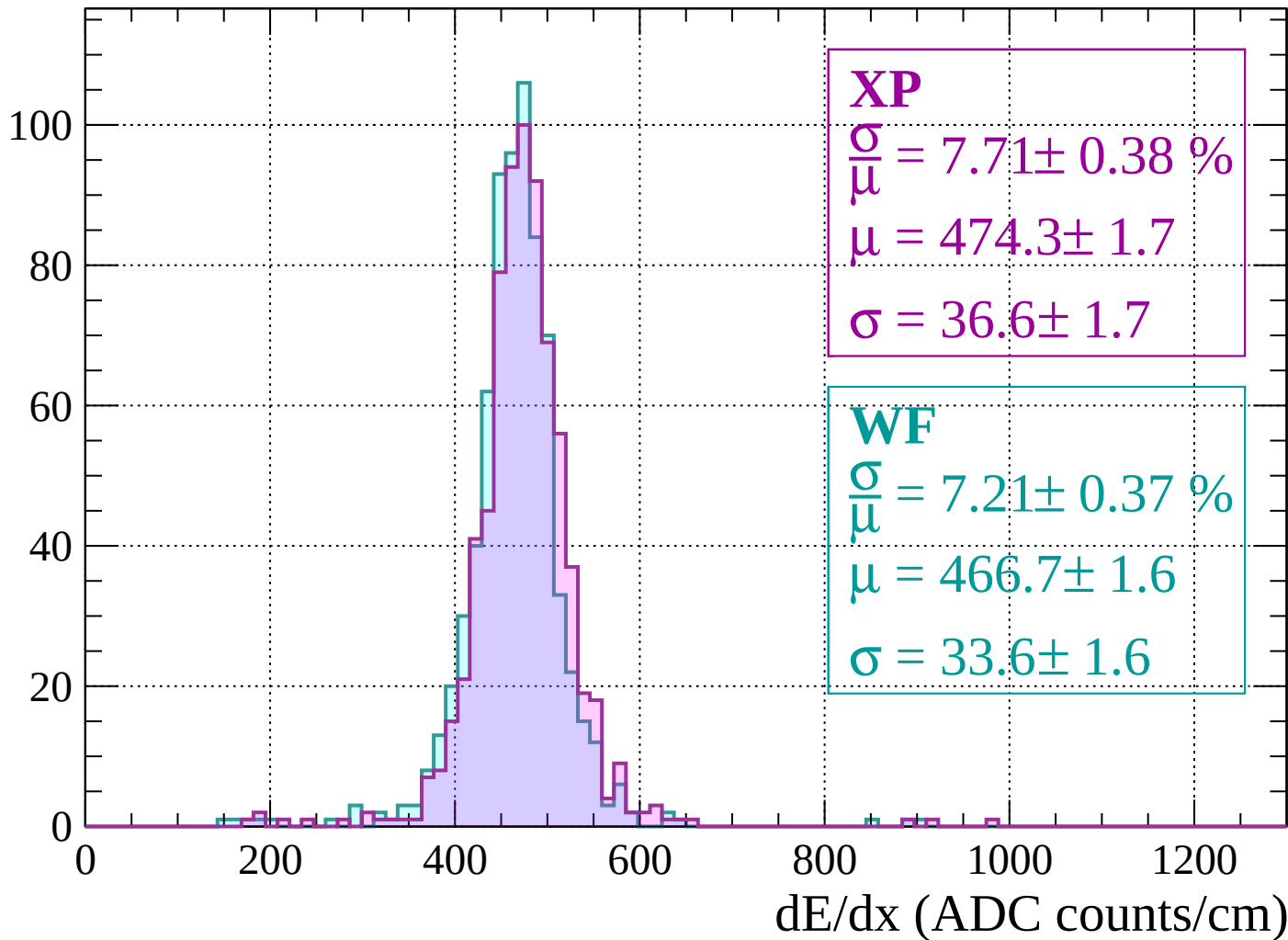
Energy loss | $-1720 < p < -1680$

Count



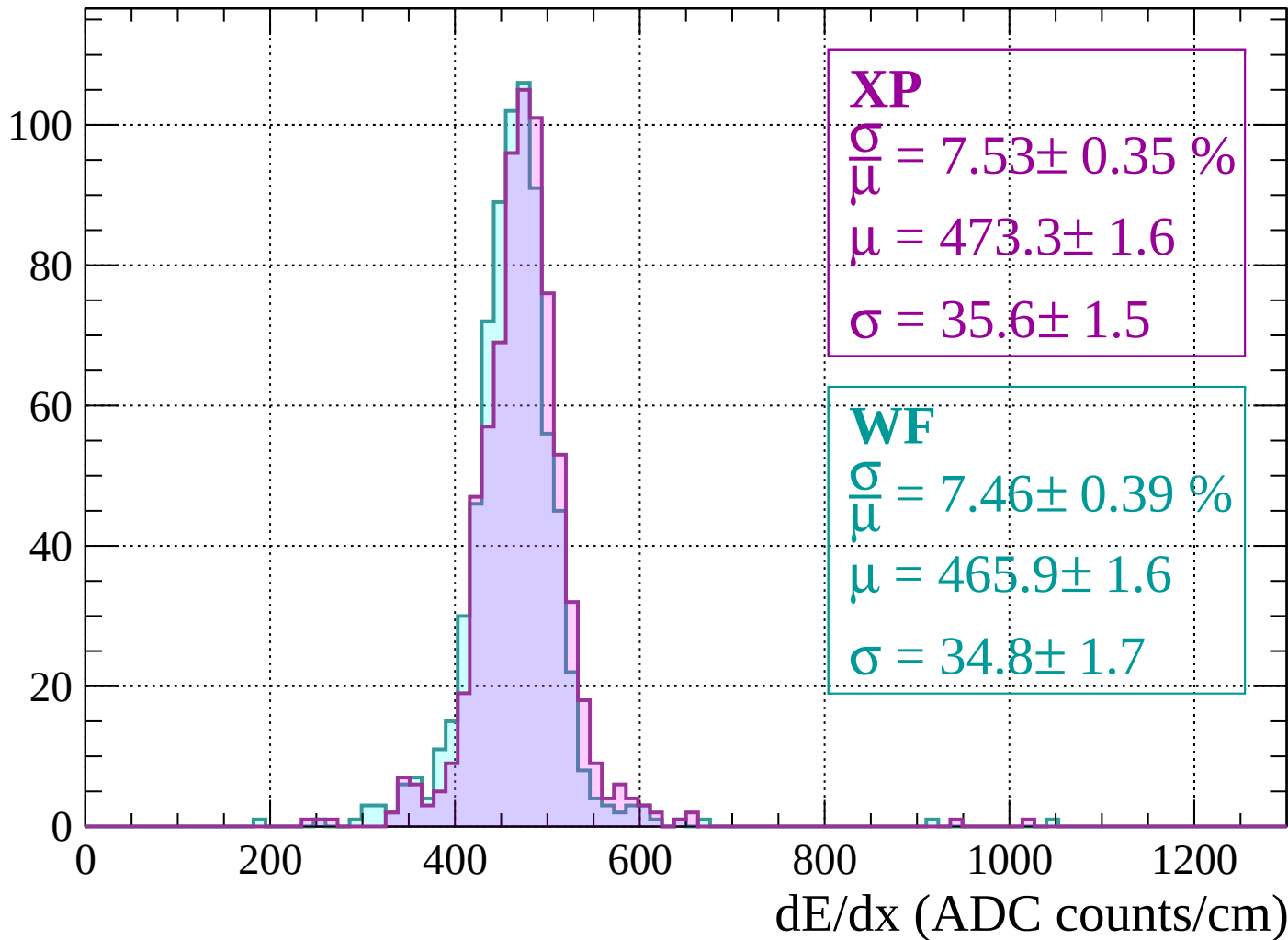
Energy loss | $-1680 < p < -1640$

Count



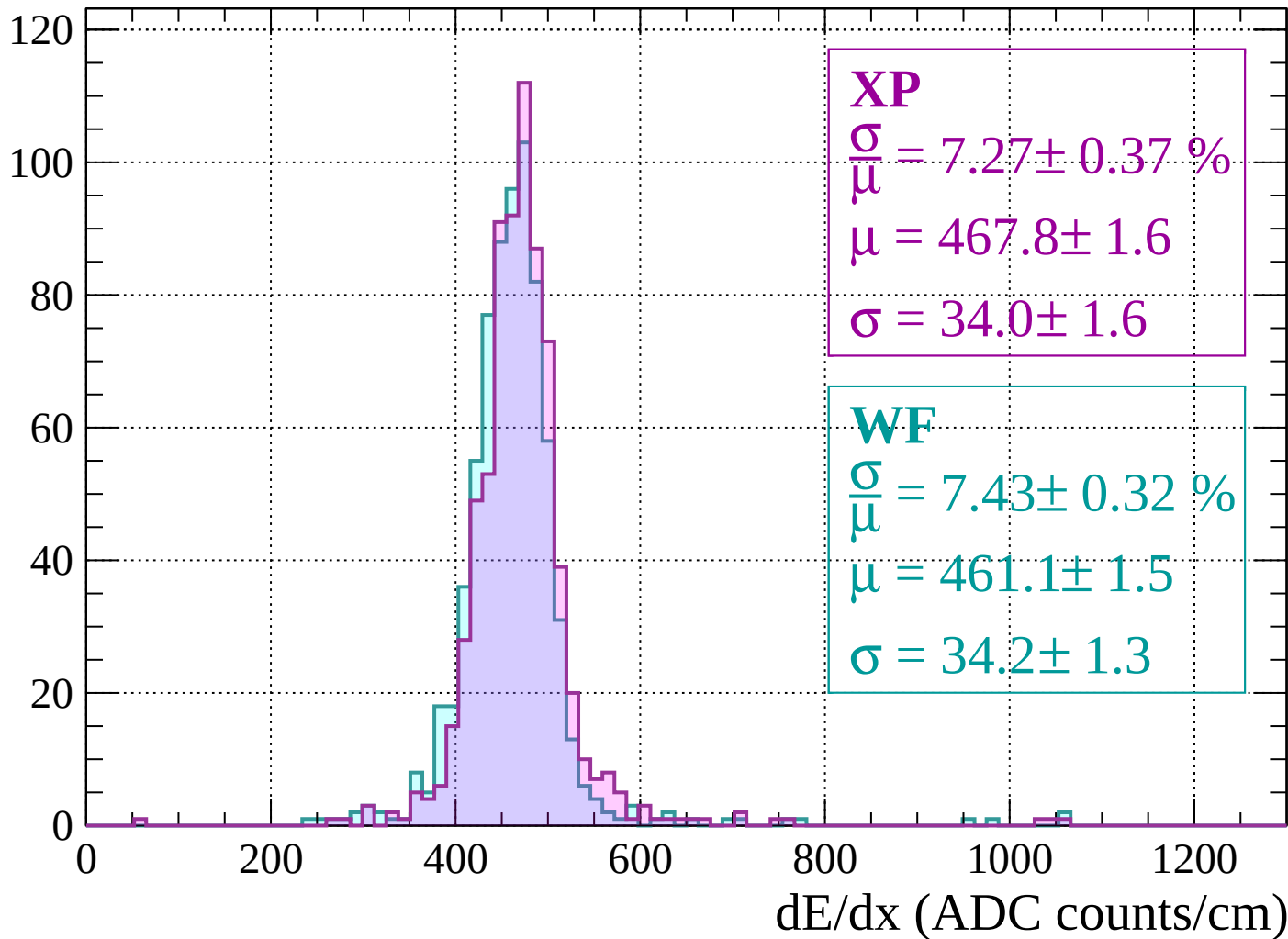
Energy loss | $-1640 < p < -1600$

Count



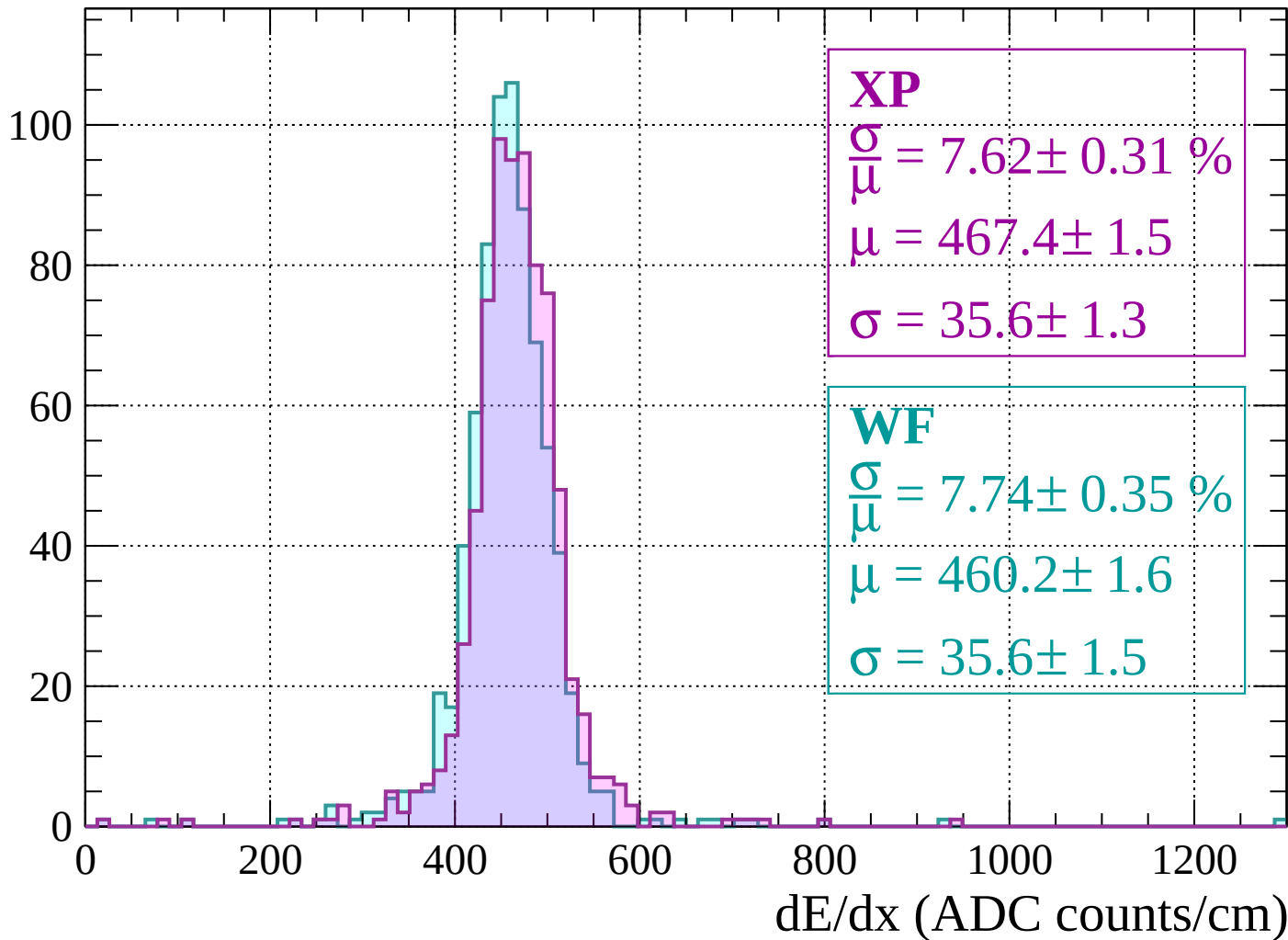
Energy loss | $-1600 < p < -1560$

Count



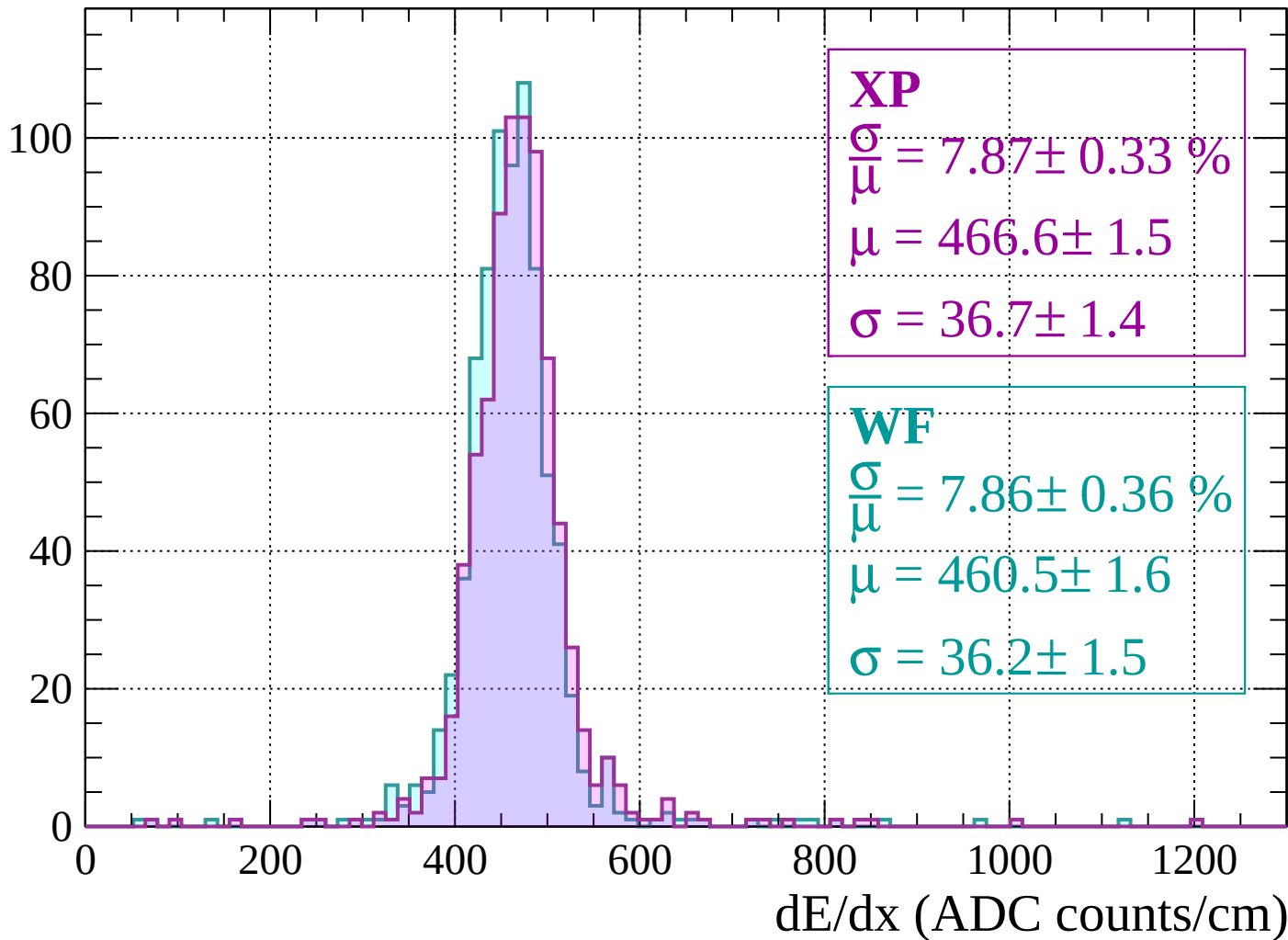
Energy loss | $-1560 < p < -1520$

Count



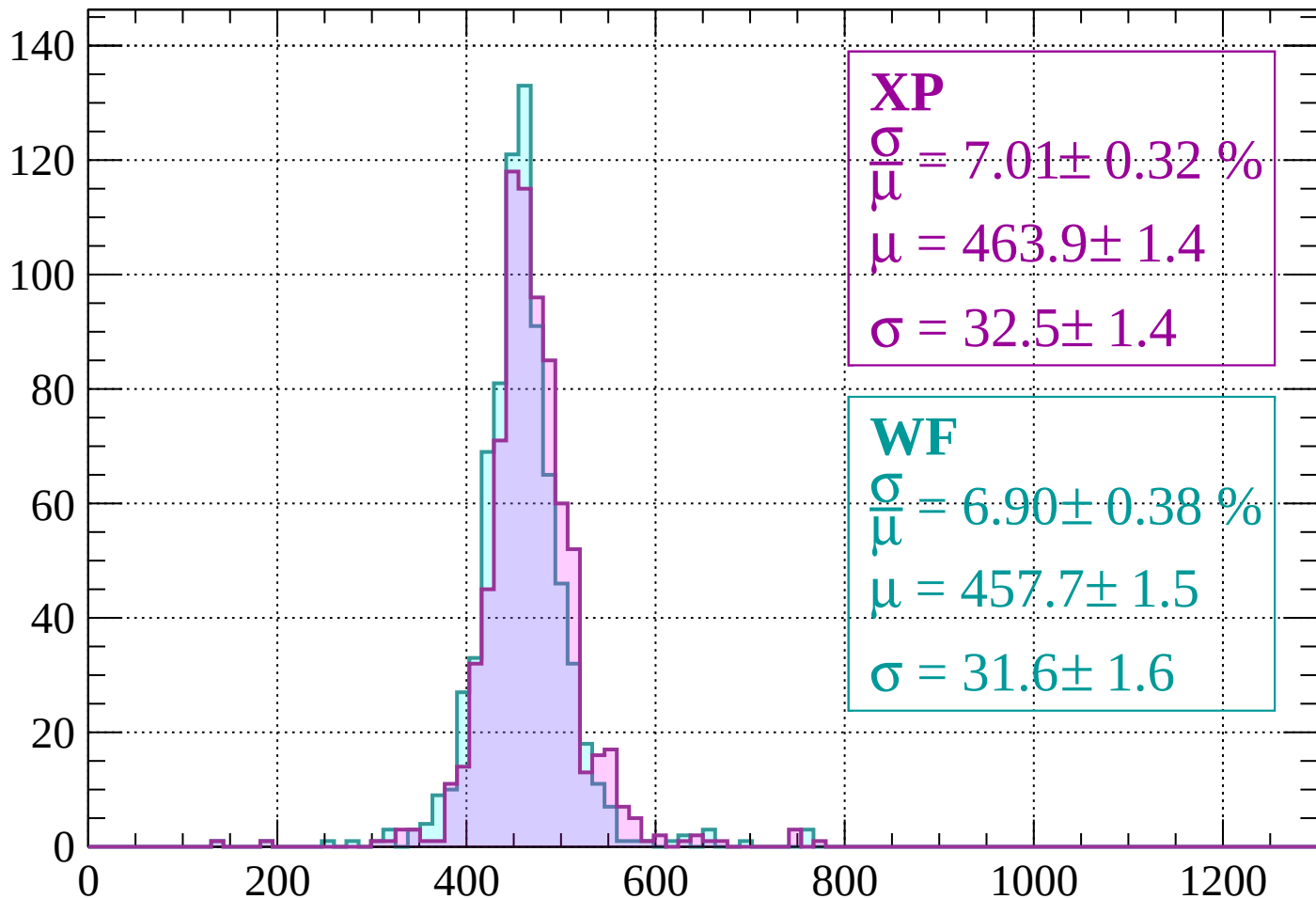
Energy loss | $-1520 < p < -1480$

Count



Energy loss | $-1480 < p < -1440$

Count



XP

$$\frac{\sigma}{\mu} = 7.01 \pm 0.32 \%$$

$$\mu = 463.9 \pm 1.4$$

$$\sigma = 32.5 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 6.90 \pm 0.38 \%$$

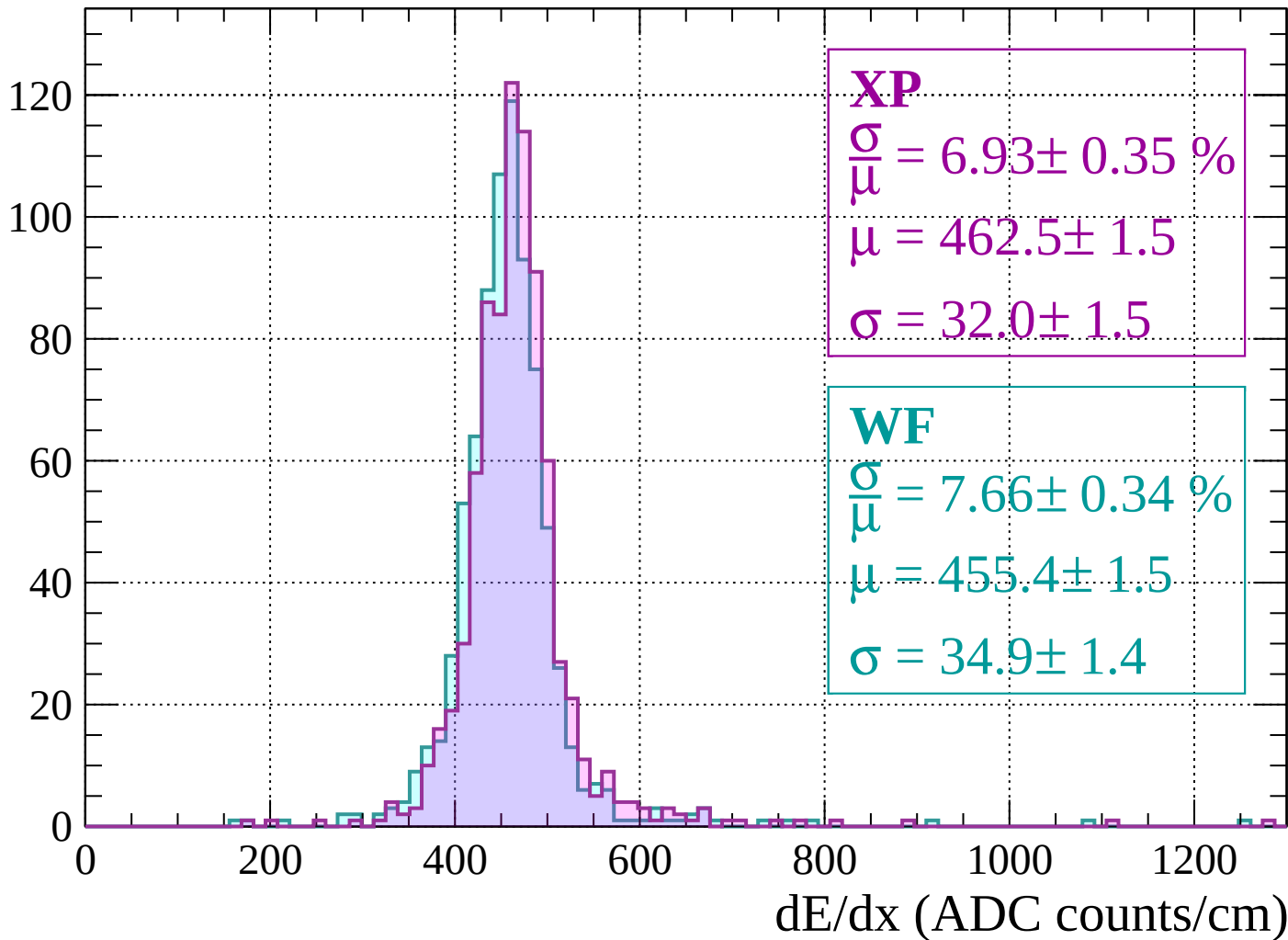
$$\mu = 457.7 \pm 1.5$$

$$\sigma = 31.6 \pm 1.6$$

dE/dx (ADC counts/cm)

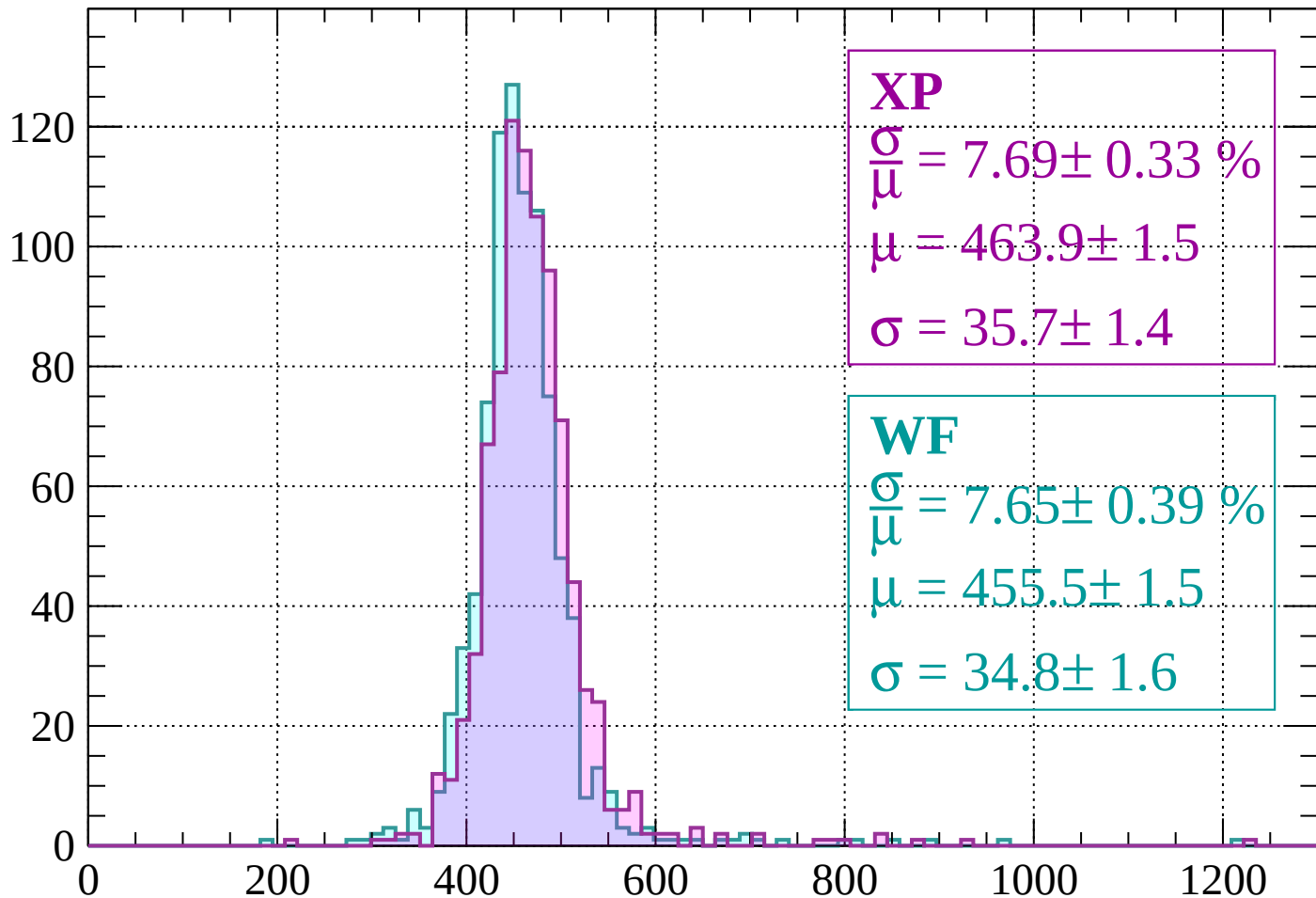
Energy loss | $-1440 < p < -1400$

Count



Energy loss | $-1400 < p < -1360$

Count



XP

$$\frac{\sigma}{\mu} = 7.69 \pm 0.33 \%$$

$$\mu = 463.9 \pm 1.5$$

$$\sigma = 35.7 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 7.65 \pm 0.39 \%$$

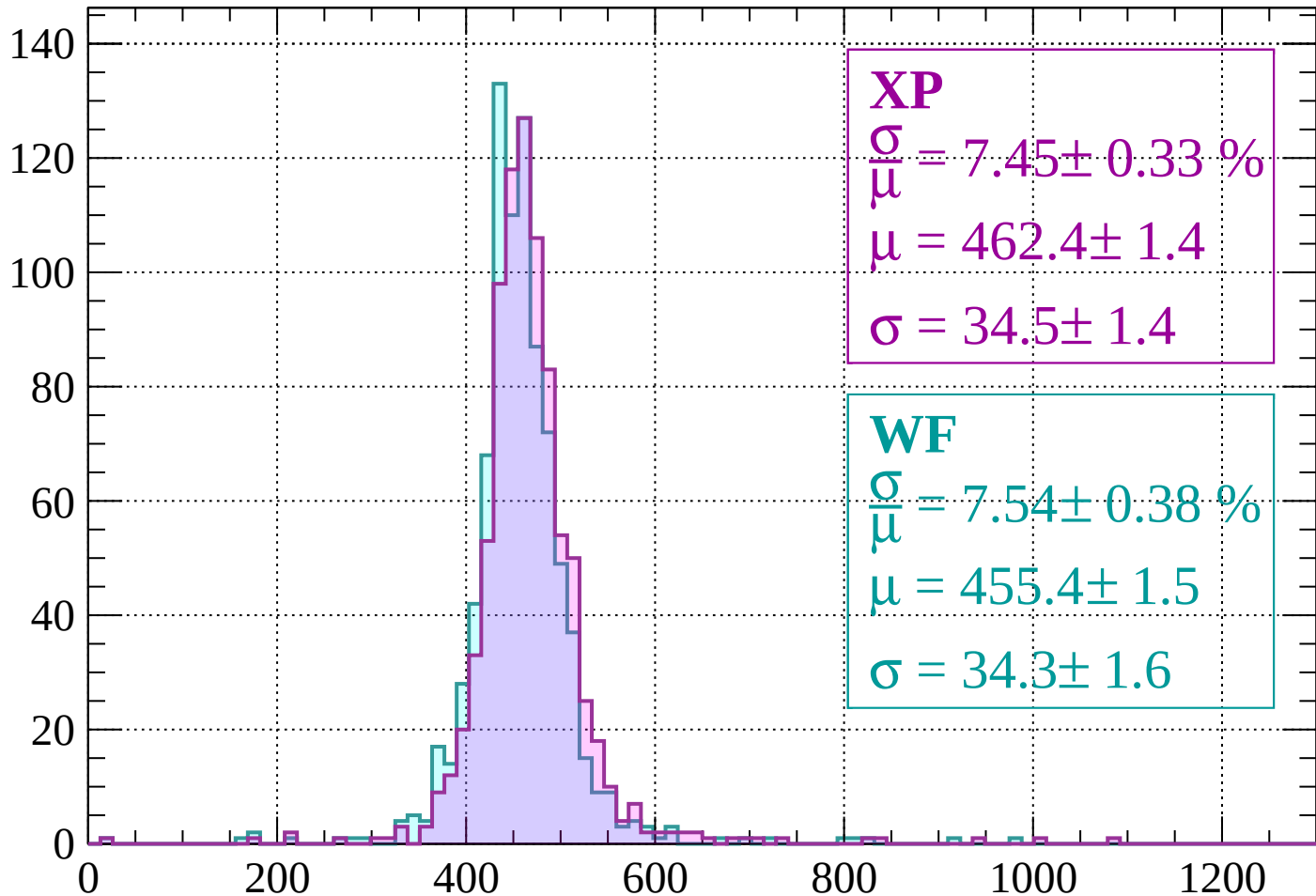
$$\mu = 455.5 \pm 1.5$$

$$\sigma = 34.8 \pm 1.6$$

dE/dx (ADC counts/cm)

Energy loss | $-1360 < p < -1320$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 7.45 \pm 0.33 \%$$

$$\mu = 462.4 \pm 1.4$$

$$\sigma = 34.5 \pm 1.4$$

WF

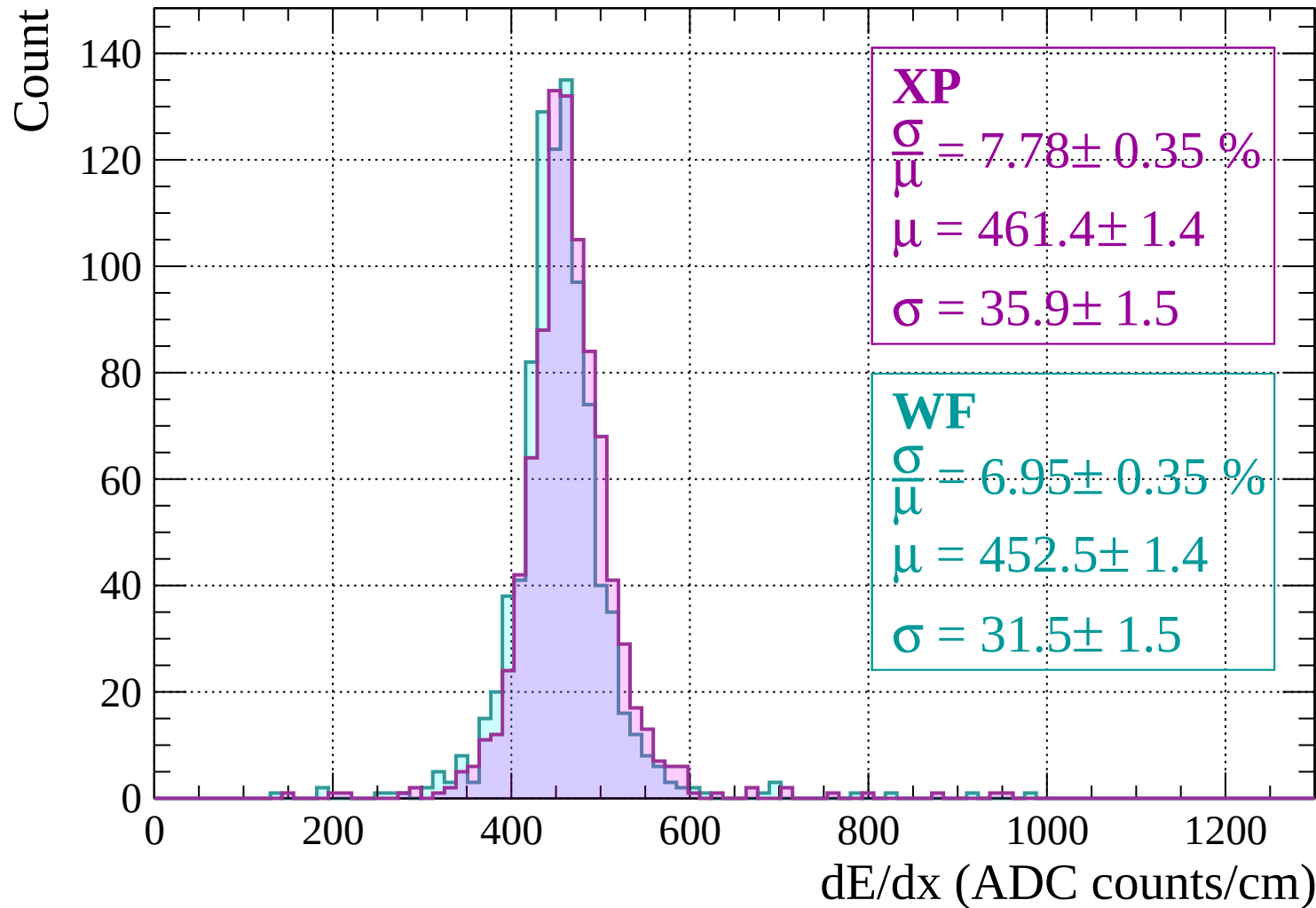
$$\frac{\sigma}{\bar{\mu}} = 7.54 \pm 0.38 \%$$

$$\mu = 455.4 \pm 1.5$$

$$\sigma = 34.3 \pm 1.6$$

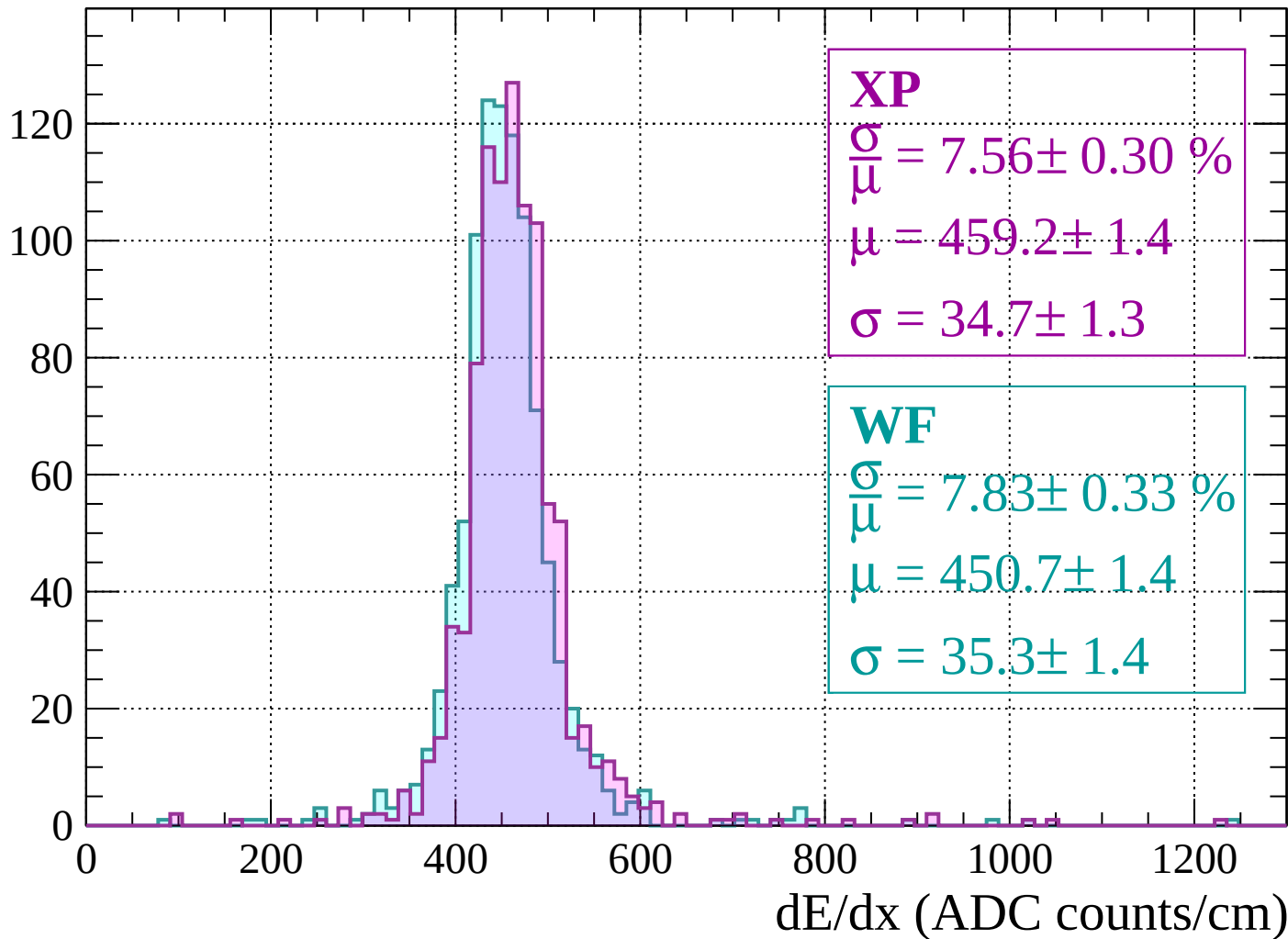
dE/dx (ADC counts/cm)

Energy loss | -1320 < p < -1280



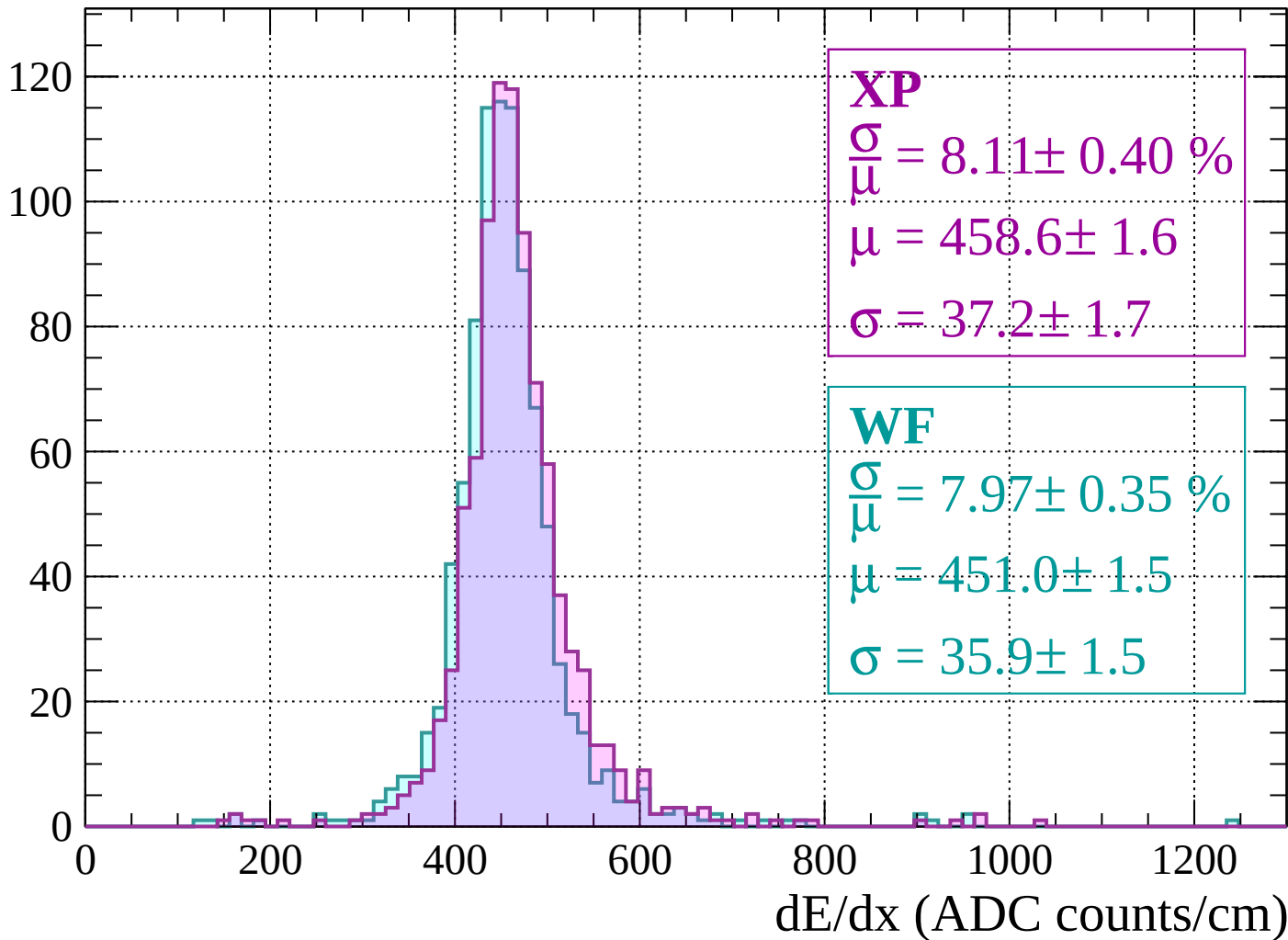
Energy loss | $-1280 < p < -1240$

Count



Energy loss | $-1240 < p < -1200$

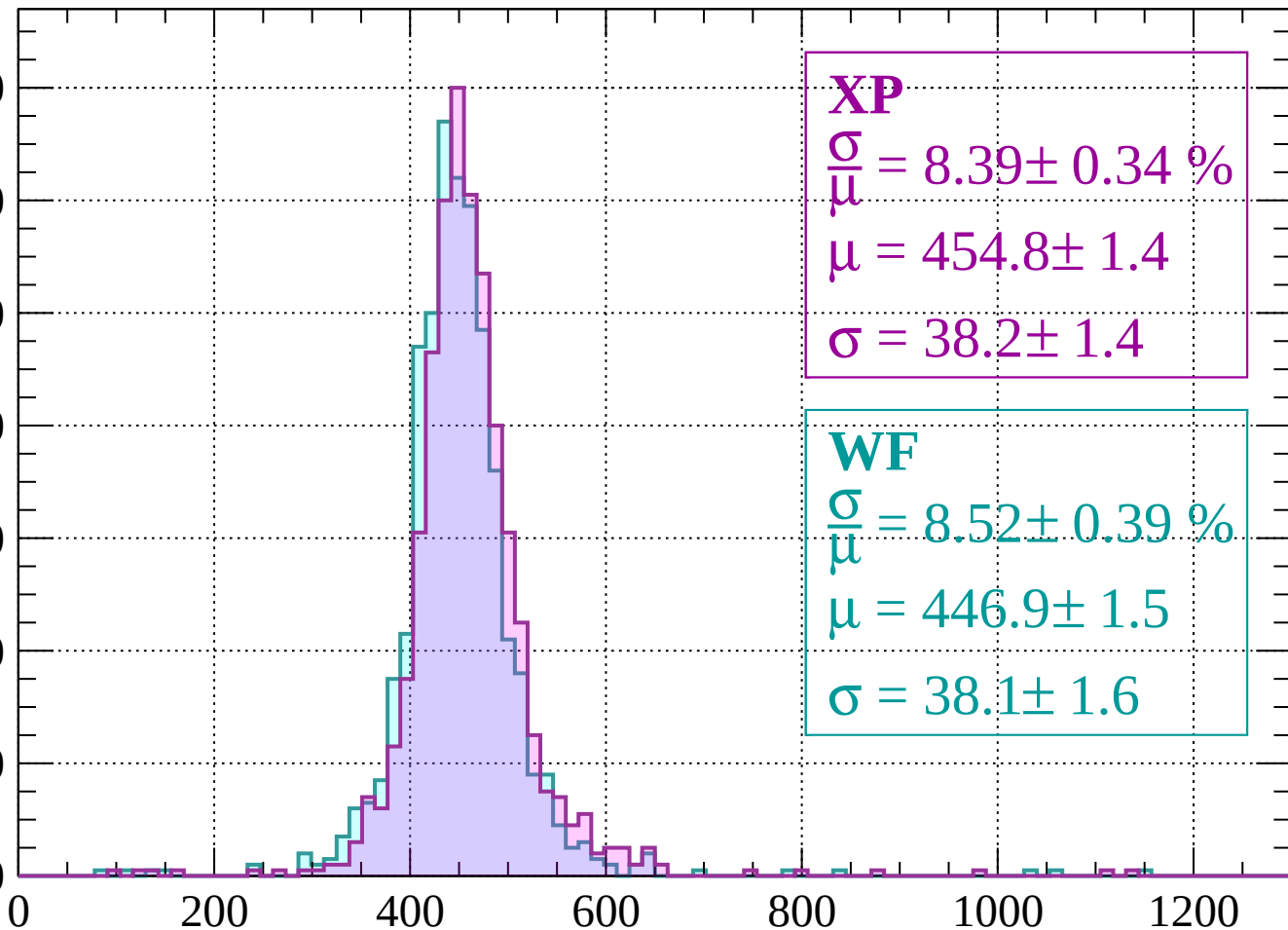
Count



Energy loss | -1200 < p < -1160

Count

140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 8.39 \pm 0.34 \%$$

$$\mu = 454.8 \pm 1.4$$

$$\sigma = 38.2 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 8.52 \pm 0.39 \%$$

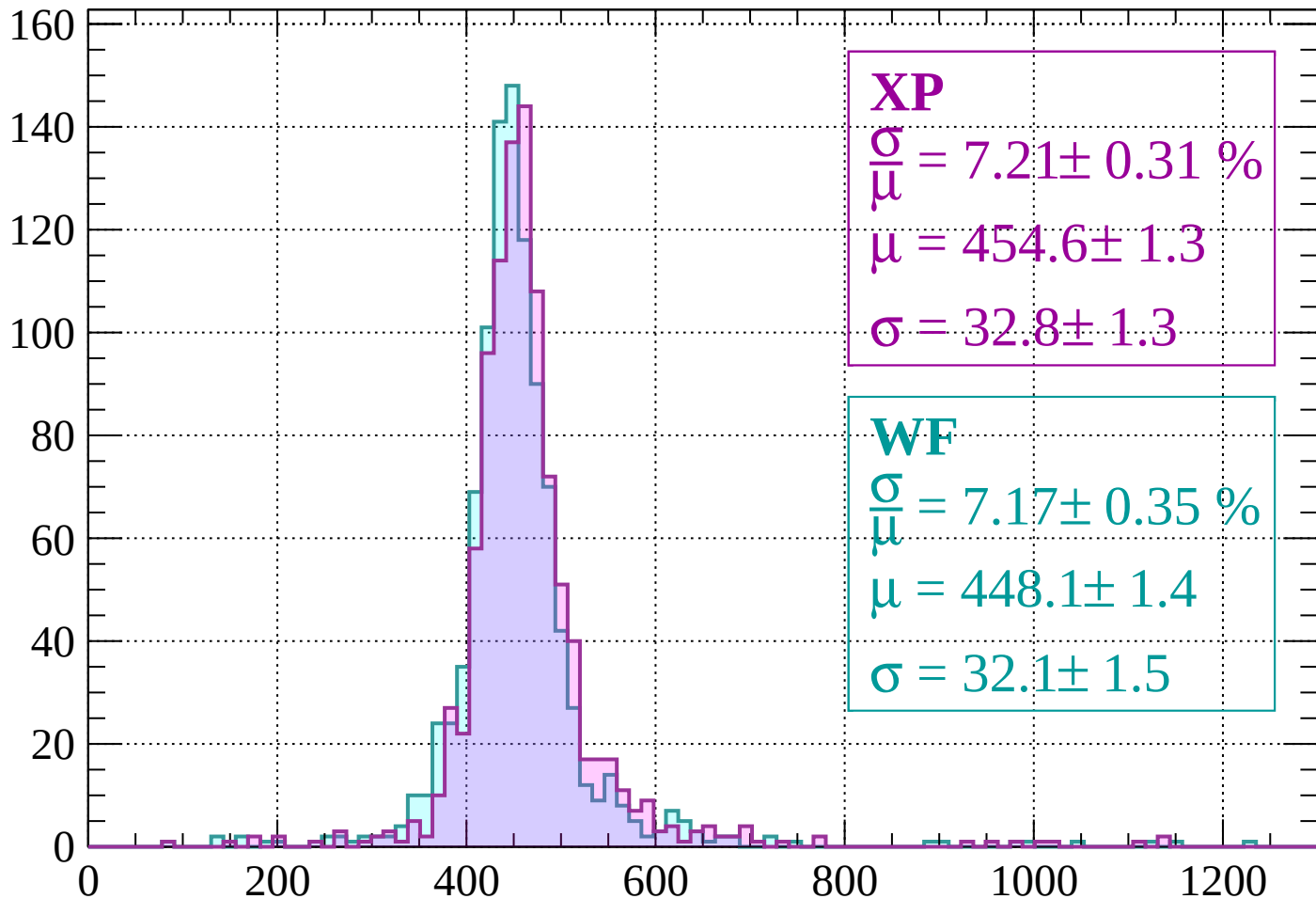
$$\mu = 446.9 \pm 1.5$$

$$\sigma = 38.1 \pm 1.6$$

dE/dx (ADC counts/cm)

Energy loss | -1160 < p < -1120

Count



XP

$$\frac{\sigma}{\mu} = 7.21 \pm 0.31 \%$$

$$\mu = 454.6 \pm 1.3$$

$$\sigma = 32.8 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 7.17 \pm 0.35 \%$$

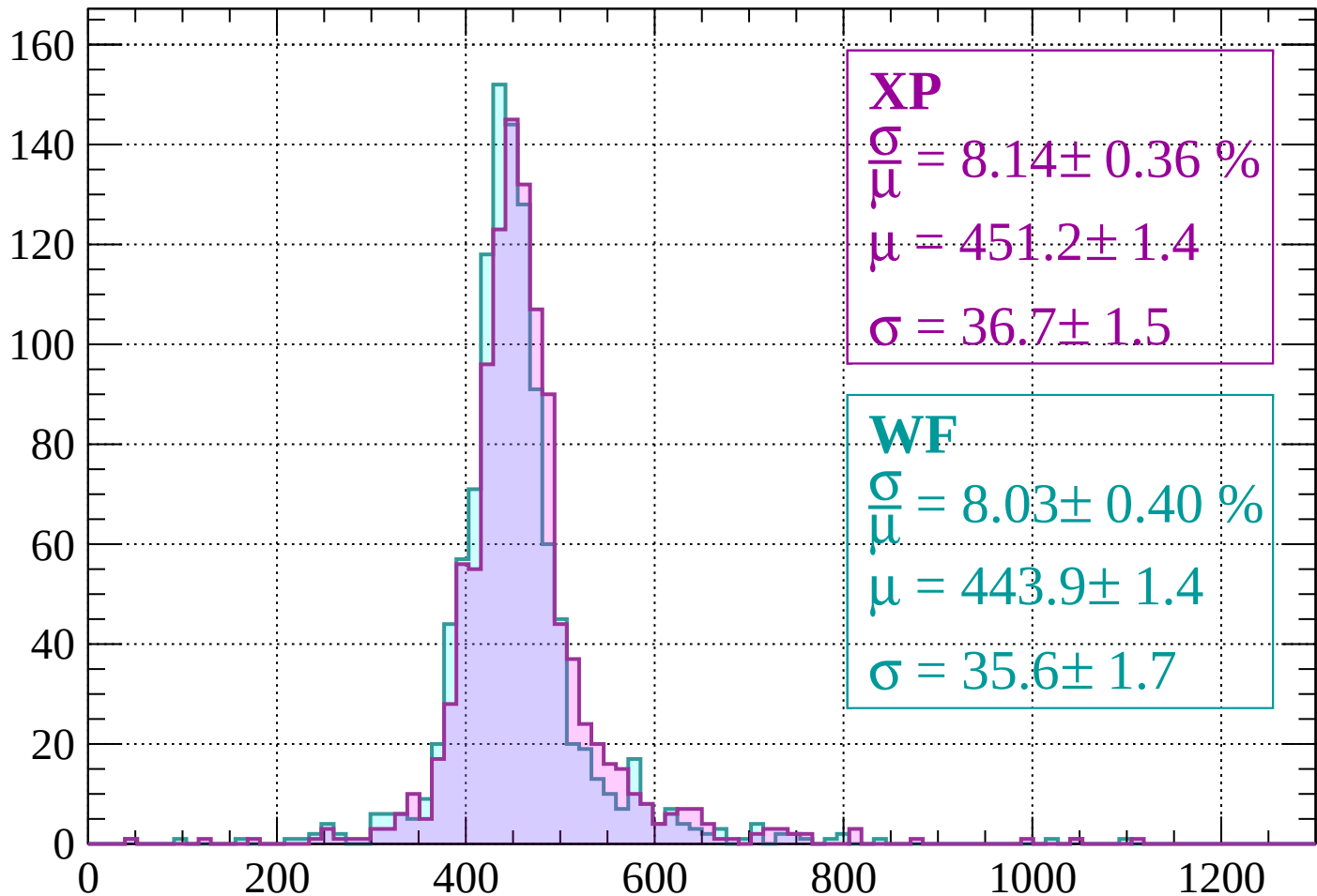
$$\mu = 448.1 \pm 1.4$$

$$\sigma = 32.1 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | -1120 < p < -1080

Count



XP

$$\frac{\sigma}{\mu} = 8.14 \pm 0.36 \%$$

$$\mu = 451.2 \pm 1.4$$

$$\sigma = 36.7 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 8.03 \pm 0.40 \%$$

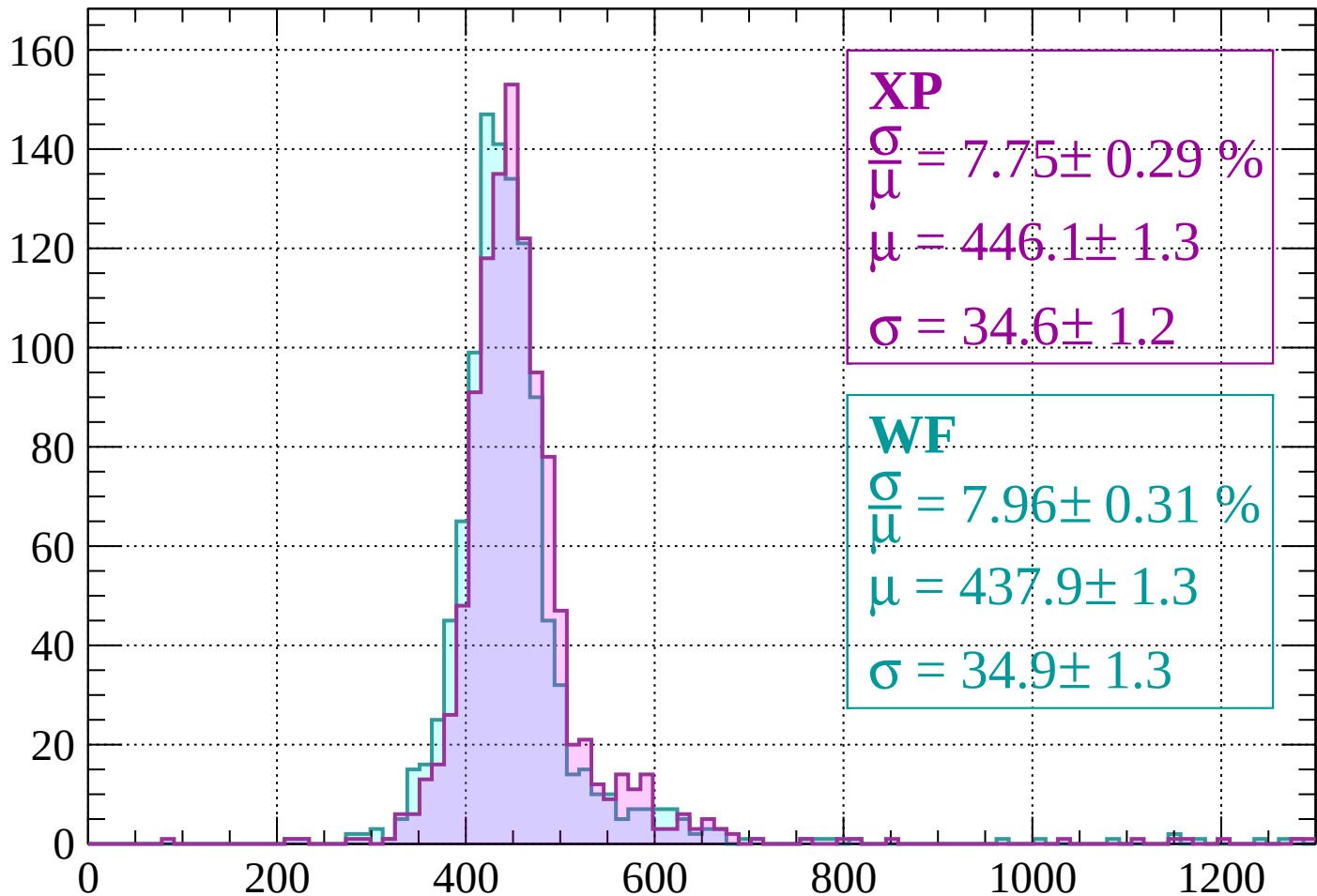
$$\mu = 443.9 \pm 1.4$$

$$\sigma = 35.6 \pm 1.7$$

dE/dx (ADC counts/cm)

Energy loss | $-1080 < p < -1040$

Count



XP

$$\frac{\sigma}{\mu} = 7.75 \pm 0.29 \%$$

$$\mu = 446.1 \pm 1.3$$

$$\sigma = 34.6 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 7.96 \pm 0.31 \%$$

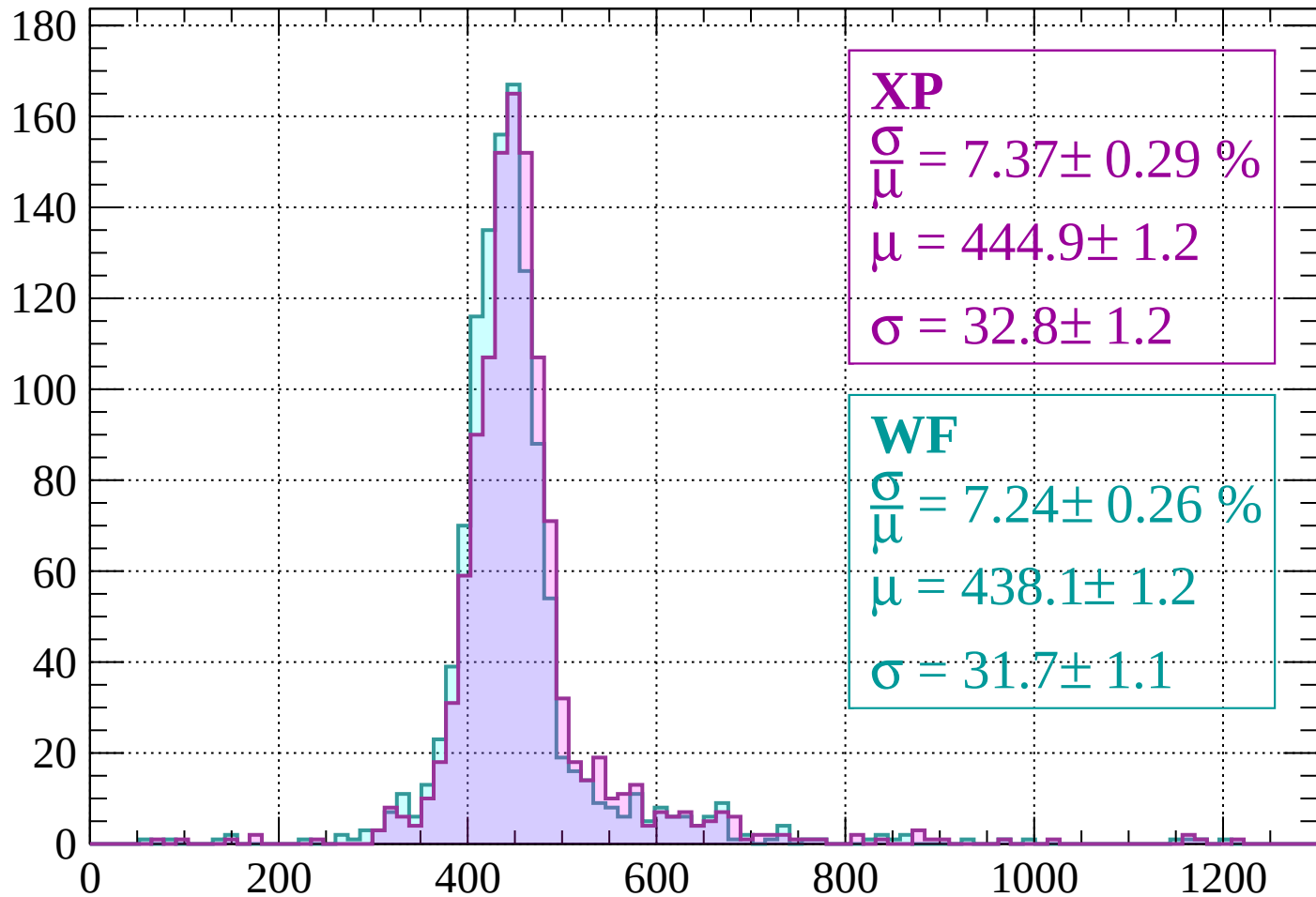
$$\mu = 437.9 \pm 1.3$$

$$\sigma = 34.9 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $-1040 < p < -1000$

Count



XP

$$\frac{\sigma}{\mu} = 7.37 \pm 0.29 \%$$

$$\mu = 444.9 \pm 1.2$$

$$\sigma = 32.8 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 7.24 \pm 0.26 \%$$

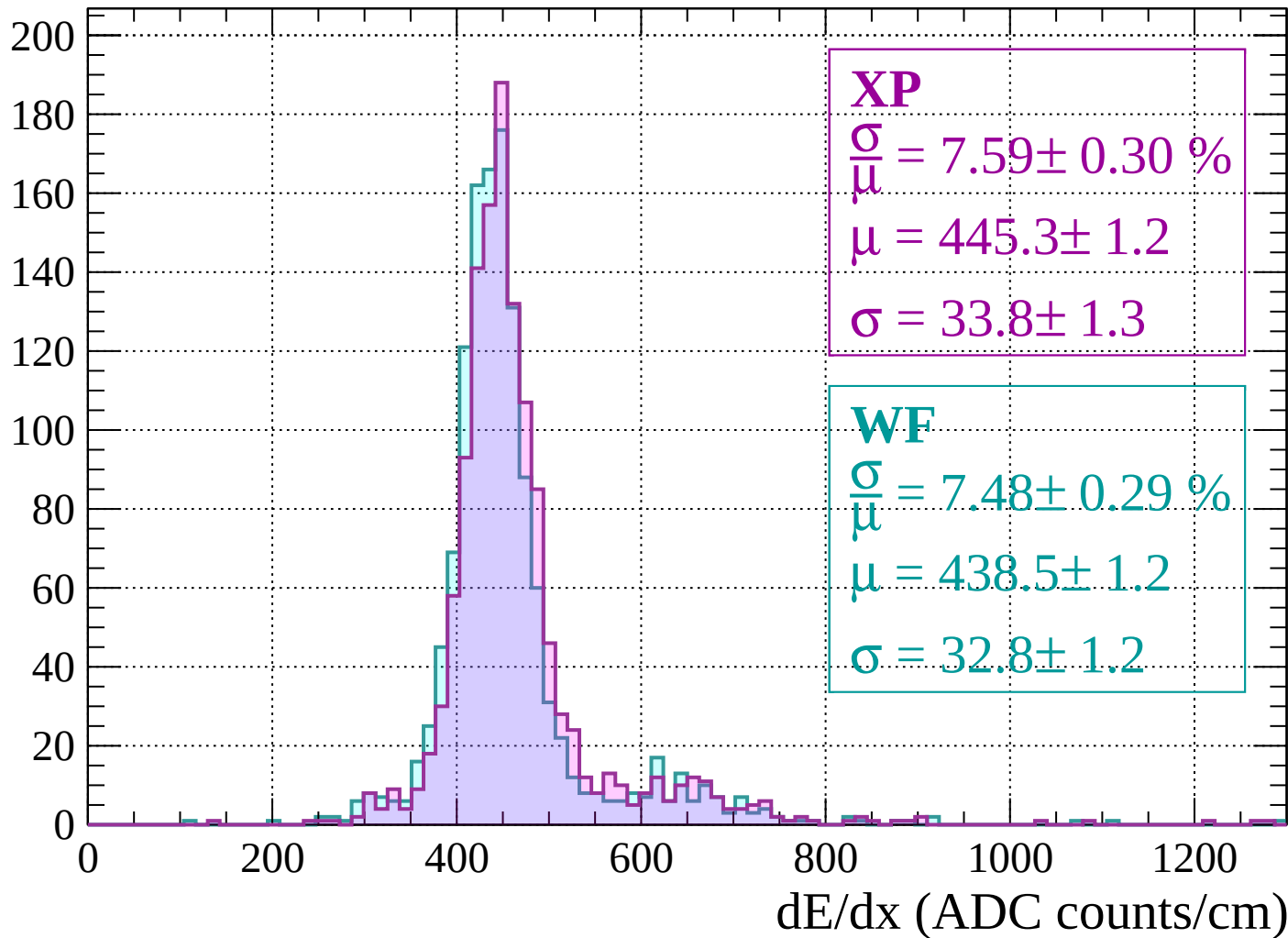
$$\mu = 438.1 \pm 1.2$$

$$\sigma = 31.7 \pm 1.1$$

dE/dx (ADC counts/cm)

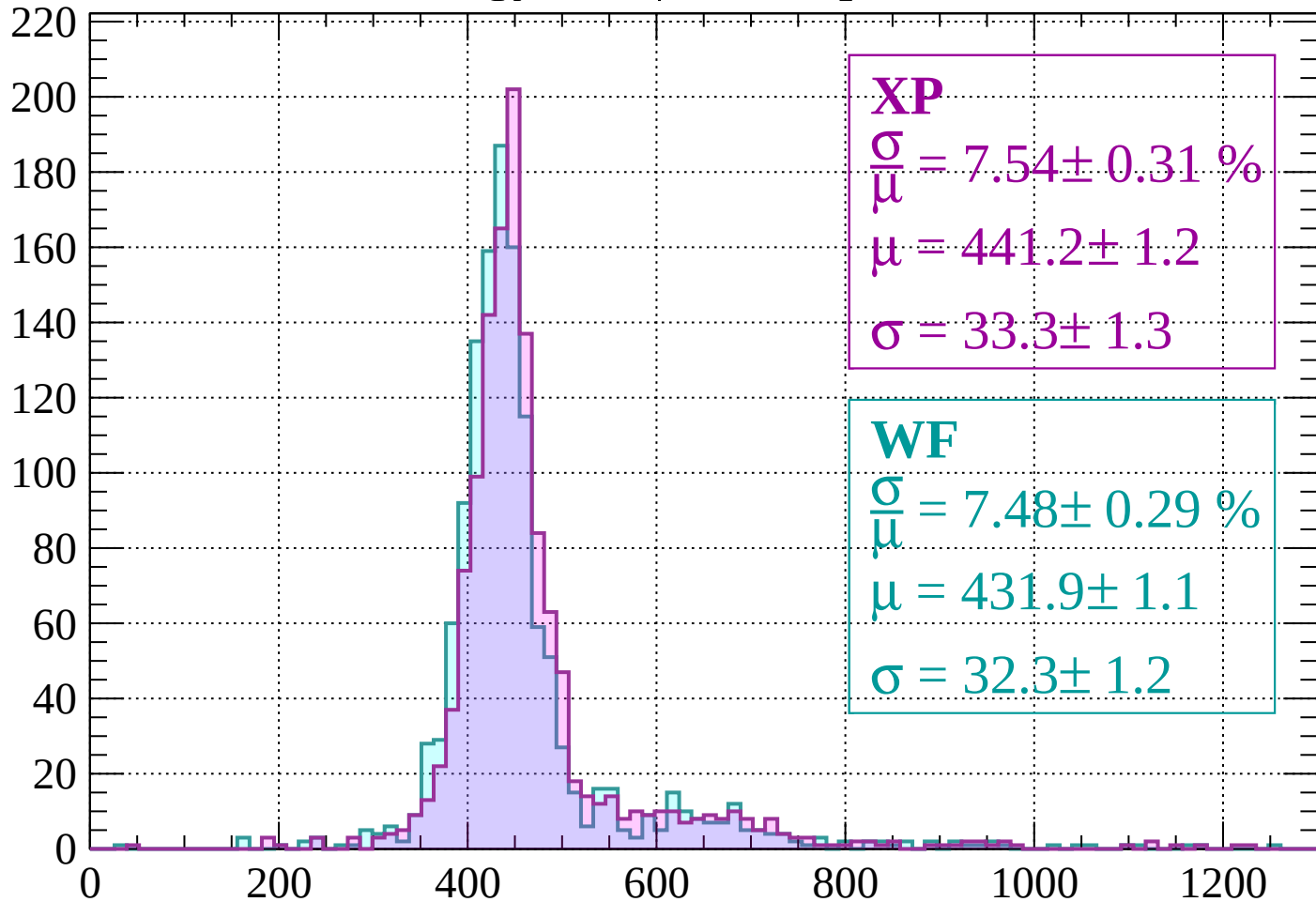
Energy loss | $-1000 < p < -960$

Count



Energy loss | $-960 < p < -920$

Count



XP

$$\frac{\sigma}{\mu} = 7.54 \pm 0.31 \%$$

$$\mu = 441.2 \pm 1.2$$

$$\sigma = 33.3 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 7.48 \pm 0.29 \%$$

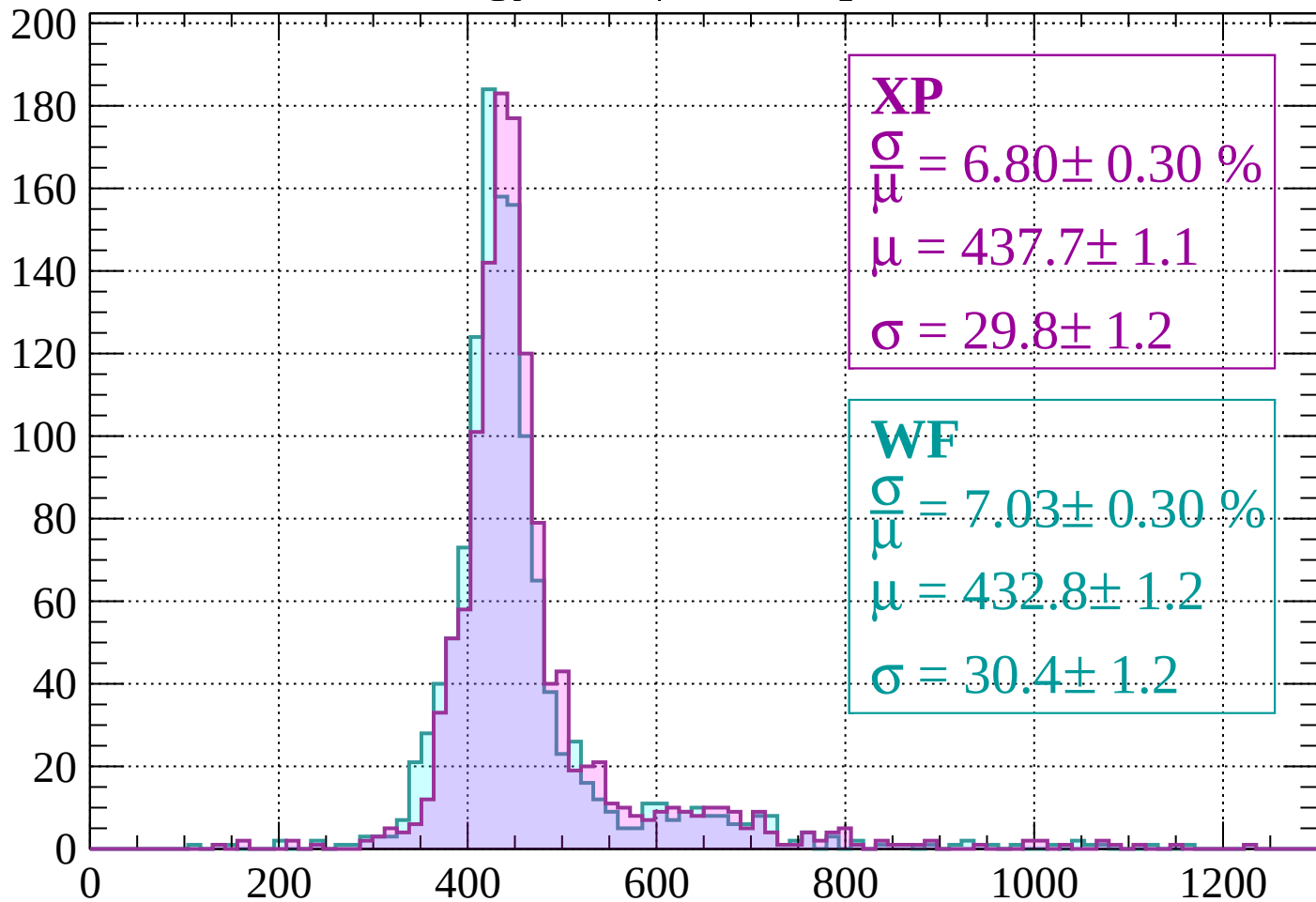
$$\mu = 431.9 \pm 1.1$$

$$\sigma = 32.3 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $-920 < p < -880$

Count



XP

$$\frac{\sigma}{\mu} = 6.80 \pm 0.30 \%$$

$$\mu = 437.7 \pm 1.1$$

$$\sigma = 29.8 \pm 1.2$$

WF

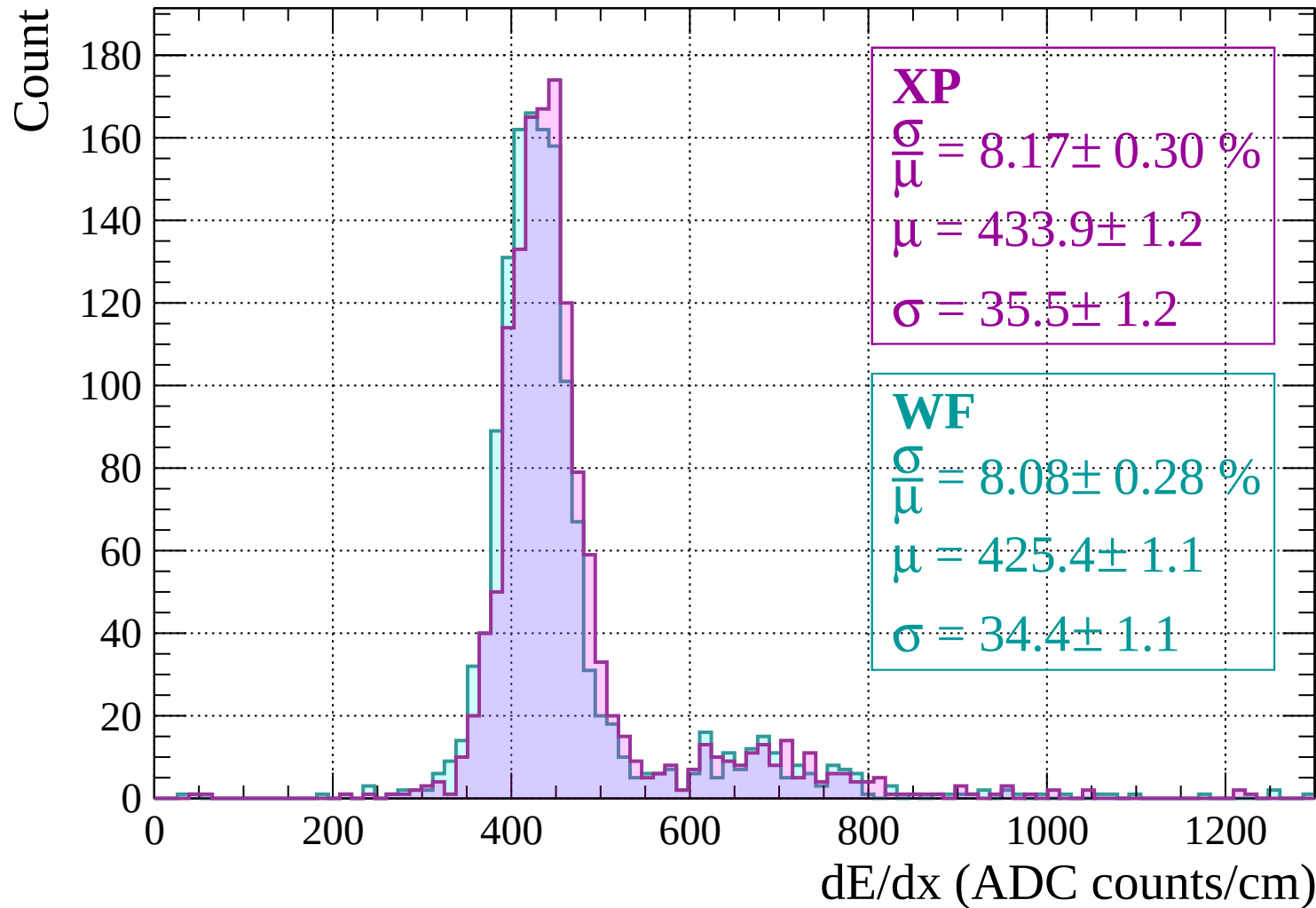
$$\frac{\sigma}{\mu} = 7.03 \pm 0.30 \%$$

$$\mu = 432.8 \pm 1.2$$

$$\sigma = 30.4 \pm 1.2$$

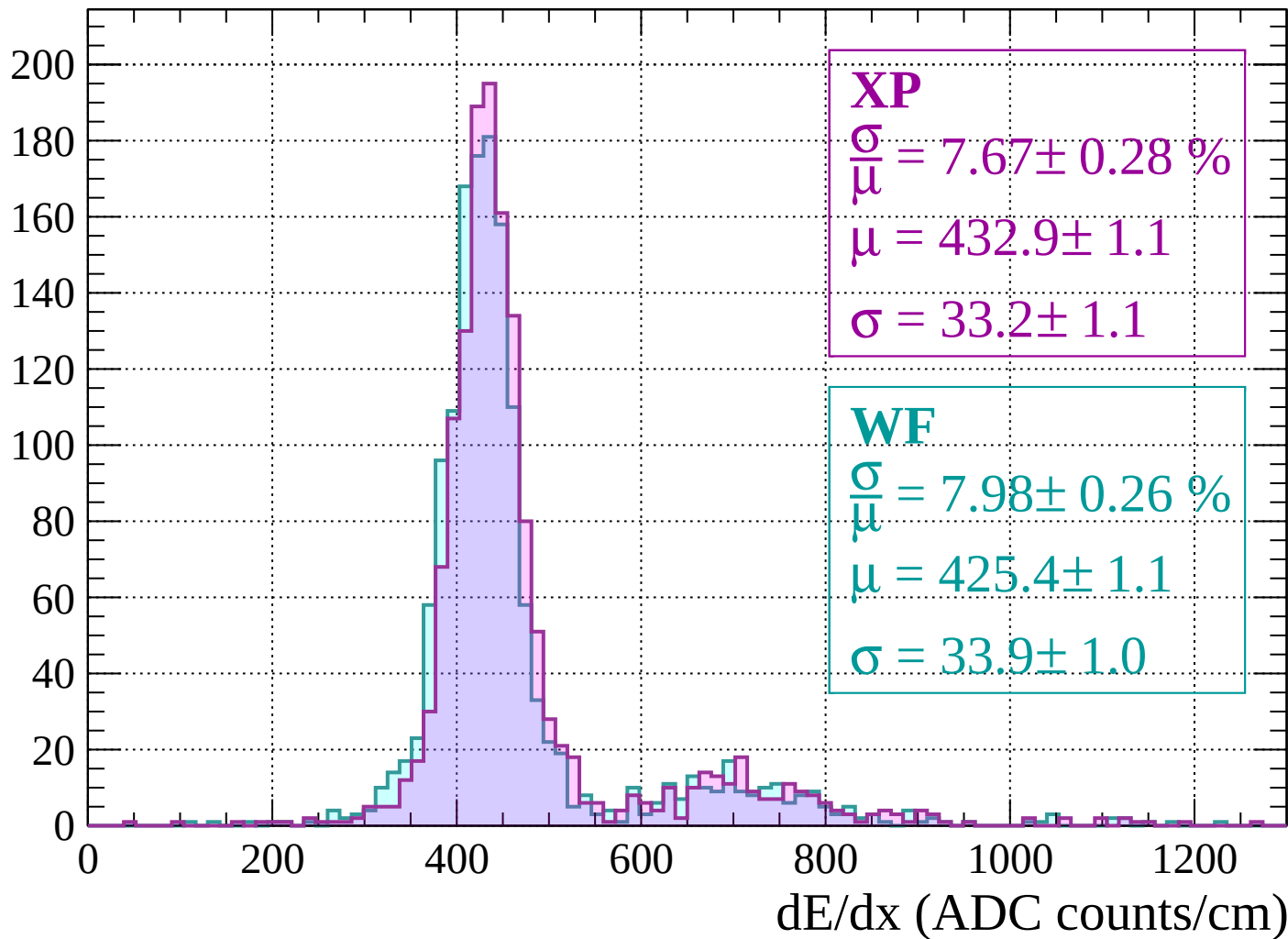
dE/dx (ADC counts/cm)

Energy loss | $-880 < p < -840$



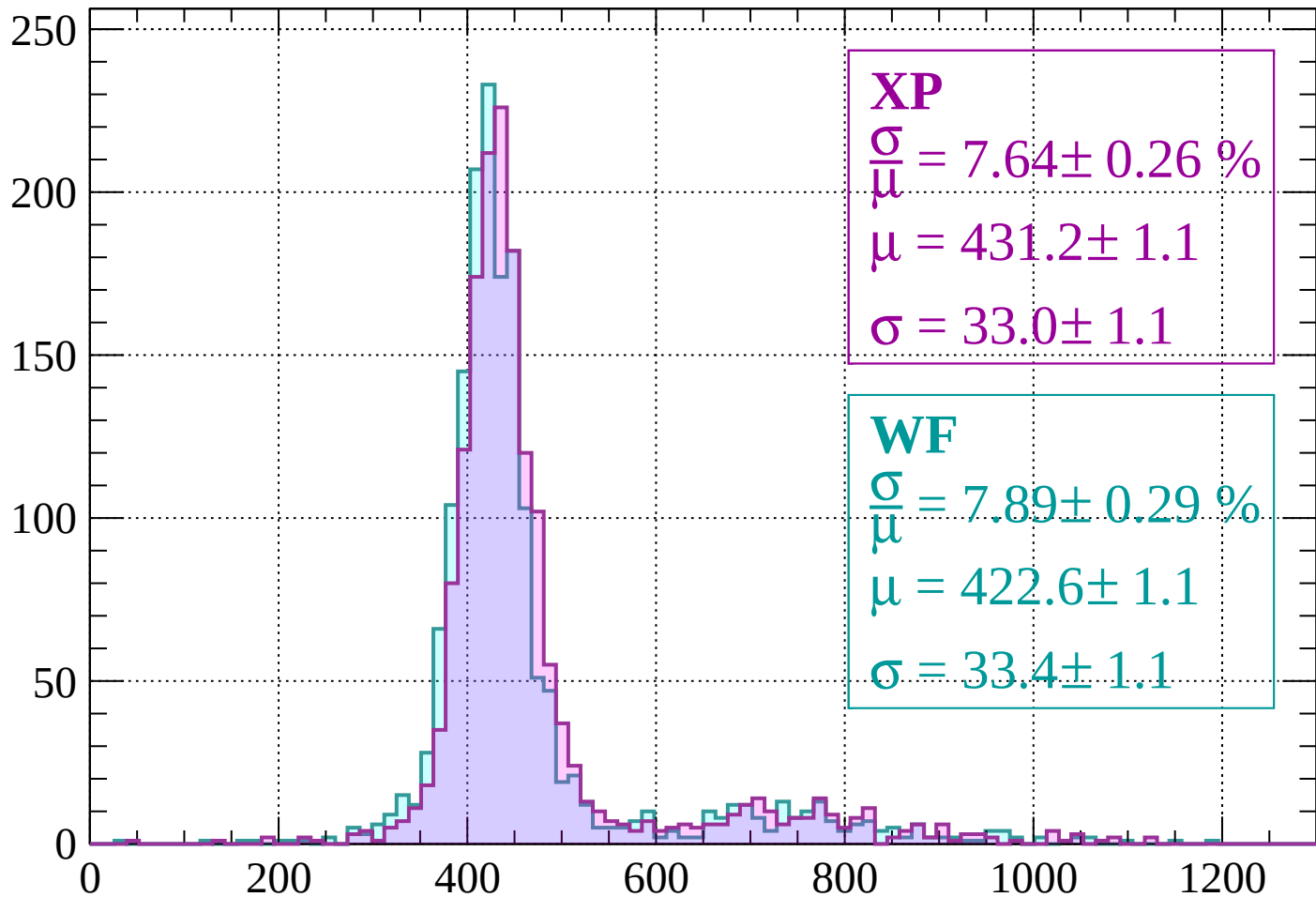
Energy loss | $-840 < p < -800$

Count



Energy loss | $-800 < p < -760$

Count



XP

$$\frac{\sigma}{\mu} = 7.64 \pm 0.26 \%$$

$$\mu = 431.2 \pm 1.1$$

$$\sigma = 33.0 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 7.89 \pm 0.29 \%$$

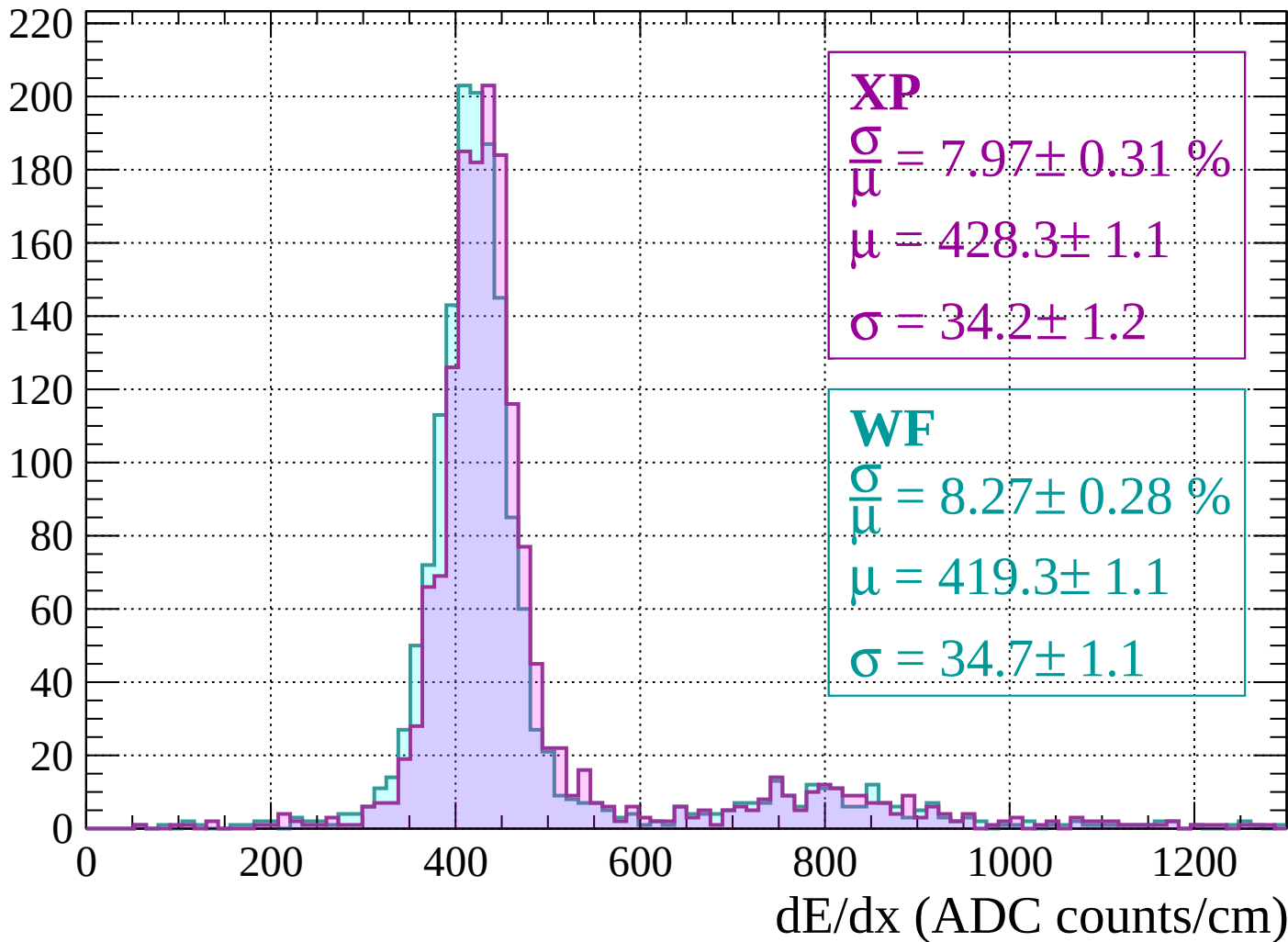
$$\mu = 422.6 \pm 1.1$$

$$\sigma = 33.4 \pm 1.1$$

dE/dx (ADC counts/cm)

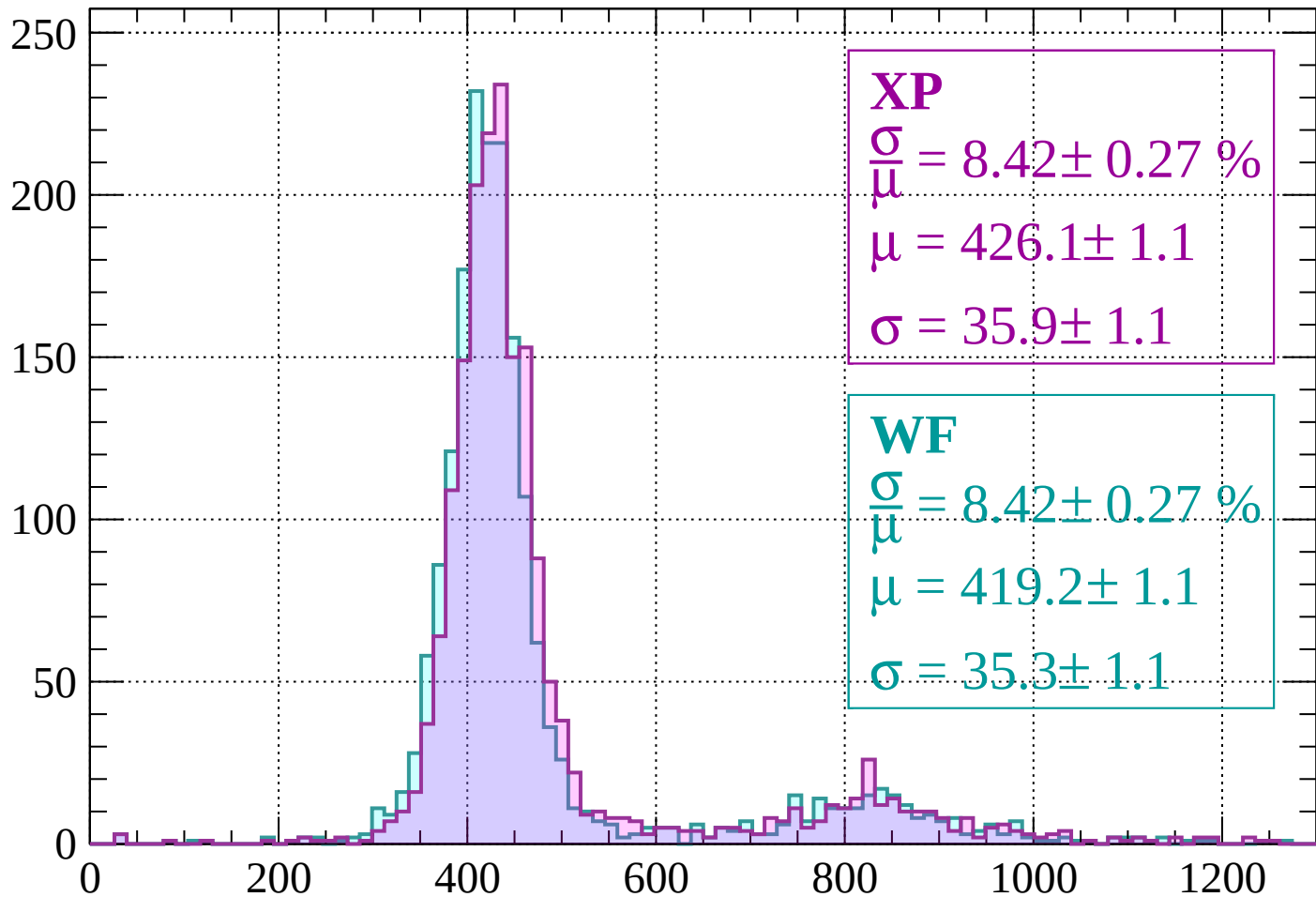
Energy loss | $-760 < p < -720$

Count



Energy loss | $-720 < p < -680$

Count



XP

$$\frac{\sigma}{\mu} = 8.42 \pm 0.27 \%$$

$$\mu = 426.1 \pm 1.1$$

$$\sigma = 35.9 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 8.42 \pm 0.27 \%$$

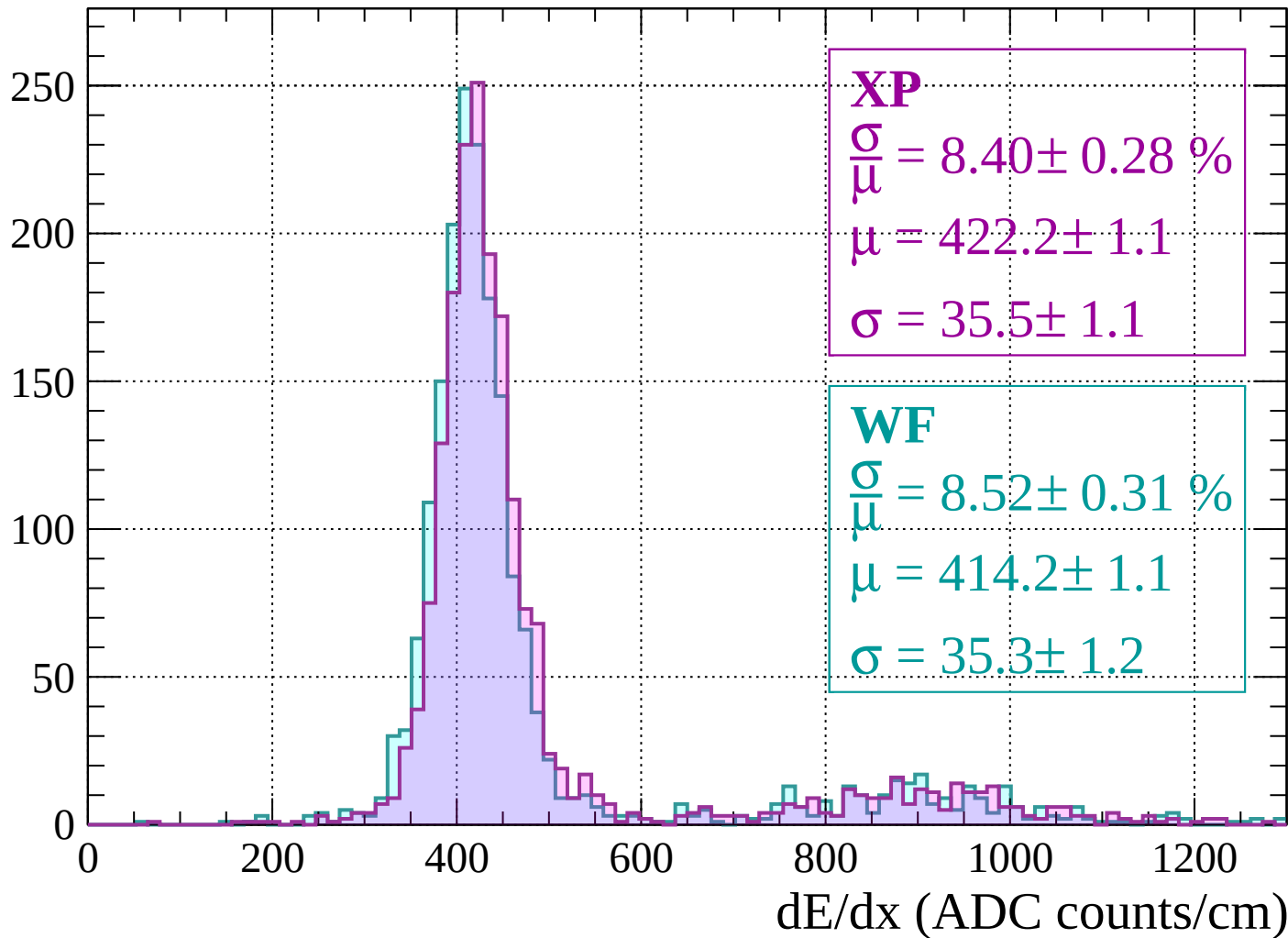
$$\mu = 419.2 \pm 1.1$$

$$\sigma = 35.3 \pm 1.1$$

dE/dx (ADC counts/cm)

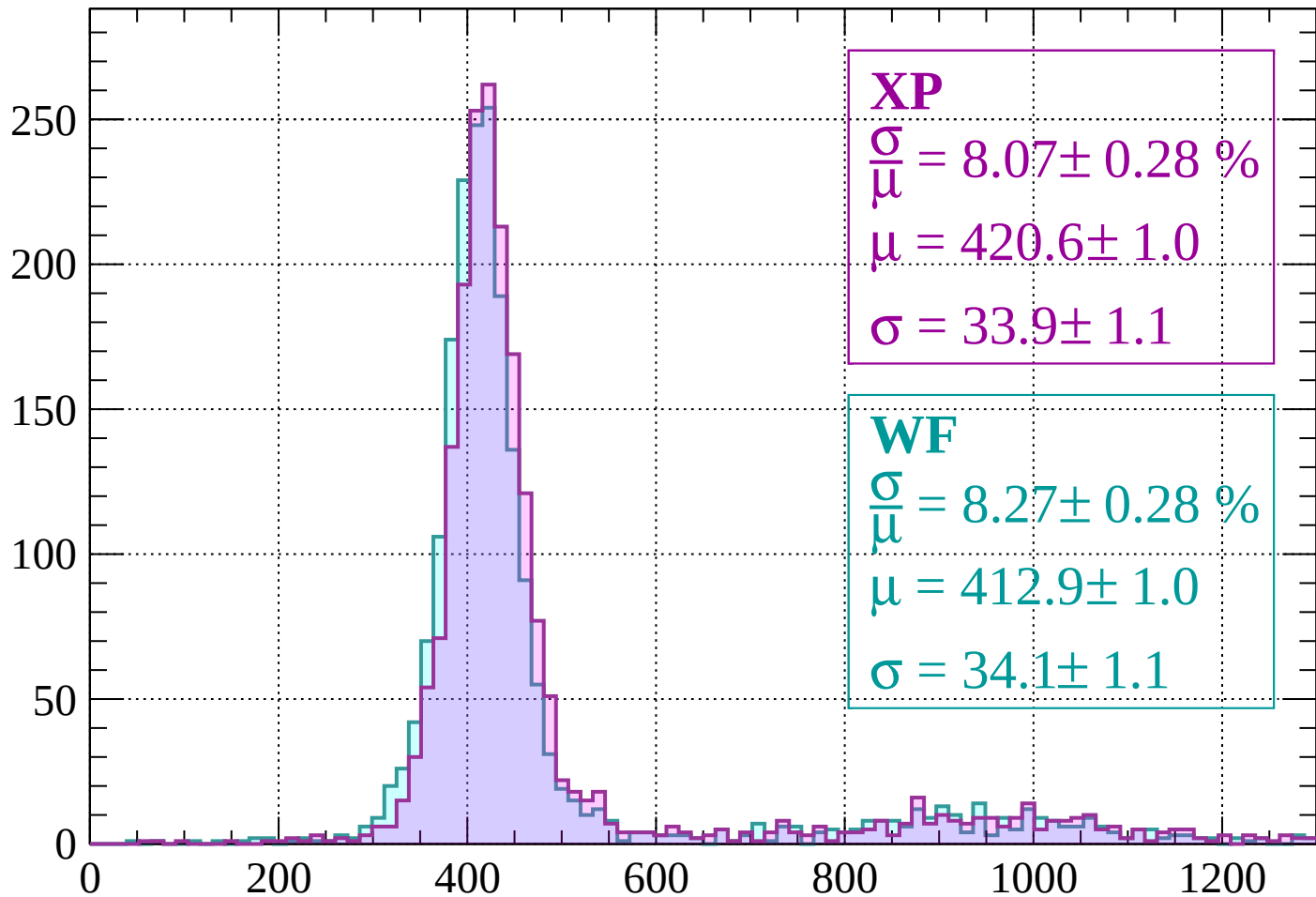
Energy loss | $-680 < p < -640$

Count



Energy loss | $-640 < p < -600$

Count



XP

$$\frac{\sigma}{\mu} = 8.07 \pm 0.28 \%$$

$$\mu = 420.6 \pm 1.0$$

$$\sigma = 33.9 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 8.27 \pm 0.28 \%$$

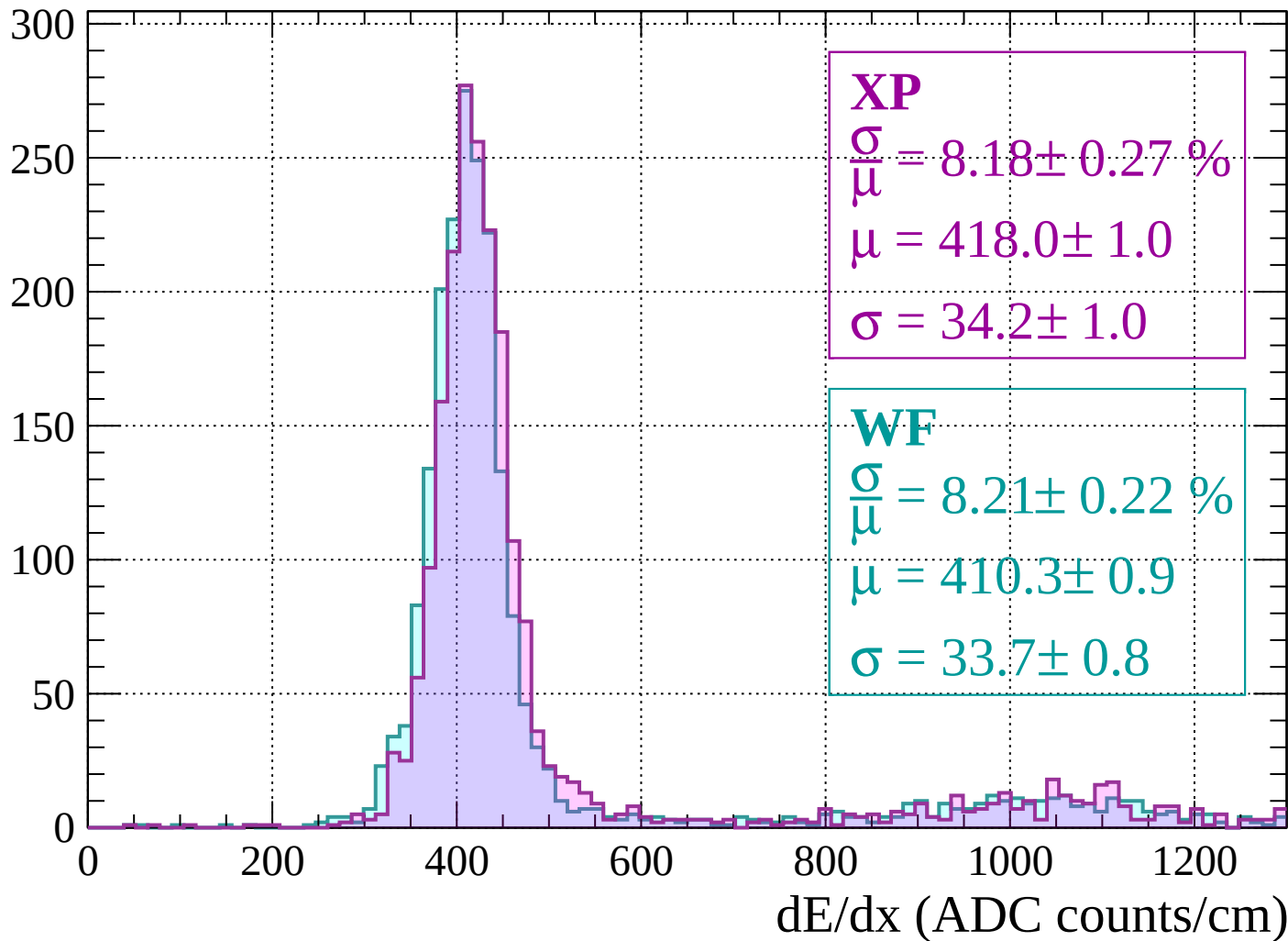
$$\mu = 412.9 \pm 1.0$$

$$\sigma = 34.1 \pm 1.1$$

dE/dx (ADC counts/cm)

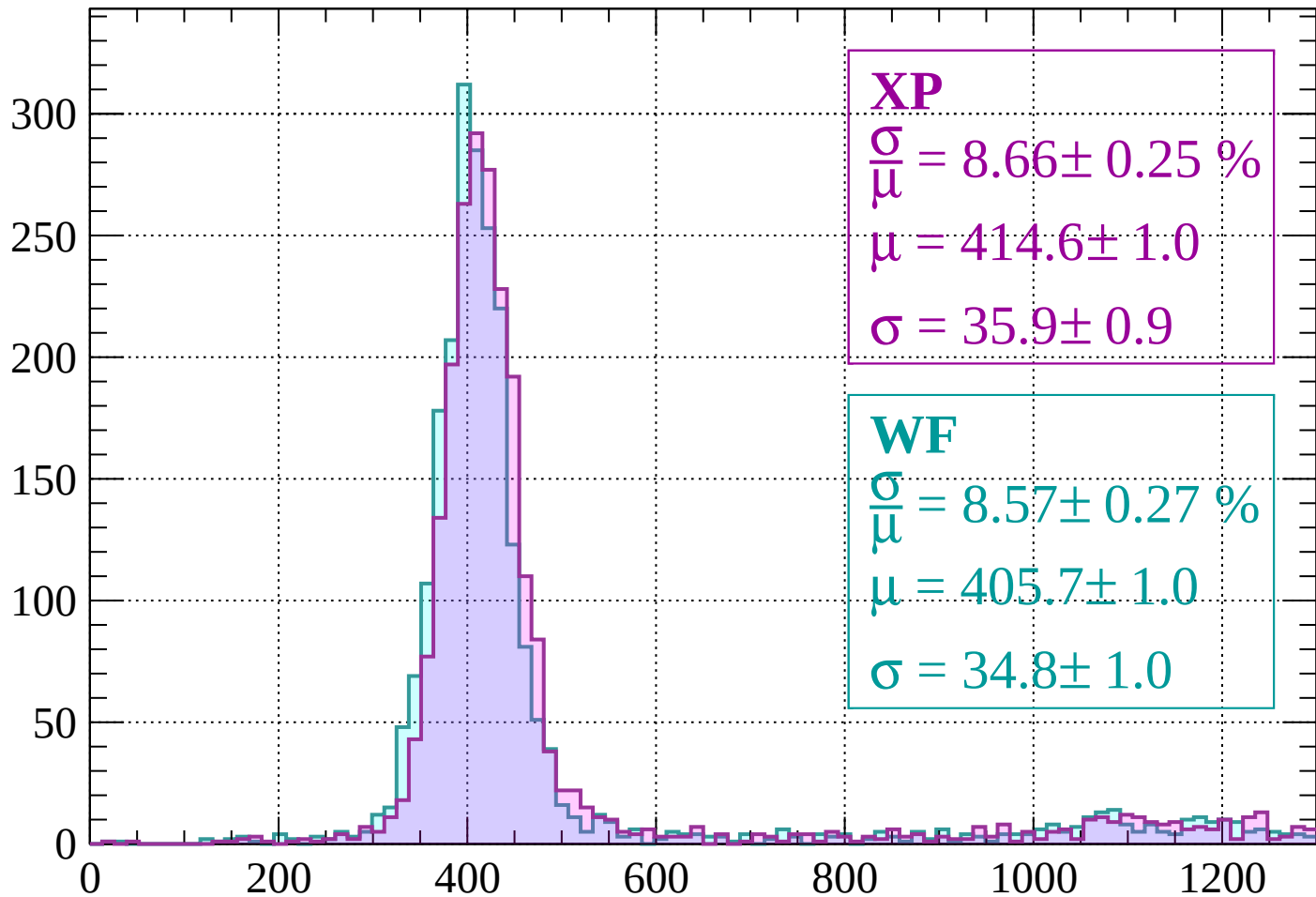
Energy loss | $-600 < p < -560$

Count



Energy loss | $-560 < p < -520$

Count



XP

$$\frac{\sigma}{\mu} = 8.66 \pm 0.25 \%$$

$$\mu = 414.6 \pm 1.0$$

$$\sigma = 35.9 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 8.57 \pm 0.27 \%$$

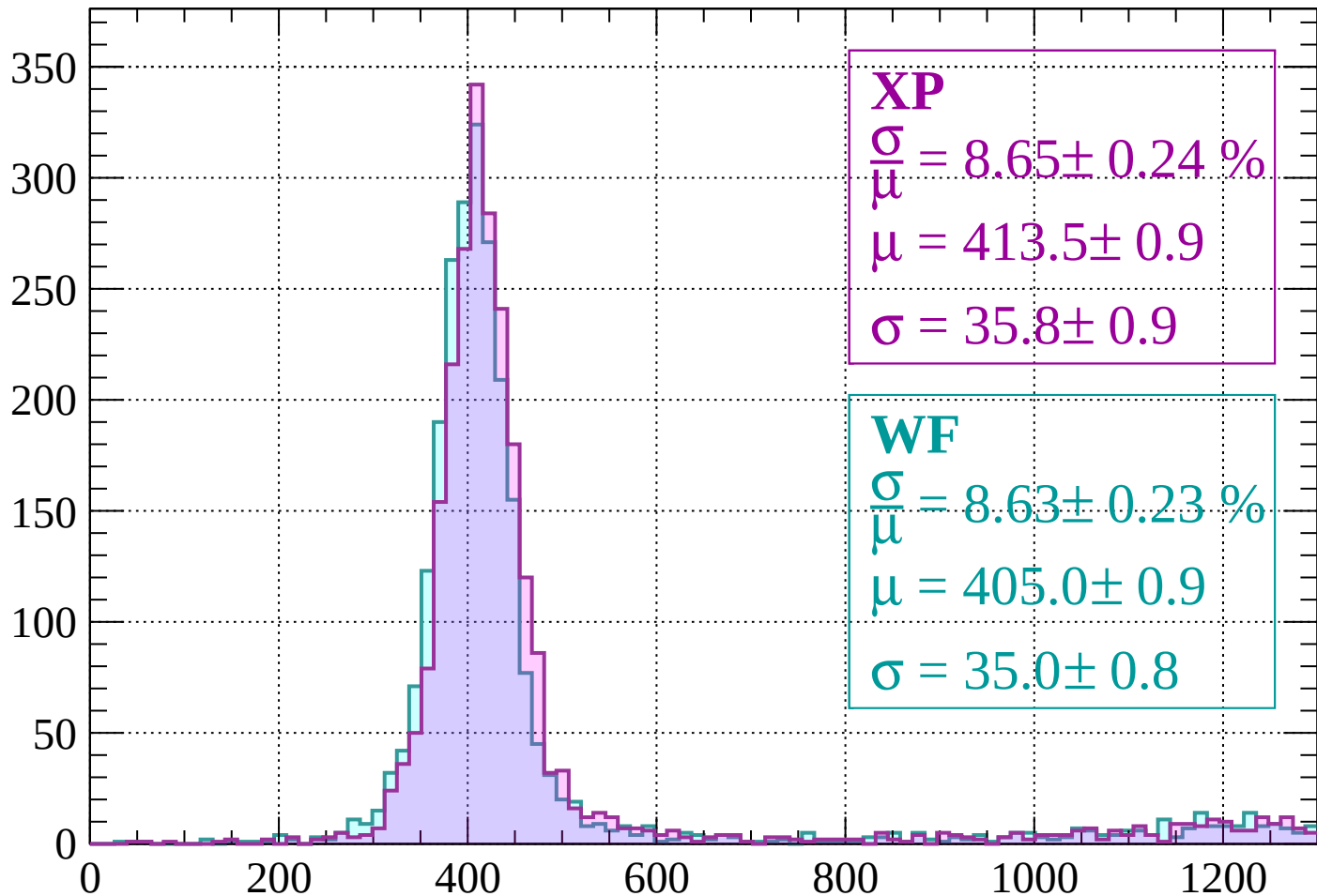
$$\mu = 405.7 \pm 1.0$$

$$\sigma = 34.8 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-520 < p < -480$

Count



XP

$$\frac{\sigma}{\mu} = 8.65 \pm 0.24 \%$$

$$\mu = 413.5 \pm 0.9$$

$$\sigma = 35.8 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 8.63 \pm 0.23 \%$$

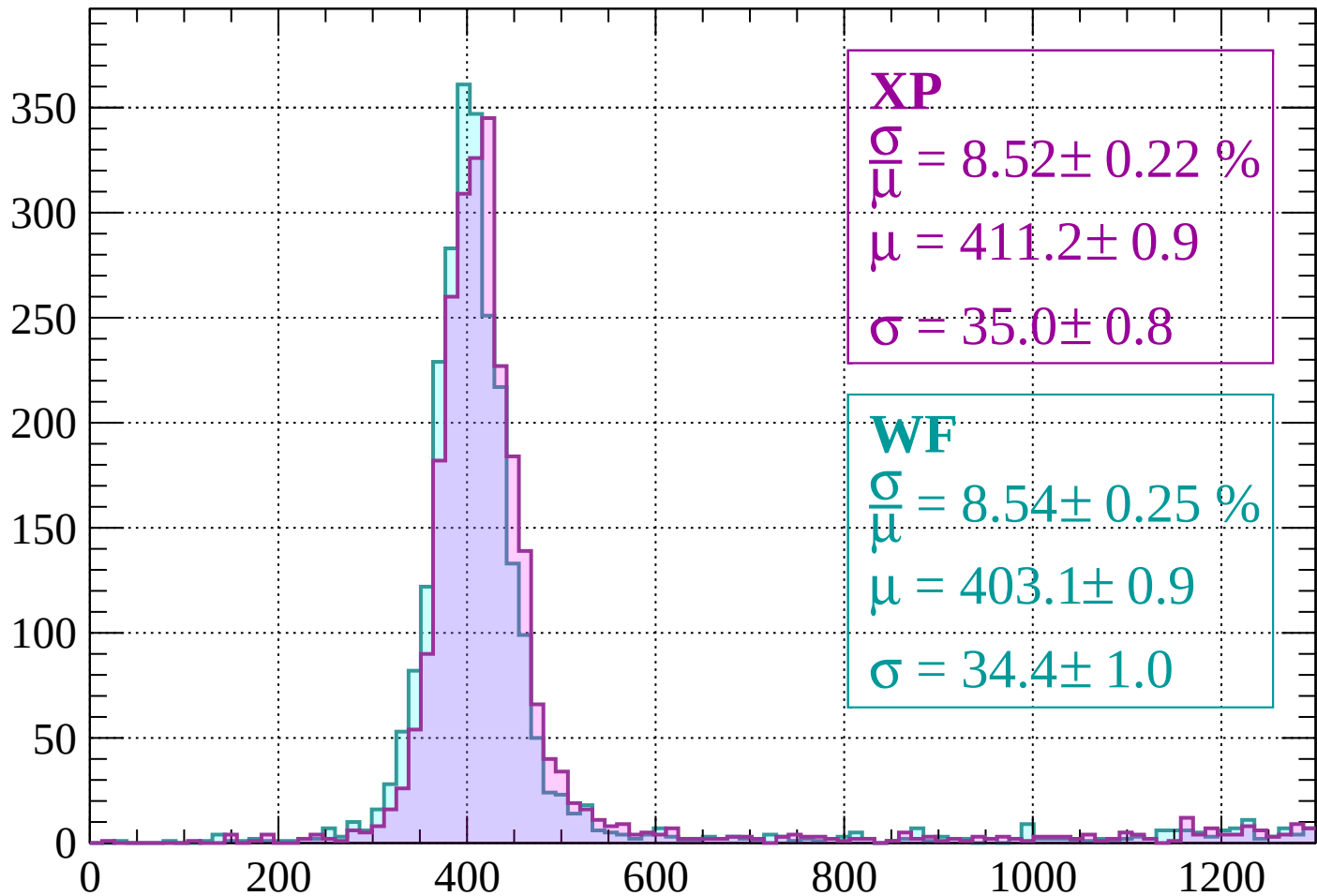
$$\mu = 405.0 \pm 0.9$$

$$\sigma = 35.0 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-480 < p < -440$

Count



XP

$$\frac{\sigma}{\mu} = 8.52 \pm 0.22 \%$$

$$\mu = 411.2 \pm 0.9$$

$$\sigma = 35.0 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 8.54 \pm 0.25 \%$$

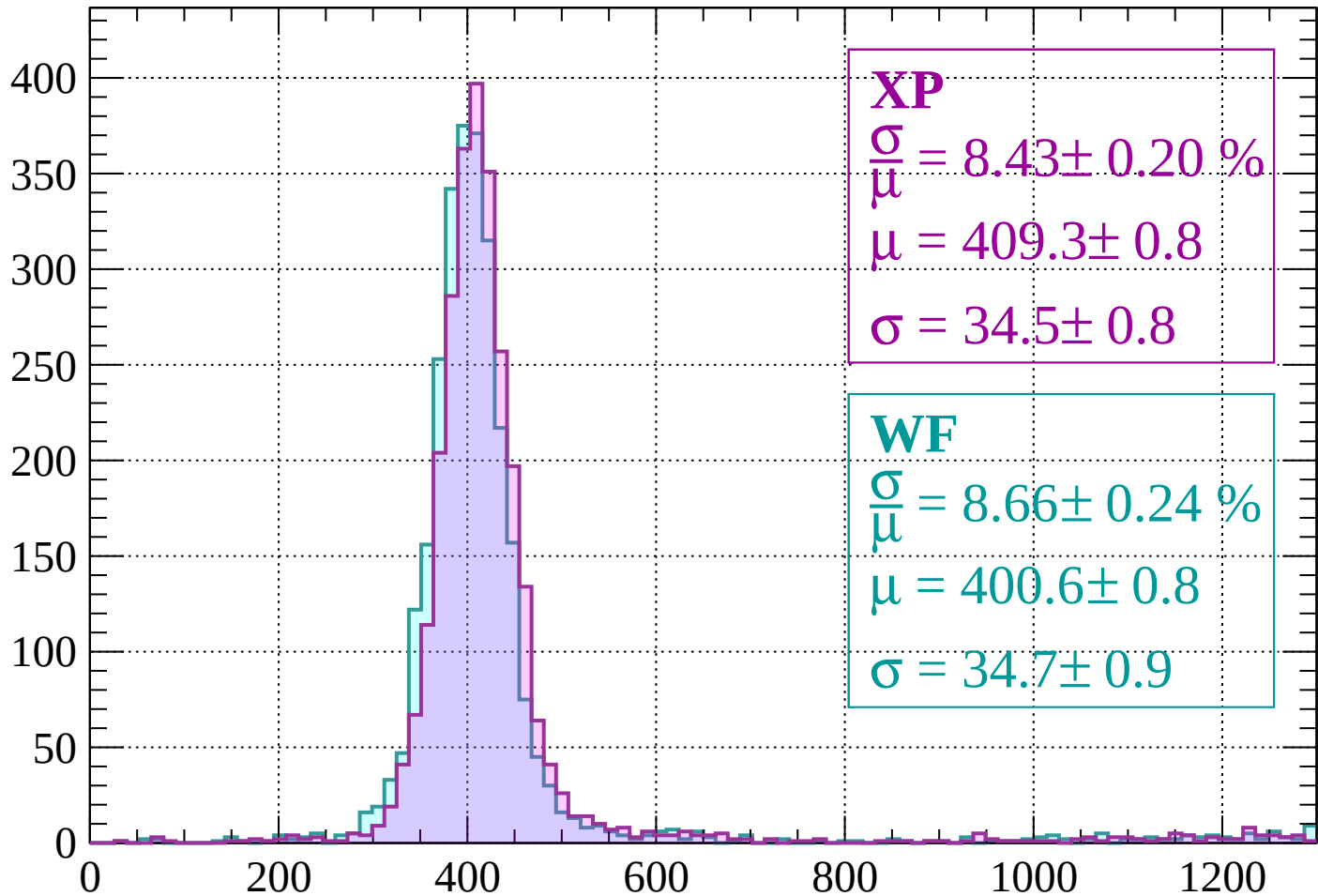
$$\mu = 403.1 \pm 0.9$$

$$\sigma = 34.4 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-440 < p < -400$

Count



XP

$$\frac{\sigma}{\mu} = 8.43 \pm 0.20 \%$$

$$\mu = 409.3 \pm 0.8$$

$$\sigma = 34.5 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 8.66 \pm 0.24 \%$$

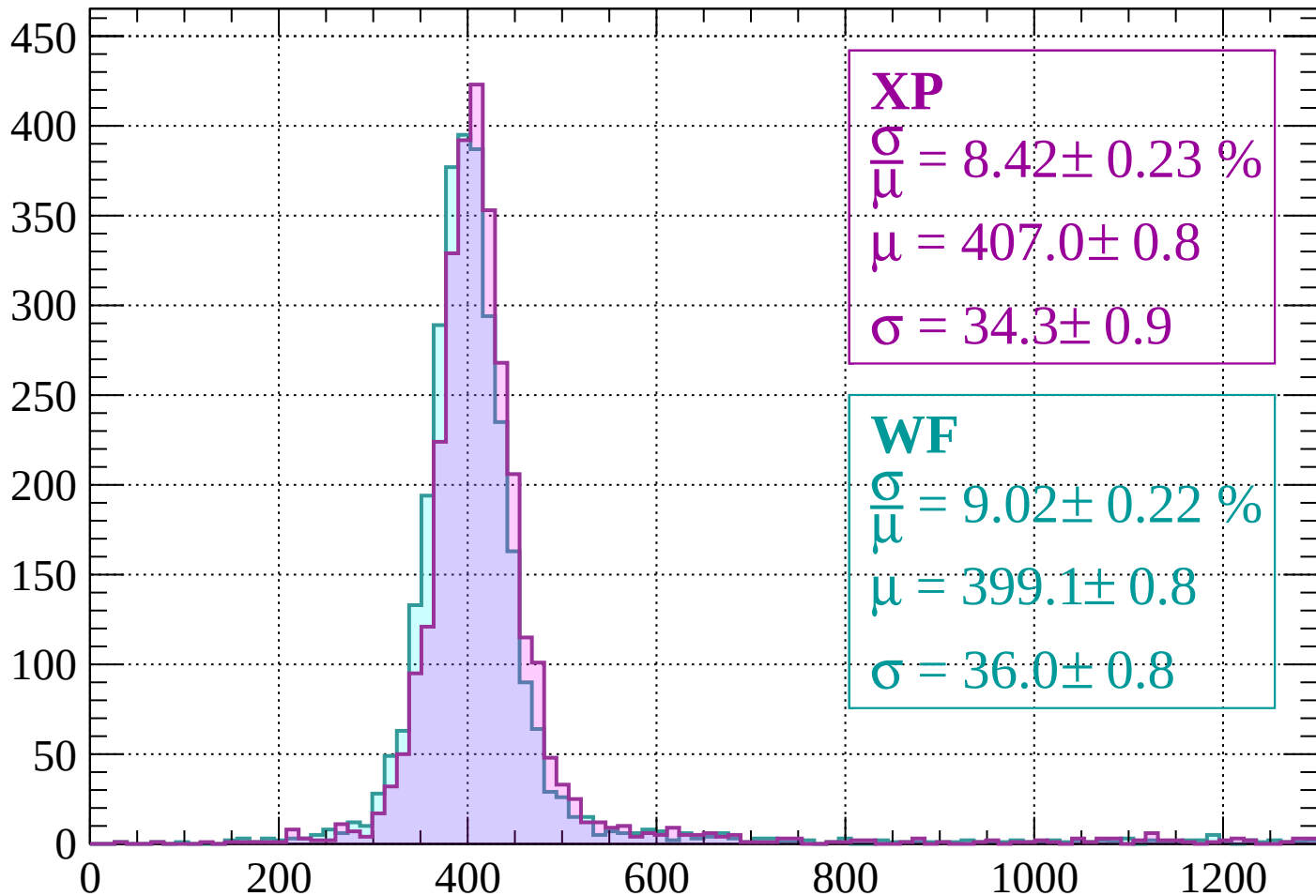
$$\mu = 400.6 \pm 0.8$$

$$\sigma = 34.7 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-400 < p < -360$

Count



XP

$$\frac{\sigma}{\mu} = 8.42 \pm 0.23 \%$$

$$\mu = 407.0 \pm 0.8$$

$$\sigma = 34.3 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.02 \pm 0.22 \%$$

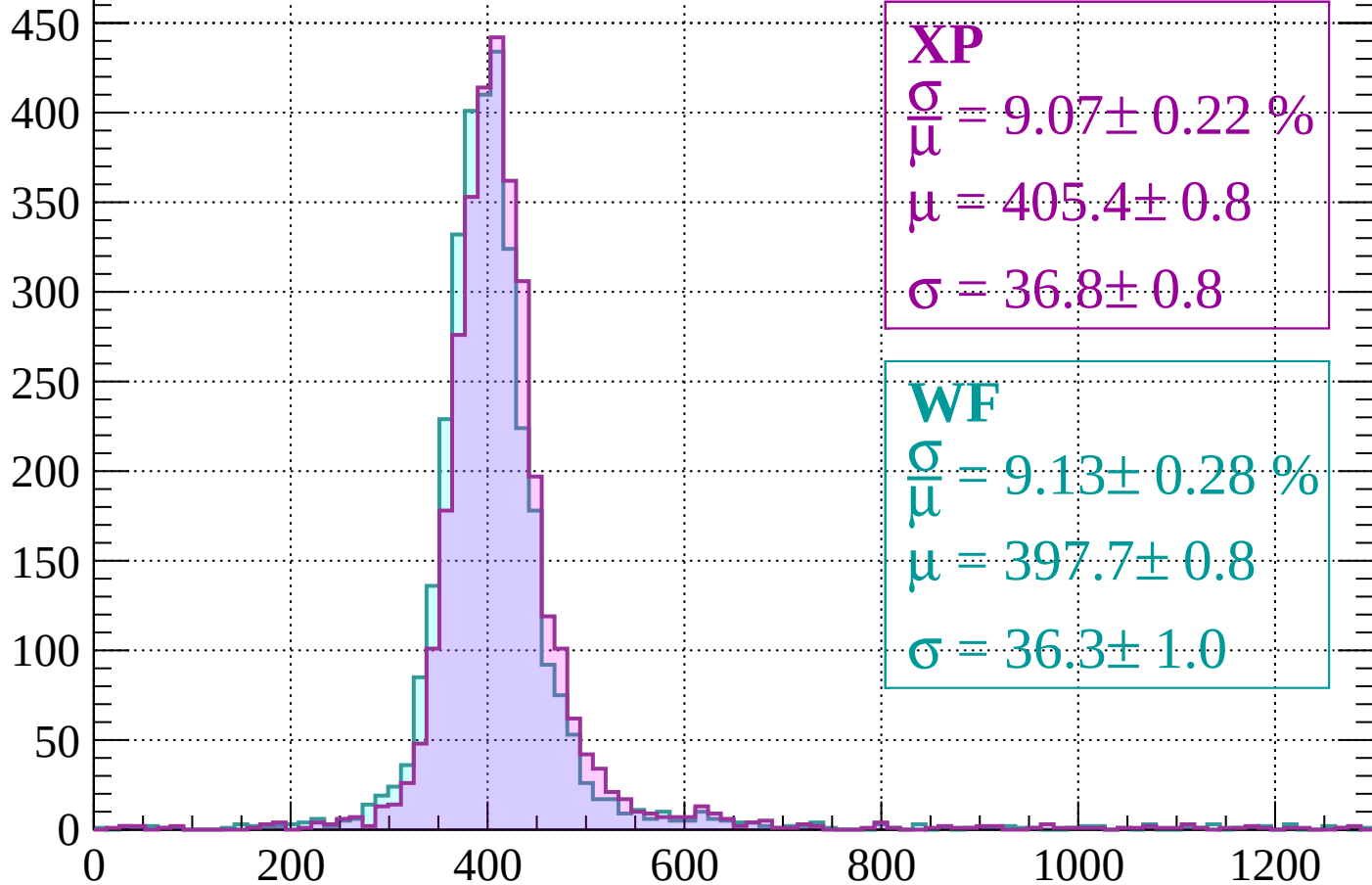
$$\mu = 399.1 \pm 0.8$$

$$\sigma = 36.0 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-360 < p < -320$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 9.07 \pm 0.22 \%$$

$$\mu = 405.4 \pm 0.8$$

$$\sigma = 36.8 \pm 0.8$$

WF

$$\frac{\sigma}{\bar{\mu}} = 9.13 \pm 0.28 \%$$

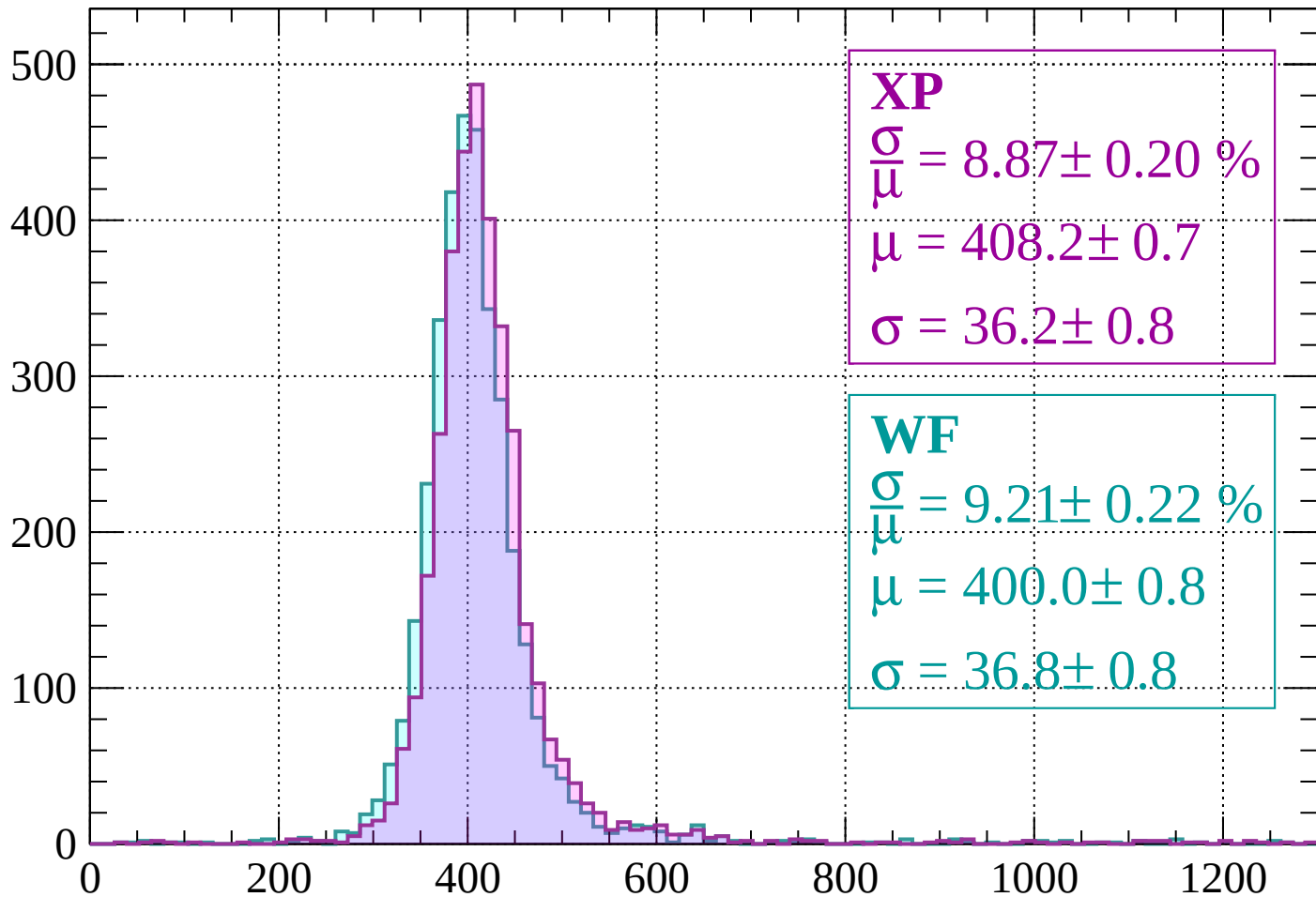
$$\mu = 397.7 \pm 0.8$$

$$\sigma = 36.3 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-320 < p < -280$

Count



XP

$$\frac{\sigma}{\mu} = 8.87 \pm 0.20 \%$$

$$\mu = 408.2 \pm 0.7$$

$$\sigma = 36.2 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.21 \pm 0.22 \%$$

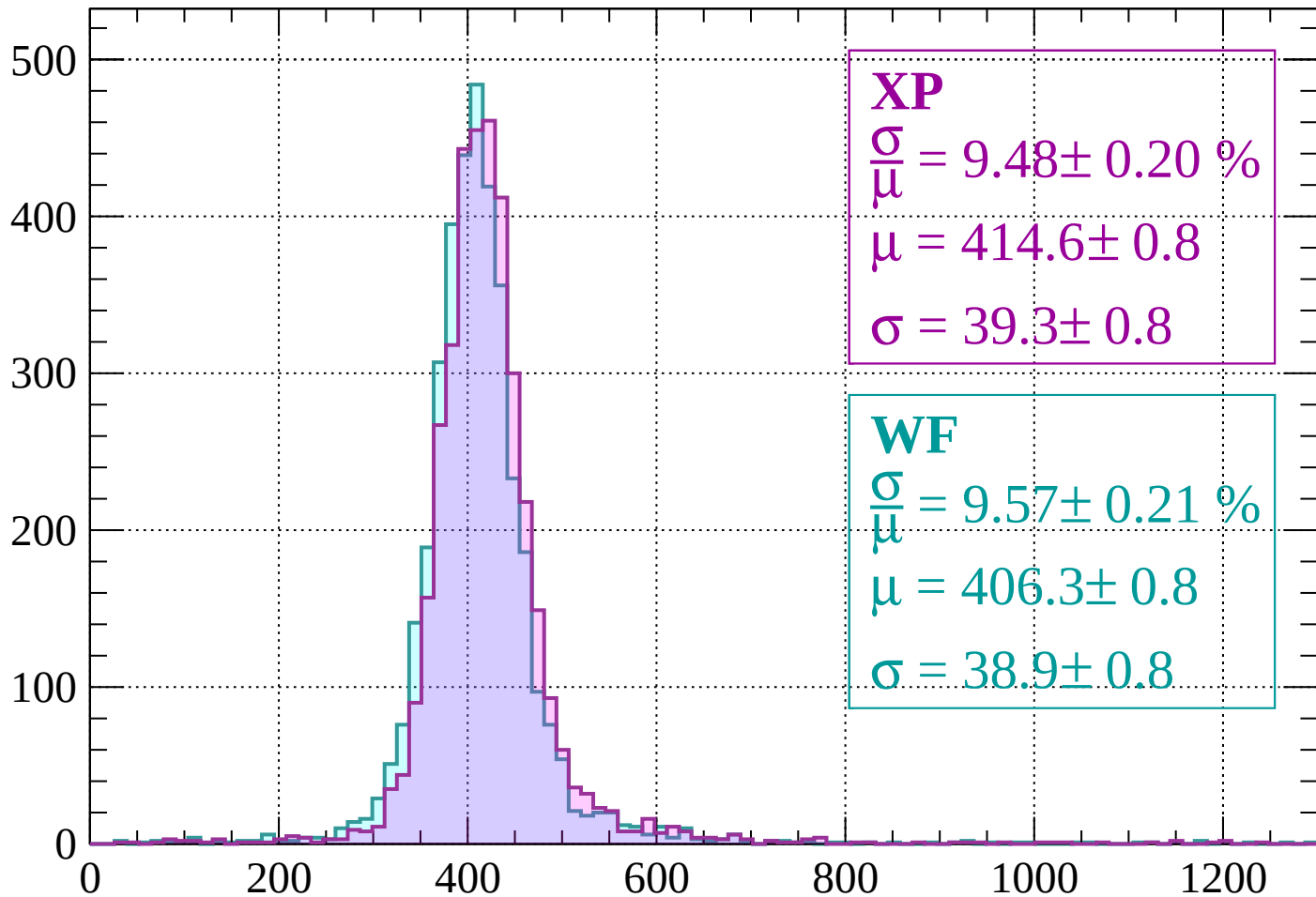
$$\mu = 400.0 \pm 0.8$$

$$\sigma = 36.8 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-280 < p < -240$

Count



XP

$$\frac{\sigma}{\mu} = 9.48 \pm 0.20 \%$$

$$\mu = 414.6 \pm 0.8$$

$$\sigma = 39.3 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.57 \pm 0.21 \%$$

$$\mu = 406.3 \pm 0.8$$

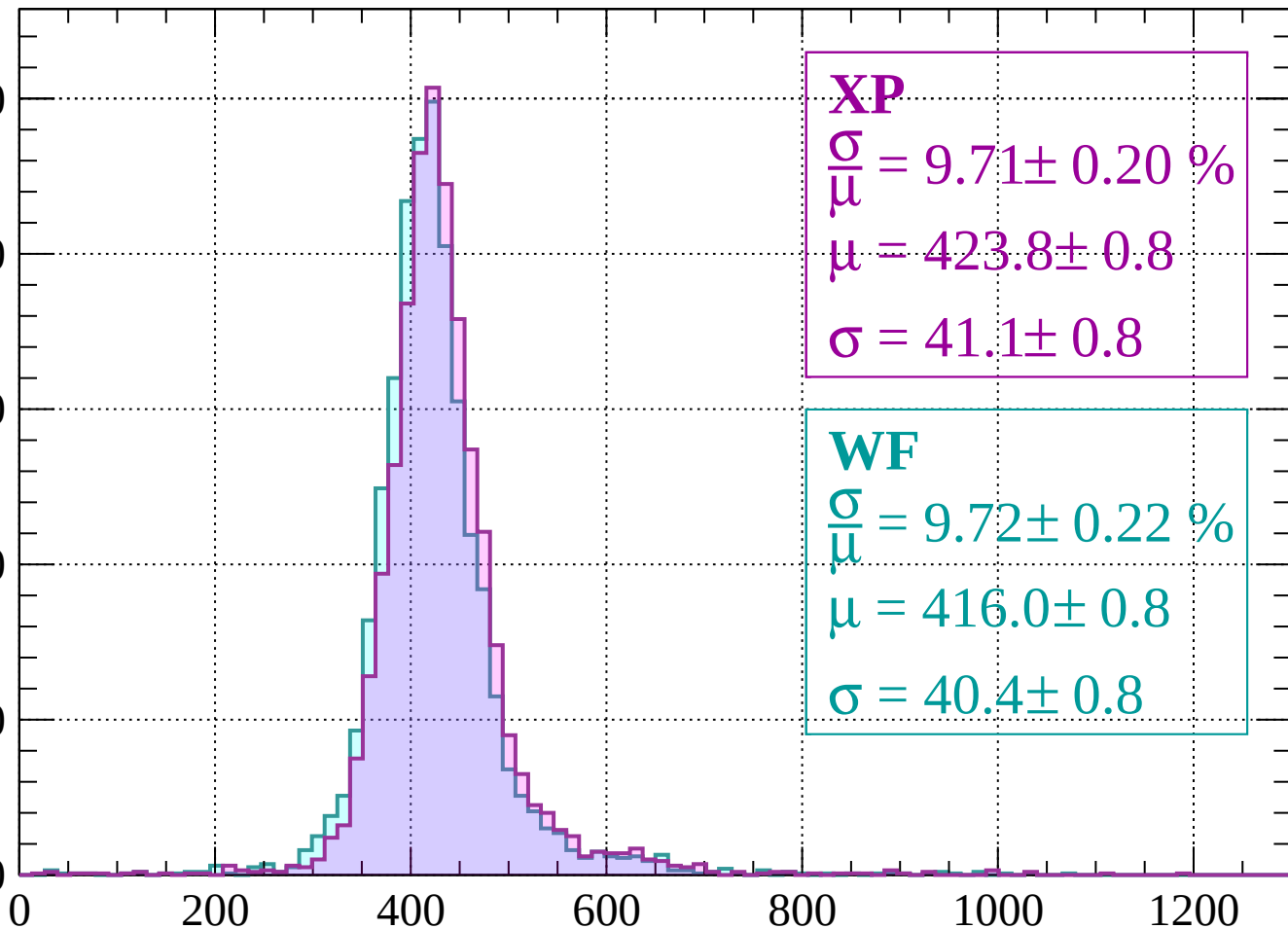
$$\sigma = 38.9 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-240 < p < -200$

Count

500
400
300
200
100
0



XP

$$\frac{\sigma}{\mu} = 9.71 \pm 0.20 \%$$

$$\mu = 423.8 \pm 0.8$$

$$\sigma = 41.1 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.72 \pm 0.22 \%$$

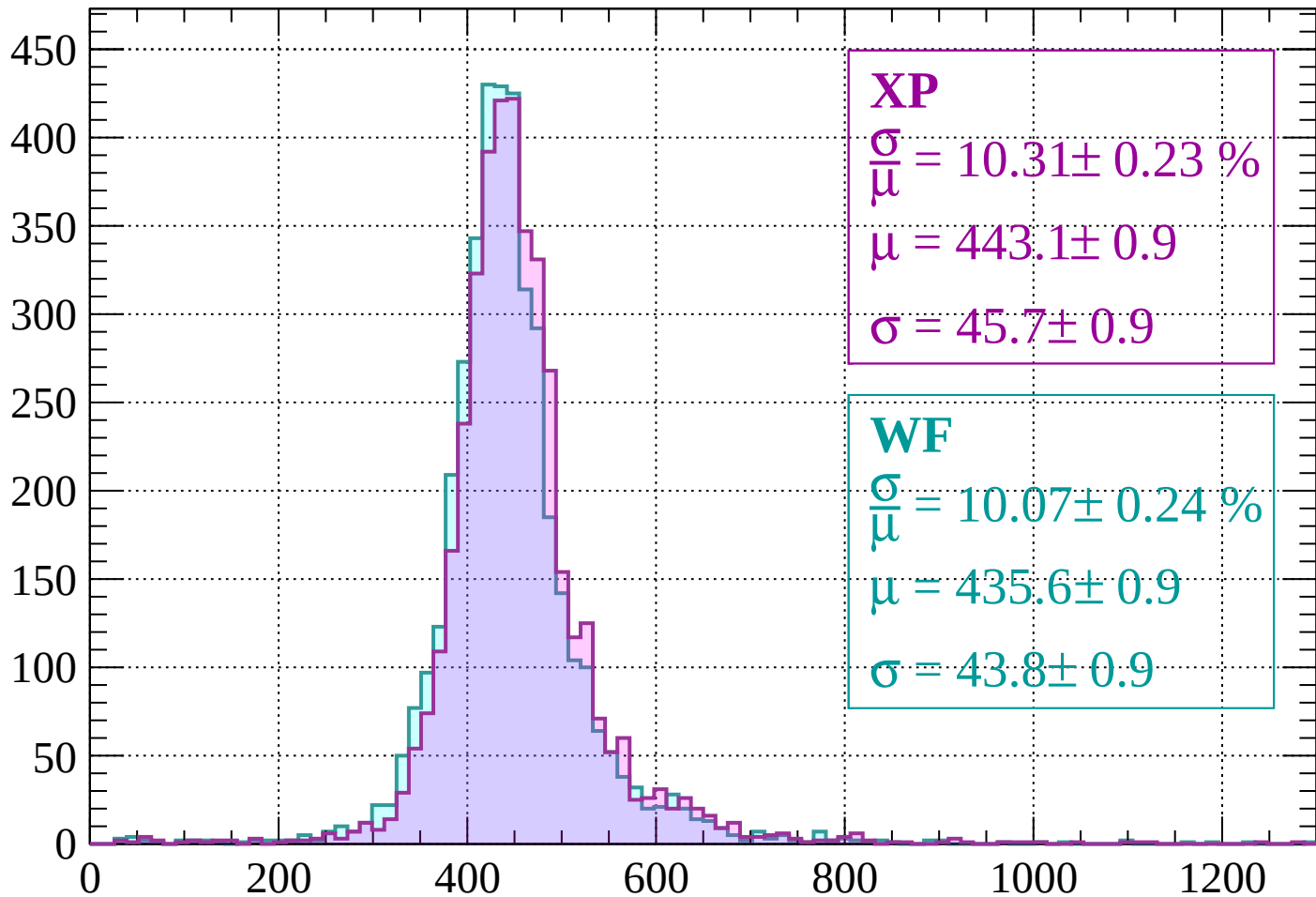
$$\mu = 416.0 \pm 0.8$$

$$\sigma = 40.4 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-200 < p < -160$

Count



XP

$$\frac{\sigma}{\mu} = 10.31 \pm 0.23 \%$$

$$\mu = 443.1 \pm 0.9$$

$$\sigma = 45.7 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 10.07 \pm 0.24 \%$$

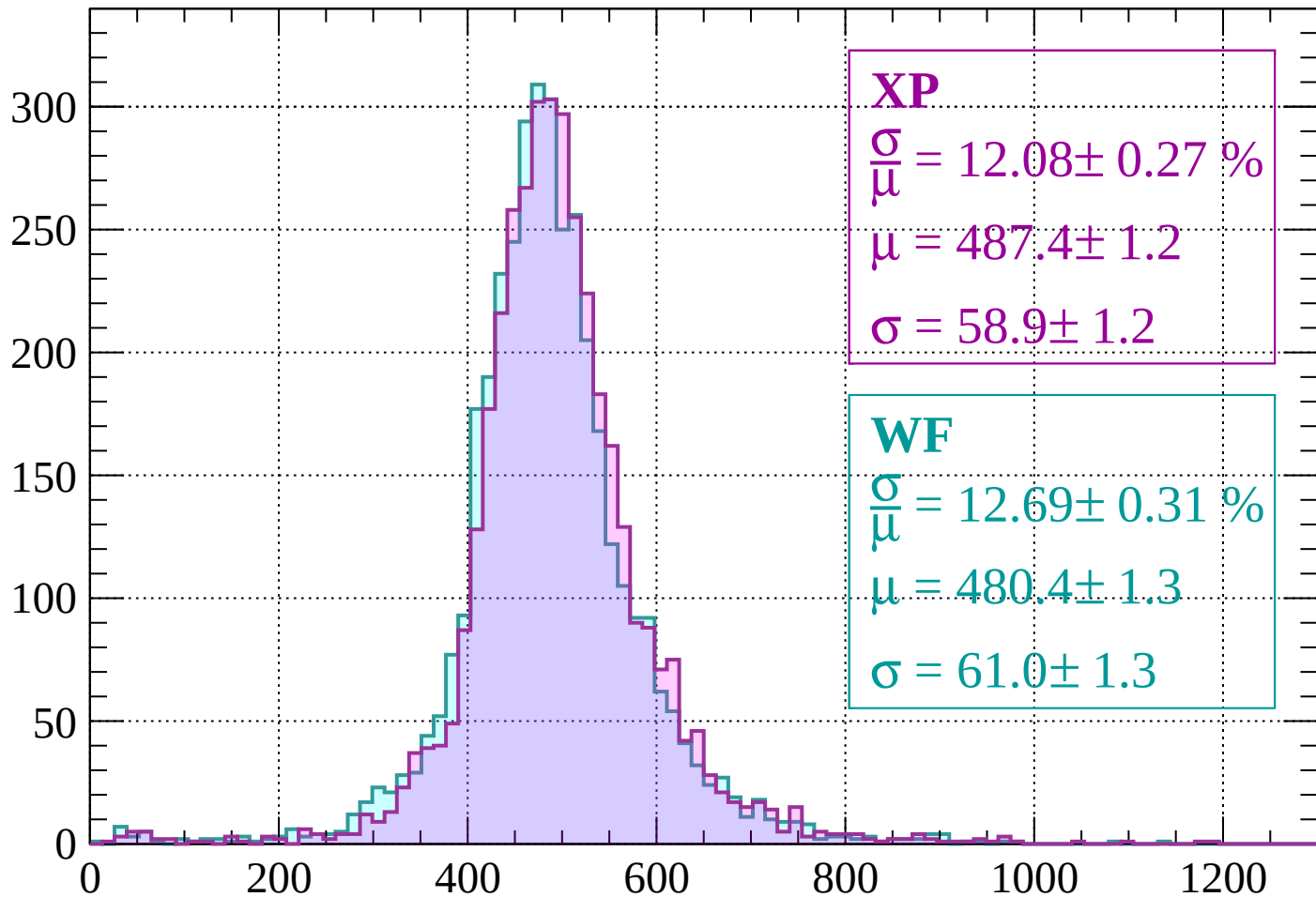
$$\mu = 435.6 \pm 0.9$$

$$\sigma = 43.8 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-160 < p < -120$

Count



XP

$$\frac{\sigma}{\mu} = 12.08 \pm 0.27 \%$$

$$\mu = 487.4 \pm 1.2$$

$$\sigma = 58.9 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 12.69 \pm 0.31 \%$$

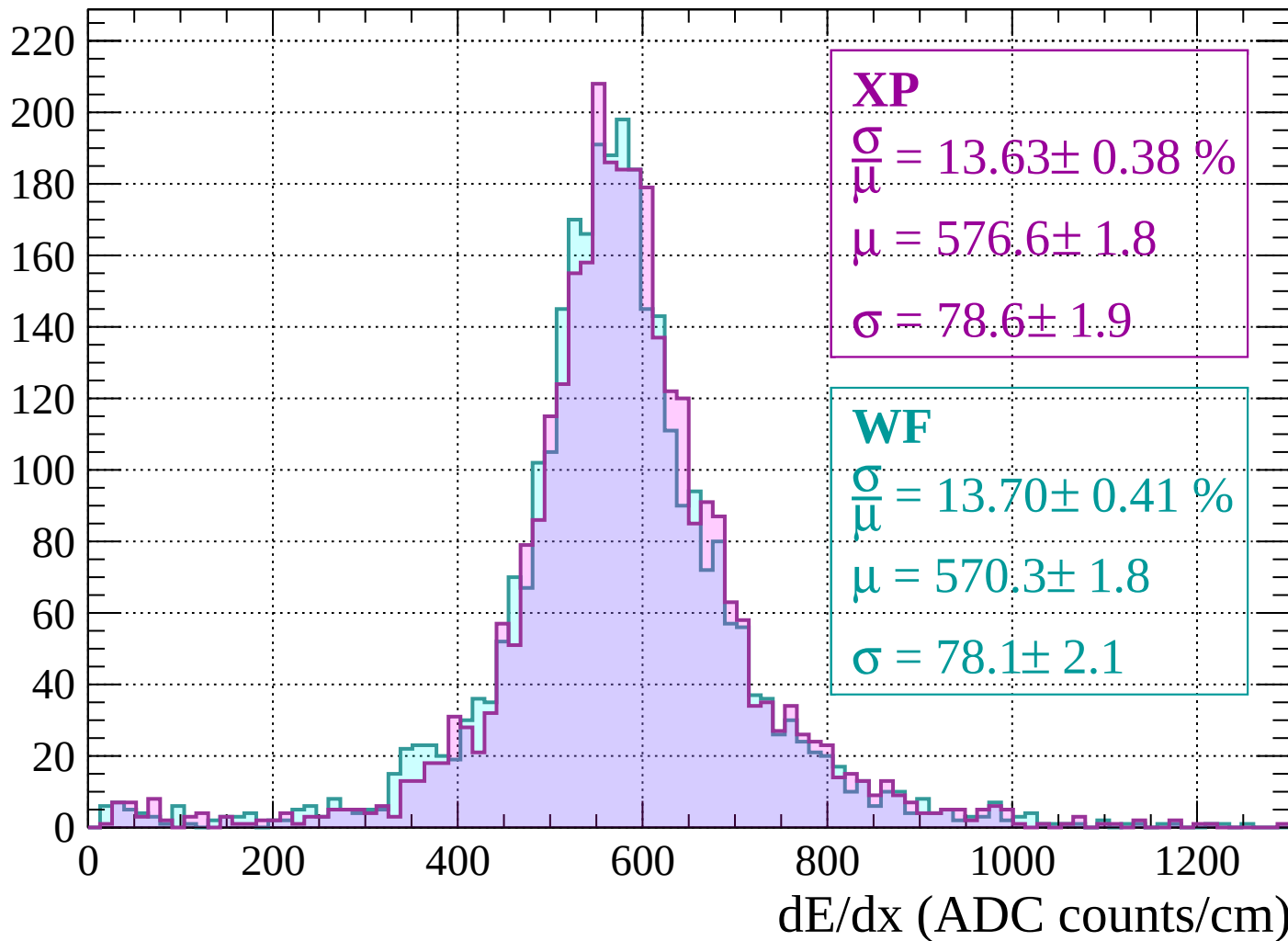
$$\mu = 480.4 \pm 1.3$$

$$\sigma = 61.0 \pm 1.3$$

dE/dx (ADC counts/cm)

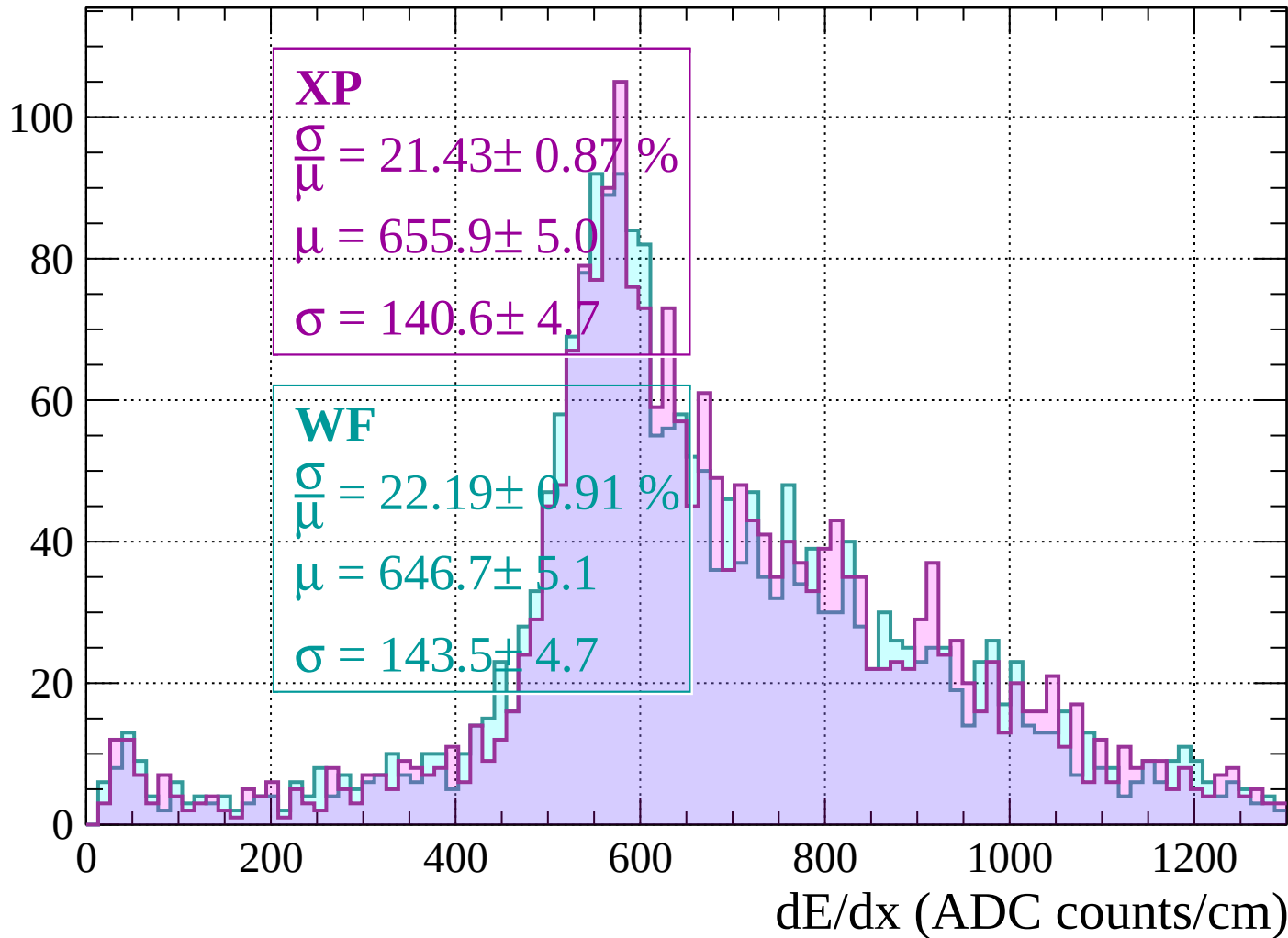
Energy loss | $-120 < p < -80$

Count



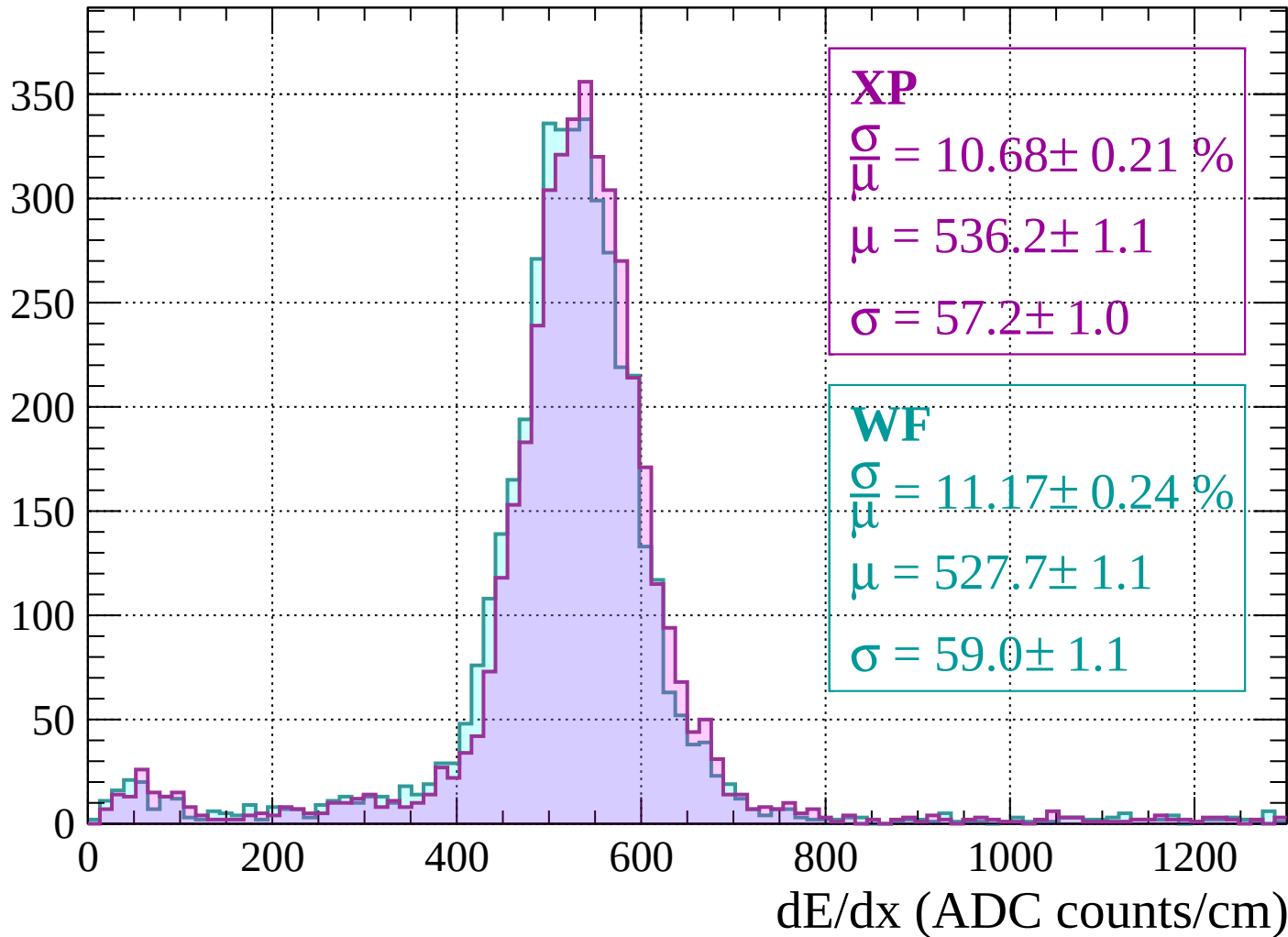
Energy loss | $-80 < p < -40$

Count



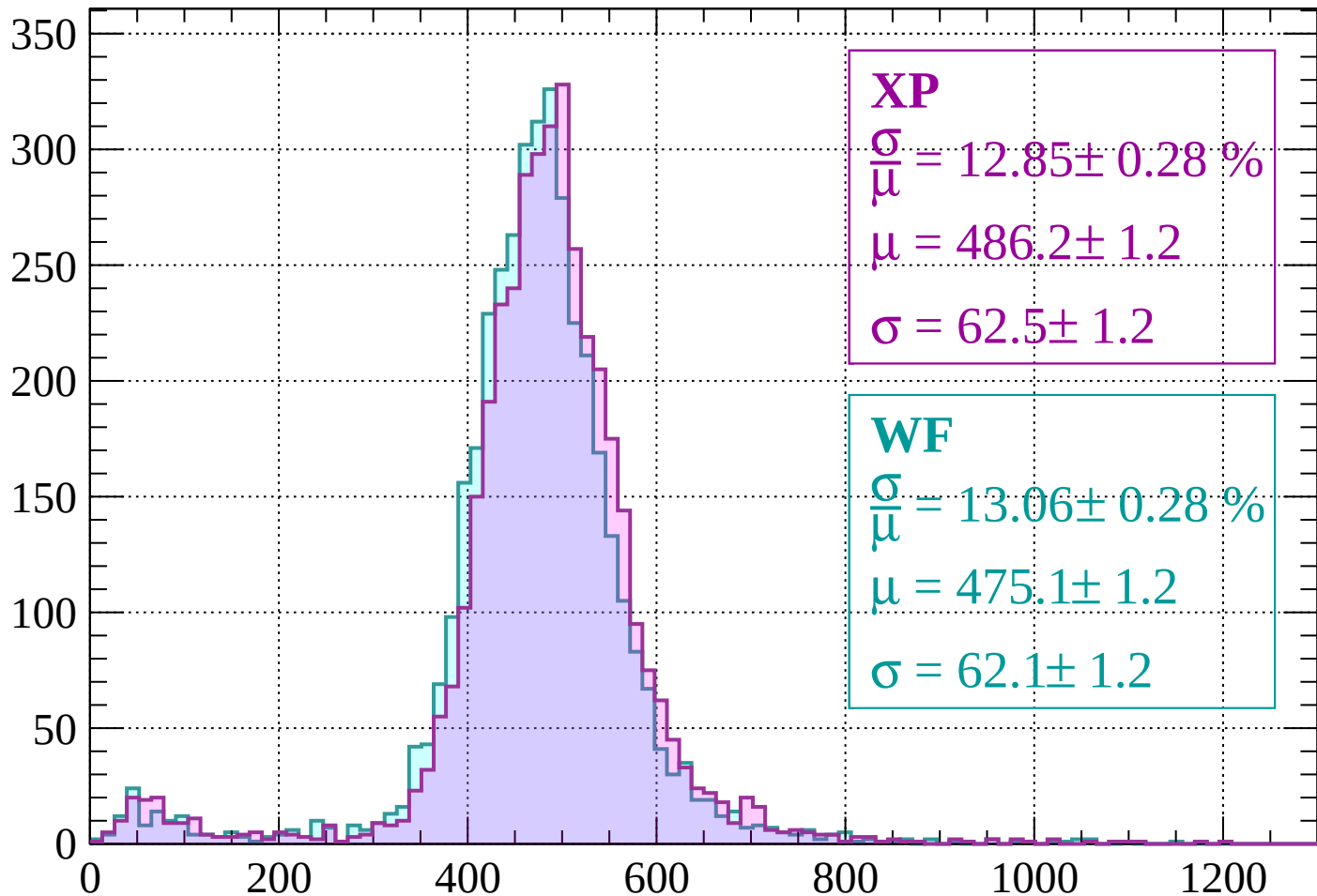
Energy loss | $-40 < p < 0$

Count



Energy loss | $0 < p < 40$

Count



XP

$$\frac{\sigma}{\mu} = 12.85 \pm 0.28 \%$$

$$\mu = 486.2 \pm 1.2$$

$$\sigma = 62.5 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 13.06 \pm 0.28 \%$$

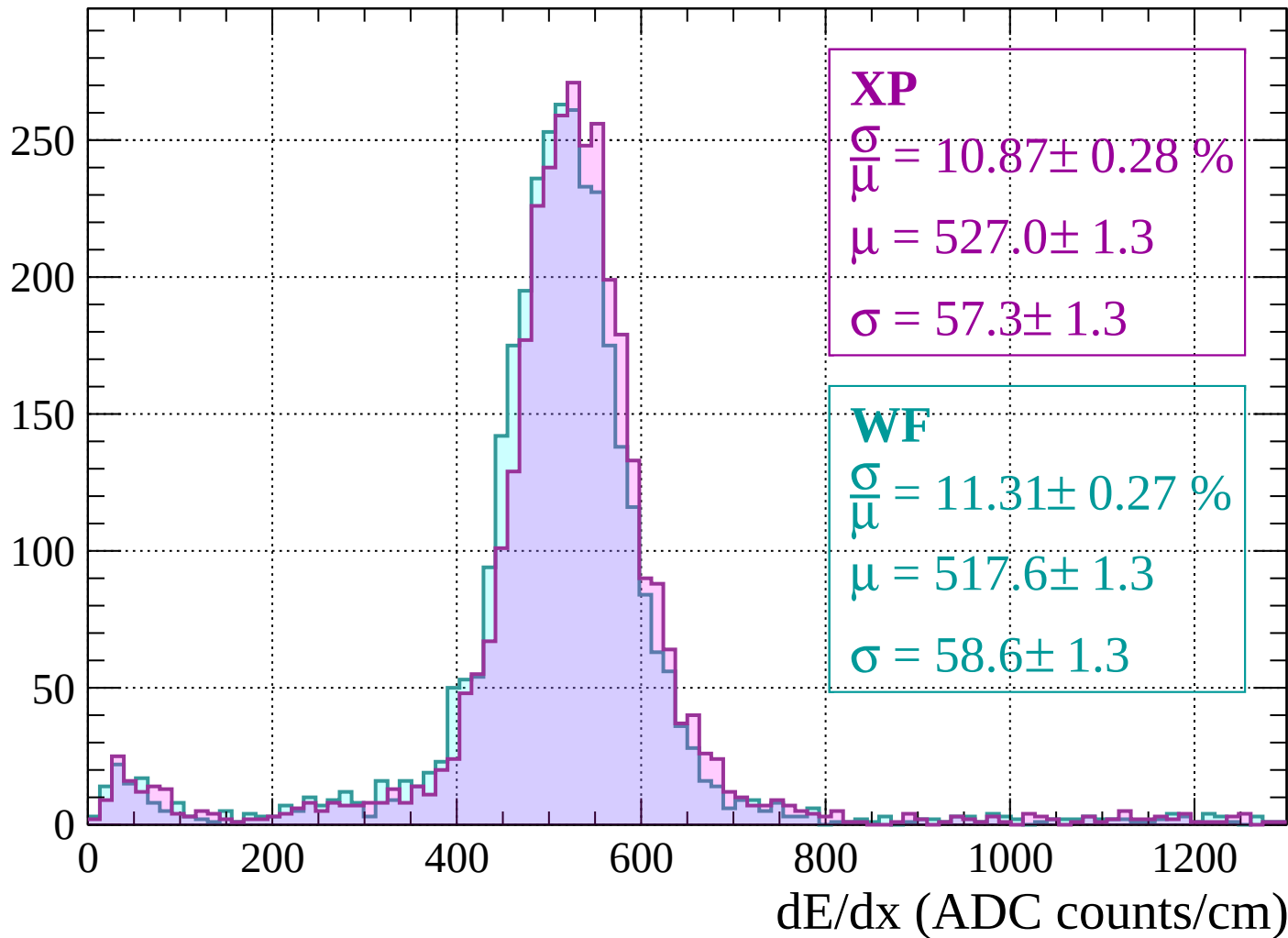
$$\mu = 475.1 \pm 1.2$$

$$\sigma = 62.1 \pm 1.2$$

dE/dx (ADC counts/cm)

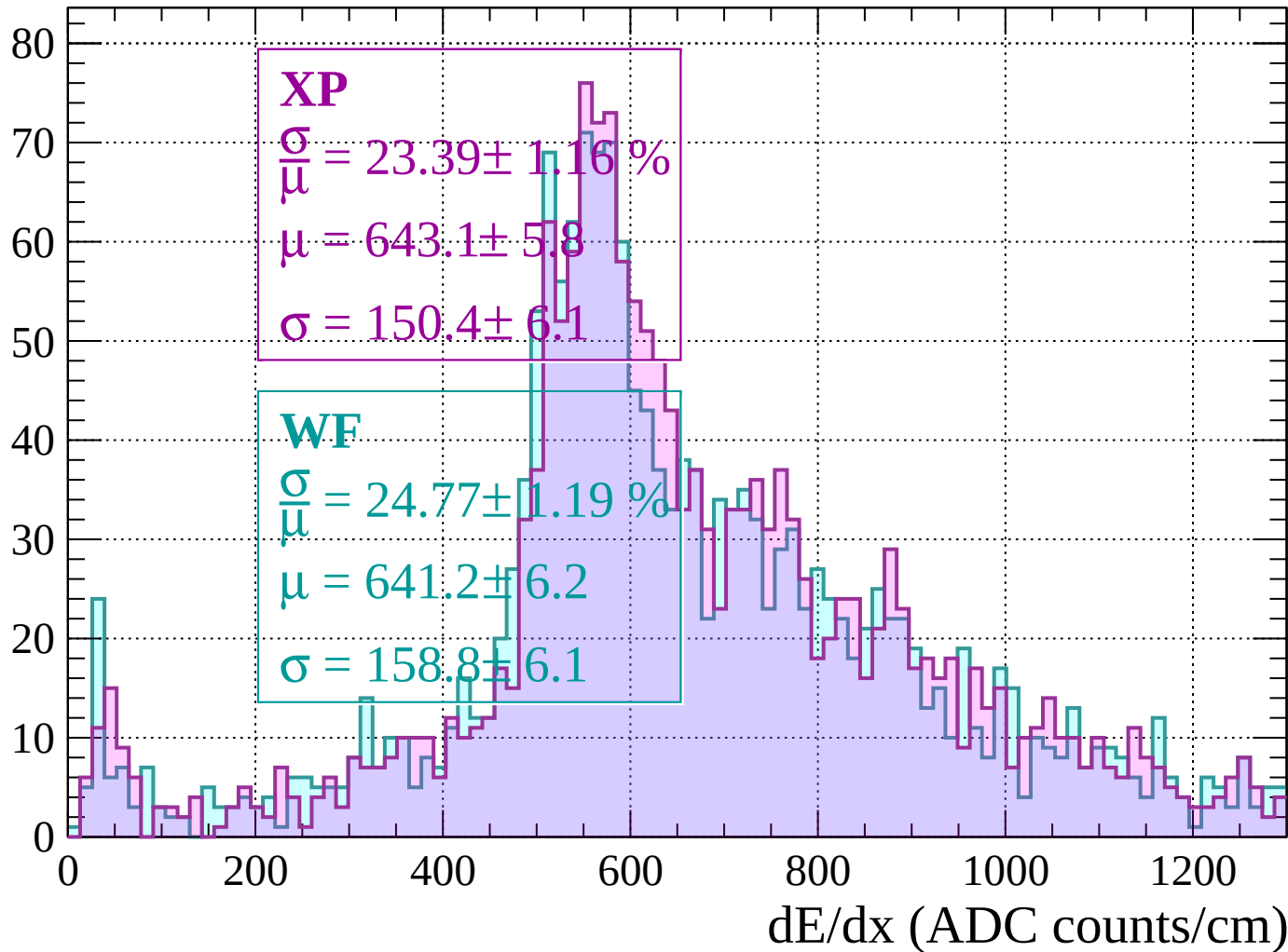
Energy loss | $40 < p < 80$

Count

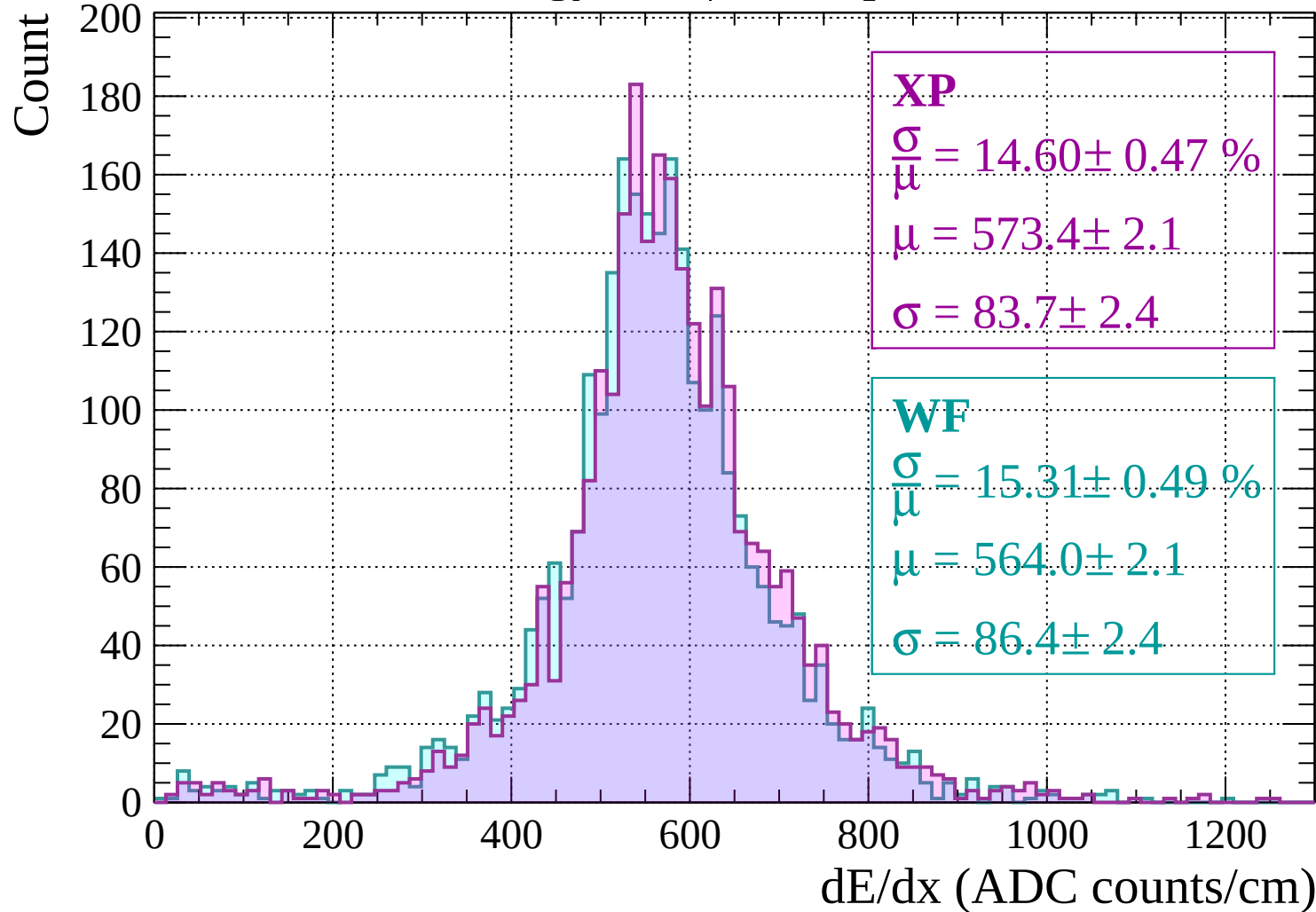


Energy loss | $80 < p < 120$

Count

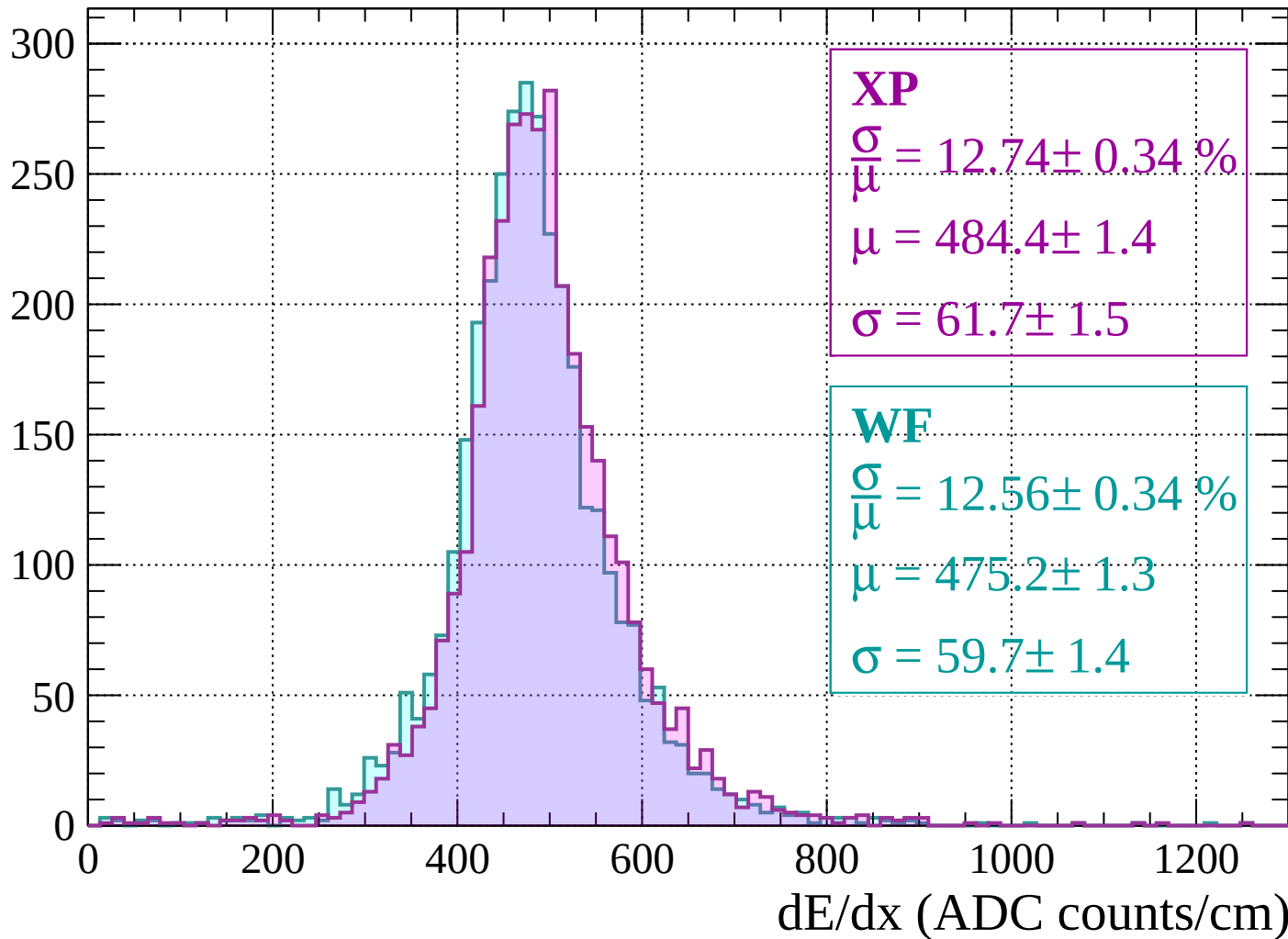


Energy loss | $120 < p < 160$



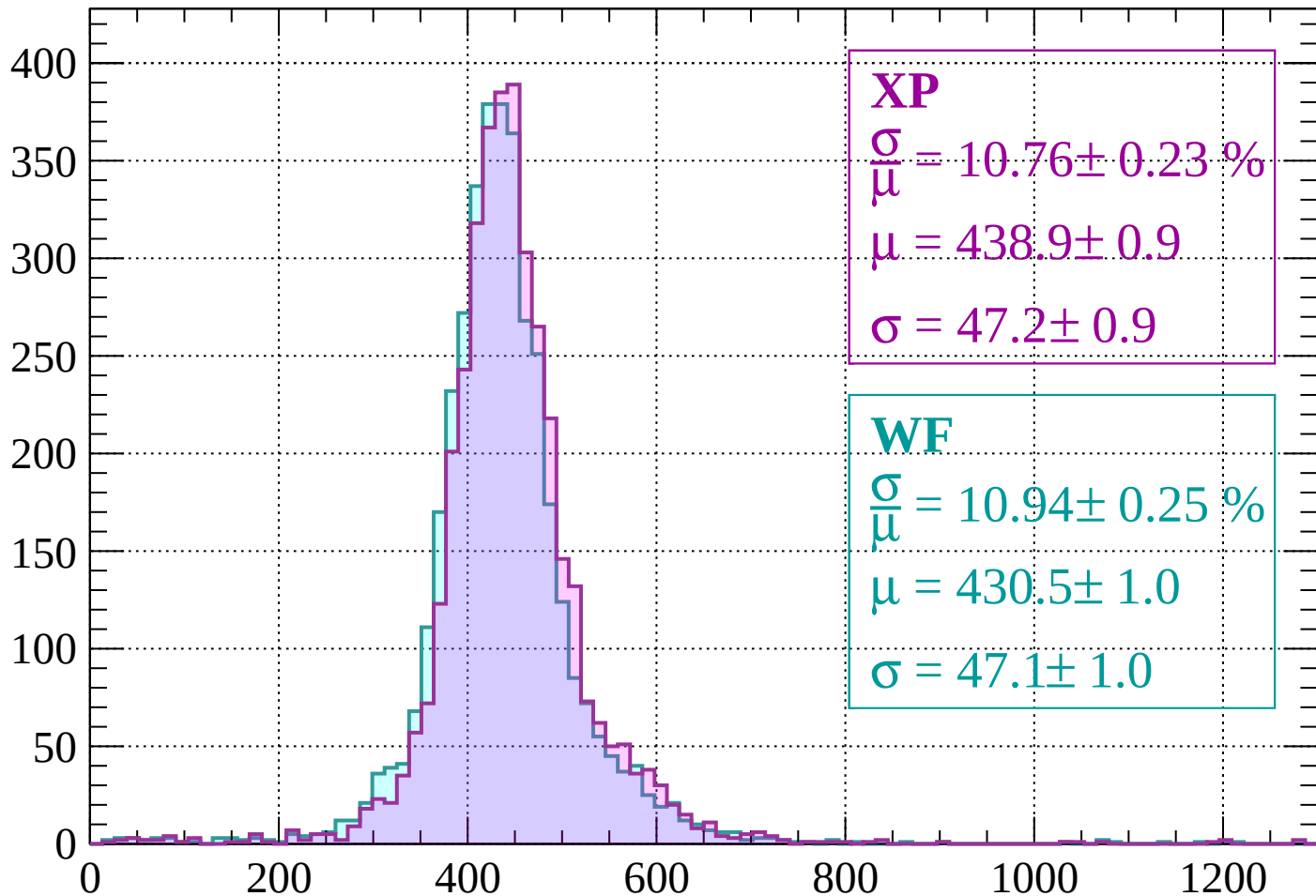
Energy loss | $160 < p < 200$

Count



Energy loss | $200 < p < 240$

Count



XP

$$\frac{\sigma}{\mu} = 10.76 \pm 0.23 \%$$

$$\mu = 438.9 \pm 0.9$$

$$\sigma = 47.2 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 10.94 \pm 0.25 \%$$

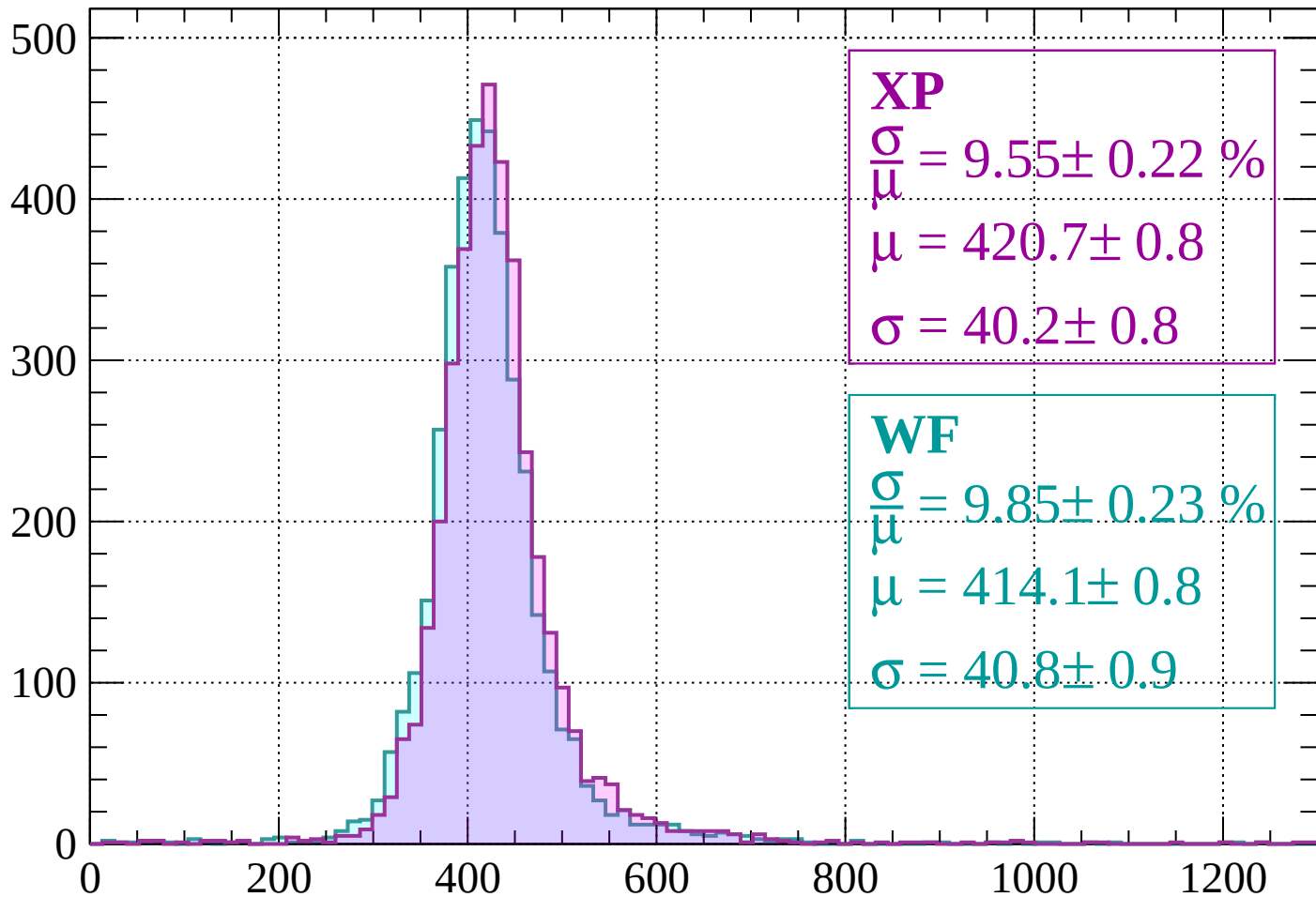
$$\mu = 430.5 \pm 1.0$$

$$\sigma = 47.1 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $240 < p < 280$

Count



XP

$$\frac{\sigma}{\mu} = 9.55 \pm 0.22 \%$$

$$\mu = 420.7 \pm 0.8$$

$$\sigma = 40.2 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.85 \pm 0.23 \%$$

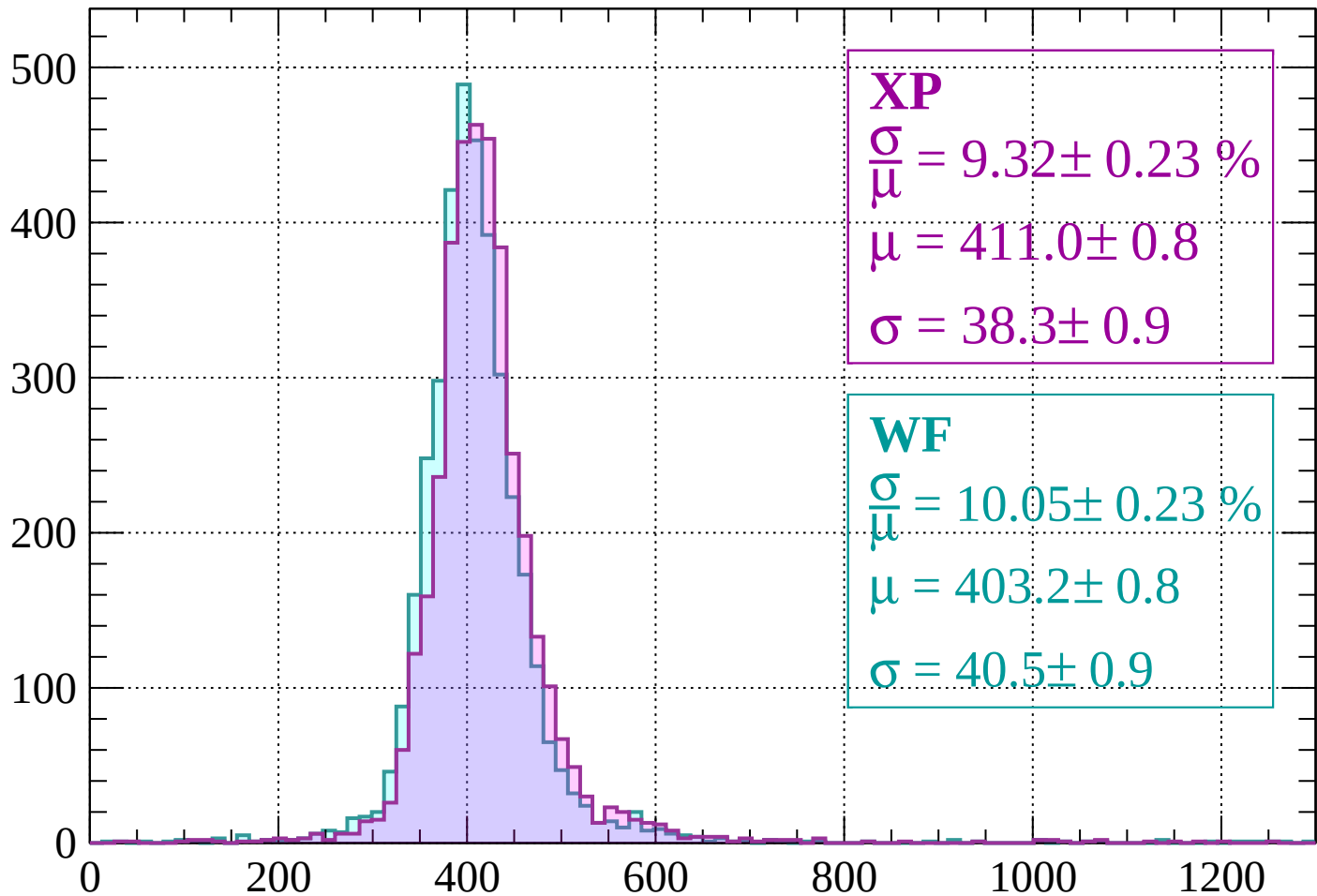
$$\mu = 414.1 \pm 0.8$$

$$\sigma = 40.8 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $280 < p < 320$

Count



XP

$$\frac{\sigma}{\mu} = 9.32 \pm 0.23 \%$$

$$\mu = 411.0 \pm 0.8$$

$$\sigma = 38.3 \pm 0.9$$

WF

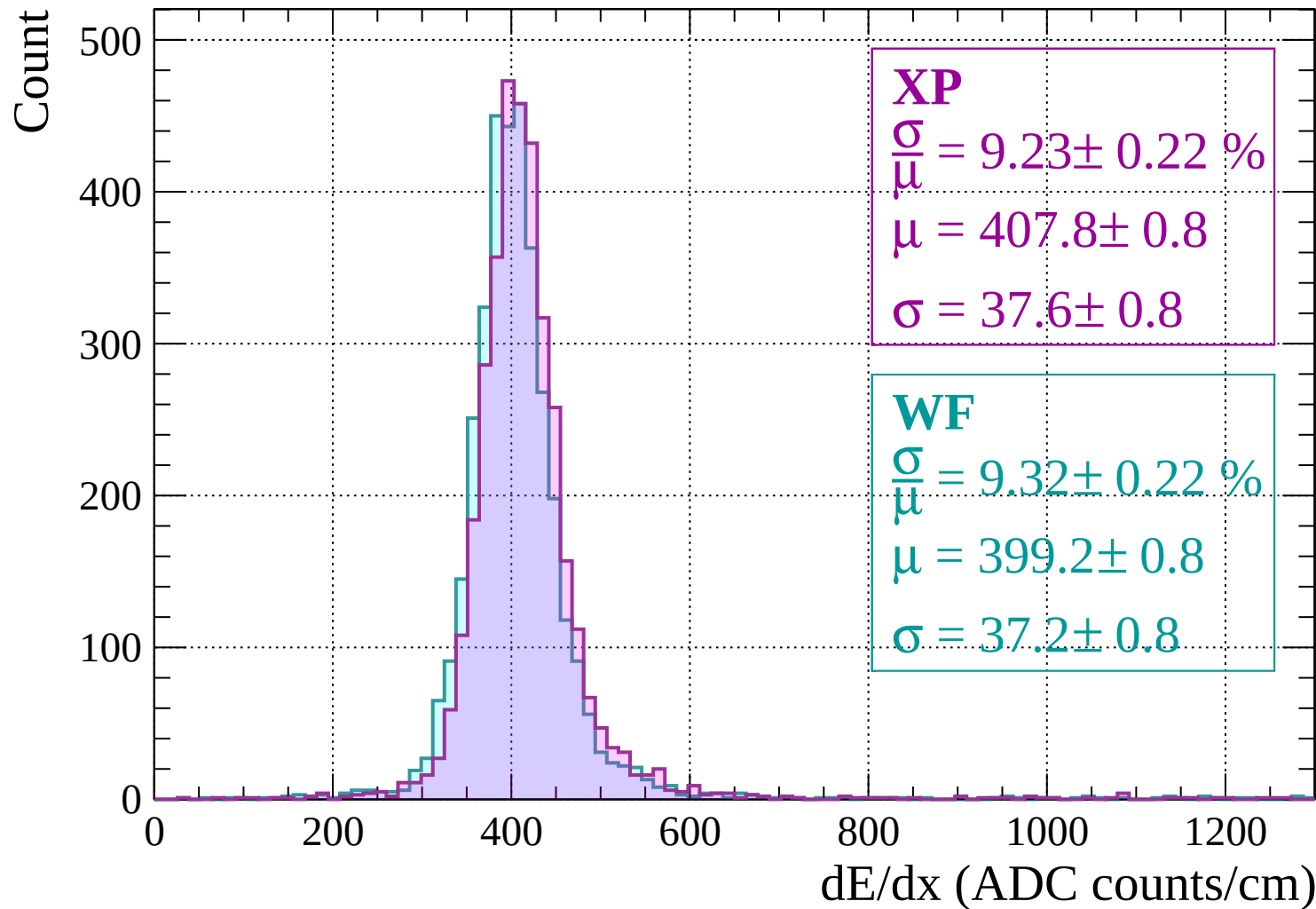
$$\frac{\sigma}{\mu} = 10.05 \pm 0.23 \%$$

$$\mu = 403.2 \pm 0.8$$

$$\sigma = 40.5 \pm 0.9$$

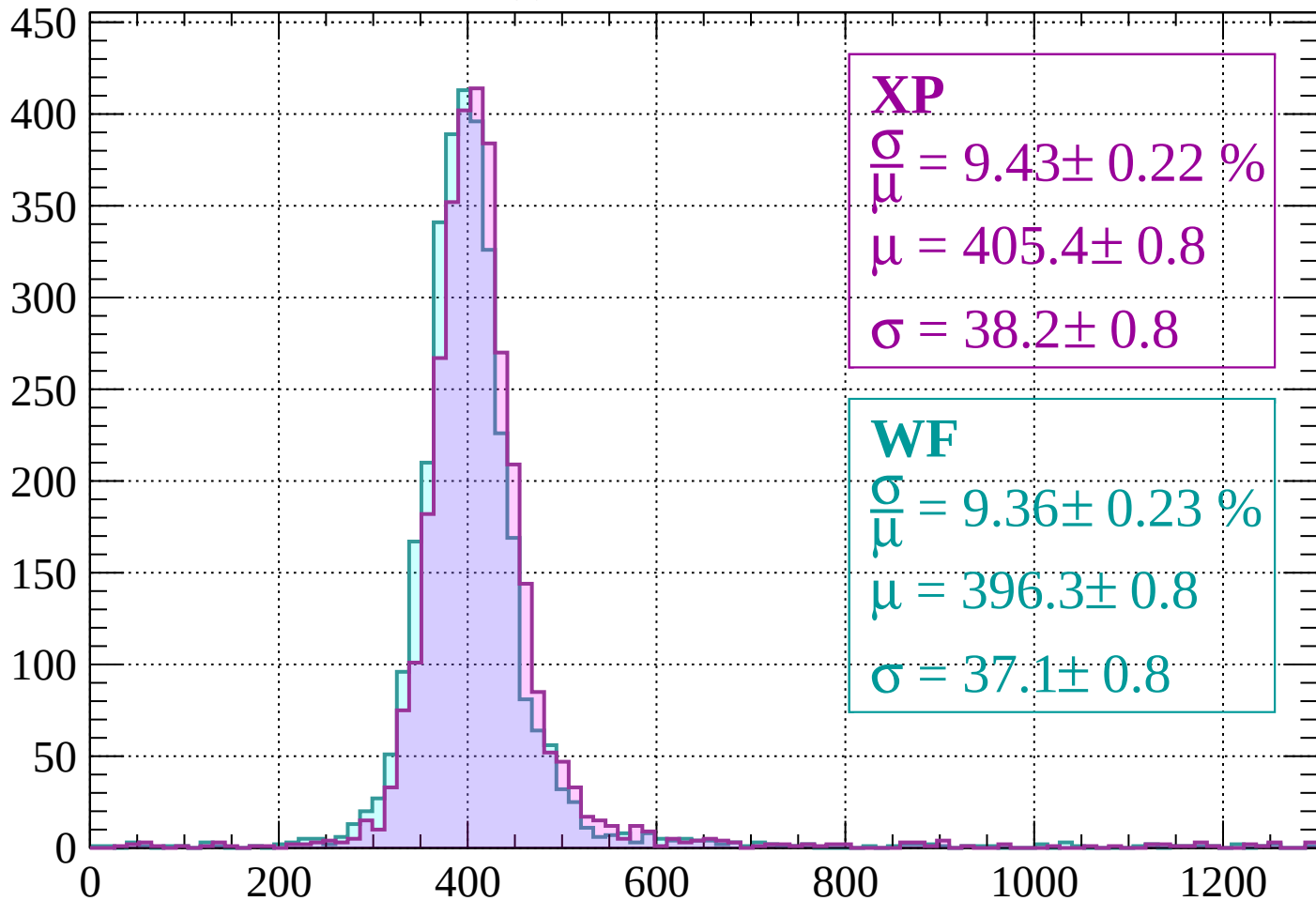
dE/dx (ADC counts/cm)

Energy loss | $320 < p < 360$



Energy loss | $360 < p < 400$

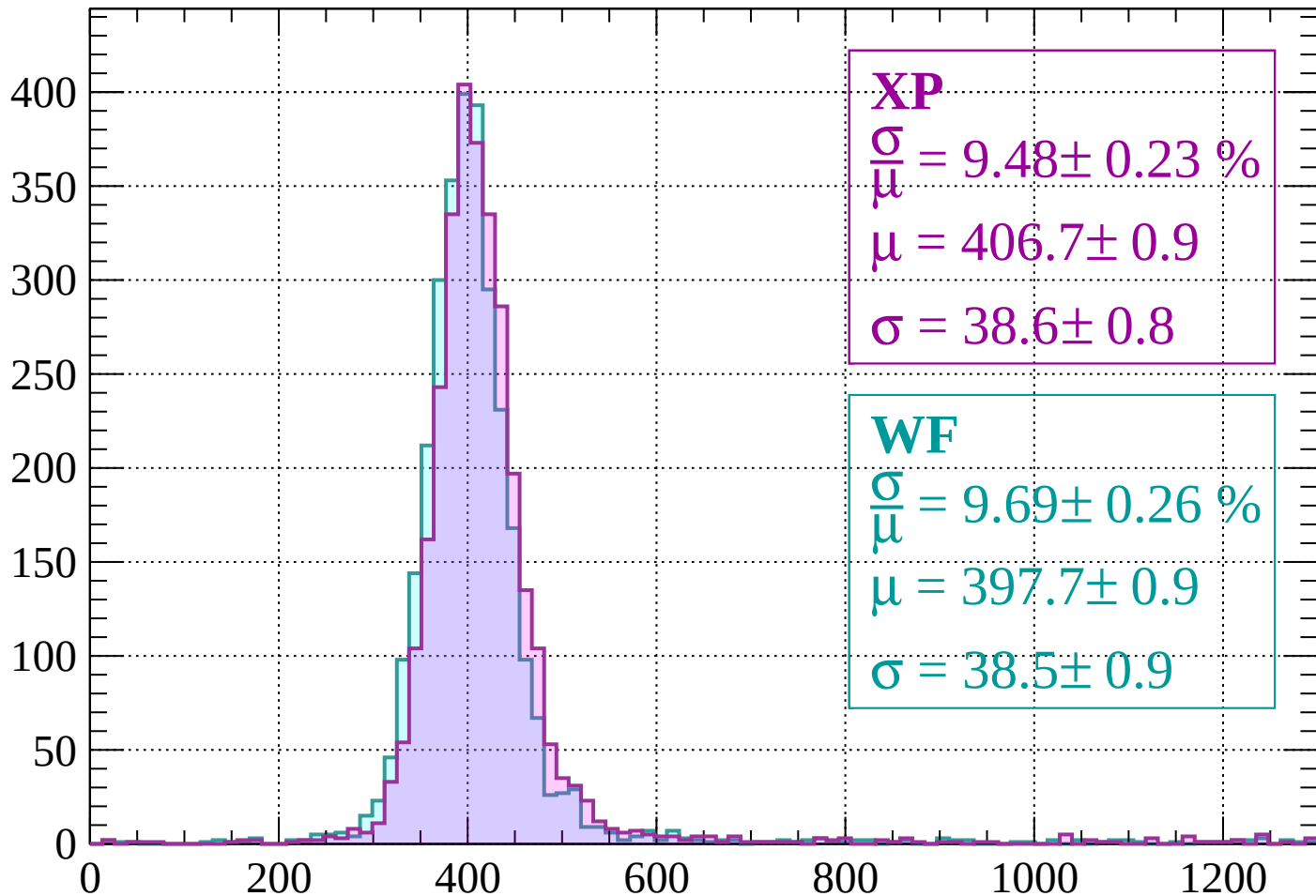
Count



dE/dx (ADC counts/cm)

Energy loss | $400 < p < 440$

Count



XP

$$\frac{\sigma}{\mu} = 9.48 \pm 0.23 \%$$

$$\mu = 406.7 \pm 0.9$$

$$\sigma = 38.6 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.69 \pm 0.26 \%$$

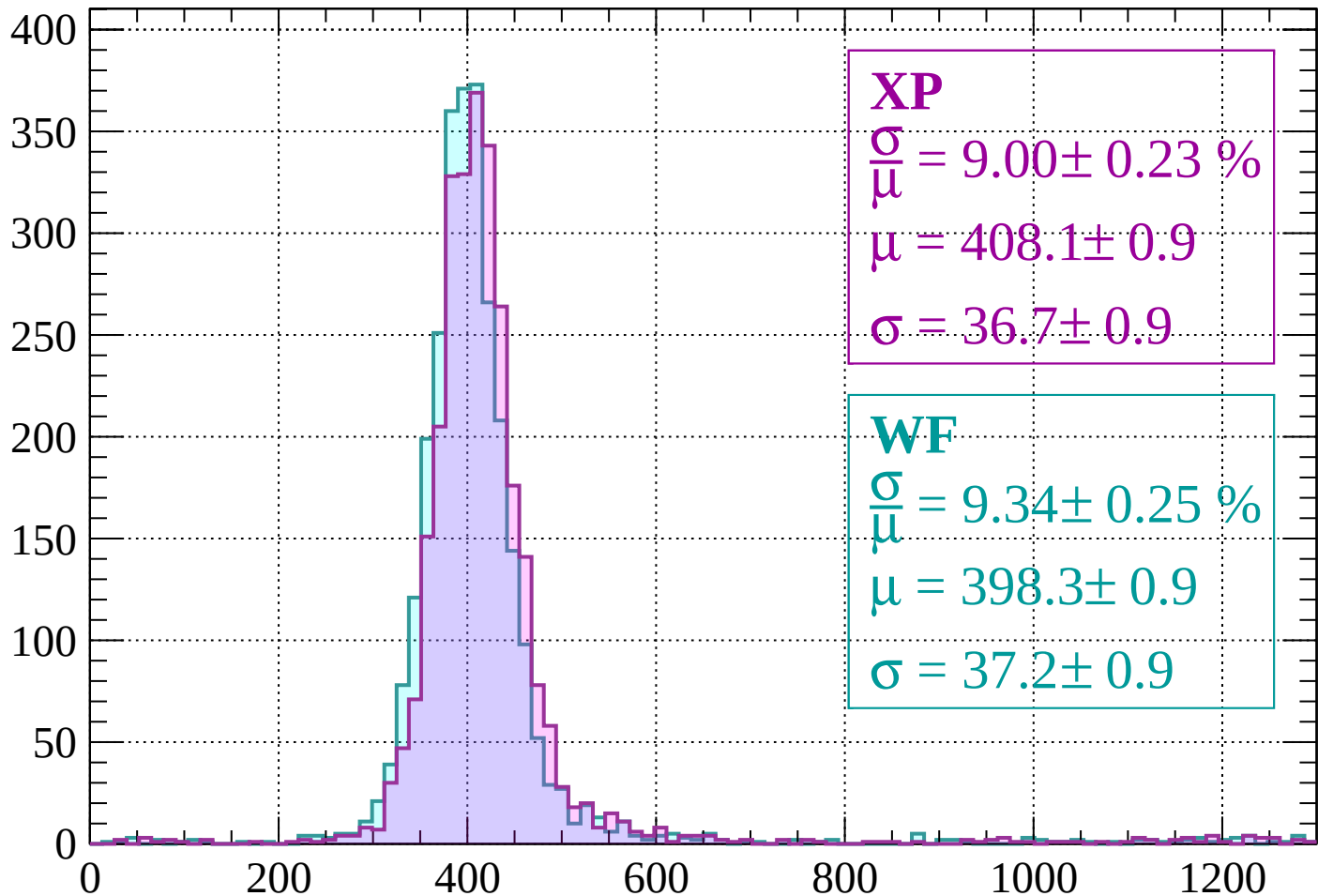
$$\mu = 397.7 \pm 0.9$$

$$\sigma = 38.5 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $440 < p < 480$

Count



XP

$$\frac{\sigma}{\mu} = 9.00 \pm 0.23 \%$$

$$\mu = 408.1 \pm 0.9$$

$$\sigma = 36.7 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.34 \pm 0.25 \%$$

$$\mu = 398.3 \pm 0.9$$

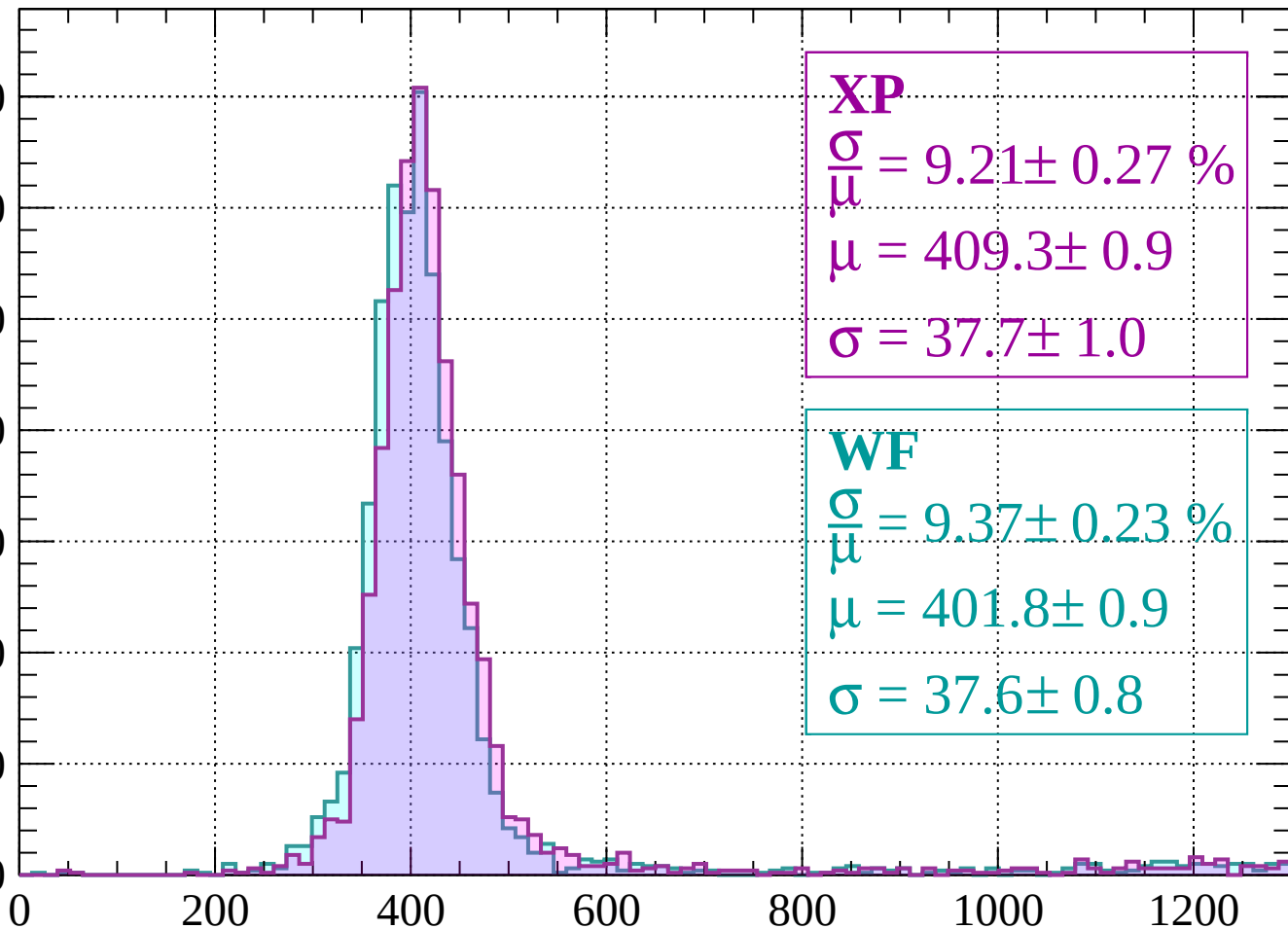
$$\sigma = 37.2 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $480 < p < 520$

Count

350
300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 9.21 \pm 0.27 \%$$

$$\mu = 409.3 \pm 0.9$$

$$\sigma = 37.7 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 9.37 \pm 0.23 \%$$

$$\mu = 401.8 \pm 0.9$$

$$\sigma = 37.6 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $520 < p < 560$

Count

300

250

200

150

100

50

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 8.58 \pm 0.22 \%$$

$$\mu = 412.2 \pm 0.9$$

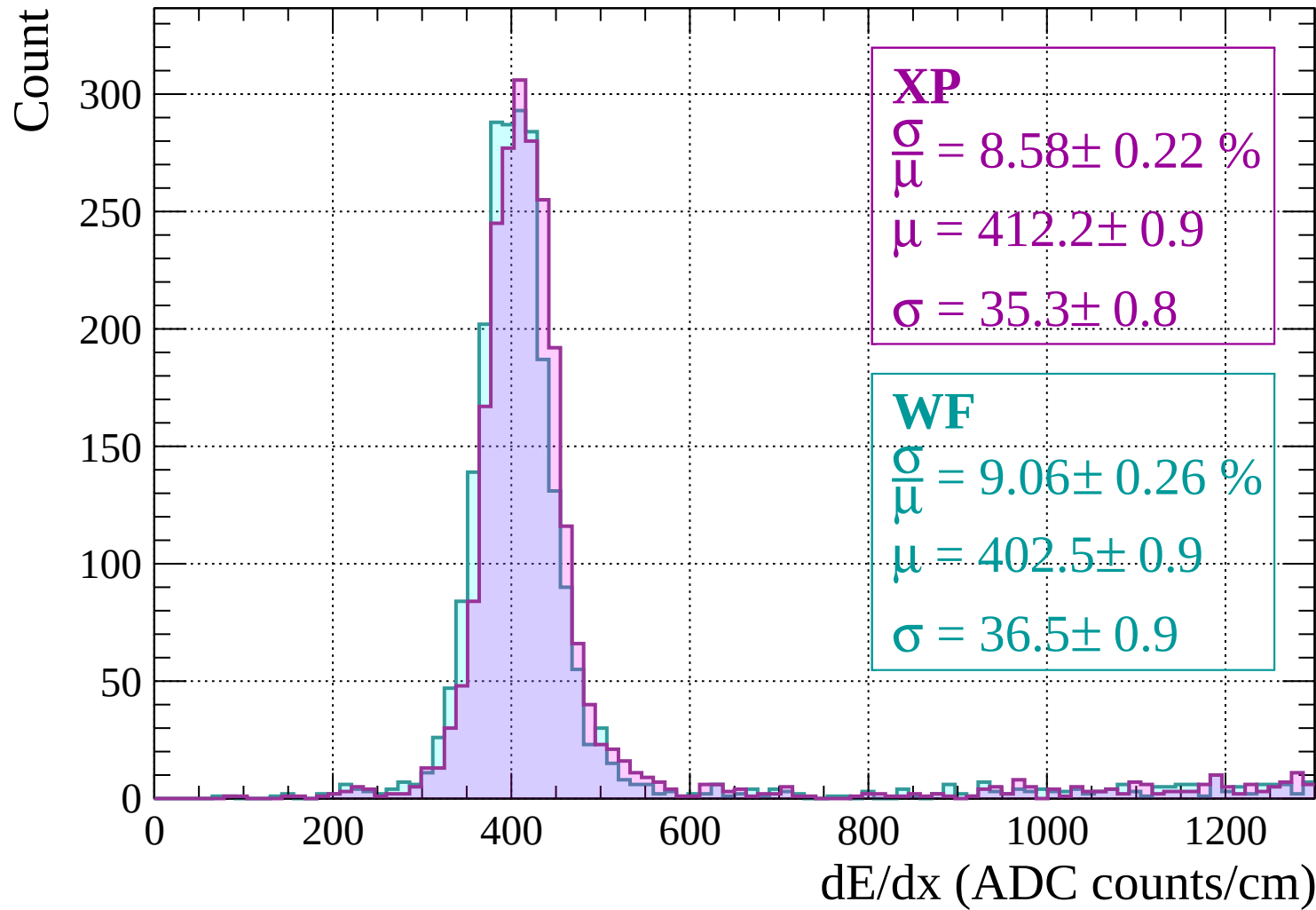
$$\sigma = 35.3 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.06 \pm 0.26 \%$$

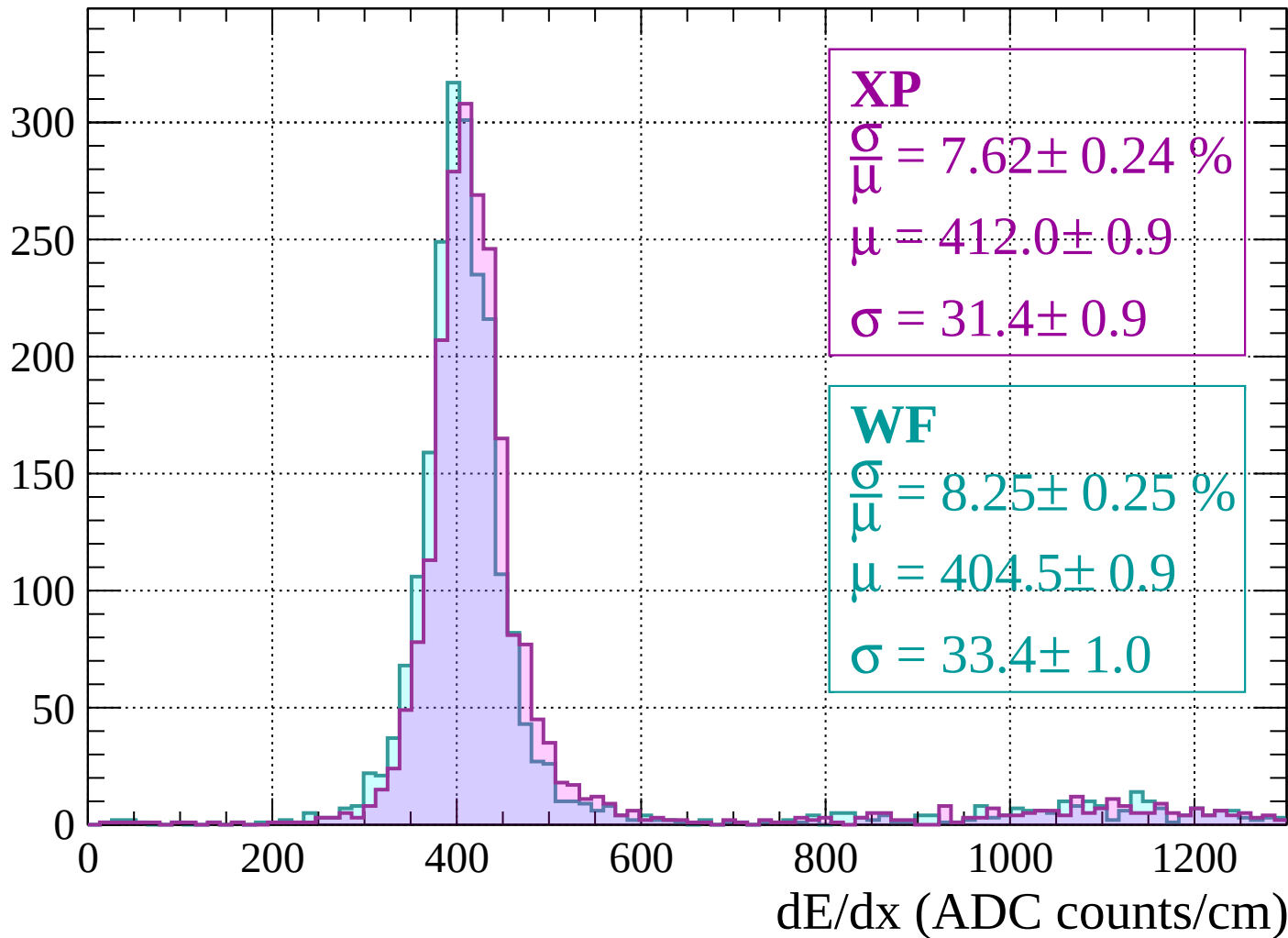
$$\mu = 402.5 \pm 0.9$$

$$\sigma = 36.5 \pm 0.9$$



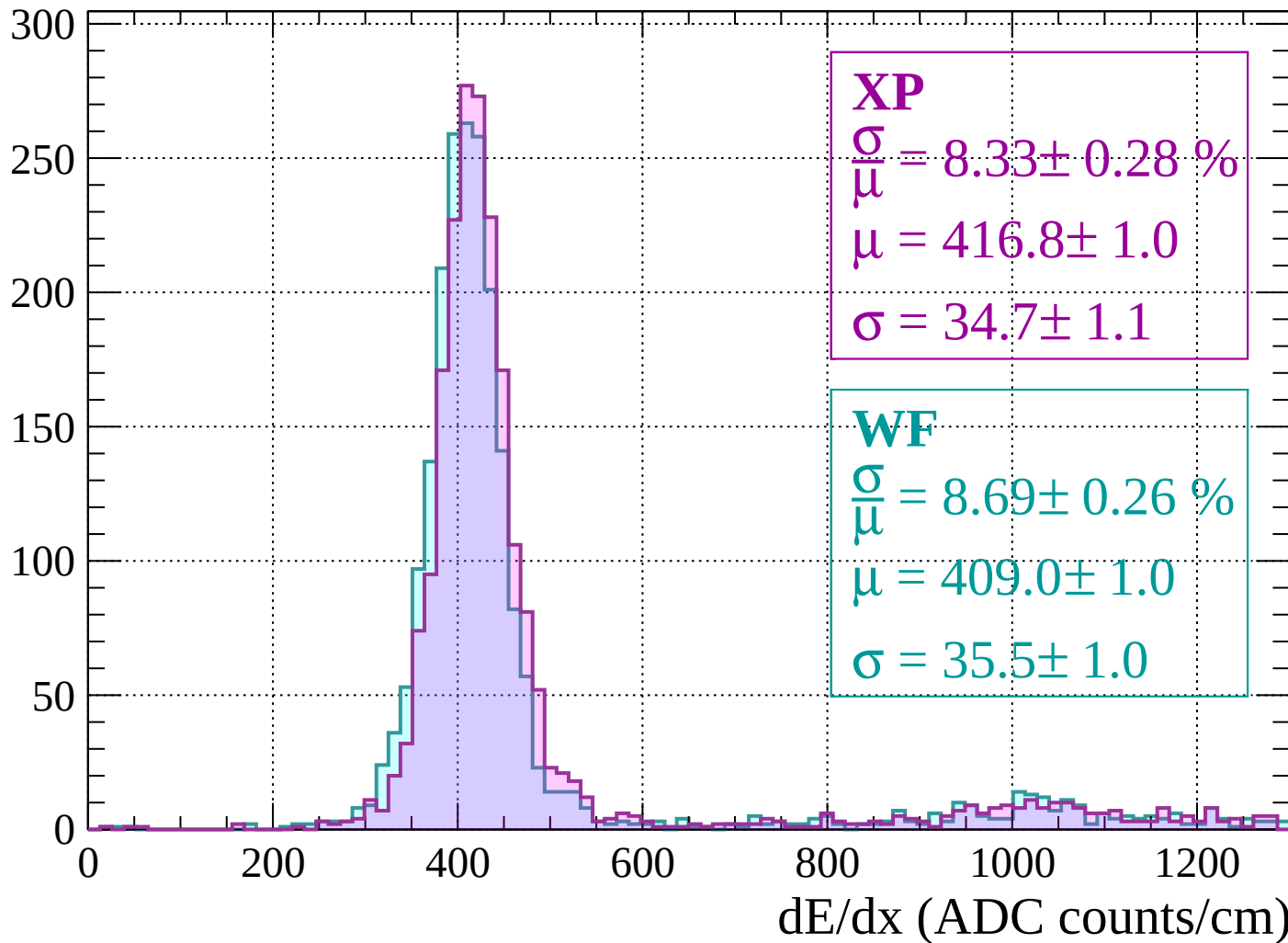
Energy loss | $560 < p < 600$

Count



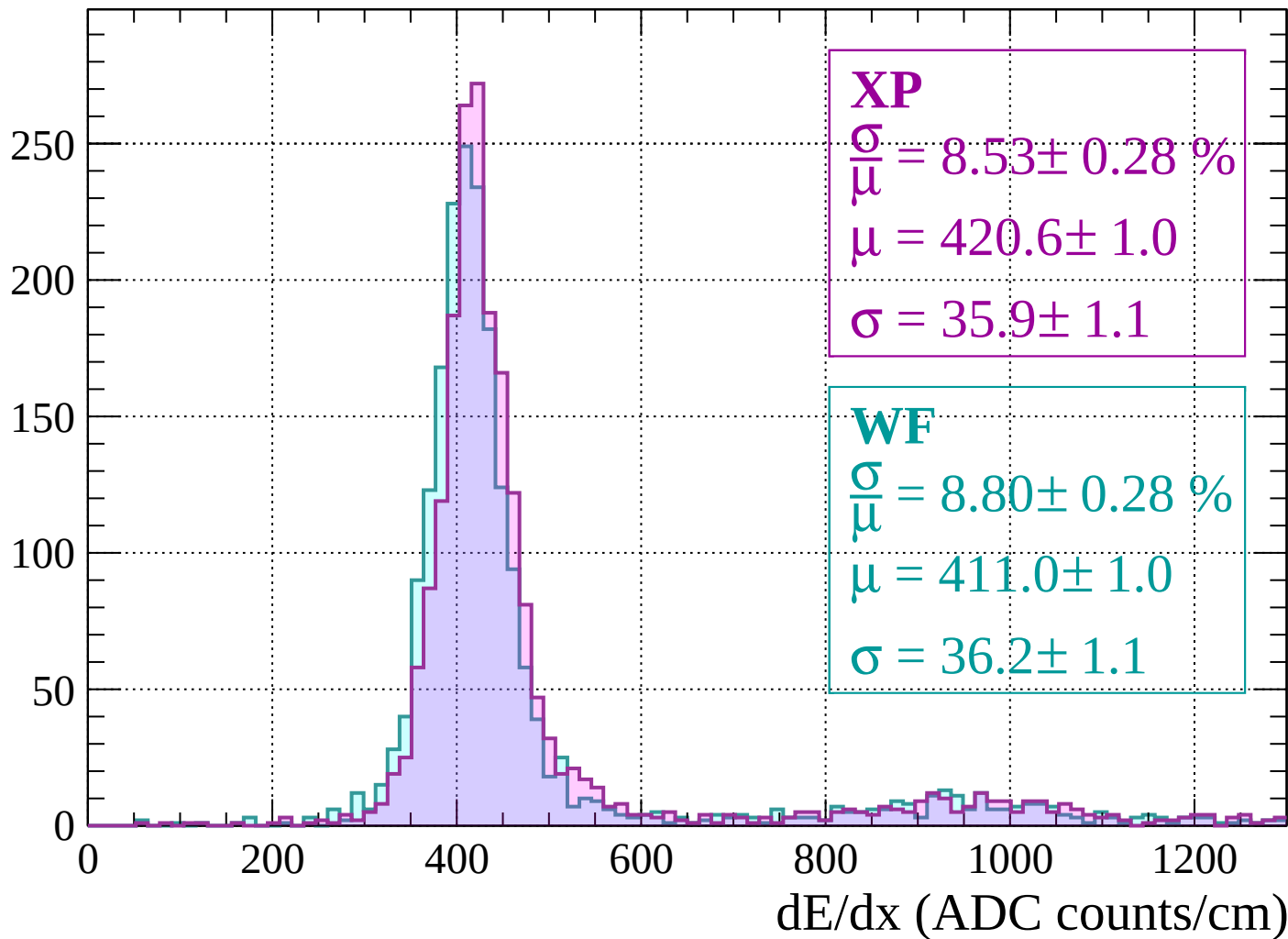
Energy loss | $600 < p < 640$

Count



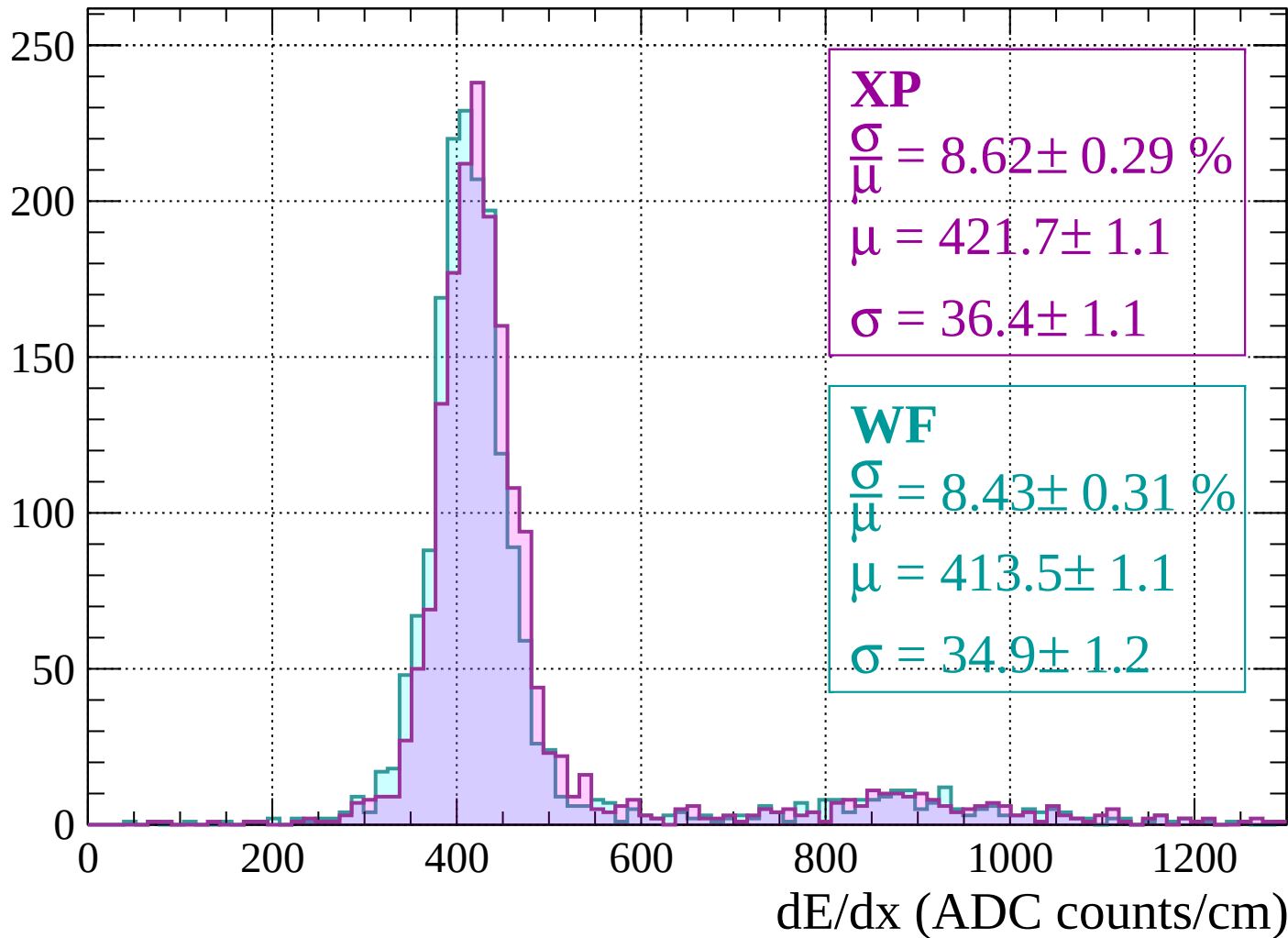
Energy loss | $640 < p < 680$

Count



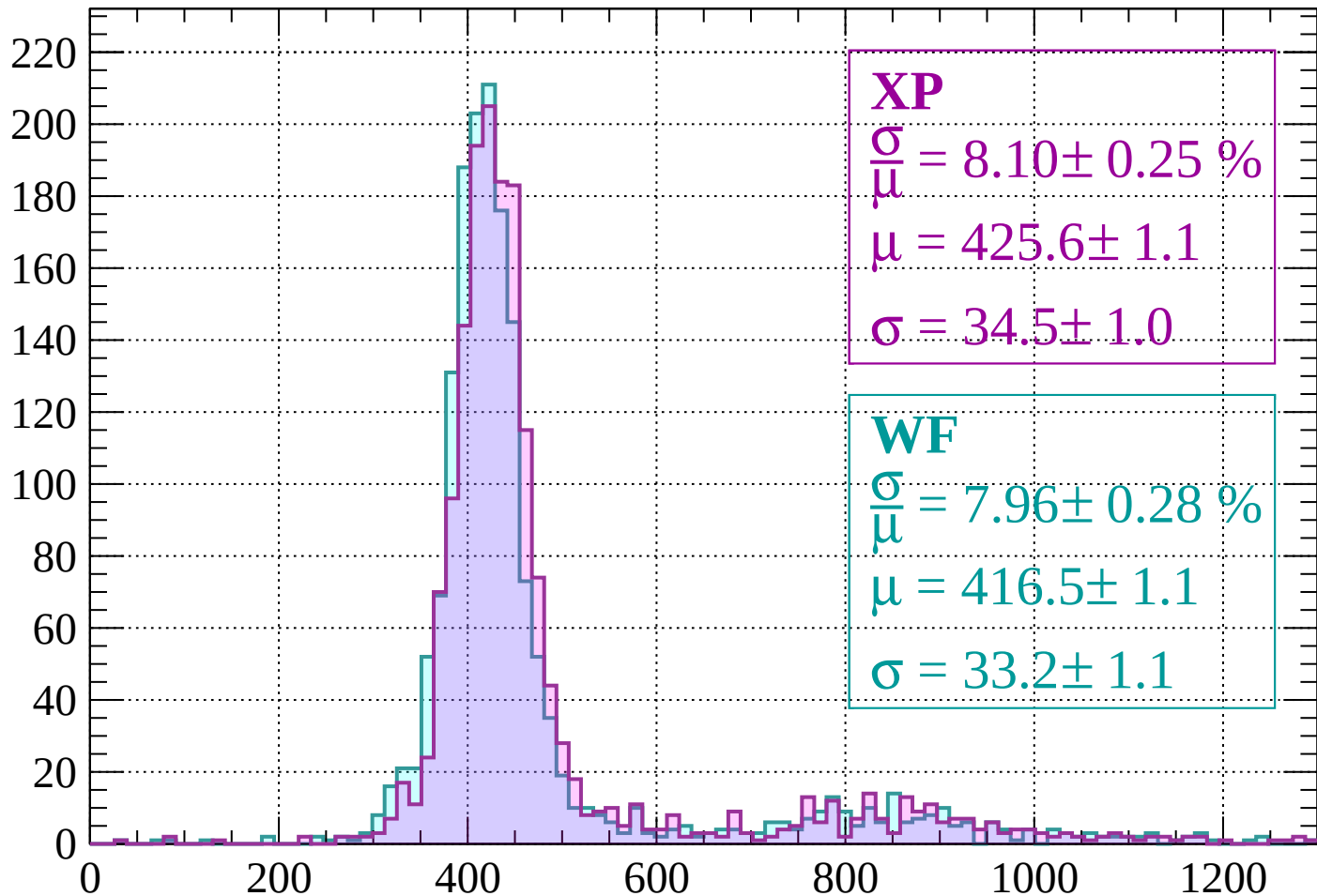
Energy loss | $680 < p < 720$

Count



Energy loss | $720 < p < 760$

Count



XP

$$\frac{\sigma}{\mu} = 8.10 \pm 0.25 \%$$

$$\mu = 425.6 \pm 1.1$$

$$\sigma = 34.5 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 7.96 \pm 0.28 \%$$

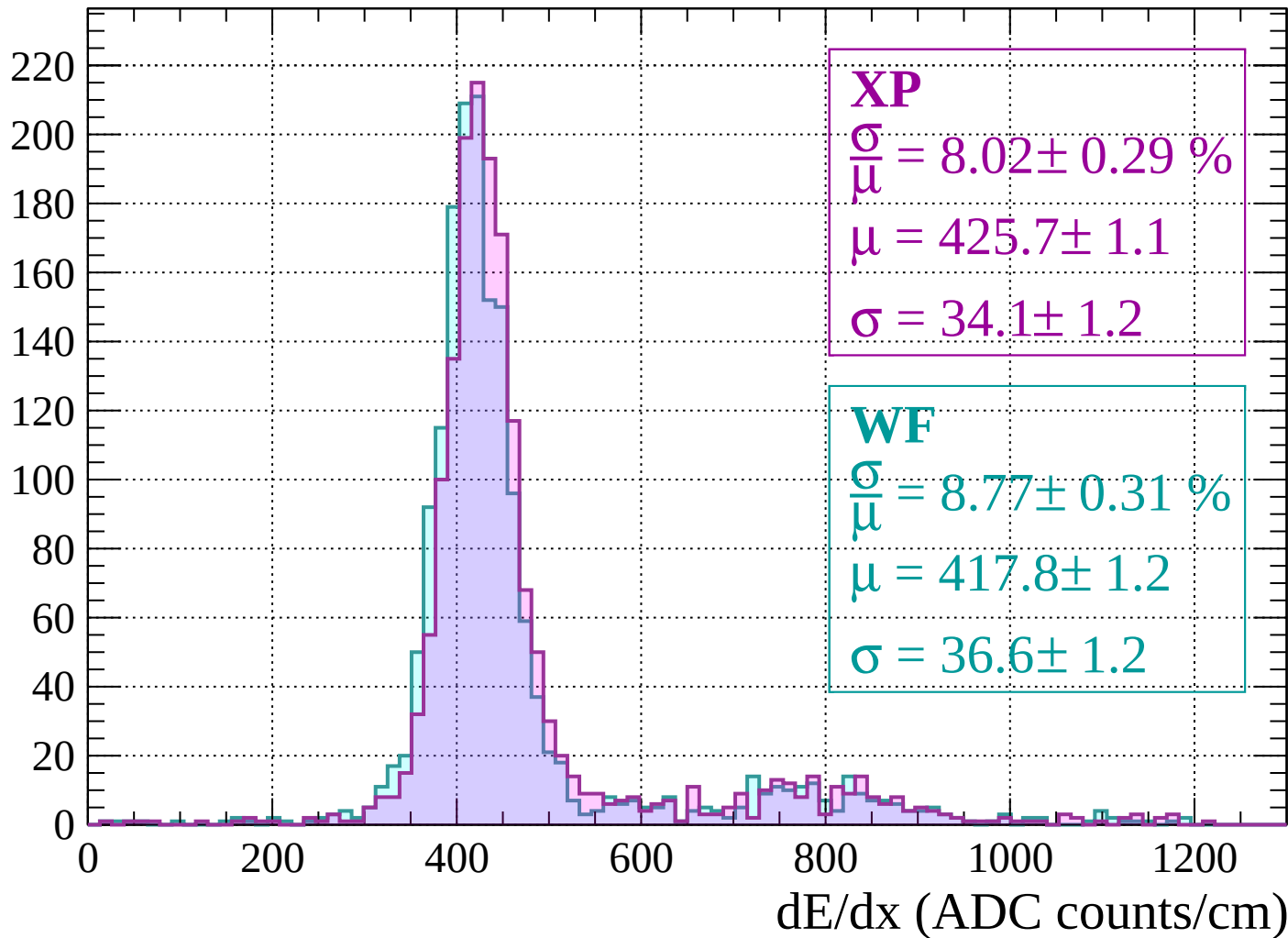
$$\mu = 416.5 \pm 1.1$$

$$\sigma = 33.2 \pm 1.1$$

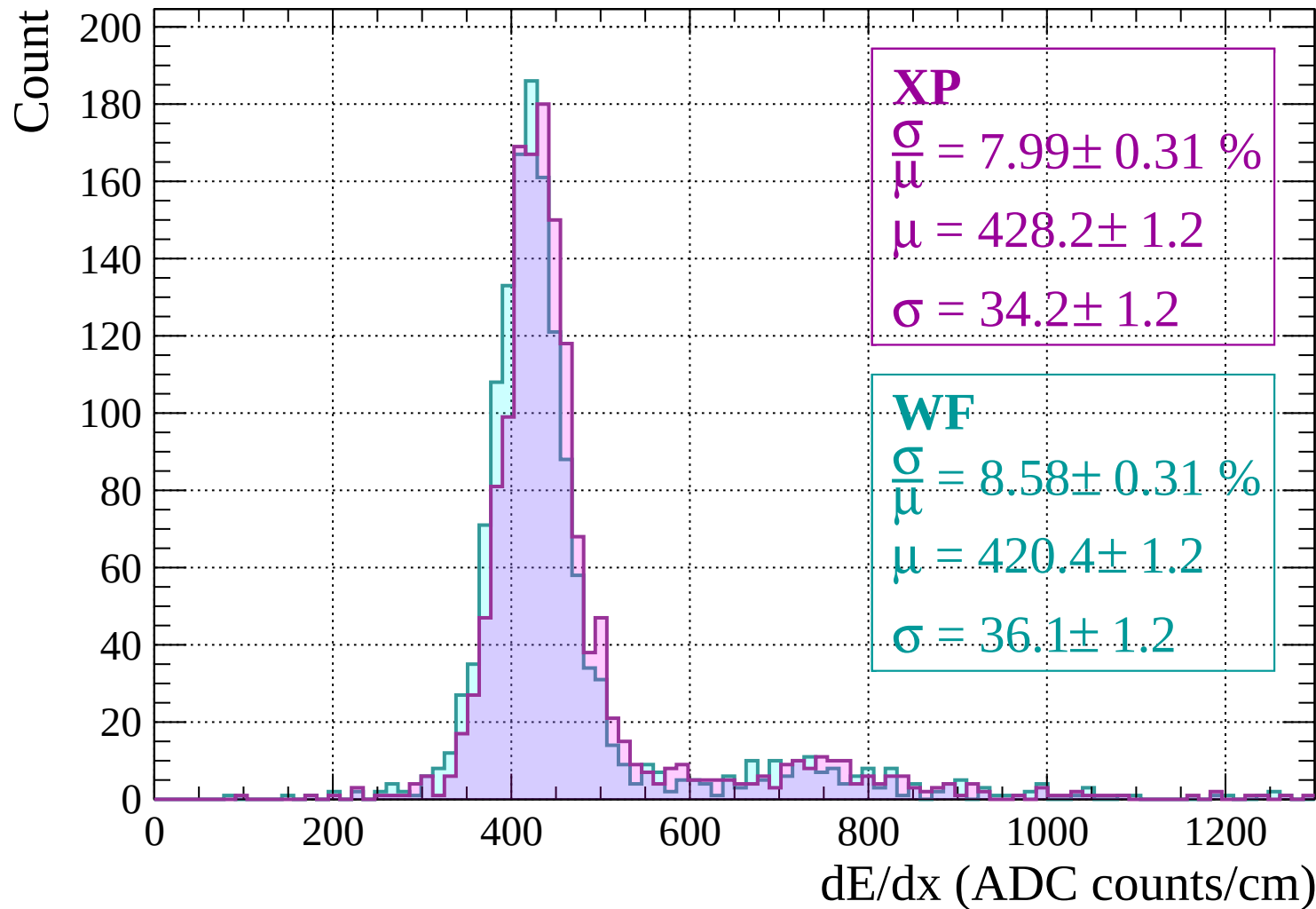
dE/dx (ADC counts/cm)

Energy loss | $760 < p < 800$

Count

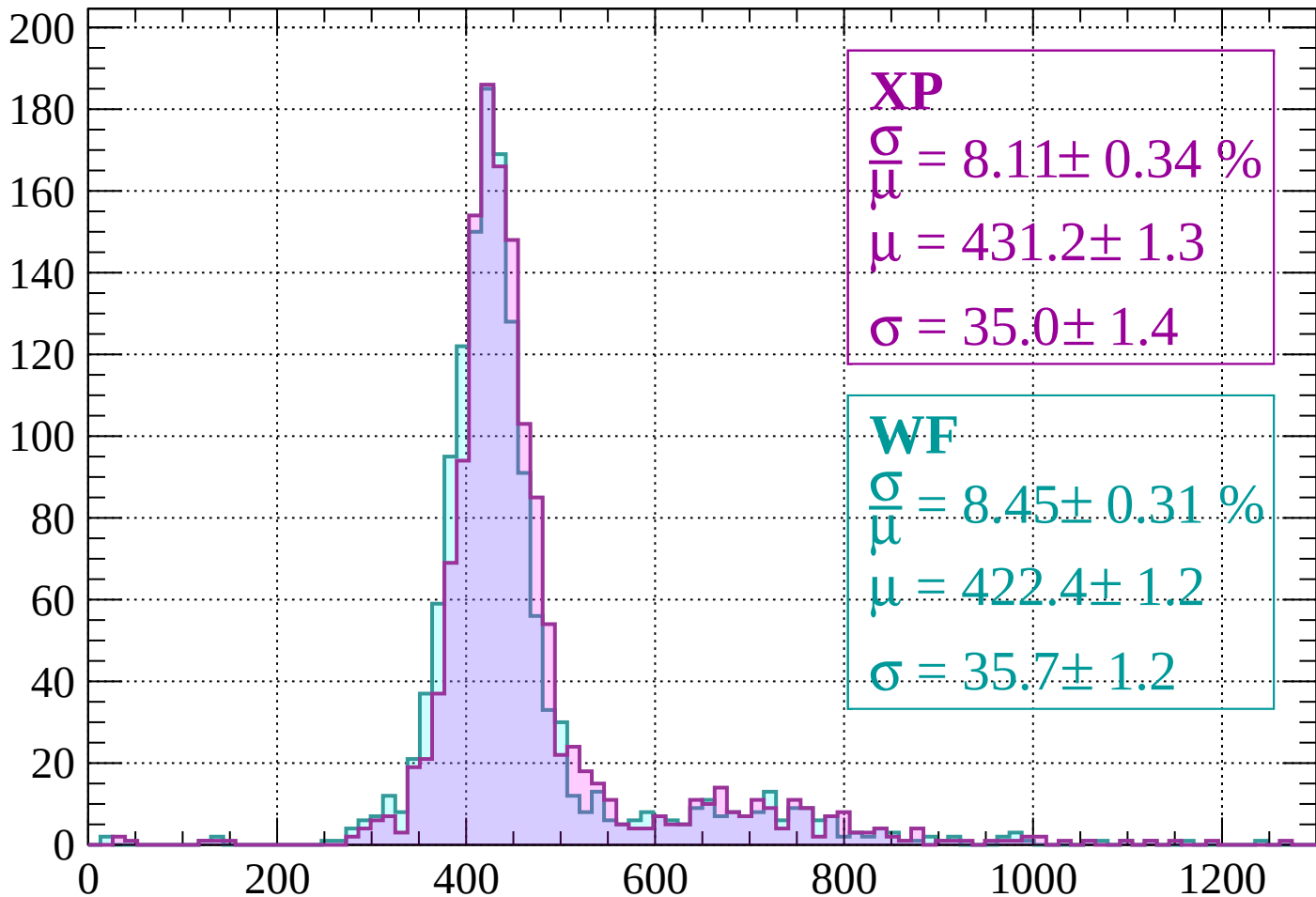


Energy loss | $800 < p < 840$



Energy loss | $840 < p < 880$

Count



XP

$$\frac{\sigma}{\mu} = 8.11 \pm 0.34 \%$$

$$\mu = 431.2 \pm 1.3$$

$$\sigma = 35.0 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 8.45 \pm 0.31 \%$$

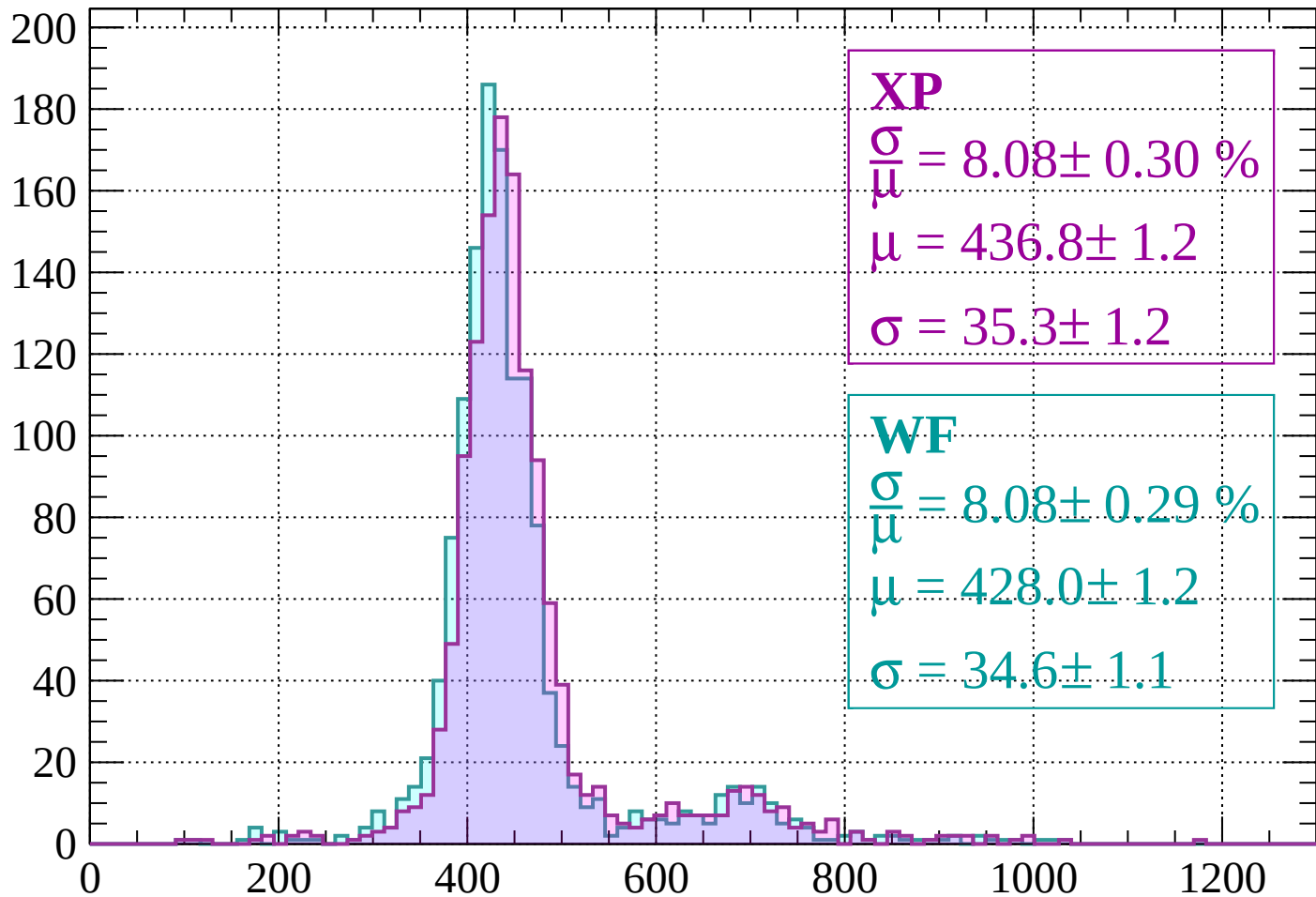
$$\mu = 422.4 \pm 1.2$$

$$\sigma = 35.7 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $880 < p < 920$

Count



XP

$$\frac{\sigma}{\mu} = 8.08 \pm 0.30 \%$$

$$\mu = 436.8 \pm 1.2$$

$$\sigma = 35.3 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 8.08 \pm 0.29 \%$$

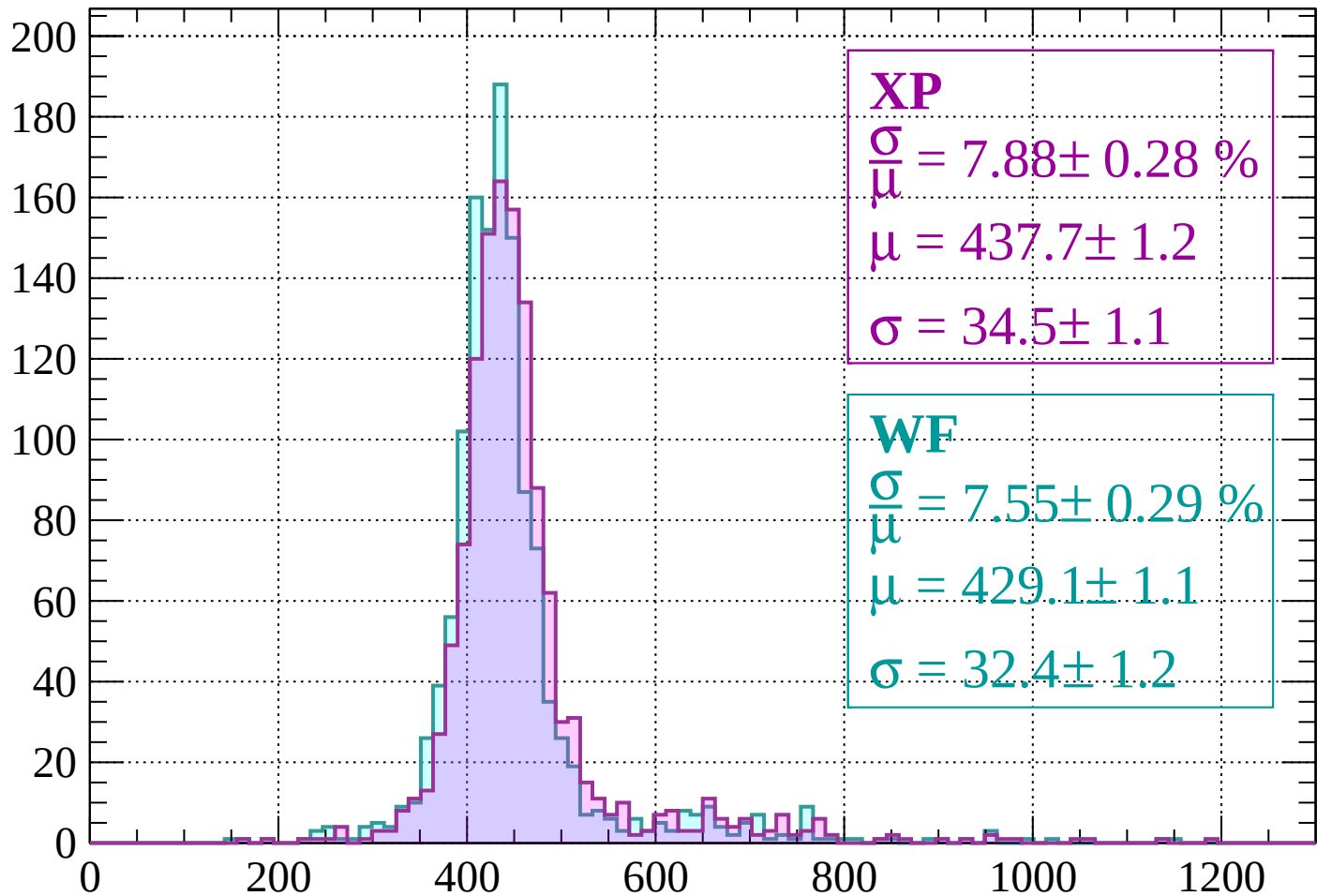
$$\mu = 428.0 \pm 1.2$$

$$\sigma = 34.6 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $920 < p < 960$

Count



XP

$$\frac{\sigma}{\mu} = 7.88 \pm 0.28 \%$$

$$\mu = 437.7 \pm 1.2$$

$$\sigma = 34.5 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 7.55 \pm 0.29 \%$$

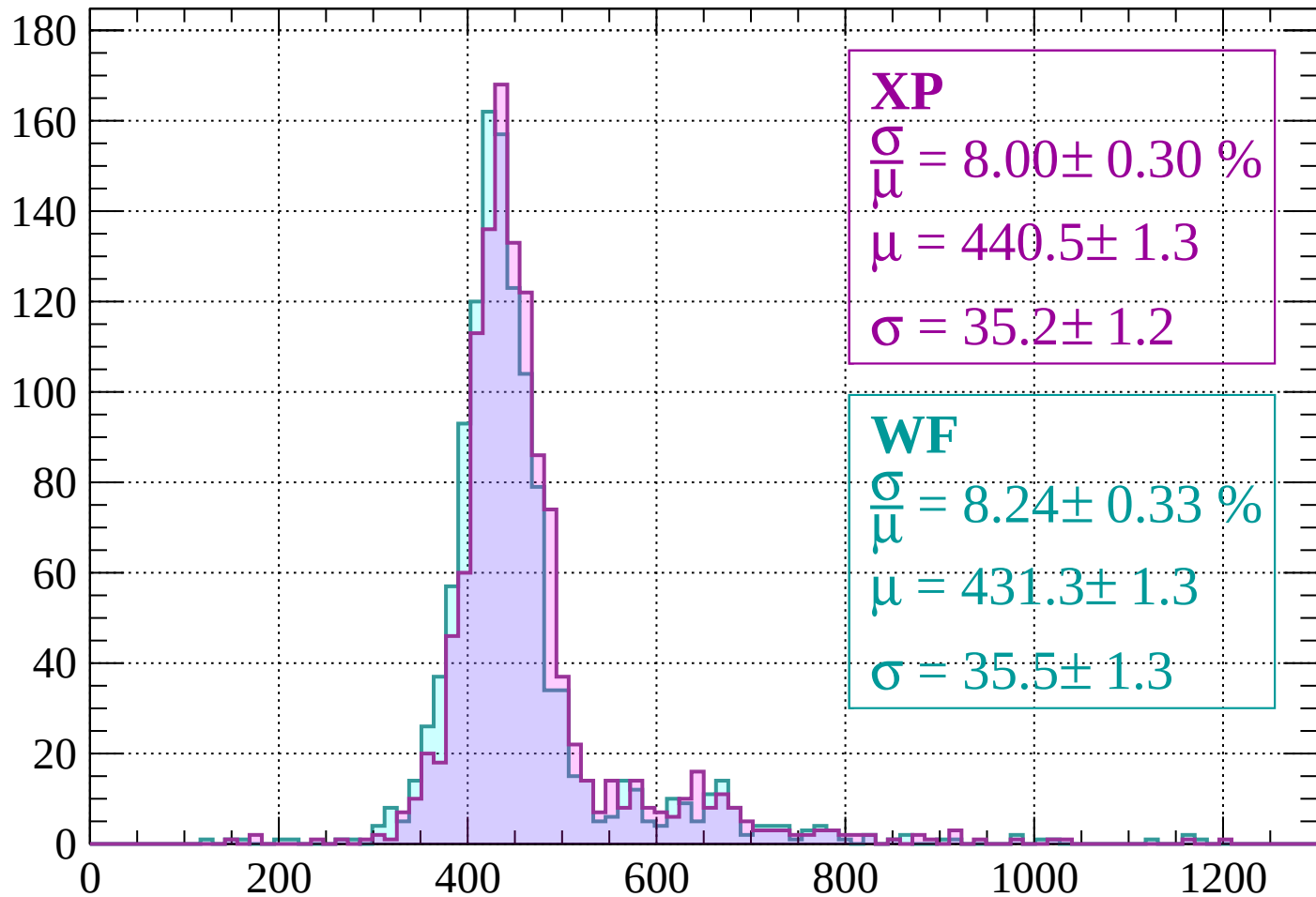
$$\mu = 429.1 \pm 1.1$$

$$\sigma = 32.4 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $960 < p < 1000$

Count



XP

$$\frac{\sigma}{\mu} = 8.00 \pm 0.30 \%$$

$$\mu = 440.5 \pm 1.3$$

$$\sigma = 35.2 \pm 1.2$$

WF

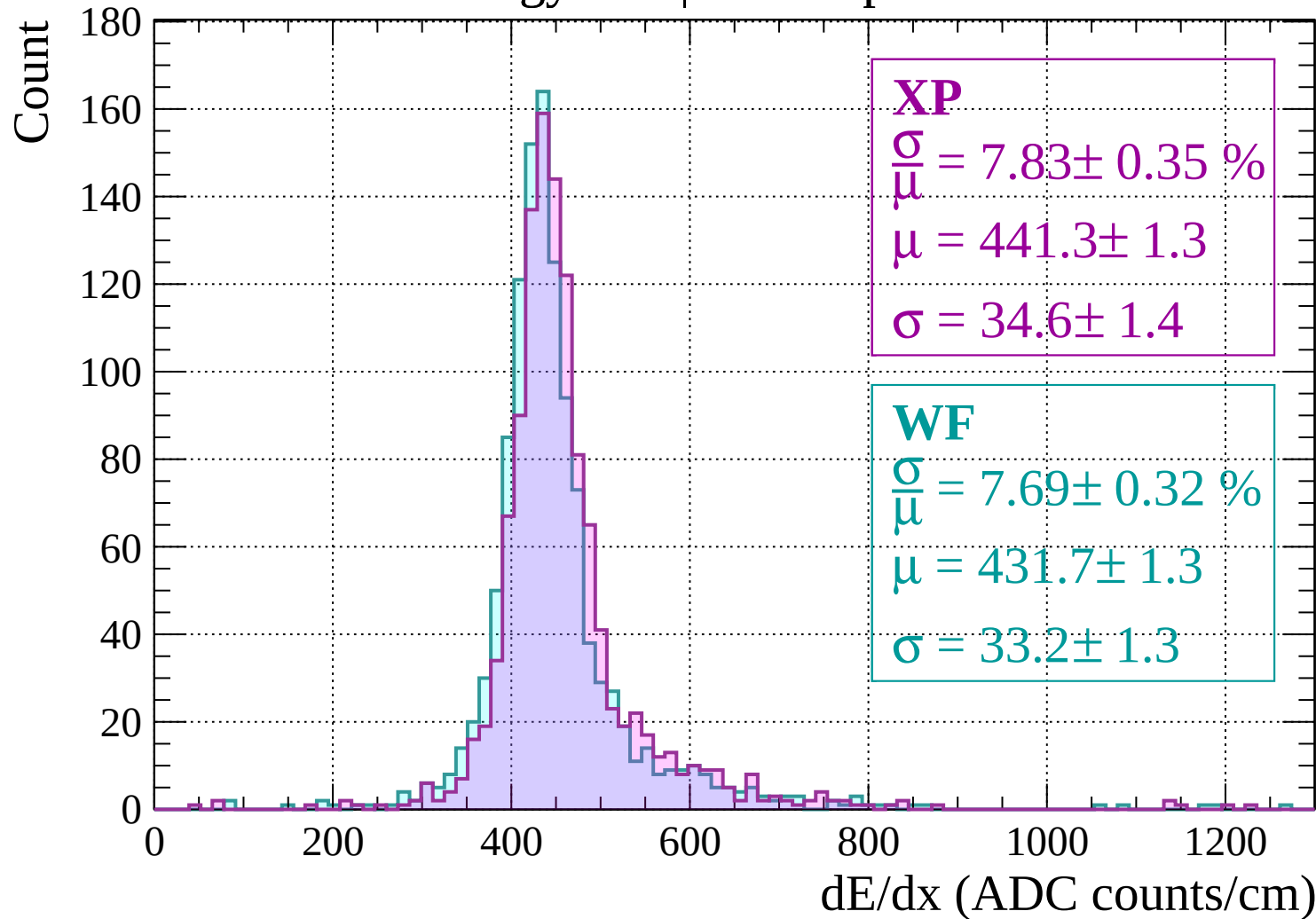
$$\frac{\sigma}{\mu} = 8.24 \pm 0.33 \%$$

$$\mu = 431.3 \pm 1.3$$

$$\sigma = 35.5 \pm 1.3$$

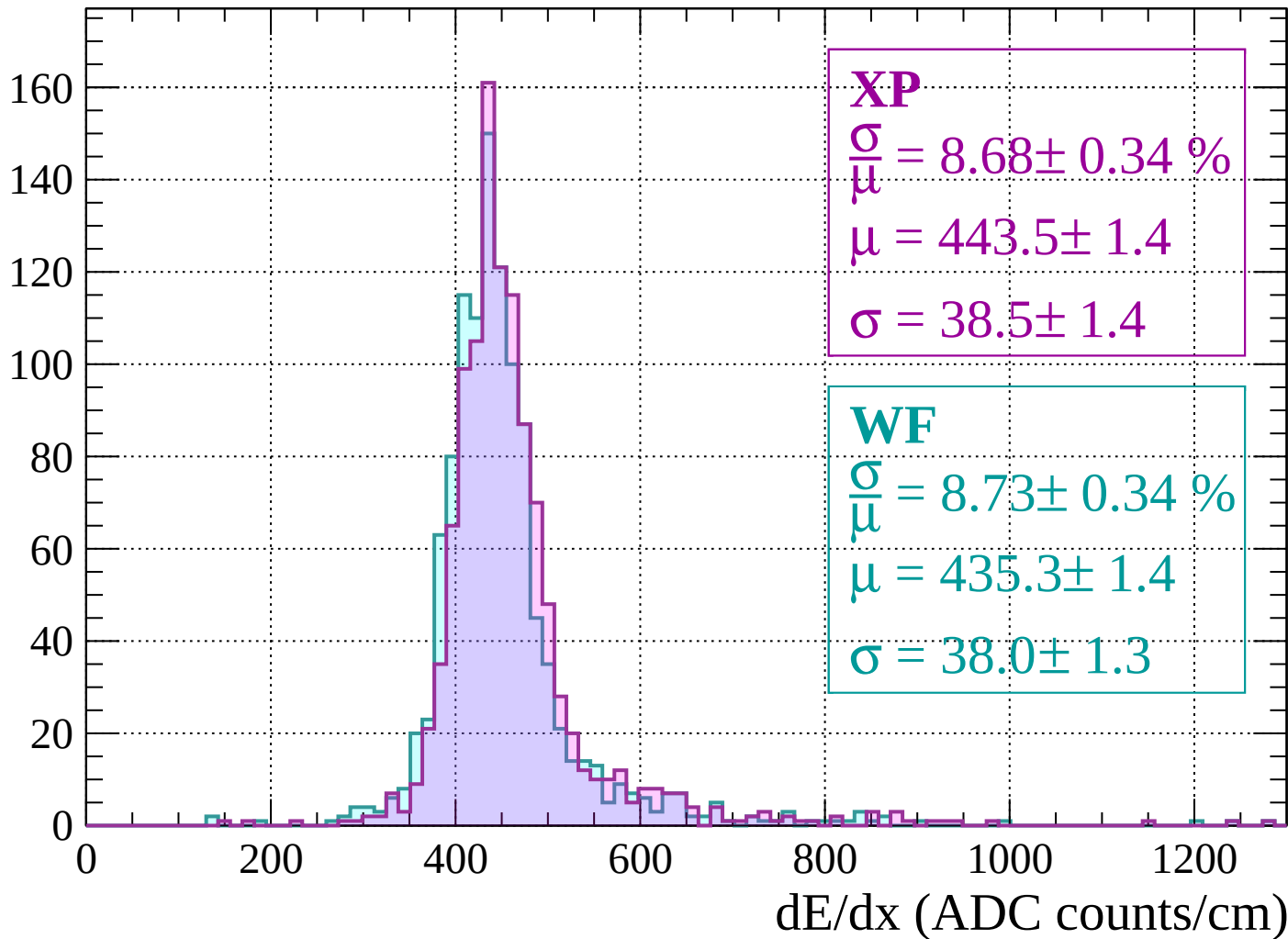
dE/dx (ADC counts/cm)

Energy loss | $1000 < p < 1040$



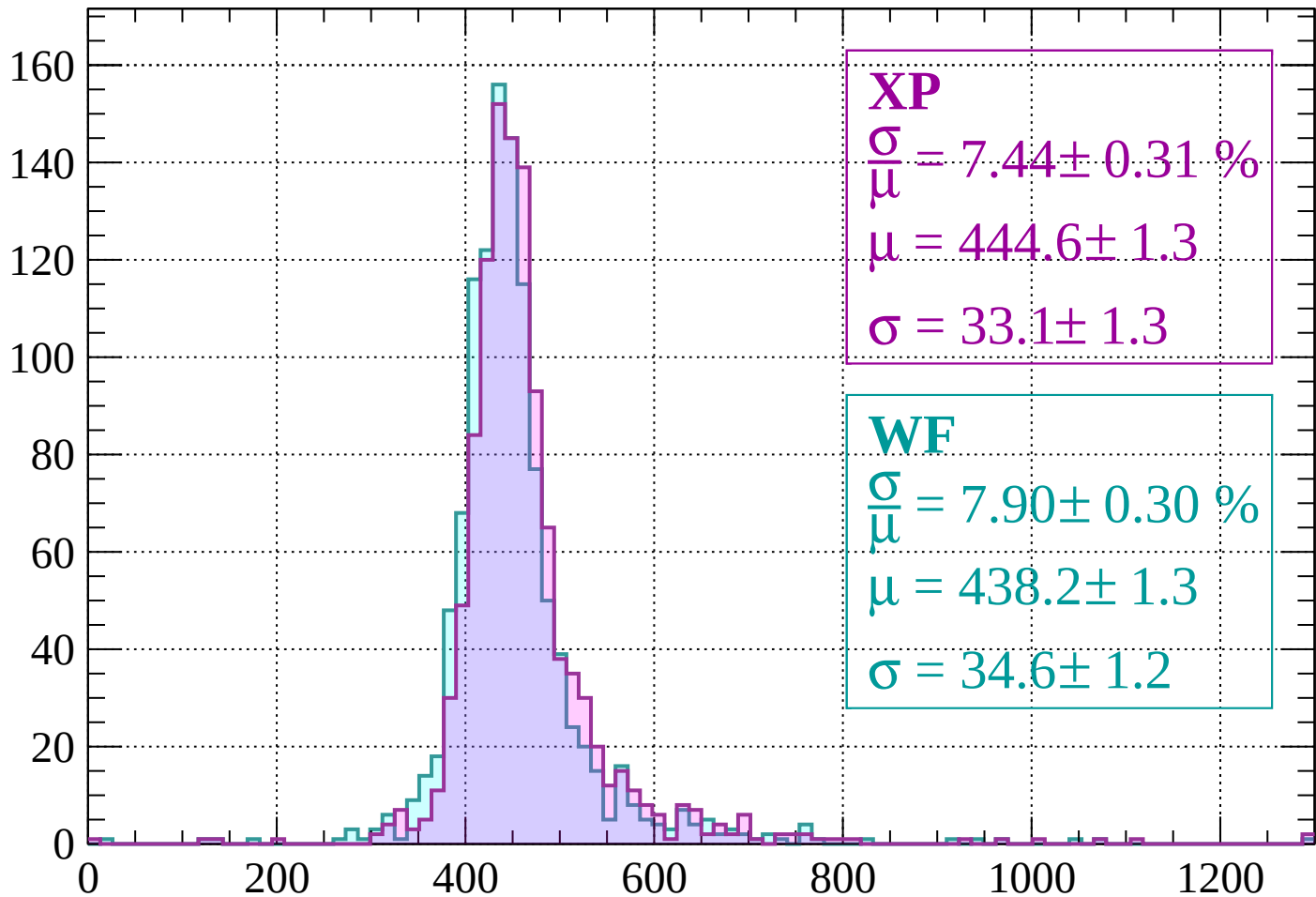
Energy loss | $1040 < p < 1080$

Count



Energy loss | $1080 < p < 1120$

Count



XP

$$\frac{\sigma}{\mu} = 7.44 \pm 0.31 \%$$

$$\mu = 444.6 \pm 1.3$$

$$\sigma = 33.1 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 7.90 \pm 0.30 \%$$

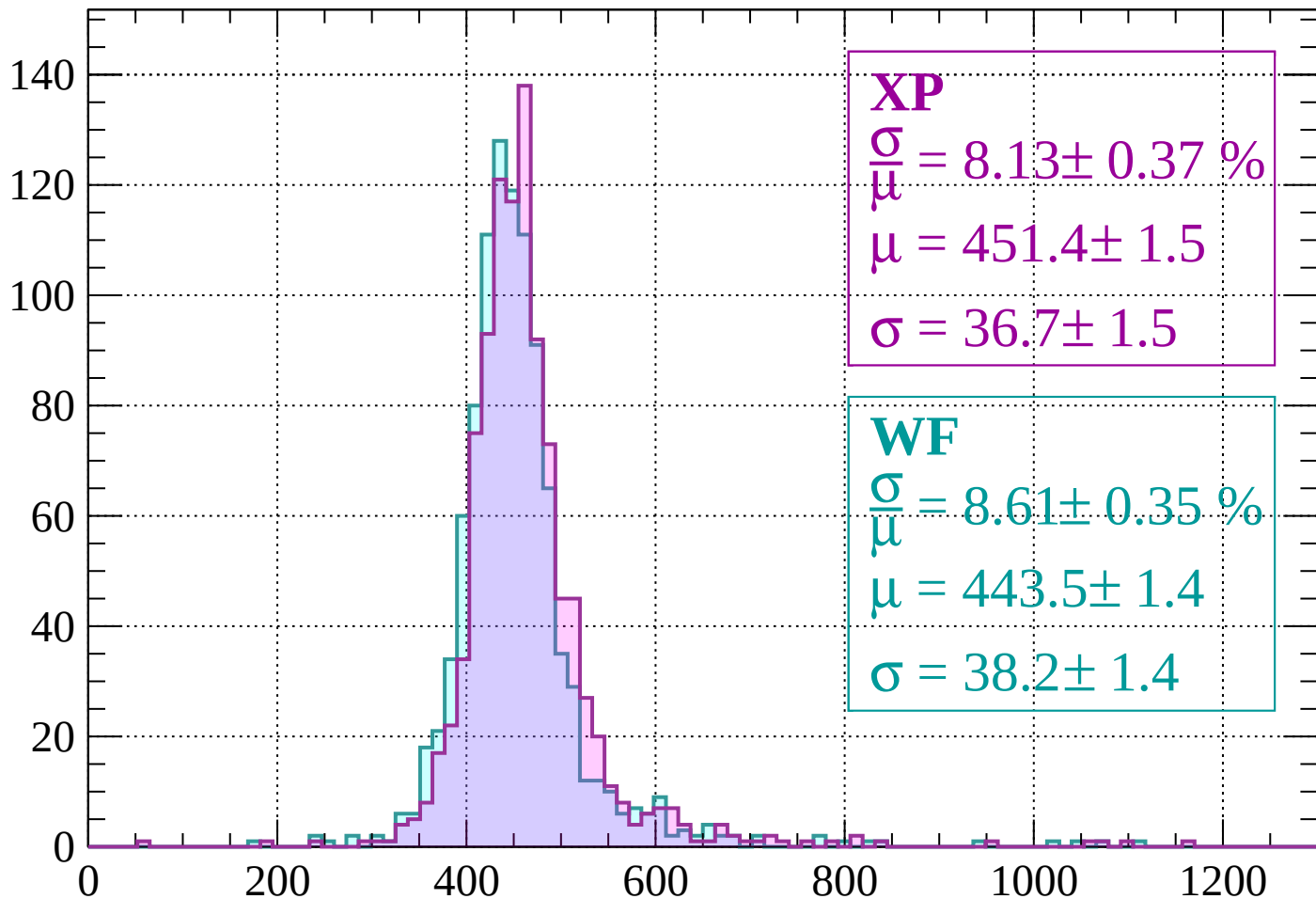
$$\mu = 438.2 \pm 1.3$$

$$\sigma = 34.6 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $1120 < p < 1160$

Count



XP

$$\frac{\sigma}{\mu} = 8.13 \pm 0.37 \%$$

$$\mu = 451.4 \pm 1.5$$

$$\sigma = 36.7 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 8.61 \pm 0.35 \%$$

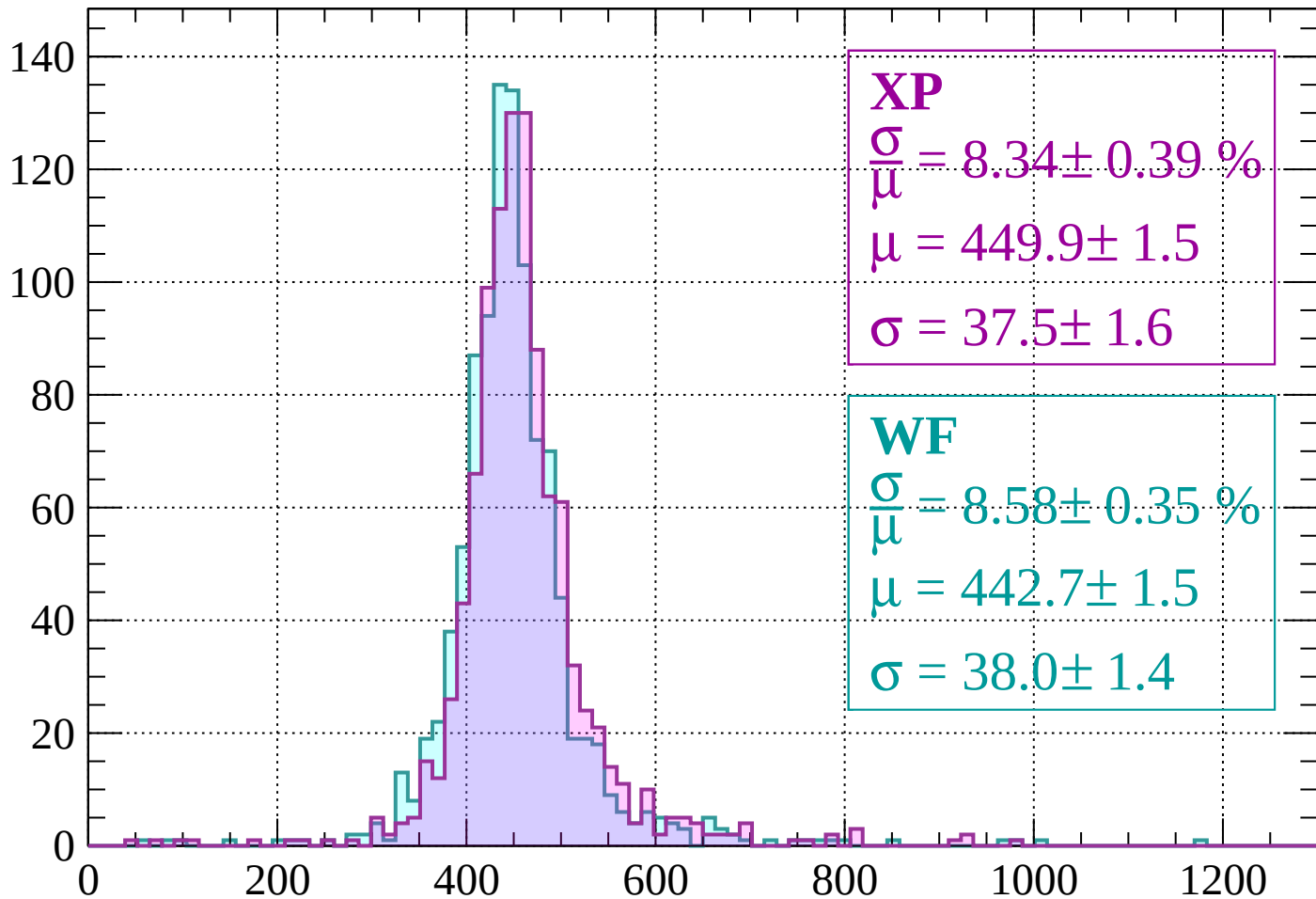
$$\mu = 443.5 \pm 1.4$$

$$\sigma = 38.2 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $1160 < p < 1200$

Count



XP

$$\frac{\sigma}{\mu} = 8.34 \pm 0.39 \%$$

$$\mu = 449.9 \pm 1.5$$

$$\sigma = 37.5 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 8.58 \pm 0.35 \%$$

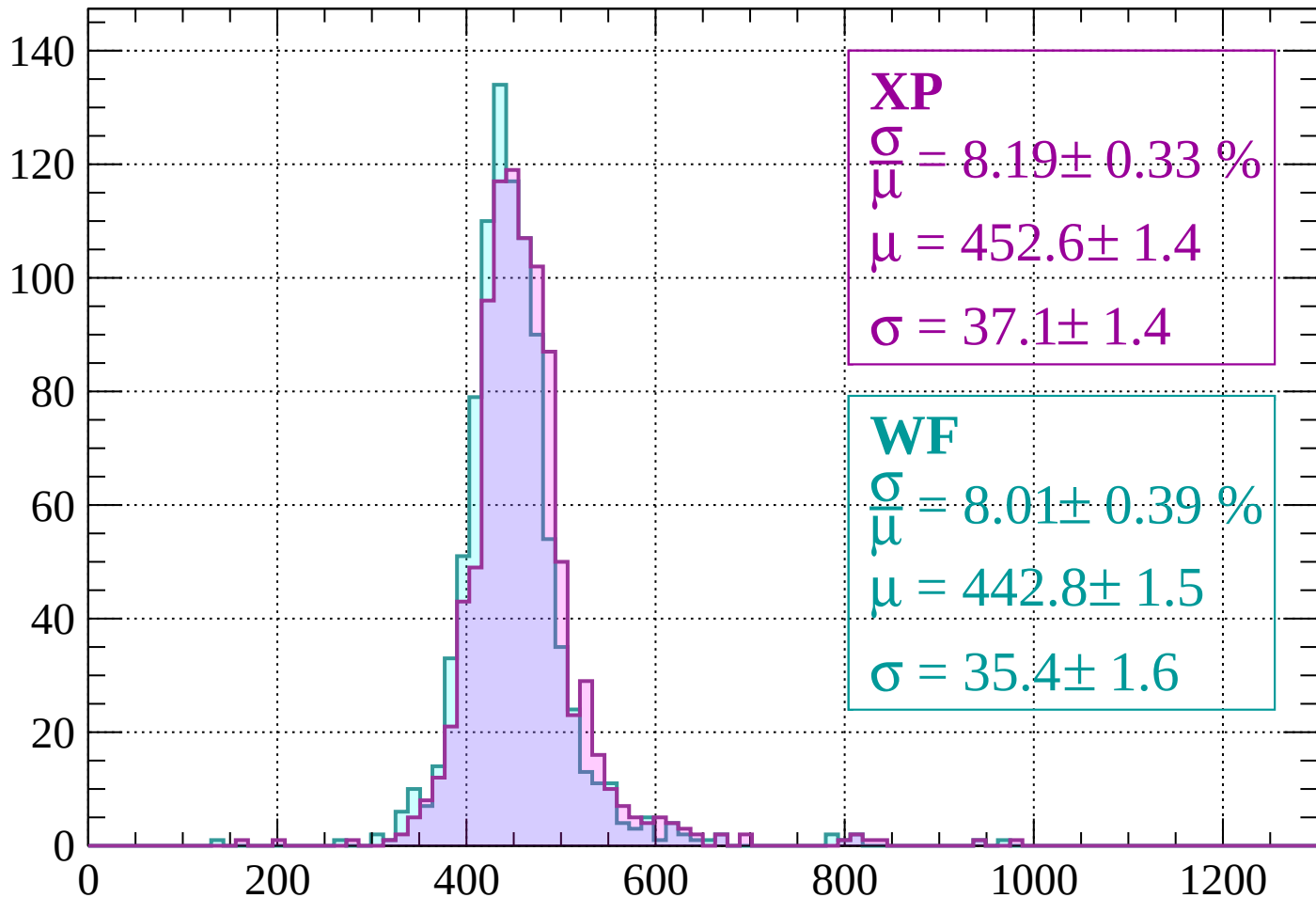
$$\mu = 442.7 \pm 1.5$$

$$\sigma = 38.0 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $1200 < p < 1240$

Count



XP

$$\frac{\sigma}{\mu} = 8.19 \pm 0.33 \%$$

$$\mu = 452.6 \pm 1.4$$

$$\sigma = 37.1 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 8.01 \pm 0.39 \%$$

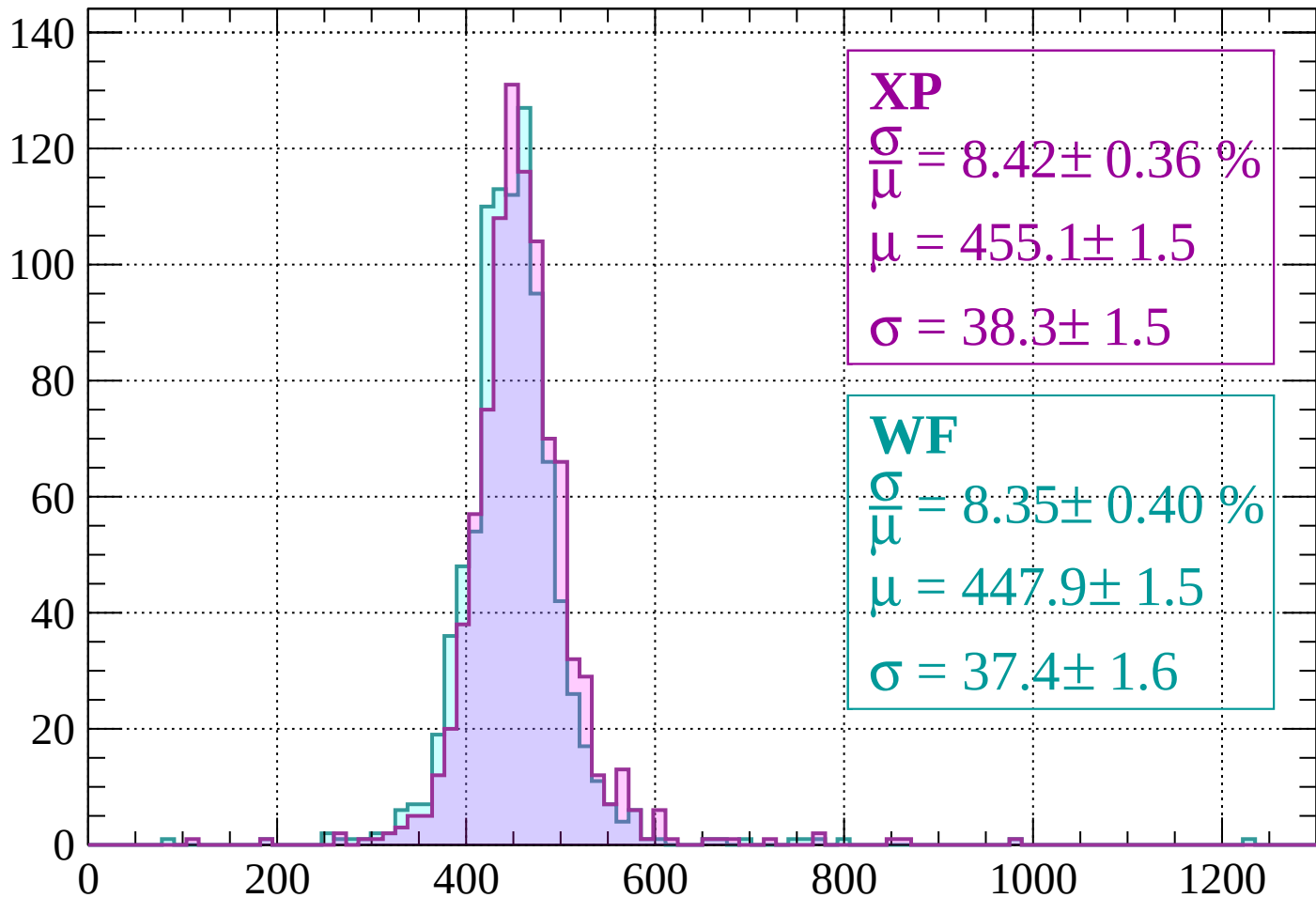
$$\mu = 442.8 \pm 1.5$$

$$\sigma = 35.4 \pm 1.6$$

dE/dx (ADC counts/cm)

Energy loss | $1240 < p < 1280$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 8.42 \pm 0.36 \%$$

$$\mu = 455.1 \pm 1.5$$

$$\sigma = 38.3 \pm 1.5$$

WF

$$\frac{\sigma}{\bar{\mu}} = 8.35 \pm 0.40 \%$$

$$\mu = 447.9 \pm 1.5$$

$$\sigma = 37.4 \pm 1.6$$

dE/dx (ADC counts/cm)

Energy loss | $1280 < p < 1320$

Count

140

120

100

80

60

40

20

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 8.36 \pm 0.41 \%$$

$$\mu = 457.3 \pm 1.6$$

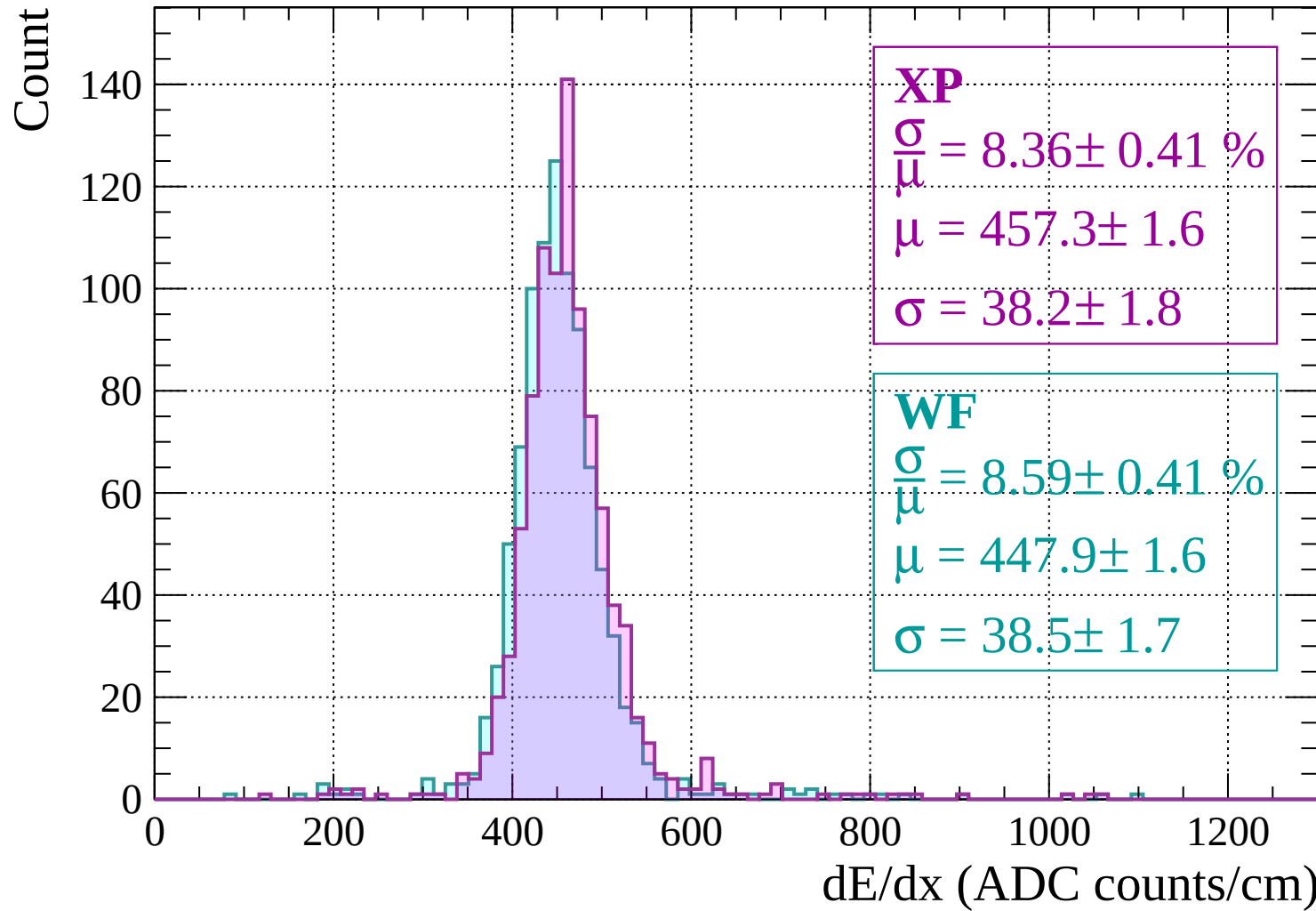
$$\sigma = 38.2 \pm 1.8$$

WF

$$\frac{\sigma}{\mu} = 8.59 \pm 0.41 \%$$

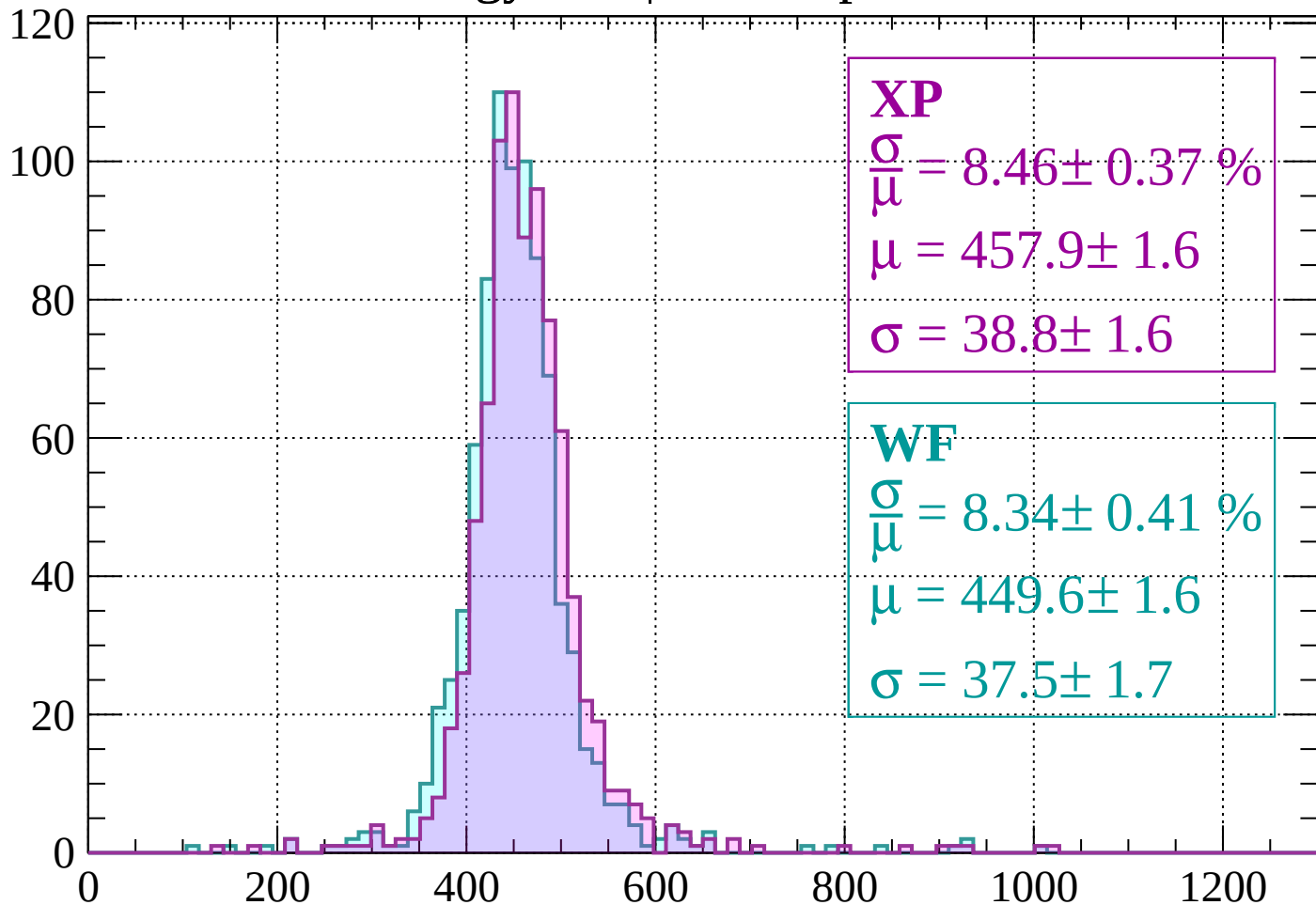
$$\mu = 447.9 \pm 1.6$$

$$\sigma = 38.5 \pm 1.7$$



Energy loss | $1320 < p < 1360$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 8.46 \pm 0.37 \%$$

$$\mu = 457.9 \pm 1.6$$

$$\sigma = 38.8 \pm 1.6$$

WF

$$\frac{\sigma}{\bar{\mu}} = 8.34 \pm 0.41 \%$$

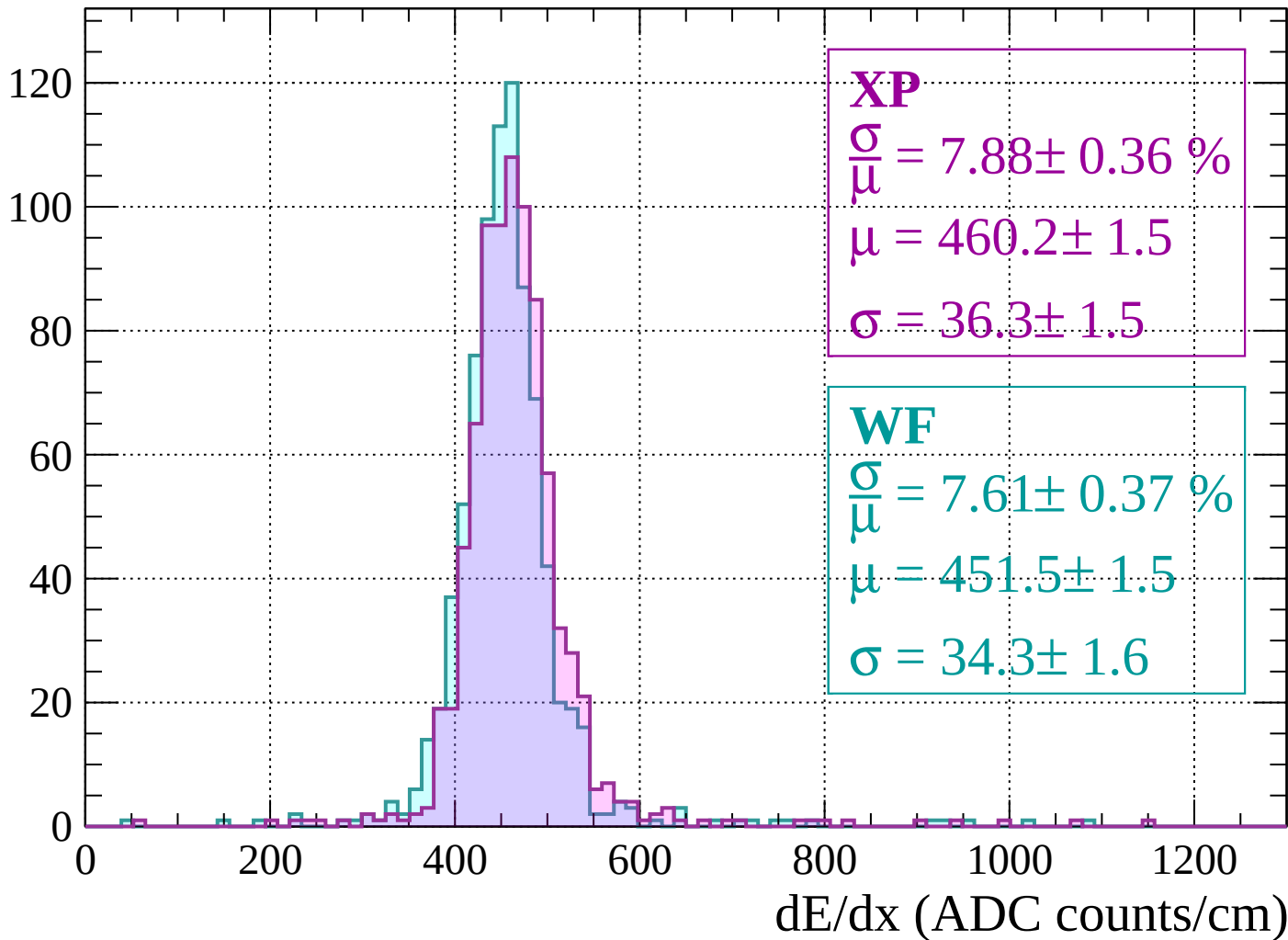
$$\mu = 449.6 \pm 1.6$$

$$\sigma = 37.5 \pm 1.7$$

dE/dx (ADC counts/cm)

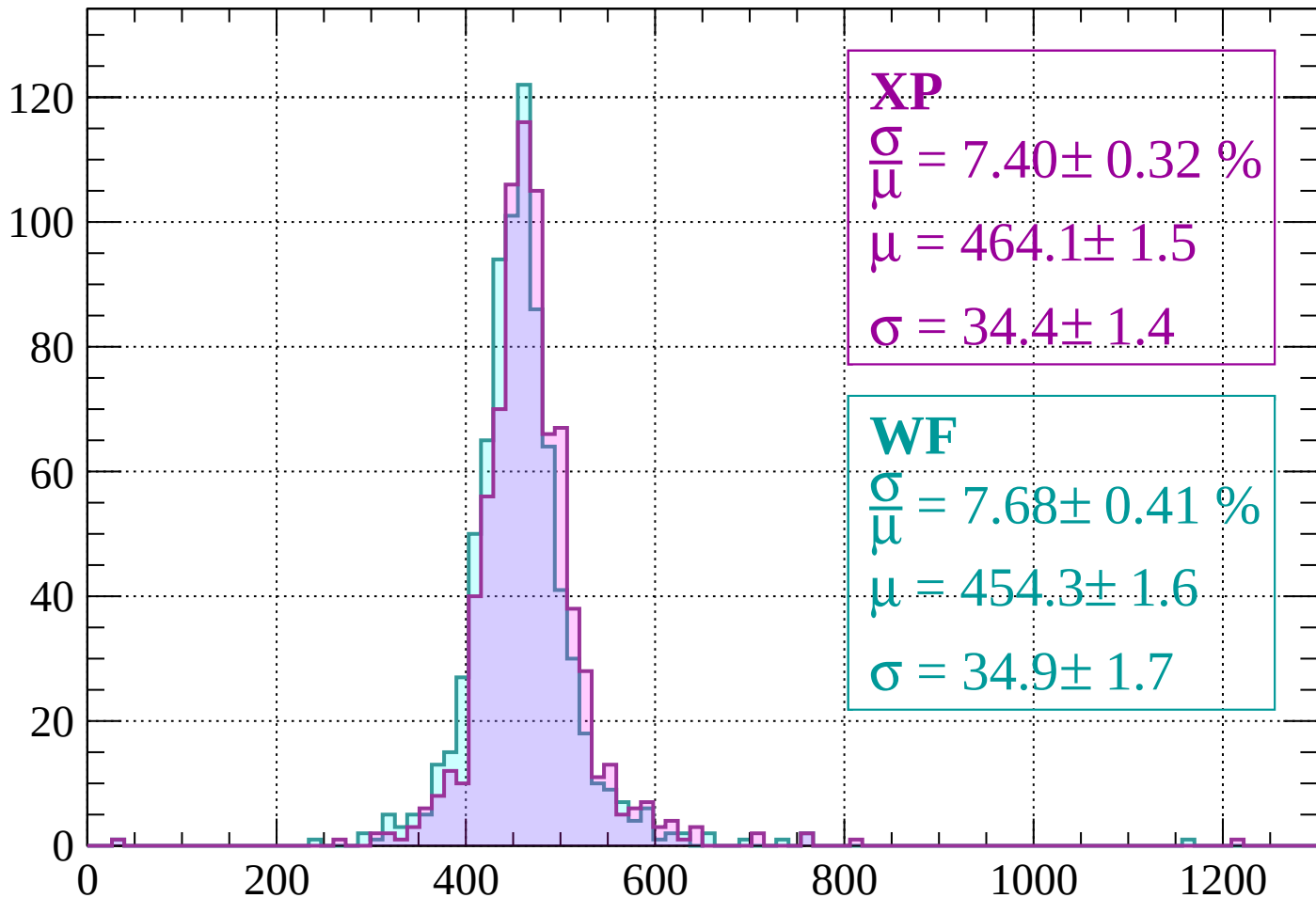
Energy loss | $1360 < p < 1400$

Count



Energy loss | $1400 < p < 1440$

Count



XP

$$\frac{\sigma}{\mu} = 7.40 \pm 0.32 \%$$

$$\mu = 464.1 \pm 1.5$$

$$\sigma = 34.4 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 7.68 \pm 0.41 \%$$

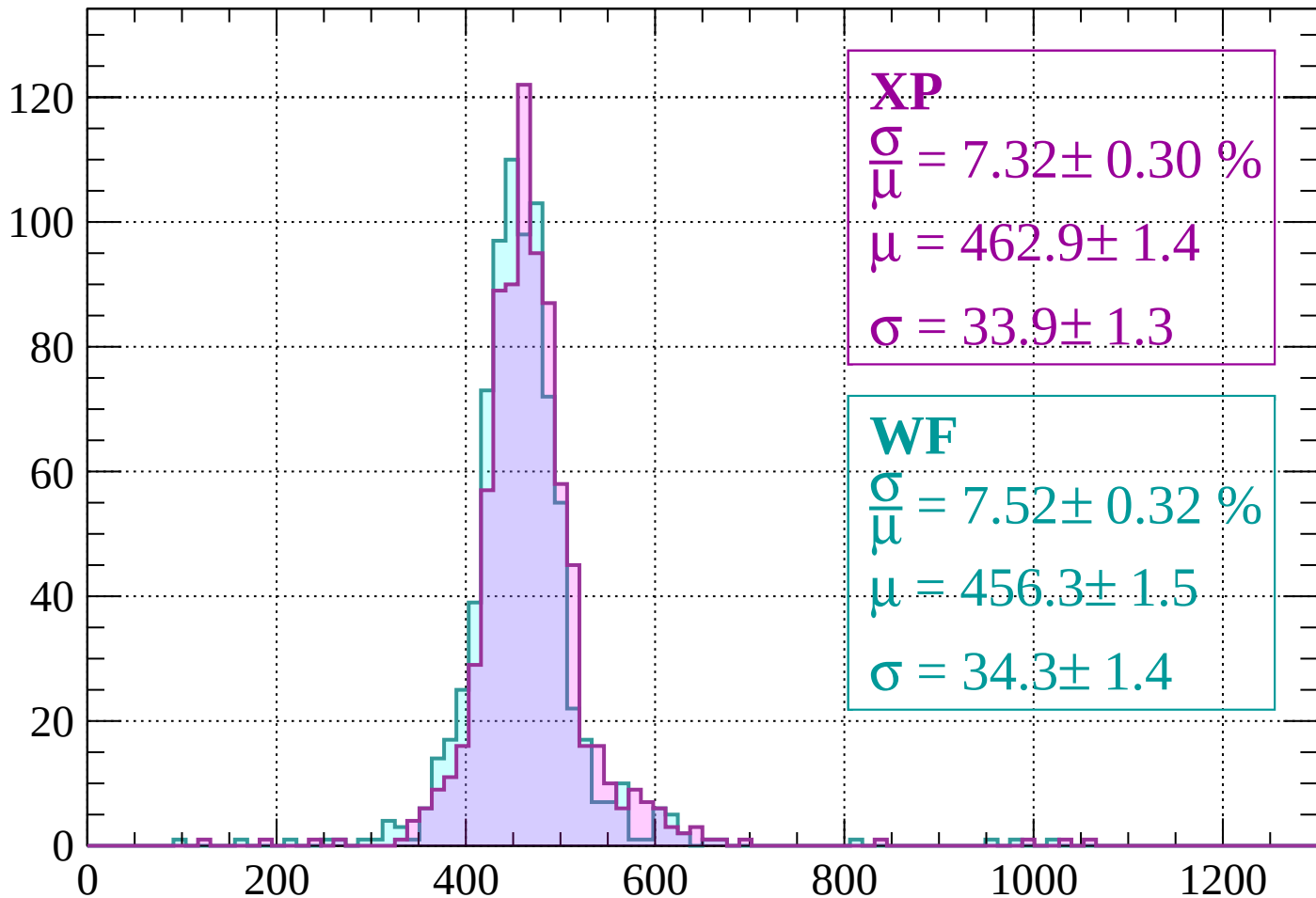
$$\mu = 454.3 \pm 1.6$$

$$\sigma = 34.9 \pm 1.7$$

dE/dx (ADC counts/cm)

Energy loss | $1440 < p < 1480$

Count



XP

$$\frac{\sigma}{\mu} = 7.32 \pm 0.30 \%$$

$$\mu = 462.9 \pm 1.4$$

$$\sigma = 33.9 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 7.52 \pm 0.32 \%$$

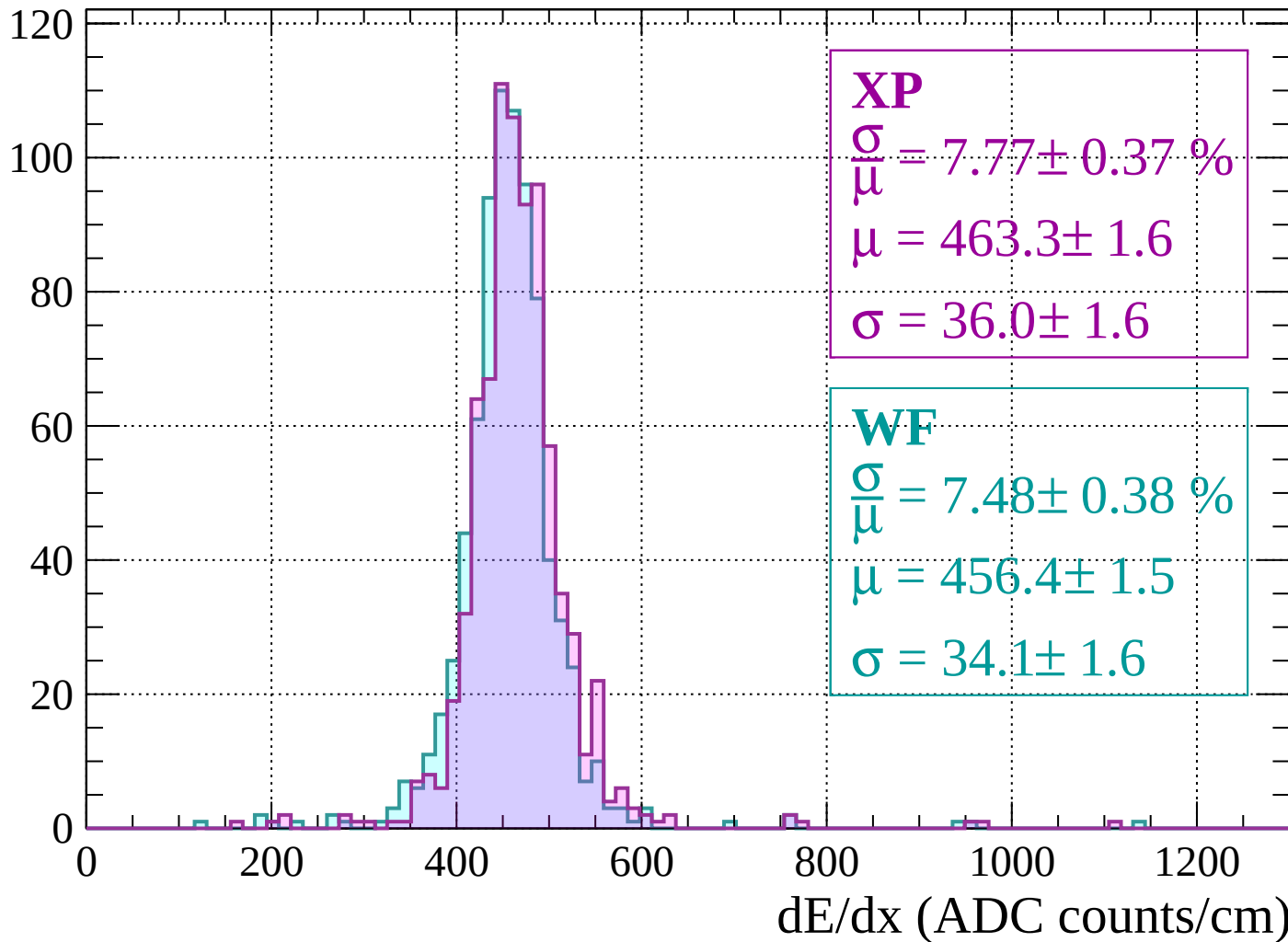
$$\mu = 456.3 \pm 1.5$$

$$\sigma = 34.3 \pm 1.4$$

dE/dx (ADC counts/cm)

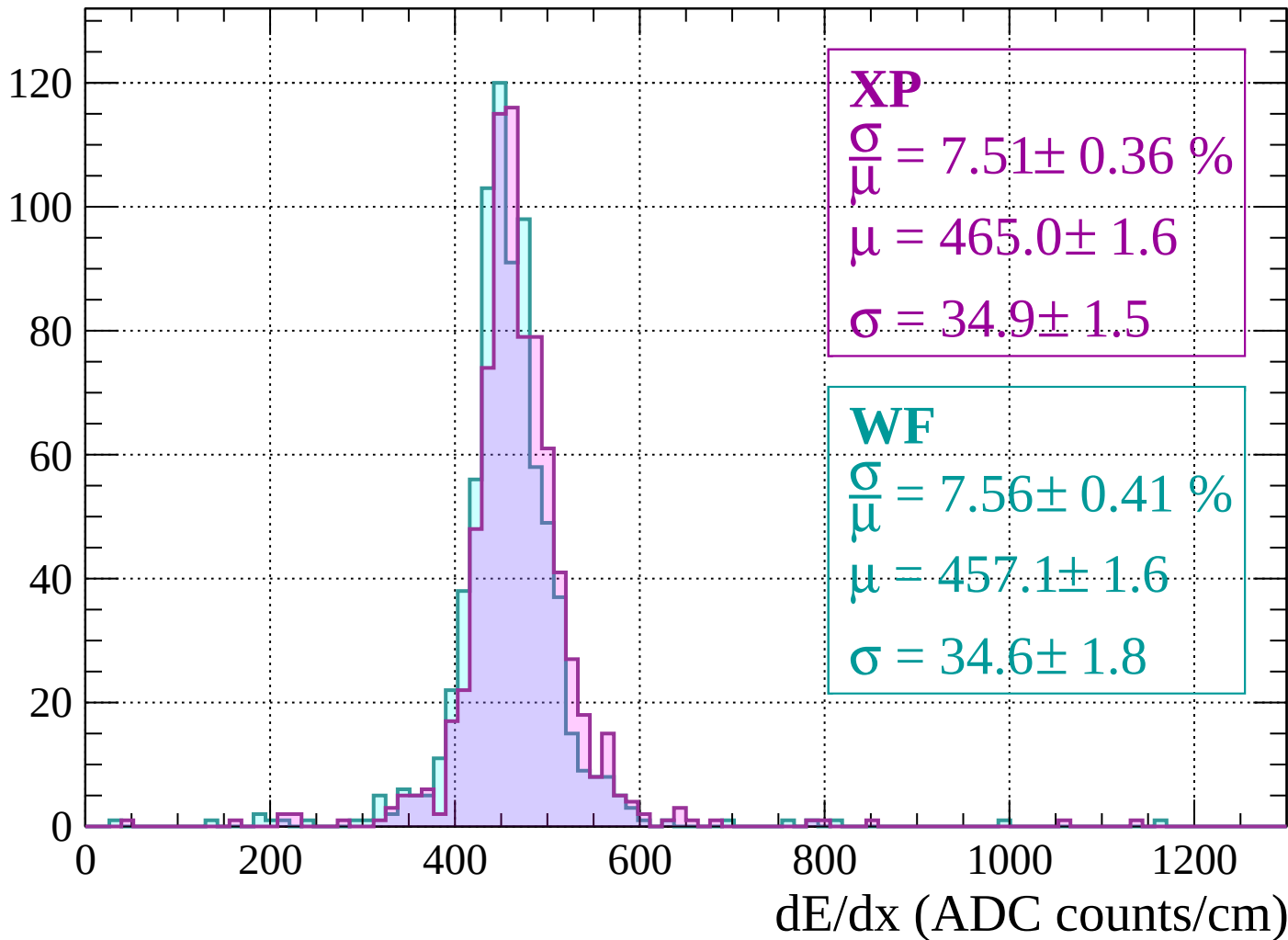
Energy loss | $1480 < p < 1520$

Count



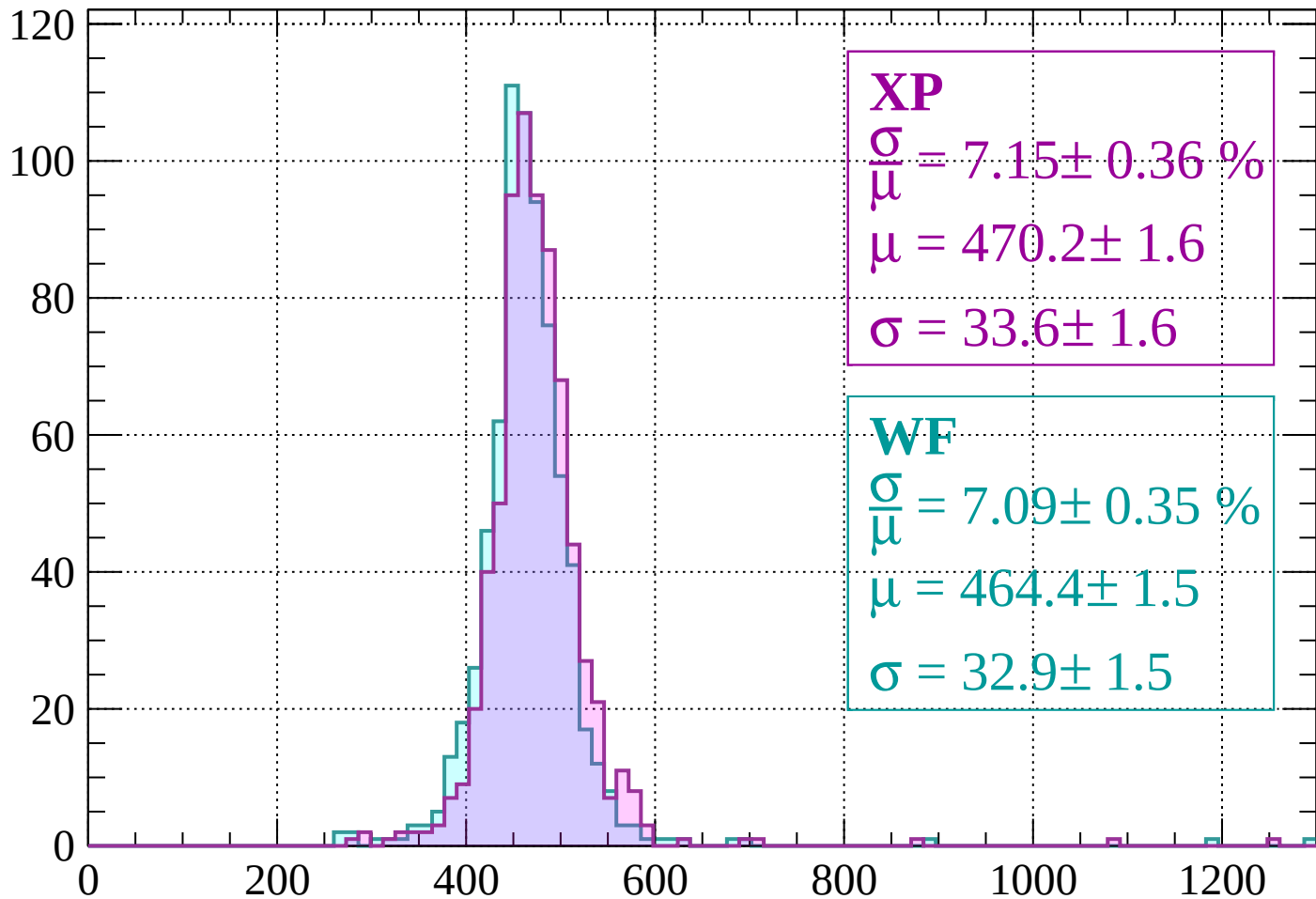
Energy loss | $1520 < p < 1560$

Count



Energy loss | $1560 < p < 1600$

Count



XP

$$\frac{\sigma}{\mu} = 7.15 \pm 0.36 \%$$

$$\mu = 470.2 \pm 1.6$$

$$\sigma = 33.6 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 7.09 \pm 0.35 \%$$

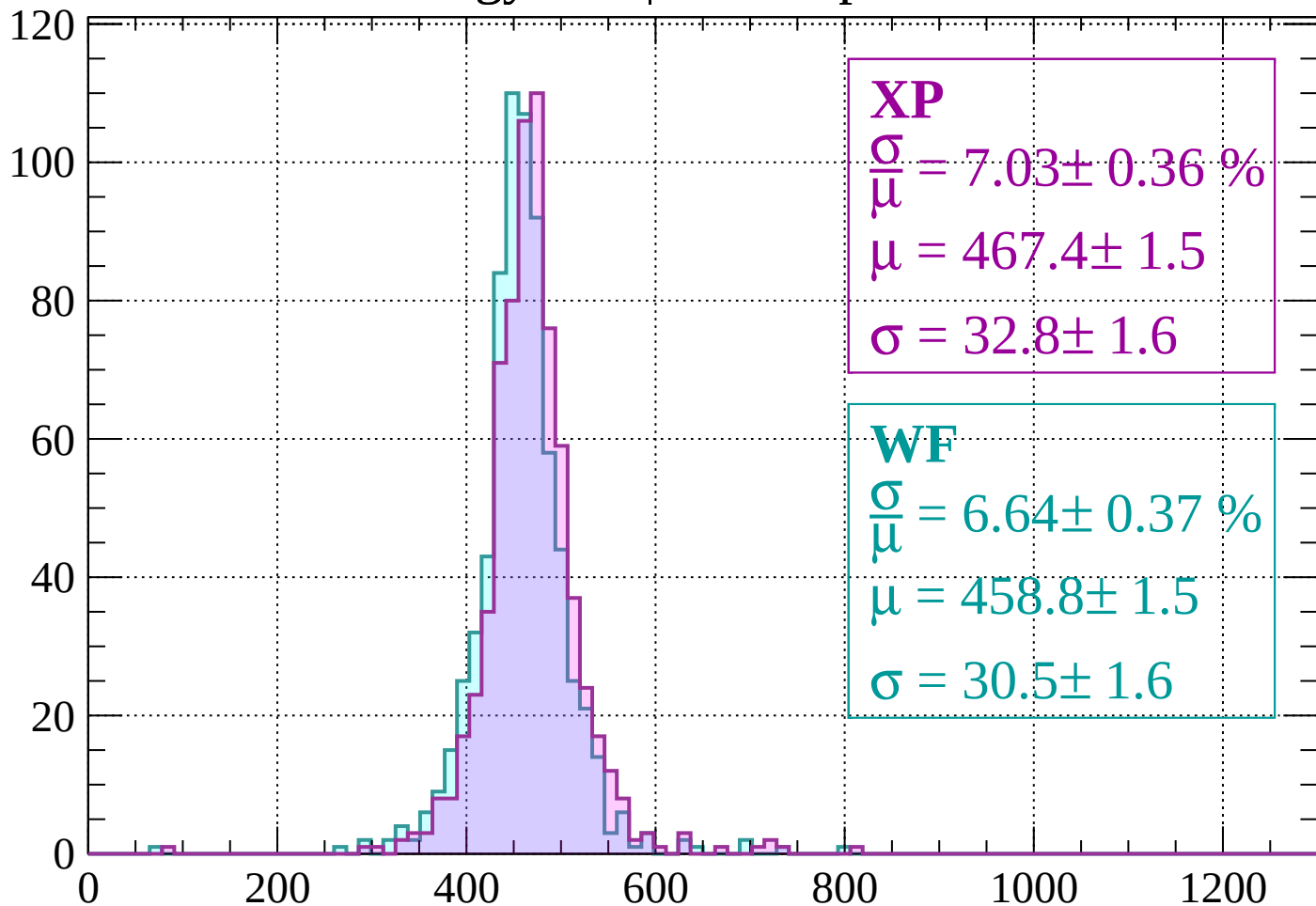
$$\mu = 464.4 \pm 1.5$$

$$\sigma = 32.9 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $1600 < p < 1640$

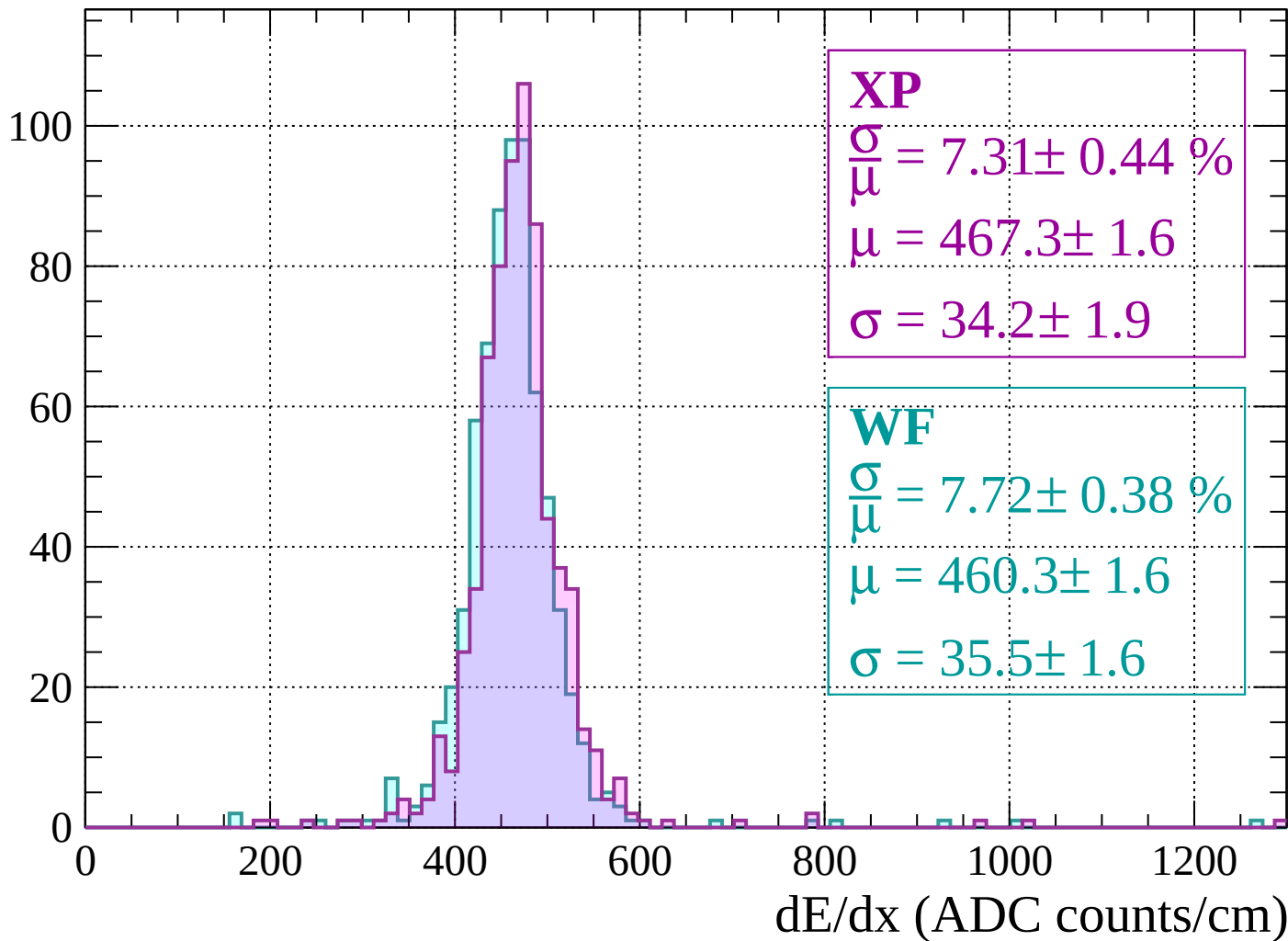
Count



dE/dx (ADC counts/cm)

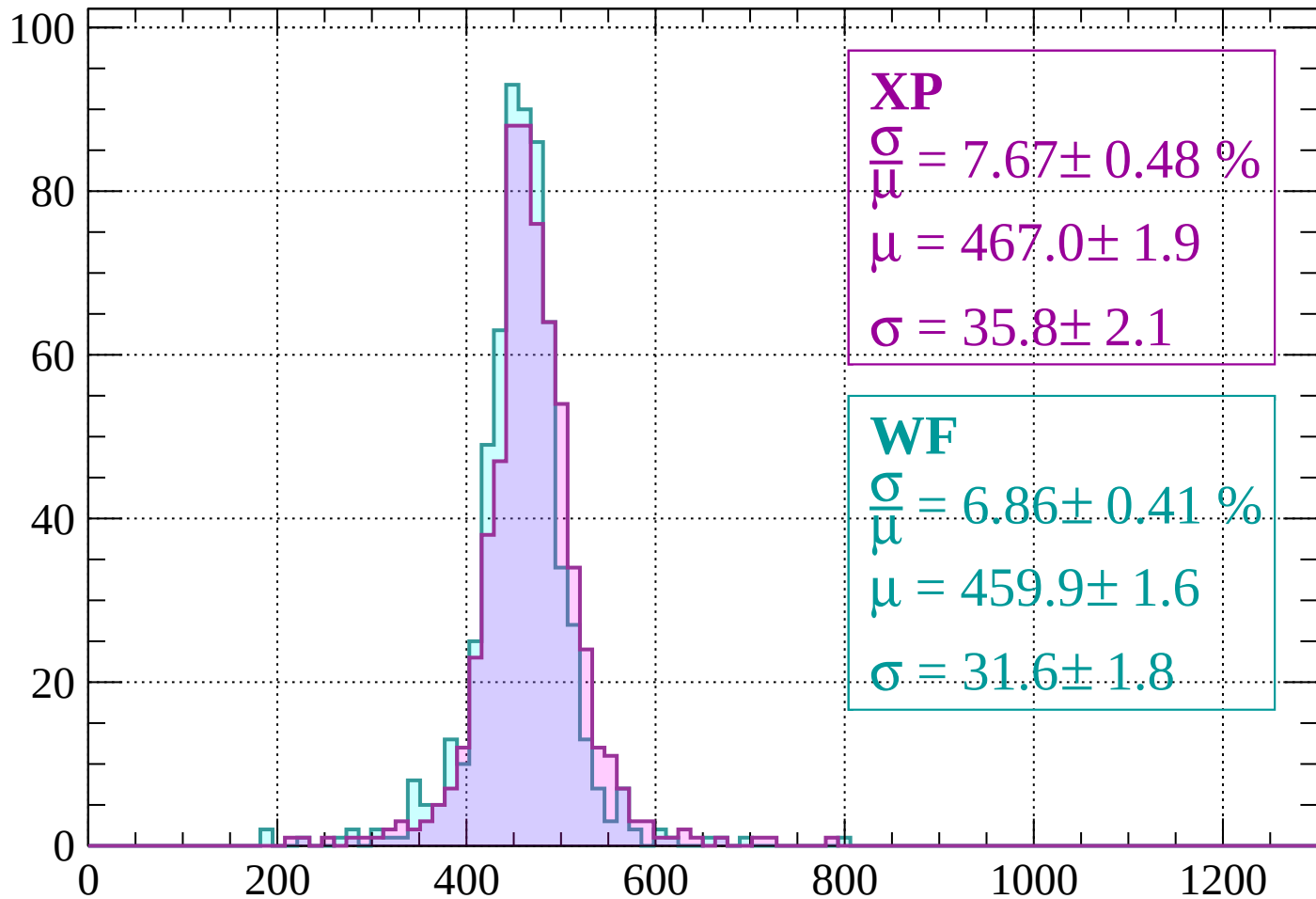
Energy loss | $1640 < p < 1680$

Count



Energy loss | $1680 < p < 1720$

Count



XP

$$\frac{\sigma}{\mu} = 7.67 \pm 0.48 \%$$

$$\mu = 467.0 \pm 1.9$$

$$\sigma = 35.8 \pm 2.1$$

WF

$$\frac{\sigma}{\mu} = 6.86 \pm 0.41 \%$$

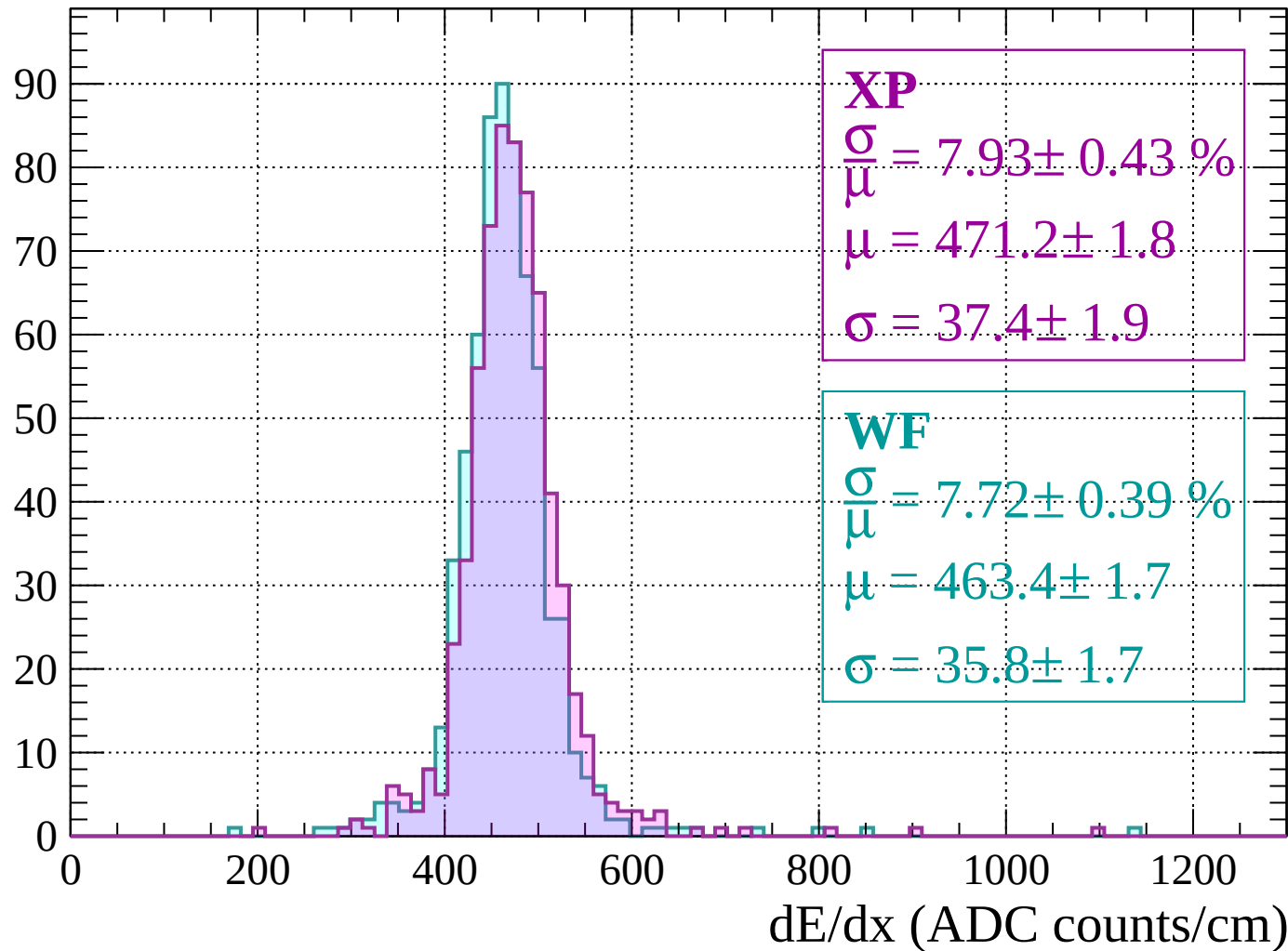
$$\mu = 459.9 \pm 1.6$$

$$\sigma = 31.6 \pm 1.8$$

dE/dx (ADC counts/cm)

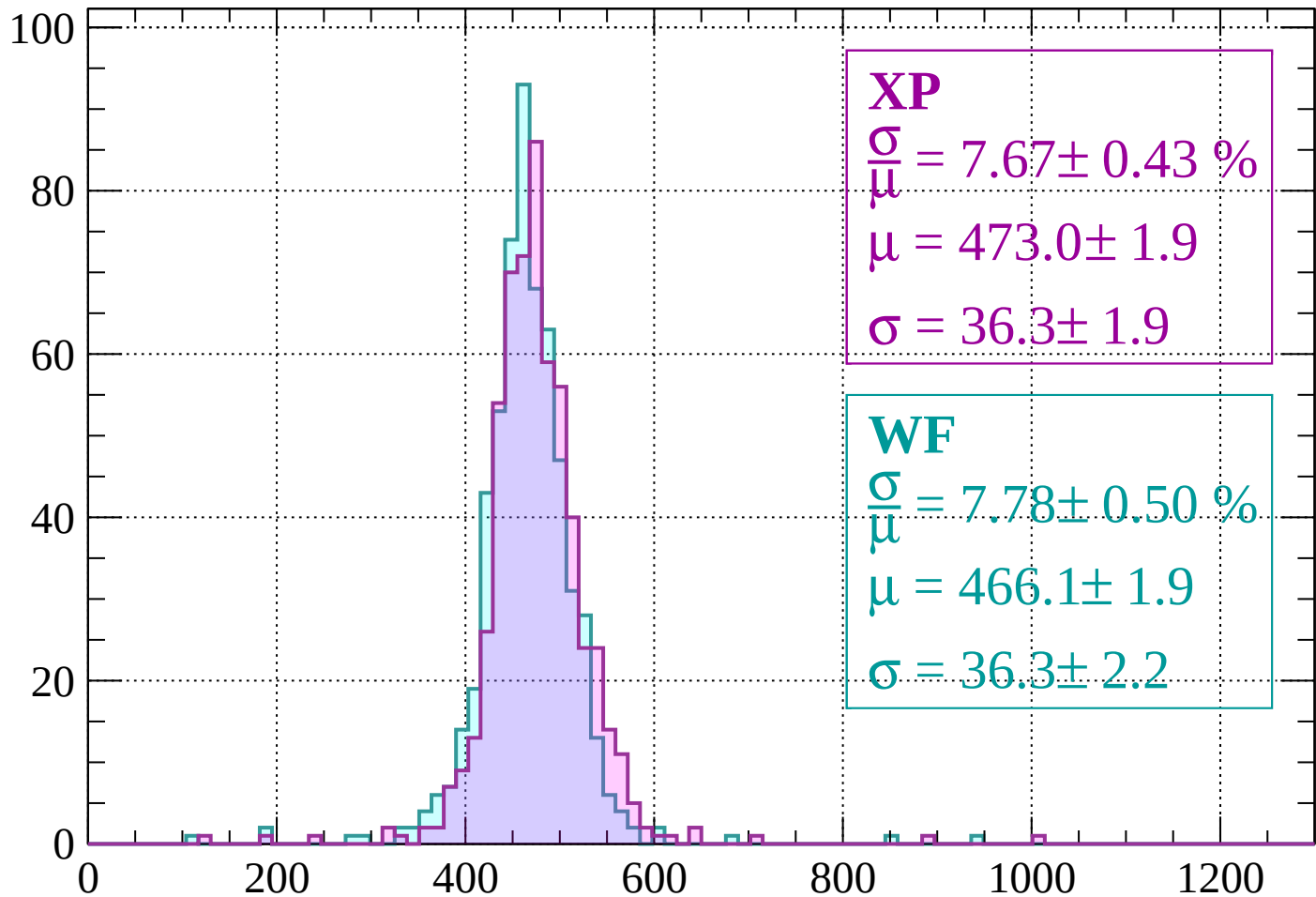
Energy loss | $1720 < p < 1760$

Count



Energy loss | $1760 < p < 1800$

Count



XP

$$\frac{\sigma}{\mu} = 7.67 \pm 0.43 \%$$

$$\mu = 473.0 \pm 1.9$$

$$\sigma = 36.3 \pm 1.9$$

WF

$$\frac{\sigma}{\mu} = 7.78 \pm 0.50 \%$$

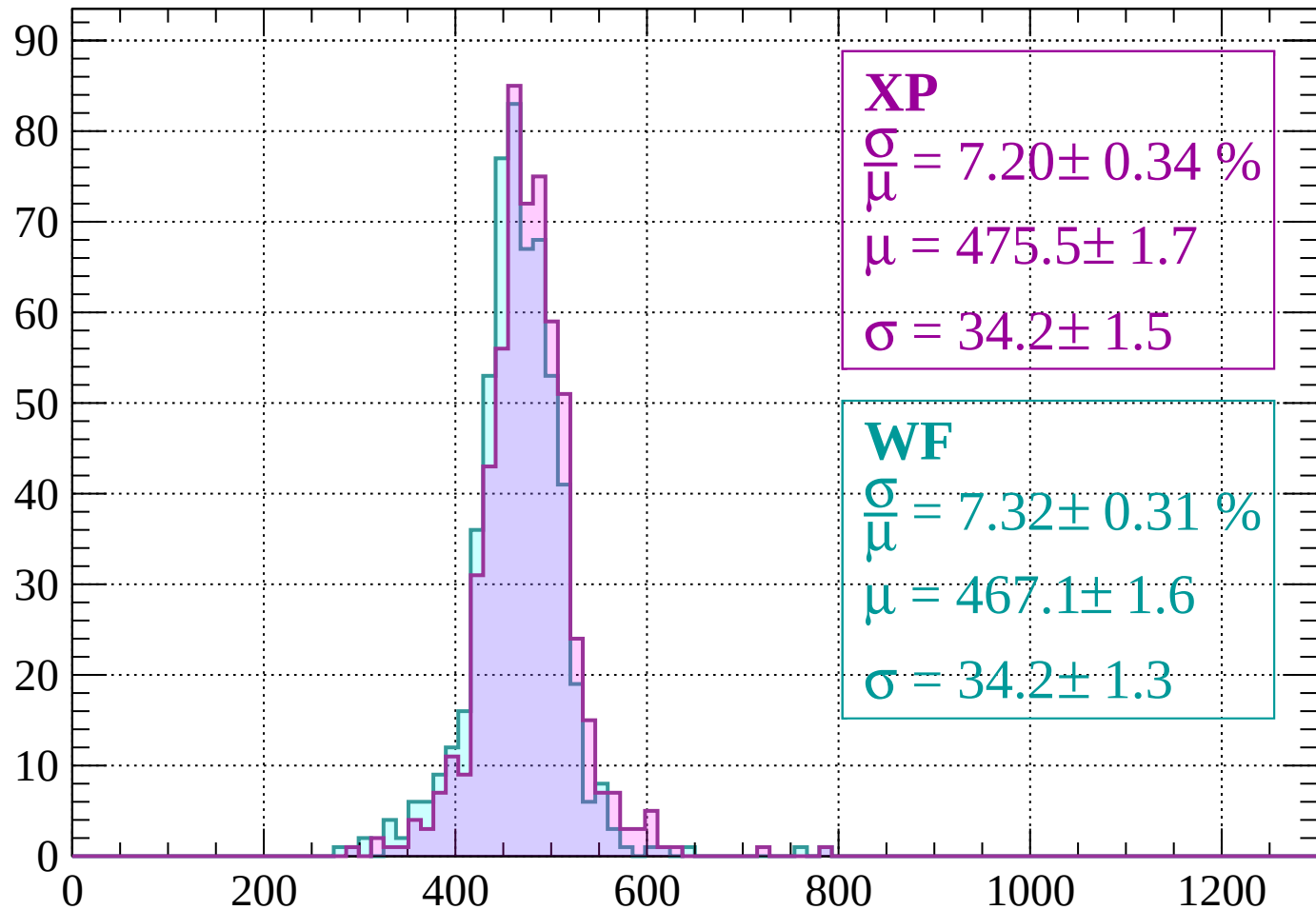
$$\mu = 466.1 \pm 1.9$$

$$\sigma = 36.3 \pm 2.2$$

dE/dx (ADC counts/cm)

Energy loss | $1800 < p < 1840$

Count



XP

$$\frac{\sigma}{\mu} = 7.20 \pm 0.34 \%$$

$$\mu = 475.5 \pm 1.7$$

$$\sigma = 34.2 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 7.32 \pm 0.31 \%$$

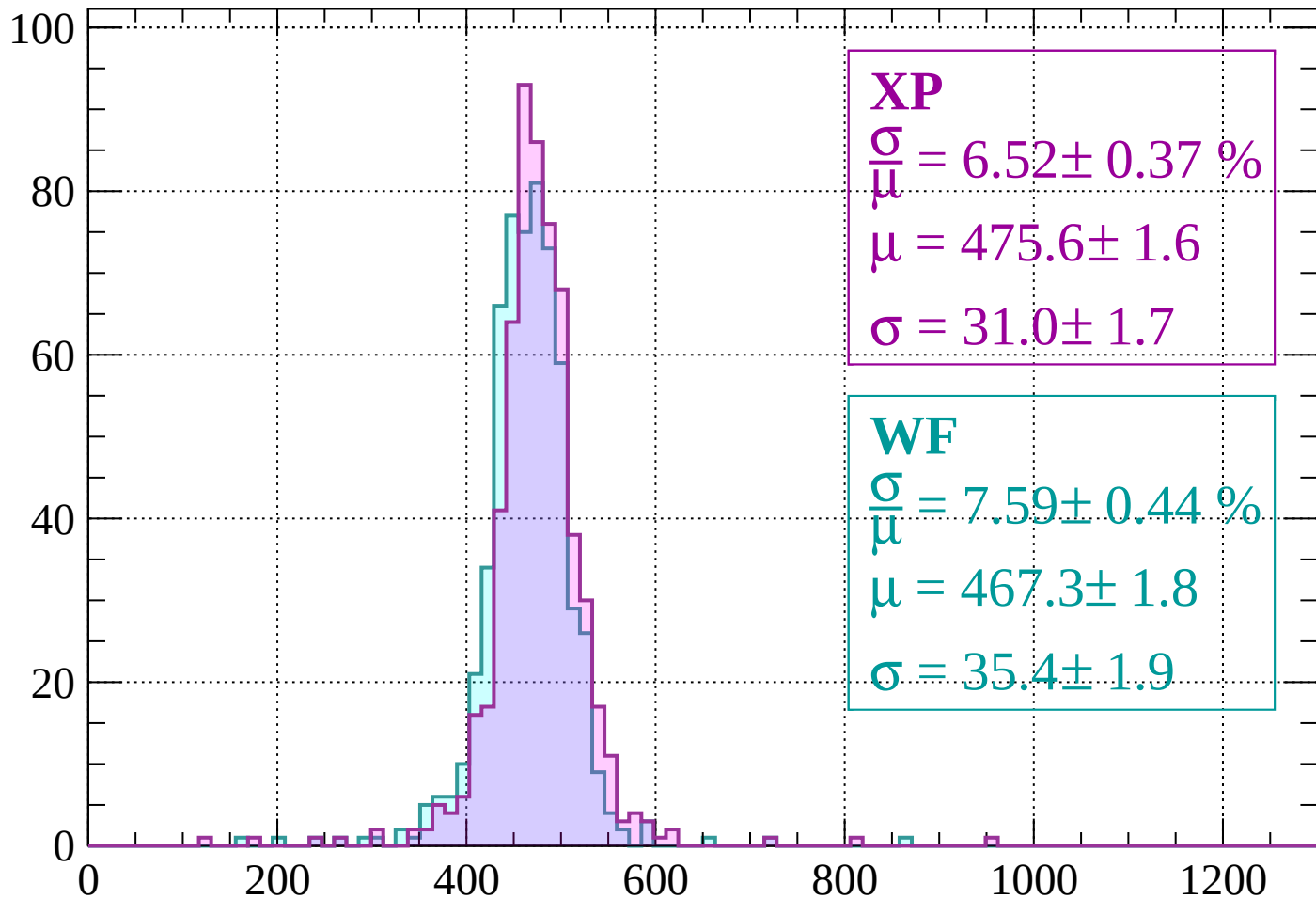
$$\mu = 467.1 \pm 1.6$$

$$\sigma = 34.2 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $1840 < p < 1880$

Count



XP

$$\frac{\sigma}{\mu} = 6.52 \pm 0.37 \%$$

$$\mu = 475.6 \pm 1.6$$

$$\sigma = 31.0 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 7.59 \pm 0.44 \%$$

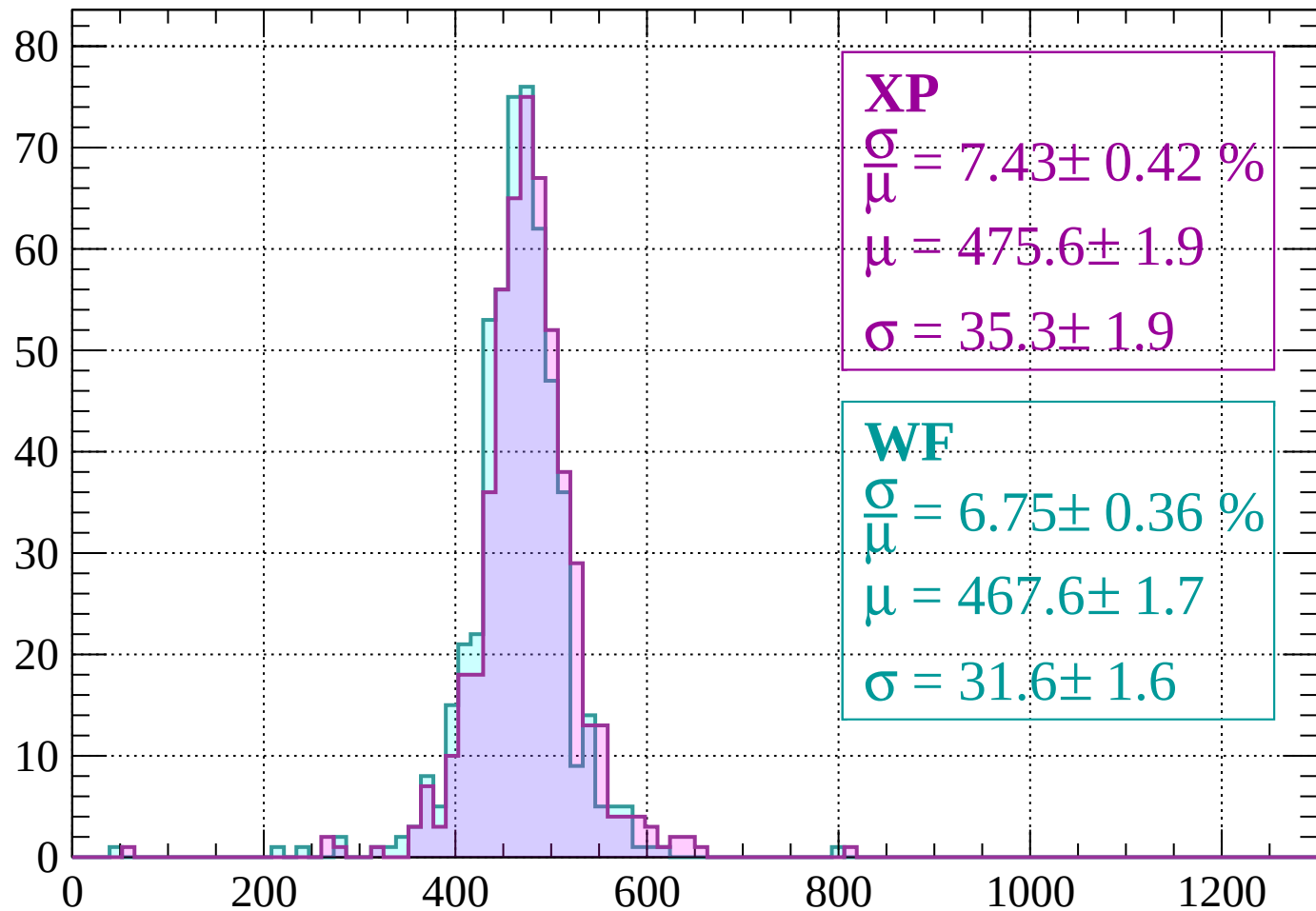
$$\mu = 467.3 \pm 1.8$$

$$\sigma = 35.4 \pm 1.9$$

dE/dx (ADC counts/cm)

Energy loss | $1880 < p < 1920$

Count



XP

$$\frac{\sigma}{\mu} = 7.43 \pm 0.42 \%$$

$$\mu = 475.6 \pm 1.9$$

$$\sigma = 35.3 \pm 1.9$$

WF

$$\frac{\sigma}{\mu} = 6.75 \pm 0.36 \%$$

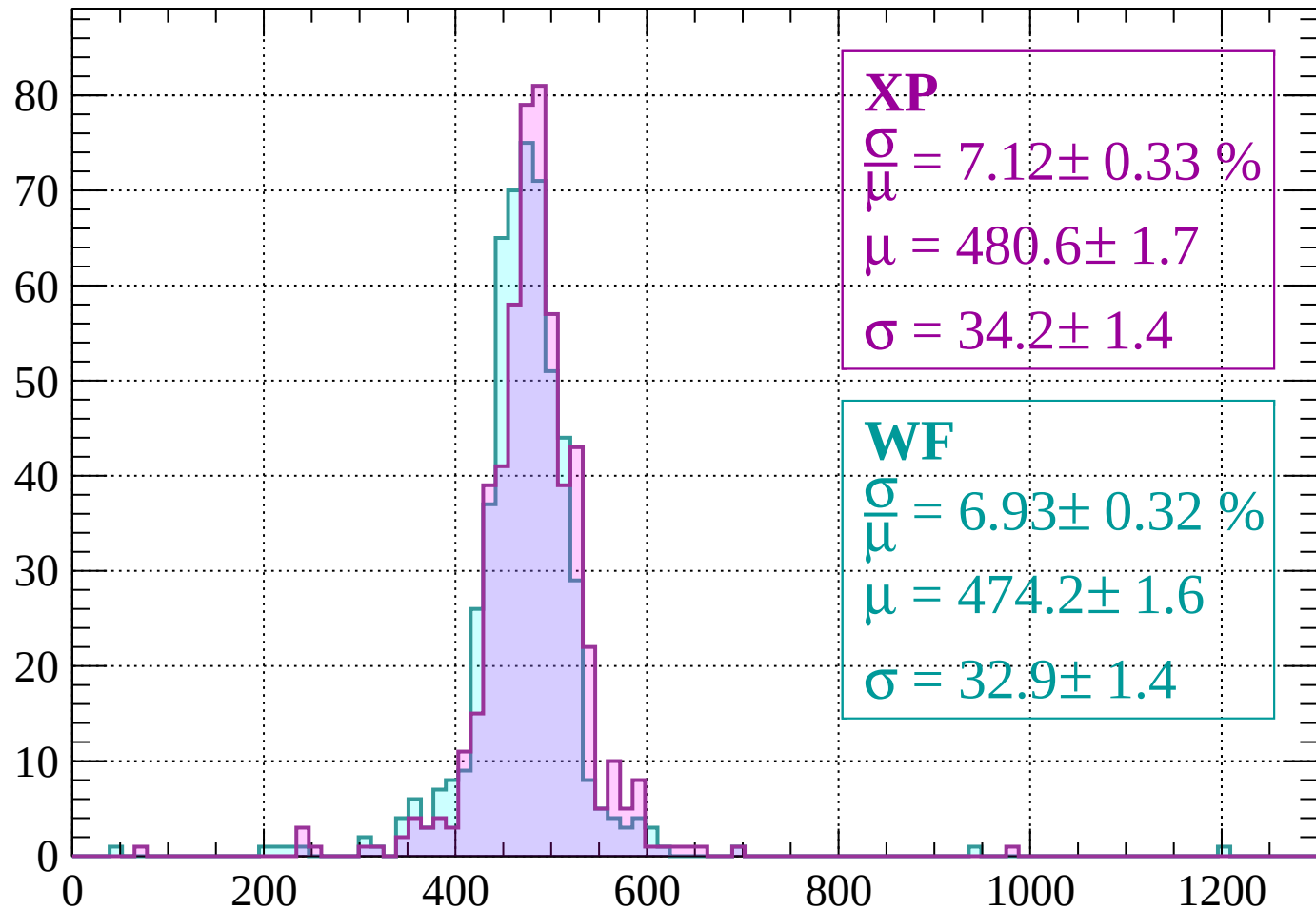
$$\mu = 467.6 \pm 1.7$$

$$\sigma = 31.6 \pm 1.6$$

dE/dx (ADC counts/cm)

Energy loss | $1920 < p < 1960$

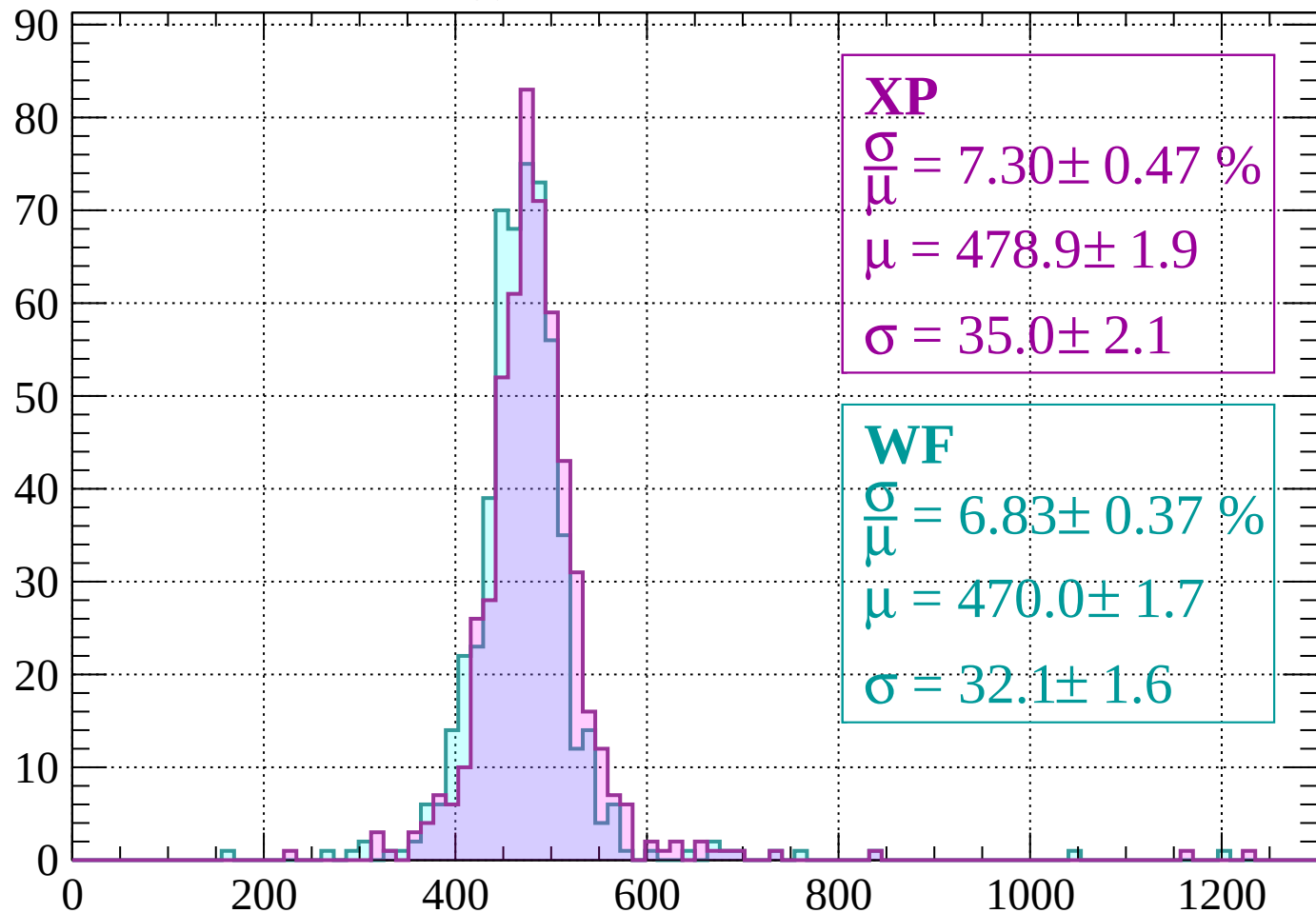
Count



dE/dx (ADC counts/cm)

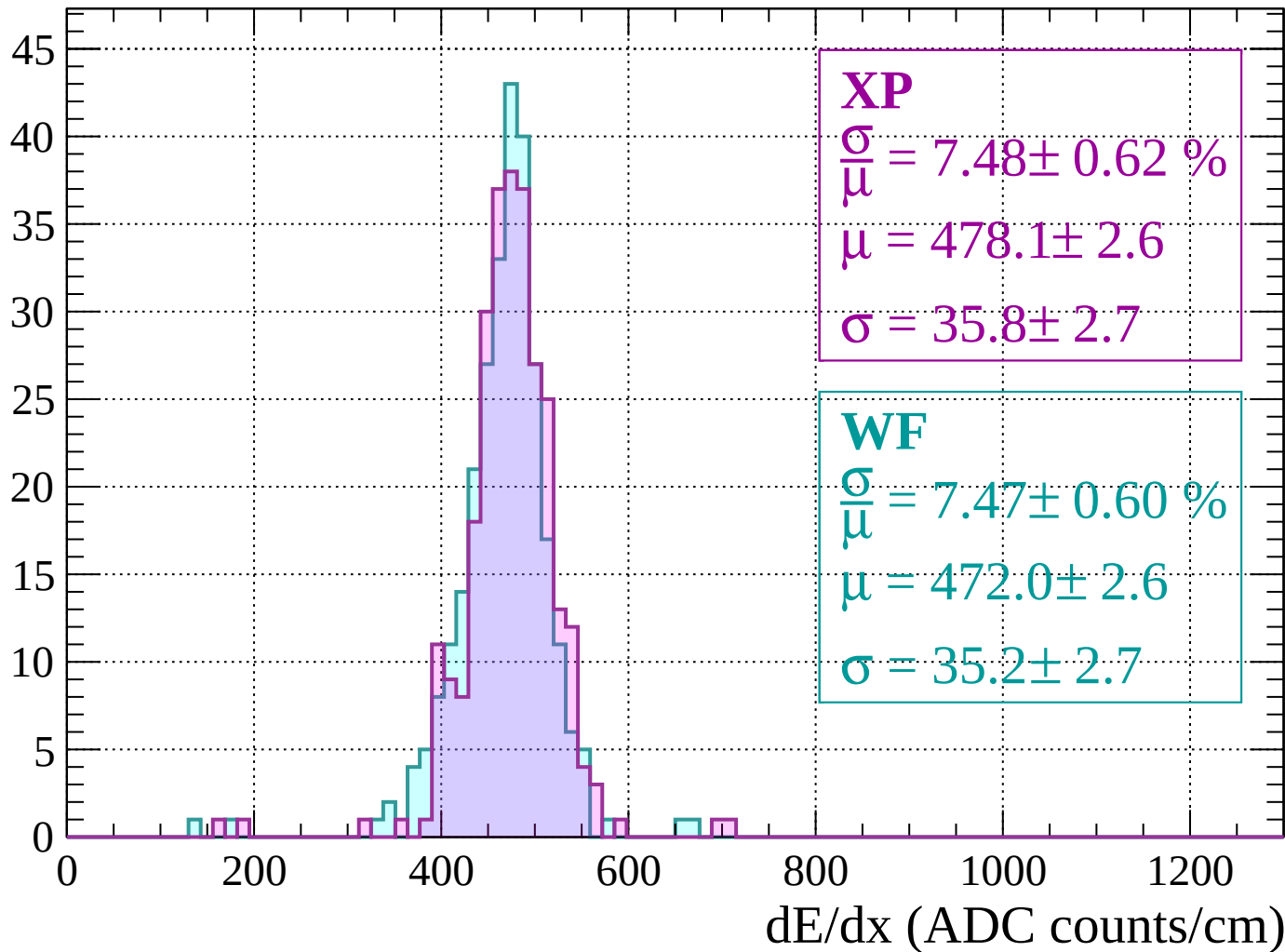
Energy loss | $1960 < p < 2000$

Count



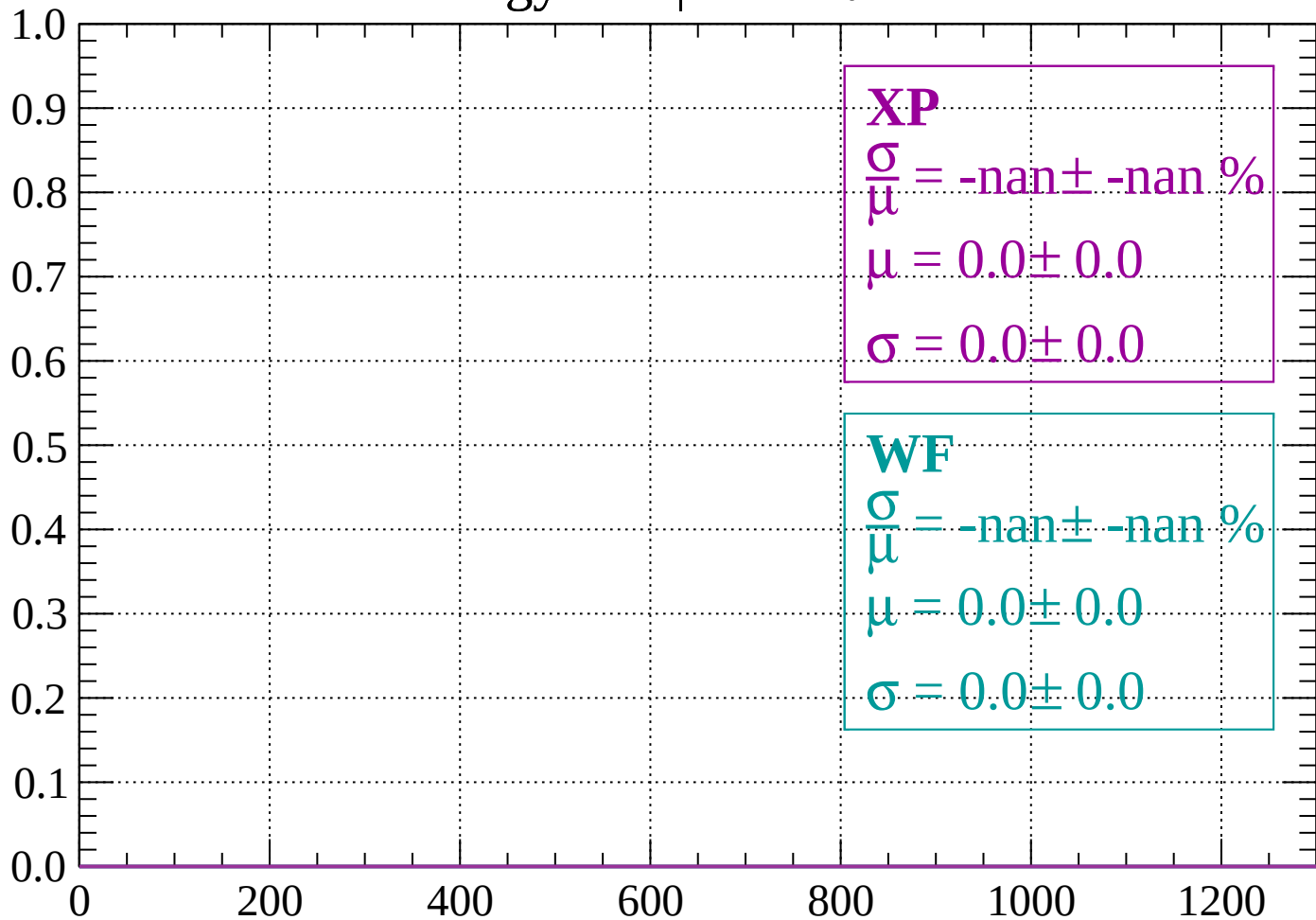
Energy loss | $2000 < p < 2040$

Count



Energy loss | $-90 < \theta < -88$

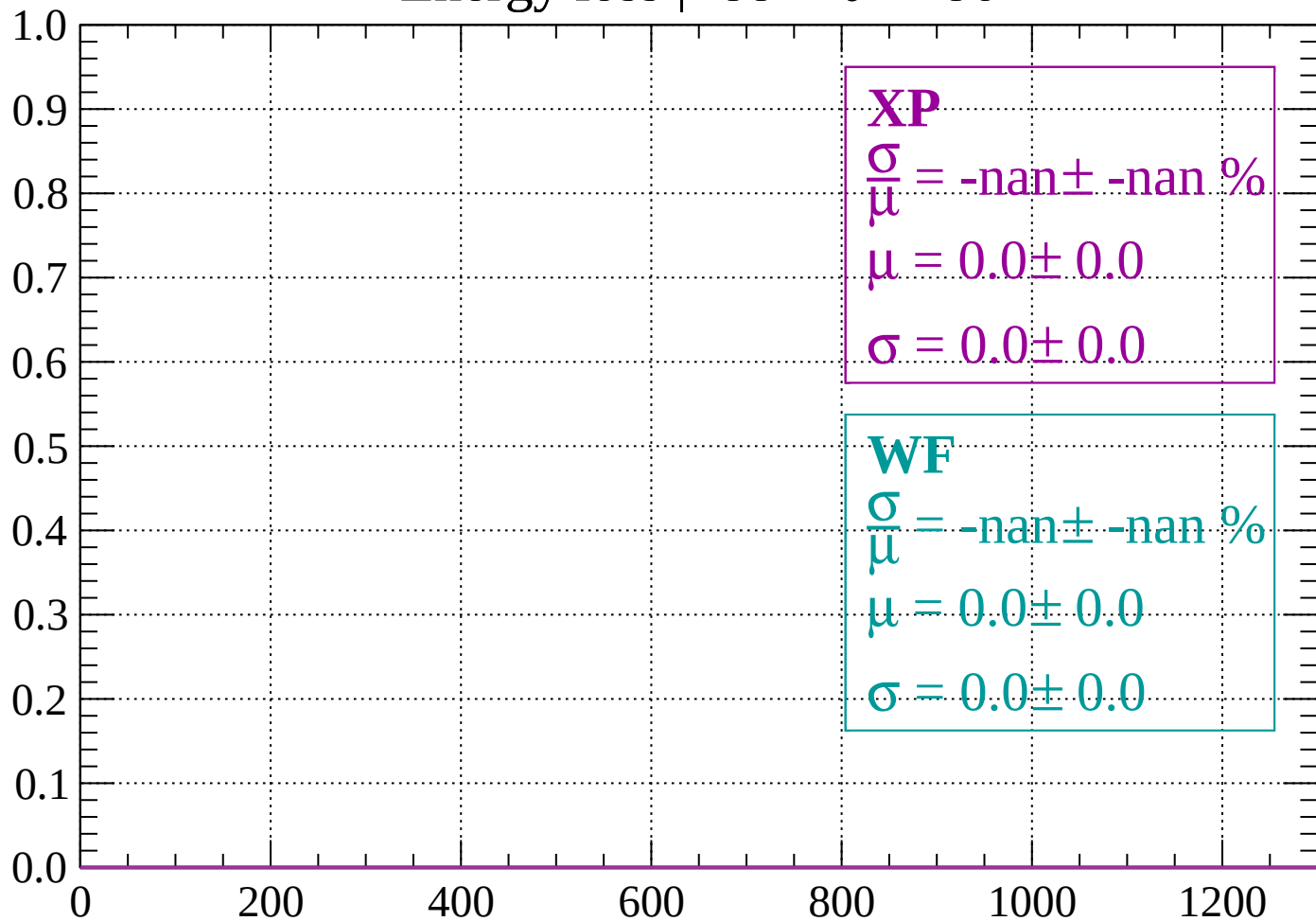
Count



dE/dx (ADC counts/cm)

Energy loss | $-88 < \theta < -86$

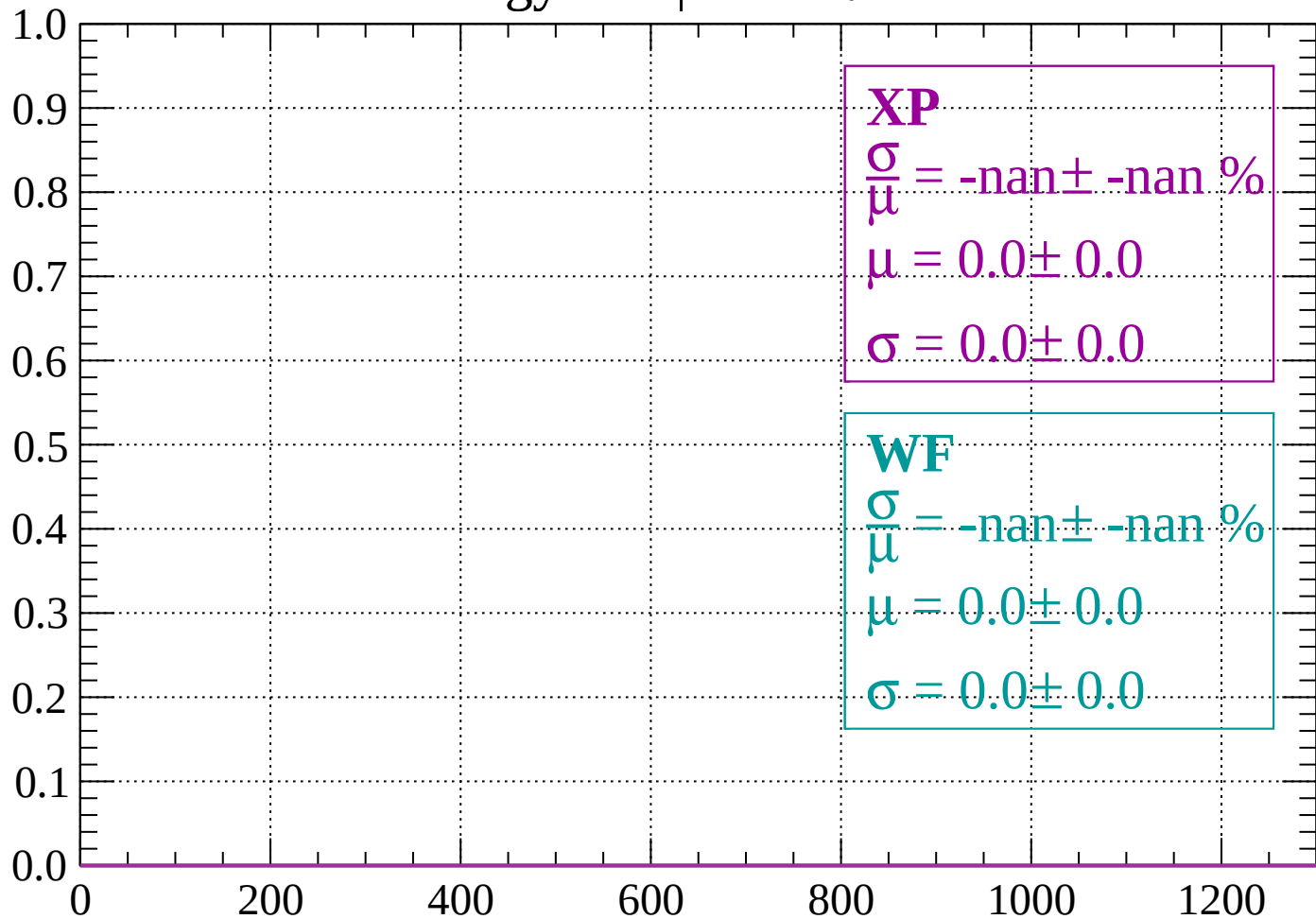
Count



dE/dx (ADC counts/cm)

Energy loss | $-86 < \theta < -84$

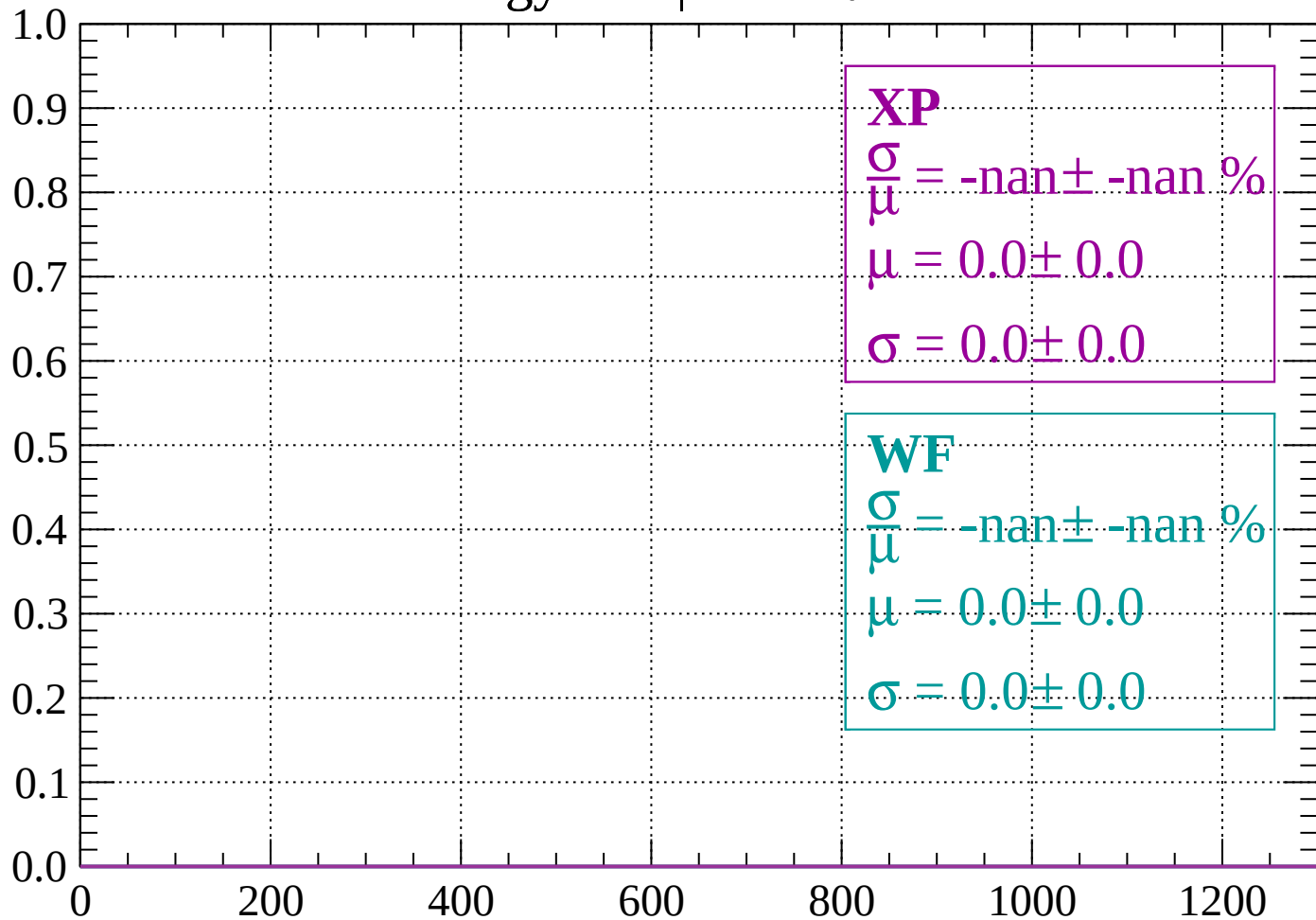
Count



dE/dx (ADC counts/cm)

Energy loss | $-84 < \theta < -82$

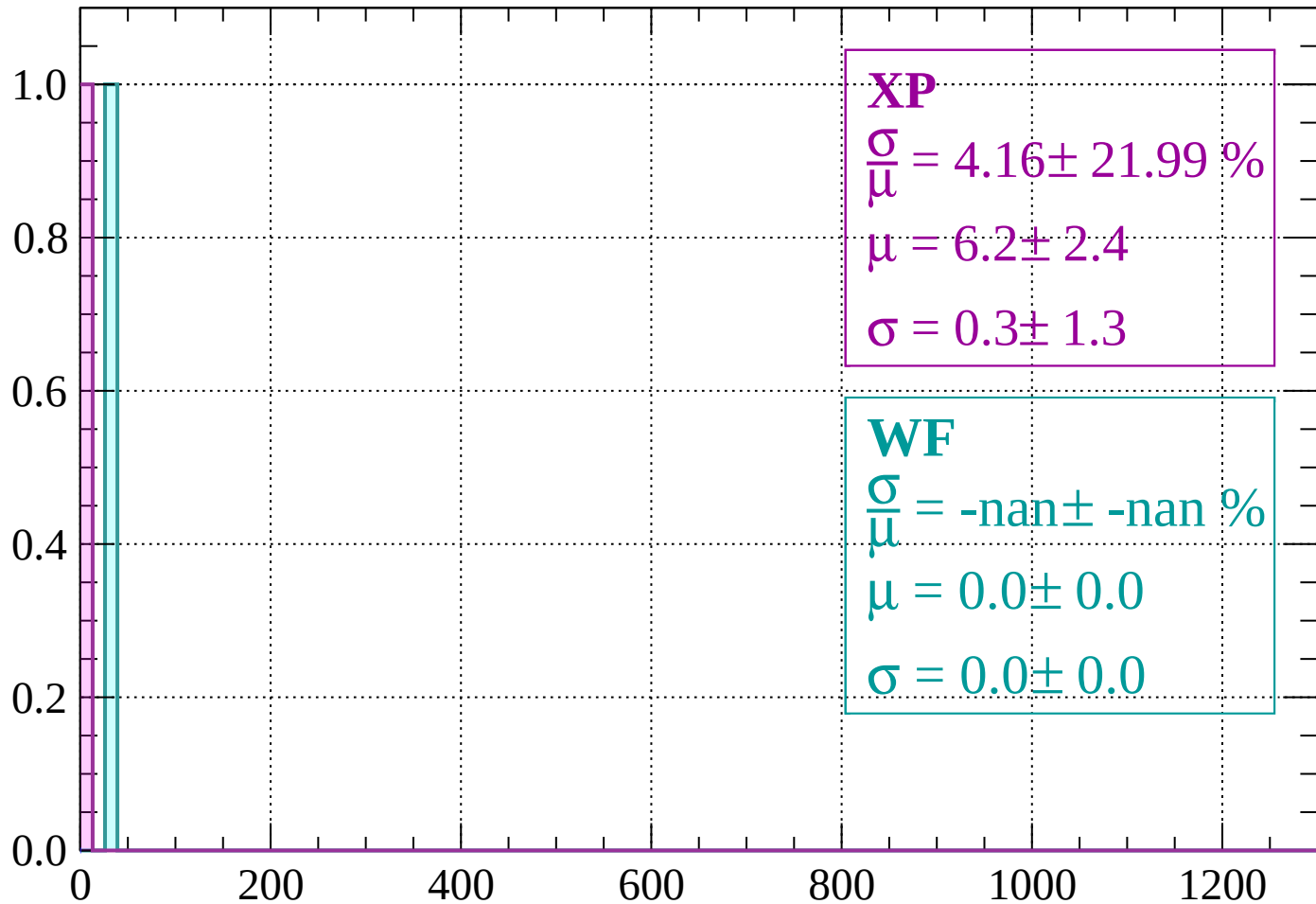
Count



dE/dx (ADC counts/cm)

Energy loss | $-82 < \theta < -80$

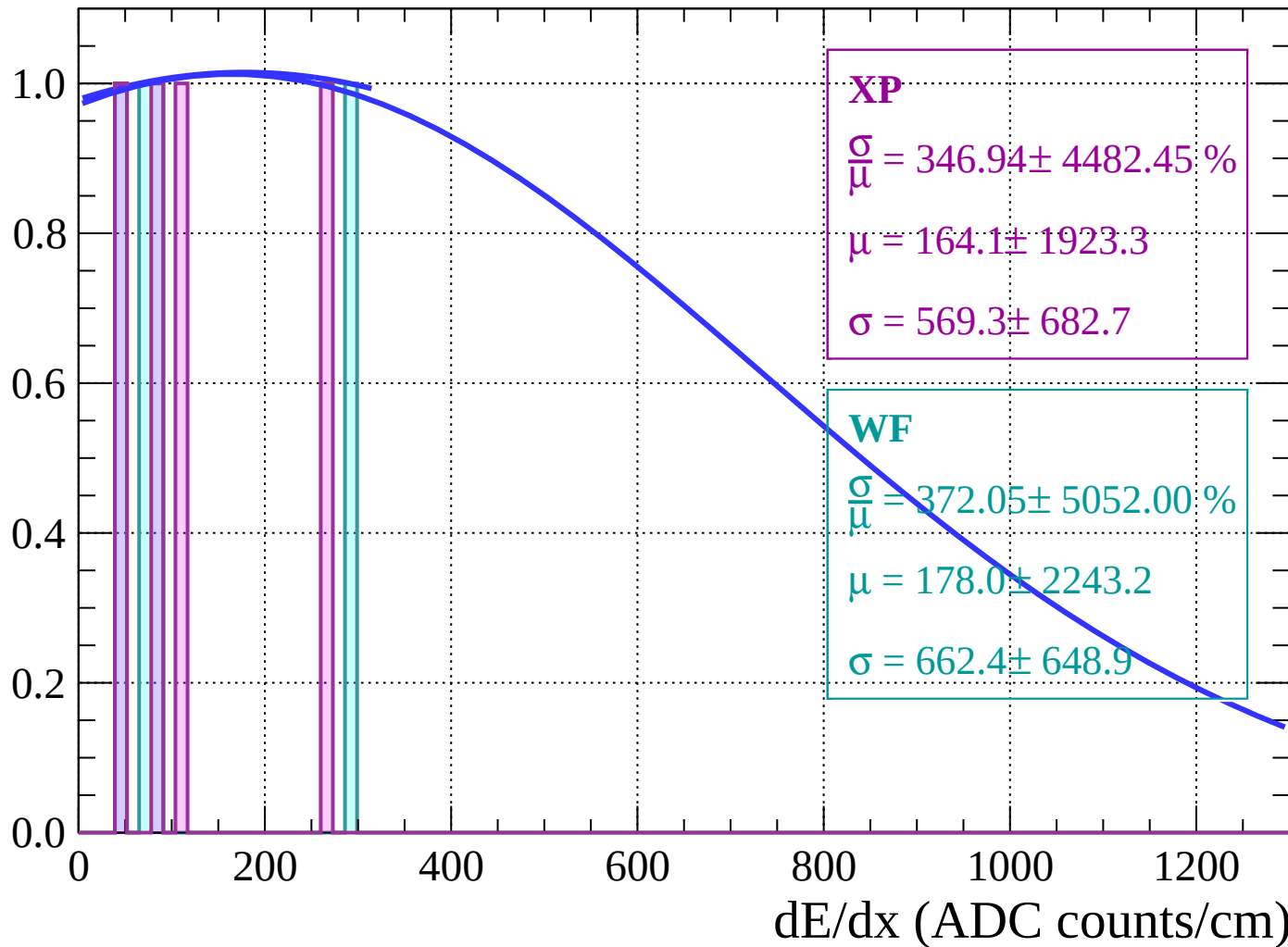
Count



dE/dx (ADC counts/cm)

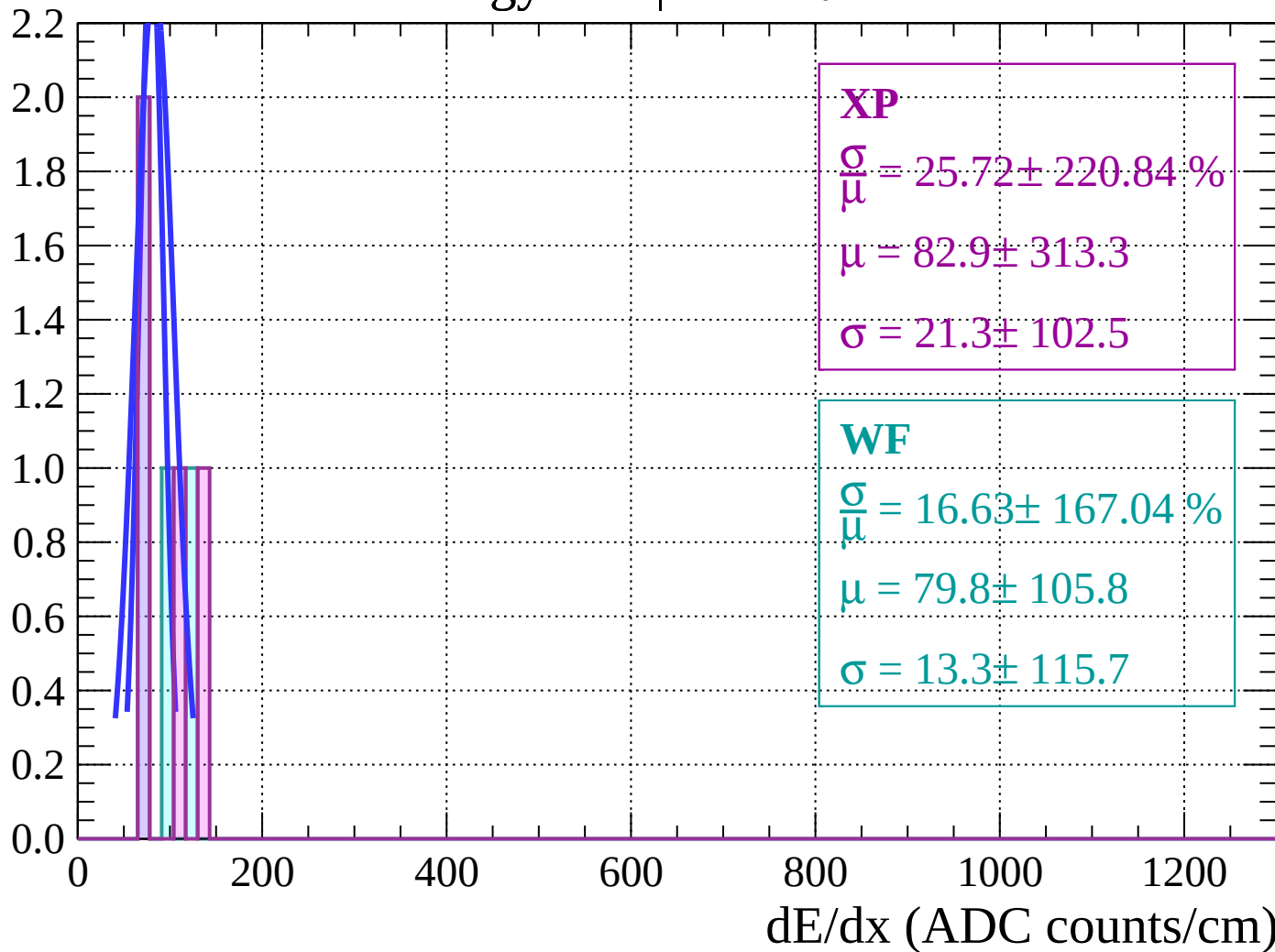
Energy loss | $-80 < \theta < -78$

Count



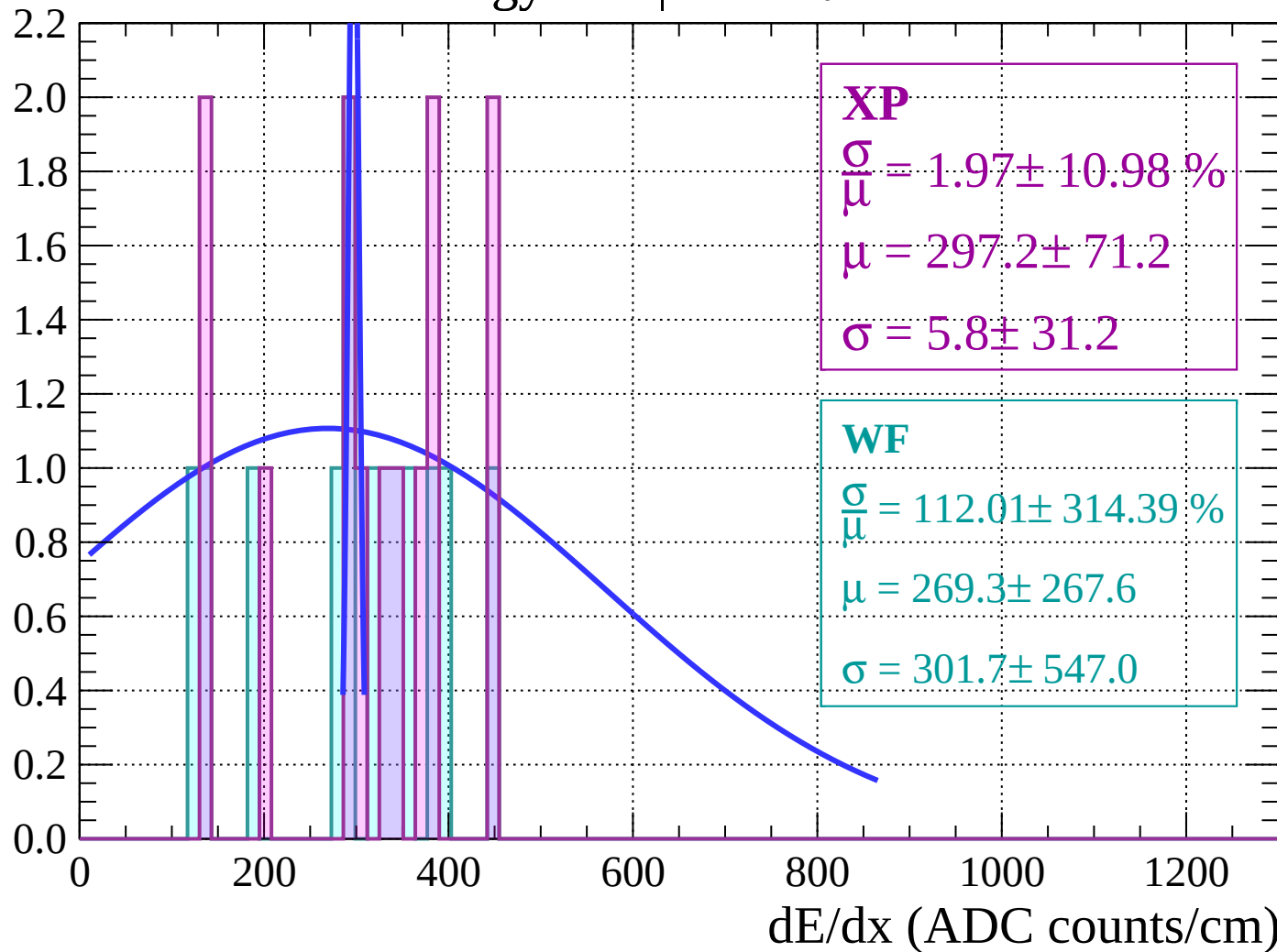
Energy loss $|-78 < \theta < -76$

Count



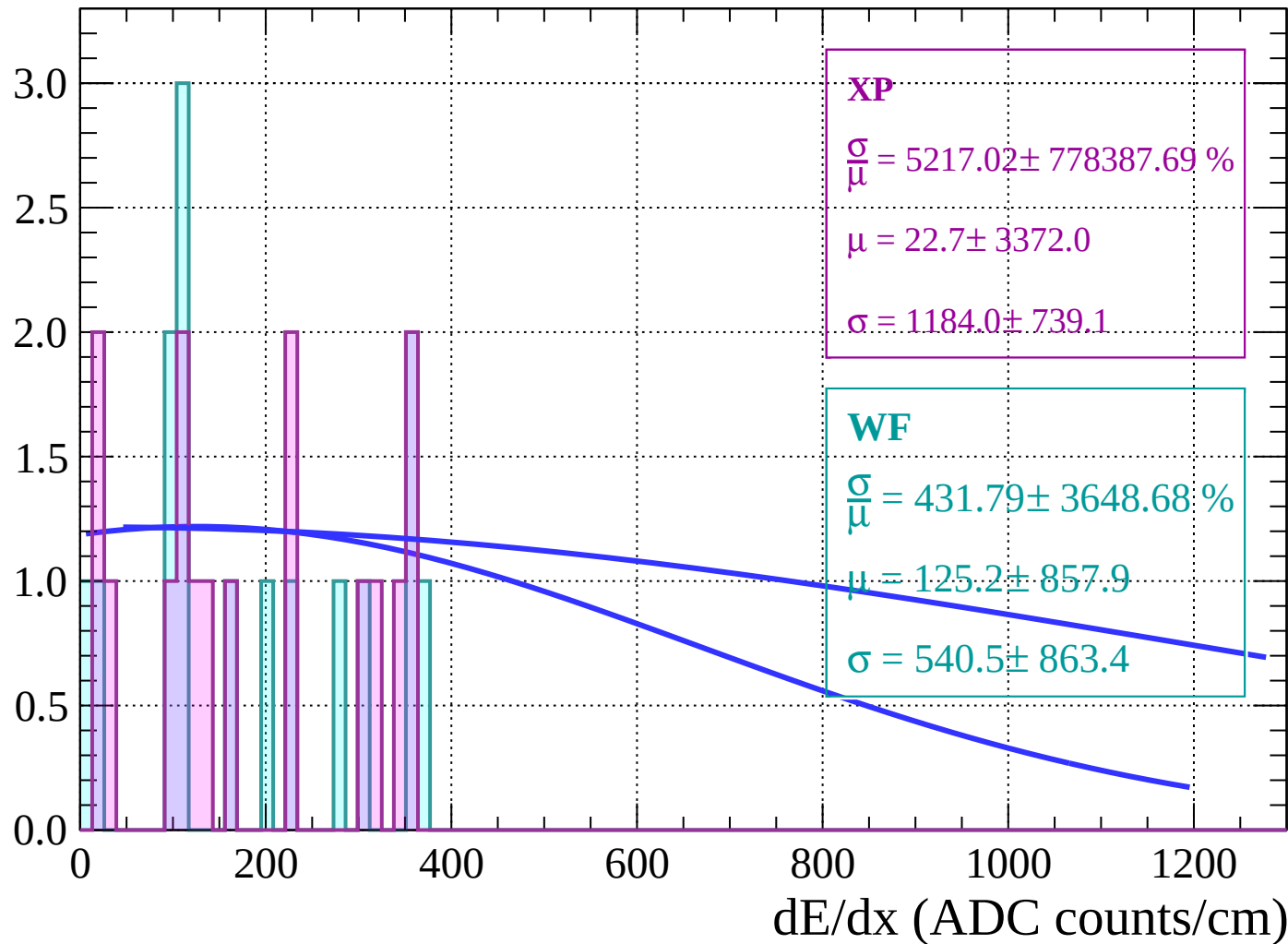
Energy loss | $-76 < \theta < -74$

Count

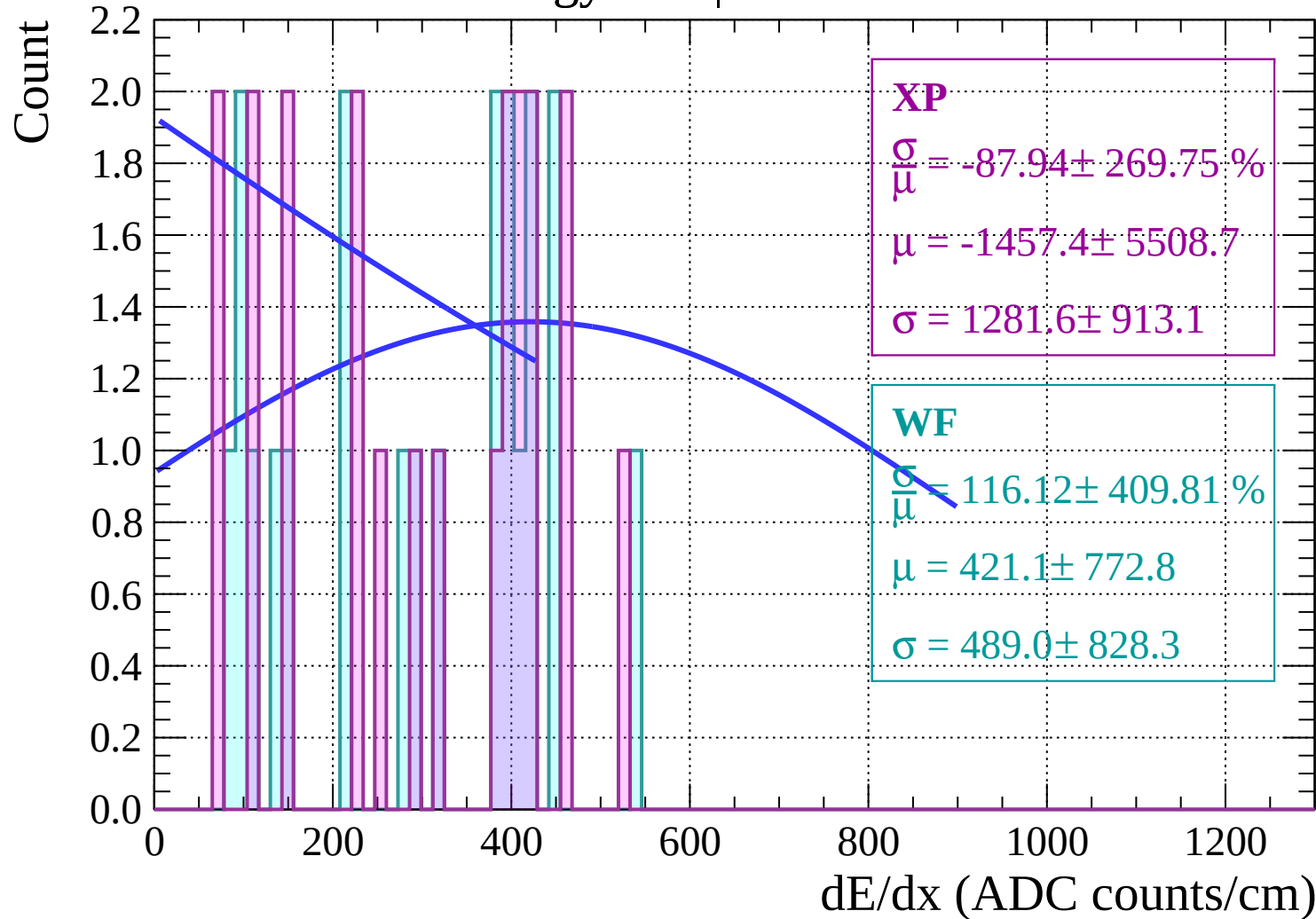


Energy loss | $-74 < \theta < -72$

Count

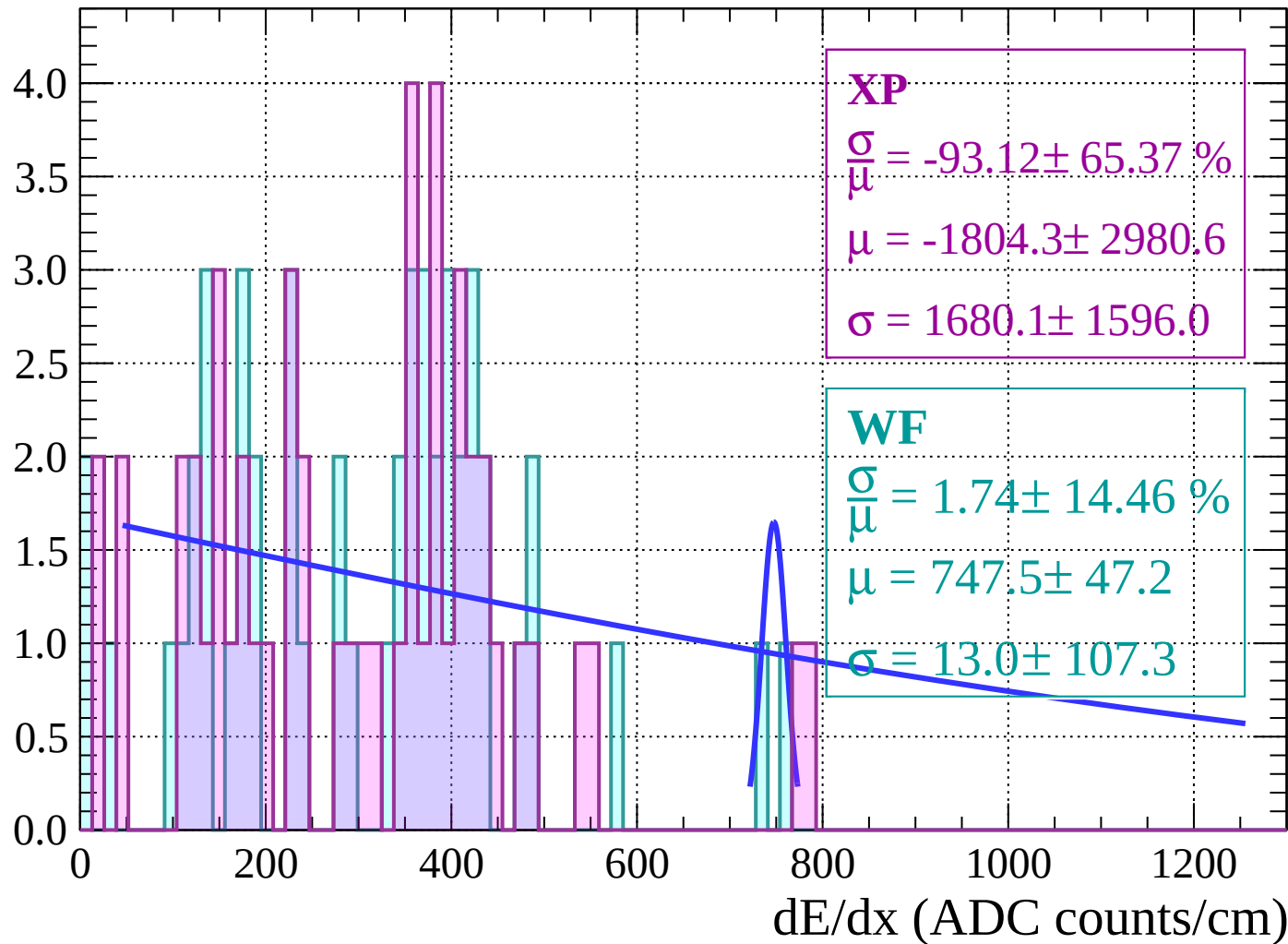


Energy loss | $-72 < \theta < -70$



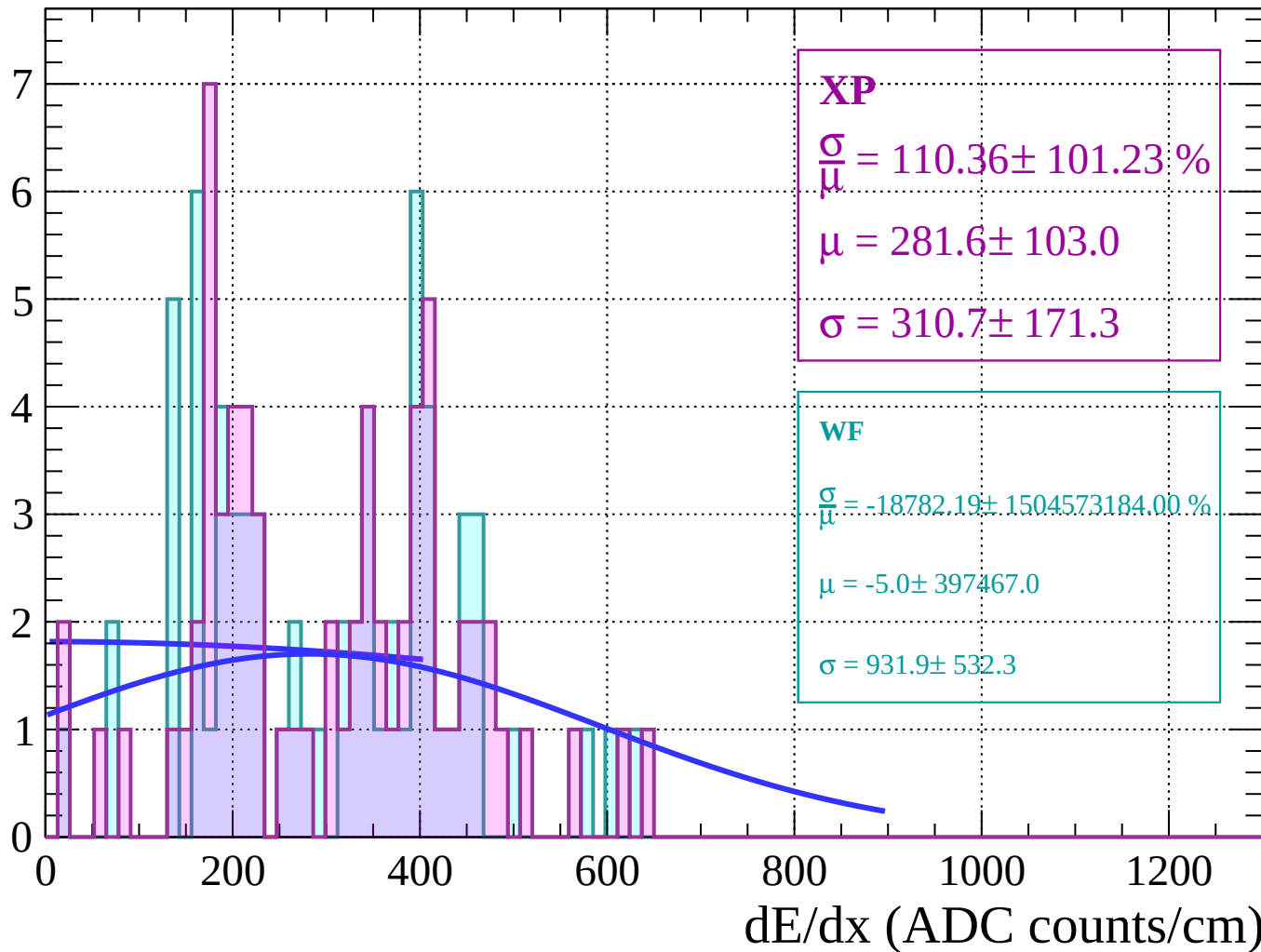
Energy loss | $-70 < \theta < -68$

Count



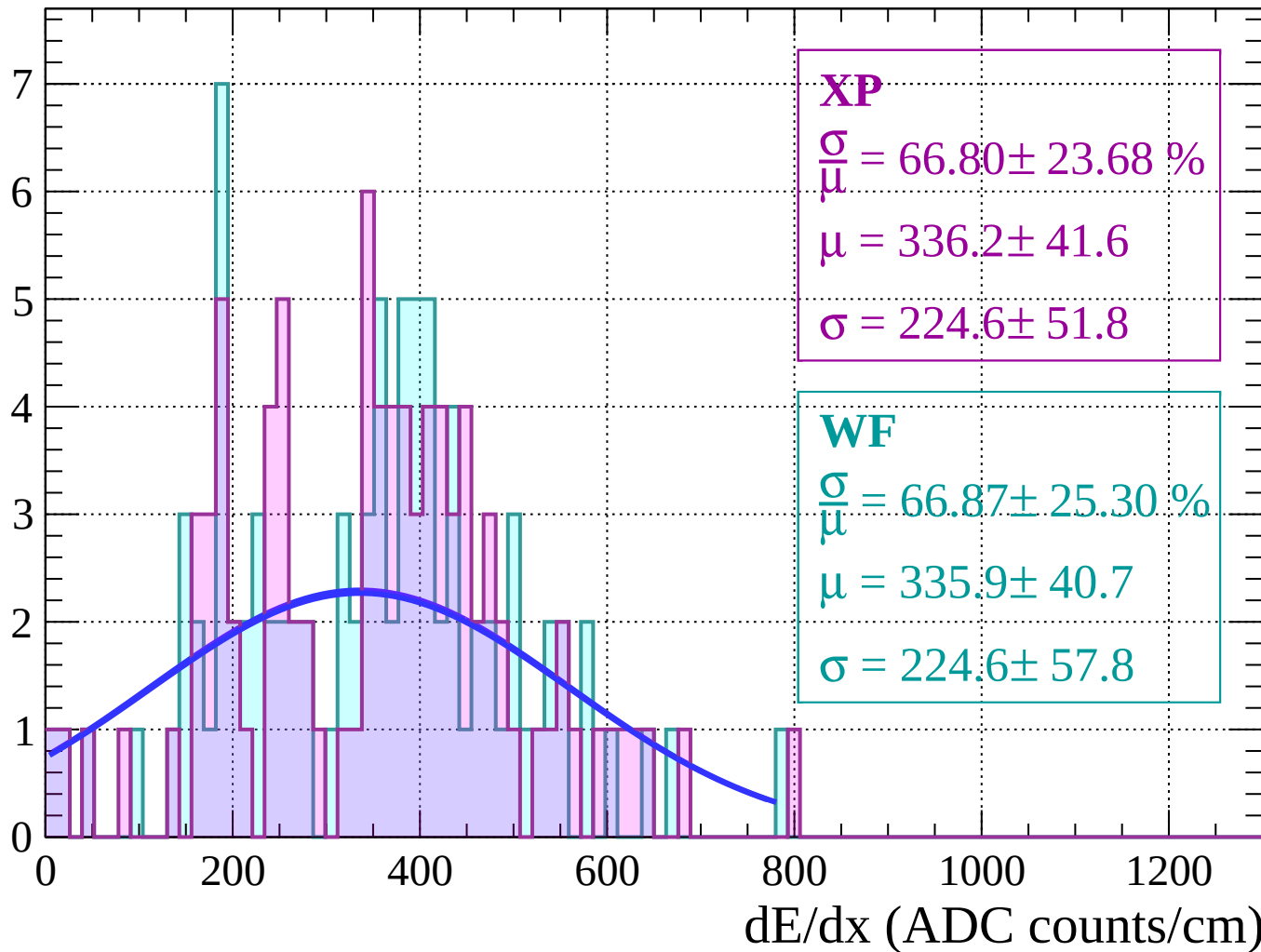
Energy loss | $-68 < \theta < -66$

Count



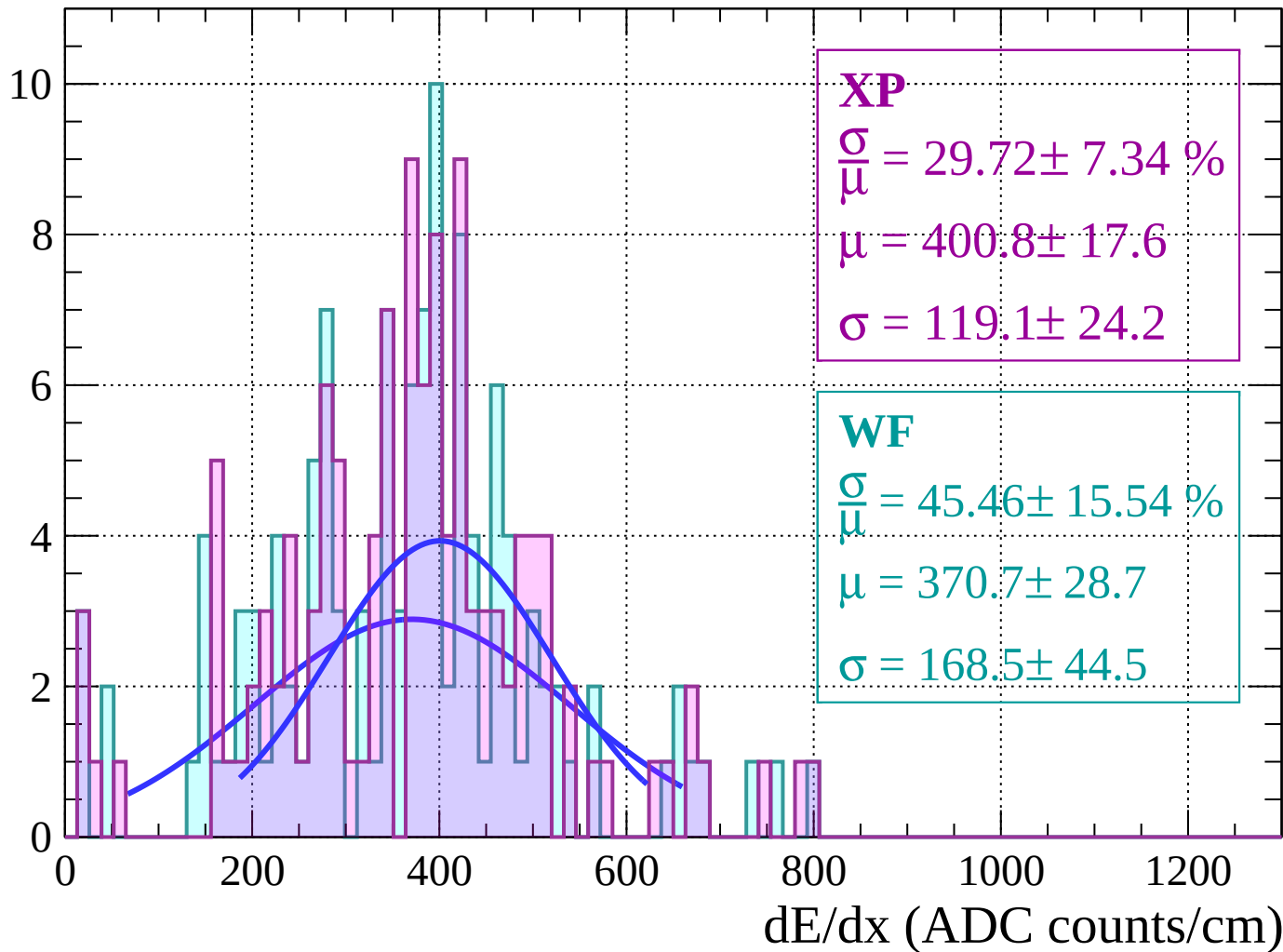
Energy loss | $-66 < \theta < -64$

Count



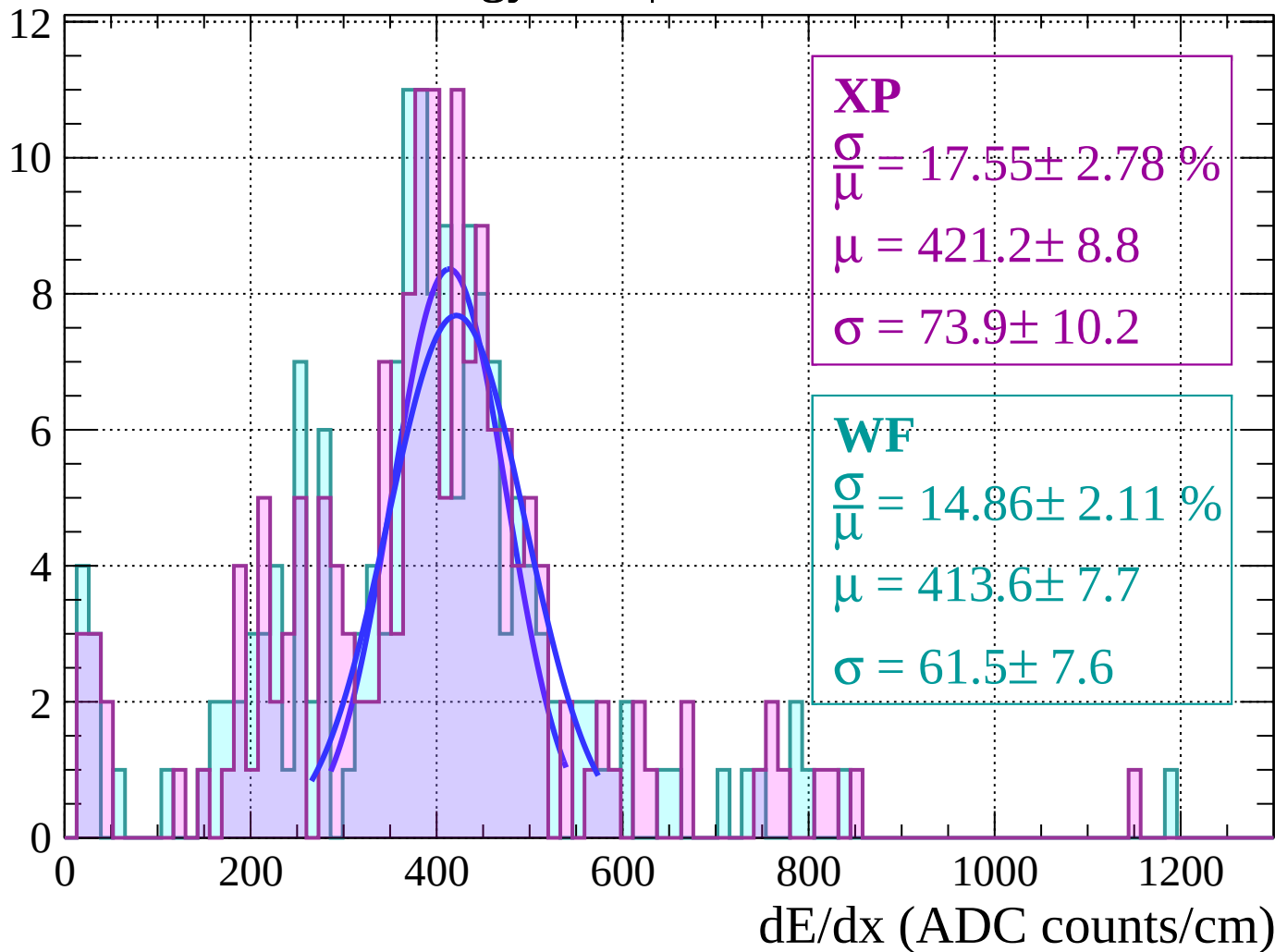
Energy loss | $-64 < \theta < -62$

Count



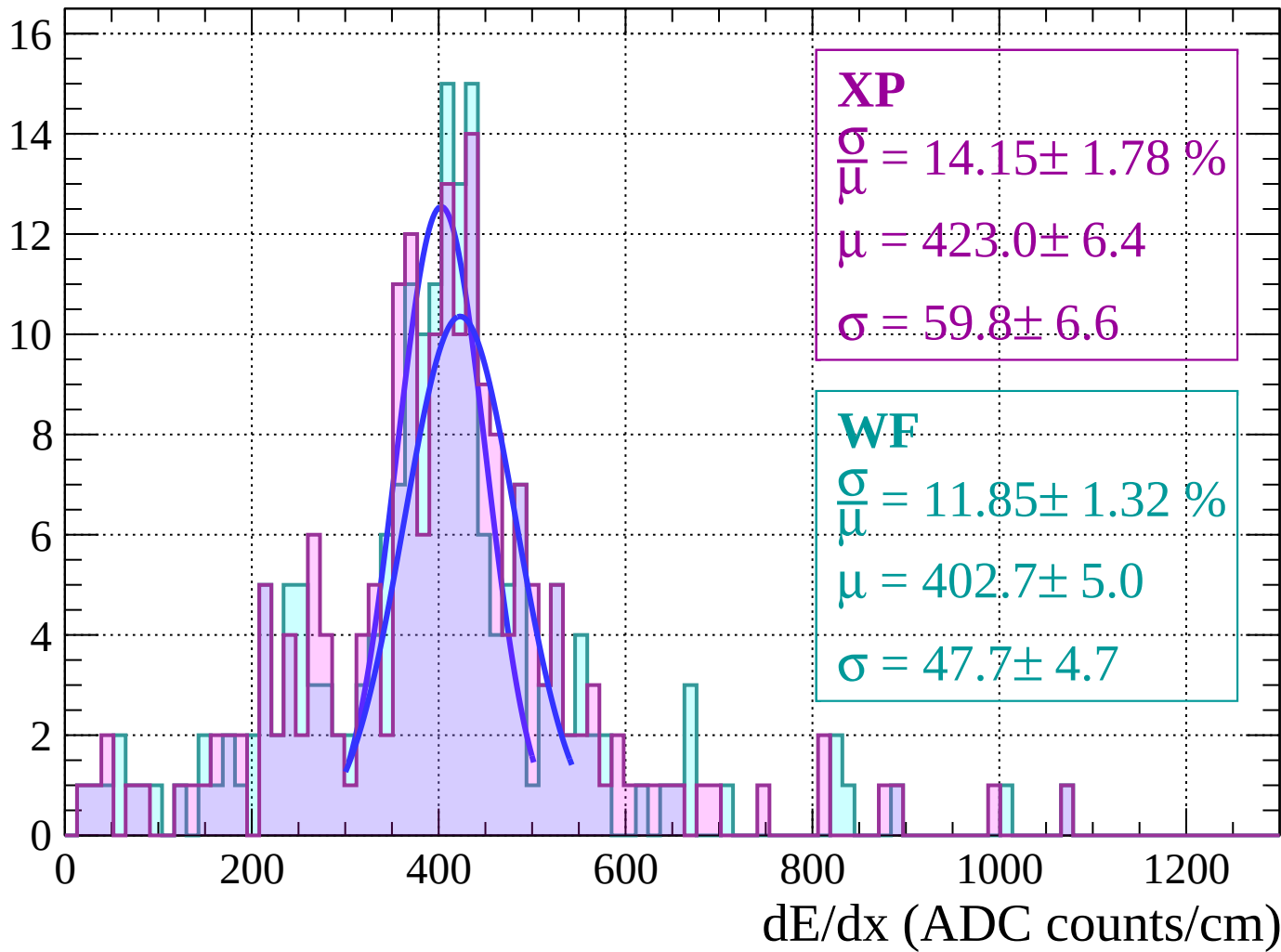
Energy loss | $-62 < \theta < -60$

Count



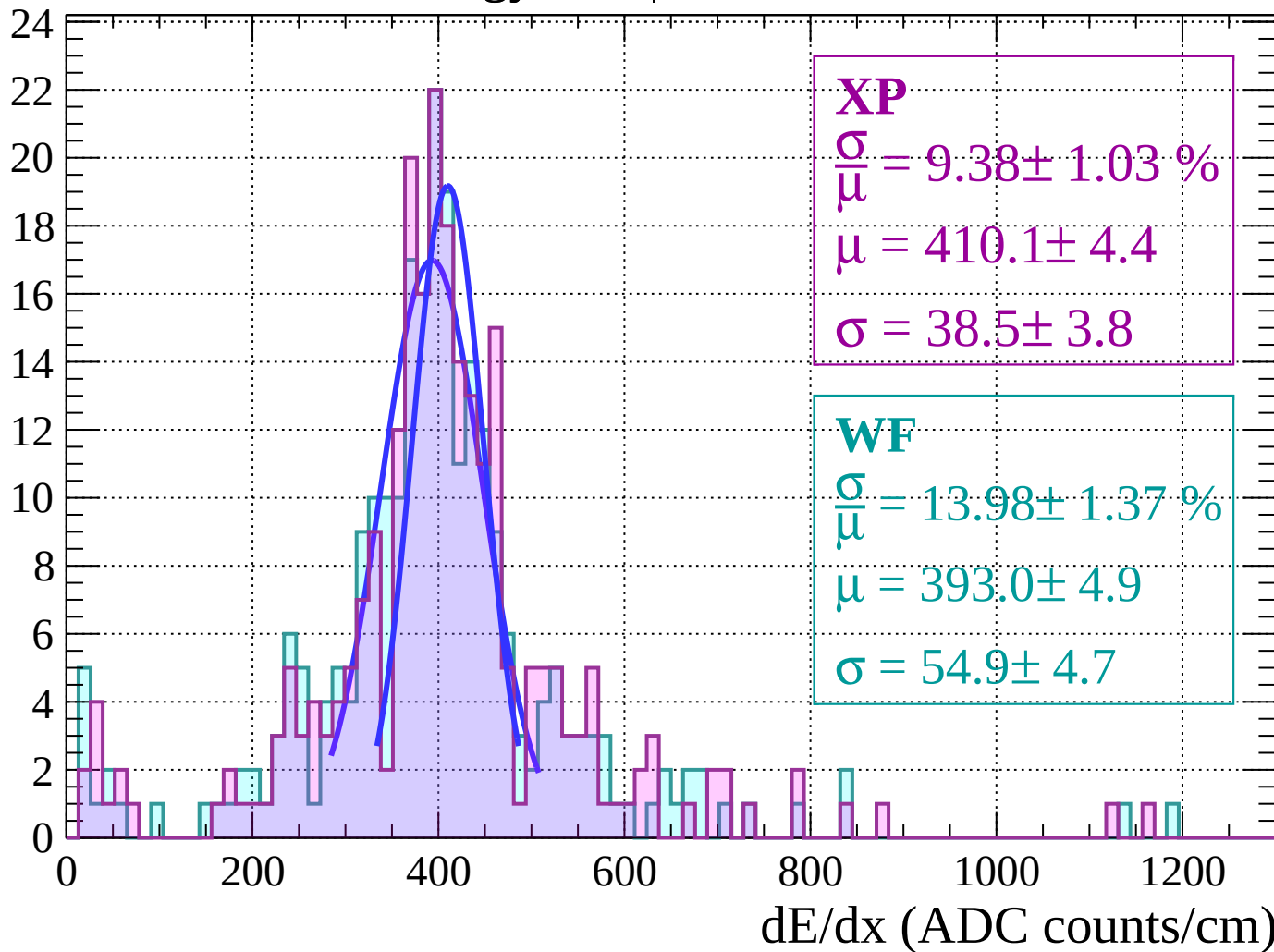
Energy loss | $-60 < \theta < -58$

Count



Energy loss $|-58 < \theta < -56$

Count



Energy loss $|-56 < \theta < -54$

Count

XP

$$\frac{\sigma}{\mu} = 15.81 \pm 1.74 \%$$

$$\mu = 418.5 \pm 5.5$$

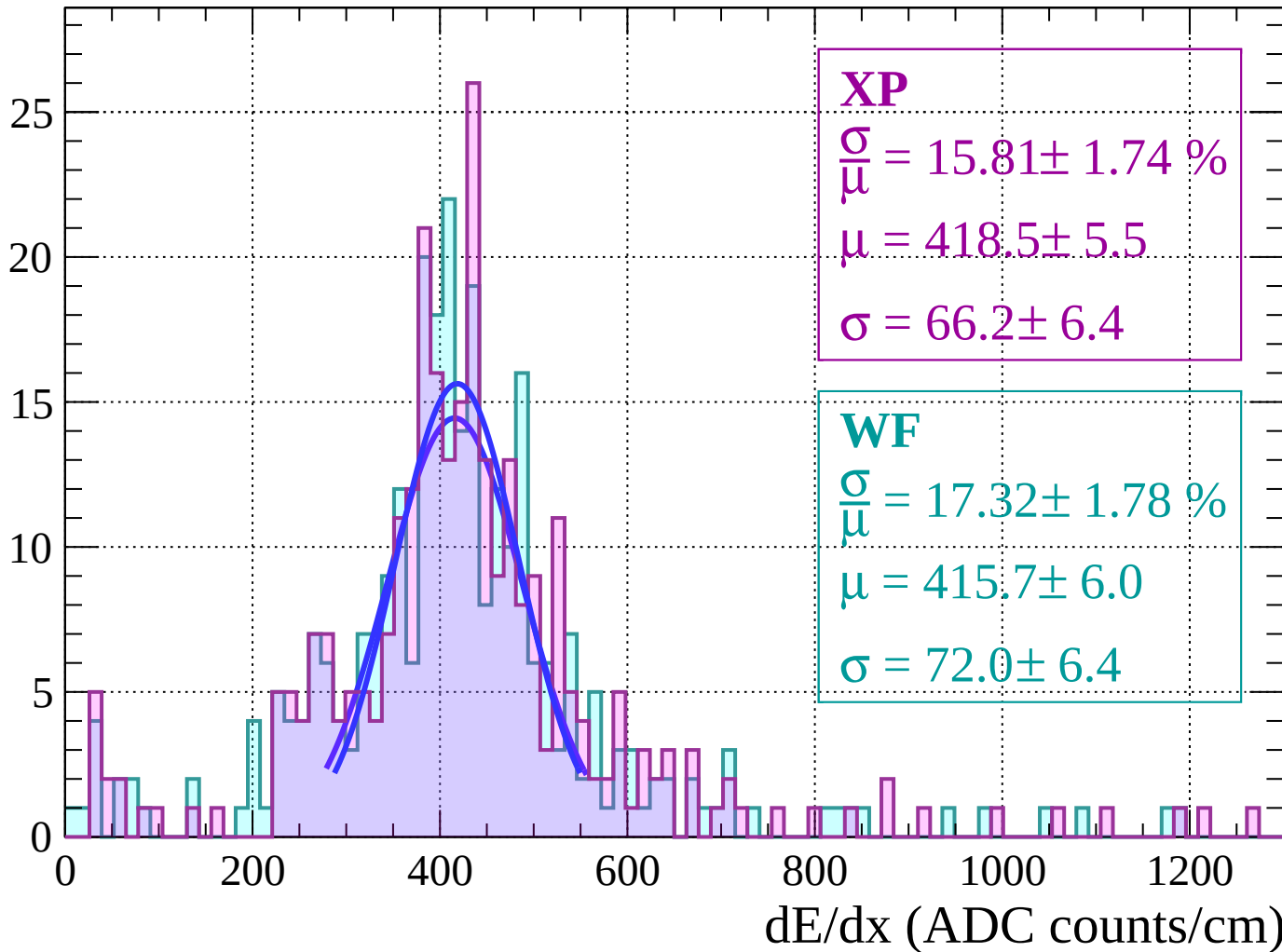
$$\sigma = 66.2 \pm 6.4$$

WF

$$\frac{\sigma}{\mu} = 17.32 \pm 1.78 \%$$

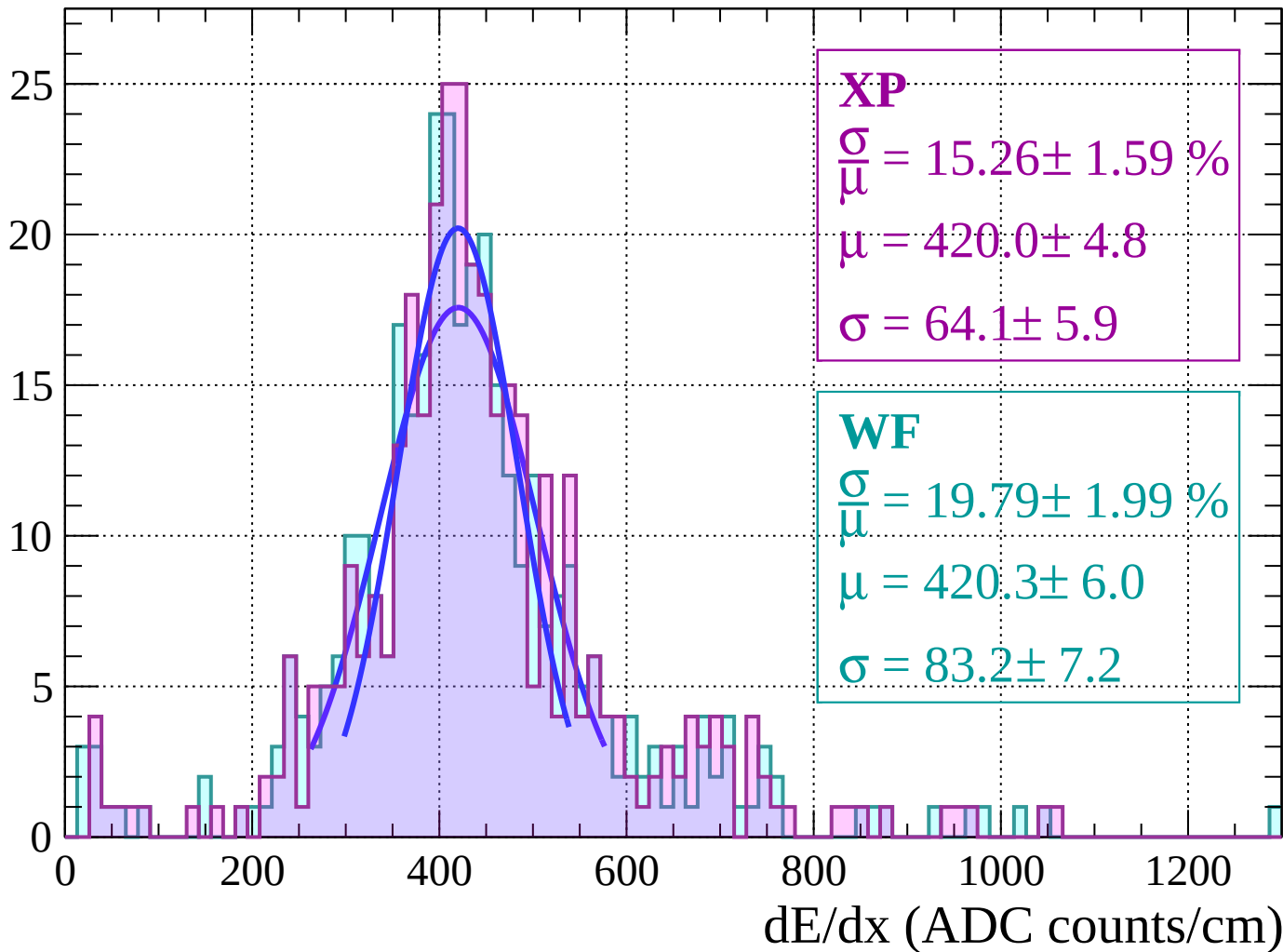
$$\mu = 415.7 \pm 6.0$$

$$\sigma = 72.0 \pm 6.4$$



Energy loss | $-54 < \theta < -52$

Count



Energy loss $|-52 < \theta < -50$

Count

XP

$$\frac{\sigma}{\mu} = 15.25 \pm 1.10 \%$$

$$\mu = 421.0 \pm 4.2$$

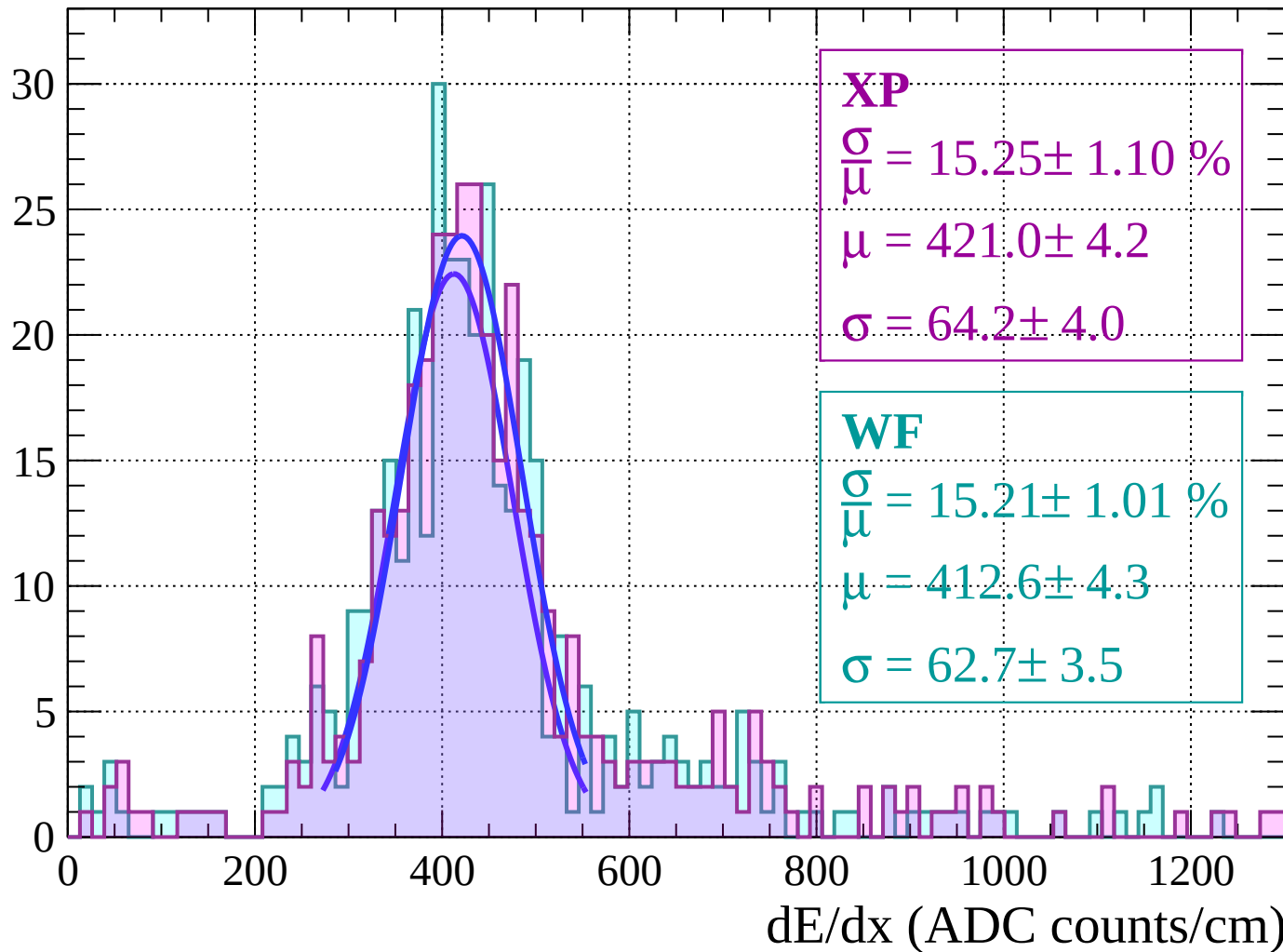
$$\sigma = 64.2 \pm 4.0$$

WF

$$\frac{\sigma}{\mu} = 15.21 \pm 1.01 \%$$

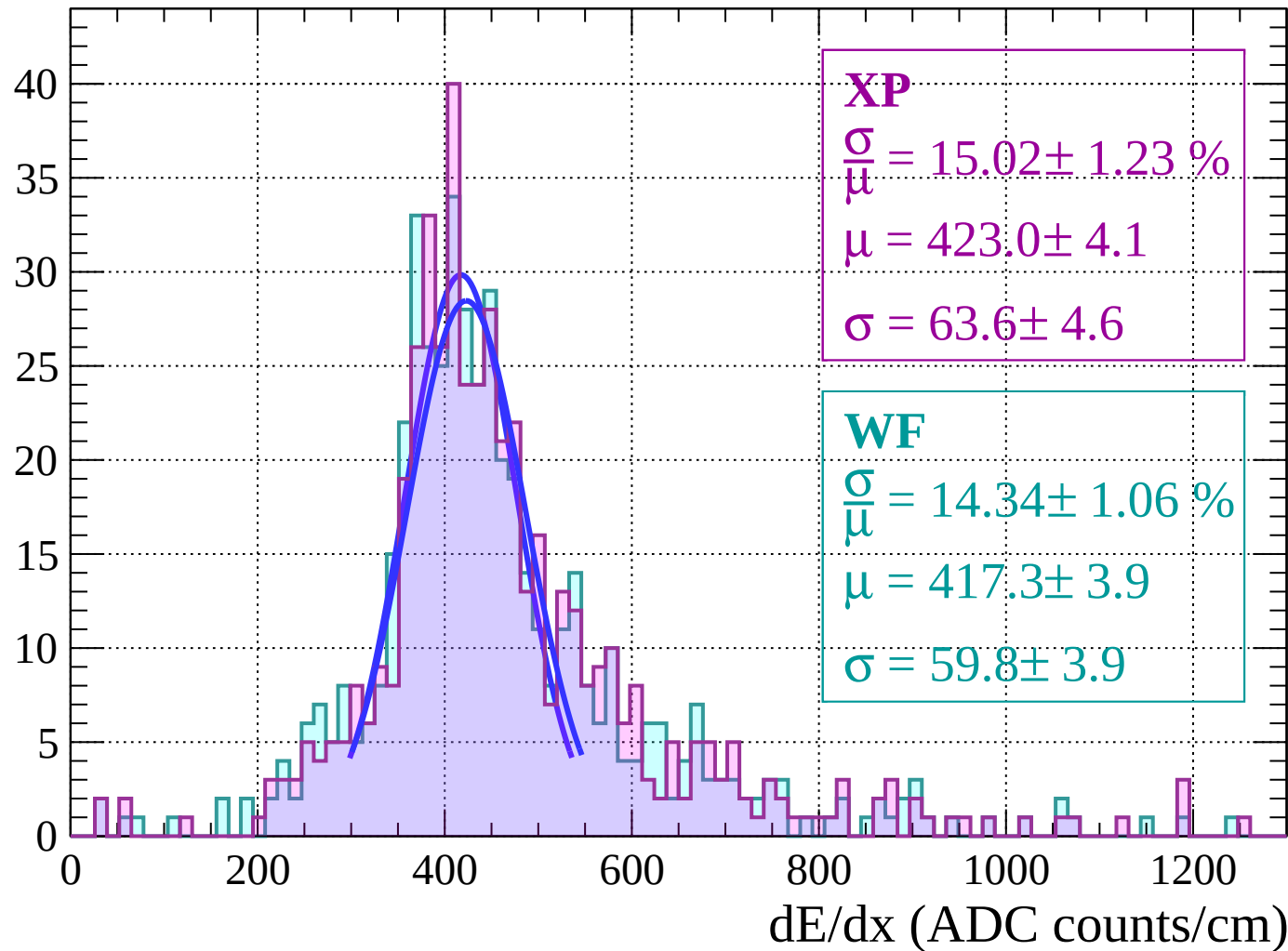
$$\mu = 412.6 \pm 4.3$$

$$\sigma = 62.7 \pm 3.5$$



Energy loss | $-50 < \theta < -48$

Count



Energy loss | $-48 < \theta < -46$

Count

XP

$$\frac{\sigma}{\mu} = 12.28 \pm 0.77 \%$$

$$\mu = 426.1 \pm 3.1$$

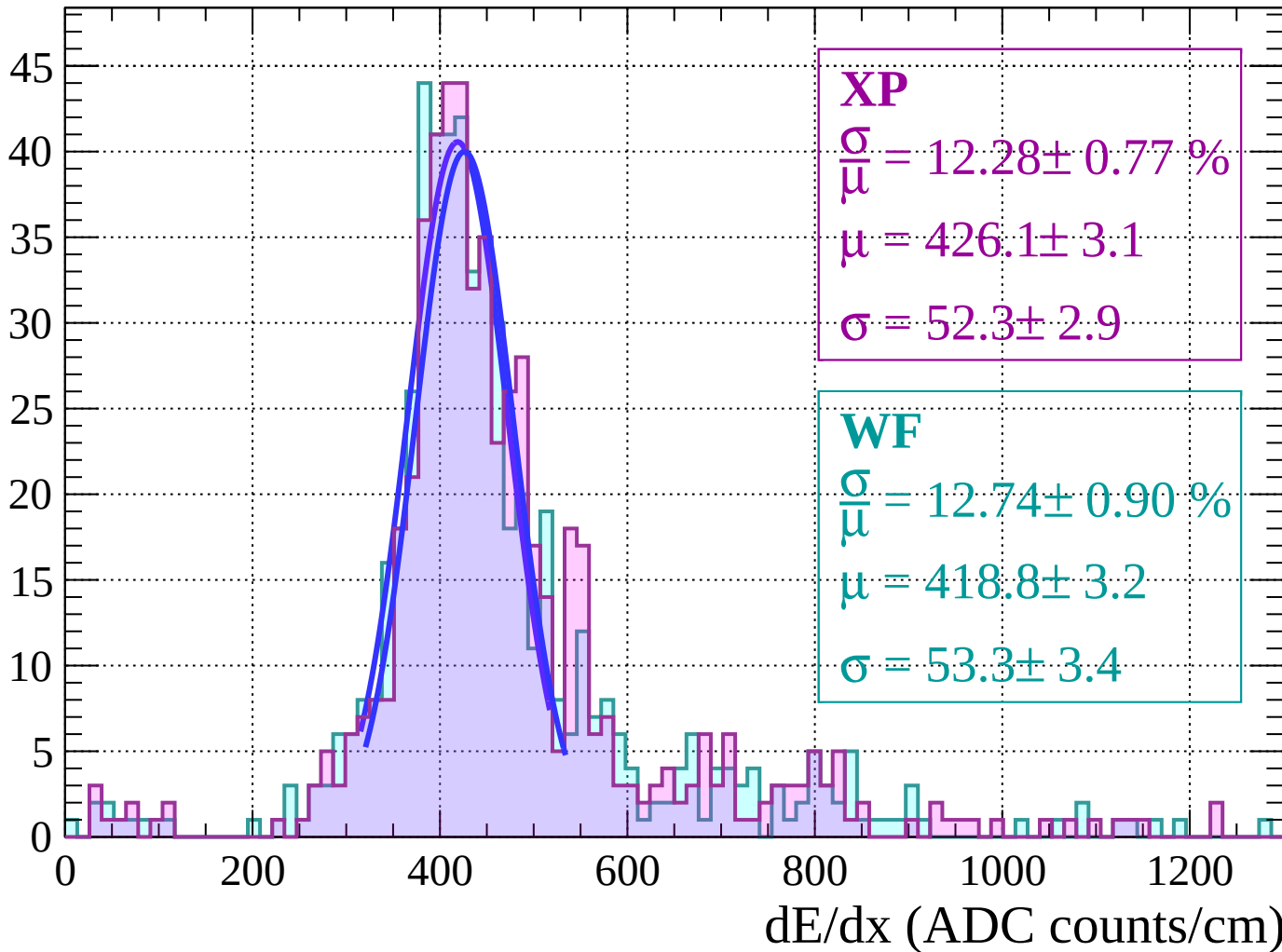
$$\sigma = 52.3 \pm 2.9$$

WF

$$\frac{\sigma}{\mu} = 12.74 \pm 0.90 \%$$

$$\mu = 418.8 \pm 3.2$$

$$\sigma = 53.3 \pm 3.4$$



Energy loss $|-46 < \theta < -44$

Count

XP

$$\frac{\sigma}{\mu} = 14.49 \pm 0.95 \%$$

$$\mu = 419.3 \pm 3.3$$

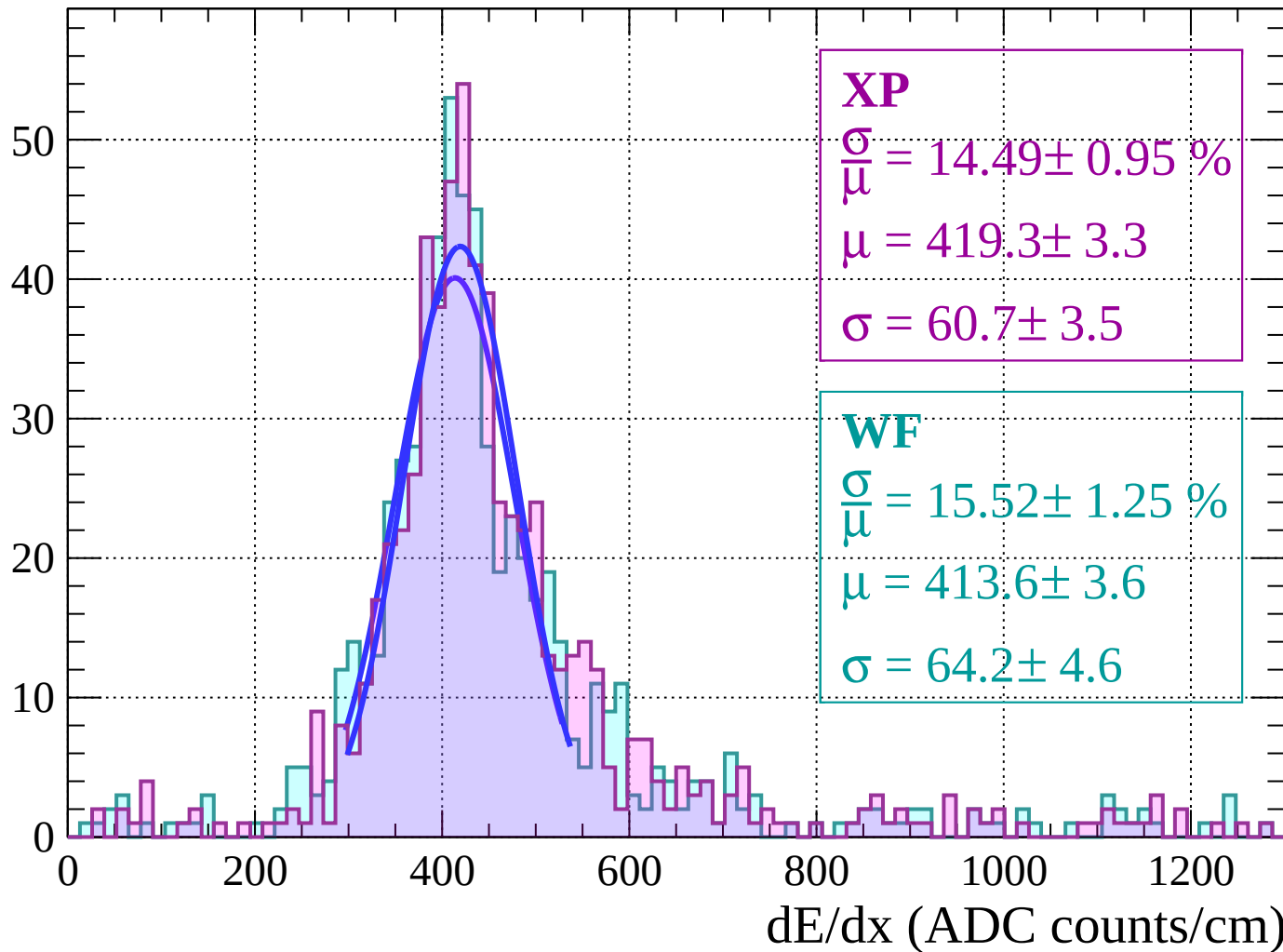
$$\sigma = 60.7 \pm 3.5$$

WF

$$\frac{\sigma}{\mu} = 15.52 \pm 1.25 \%$$

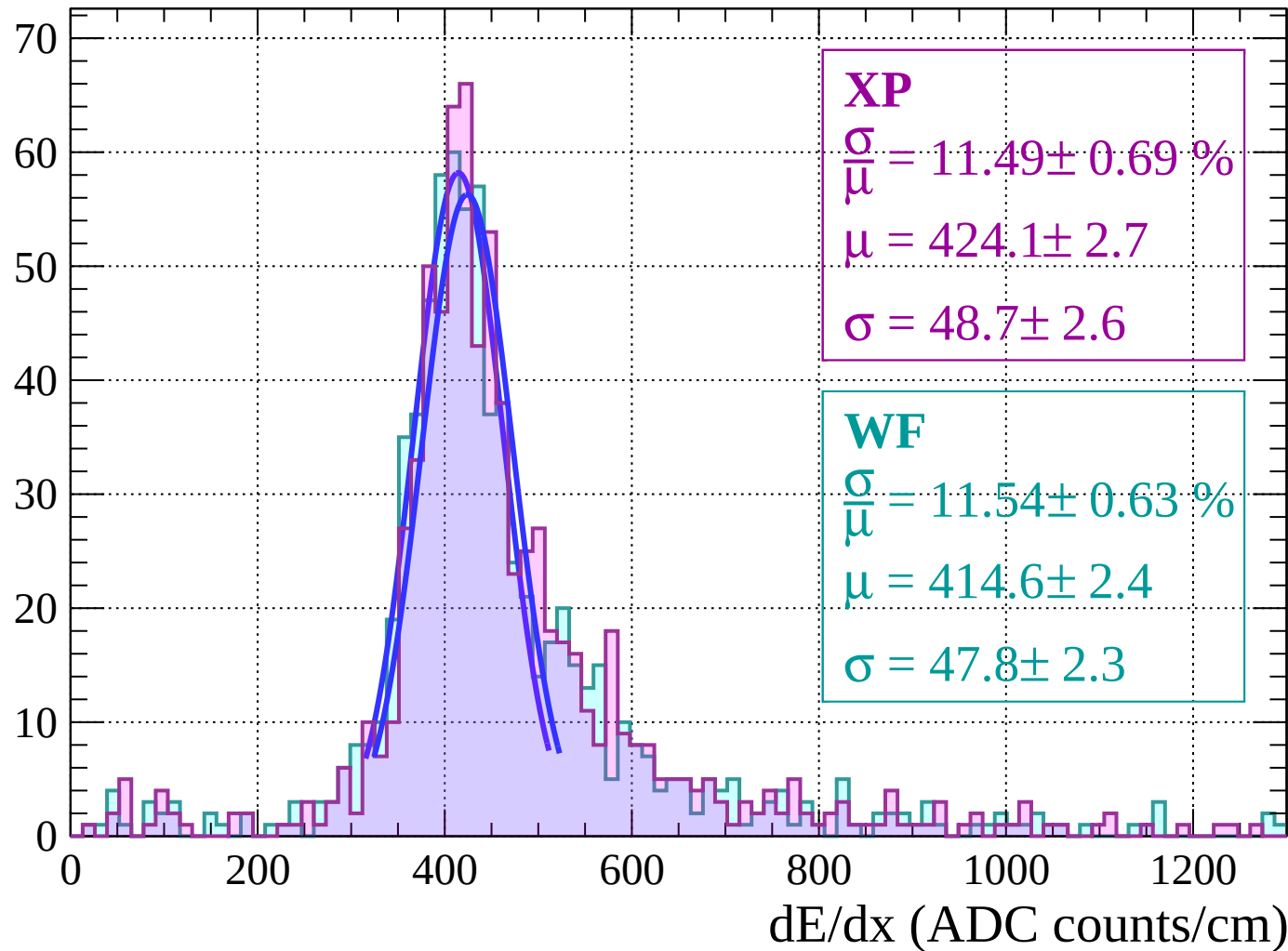
$$\mu = 413.6 \pm 3.6$$

$$\sigma = 64.2 \pm 4.6$$



Energy loss | $-44 < \theta < -42$

Count



Energy loss $|-42 < \theta < -40|$

Count

XP

$$\frac{\sigma}{\mu} = 13.21 \pm 0.79 \%$$

$$\mu = 424.6 \pm 2.7$$

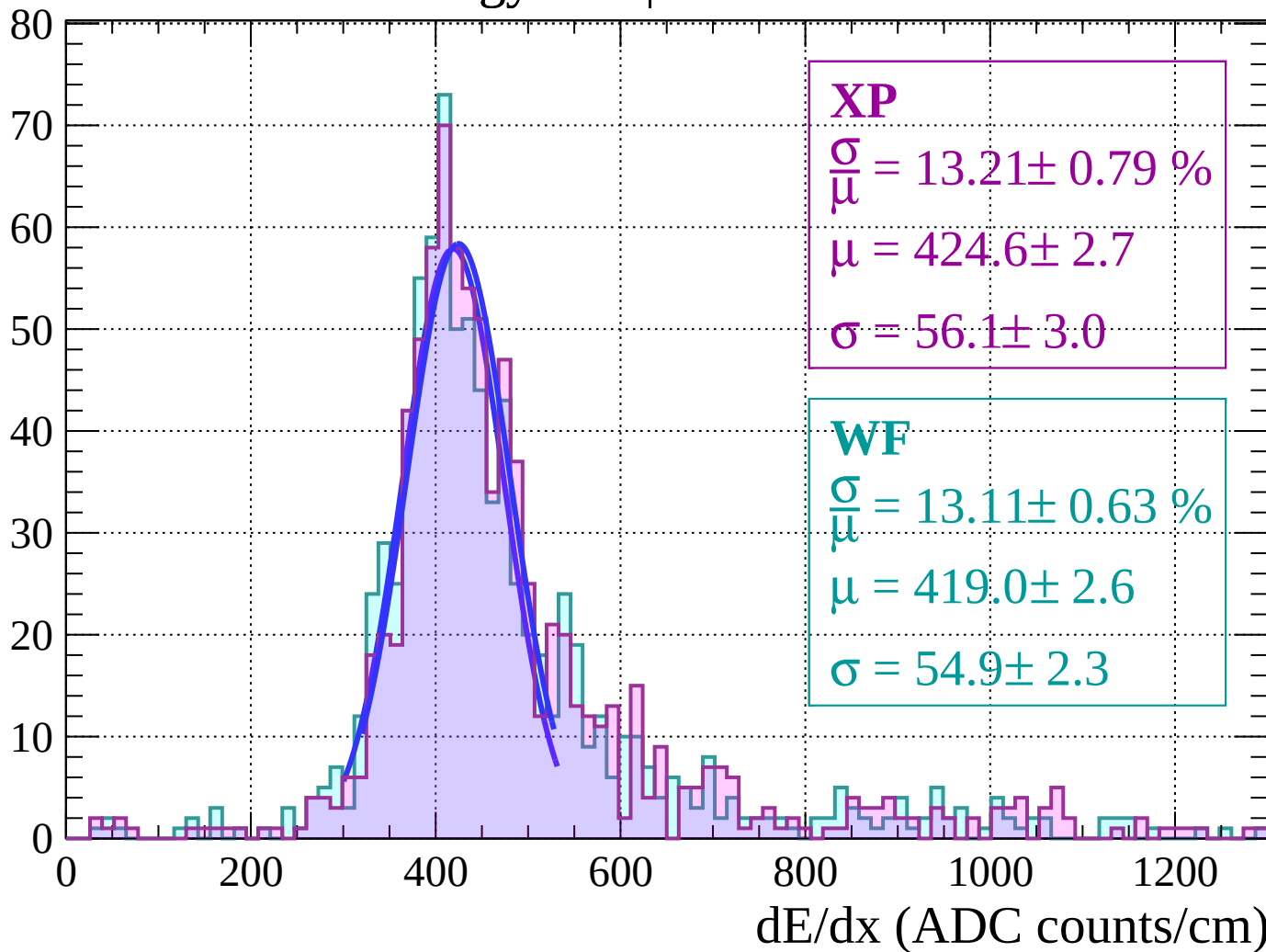
$$\sigma = 56.1 \pm 3.0$$

WF

$$\frac{\sigma}{\mu} = 13.11 \pm 0.63 \%$$

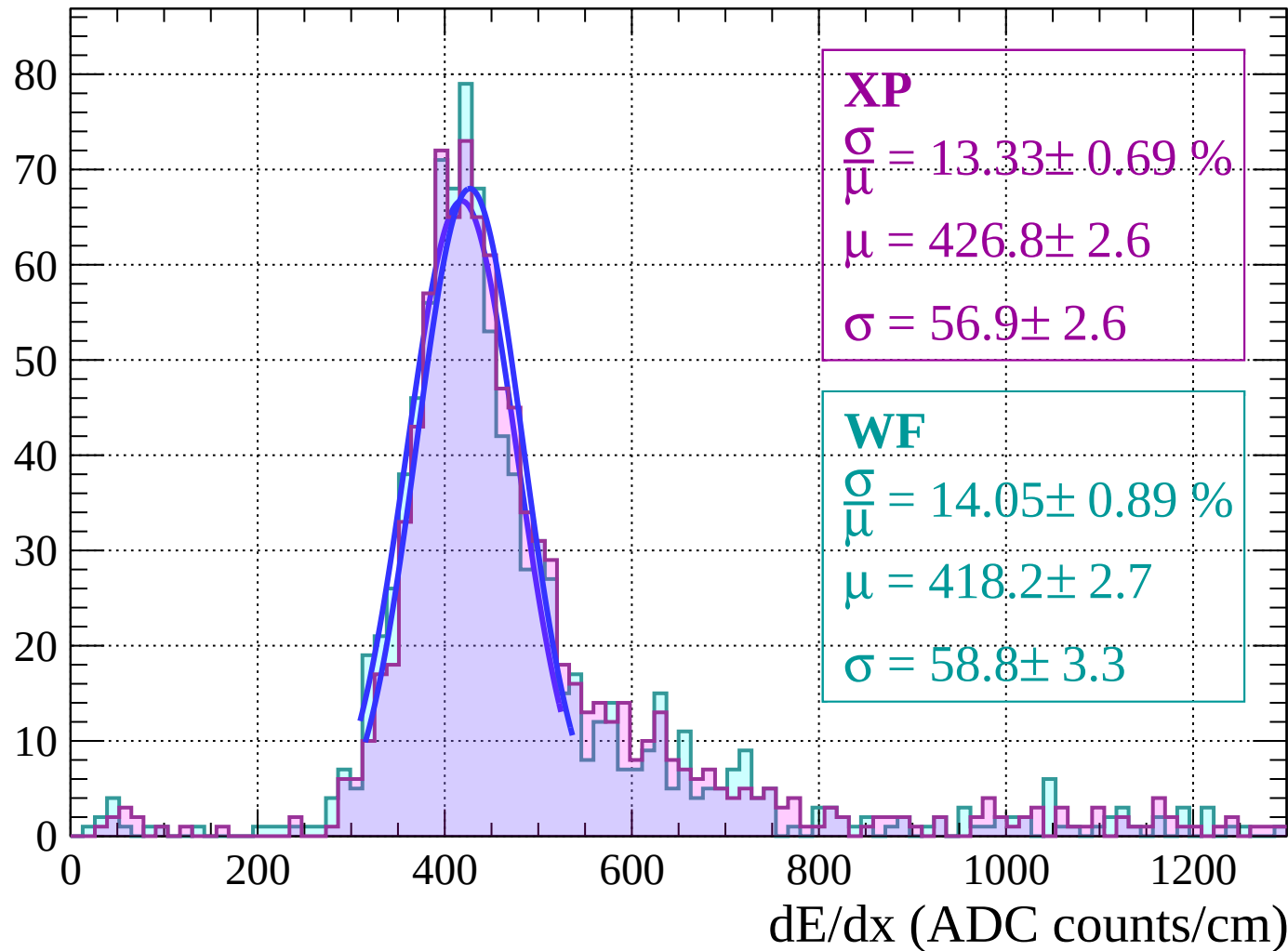
$$\mu = 419.0 \pm 2.6$$

$$\sigma = 54.9 \pm 2.3$$



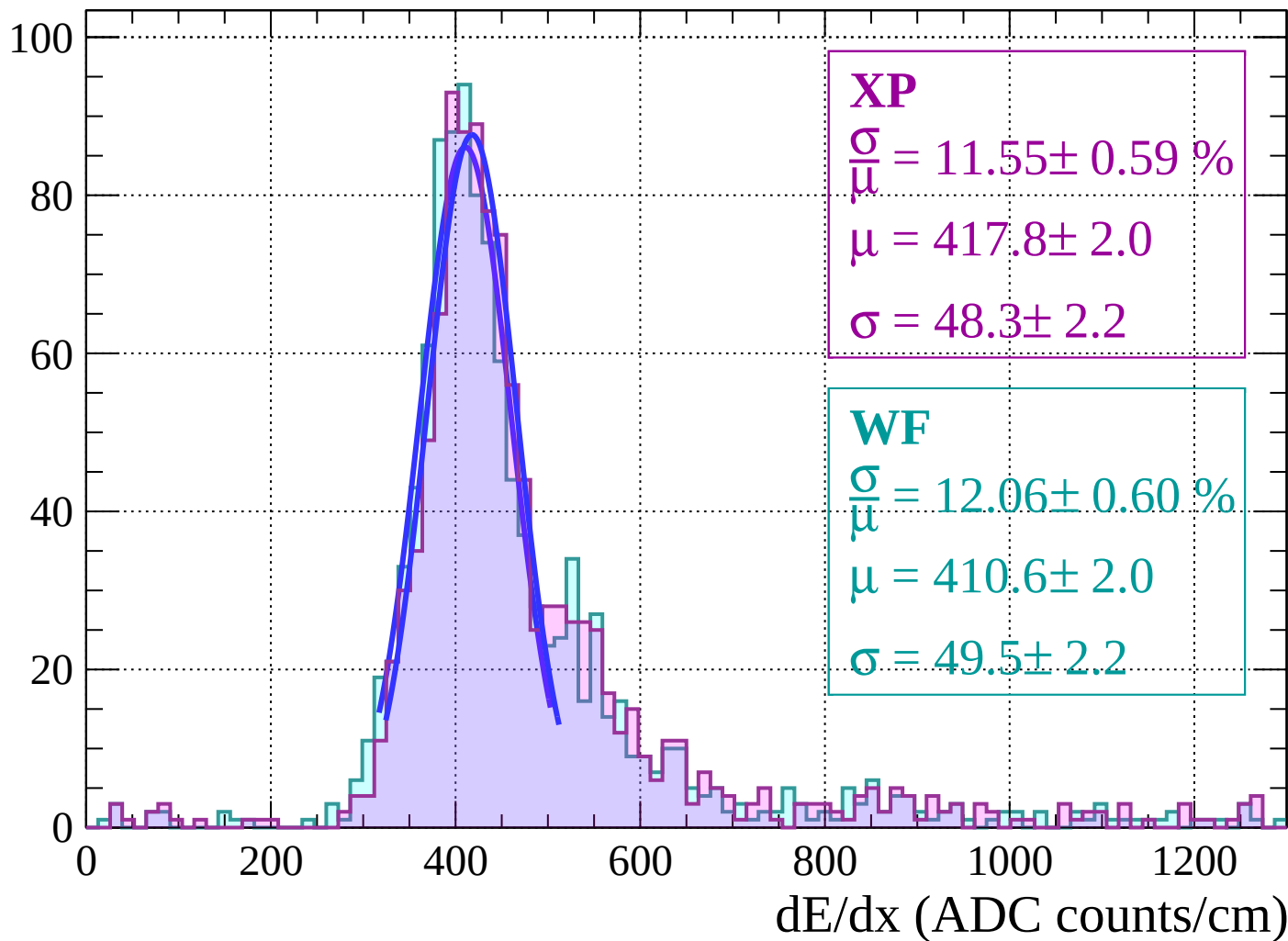
Energy loss | $-40 < \theta < -38$

Count



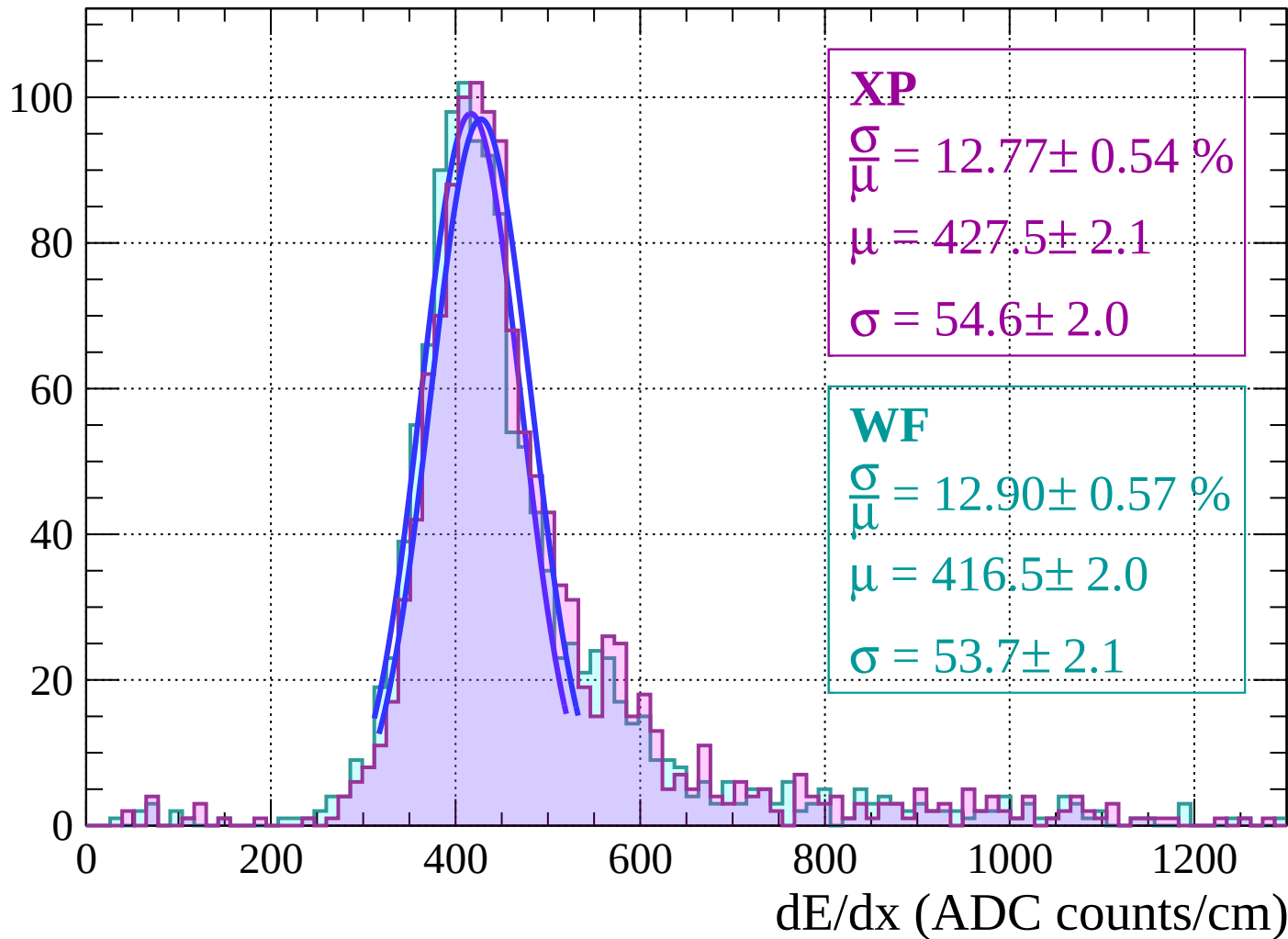
Energy loss $|-38 < \theta < -36$

Count

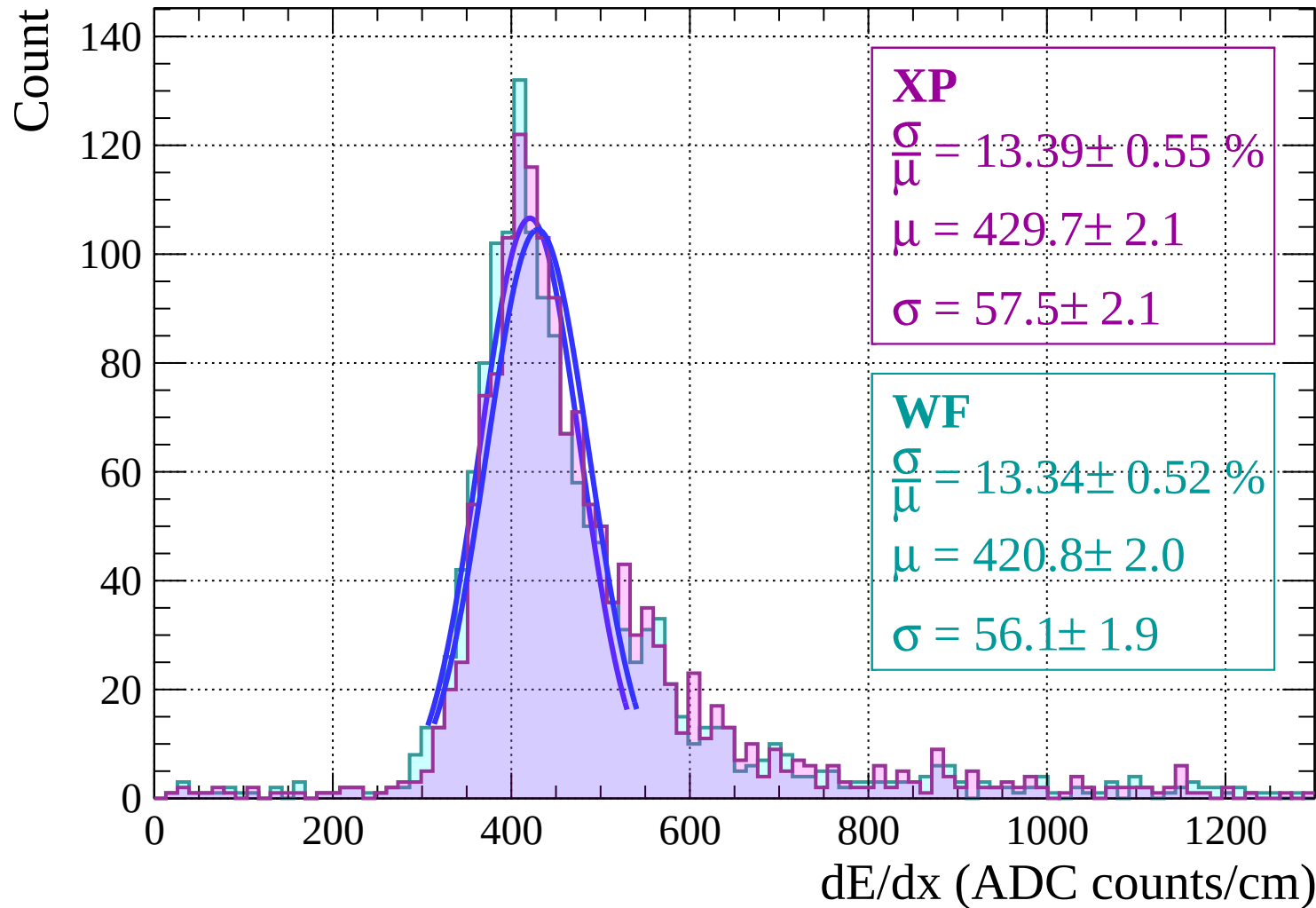


Energy loss $|-36 < \theta < -34$

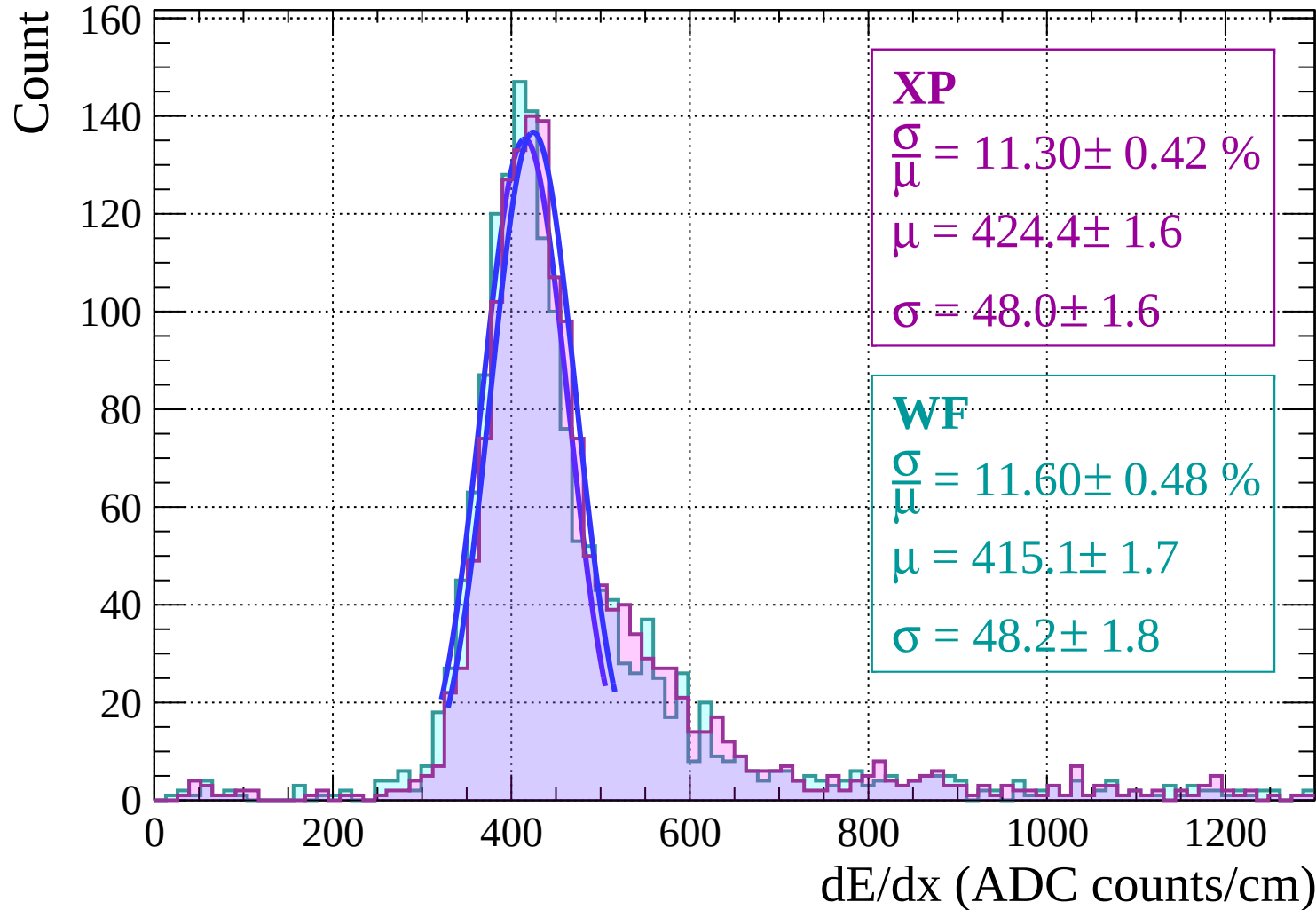
Count



Energy loss $|-34 < \theta < -32$

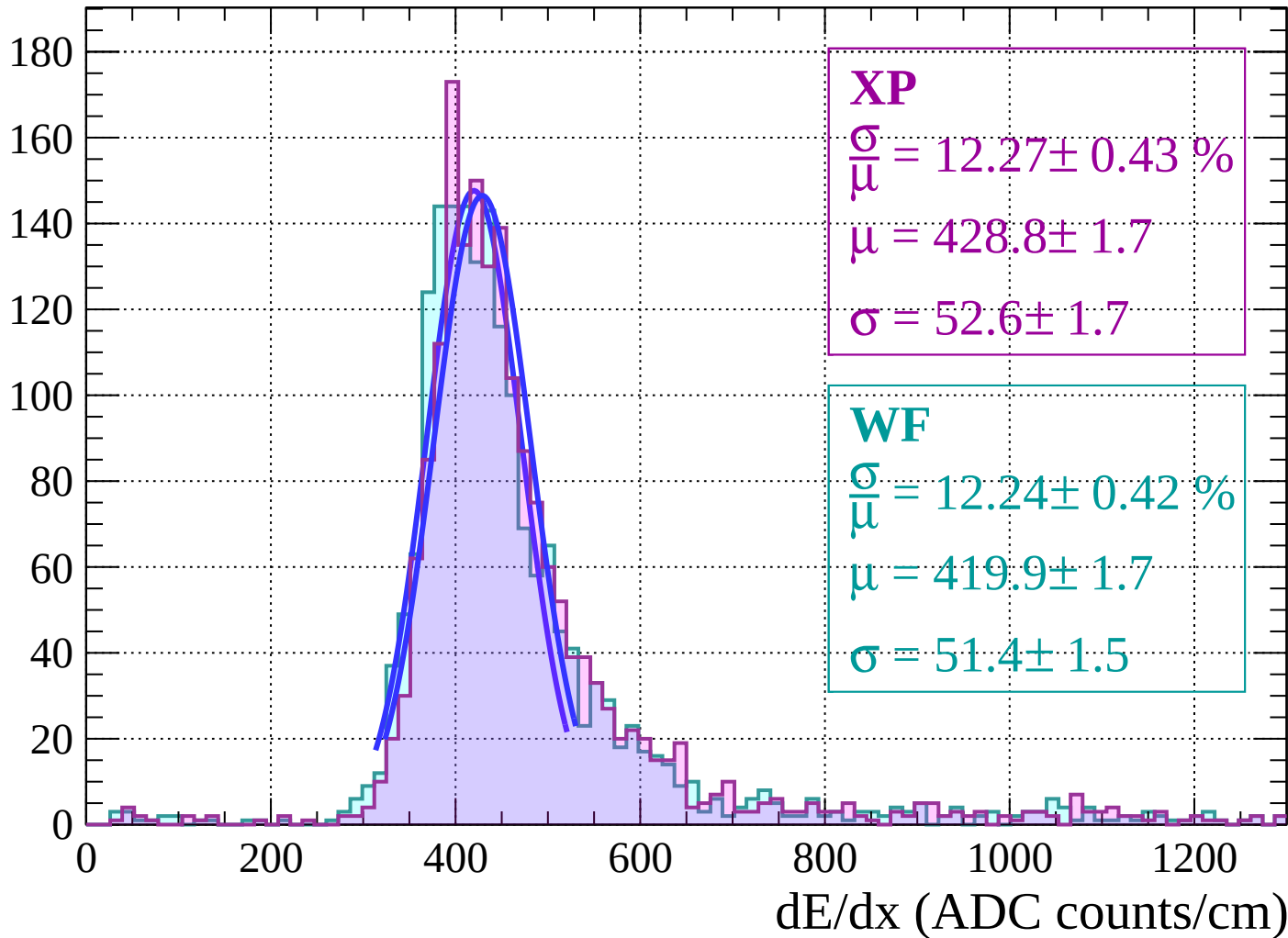


Energy loss $|-32 < \theta < -30$

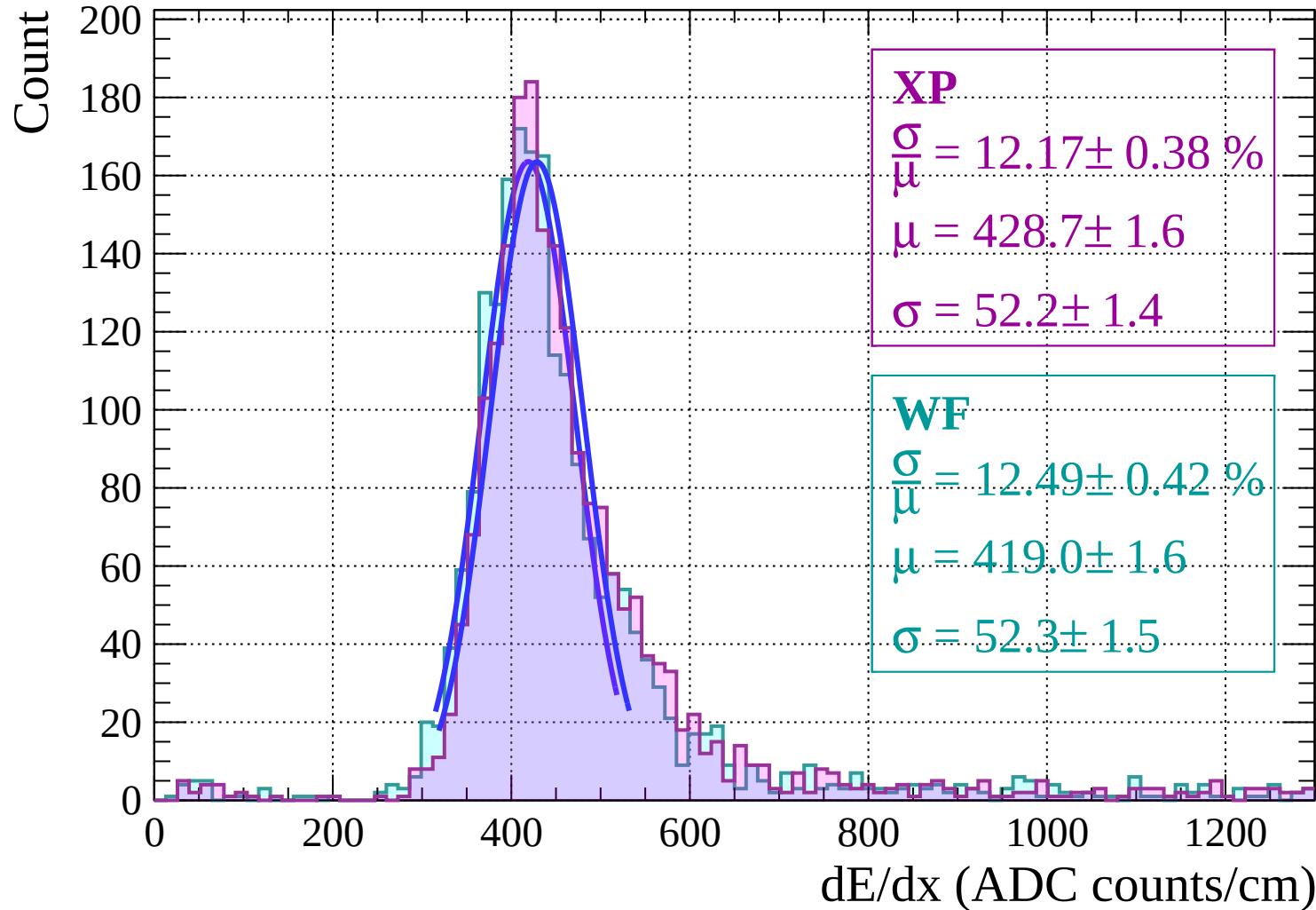


Energy loss | $-30 < \theta < -28$

Count

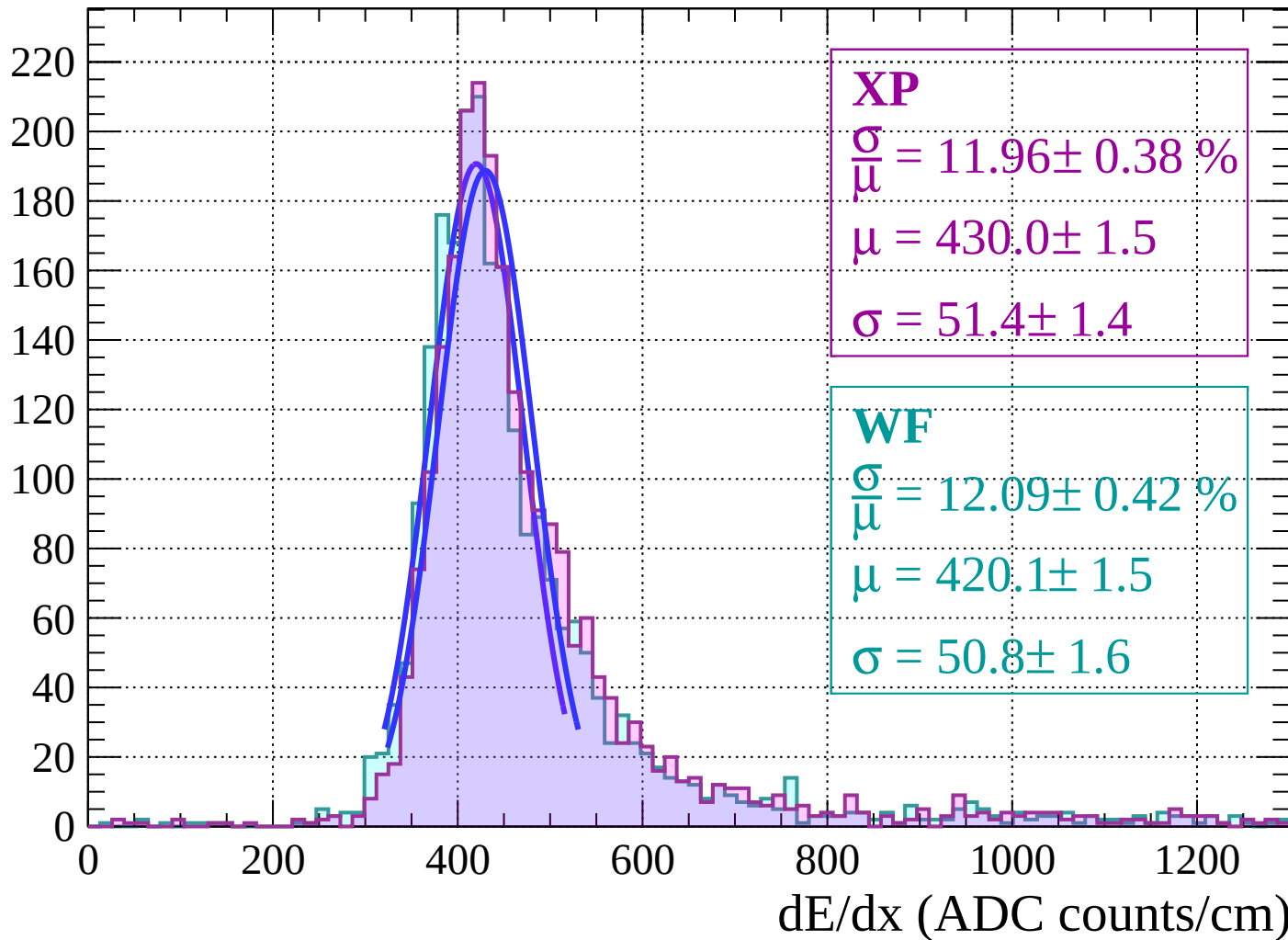


Energy loss | $-28 < \theta < -26$

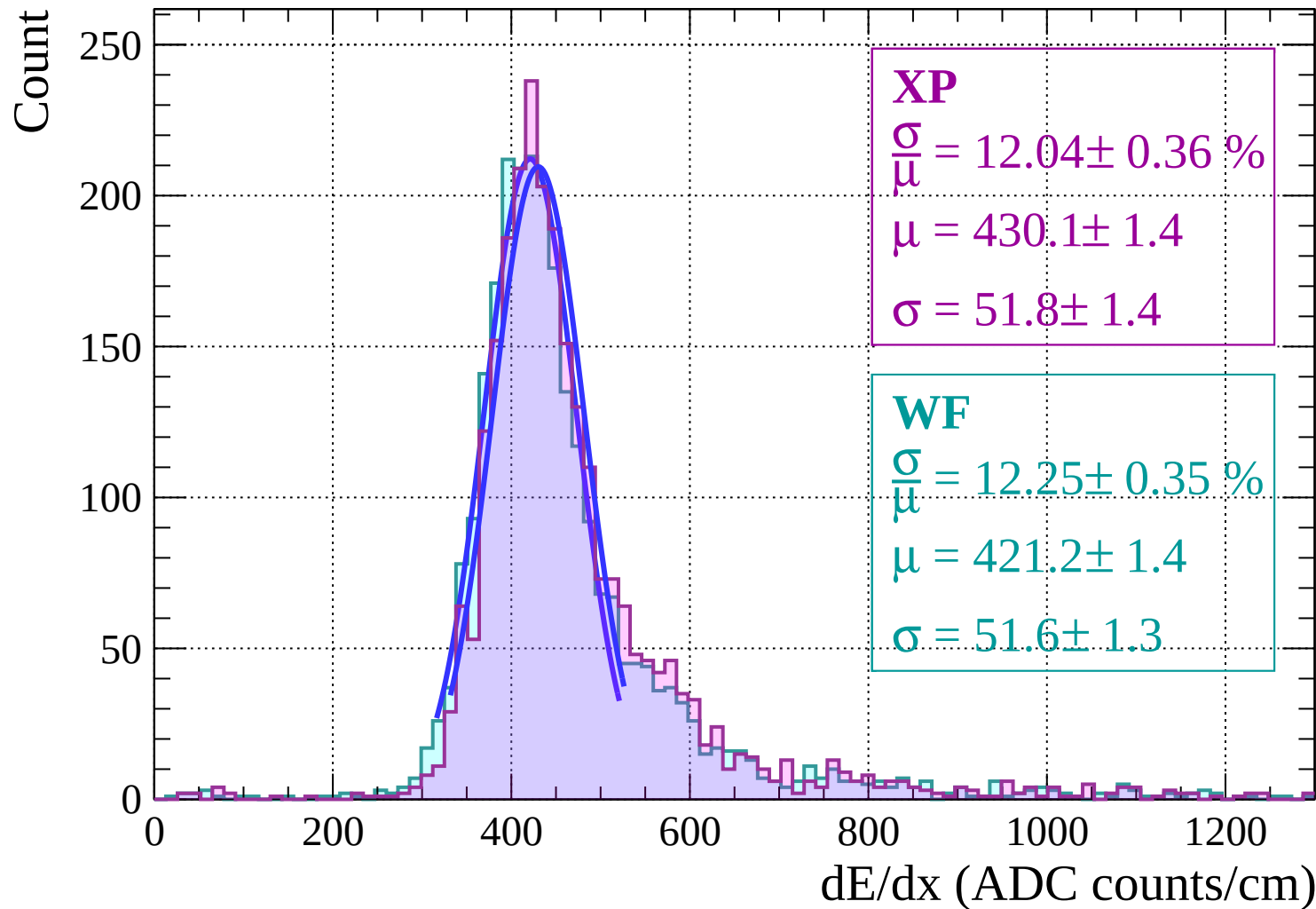


Energy loss | $-26 < \theta < -24$

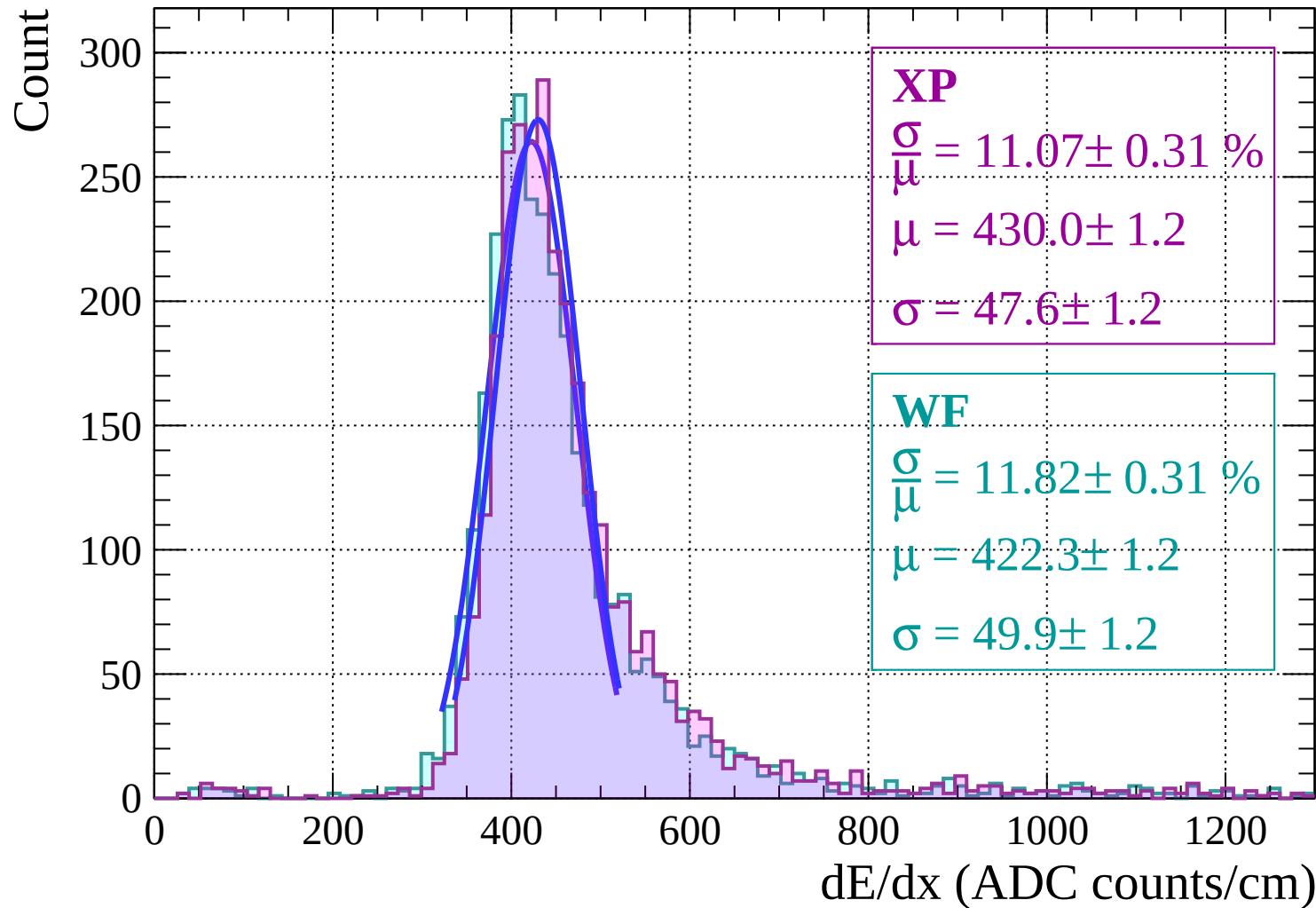
Count



Energy loss | $-24 < \theta < -22$

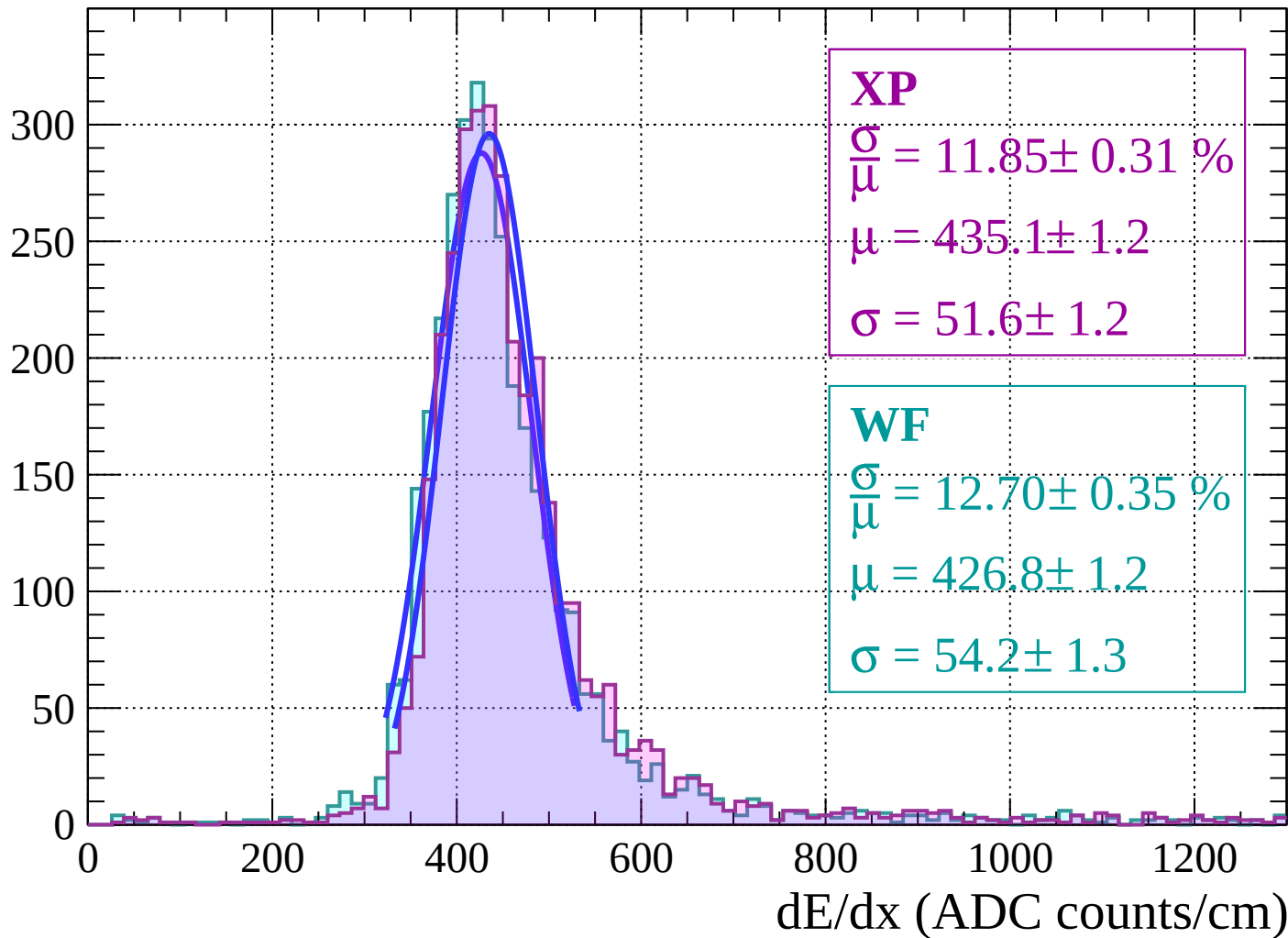


Energy loss | $-22 < \theta < -20$

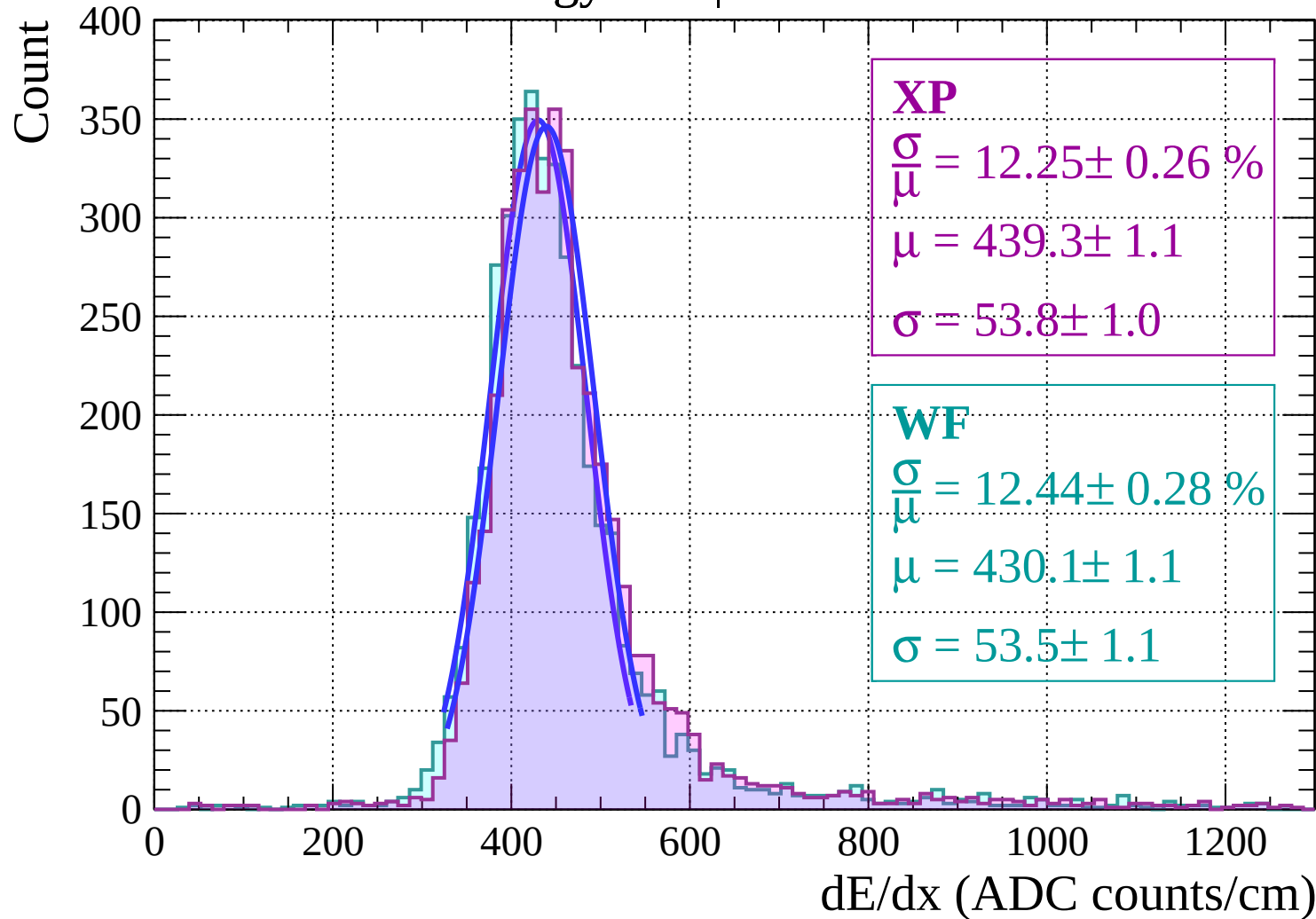


Energy loss | $-20 < \theta < -18$

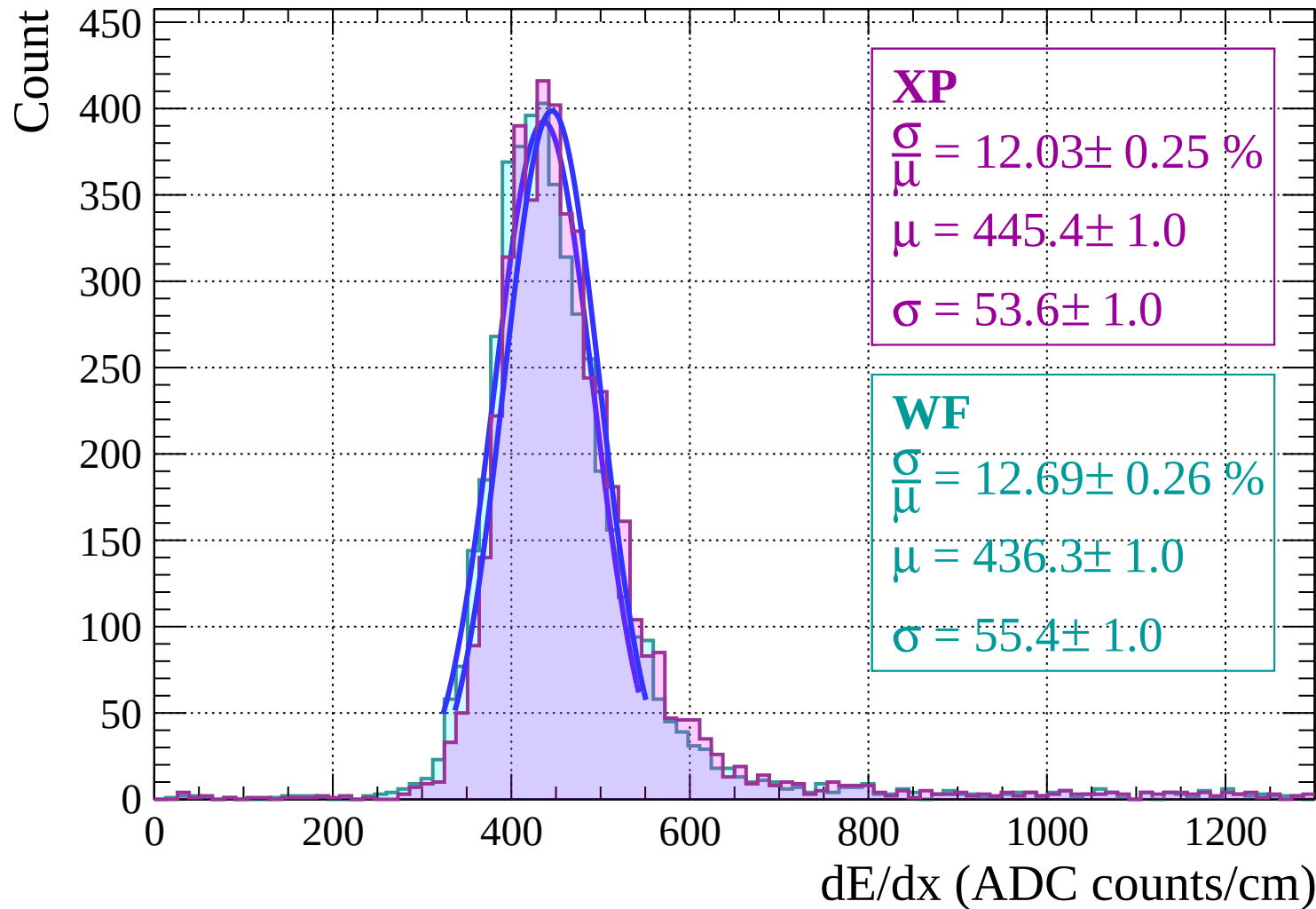
Count



Energy loss | $-18 < \theta < -16$

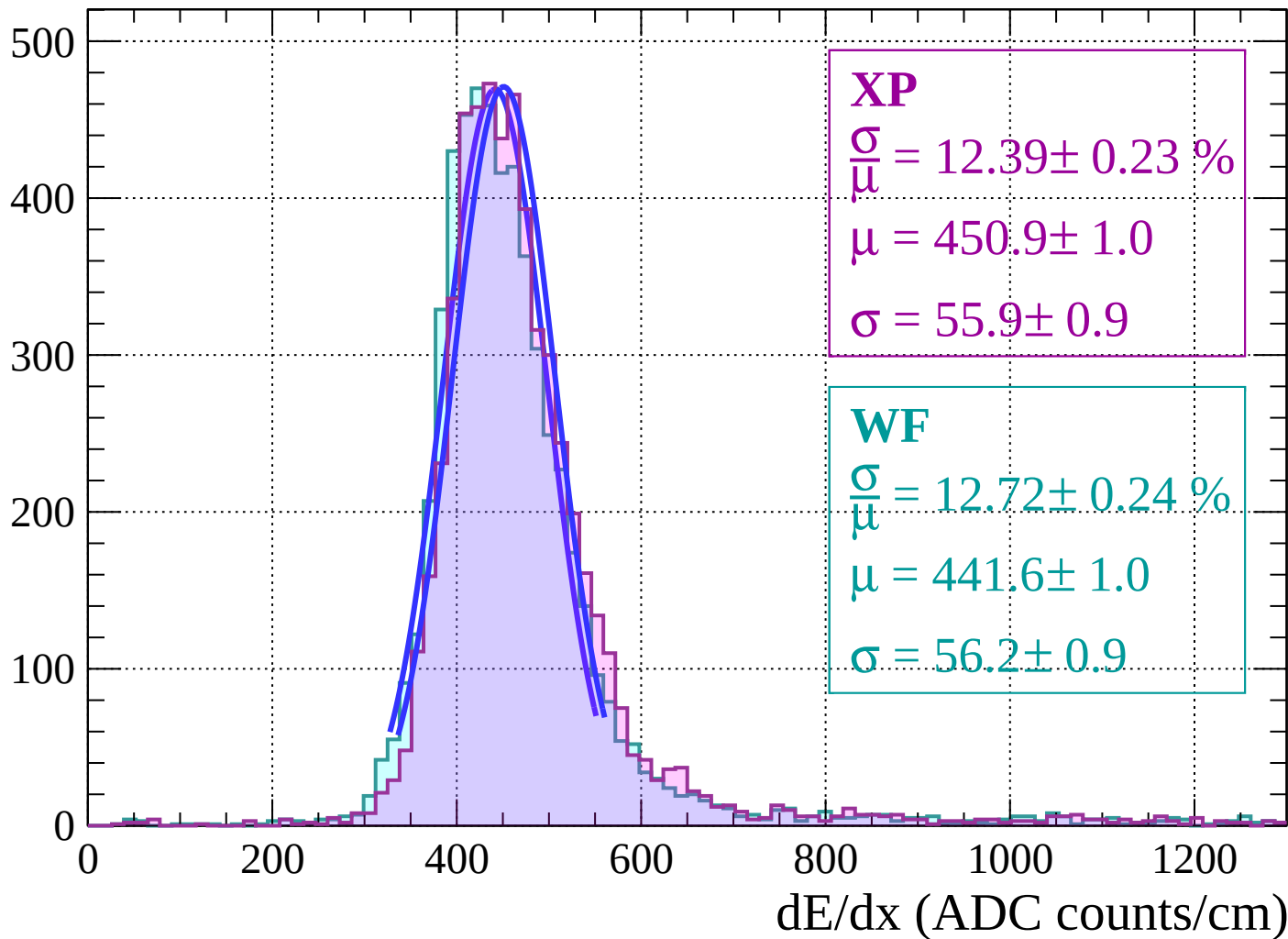


Energy loss | $-16 < \theta < -14$

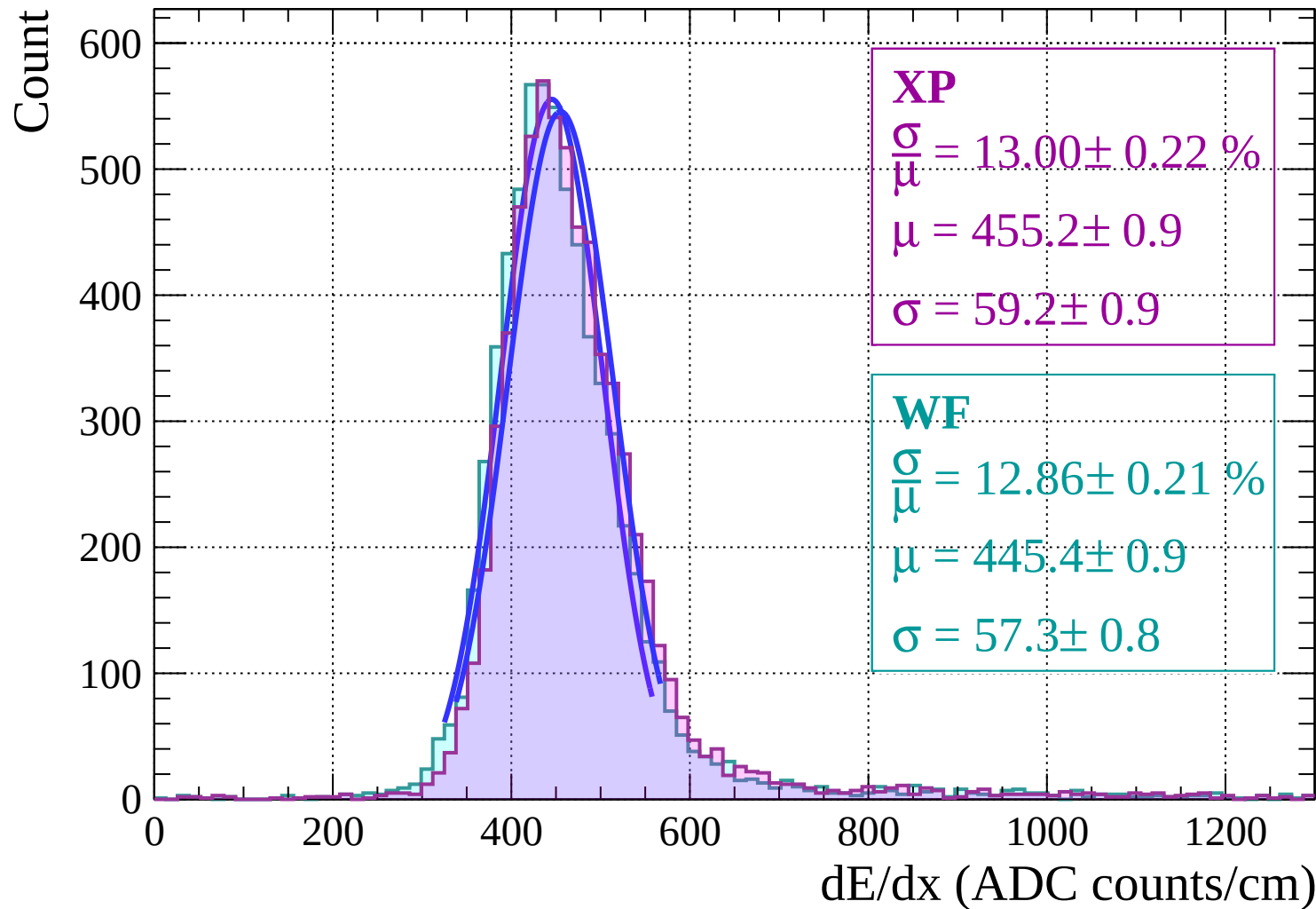


Energy loss $|-14 < \theta < -12$

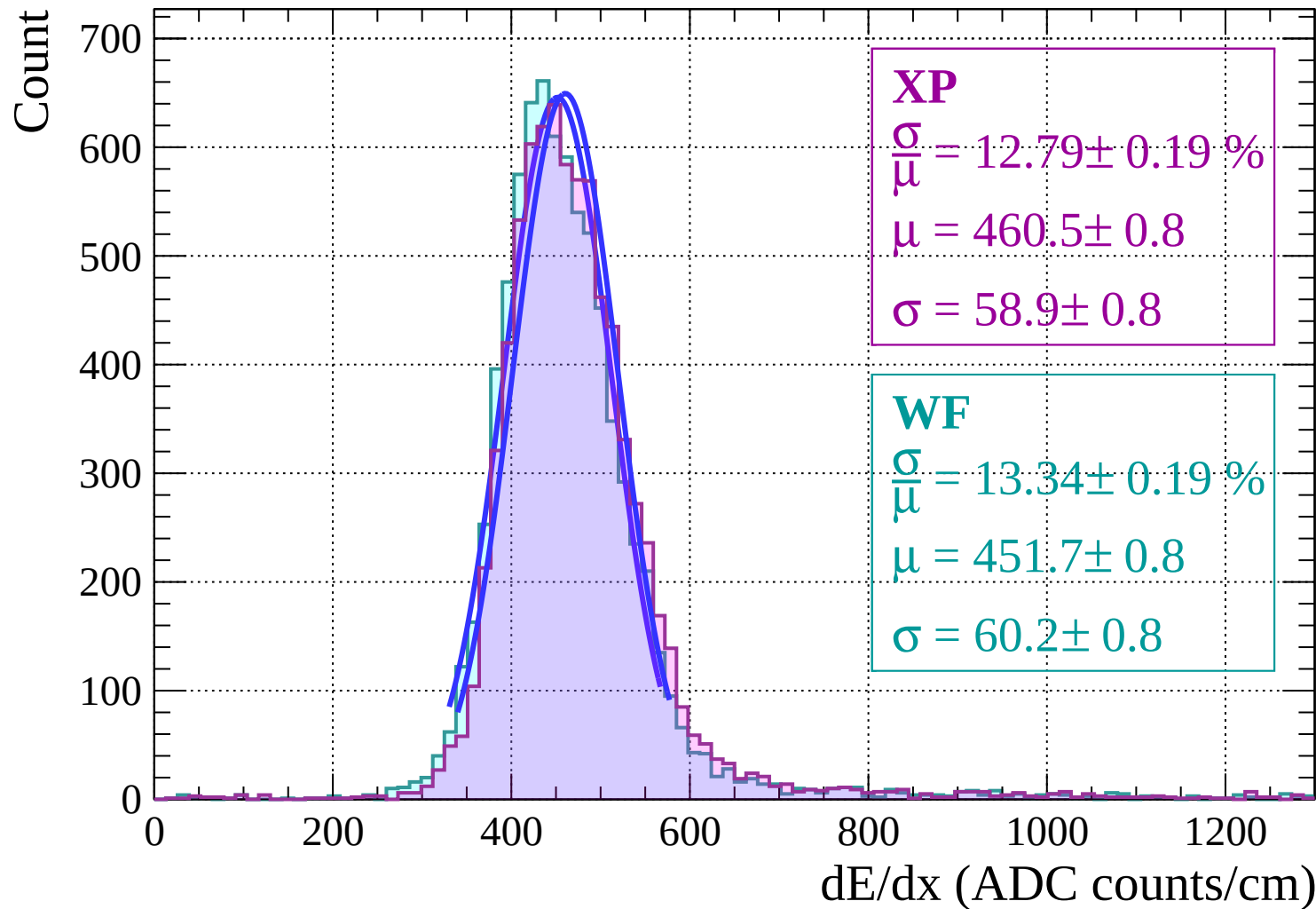
Count



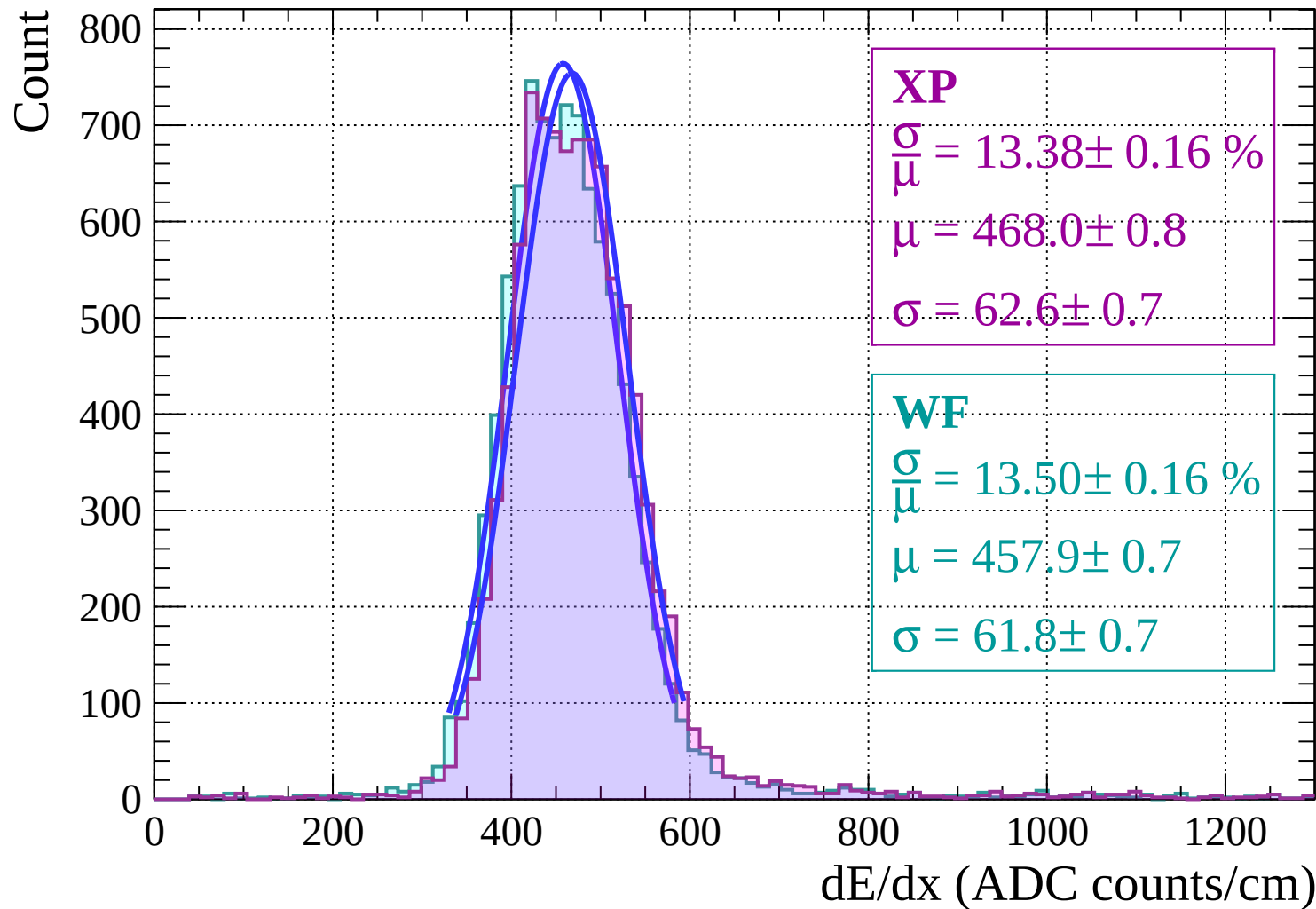
Energy loss | $-12 < \theta < -10$



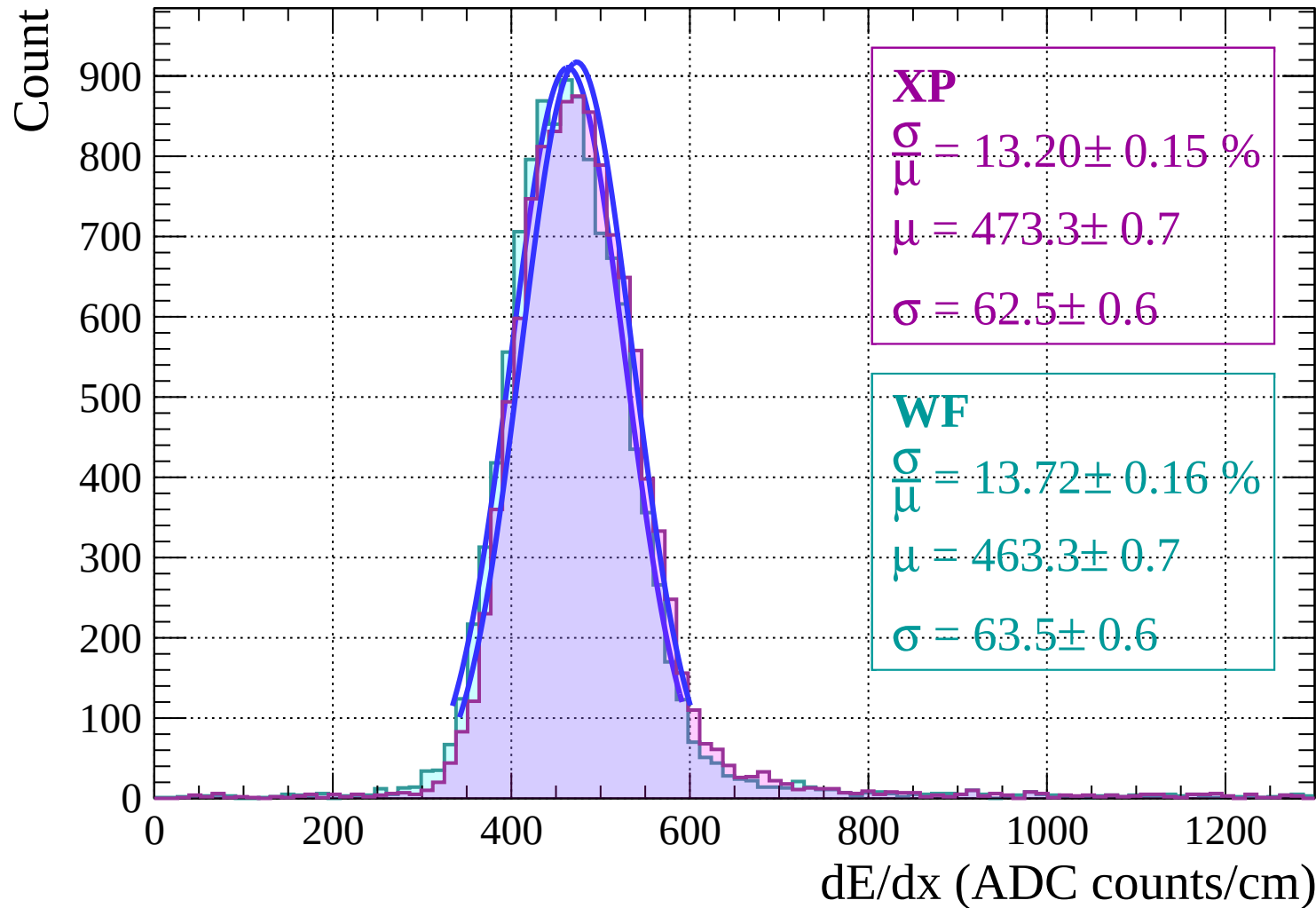
Energy loss | $-10 < \theta < -8$



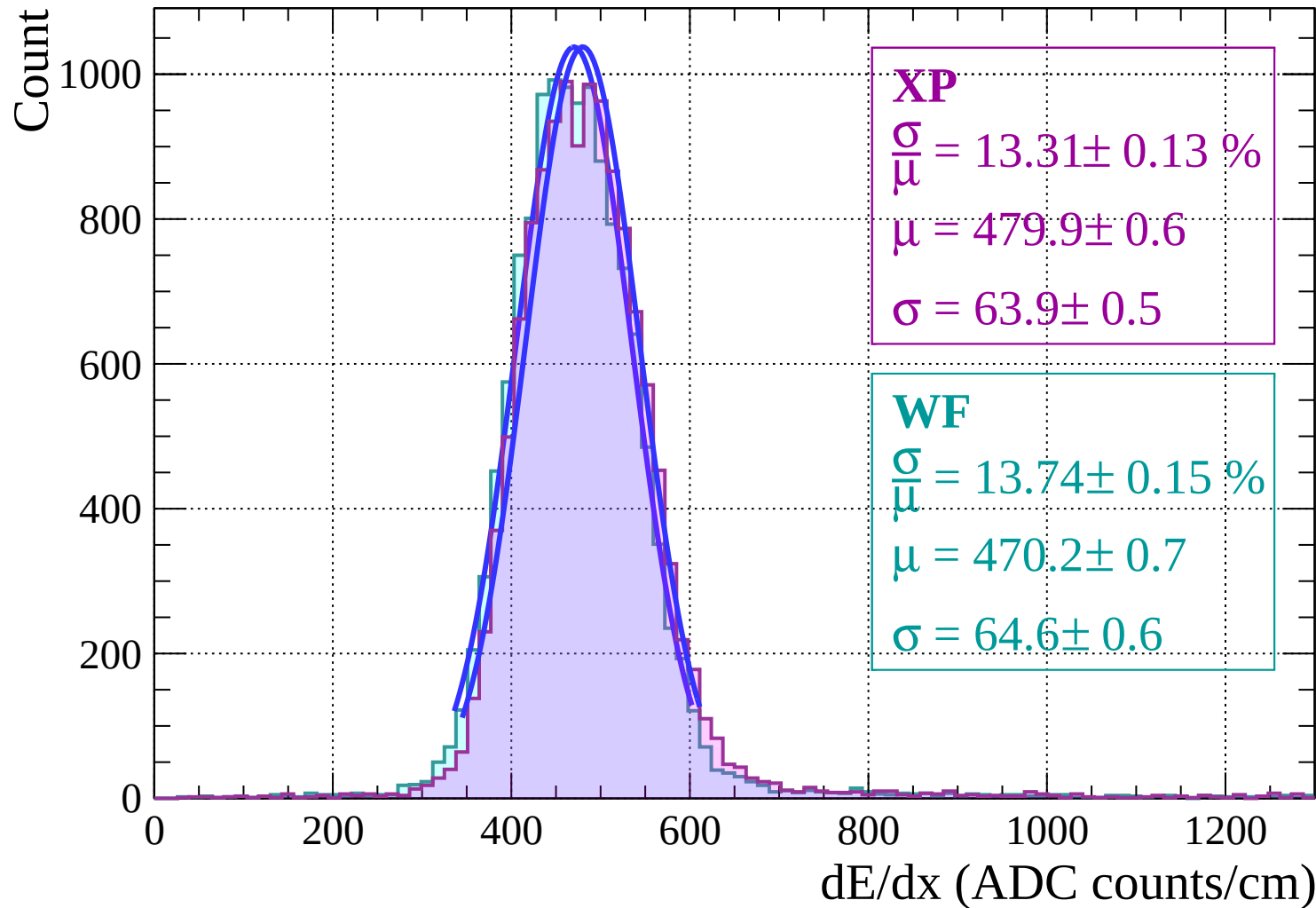
Energy loss | $-8 < \theta < -6$



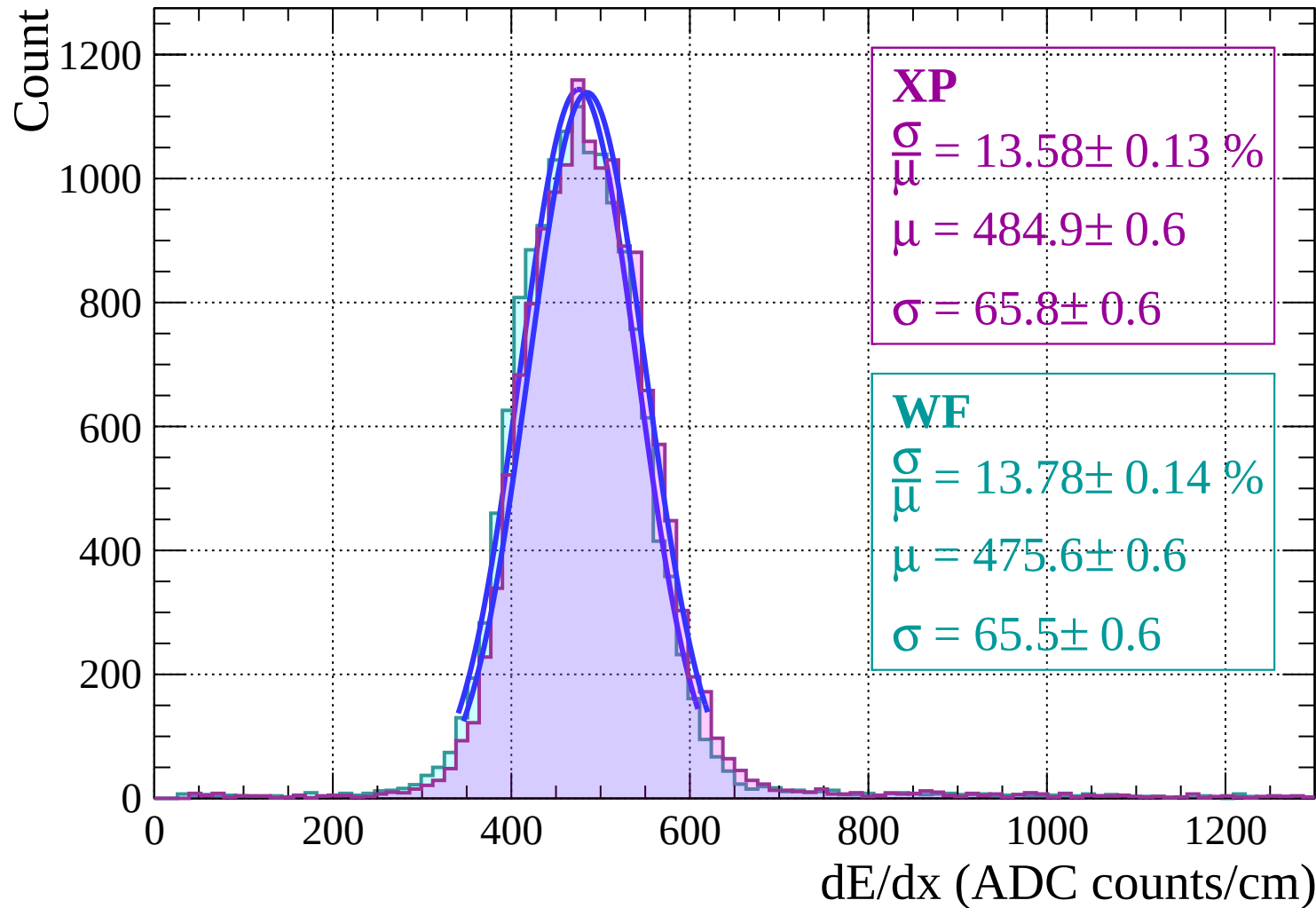
Energy loss | $-6 < \theta < -4$



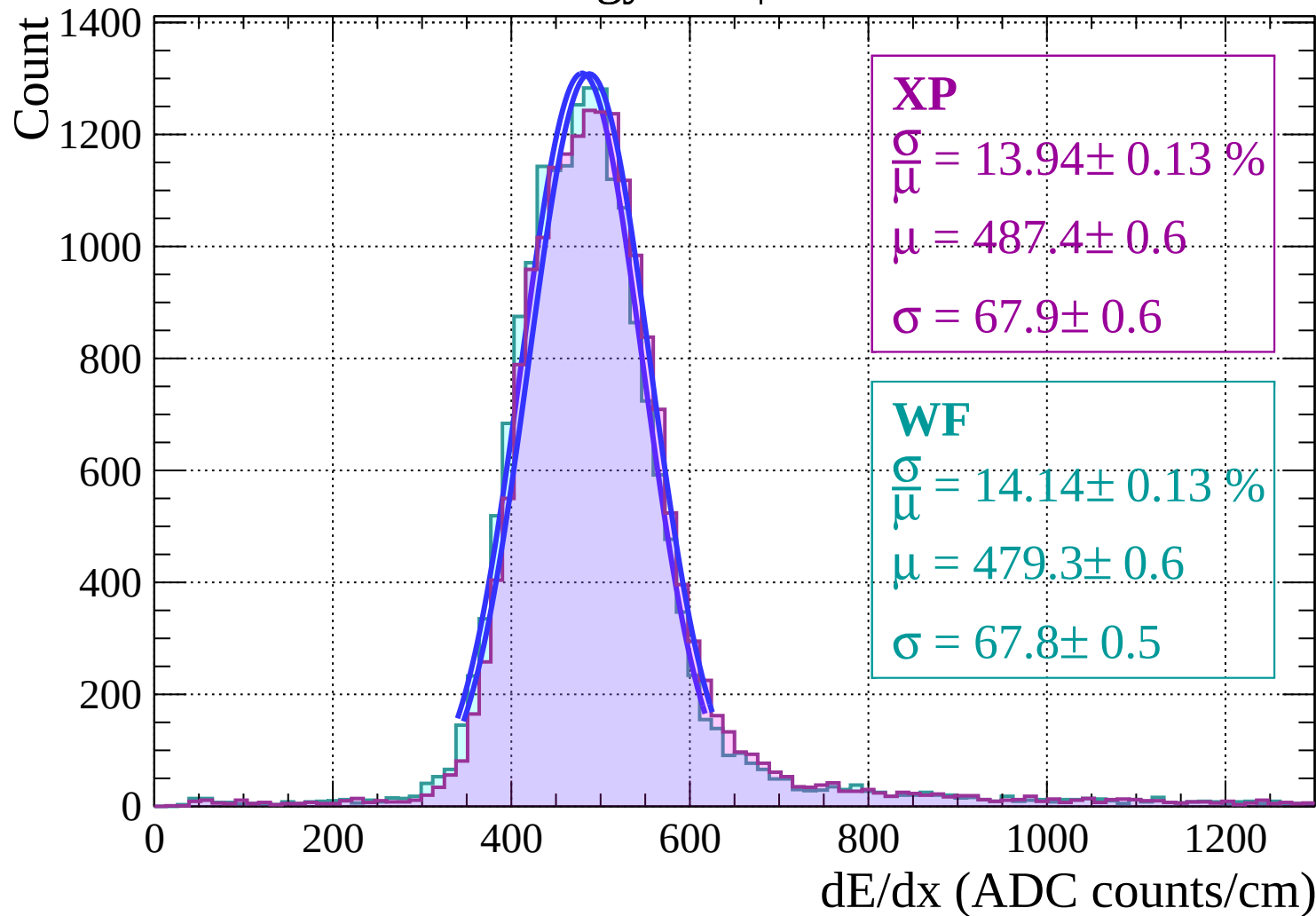
Energy loss | $-4 < \theta < -2$



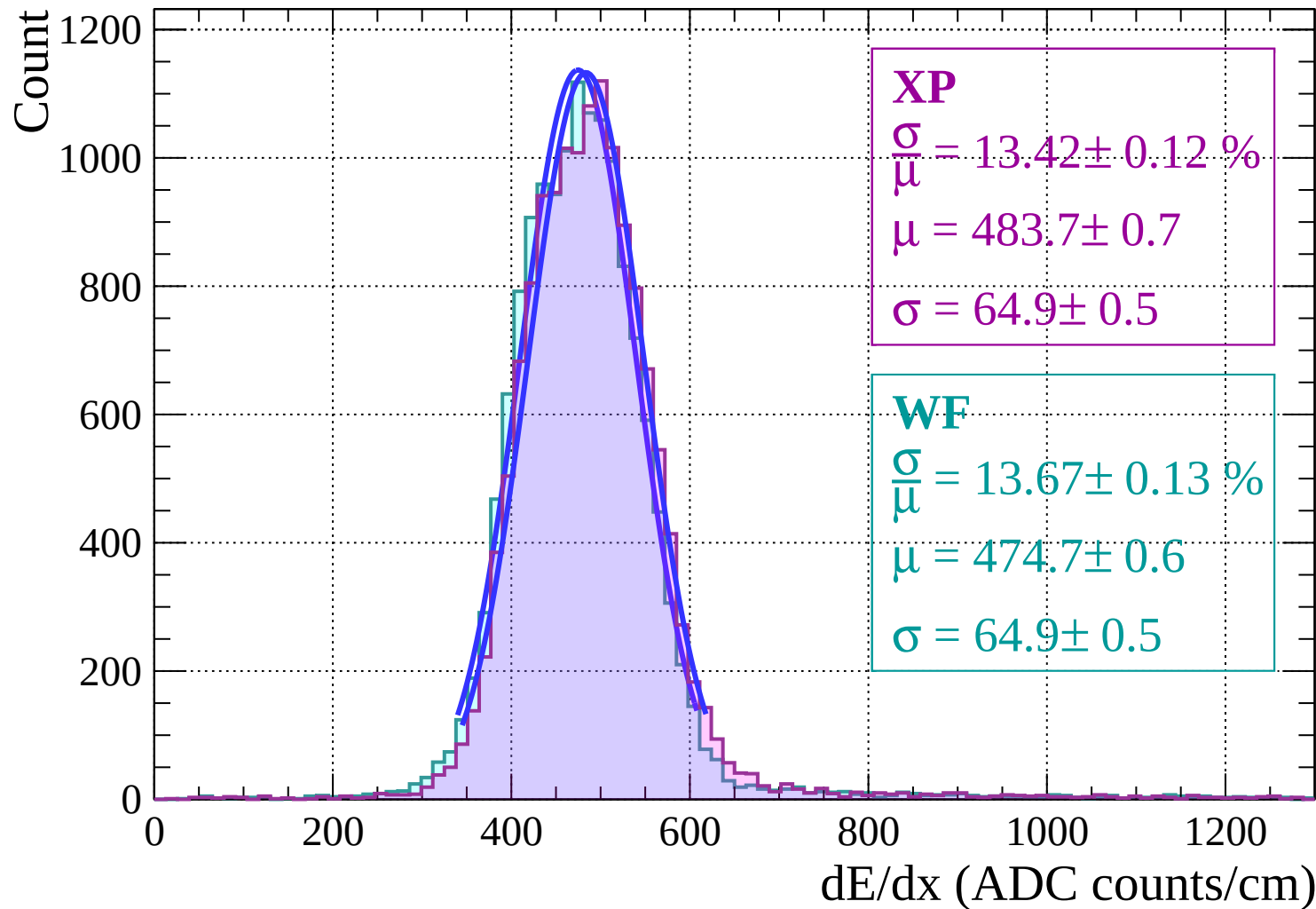
Energy loss | $-2 < \theta < 0$



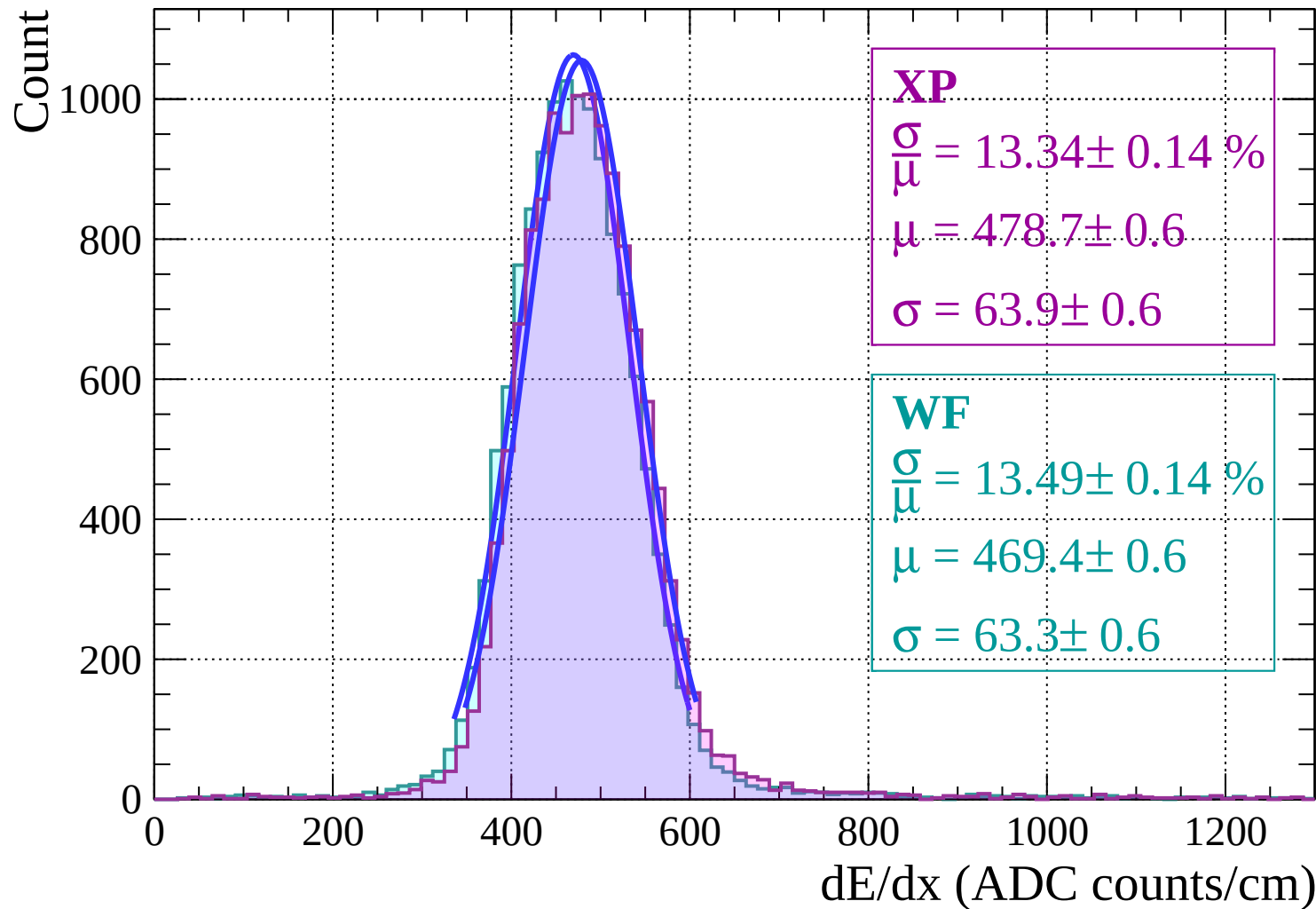
Energy loss | $0 < \theta < 2$



Energy loss | $2 < \theta < 4$

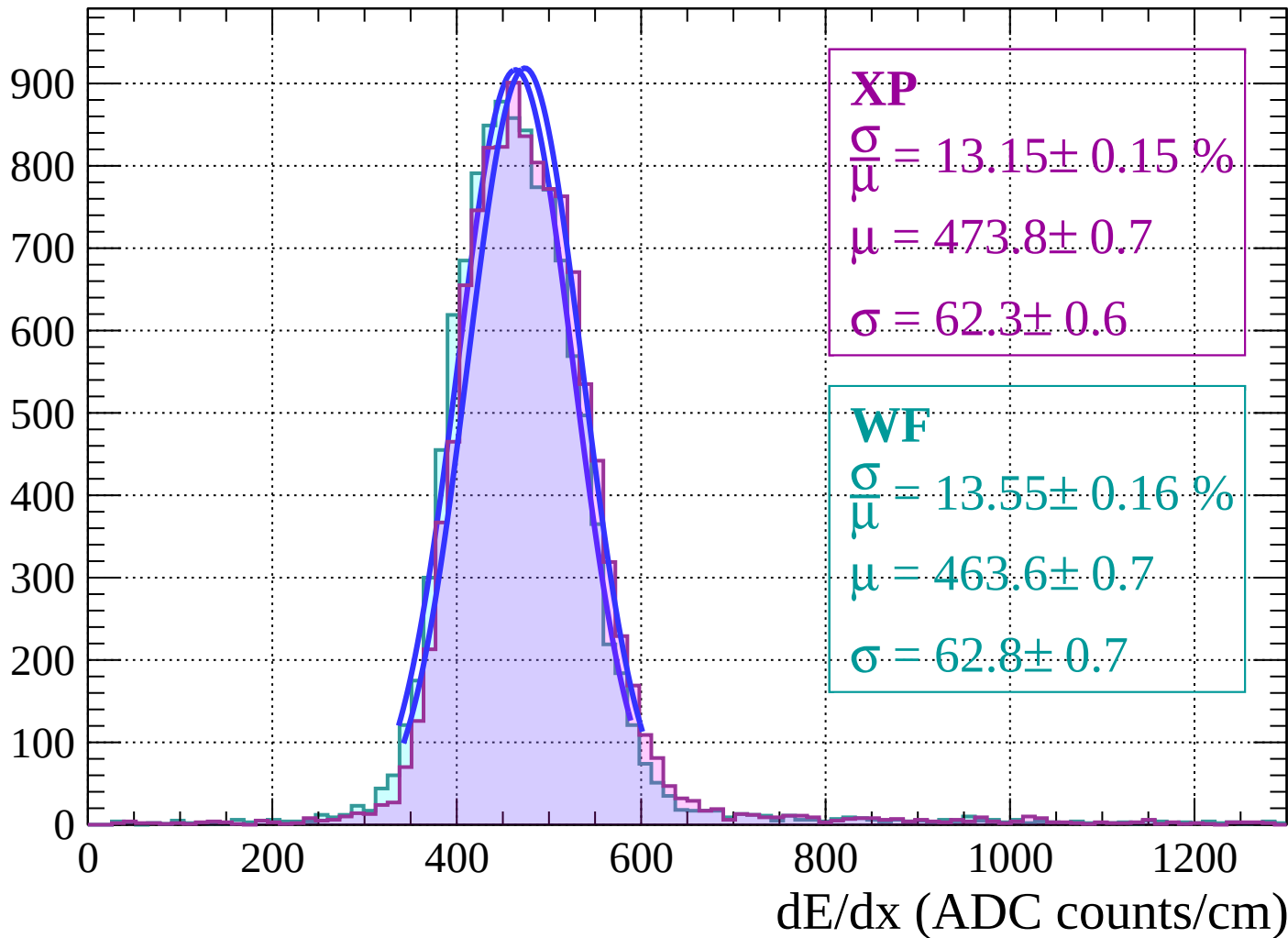


Energy loss | $4 < \theta < 6$

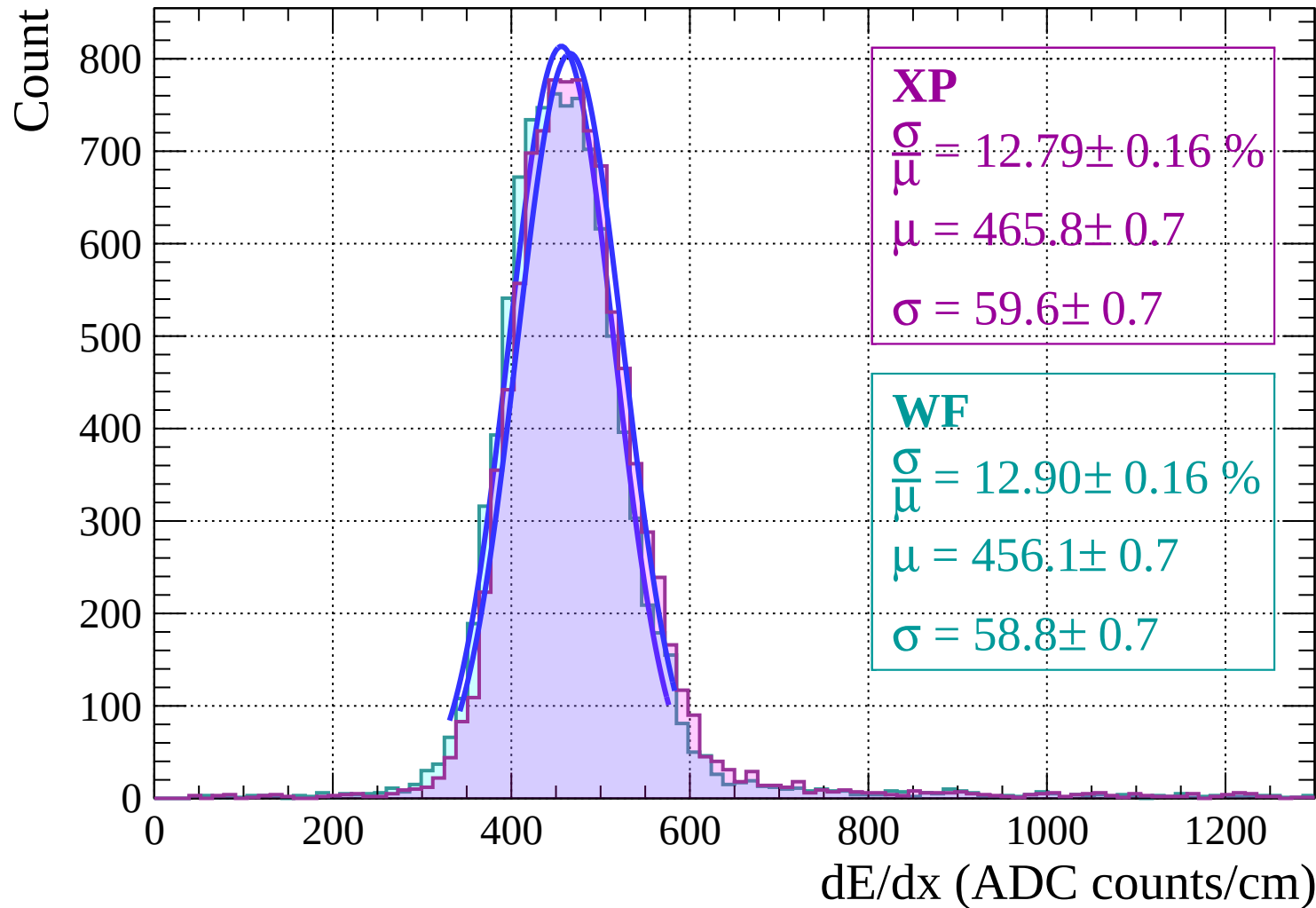


Energy loss | $6 < \theta < 8$

Count

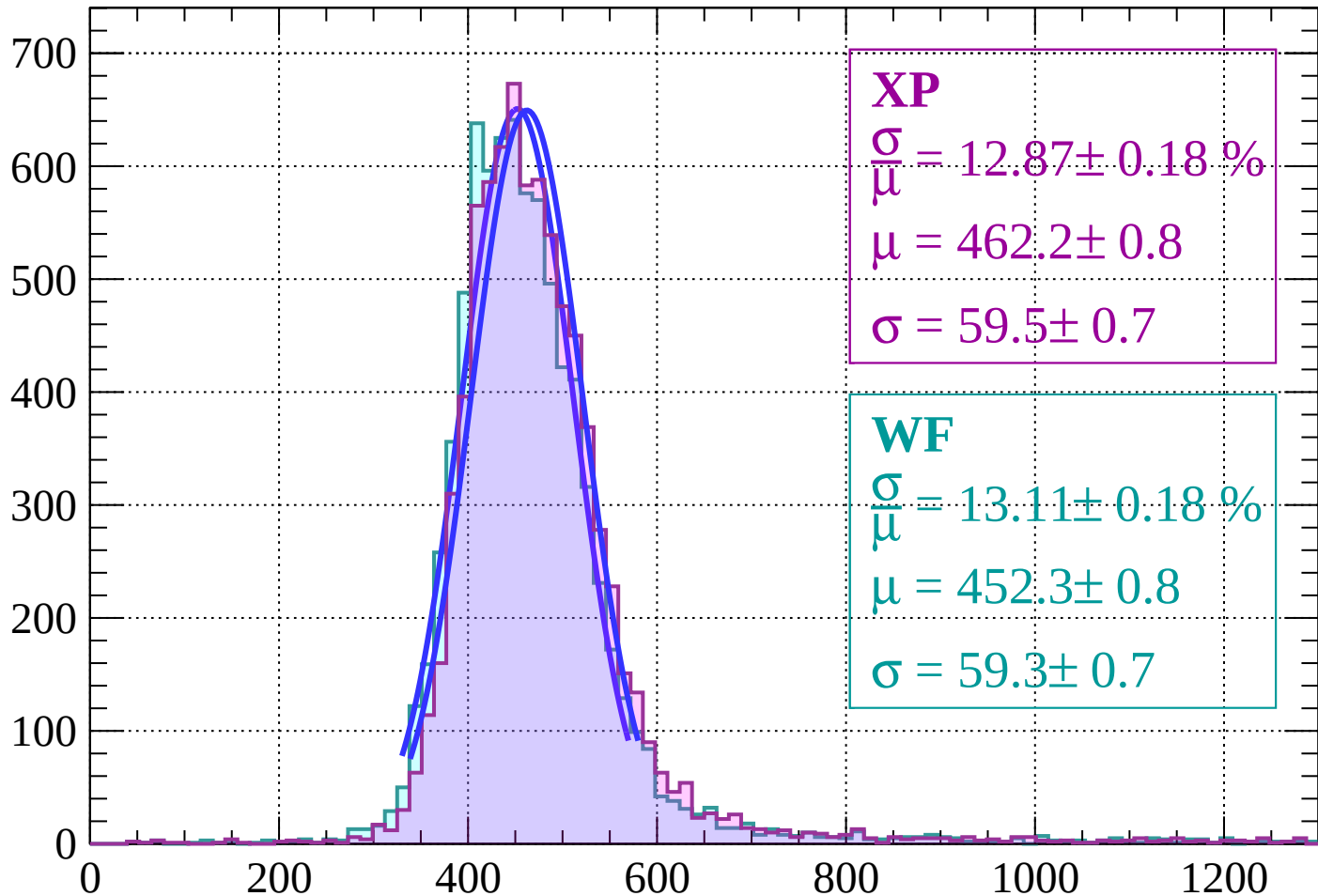


Energy loss | $8 < \theta < 10$



Energy loss | $10 < \theta < 12$

Count



XP

$$\frac{\sigma}{\mu} = 12.87 \pm 0.18 \%$$

$$\mu = 462.2 \pm 0.8$$

$$\sigma = 59.5 \pm 0.7$$

WF

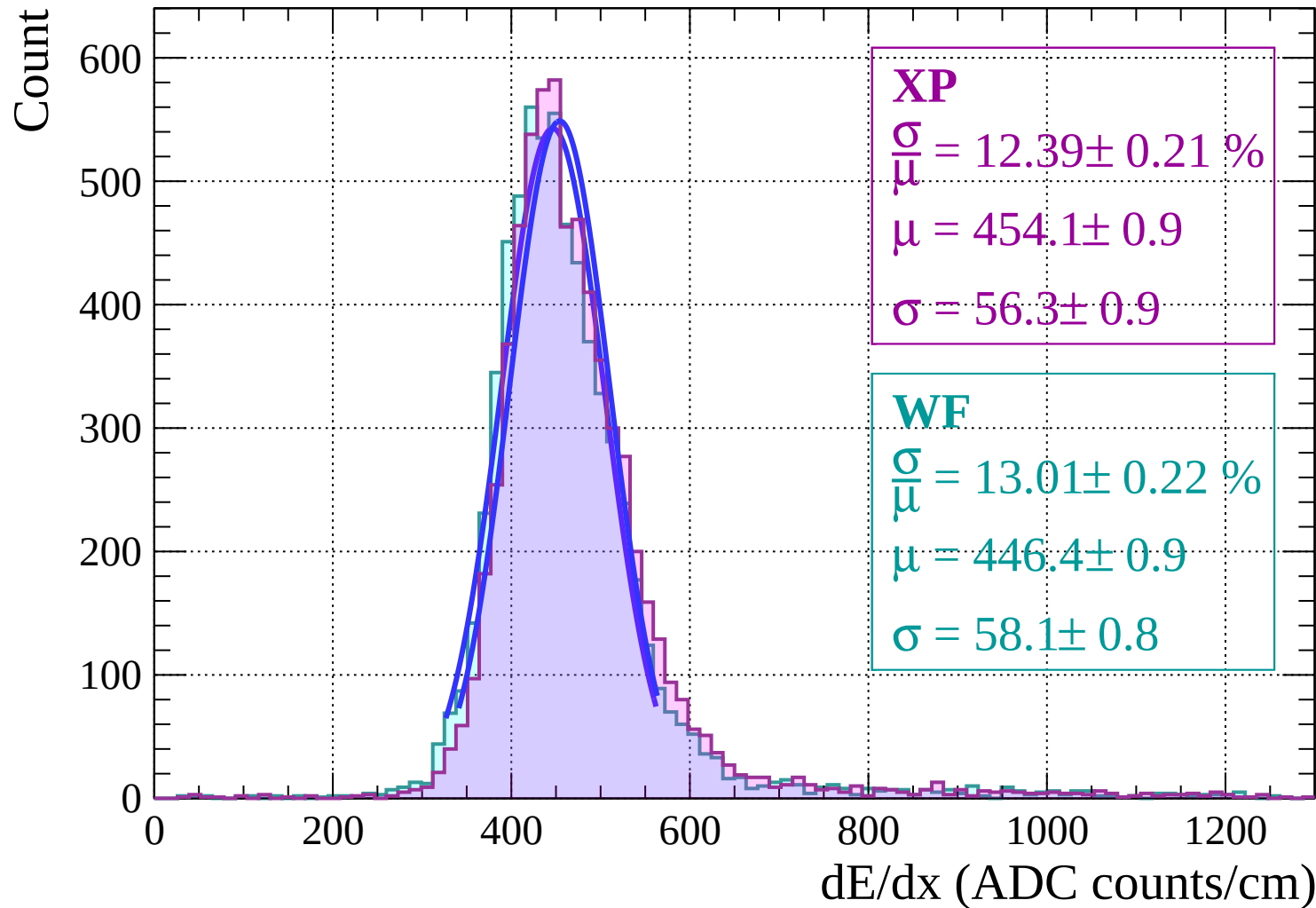
$$\frac{\sigma}{\mu} = 13.11 \pm 0.18 \%$$

$$\mu = 452.3 \pm 0.8$$

$$\sigma = 59.3 \pm 0.7$$

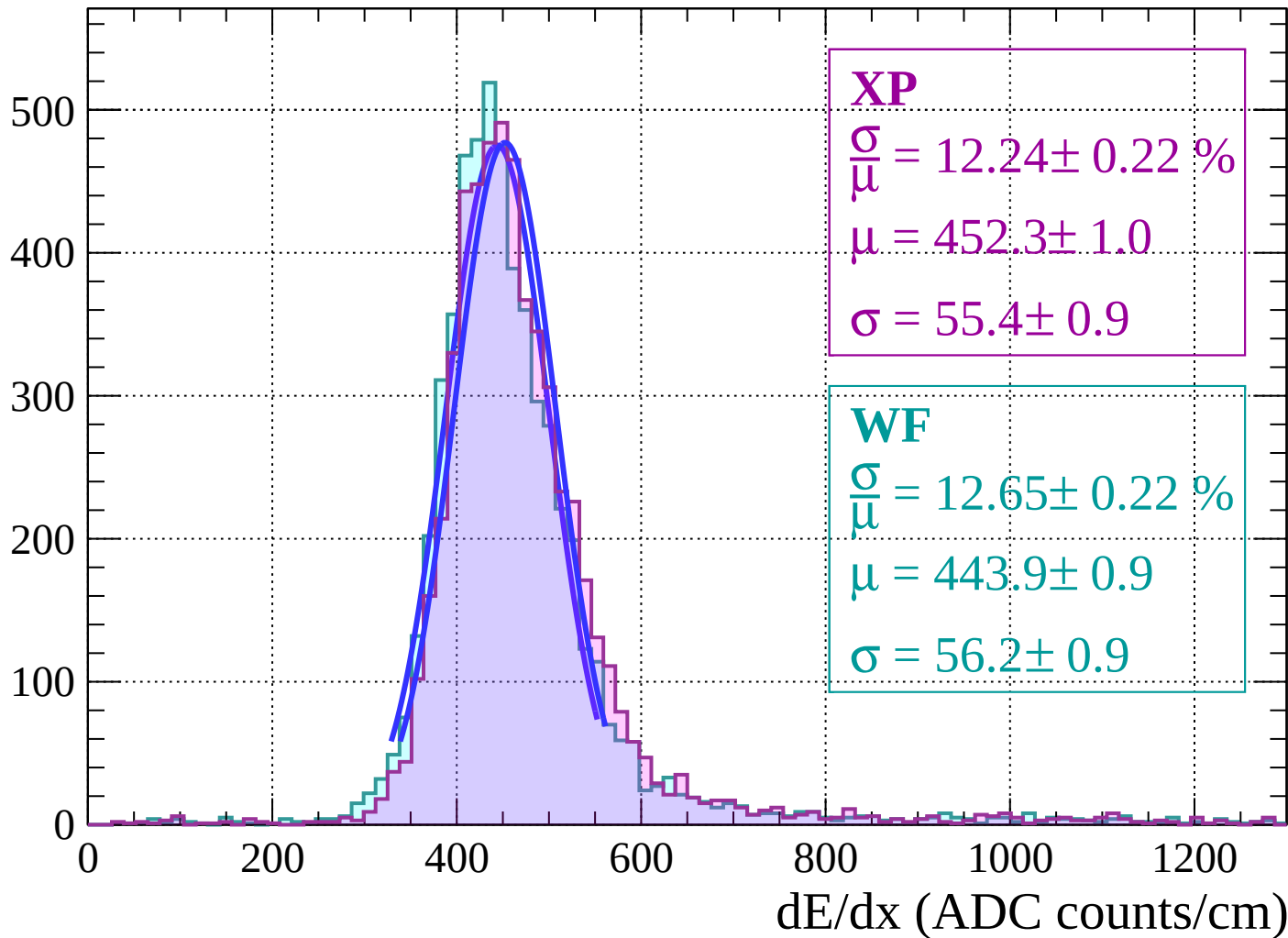
dE/dx (ADC counts/cm)

Energy loss | $12 < \theta < 14$

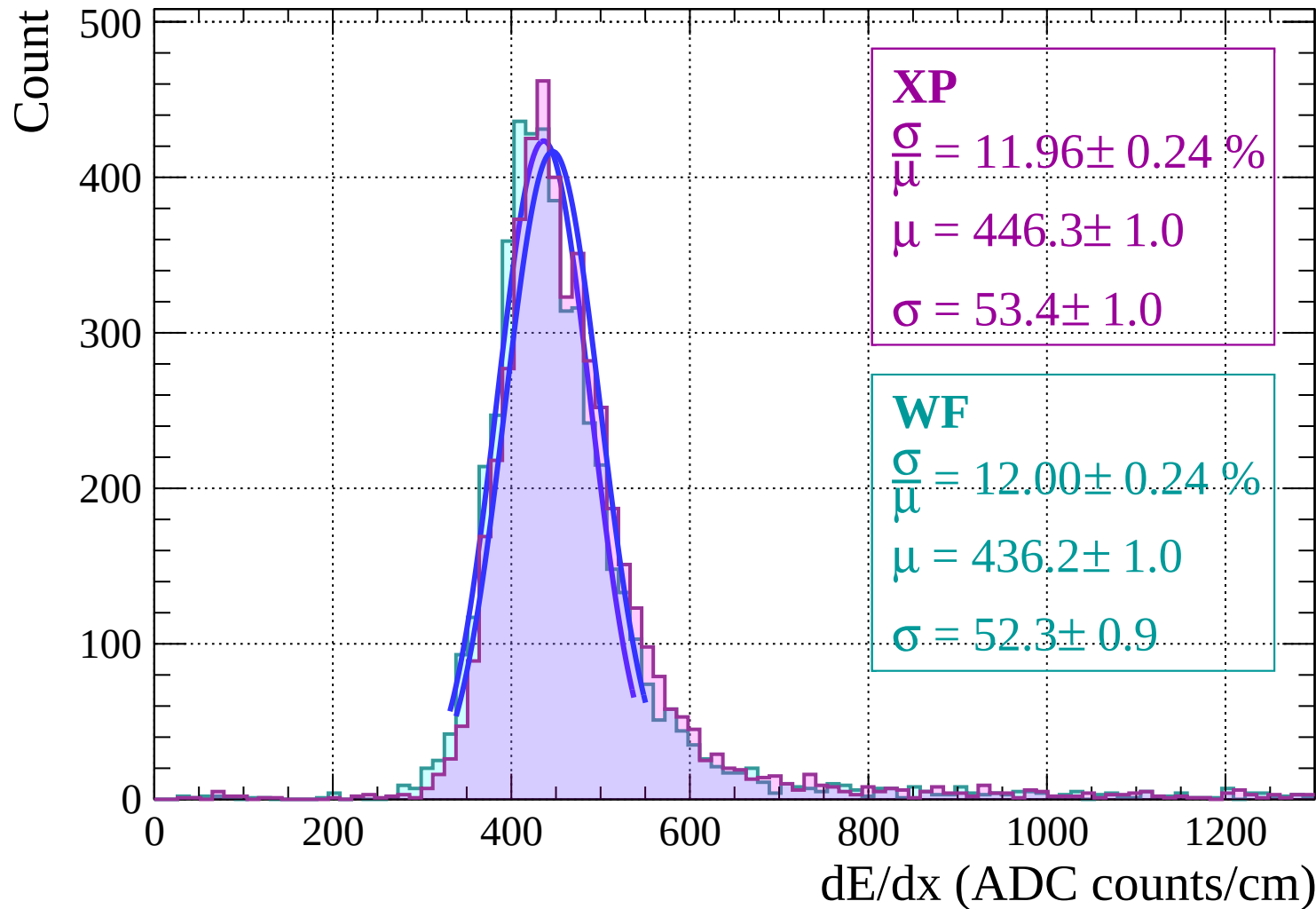


Energy loss | $14 < \theta < 16$

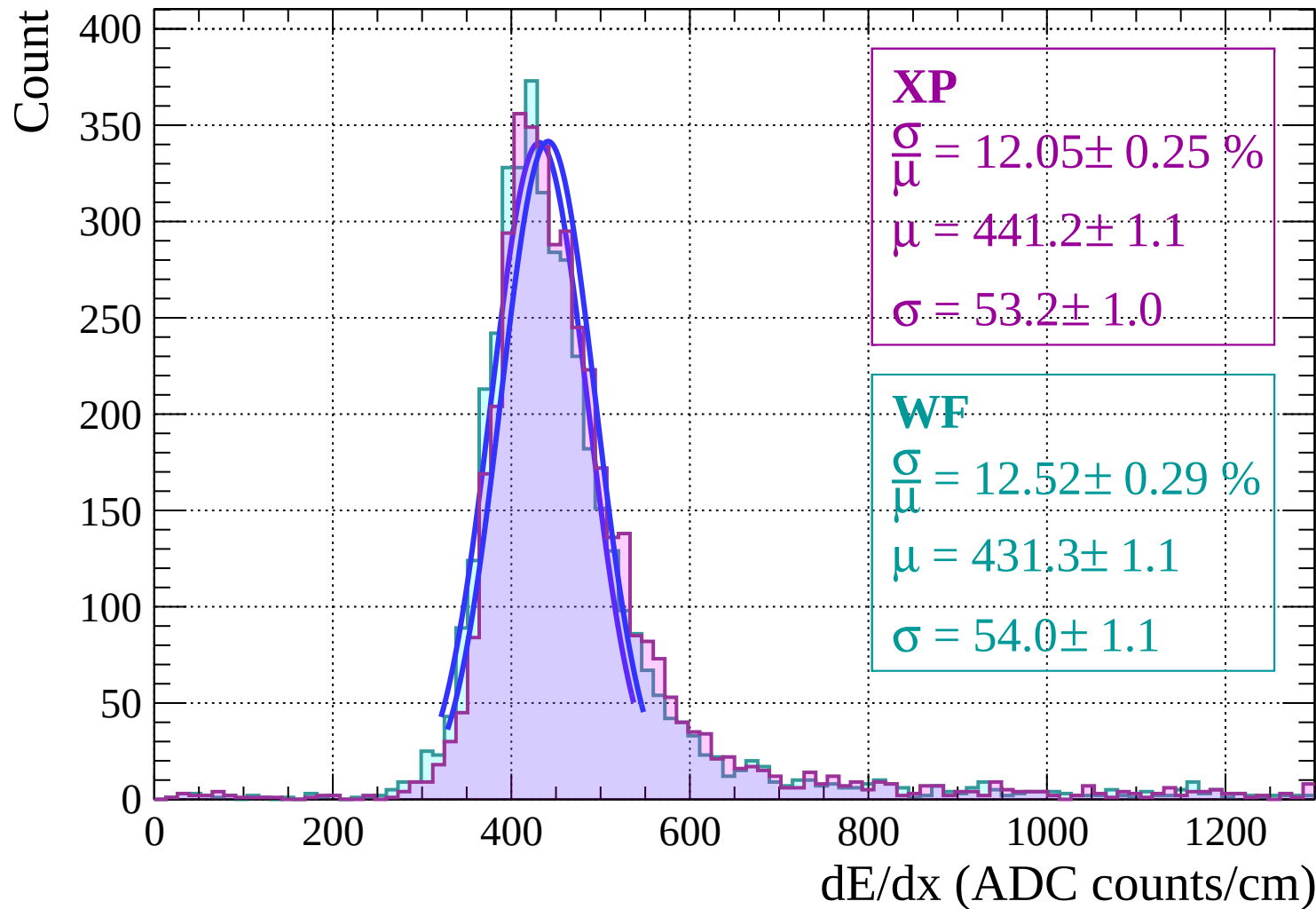
Count



Energy loss | $16 < \theta < 18$

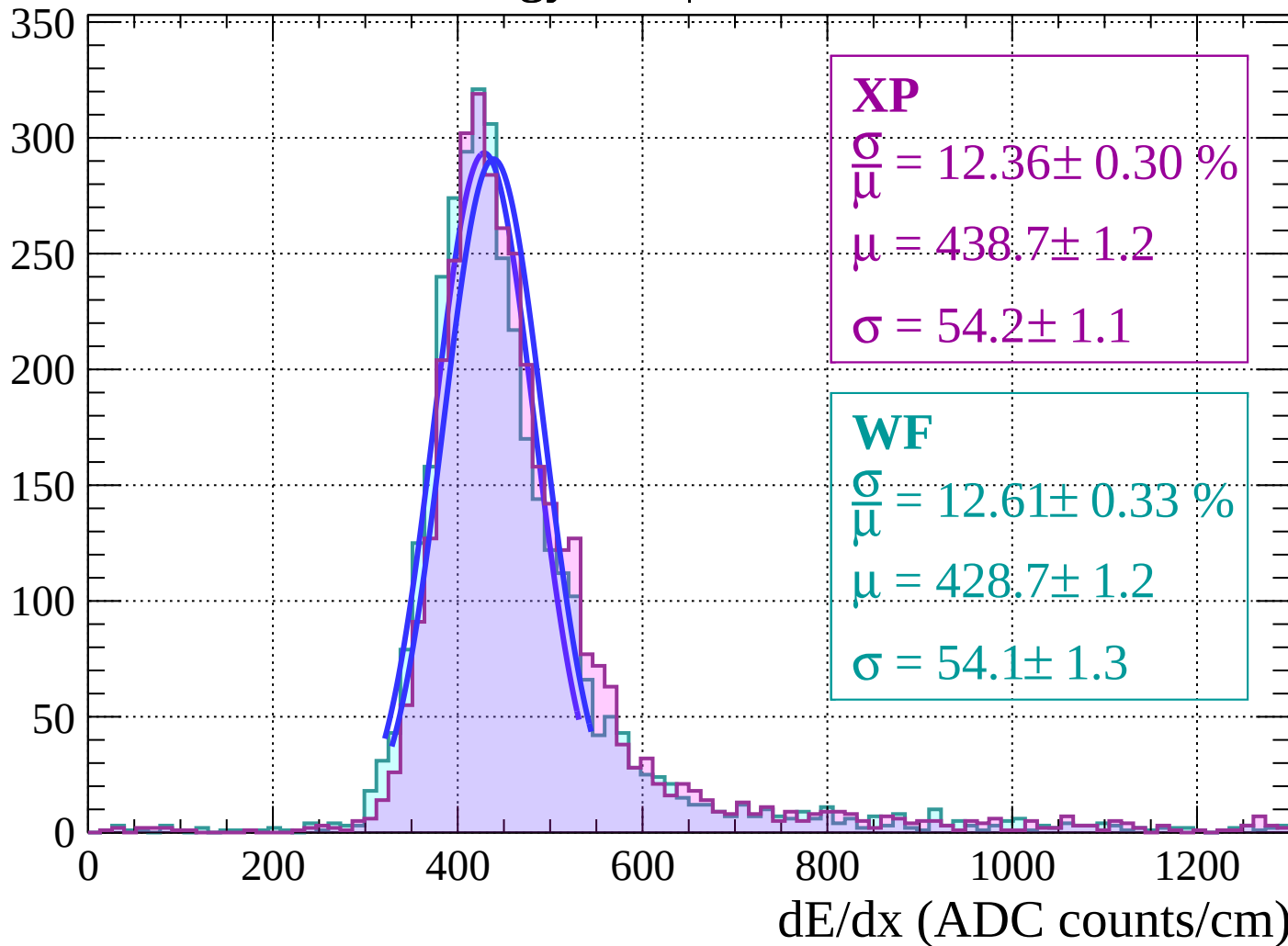


Energy loss | $18 < \theta < 20$

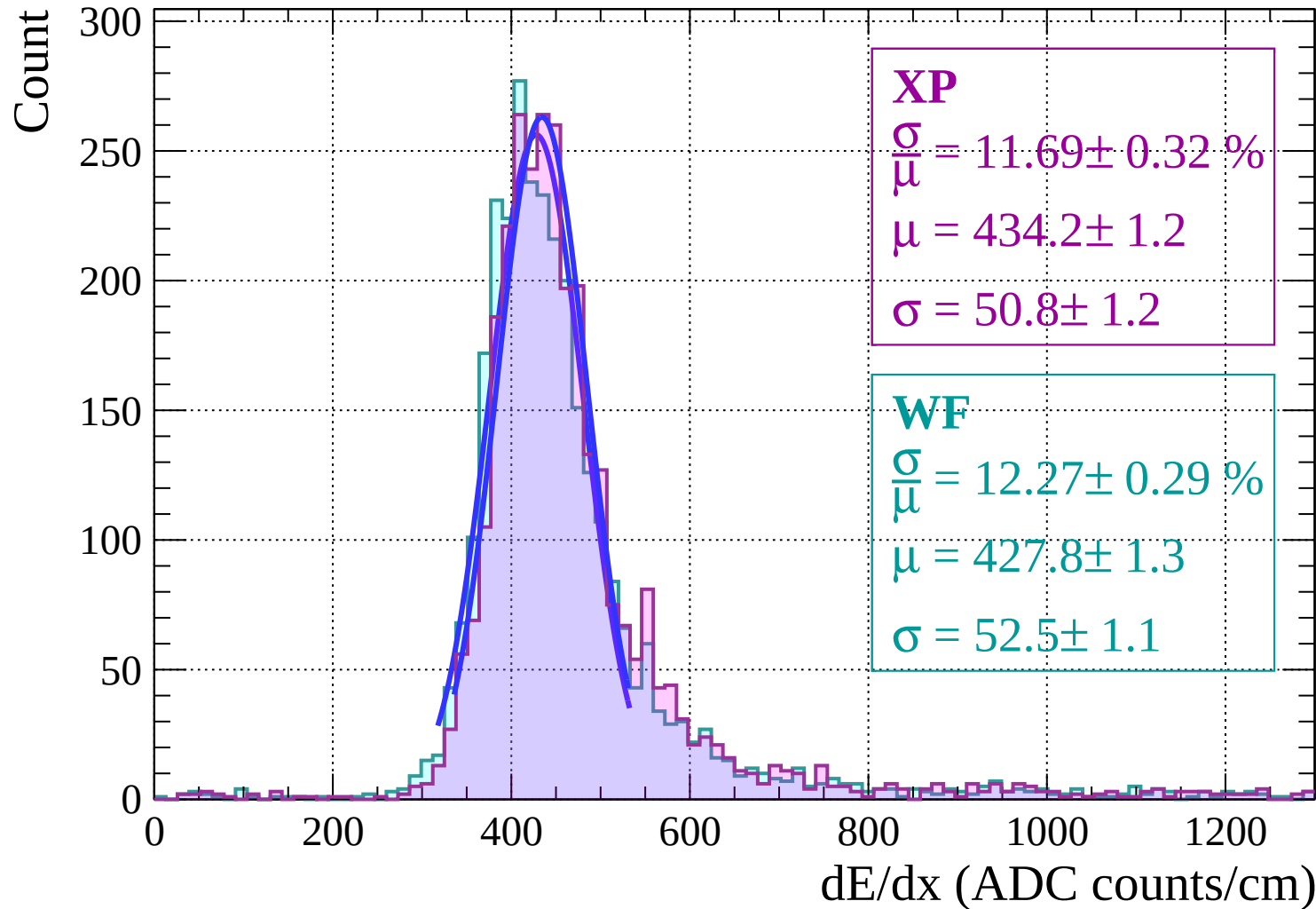


Energy loss | $20 < \theta < 22$

Count

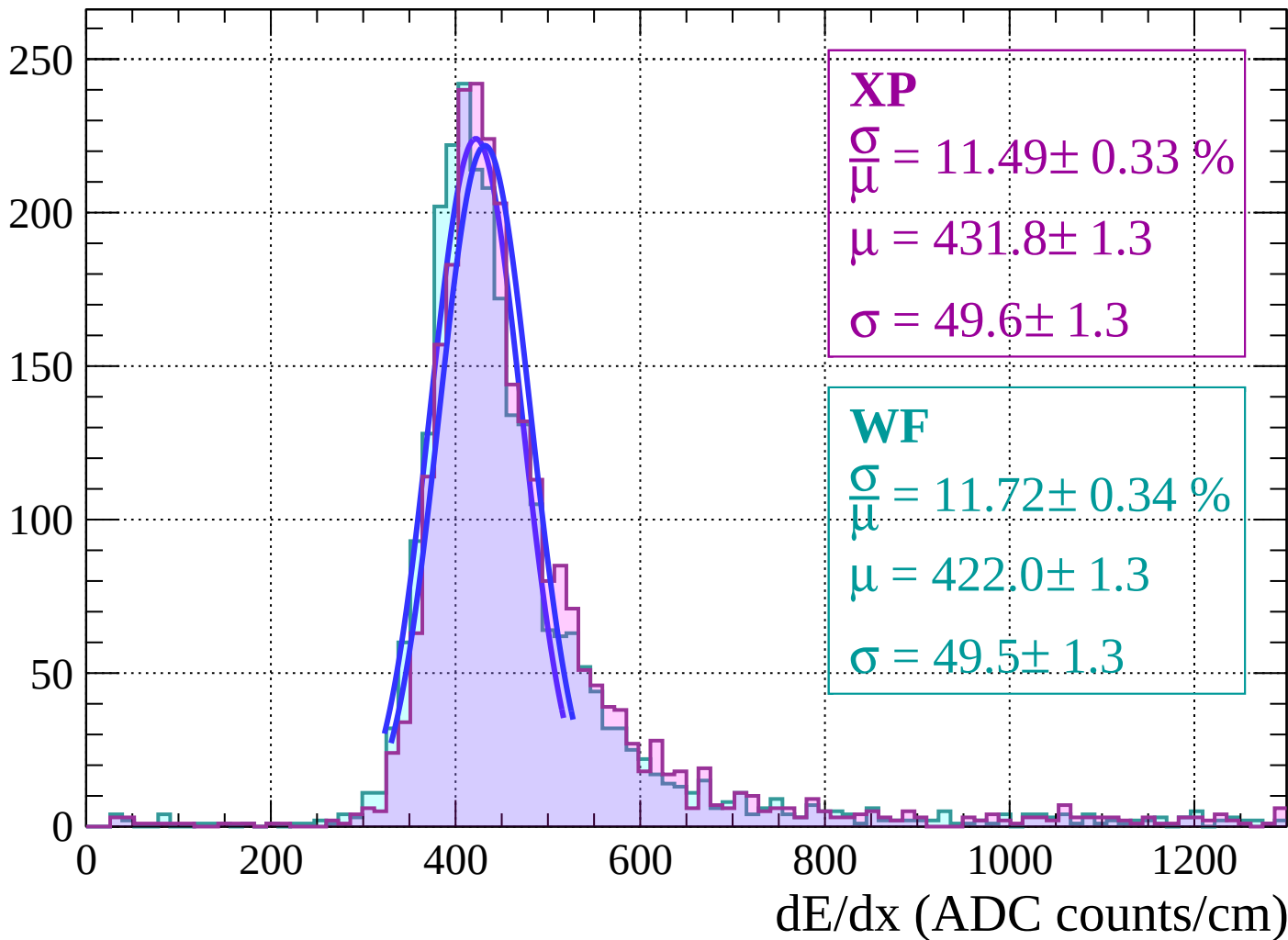


Energy loss | $22 < \theta < 24$



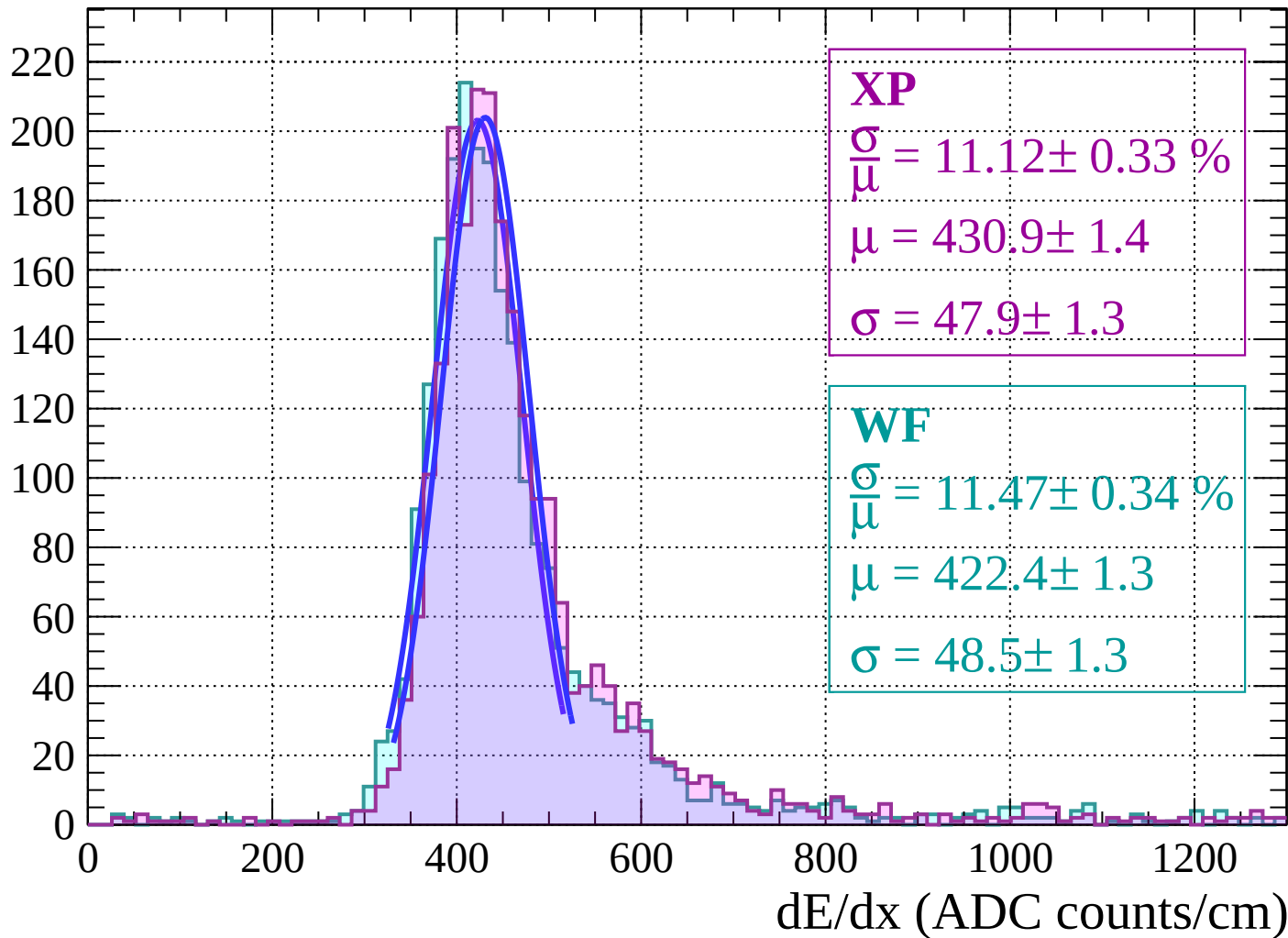
Energy loss | $24 < \theta < 26$

Count



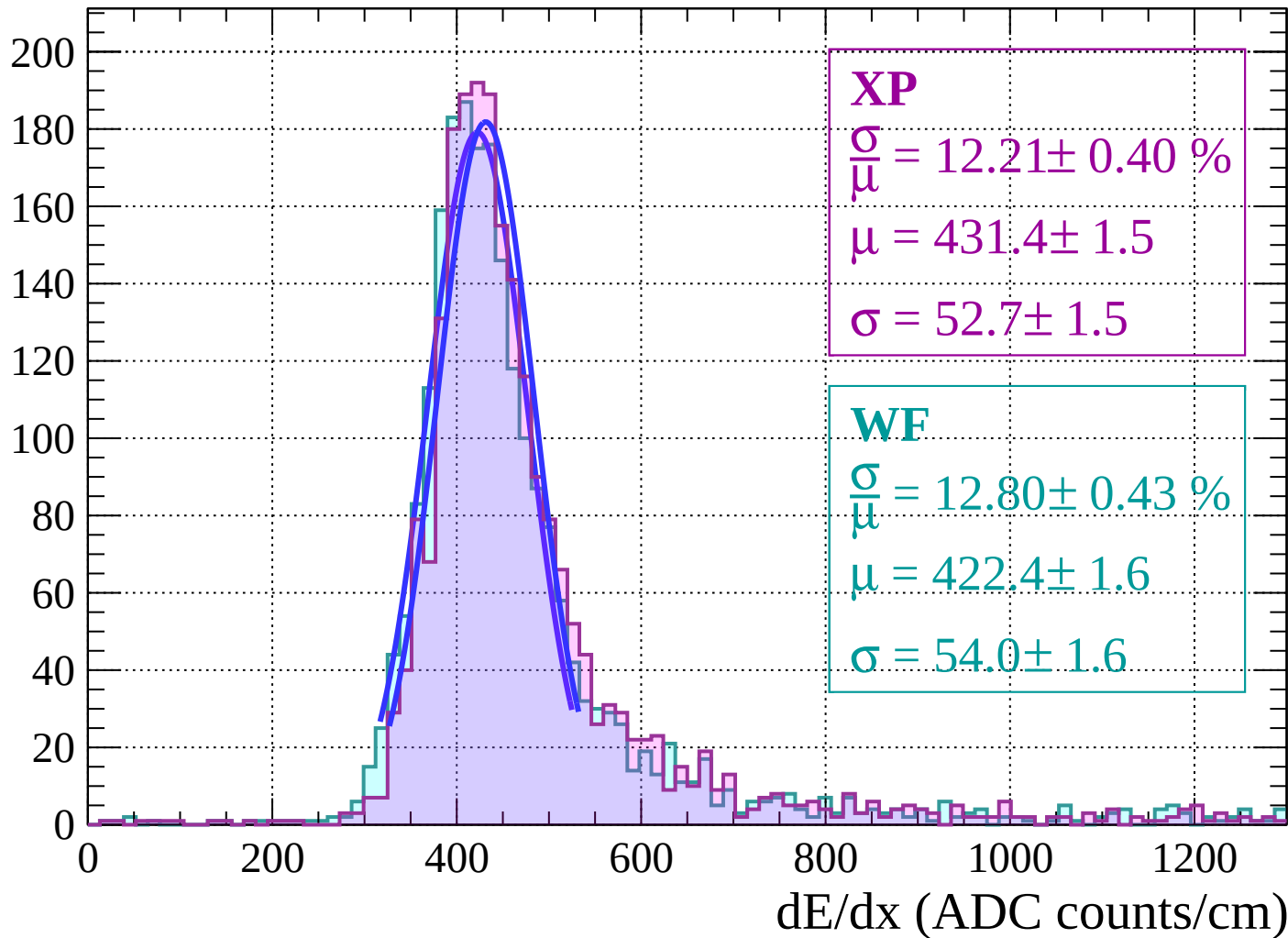
Energy loss | $26 < \theta < 28$

Count



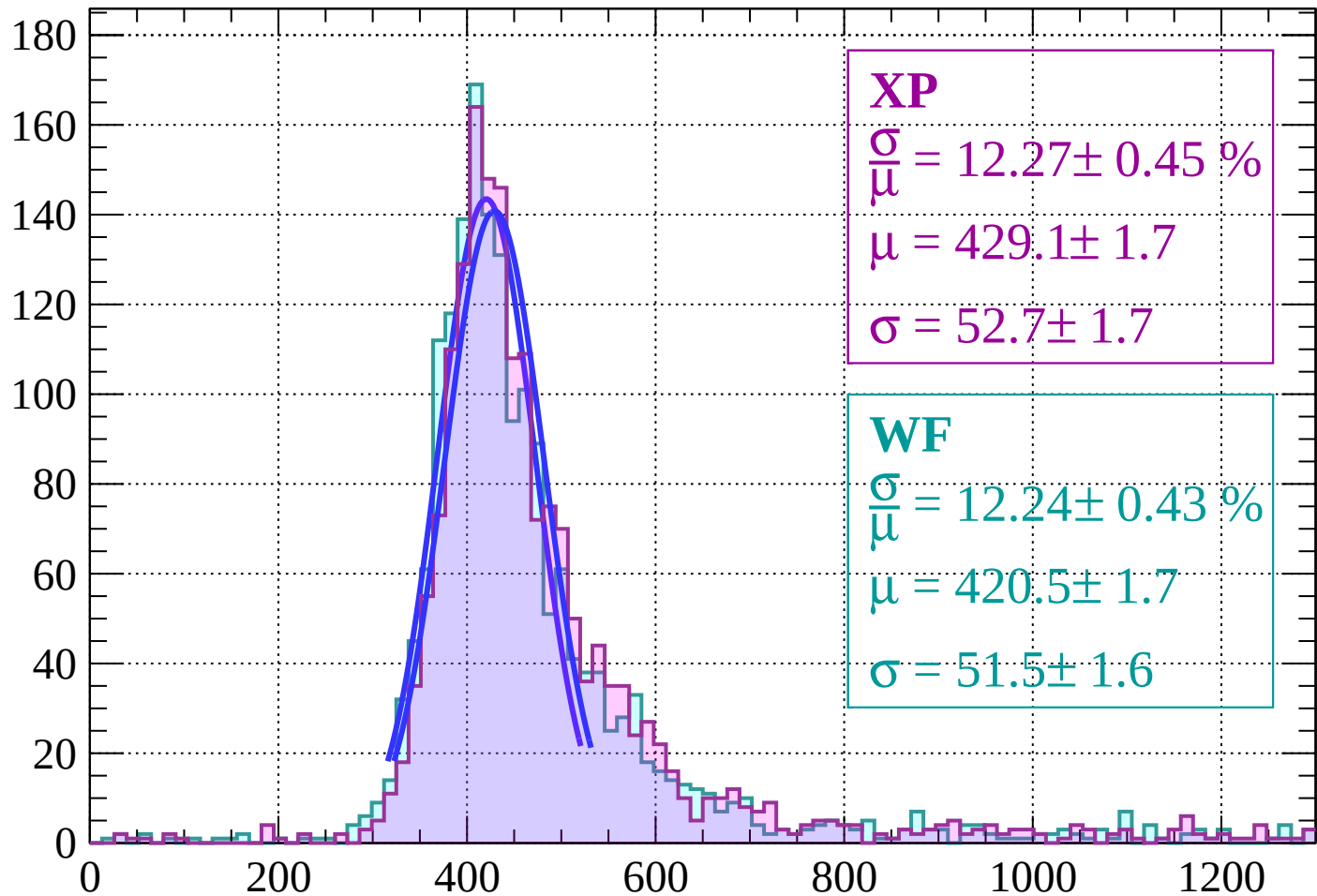
Energy loss | $28 < \theta < 30$

Count



Energy loss | $30 < \theta < 32$

Count



XP

$$\frac{\sigma}{\mu} = 12.27 \pm 0.45 \%$$

$$\mu = 429.1 \pm 1.7$$

$$\sigma = 52.7 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 12.24 \pm 0.43 \%$$

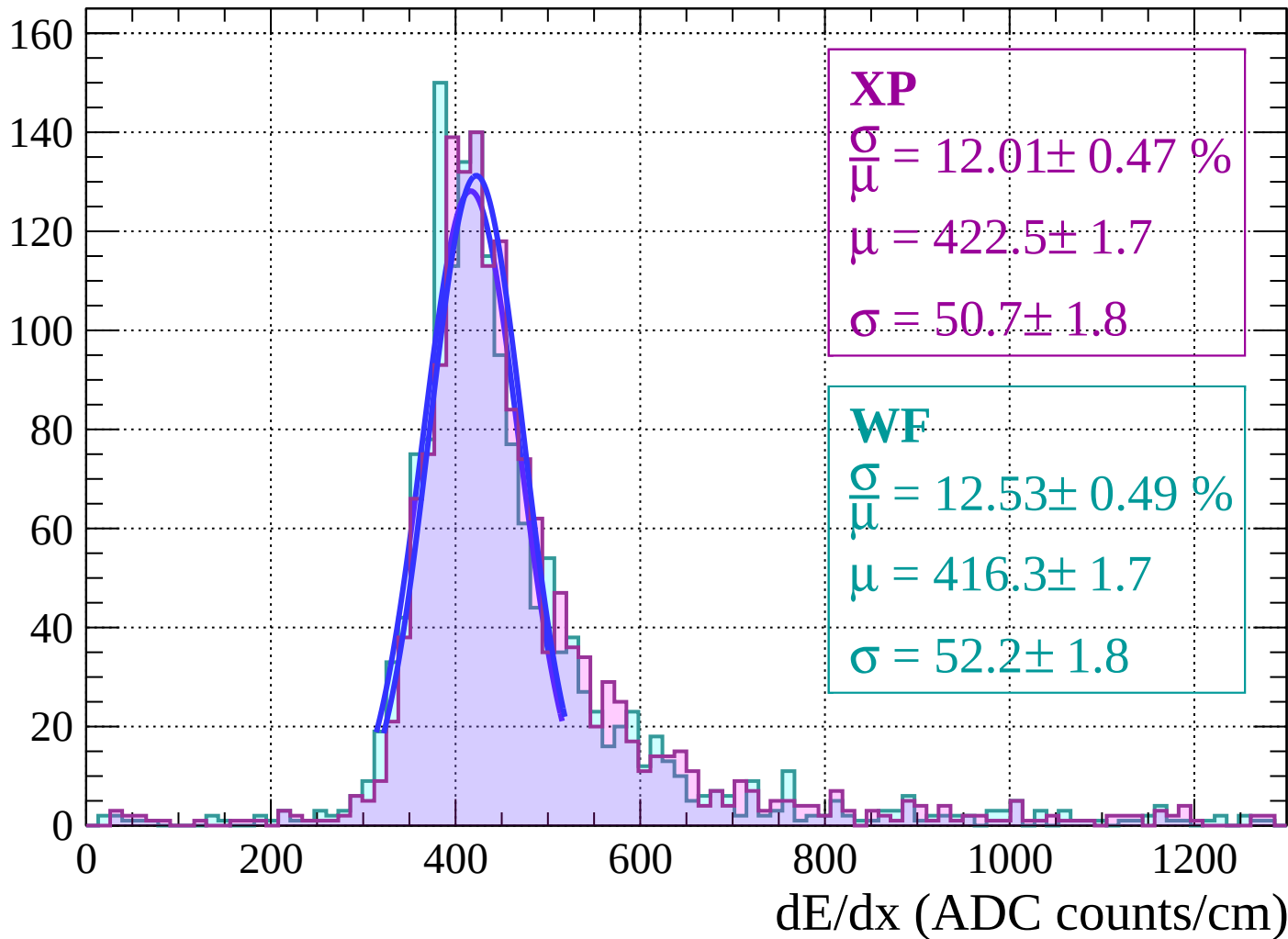
$$\mu = 420.5 \pm 1.7$$

$$\sigma = 51.5 \pm 1.6$$

dE/dx (ADC counts/cm)

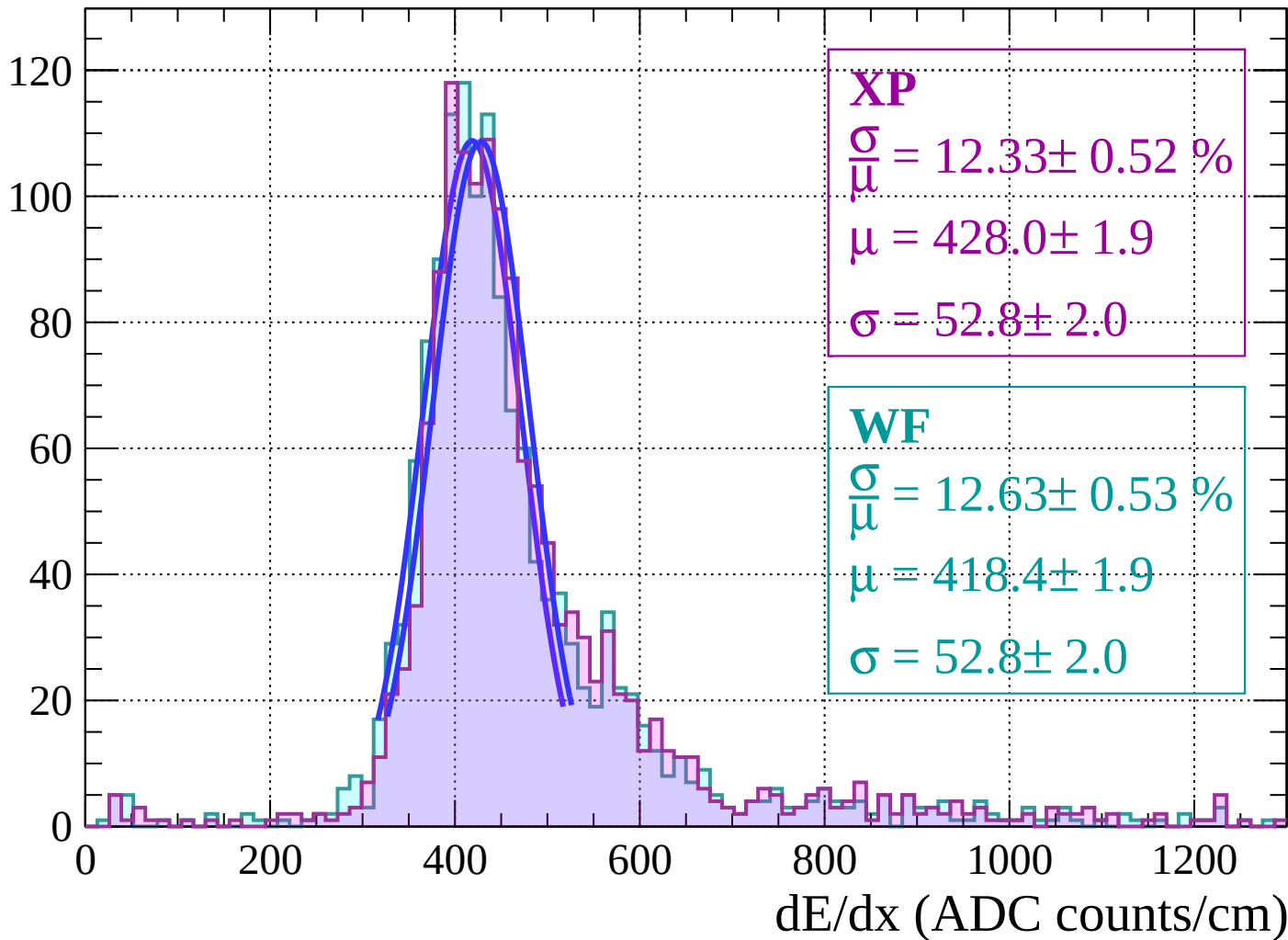
Energy loss | $32 < \theta < 34$

Count



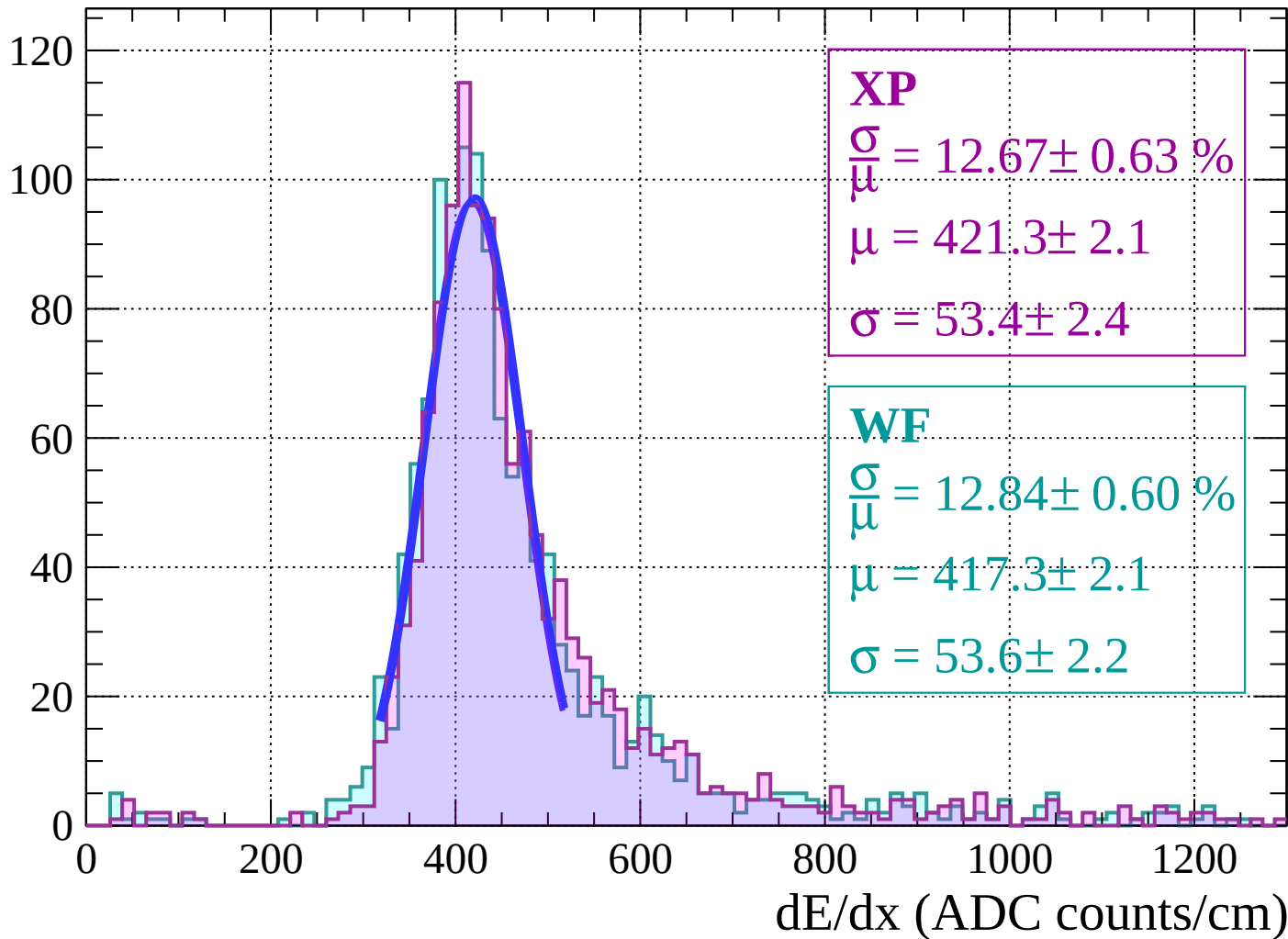
Energy loss | $34 < \theta < 36$

Count



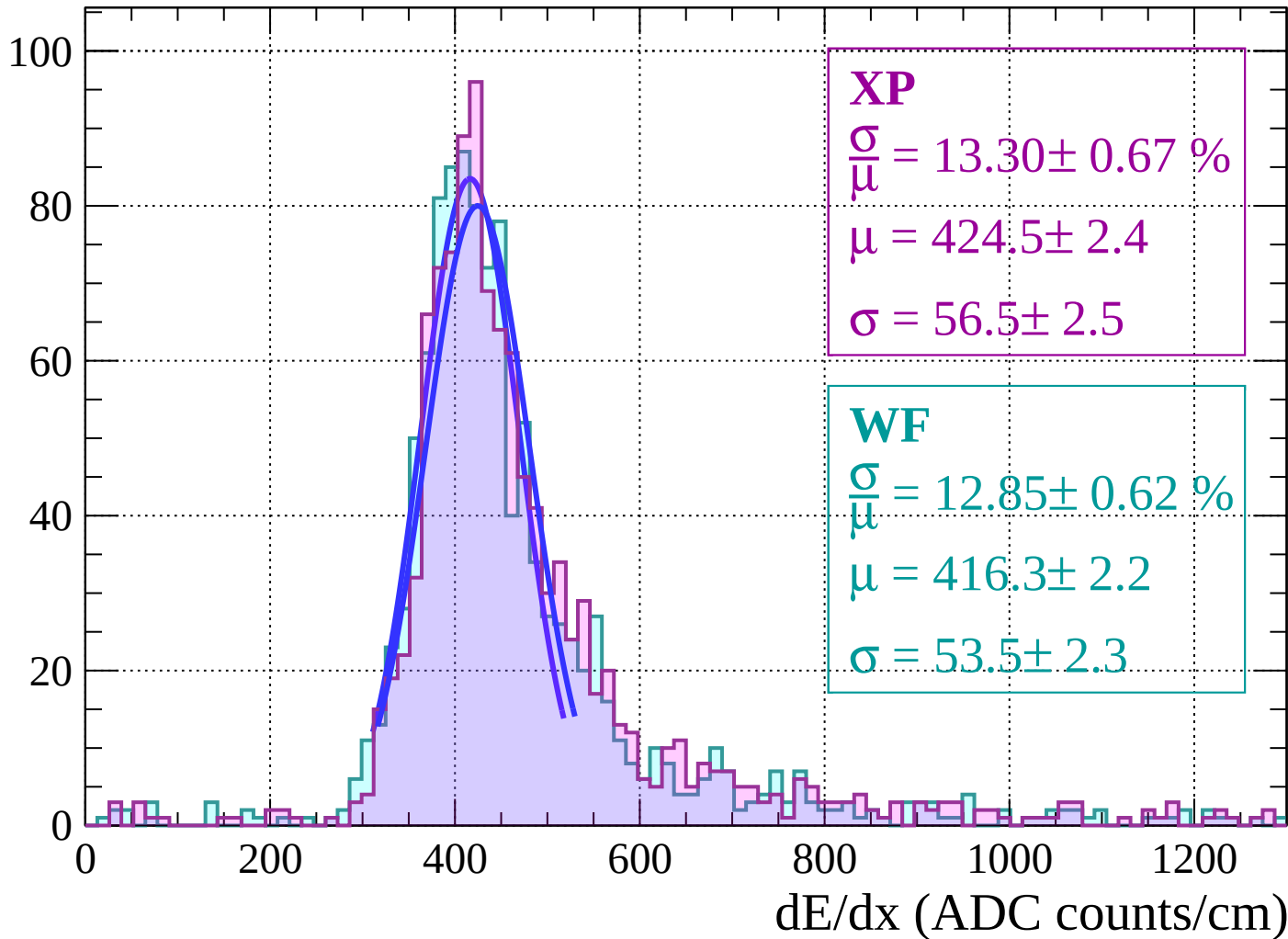
Energy loss | $36 < \theta < 38$

Count



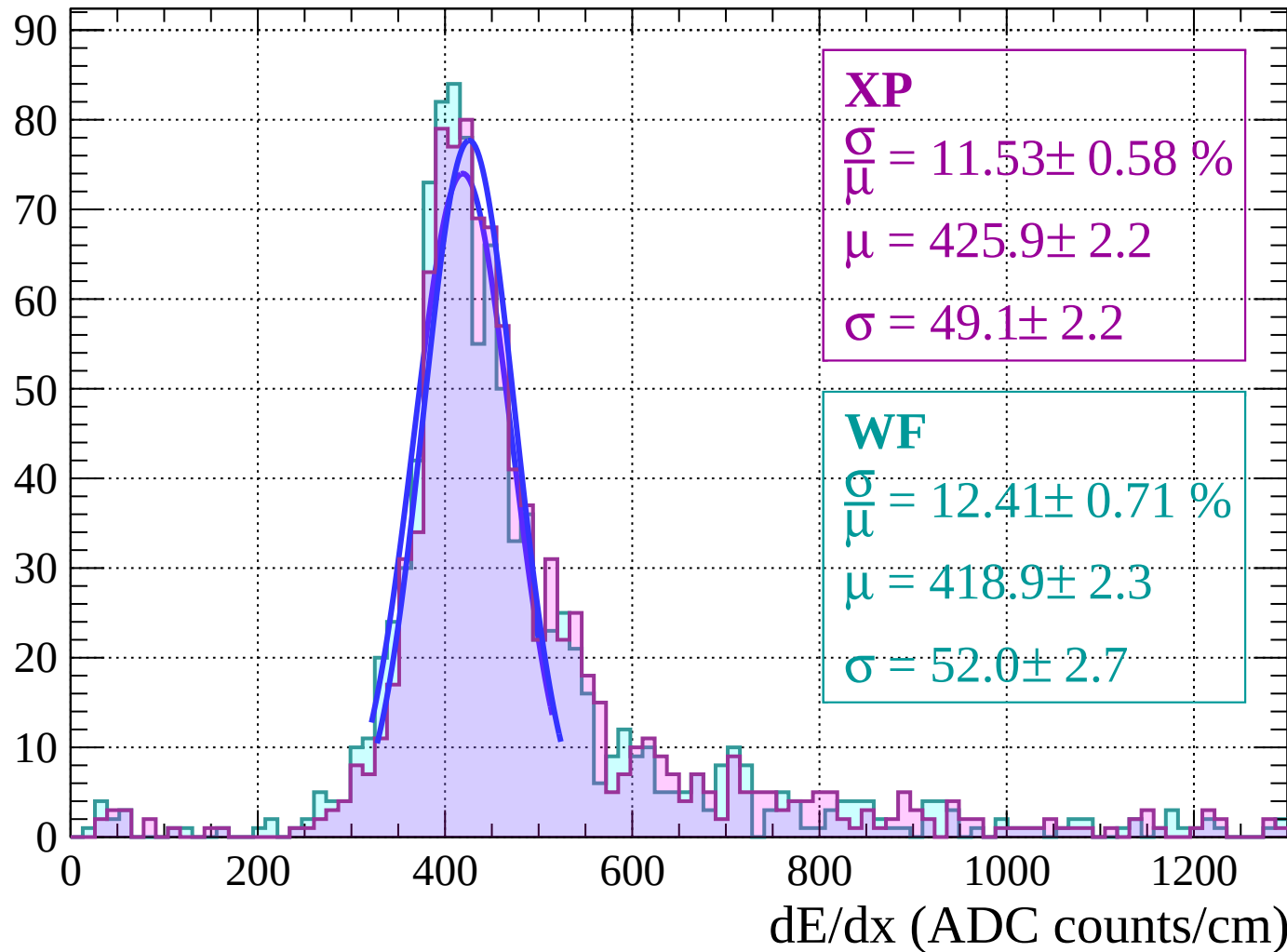
Energy loss | $38 < \theta < 40$

Count



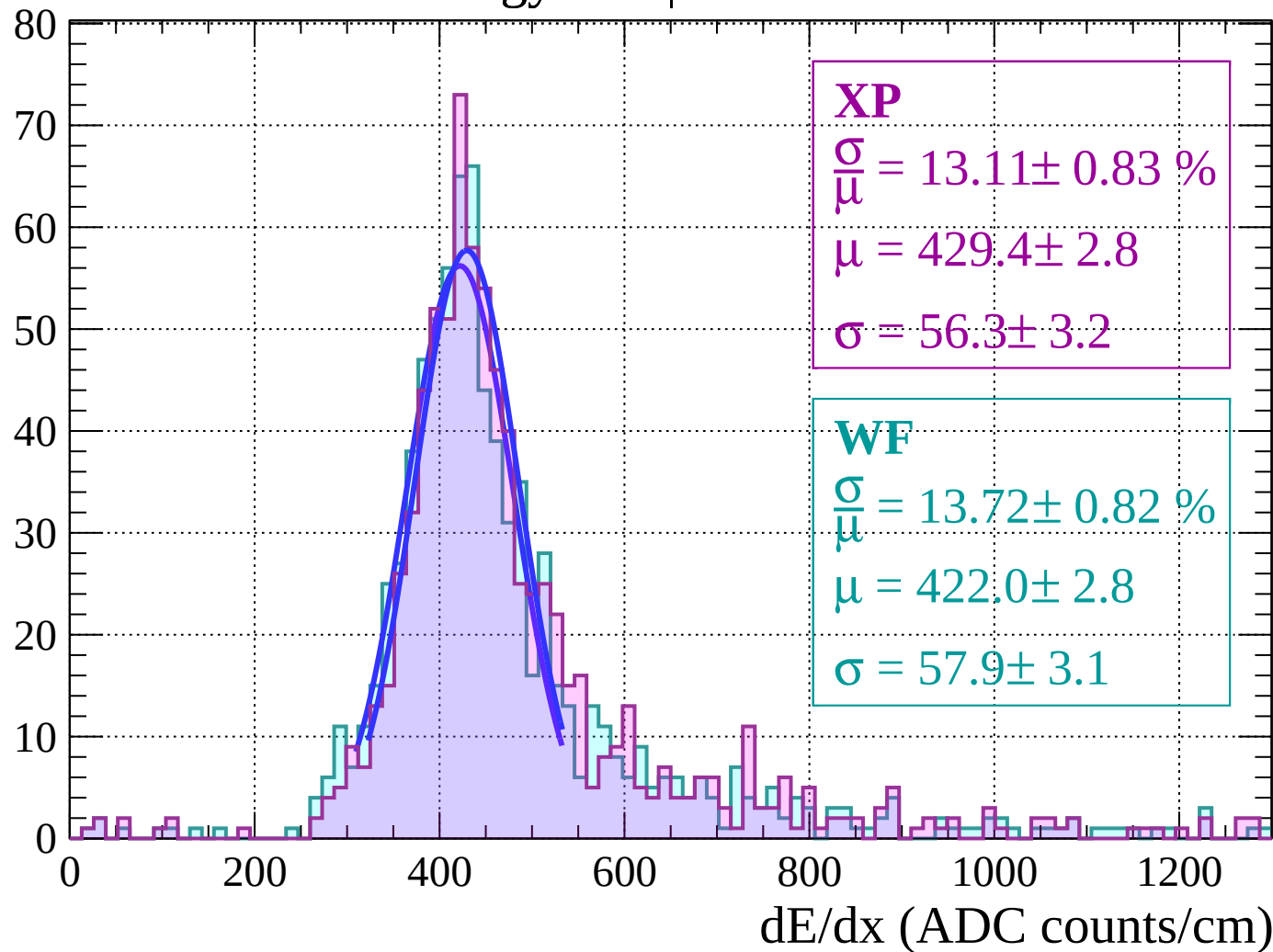
Energy loss | $40 < \theta < 42$

Count



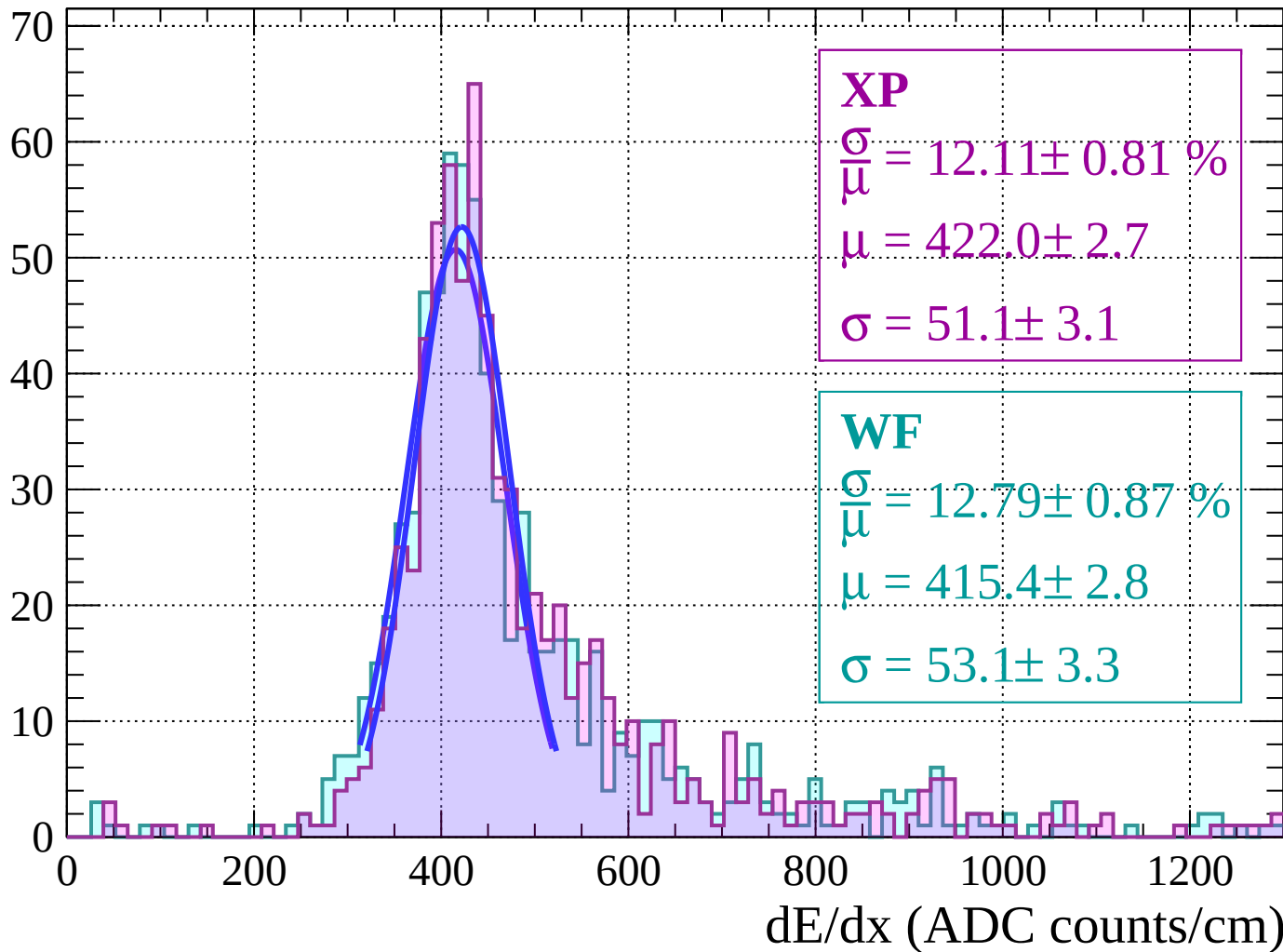
Energy loss | $42 < \theta < 44$

Count



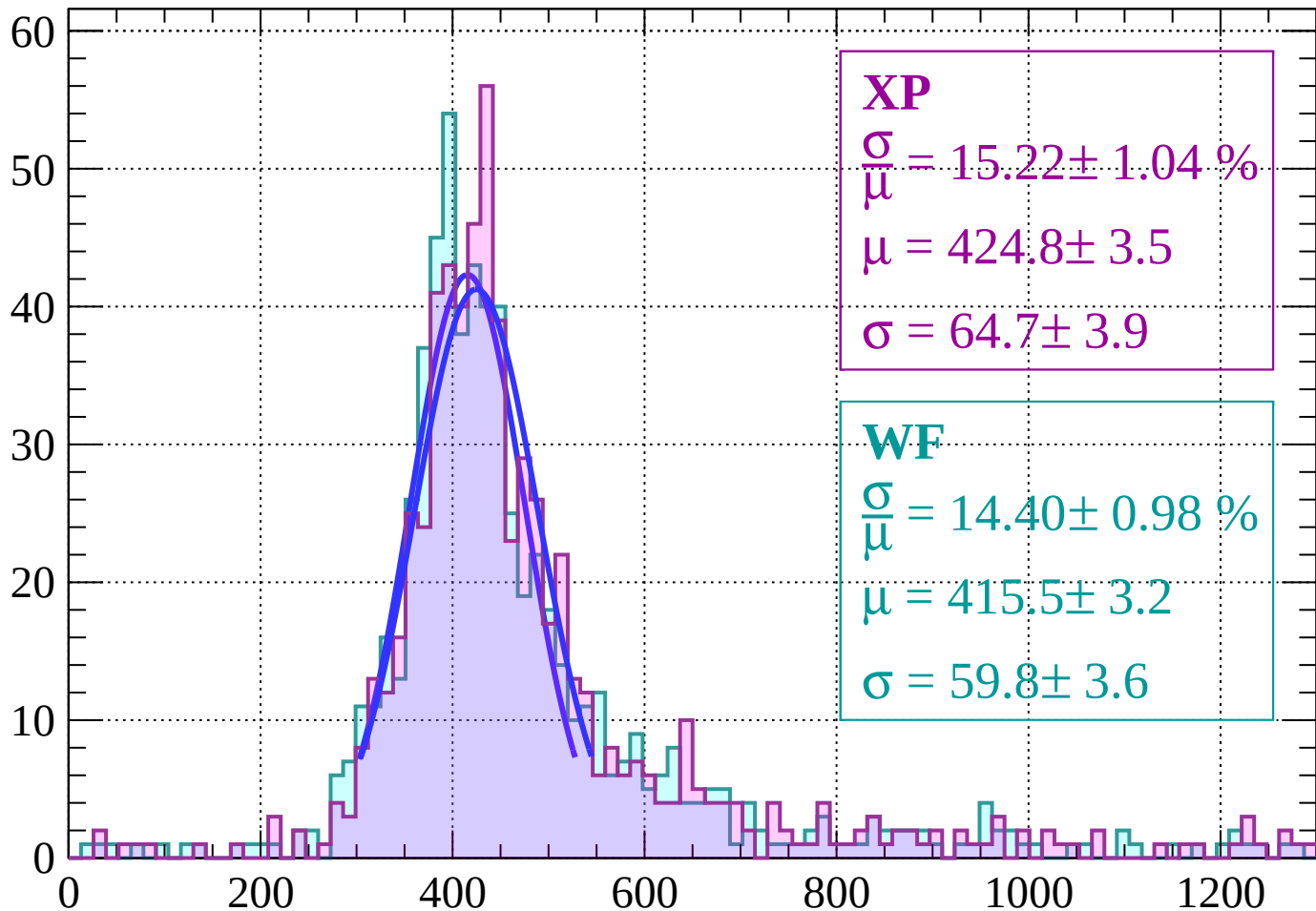
Energy loss | $44 < \theta < 46$

Count



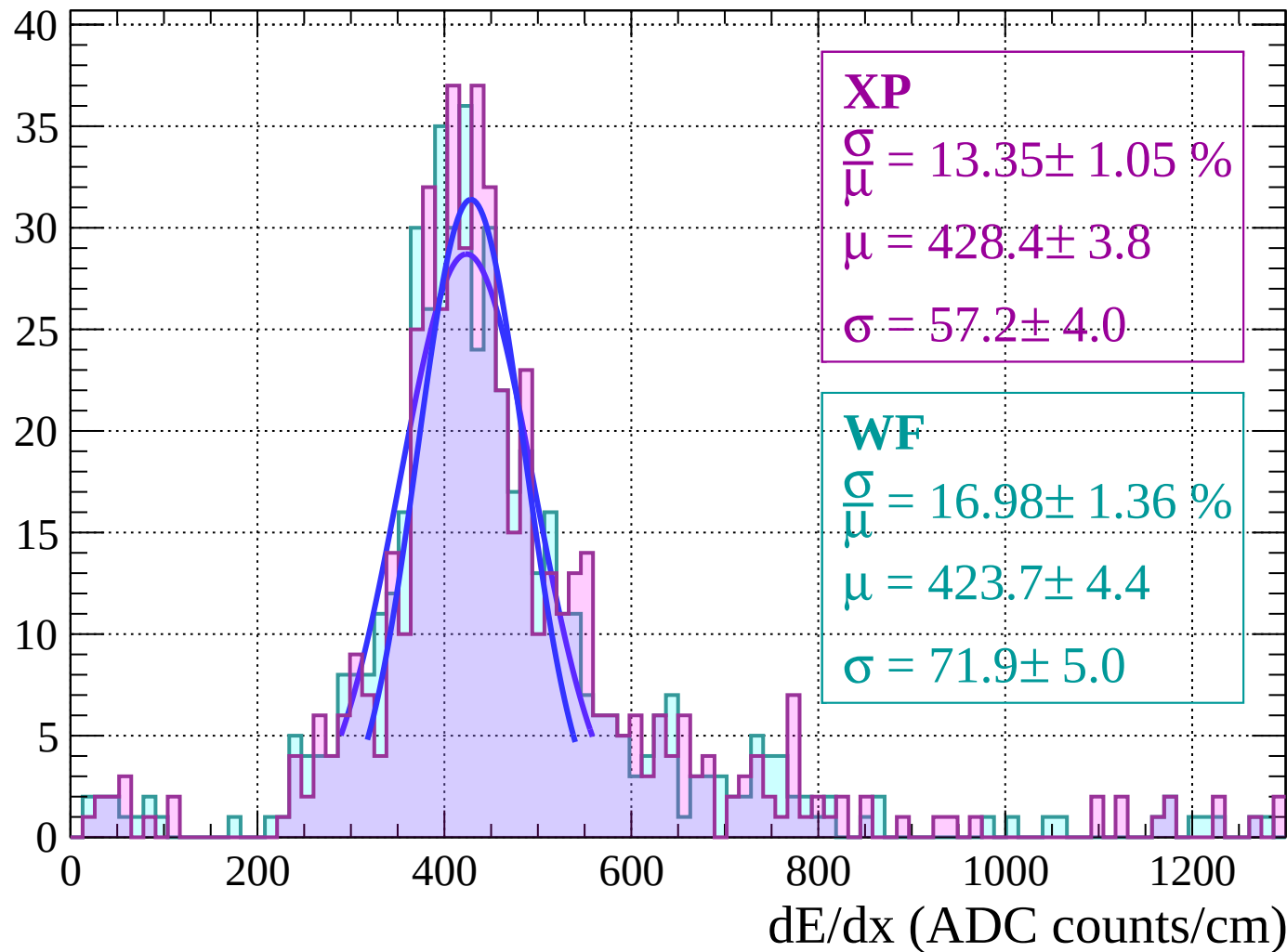
Energy loss | $46 < \theta < 48$

Count



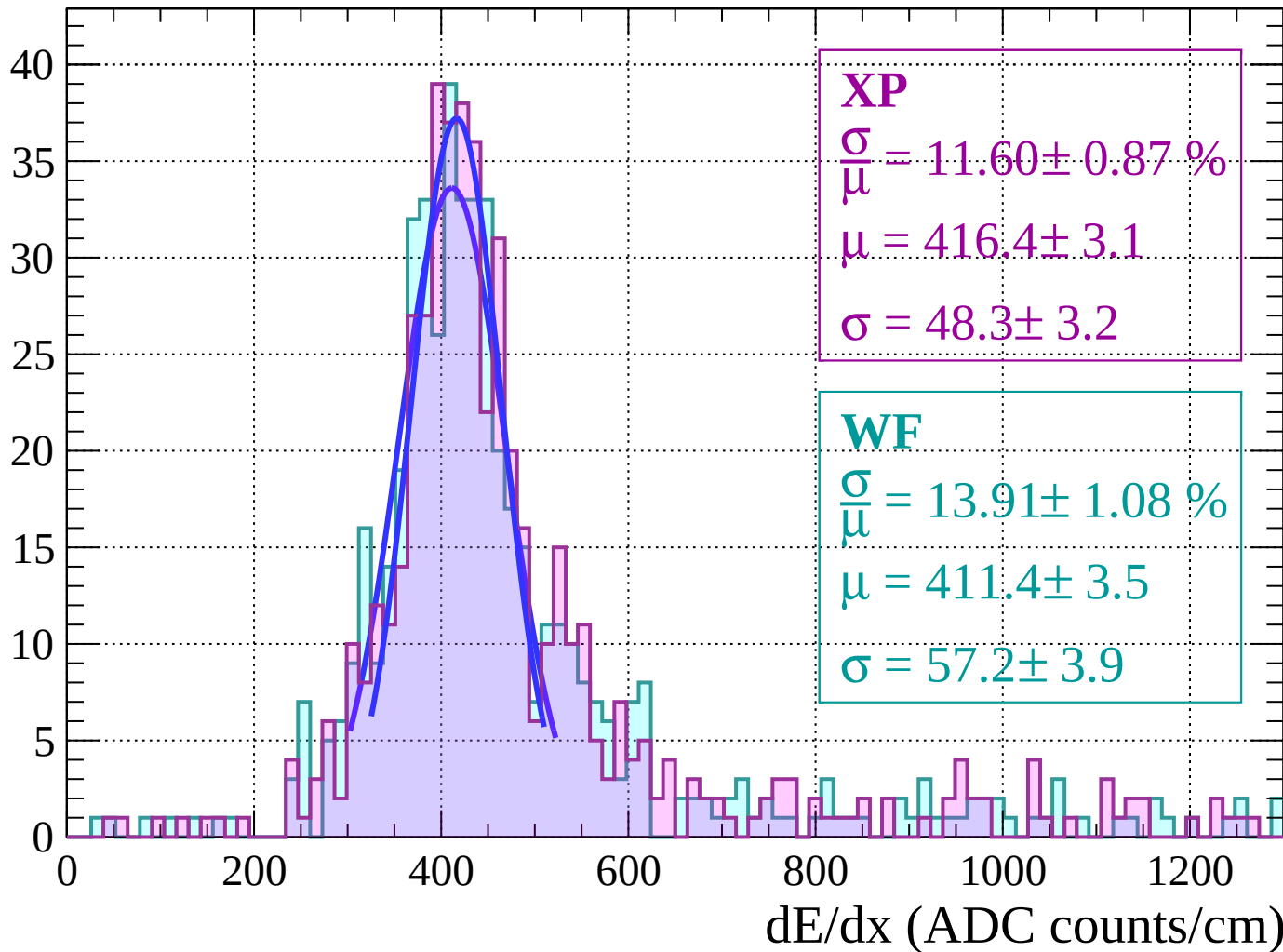
Energy loss | $48 < \theta < 50$

Count



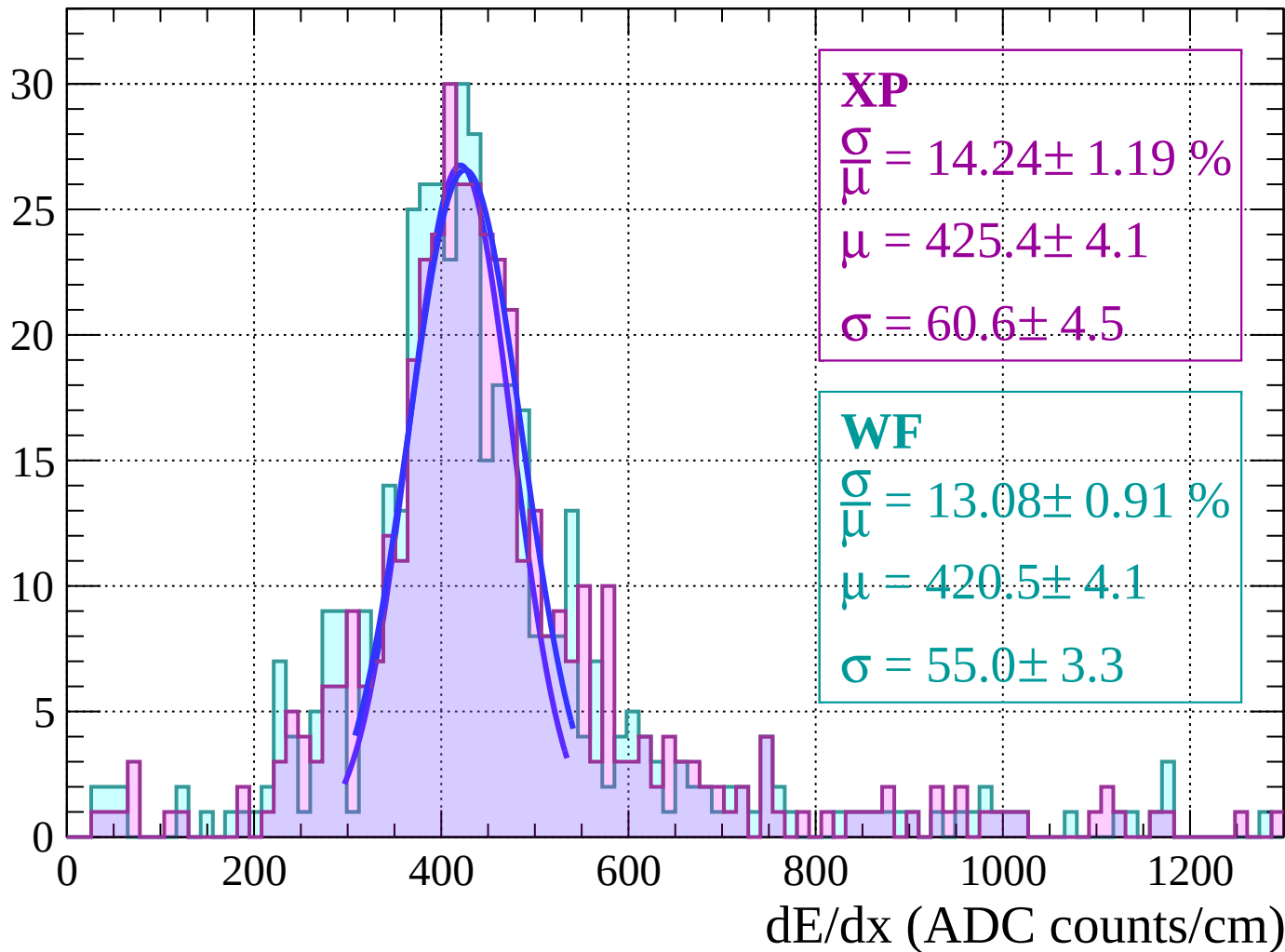
Energy loss | $50 < \theta < 52$

Count



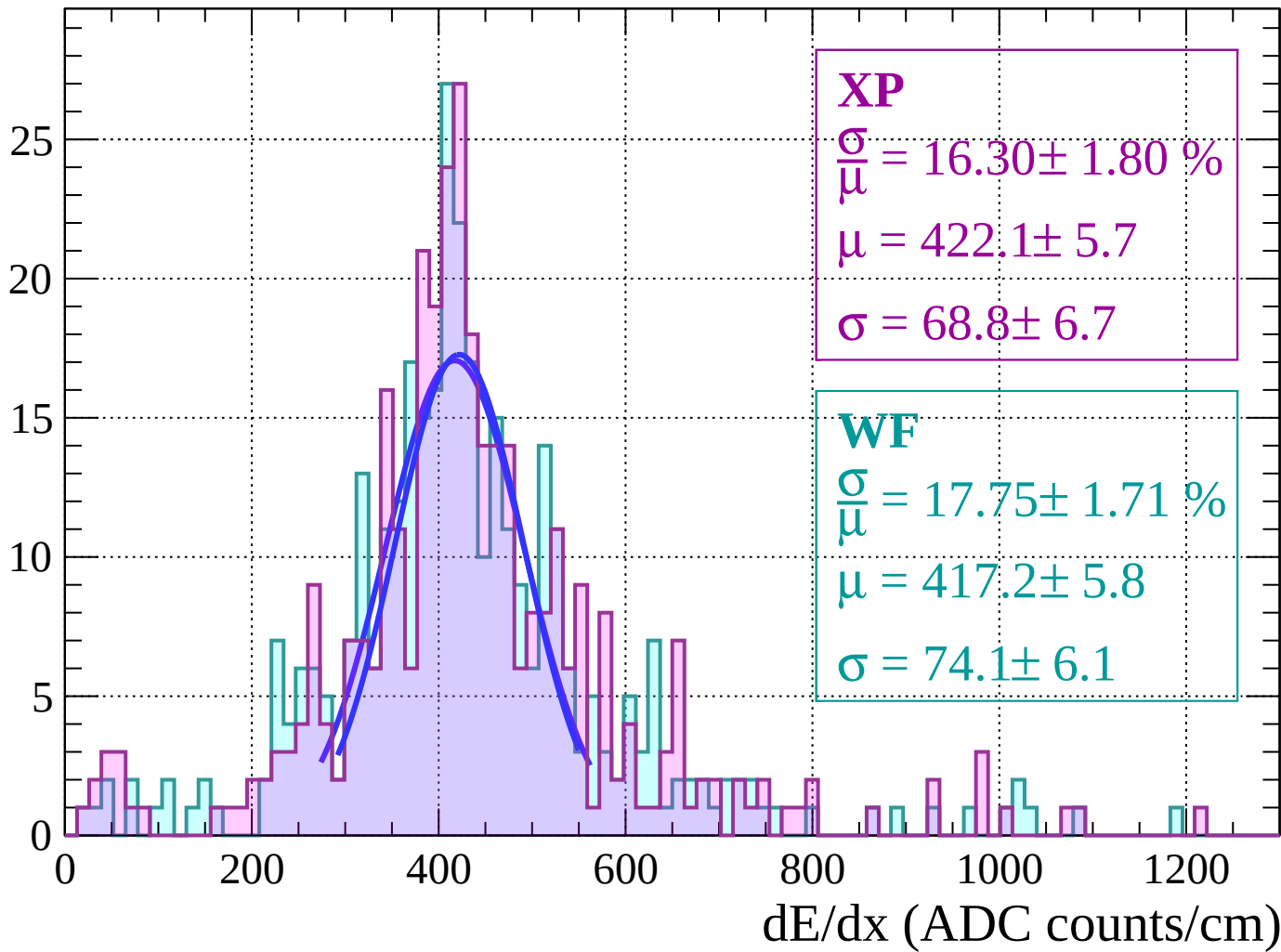
Energy loss | $52 < \theta < 54$

Count



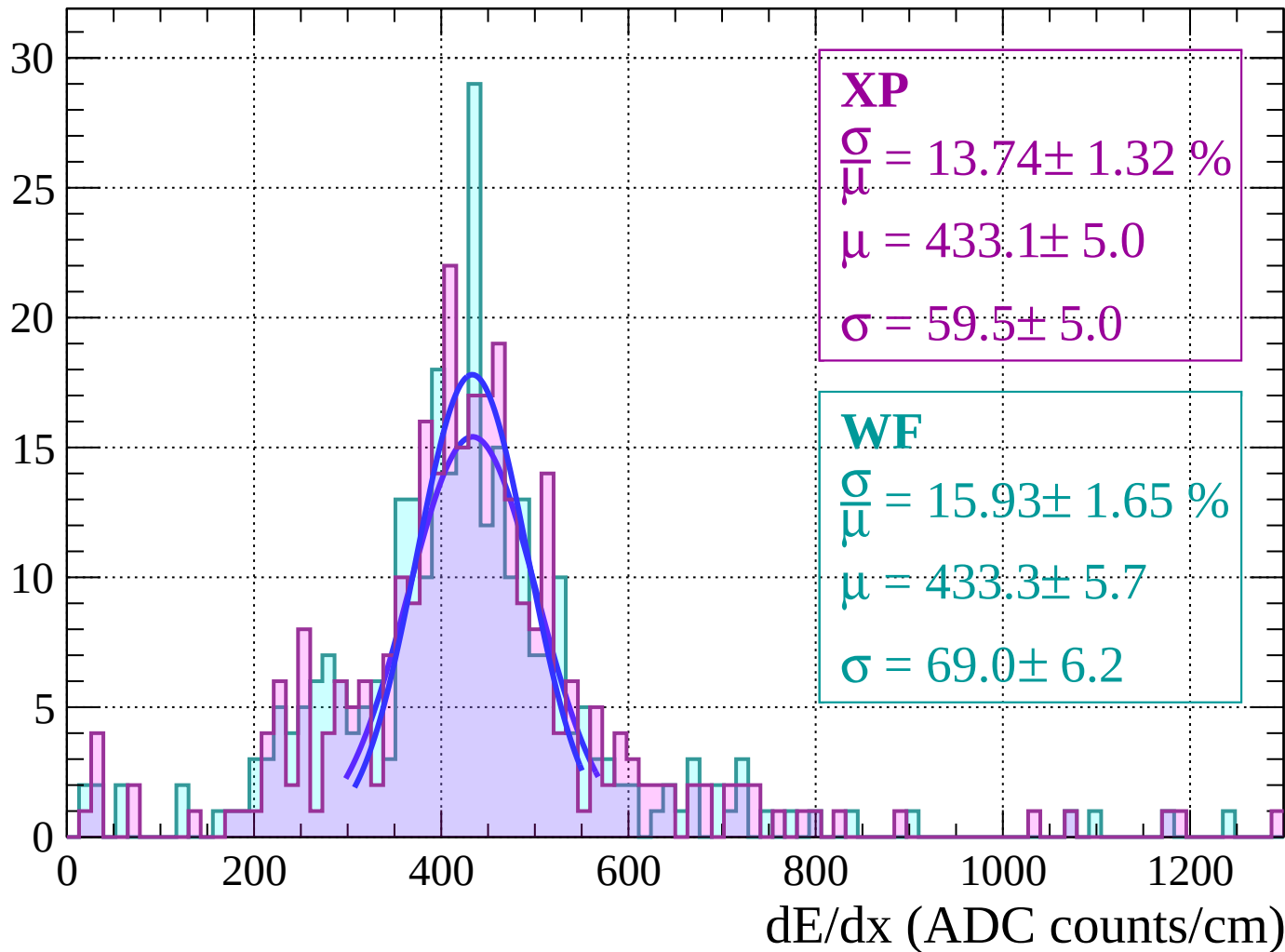
Energy loss | $54 < \theta < 56$

Count



Energy loss | $56 < \theta < 58$

Count



Energy loss | $58 < \theta < 60$

Count

XP

$$\frac{\sigma}{\mu} = 20.02 \pm 2.13 \%$$

$$\mu = 434.5 \pm 7.4$$

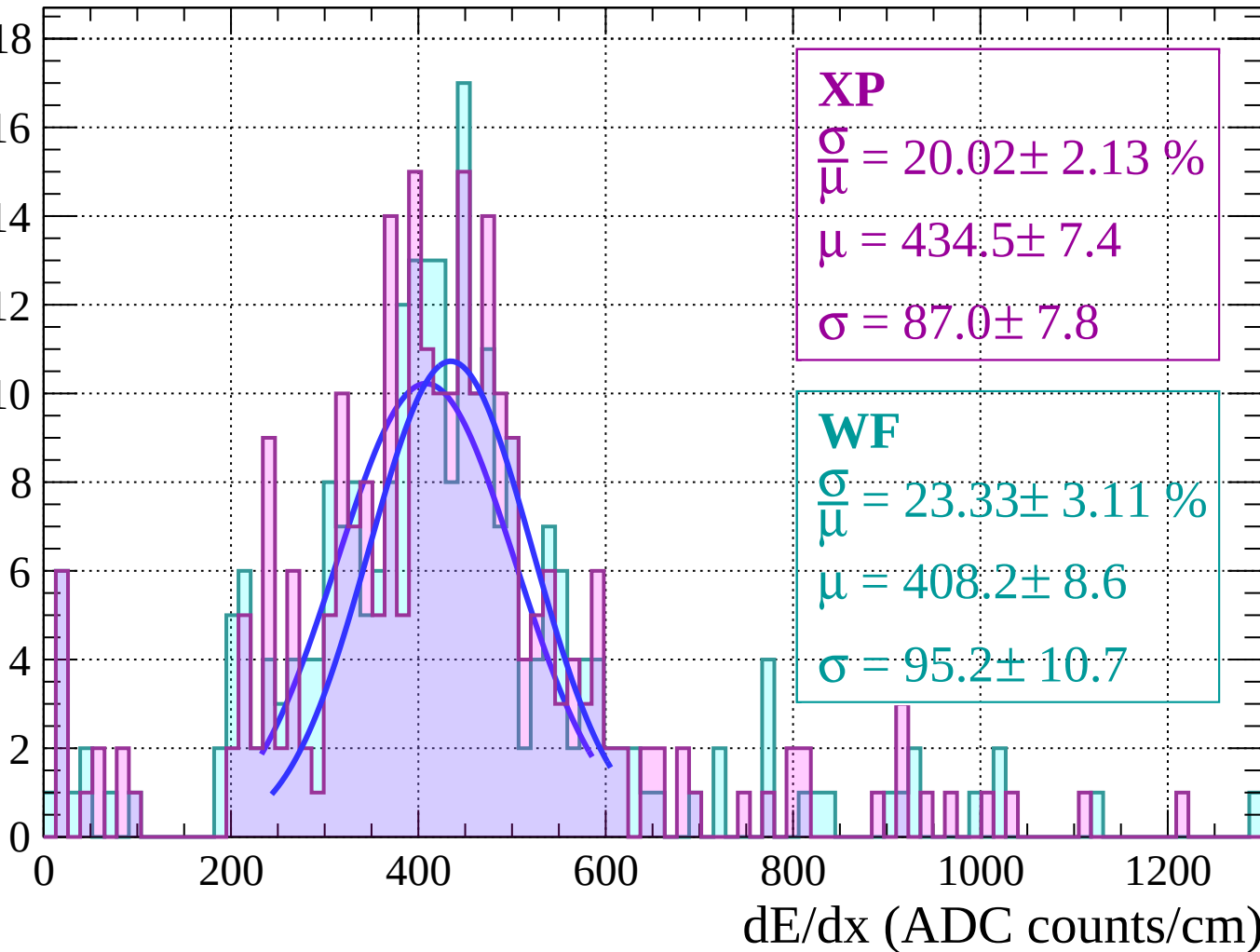
$$\sigma = 87.0 \pm 7.8$$

WF

$$\frac{\sigma}{\mu} = 23.33 \pm 3.11 \%$$

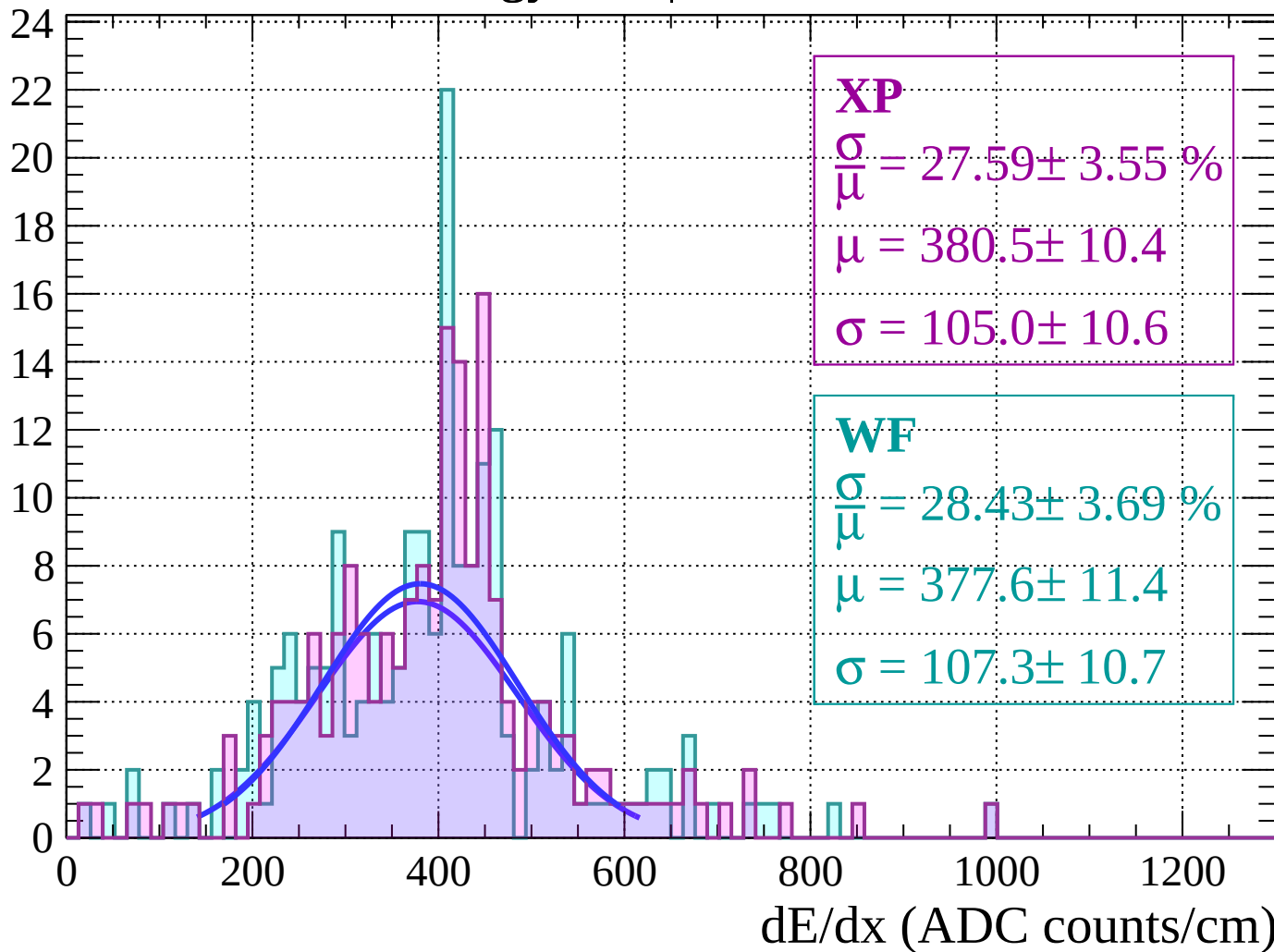
$$\mu = 408.2 \pm 8.6$$

$$\sigma = 95.2 \pm 10.7$$



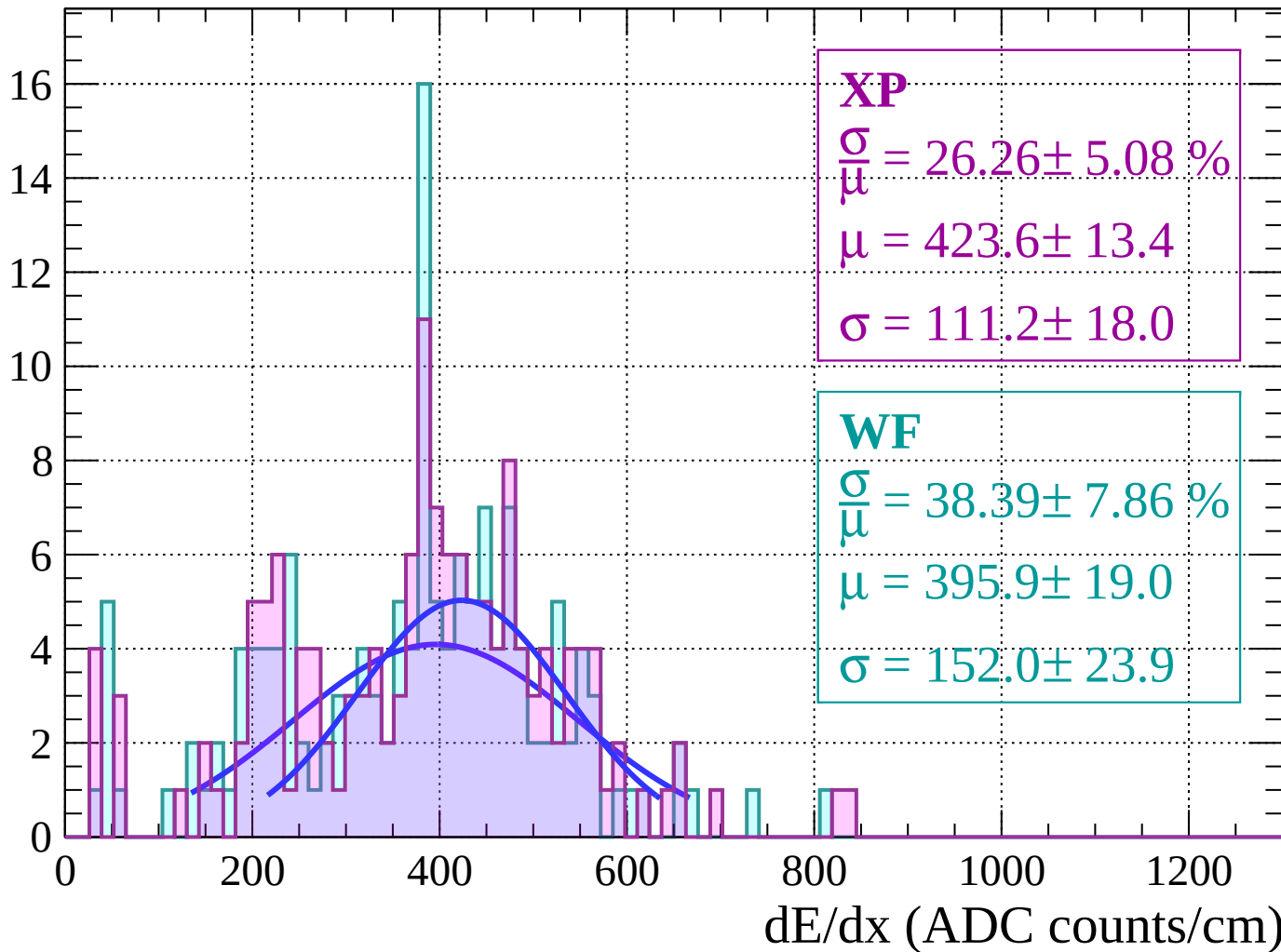
Energy loss | $60 < \theta < 62$

Count



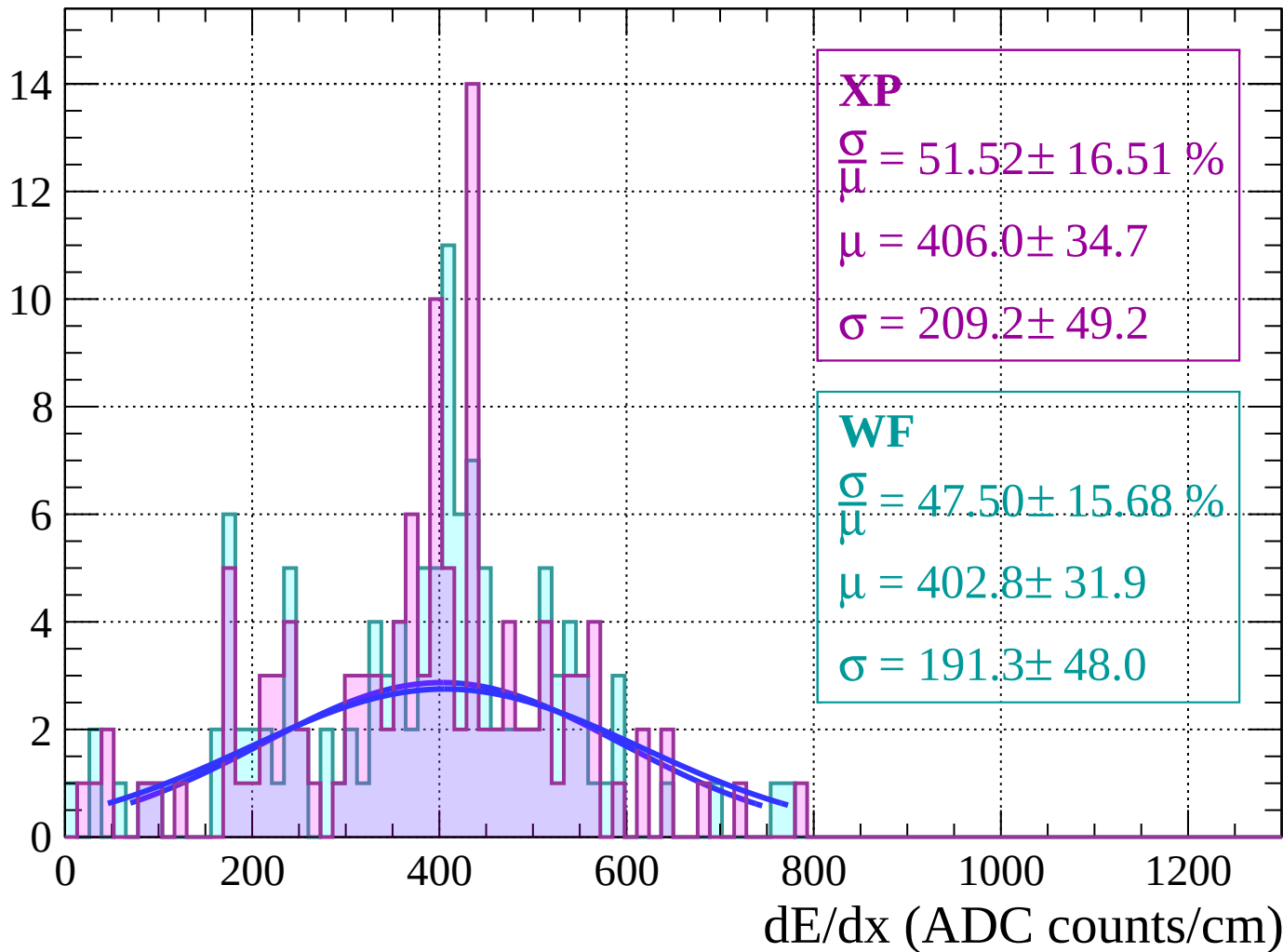
Energy loss | $62 < \theta < 64$

Count



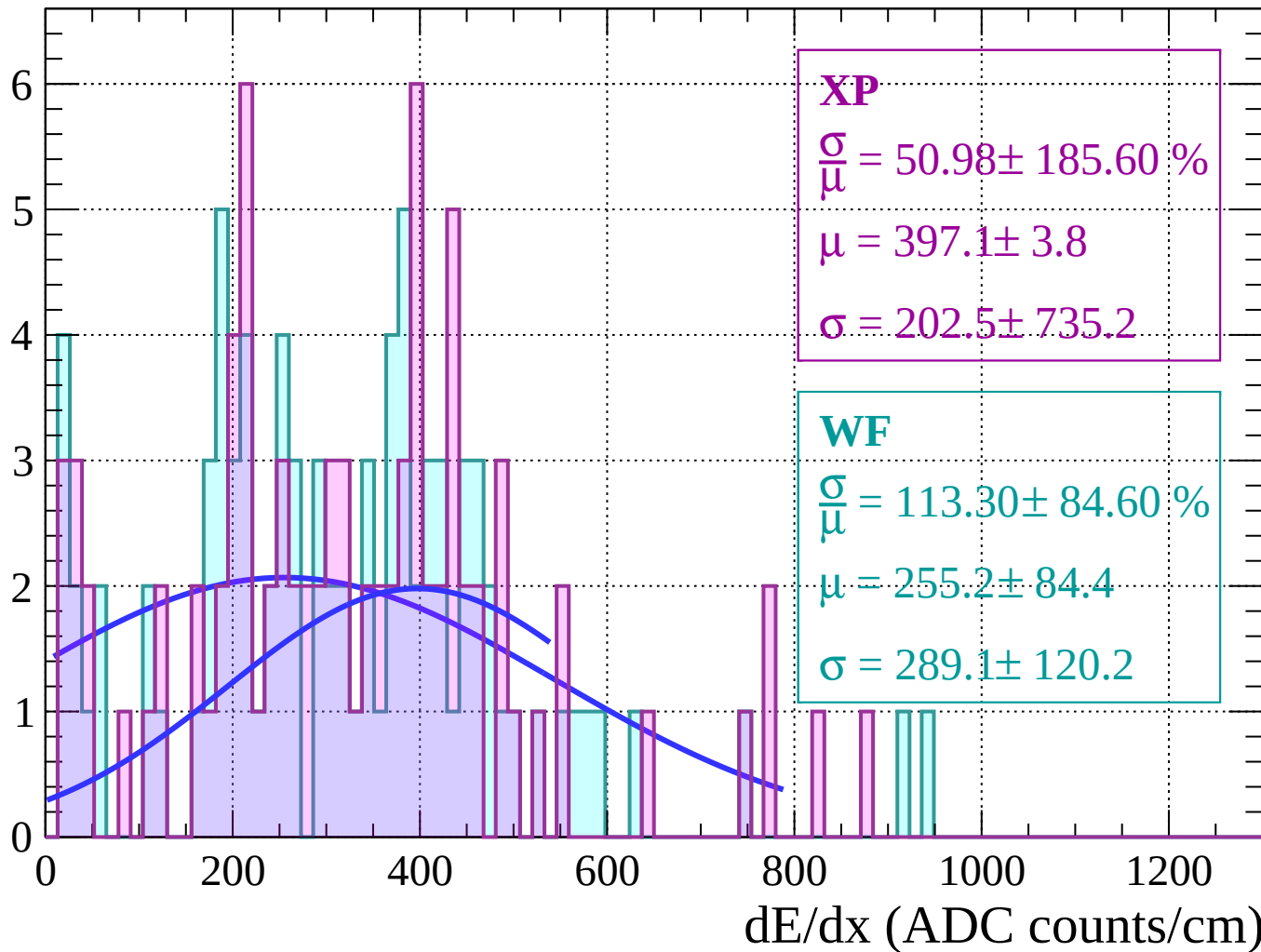
Energy loss | $64 < \theta < 66$

Count



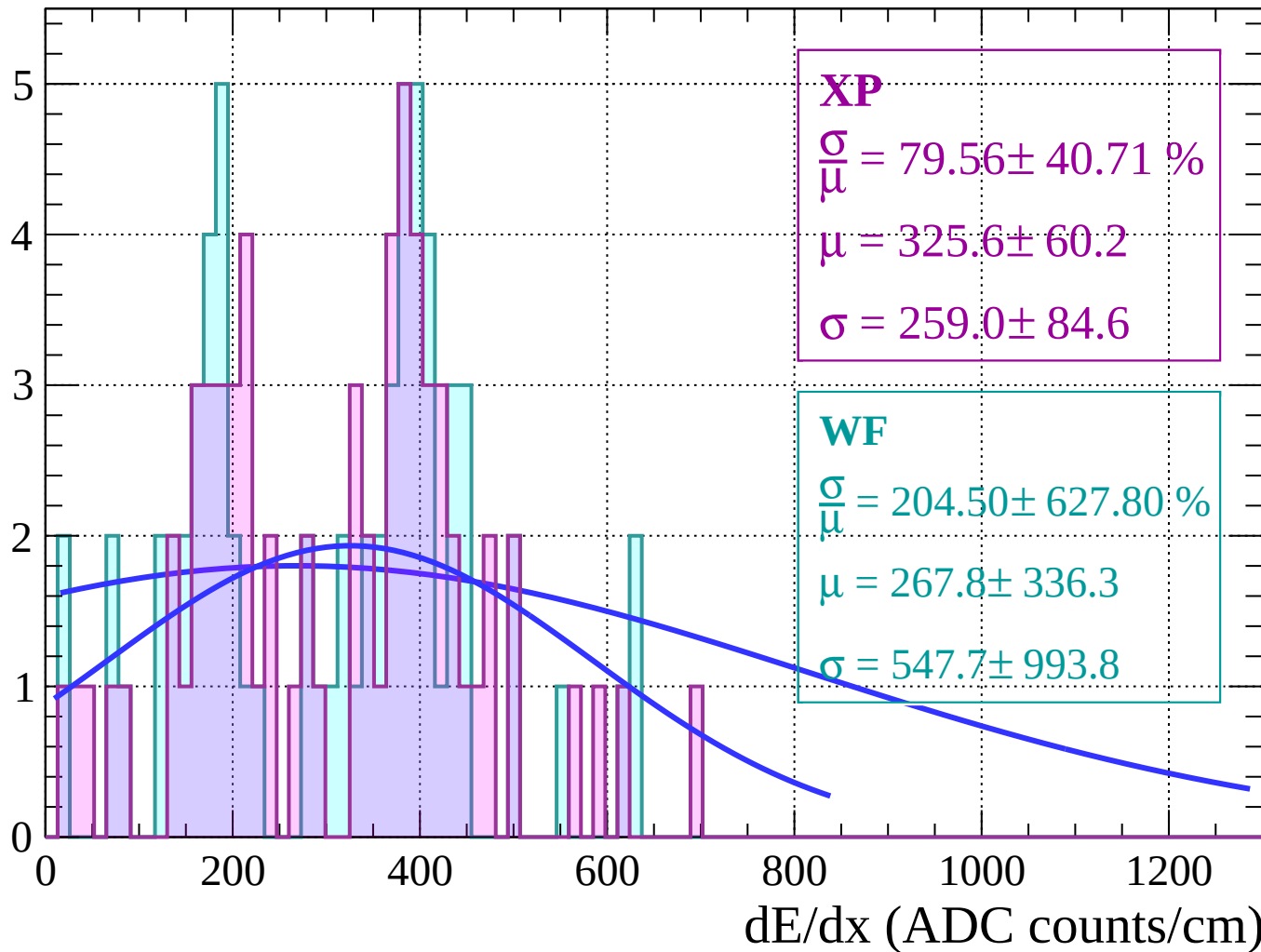
Energy loss | $66 < \theta < 68$

Count



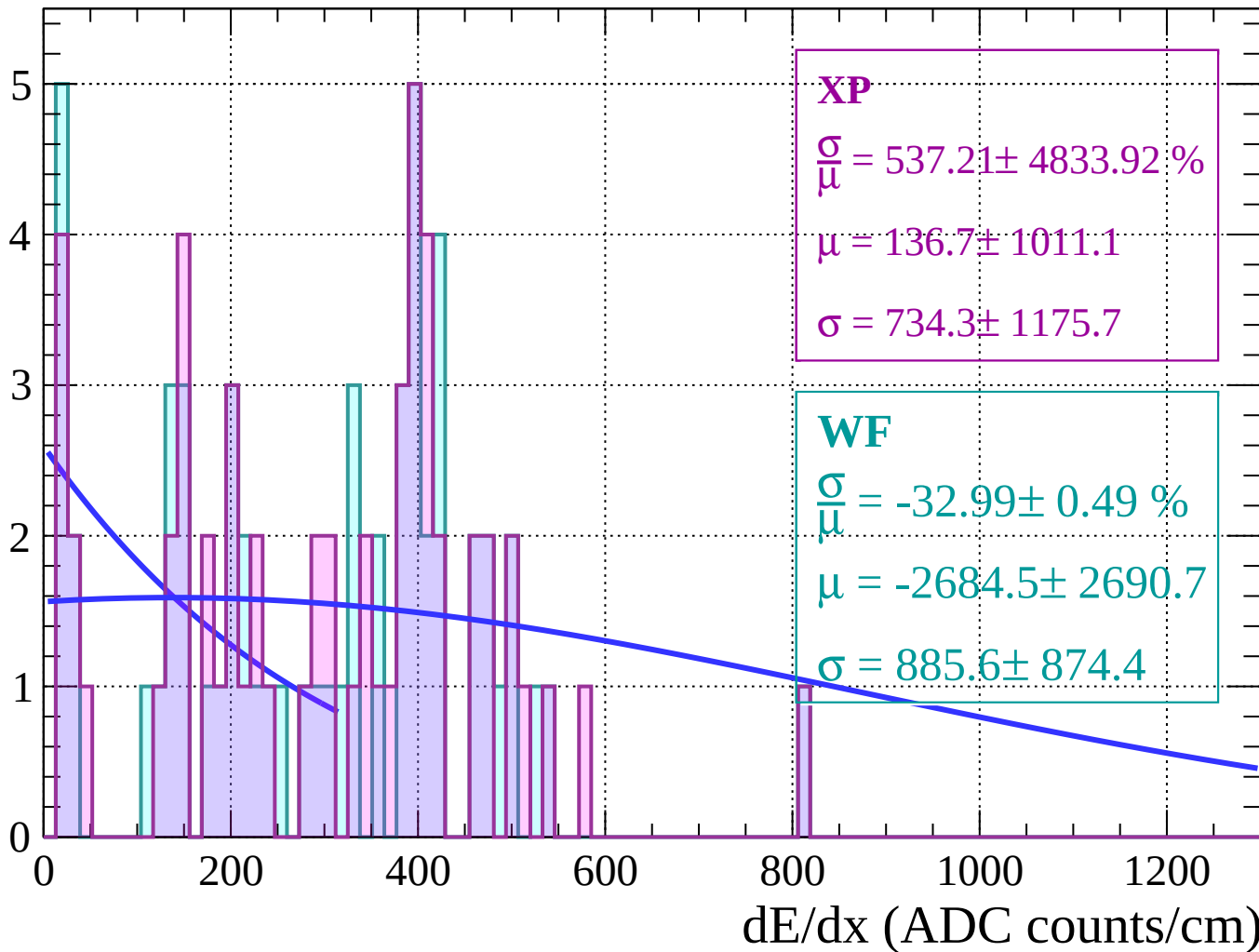
Energy loss | $68 < \theta < 70$

Count



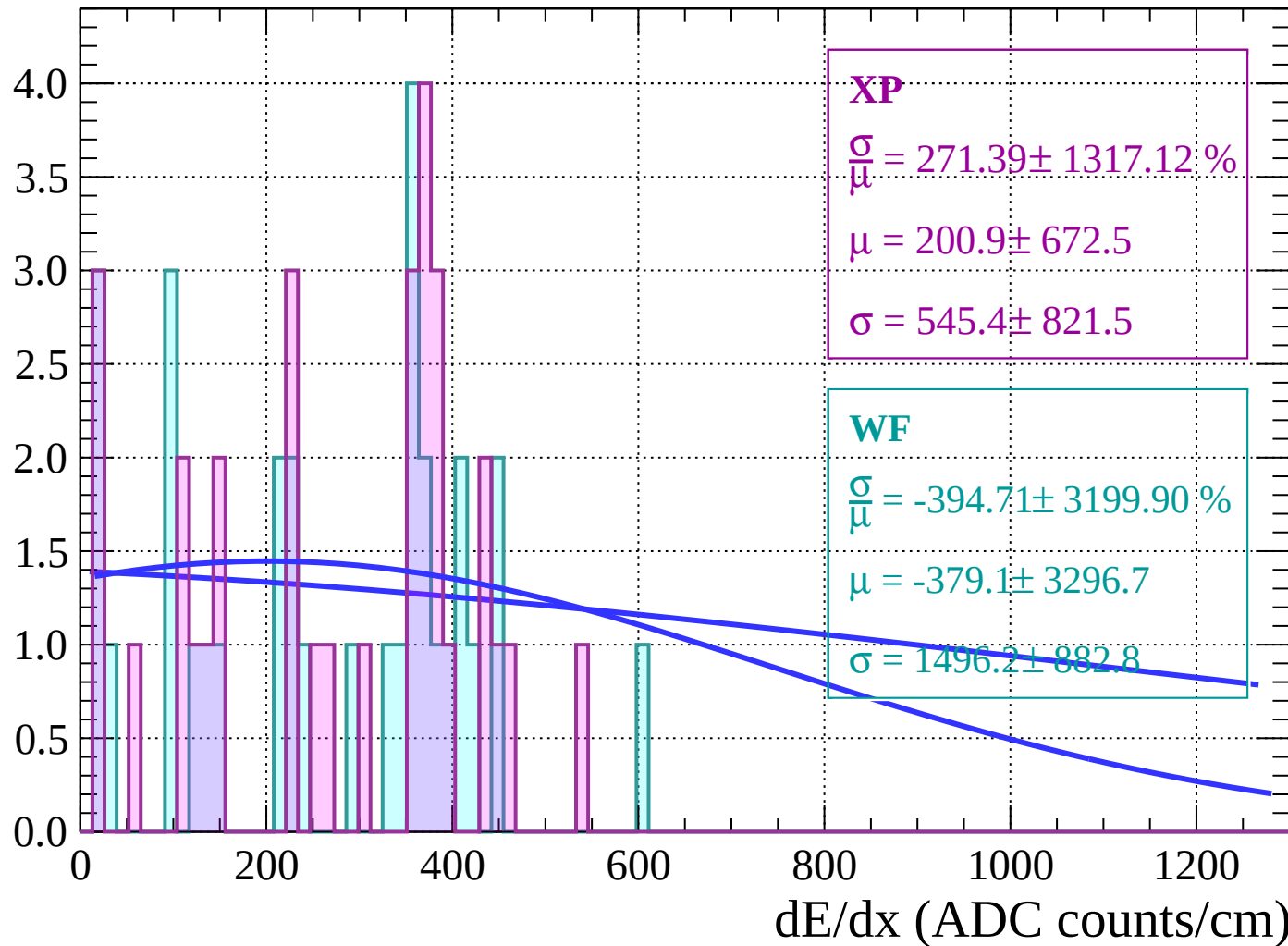
Energy loss | $70 < \theta < 72$

Count



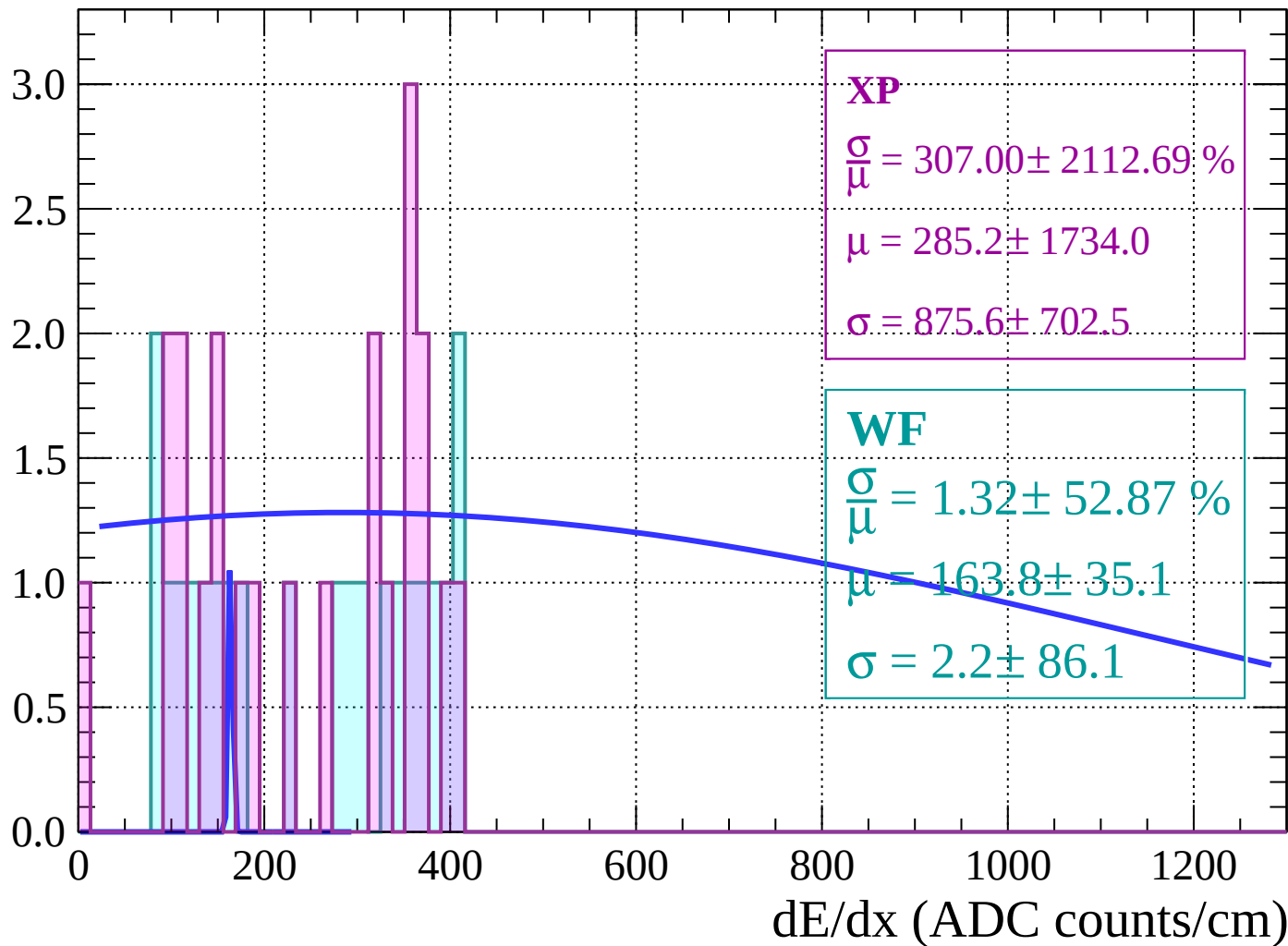
Energy loss | $72 < \theta < 74$

Count



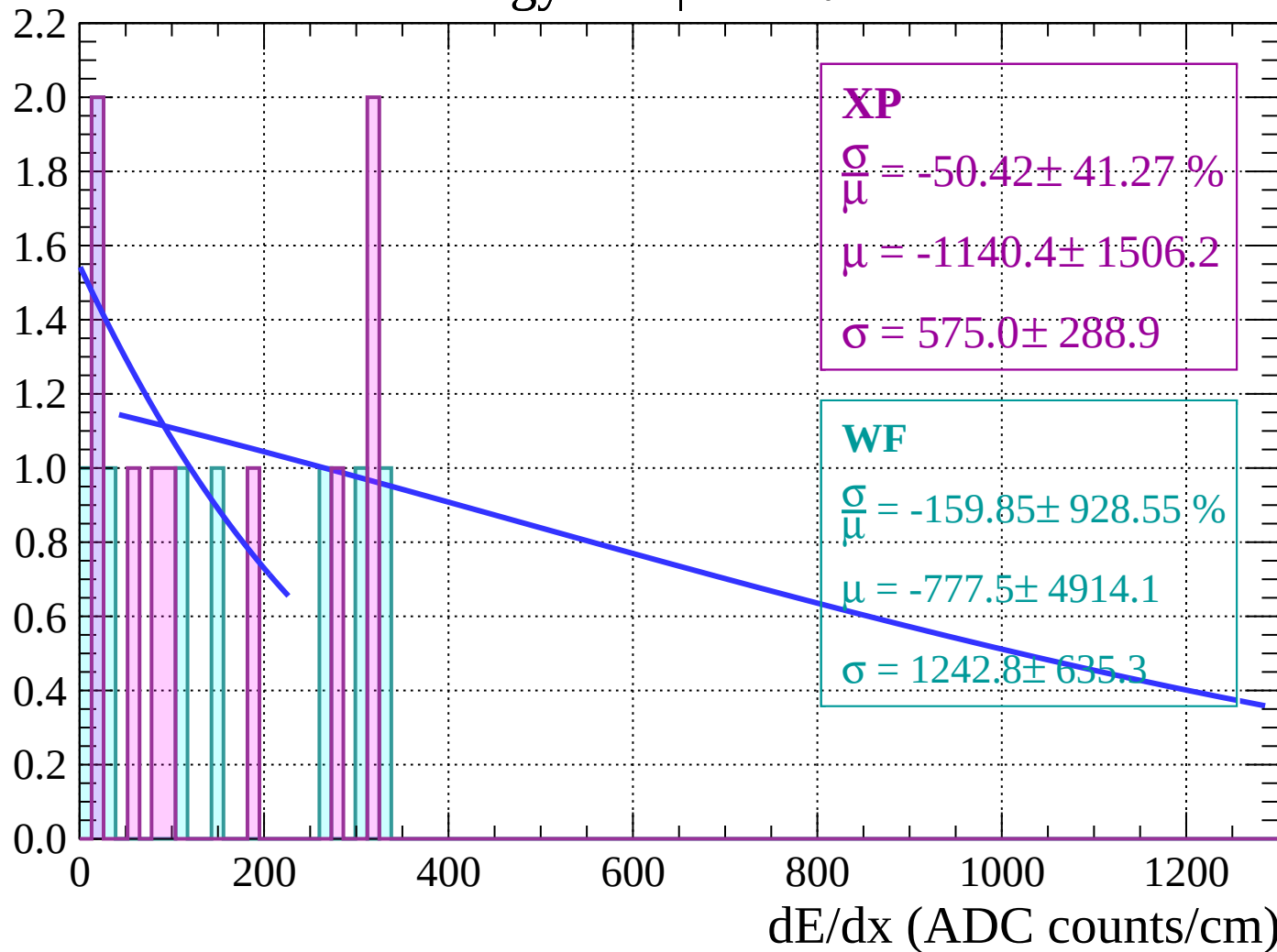
Energy loss | $74 < \theta < 76$

Count



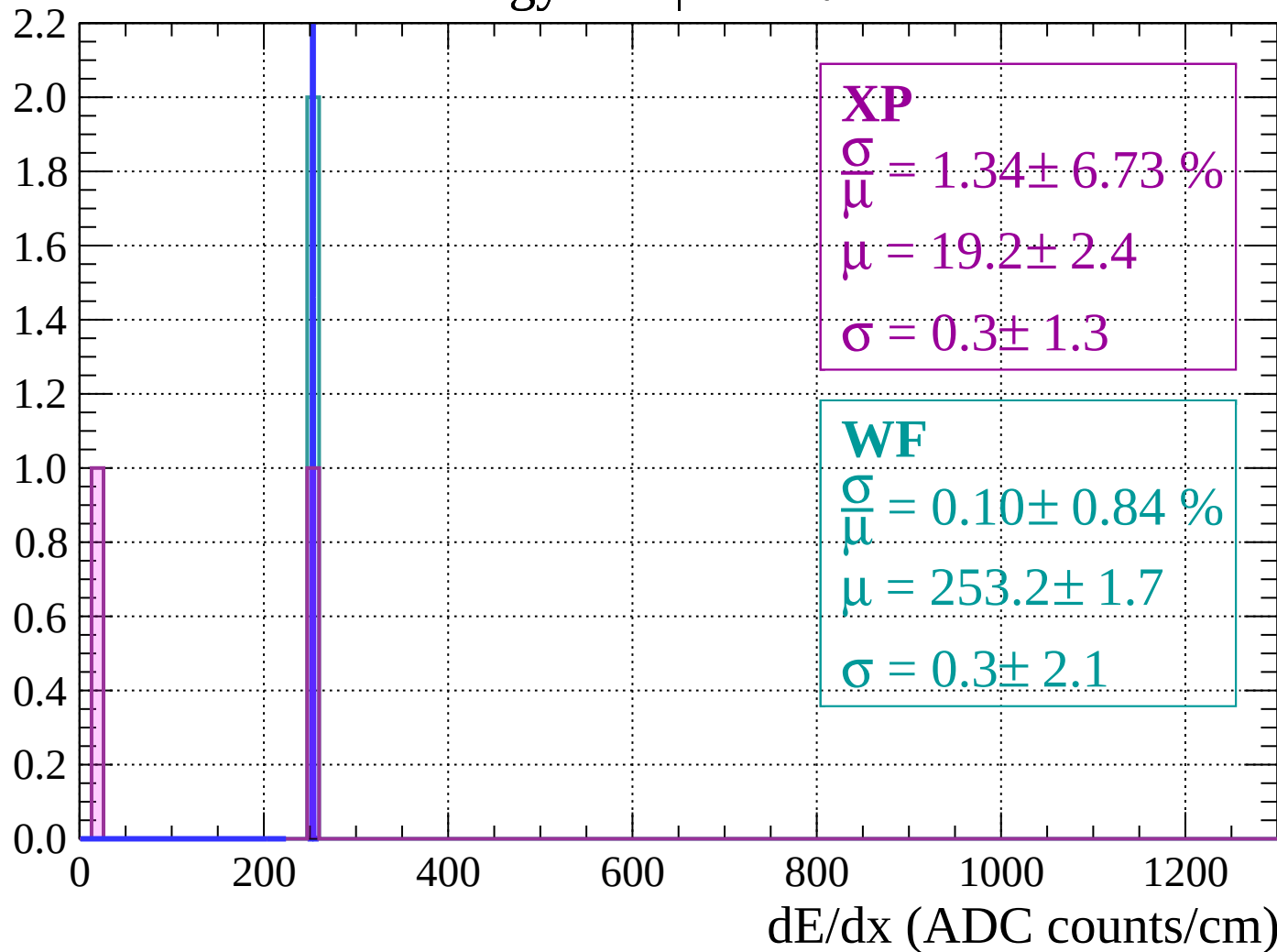
Energy loss | $76 < \theta < 78$

Count



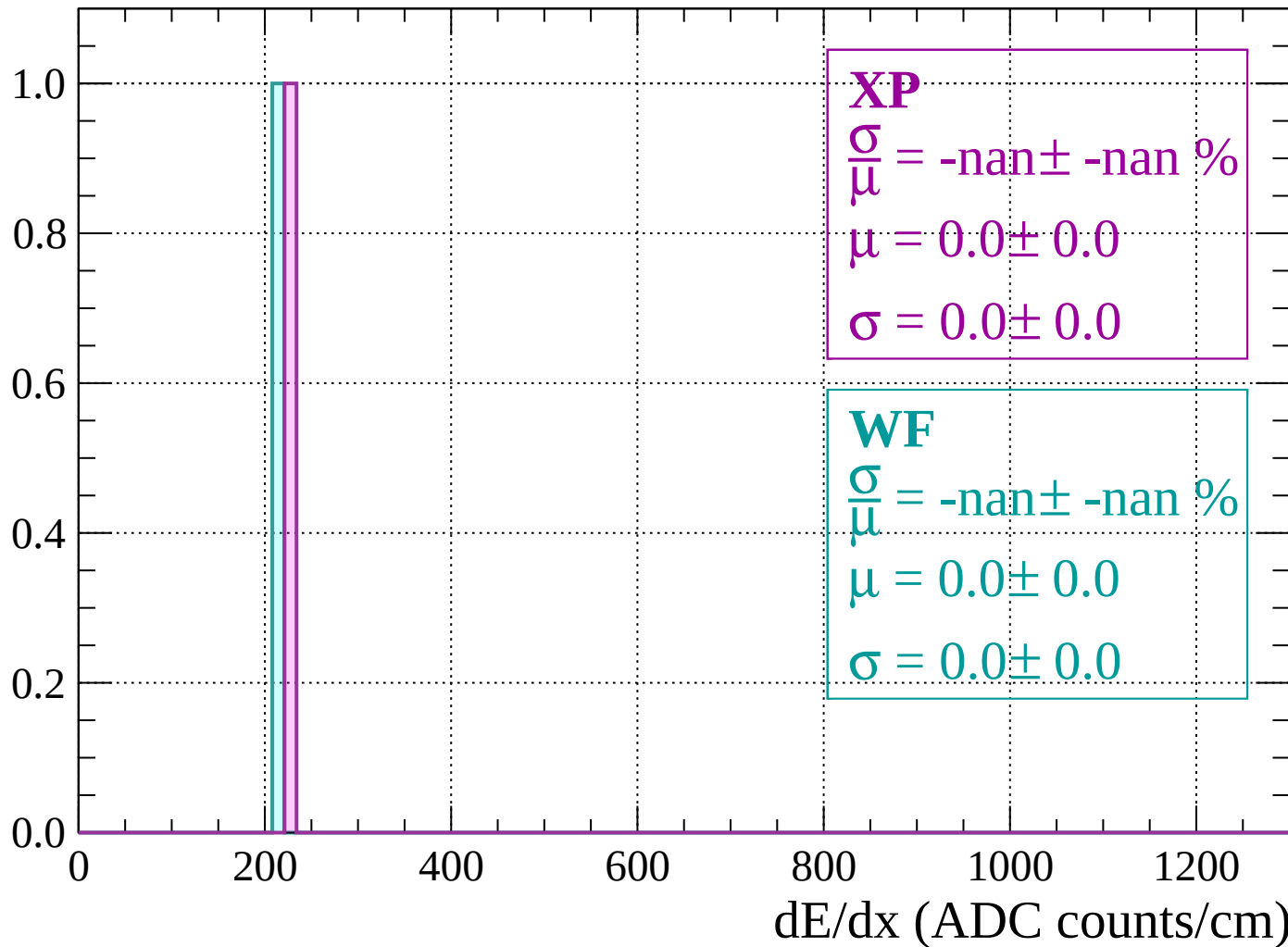
Energy loss | $78 < \theta < 80$

Count



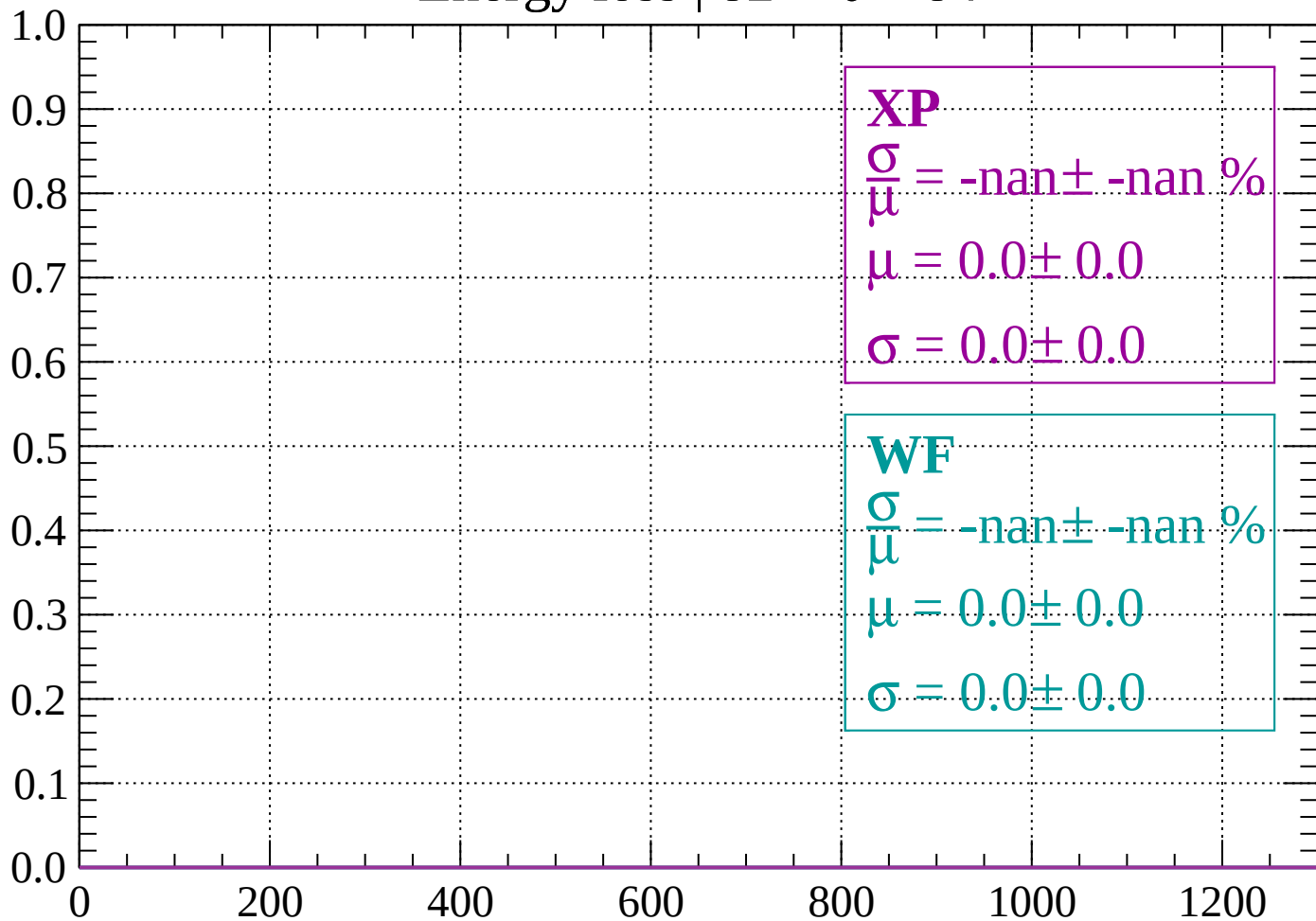
Energy loss | $80 < \theta < 82$

Count



Energy loss | $82 < \theta < 84$

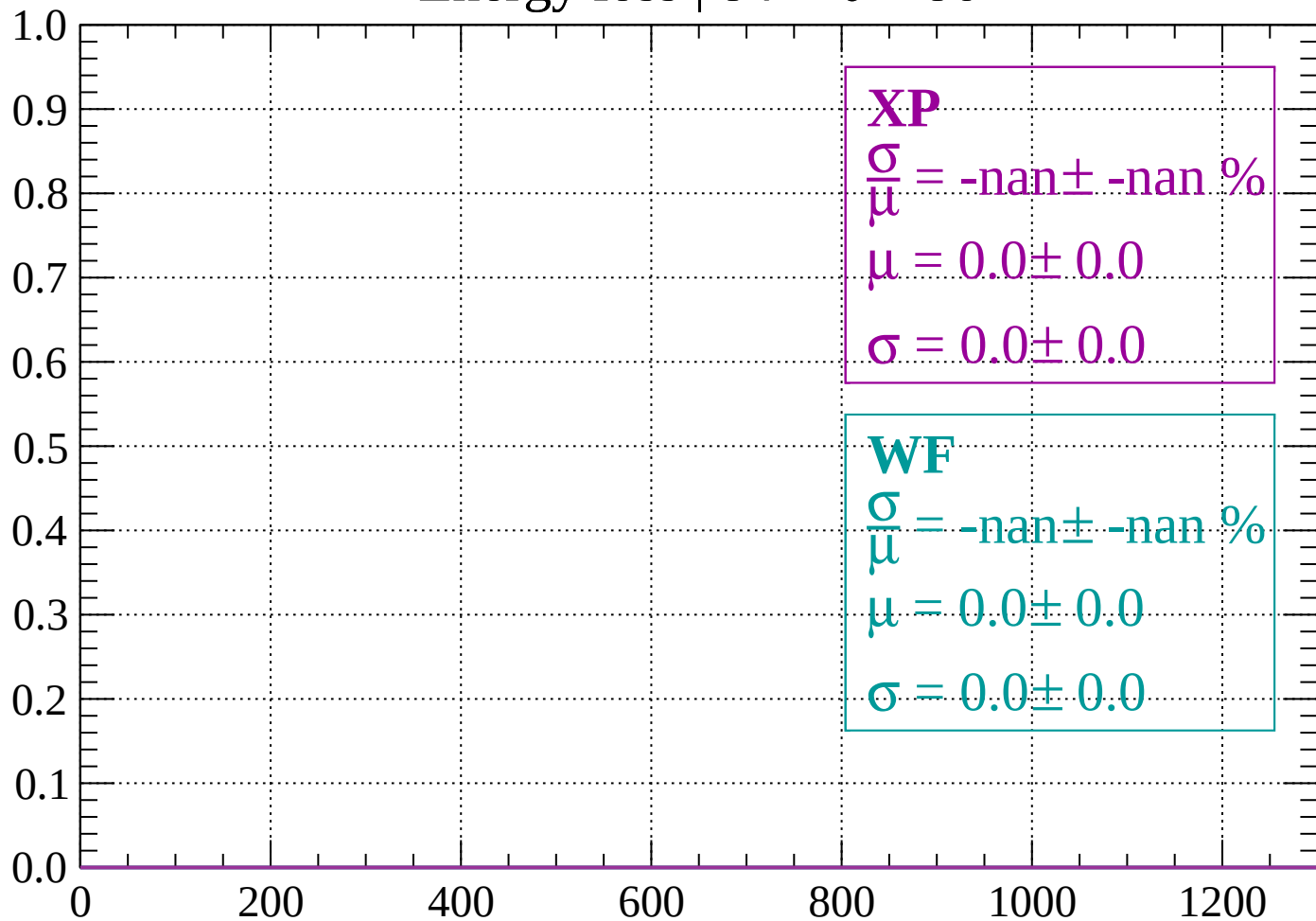
Count



dE/dx (ADC counts/cm)

Energy loss | $84 < \theta < 86$

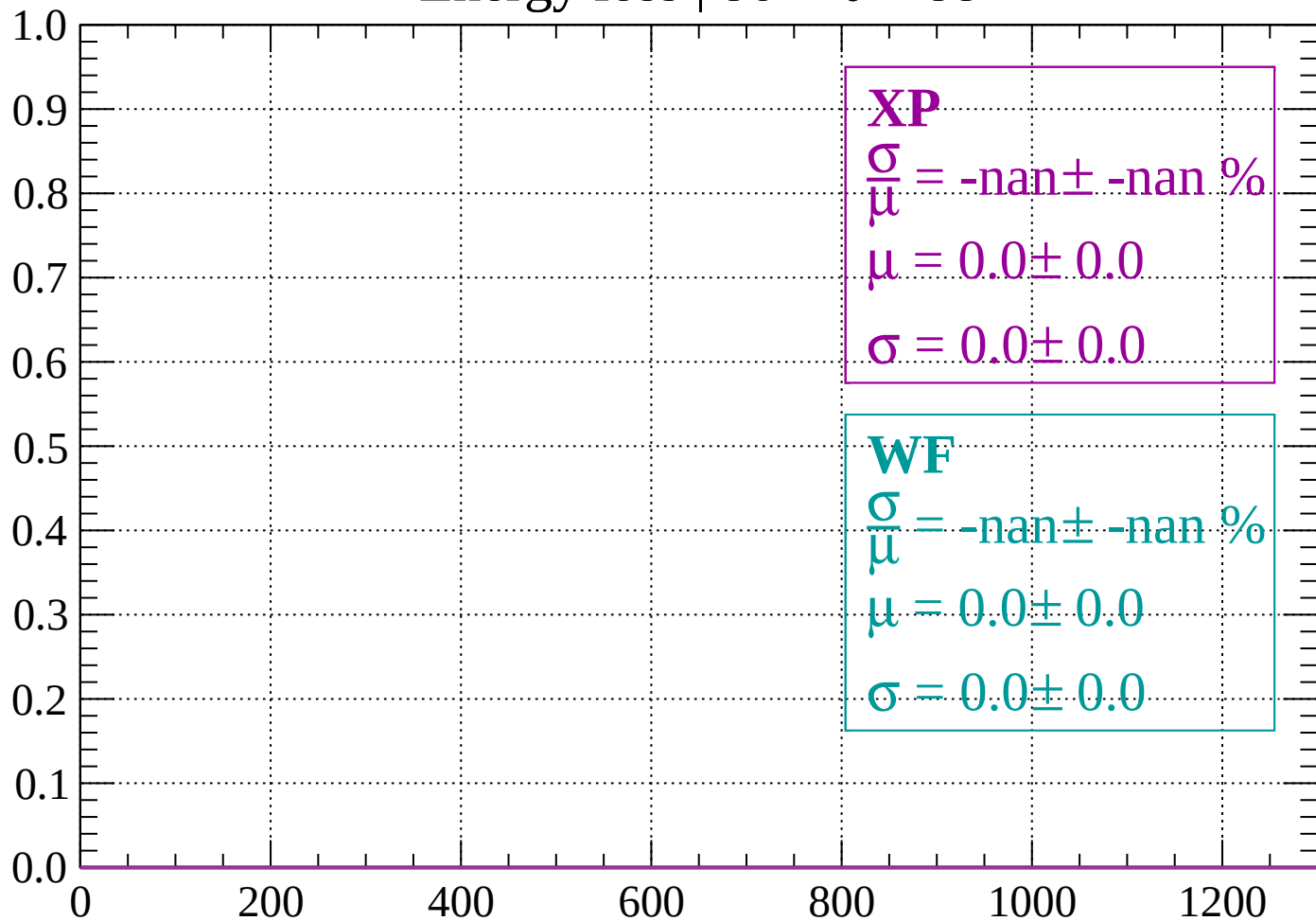
Count



dE/dx (ADC counts/cm)

Energy loss | $86 < \theta < 88$

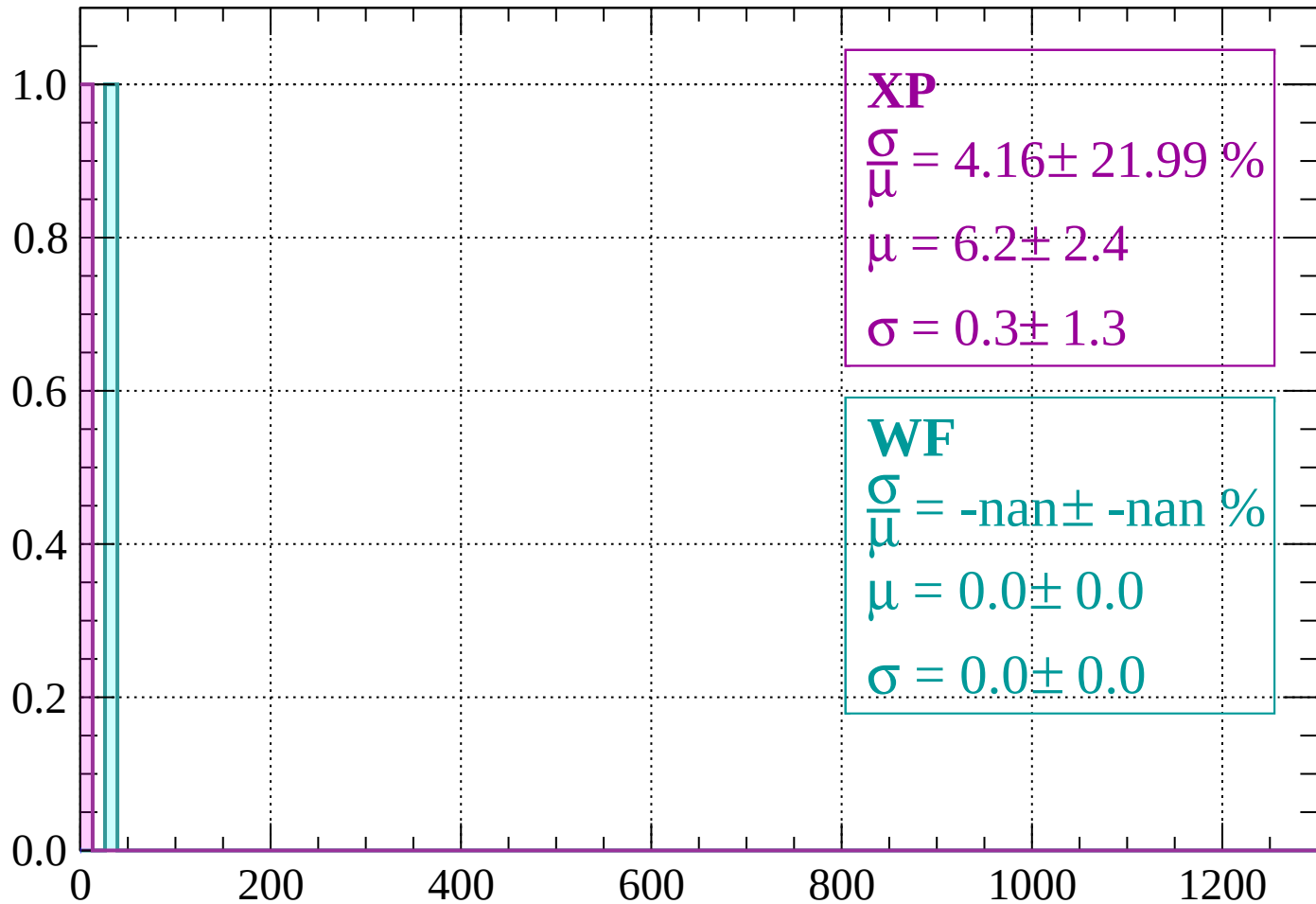
Count



dE/dx (ADC counts/cm)

Energy loss | $88 < \theta < 90$

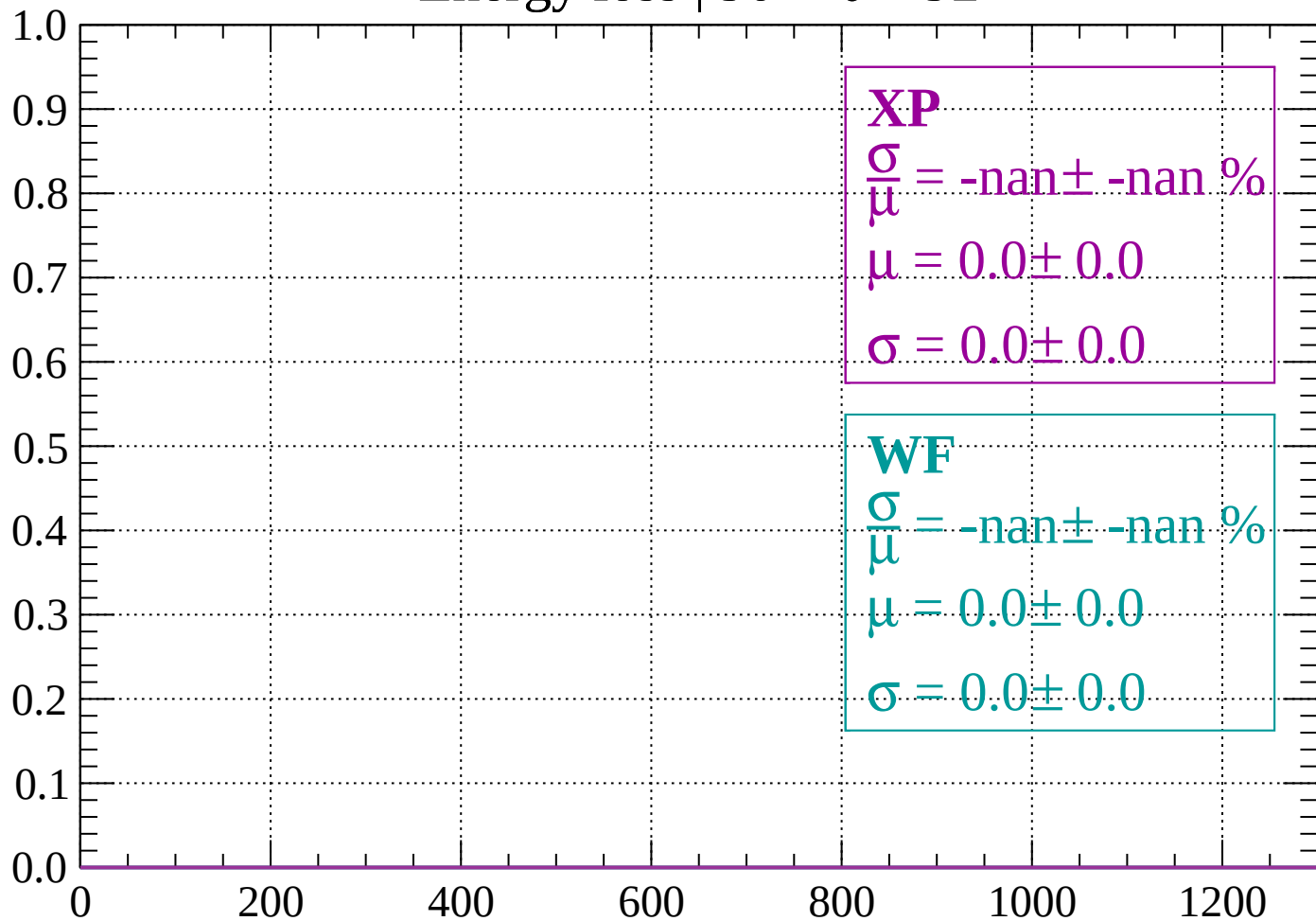
Count



dE/dx (ADC counts/cm)

Energy loss | $90 < \theta < 92$

Count



dE/dx (ADC counts/cm)

